DE GIORGI SYMMETRY FOR MATCHED p-LAPLACE TRANSITIONS WITH $p > 2$ IN DIMENSIONS EIGHT

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ABSTRACT. Fix $p > 2$. We study the matched equation $-\Delta_p u = u(1-u^2)^{p-1}$. We prove that every scalar-stable entire weak solution in $\mathbb{R}^7$ with values in $[-1,1]$ is a constant well ± 1 or a planar transition $\tanh(a_p(\omega \cdot x - t_0))$. Here $a_p = [2(p-1)]^{-1/p}$, $|\omega| = 1$, and $t_0 \in \mathbb{R}$. An endpoint-profile theorem of Farina and Valdinoci then gives the same planar form for entire weak solutions $u : \mathbb{R}^N \to (-1,1)$ with $N \leq 8$ and $\partial_{x_N} u > 0$. In this case, $\omega_N > 0$. No energy-growth bound or directional limits are assumed. Weighted Jacobi tests handle the degeneracy at critical gradients. Sharp propagation of the phase discrepancy controls errors in normal tubes. A mean-curvature estimate retains a negative curvature term. In regions of small discrepancy, stability tests adapted to the transition profile transfer ambient stability to regular zero sheets. They retain half the sum of a normalized normal integral of the discrepancy and the squared sheet curvature. The losses can be made arbitrarily small, and the cutoff errors are controlled. These estimates give quantitative layer preparation and classify solutions with finite normalized energy density. Penalized reach estimates and fourth-moment bounds then allow a common radial stability test to exclude unbounded energy density. Constants and preparation scales are uniform when p ranges over a fixed compact subinterval of $(2, \infty)$.

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1. Introduction

1.1. **Main result.** Our aim is to treat the full range $p > 1$. The present paper proves the classification and symmetry theorem for $p > 2$. Some auxiliary results hold on the larger ranges stated below. Our companion preprint proves a symmetry result for $p < 2$ sufficiently close to 2, using estimates uniform near 2.

We consider the energy

$$E_p(u;\Omega) = \int_\Omega \left(\frac{|\nabla u|^p}{p} + W_p(u) \right) dx, \qquad W_p(t) = \frac{|1-t^2|^p}{2p}, \qquad p > 1. \tag{1.1}$$

Let $u \in W^{1,p}_{\mathrm{loc}}(\mathbb{R}^n)$ with $|u| \le 1$. The energy is finite on each bounded open set Ω. For $\phi \in C^1_c(\Omega)$, the first variation is

$$\frac{d}{dt} E_p(u + t\phi;\Omega)\Big|_{t=0} = \int_\Omega \left(|\nabla u|^{p-2} \nabla u \cdot \nabla \phi + W_p'(u)\phi \right) dx.$$

Set $K = \operatorname{supp} \phi$. For $|t| \le 1$, the gradient part of the integrand has derivative bounded in absolute value by

$$|\nabla u + t\nabla\phi|^{p-1} |\nabla \phi| \le C_p (|\nabla u|^{p-1} + |\nabla \phi|^{p-1}) |\nabla \phi|.$$

This bound is integrable on K by Hölder's inequality. Also, $u + t\phi$ lies in the fixed compact interval $[-1 - \|\phi\|_\infty, 1 + \|\phi\|_\infty]$. Since $W_p \in C^1(\mathbb{R})$ for every $p > 1$, $|W_p'(u + t\phi)\phi| \le C_{p,\phi} |\phi|$ on K. This is another integrable bound independent of t. The fundamental theorem of calculus gives the same

bounds for the difference quotients. Dominated convergence therefore proves the first-variation formula. The variations need not stay in $[-1, 1]$; the potential is defined on all of $\mathbb{R}$. On $[-1, 1]$, one has $W_p'(t) = -t(1 - t^2)^{p-1}$, with value zero at the wells. We call u an entire weak solution when this first variation vanishes for every compactly supported test. This gives the weak form of

$$-\Delta_p u = u(1 - u^2)^{p-1}, \qquad \Delta_p u = \operatorname{div}(|\nabla u|^{p-2}\nabla u). \tag{1.2}$$

The potential vanishes to order p at both wells, matching the gradient exponent. We use the increasing one-dimensional profile

$$g_p(s) = \tanh(a_p s), \qquad a_p = [2(p - 1)]^{-1/p}. \tag{1.3}$$

Indeed,

$$g_p' = a_p(1 - g_p^2), \qquad 2(p - 1)a_p^p = 1.$$

These identities give its equation and the first integral $(p-1)|g_p'|^p/p = W_p(g_p)$. It is normalized by $g_p(0) = 0$ and tends to ± 1 as $s \to \pm\infty$. Both the phase estimates and the interaction law use this matching.

The scalar stability framework used in the global proof is for $p > 2$. For $\xi \neq 0$, set

$$A_p(\xi) = |\xi|^{p-2}I + (p - 2)|\xi|^{p-4}\xi \otimes \xi, \qquad A_p(0) = 0.$$

Whenever $p = 2$ is allowed below, use $A_2 = I$. For $1 < p < 2$, use the formula only at nonzero gradients, unless a singular weighted form domain is specified. For $p > 2$, an entire weak solution of (1.2) is *scalar-stable* if

$$Q_u(\phi) = \int_{\mathbb{R}^n} \left(A_p(\nabla u)\nabla\phi \cdot \nabla\phi + W_p''(u)\phi^2 \right) dx \geq 0 \qquad (\phi \in C_c^1(\mathbb{R}^n)). \tag{1.4}$$

We use the local gradient regularity theorem of DiBenedetto [14] and Tolksdorf [42, Theorem 1, p. 127]. For each fixed $p > 1$, a locally bounded weak solution $v \in W^{1,p}_{\mathrm{loc}}(O)$ of $-\Delta_p v = F$, with $F \in L^\infty_{\mathrm{loc}}(O)$, has a representative in $C^{1,\eta}_{\mathrm{loc}}(O)$ for some $\eta > 0$. We check these hypotheses on each ball $B_{2r} \Subset \mathbb{R}^n$. Our solution belongs to $W^{1,p}(B_{2r})$, has $|u| \leq 1$, and satisfies the weak equation for every smooth compact test. Its right-hand side is the fixed measurable function $F(x) = u(x)(1-u(x)^2)^{p-1}$, with $|F| \leq 1$. The principal flux $a(\xi) = |\xi|^{p-2}\xi$ is continuous everywhere and C^1 away from zero. For $\xi \neq 0$,

$$\min\{1, p - 1\}|\xi|^{p-2}I \leq Da(\xi) \leq \max\{1, p - 1\}|\xi|^{p-2}I.$$

It is independent of x and the solution value. These are the p-growth and ellipticity conditions of the cited local theorem; the frozen source satisfies its bounded-source condition. Hence $u \in C^{1,\eta}(B_r)$ after choosing this representative. In particular, ∇u is bounded on each fixed compact set. This argument is for a fixed p and asserts no uniform exponent here. The matrix A_p is the Hessian of $\xi \mapsto |\xi|^p/p$. For $p > 2$, it extends continuously to zero with value zero. For the potential, the corresponding coefficient is

$$W_p''(t) = (1 - t^2)^{p-2}((2p - 1)t^2 - 1) \quad (-1 < t < 1), \qquad W_p''(\pm 1) = 0.$$

Both energy densities are C^2 in this range of exponents. Let $K = \operatorname{supp}\phi$ and let a bounded open set Ω contain K. The preceding local estimate bounds ∇u on this fixed compact set. For $|t| \leq 1$, the absolute value of the second derivative of the gradient energy density is at most $C_p(|\nabla u| + |\nabla\phi|)^{p-2}|\nabla\phi|^2$. This is bounded and supported in K. The potential second derivative is at most $C_{p,\phi}\phi^2$, since W_p'' is bounded on the compact range of $u + t\phi$. Both bounds are integrable. Thus, for $p > 2$, dominated convergence justifies two derivatives under the integral and gives

$$\frac{d^2}{dt^2} E_p(u + t\phi; \Omega)\bigg|_{t=0} = Q_u(\phi).$$

Thus (1.4) is the usual scalar second-variation condition.

Theorem 1.1 (Stable classification and monotone symmetry)**.** *Let $p > 2$. Every scalar-stable entire weak solution $v : \mathbb{R}^7 \to [-1, 1]$ of (1.2) is a constant well $v \equiv \pm 1$ or a planar transition*

$$v(x) = \tanh(a_p(\omega \cdot x - t_0)), \qquad |\omega| = 1, \quad t_0 \in \mathbb{R}.$$

Consequently, if $N \le 8$ and an entire weak solution $u : \mathbb{R}^N \to (-1, 1)$ of (1.2) satisfies $\partial_{x_N} u > 0$, then

$$u(x) = \tanh(a_p(\omega \cdot x - t_0)), \qquad |\omega| = 1, \quad \omega_N > 0. \tag{1.5}$$

The result holds for each fixed finite $p > 2$. The uniform estimates stated below concern the normalized equation (1.2), with p in a fixed compact interval $\mathcal{P} \Subset (2, \infty)$. Their constants and preparation scales may depend on $\mathcal{P}$ and on the fixed tolerances specified in each statement. This convention does not assert uniformity as $p \downarrow 2$ or $p \to \infty$.

For the potential $C|1 - t^2|^p$, fix $C > 0$ and set $\lambda = (2pC)^{1/p}$. If v is a weak solution of (1.2), define $u(x) = v(\lambda x)$. Dilation preserves $W^{1,p}_{\mathrm{loc}}$, and $\nabla u(x) = \lambda \nabla v(\lambda x)$ almost everywhere. For a compactly supported C^1 test ϕ, put $\chi(y) = \phi(y/\lambda)$. Changing variables gives

$$\begin{aligned} &\int \left\{ |\nabla u|^{p-2} \nabla u \cdot \nabla \phi + \lambda^p W_p'(u)\phi \right\} dx \\ &\qquad = \lambda^{p-n} \int \left\{ |\nabla v|^{p-2} \nabla v \cdot \nabla \chi + W_p'(v)\chi \right\} dy = 0. \end{aligned}$$

Here the Sobolev chain rule applies to the invertible linear map $x \mapsto \lambda x$. The same computation with its inverse proves the converse. Since $\lambda^p = 2pC$, the rescaled equation is

$$-\Delta_p u = 2pC\, u(1 - u^2)^{p-1}.$$

Let $Q_{C,u}$ denote the second-variation form for this potential. Set $y = \lambda x$ and $\chi(y) = \phi(y/\lambda)$. The identities $A_p(\lambda\xi) = \lambda^{p-2} A_p(\xi)$ and $C|1 - t^2|^p = \lambda^p W_p(t)$ give

$$Q_{C,u}(\phi) = \lambda^{p-n} Q_v(\phi(\cdot/\lambda)).$$

The factor is positive, so stability is preserved in both directions. Also $\partial_{x_i} u(x) = \lambda \partial_{y_i} v(\lambda x)$, so strict monotonicity in that direction is preserved in both directions. For fixed $C > 0$, the continuous positive function $p \mapsto (2pC)^{1/p}$ has a positive minimum and a finite maximum on $\mathcal{P}$. Thus every estimate already uniform for the normalized equation transfers under this dilation, with its spatial scales divided by λ and its derivatives multiplied by the appropriate power of λ. The resulting constants may also depend on this fixed C. Uniformity over varying C requires a fixed compact interval in $(0, \infty)$ for C. Under the same dimension and stability or monotonicity hypotheses as Theorem 1.1, the classified nonconstant profiles become

$$u(x) = \tanh\left[\left(\frac{pC}{p-1} \right)^{1/p} (\omega \cdot x - t_0) \right].$$

1.2. **Related work.** For the classical equation $-\Delta u = u - u^3$, De Giorgi's conjecture concerns entire C^2 solutions $u : \mathbb{R}^N \to (-1, 1)$. It asks whether strict monotonicity in one direction forces one-dimensionality when $N \le 8$. No directional limits are assumed. Ghoussoub–Gui proved the two-dimensional case [22]. Ambrosio–Cabré proved the three-dimensional case [3]. For $N \ge 3$, Barlow–Bass–Gui proved one-dimensional symmetry under uniform directional well limits [4, Theorem 0.1]. Here uniformity is with respect to all transverse variables. Ghoussoub–Gui also obtained symmetry in dimensions four and five under additional assumptions [23, Theorem 1.2]. Their theorem assumes strict monotonicity and pointwise well limits. It also requires the antisymmetry $u(y, z) = -u(y, -z)$ after a rigid change of coordinates. Here $(y, z) \in \mathbb{R}^{N-k} \times \mathbb{R}^k$ for some $1 \le k \le N$. Gui also studied uniqueness of Allen–Cahn minimizers and nonexistence of monotone stationary solutions when the two wells have different depths [25].

Savin used minimizer regularity to obtain symmetry through dimension eight under the additional pointwise limits $u(x', t) \to \pm 1$ as $t \to \pm\infty$ [37]. Wang gave a new proof of Savin's classification of entire minimizers in dimensions at most seven, using tilt-excess decay and harmonic approximation [44, Theorem 1.1]. Del Pino–Kowalczyk–Wei constructed nonplanar strictly monotone solutions in every dimension $N \geq 9$ [13]. Thus the dimension bound is sharp. Pacard–Wei constructed bounded stable entire solutions with nonplanar zero sets already in dimension eight [36]. Liu–Wang–Wei constructed nonplanar global minimizers in every dimension $N \geq 8$ [31].

Wang–Wei proved that finite Morse index solutions $u : \mathbb{R}^2 \to (-1, 1)$ of the classical Allen–Cahn equation have finitely many ends and linear energy growth [46, Theorem 2.2]. Their proof obtains curvature decay through a Toda-system analysis of clustering interfaces. They also proved uniform $C^{2,\alpha}$ estimates for interior transition level sets of the singularly perturbed Allen–Cahn equation in dimensions at most ten [47]. Here $0 < \alpha < 1$ is fixed. The solutions must be stable, and the perturbation parameter must be sufficiently small. These estimates assume a nonzero gradient and uniform bounds for both the level-set curvature and the tangential derivative of $\log |\nabla u_\varepsilon|$ in a fixed central transition region. The layer analysis below develops weighted estimates for the degenerate p-equation.

Two September 2026 preprints classify bounded stable entire classical solutions in $\mathbb{R}^3$ and deduce monotone symmetry in $\mathbb{R}^4$. They are due to Liu–Luo–Wang–J. Wei–Y. Wei–Wu [29, Theorems 1.1–1.2] and Chan–Fernández-Real–Figalli–Florit-Simon–Serra [10, Theorem 1.1 and Corollary 1.2]. An October 2026 preprint by Liu and coauthors proves one-dimensionality through dimension seven for entire C^2 solutions [30, Theorem 1.2]. It assumes stability, values in $(-1, 1)$, and an $O(R^{N-1})$ energy-growth bound.

OpenAI's manuscript of September 26, 2026 presents a proposed proof of the classical conjecture in dimension eight [35]. Its Theorem 1.2 classifies stable C^2 solutions $v : \mathbb{R}^7 \to [-1, 1]$ as constant wells or planar heteroclinics. It assumes no energy-growth bound. Its Theorem 1.1 treats C^2 solutions $u : \mathbb{R}^8 \to (-1, 1)$ with $\partial_8 u > 0$. No directional limits or minimizing hypothesis are imposed. The work is available as an OpenAI Math Release preprint. The repository identifies its manuscripts as outputs of an unnamed internal model. They are at different stages of verification. The release is described as raw research data rather than polished papers. The proof follows the geometric strategy in [35, Sections 5–10]. We adapt its small-discrepancy limits, mean-curvature and mass comparisons, finite-density argument, penalized reach, fourth-moment estimates, and common radial test. The corresponding constructions are credited again where they enter the proof. The arguments below retain the weights and coefficients of the degenerate p-equation. We derive the required weighted estimates and keep their dependence on the exponent explicit.

The fractional analogue is

$$(-\Delta)^s u = u - u^3 \quad \text{in } \mathbb{R}^N, \qquad 0 < s < 1. \tag{1.6}$$

We use the Fourier symbol $|\xi|^{2s}$. The symmetry question concerns bounded entire classical solutions with $\partial_{x_N} u > 0$. Its answer depends on both N and s. Gui–Li obtained BV and energy estimates for stable solutions in every dimension [26, Proposition 1.7]. They used these estimates to give another proof of one-dimensionality for stable entire solutions in $\mathbb{R}^2$ for every $0 < s < 1$ [26, Theorem 3.7]. In dimension three, Cabré–Cinti proved symmetry for $1/2 \leq s < 1$ [6, Theorem 1.5]. Dipierro–Farina–Valdinoci treated $0 < s < 1/2$ [15, Theorem 1.1]. Neither result prescribes directional limits. Figalli–Serra classified bounded stable entire solutions in $\mathbb{R}^3$ at $s = 1/2$. They thereby obtained the monotone result in dimension four [19, Theorem 1.5 and Corollary 1.6]. With pointwise directional well limits ± 1, Savin obtained symmetry through dimension eight for $1/2 \leq s < 1$ [38, 39, Theorem 1.3]. Cabré–Cinti–Serra obtained the four-dimensional monotone conclusion for $s < 1/2$ sufficiently close to $1/2$ [7, Corollary 1.11]. Their operator order and nonlocal perimeter order both equal $2s$ in our notation.

The Fractional De Giorgi Conjecture Results repository contains five further manuscripts. As of November 1, 2026, these texts remain candidate proofs under verification. We cite the versions consulted as repository drafts. The two dimension-four manuscripts concern bounded entire classical solutions with $\partial_4 u > 0$ and no prescribed directional limits. The below-half draft treats every $0 < s < 1/2$ [52, Theorem 1.2]. Its cone theorem states halfspace rigidity for nontrivial stationary stable nonlocal perimeter cones in $\mathbb{R}^3$, with perimeter order $2s$ [52, Theorem 1.3]. The boundary must be smooth and locally one-sided away from the vertex. The spherical link must be compact and smooth, but it may be disconnected. Stability is tested by compact variations away from the vertex. For $1/2 < s < 1$, the above-half draft first states classification of bounded stable entire classical solutions in $\mathbb{R}^3$ as wells or planar layers. It then deduces the dimension-four monotone conclusion [49, Theorems 1.1–1.2].

Two further drafts treat dimension eight near one and one half. The near-one manuscript states symmetry for bounded entire solutions with $\partial_8 u > 0$ when $s_0 < s < 1$, for some $s_0 \in (1/2, 1)$ [50, Theorem 1.1]. Its theorem is explicitly conditional on two classical conclusions: stable classification in $\mathbb{R}^7$ from [35, Theorem 1.2], and the stable Bernstein result in the preprint of Hong–Li–Wang [27, Theorem 1.1]. The latter concerns smooth, complete, connected, boundaryless, two-sided stable minimal immersions $M^6 \to \mathbb{R}^7$. It imposes no properness or area-growth assumption. The fractional statement assumes no energy-growth bound, prescribed end states, or bound on the number of transition components.

The near-half manuscript treats bounded entire classical solutions in $\mathbb{R}^8$ with $\partial_8 u > 0$ when $s_0 < s < 1/2$, for some $s_0 \in (0, 1/2)$ [48, Theorem 1.1]. It prescribes no directional limits. That threshold depends only on the dimension and the fixed cubic potential. Neither manuscript gives a numerical threshold.

The fifth draft treats the full interval $1/2 < s < 1$ under a growth hypothesis [51, Theorem 1.1]. For each fixed s, it assumes a bound for every bounded stable entire classical solution w of (1.6) in each $\mathbb{R}^n$, $1 \le n \le 7$:

$$\int_{B_R(z)} |\nabla w|\, dx \le C_w R^{n-1} \qquad (z \in \mathbb{R}^n,\ R \ge 1).$$

The finite constant C_w may depend on w and s. It is independent of the center z and radius R. Under this hypothesis, the draft states symmetry for every bounded entire classical strictly monotone solution in dimensions $N \le 8$. It imposes no directional limits on that solution. The growth hypothesis concerns the full stable class in the lower dimensions.

These fractional results and drafts concern the cubic nonlocal equation (1.6). Our equation is local, with a degenerate gradient operator. We develop its estimates directly for the matched potential (1.1).

Valdinoci–Sciunzi–Savin established flat-level-set regularity for p-Laplace minimizers [43]. Their Theorem 1.3 classifies Class A minimizers in dimensions at most seven. Their Theorem 1.4 gives monotone symmetry through dimension eight with pointwise directional limits -1 and 1. The results allow every $p > 1$ and apply to (1.1). For the stable problem, we use their weak-solution contact barriers. Under a unit ball-energy cap, we replace the minimizing energy input and prove contact projection and flatness improvement. These estimates give the unit-density rigidity theorem in Section 5.

Farina–Sciunzi–Valdinoci developed geometric Poincaré inequalities for singular and degenerate equations [16, 17]. For $p > 2$, the derivative spaces in the later paper use the weight $|\nabla u|^{p-2}$, which vanishes at critical gradients. Their calculus and the rigidity results of Cozzi–Farina–Valdinoci [11] provide the analytic setting. The final passage to monotone symmetry uses the endpoint reduction of Farina–Valdinoci [18, Theorem 1.1]. The theorem holds for every $p > 1$ under its stated double-well hypotheses. Strict monotonicity and two endpoint profiles that are at most two-dimensional yield the well limits and global minimality. For $N \le 8$, the solution is also one-dimensional.

To apply this reduction, first add passive coordinates to work in $\mathbb{R}^8$ when needed. Strict monotonicity keeps the gradient nonzero, so local elliptic regularity makes u smooth. The positive Jacobi field u_{x_8} then gives scalar stability; see [16, Lemma 7.1, p. 785]. Bounded monotonicity gives the directional limits $v_\pm(x') = \lim_{t\to\pm\infty} u(x', t)$ on $\mathbb{R}^7$. Local C^1 compactness of translates passes the equation and stability to the limits in $\mathbb{R}^8$. Here $p > 2$ ensures continuity of both A_p at critical gradients and W_p'' at the wells. The limits are independent of x_8. Long vertical cutoffs then give stability on $\mathbb{R}^7$; the full argument is in Section 8. The stable part of Theorem 1.1 then makes each endpoint a well or a planar layer. The normals of the nonconstant profiles span a subspace of dimension at most two. Thus both profiles depend on at most two common orthogonal coordinates. The same final argument checks all potential hypotheses of [18, Theorem 1.1] before applying it. This explains why stable classification in one lower dimension is the main task.

1.3. **Outline of the proof.** We first separate the constant wells. For $p > 2$, the constant zero is unstable, since $Q_0(\phi) = -\int \phi^2 < 0$ for every nonzero test. For the stable classification, take a nonconstant scalar-stable solution $u : \mathbb{R}^7 \to [-1, 1]$. The gradient bound and well rigidity give $-1 < u < 1$; see Proposition 2.1 and Remark 2.2. The phase is therefore well defined. Lemma 2.3 also gives a nonempty zero set. Set $c = p - 1$, $a = a_p$, and $\beta = pa$, and write

$$q = 1 - u^2, \qquad U = a^{-1} \operatorname{arctanh} u, \qquad Z = \nabla U, \qquad D = 1 - |Z|^p.$$

The gradient bound gives $D \geq 0$. For a planar transition, $D = 0$. The weighted Jacobi calculus makes the derivative tests admissible even where the original gradient operator degenerates. This calculus transfers ambient stability to estimates on regular zero sheets.

Only regular parts of the zero set are used in the surface estimates. The bound $D \leq \delta < 1$ gives $|Z| \geq (1 - \delta)^{1/p} > 0$ there. On a regular zero sheet Σ, choose the phase normal $\nu = Z/|Z|$ and use $B = d\nu|_{T\Sigma}$ and $H = \operatorname{tr} B$. Let I_P be the normalized normal mass in (2.1), with window $[-P, P]$. In a regular normal tube, define the field e by

$$e(z + s\nu(z)) = \nu(z), \qquad u_e(z + s\nu(z)) = \nabla u(z + s\nu(z)) \cdot \nu(z).$$

Thus the normal varies with the sheet point and is constant along each normal ray. The physical branch is $h = (q + a^{-1}u_e)/2$. Its target surface mass is

$$S = \frac{I_P + |B|^2}{2}. \tag{1.7}$$

For small zero discrepancy and fixed P, the core calculation retains

$$\frac{1}{2} I_P + \frac{J_{p,P}}{2J_p} |B|^2 + o_P(D(z)), \qquad J_{p,P} = \int_{-P}^{P} \operatorname{sech}^{2p}(a_p s)\, ds, \qquad J_p = \int_{\mathbb{R}} \operatorname{sech}^{2p}(a_p s)\, ds.$$

Here the remainder is taken as the zero discrepancy tends to zero at fixed P. The finite-window fraction tends to one as P increases. The full estimate also includes controlled cap errors. More precisely, for $0 < \eta < 1/10$ and $Q > 2$, Proposition 3.13 chooses P, A_0, C, δ_0 uniformly on each compact exponent interval $\mathcal{P}$. For a compactly supported test $\psi \in C_c^1(\Sigma)$, if $0 < \delta \leq \delta_0$ and every zero in the ambient $10A_0 \log(1/\delta)$-neighborhood of $\operatorname{supp} \psi$ has discrepancy at most δ, then

$$(1 - \eta) \int_\Sigma S\psi^2 \leq (1 + \eta) \int_\Sigma |\nabla_\Sigma \psi|^2 + C\delta^Q \int_\Sigma \psi^2 \qquad (\psi \in C_c^1(\Sigma)).$$

For every $\varepsilon_I > 0$, Corollary 3.5 also chooses a sufficiently large finite P, then a sufficiently small discrepancy threshold, so that

$$I_P \geq (1 - \varepsilon_I)|B|^2$$

at zeros below that threshold. Use the prescribed lower bound in Proposition 3.13 to choose its window above the mass-comparison window. Positivity of the normal-mass integrand preserves the latter lower bound. Then decrease the common discrepancy threshold. Together these estimates recover the minimal-surface stability coefficient one with arbitrarily small losses.

Two estimates are needed to carry out this argument for every fixed $p > 2$. For a noncritical base point z, set $e_0 = Z(z)/|Z(z)|$. For every $\varepsilon > 0$, the normal propagation theorem gives

$$D(z + se_0) \le C_\varepsilon D(z) e^{(2\beta+\varepsilon)|s|} \qquad (s \in \mathbb{R}).$$

The same bound holds along unit directions in a sufficiently small cone about e_0; see Theorem 2.20 and Corollary 2.21. The cone width and constants can depend on the rate loss, the fixed dimension, and $\mathcal{P}$. Combining propagation with sharp clearing and the weighted current estimate makes the normal currents integrable on logarithmic tubes; see (3.30) and (3.31).

The second estimate controls mean curvature while retaining a negative curvature term. For every $\varepsilon > 0$, Theorem 3.7 chooses a finite window P and then a small threshold δ, uniformly on $\mathcal{P}$. If a regular zero-sheet patch has $D < \delta$, each compactly supported C^1 test φ on that patch satisfies

$$\beta \int_\Sigma |H|\varphi^2 \le (1/2+\varepsilon) \int_\Sigma I_P \varphi^2 - (1/2-\varepsilon) \int_\Sigma |B|^2 \varphi^2 + C_{\mathcal{P}} \int_\Sigma I_P |\nabla_\Sigma \varphi|^2. \tag{1.8}$$

The constant is independent of the discrepancy threshold. Since $I_P \le C_P D \le C_P \delta$, decreasing δ at fixed P makes the last term an arbitrarily small multiple of the unweighted Dirichlet energy. Its weighted form is retained near the edge of the prepared region; see (3.16). The curvature subtraction remains in the transfer of mean-curvature terms between sheets and supplies the margin in the reach multiplier.

Theorem 5.25 classifies scalar-stable entire weak solutions with finite normalized energy density. In $\mathbb{R}^7$, this means

$$\sup_{R \ge 1} R^{-6} E_p(u; B_R(0)) < \infty.$$

The stress law and weighted Toda stability give quantitative preparation before the finite-density argument is applied. For the gap comparison, on each compact $\mathcal{P} \Subset (2, \infty)$ we choose

$$\frac{1}{2} < \gamma < \min_{p \in \mathcal{P}} \left(\frac{1}{2} + \frac{\sqrt{p-1}}{p} \right), \qquad 1 < \alpha < 2\gamma < 2.$$

The upper bound $\gamma < 1$ follows from $\sqrt{p-1}/p < 1/2$ for $p > 2$. Thus the transform $t + t^{2-\alpha}$ is increasing and strictly concave for $t \ge 1$. Its use in the gap comparison is given in (5.78).

Reach and global physical stability then give the fourth-moment estimate (7.47) without a density bound. A common compact radial amplitude permits the reach, mean-curvature, and physical stability estimates to be combined. The area bound (8.48) supplies enough regular zeros. Jointly injective normal tubes remove the negligible large-reach part of this area; see Remark 6.2. The radial weight has a positive lower bound of the required scale on the remaining area. This gives a lower bound for the same integral as in (8.47), contradicting its radial upper bound. The outer cutoff is removed before the inner radius tends to infinity. Proposition 8.3 therefore excludes unbounded normalized density. The endpoint reduction described above then completes monotone symmetry.

The phase estimates and weighted Jacobi calculus are developed in Section 2. They lead to the mean-curvature and physical surface stability estimates in Section 3. The interaction laws and quantitative preparation in Section 4 support the finite-density classification in Section 5. Reach and global stability are established in Section 6. The curvature moments in Section 7 then support the radial contradiction in Section 8, which completes the proof of Theorem 1.1.

2. Analytic preliminaries

We work on a compact exponent interval $\mathcal{P} \Subset (2, \infty)$ unless another range is stated. The constants can depend on $\mathcal{P}$ and on fixed geometric tolerances. They are independent of the solution and of radii that tend to infinity. The notation below remains in force throughout the paper unless stated otherwise. Set

$$c = p-1, \quad a = a_p, \quad \beta = pa, \quad q = 1-u^2, \quad U = a^{-1} \operatorname{arctanh} u, \quad Z = \nabla U, \quad r = |Z|, \quad D = 1-r^p.$$

The phase variables are used only for solutions with $-1 < u < 1$. Use the same phase variables when the solution is denoted by v. By Proposition 2.1 and Remark 2.2, every nonconstant entire solution with values in $[-1, 1]$ has this strict range. On the regular part of $\Sigma = \{u = 0\}$, define $\nu = Z/r$, $B = d\nu|_{T\Sigma}$, $b = |B|$, and $H = \operatorname{tr} B$. This fixes the sign of H; reversing the orientation leaves $|H|$ unchanged. We also set

$$G_p(s) = \operatorname{sech}^2(a_p s), \qquad J_p = \int_{\mathbb{R}} G_p^p \, ds, \qquad \sigma_q = a_p^{p-2} J_p, \qquad \sigma_E = a_p^p J_p.$$

Choose a fixed normal width P and define

$$I(z) = \sigma_q^{-1} \int_{-P}^{P} q^{p+1} D(z + s\nu(z)) \, ds. \tag{2.1}$$

We also write I_P when the window must be specified. We always choose the normal window before the discrepancy threshold. On that window the frozen-profile mass approaches (2.1) with relative error tending to zero as the zero discrepancy tends to zero. The comparison follows from the relative phase estimates below; see the remark after Corollary 2.13. A planar layer has energy σ_E per unit area. Indeed, the first integral in Section 1 gives $|g_p'|^p/p + W_p(g_p) = |g_p'|^p = a_p^p G_p^p$. Later density arguments denote this constant by σ_p.

2.1. **Gradient bounds and clearing.** For each fixed $p > 1$, local regularity gives a locally Lipschitz representative of every bounded weak solution with bounded source [14, 42]. The uniform constants used below also follow from [28, Theorem 3.2, p. 303]. To check its hypotheses, allow a compact interval $2 \le p_- \le p \le p_+ < \infty$ and put $\mathfrak{g}_p(t) = t^{p-1}$. Then

$$p_- - 1 \le \frac{t\mathfrak{g}_p'(t)}{\mathfrak{g}_p(t)} = p - 1 \le p_+ - 1 \qquad (t > 0).$$

At a nonzero vector ξ, the eigenvalues of $A_p(\xi)$ are $|\xi|^{p-2}$ and $(p-1)|\xi|^{p-2}$. Thus Lieberman's ellipticity bounds hold with lower coefficient one and upper coefficient $p_+ - 1$. The flux is C^1 at zero as well, including when $p = 2$. It is independent of the position and the solution value, so the coefficient-continuity condition has zero left side. The lower-order term is bounded because

$$|W_p'(t)| = |t|(1 - t^2)^{p-1} \le 1 \qquad (-1 \le t \le 1).$$

Extend this term by zero outside $[-1, 1]$ when applying the theorem. One may therefore take its continuity exponents to be one and its source constant to be one. The theorem gives one positive gradient Hölder exponent and one local Hölder bound for the whole interval. This bound does not depend on the initial Lipschitz norm. A difference quotient also gives the full local $C^{1,\alpha}$ bound: if e is a unit vector and $[x, x + he]$ lies in the larger ball, then

$$|\nabla u(x) \cdot e| \le \frac{\operatorname{osc} u}{h} + \frac{[\nabla u]_{C^{0,\alpha}}}{1 + \alpha} h^\alpha.$$

Choose h to be a fixed fraction of the distance to the larger ball's boundary and use $\operatorname{osc} u \le 2$. These estimates are uniform under translation and include the endpoint $p = 2$. After a subsequence, bounded entire solutions with $p_j \to p \in \mathcal{P}$ therefore converge in C^1_{loc}. Joint continuity of $(p, \xi) \mapsto |\xi|^{p-2}\xi$ gives uniform convergence of their fluxes on each compact set. The reactions converge there as well, so the weak equation passes to the limit. For stability, we also need convergence of the differentiated coefficients. The maps $(p, \xi) \mapsto A_p(\xi)$ and $(p, t) \mapsto W_p''(t)$ are jointly continuous on the relevant compact sets when $\mathcal{P} \Subset (2, \infty)$. Thus the coefficients in the second variation converge uniformly on the support of each fixed test. Its integrals converge term by term, which passes (1.4) to the limit. All these conclusions apply to translates of the solutions. The weak-equation compactness argument alone also allows a limit exponent two. We do not use the preceding stability argument at that endpoint.

We also record a local regularity observation. Let $O \subset \mathbb{R}^n$ be open. Fix $p > 1$ and let v be a locally bounded weak solution of $-\Delta_p v = \mathfrak{f}(v)$ on O, where $\mathfrak{f}$ is C^1 on a neighborhood of the local range. No entire-solution or stability assumption is needed here. The source is locally bounded, so fixed-p regularity gives $v \in C^{1,\alpha}_{\mathrm{loc}}(O)$. On a relatively compact noncritical region, choose small buffered balls where the gradients stay in a compact convex neighborhood avoiding zero. The flux $\Phi_p(\xi) = |\xi|^{p-2}\xi$ is smooth there. The eigenvalues of $D\Phi_p(\xi)$ lie between

$$\min\{1, p-1\}|\xi|^{p-2} \quad \text{and} \quad \max\{1, p-1\}|\xi|^{p-2}.$$

They therefore have positive uniform bounds on each such ball. Difference quotients of v satisfy linear divergence equations. Their principal matrix is the segment average of $D\Phi_p$, with uniform $C^{0,\alpha}$ bounds. The right side is a bounded reaction secant times the difference quotient, and is uniformly bounded. The interior estimate in [24, Corollary 8.36, p. 212] allows a bounded scalar right side and gives uniform $C^{1,\eta}$ bounds for these weak difference-quotient solutions on smaller balls, for some fixed $0 < \eta < \alpha$. Passing to the limit gives $v \in C^{2,\eta}_{\mathrm{loc}}$ on the noncritical region. This conclusion requires only $\mathfrak{f} \in C^1$. If $\mathfrak{f}$ is smooth on the retained value interval, differentiation and successive interior Schauder estimates give $v \in C^\infty$ there. We make the finite-order uniformity explicit. Fix $k \geq 2$, $0 < \eta < \alpha < 1$, and nested balls $B_r \subset B_{2r} \Subset O$. Take p in a compact interval $\mathcal{P} \Subset (1, \infty)$. Assume a common $C^{1,\alpha}(B_{2r})$ bound M, a lower bound $|\nabla v| \geq \lambda > 0$, and a common C^{k-1} bound for the reactions on a fixed neighborhood of the compact value interval I. These data determine small balls on which all gradients lie in one compact convex set avoiding zero. Indeed, choose their radius ρ so that $M(4\rho)^\alpha \leq \lambda/4$ and keep a fixed boundary buffer. On these gradient sets, the matrices $A = D\Phi_p(\nabla v)$ have common positive ellipticity bounds. All fixed derivatives of Φ_p are bounded uniformly in $p \in \mathcal{P}$.

Set

$$\eta_m = \eta + 2^{1-m}(\alpha - \eta), \qquad 2 \leq m \leq k.$$

The difference-quotient argument above gives a common C^{2,η_2} bound on smaller balls. To obtain this prescribed exponent, take a finite q with $1 - n/q > \eta_2$ in the bounded-source interior estimate. The secant matrices have common $C^{0,\alpha}$ bounds, and the scalar sources are bounded by the C^1 reaction norm and M.

Suppose the C^{m,η_m} bound has been obtained, where $2 \leq m < k$. Put $w_i = \partial_i v$. Its linear equation is

$$-\operatorname{div}(A\nabla w_i) = g_i, \qquad g_i = \mathfrak{f}'(v)w_i.$$

The chain rule gives common C^{m-1,η_m} bounds for A. It also gives common C^{m-2,η_m} bounds for g_i. For the latter claim, the highest reaction derivative that occurs is $\mathfrak{f}^{(m-1)}$. Its Lipschitz bound follows from the bound for $\mathfrak{f}^{(m)}$. Thus the assumed C^{k-1} norm suffices, even at the last step $m = k - 1$.

For a multiindex γ with $|\gamma| = m - 2$, differentiation gives

$$-\operatorname{div}(A\nabla\partial^\gamma w_i) = \partial^\gamma g_i + \operatorname{div} F_{i,\gamma}, \qquad F_{i,\gamma} = \sum_{0<\delta\leq\gamma} \binom{\gamma}{\delta} (\partial^\delta A)\nabla\partial^{\gamma-\delta} w_i.$$

The sum is zero when $\gamma = 0$. Each vector $F_{i,\gamma}$ has a common C^{1,η_m} bound. In fact, differentiating one of its products once uses at most $m-1$ derivatives of A and $m-1$ derivatives of w_i, which the induction hypothesis supplies. Write $z = \partial^\gamma w_i$. One more weak derivative satisfies

$$-\operatorname{div}(A\nabla\partial_j z) = \operatorname{div}\left(e_j\partial^\gamma g_i + \partial_j F_{i,\gamma} + (\partial_j A)\nabla z\right).$$

All vector data on the right have common C^{0,η_m} bounds. Difference quotients first give the required local H^1 membership of $\partial_j z$. In that energy estimate, the scalar derivative is written as $\operatorname{div}(e_j\partial^\gamma g_i)$, so no additional derivative of g_i is needed. The interior divergence estimate then gives a common $C^{1,\eta_{m+1}}$ bound for $\partial_j z$ on a smaller ball. This proves the $C^{m+1,\eta_{m+1}}$ bound for v.

Use finitely many nested balls between B_r and B_{2r}, with their number and buffers fixed by k. A finite cover by the radius-ρ patches gives the requested $C^{k,\eta}(B_r)$ estimate, since $\eta_k > \eta$. Every intermediate coefficient bound and interior constant depends only on the stated data, k, and the fixed exponent margins. The earlier uniform theorem supplies the initial $C^{1,\alpha}$ bound for the matched family in the standing compact range. Its reaction has all these derivative bounds on every $I \Subset (-1, 1)$. Thus the local maximum principle, smooth unit-gradient flow, and differentiated phase identities below are justified on their stated noncritical strict-range regions. No bound across critical points or attained wells is used.

We use the gradient bound and equality theorem of Caffarelli–Garofalo–Segala [8, Theorems 1.6 and 5.1].

Proposition 2.1. *Suppose $p > 1$ and u is a bounded entire weak solution taking values in $[-1, 1]$ of*

$$\operatorname{div}(|\nabla u|^{p-2}\nabla u) = W_p'(u), \qquad W_p(t) = \frac{(1-t^2)^p}{2p} \quad (-1 \le t \le 1).$$

Then

$$\frac{p-1}{p}|\nabla u|^p \le W_p(u) \tag{2.2}$$

at every point. If equality occurs at a point with $-1 < u < 1$, some $\nu \in \mathbb{S}^{n-1}$ and $b \in \mathbb{R}$ satisfy

$$u(x) = \tanh\big(a(\nu \cdot x + b)\big), \qquad a = [2(p-1)]^{-1/p}. \tag{2.3}$$

Proof. In the notation of [8], take

$$\Phi(s) = \frac{2}{p}s^{p/2}, \qquad F(t) = \frac{|1-t^2|^p}{2p}.$$

The function Φ satisfies their structural assumption (A), with ellipticity eigenvalues at nonzero gradients equal to $|\nabla u|^{p-2}$ and $(p-1)|\nabla u|^{p-2}$. For $p \ge 2$, $F \in C^2(\mathbb{R})$ and $F \ge 0$. Its zeros are $r = \pm 1$, and $F(t) = O(|t-r|^p)$ near each zero. This also verifies the well-order hypothesis in their equality theorem. Local regularity gives a globally bounded gradient. Thus [8, Theorem 1.6] applies. Its conclusion is $2s\Phi'(s) - \Phi(s) \le 2F(u)$, with $s = |\nabla u|^2$. Here $2s\Phi'(s) - \Phi(s) = 2(p-1)|\nabla u|^p/p$, which gives (2.2).

For $1 < p < 2$, F is only C^1 at the wells. The translation argument in [8, pp. 1464–1465] still applies with the following local observation. Set $P = (p-1)|\nabla u|^p/p - W_p(u)$. For each fixed p, the bounded continuous reaction gives local $C^{1,\alpha}$ estimates uniform under translation, and hence C^1_{loc} compactness for translates. If $\sup P = P_* > 0$, a translate limit v attains this maximum. At every maximizing point, $\nabla v \ne 0$, since $W_p \ge 0$. Such a point also has $-1 < v < 1$: a C^1 function with values in $[-1, 1]$ has zero gradient wherever it attains either endpoint. A small ball about the point is therefore noncritical and has range compactly contained in $(-1, 1)$. The potential is smooth on that range, so the local strong maximum principle in [8, Theorem 2.2] applies. It makes $P(v; \cdot) = P_*$ throughout that ball. The nonempty maximum set is consequently both open and closed in $\mathbb{R}^n$. Connectedness gives $P(v; \cdot) = P_*$ everywhere. Since $W_p \ge 0$, this gives a positive global lower bound for $|\nabla v|$. Along the complete unit-gradient flow, v then increases at a fixed positive rate, contradicting boundedness. More explicitly, the unit-gradient field is locally smooth and has length one. Its integral curve stays in a compact ball on each finite time interval, so the curve extends for all positive time. Along it,

$$\frac{d}{dt}v(x(t)) = |\nabla v(x(t))| \ge \left(\frac{pP_*}{p-1}\right)^{1/p}.$$

Integrating this inequality contradicts $|v| \le 1$.

For equality, use the rigidity argument in [8, Theorem 5.1, pp. 1470–1472]. At the equality point the gradient is nonzero, because $W_p(u) > 0$. The same local maximum principle makes $P = 0$ on

its connected regular component: the gradient bound gives $P \le 0$, so each equality point is a local maximum. The potential is smooth on each small noncritical ball in this component. The phase substitution there gives

$$\Delta_p u = a^{p-1} q^{p-1} \left[\Delta_p U - 2cau|\nabla U|^p\right], \qquad \Delta_p U = -2cau(1 - |\nabla U|^p).$$

Here division is allowed because $q > 0$, and $a^{1-p} = 2ca$.

There the phase $U = a^{-1} \operatorname{arctanh} u$ satisfies

$$|\nabla U| = 1, \qquad \Delta U = 0.$$

The first identity is equality in the gradient bound. To obtain the second, differentiate the first to get $D^2U\nabla U = 0$. Substitution in the matched equation gives $\Delta_p U = 0$. Since $|\nabla U| = 1$, this reduces to $\Delta U = 0$. Hence $D^2U = 0$ by $\Delta|\nabla U|^2 = 2|D^2U|^2$. Thus $U = \nu\cdot x + b$, with $|\nu| = 1$, on that component. At any finite boundary point this formula keeps u in $(-1, 1)$ and $|\nabla u| = a(1-u^2) > 0$. Such a point cannot be on the boundary of the regular component. The component is all of $\mathbb{R}^n$, and (2.3) follows. This local argument also covers $1 < p < 2$. □

Remark 2.2. Attaining a well forces constancy for solutions with values in $[-1, 1]$. The gradient estimate yields

$$|\nabla(1 - u)| \le a(1 - u^2) \le 2a(1 - u), \qquad |\nabla(1 + u)| \le 2a(1 + u).$$

If one of these nonnegative functions vanishes at x_0, apply Gronwall's inequality on the segment from x_0 to an arbitrary x. It vanishes on that segment. Since x was arbitrary, u is the corresponding constant well. Every nonconstant solution therefore has strict range. This includes each nonconstant limit obtained by translation.

For the clearing argument use

$$f_p(t) = t(1 - t^2)^{p-1}, \qquad a_p = [2(p - 1)]^{-1/p}, \qquad g_p(s) = \tanh(a_p s).$$

Here the compact exponent set may be $\mathcal{P} \Subset [2, \infty)$. The estimates are uniform as p varies in $\mathcal{P}$.

Lemma 2.3 (Uniform clearing for $p \ge 2$)**.** *Let $\mathcal{P} \Subset [2, \infty)$. Given the ambient dimension n and $b \in (0, 1)$, there exists $R_b = R_b(n, \mathcal{P}, b) < \infty$ with the following property. If a solution*

$$-\Delta_p u = f_p(u), \qquad -1 < u < 1,$$

has no zero in $\overline{B_{R_b}(x)}$, then $|u(x)| \ge b$. In particular, an entire solution of one strict sign must be a pure well when the allowed range is $[-1, 1]$.

Proof. Reflection reduces the proof to a ball where $0 < u < 1$. Consider

$$F_p(t) := \frac{f_p(t)}{t^{p-1}} = t^{2-p}(1 - t^2)^{p-1}.$$

It decreases strictly on $(0, 1)$ for $p \ge 2$, since

$$\frac{d}{dt} \log F_p(t) = \frac{2 - p}{t} - \frac{2(p - 1)t}{1 - t^2} < 0.$$

Use $\psi(x) = (1 - |x|^2)^2$ in the unit ball. Radial differentiation, for $0 < r = |x| < 1$, gives

$$-\Delta_p \psi = 4^{p-1} r^{p-2} (1 - r^2)^{p-2} \big[(n + p - 2) - (n + 3p - 4)r^2\big].$$

The positive part is supported on

$$1 - r^2 \ge \frac{2(p - 1)}{n + 3p - 4}.$$

The bound $r^{p-2} \le 1$ now implies

$$-\Delta_p \psi \le C_{n,p} \psi^{p-1}, \qquad C_{n,p} := 4^{p-1}(n + p - 2) \left[\frac{n + 3p - 4}{2(p - 1)}\right]^p.$$

Extend ψ by zero. Its normal flux vanishes on the sphere. The radial flux also tends to zero at the center. Thus integration by parts adds no measure at either place, and $-\Delta_p \psi \le C_{n,p}\psi^{p-1}$ holds weakly after extension. This argument includes $p = 2$.

Rescale this barrier to B_R by setting $v(x) = b\psi(x/R)$. Under the radius condition

$$R^p \ge \frac{C_{n,p}}{F_p(b)},$$

we have

$$-\Delta_p v \le F_p(b)v^{p-1} \le f_p(v).$$

We prove $v \le u$ on B_R by Picone comparison. The function u is positive on the closed ball, whereas $v = 0$ on the boundary. If $E = \{v > u\}$ is nonempty, use $(v^p - u^p)_+/v^{p-1}$ as a test in the inequality for v. Use $(v^p - u^p)_+/u^{p-1}$ in the equation for u. Let $m = \min_{\overline{B_R}} u > 0$. Near the boundary $v < m$, so both tests vanish there. On E, one has $v > u \ge m$. The quotients, extended by zero outside E, are therefore compactly supported Lipschitz tests. Nonnegative smooth approximation makes them admissible in the weak equation and subsolution inequality. Subtract the two tested inequalities. Their right-hand sides differ by $f_p(v)(v^p - u^p)/v^{p-1} - f_p(u)(v^p - u^p)/u^{p-1}$ on E. The p-Laplacian Picone identity of Allegretto–Huang [2, Theorem 1.1, p. 820], applied in both orders on E, makes the gradient contribution nonnegative. Its hypotheses require positivity of the denominator, with no restriction on the gradients. The needed pointwise inequality can also be checked directly. For positive v, u on E, put

$$\begin{aligned}\mathscr{L}(v,u) &:= |\nabla v|^p - |\nabla u|^{p-2}\nabla u \cdot \nabla\left(\frac{v^p}{u^{p-1}}\right)\\ &= |\nabla v|^p + (p-1)(v/u)^p|\nabla u|^p - p(v/u)^{p-1}|\nabla u|^{p-2}\nabla u \cdot \nabla v \ge 0.\end{aligned}$$

The last step is Young's inequality with $X = |\nabla v|$ and $Y = (v/u)|\nabla u|$. The tested gradient difference is exactly $\mathscr{L}(v,u)+\mathscr{L}(u,v)$, so it is nonnegative also at zero gradients. All denominators are bounded away from zero on E. We obtain

$$0 \le \int_E (\mathscr{L}(v,u) + \mathscr{L}(u,v)) \le \int_E (F_p(v) - F_p(u))(v^p - u^p).$$

If E is nonempty, it is open and has positive measure. Strict decrease of F_p makes the right-hand side negative. Thus E is empty, and $u(0) \ge v(0) = b$. One may choose

$$R_b = \max\left\{1, \sup_{p\in\mathcal{P}}\left(\frac{C_{n,p}}{F_p(b)}\right)^{1/p}\right\}.$$

This is finite because the expression is continuous on the compact exponent set and $F_p(b) > 0$ there. For a positive entire solution valued in $[-1, 1]$, an attained well gives constancy by Remark 2.2. Otherwise its range is $(0, 1)$. Apply clearing at each point and let $b \uparrow 1$ to rule out this remaining case. Reflection gives the negative case. □

Clearing and the equality case give uniform preparation on fixed balls. The gradient estimate gives $D \ge 0$.

Lemma 2.4 (Preparation from small zero defect)**.** *Here $\mathcal{P} \Subset [2, \infty)$ is allowed. Let $p_j \in \mathcal{P}$ and let $u_j : \mathbb{R}^n \to (-1, 1)$ be entire solutions of (1.2), with nonempty zero sets, such that*

$$\delta_j := \sup_{\{u_j=0\}} D_j \longrightarrow 0.$$

For each sufficiently large j and each zero z, choose an orthogonal map $R_{j,z}$ with $R_{j,z}e_n = \nabla u_j(z)/|\nabla u_j(z)|$. For every fixed $L < \infty$ and finite k,

$$\sup_{u_j(z)=0} \|u_j(z + R_{j,z}\,\cdot) - g_{p_j}(x_n)\|_{C^k(B_L)} \longrightarrow 0.$$

The normal is well defined for all sufficiently large j. For each $b < 1$ it also holds uniformly in $\{|u_j| \le b\}$ after the appropriate normal translation. This fixed core is noncritical for all sufficiently large j.

Proof. At a zero, the definition of discrepancy gives

$$|\nabla u_j(z)| = a_{p_j}(1 - D_j(z))^{1/p_j}.$$

Since $0 \le D_j(z) \le \delta_j \to 0$, every zero is regular for all sufficiently large j. If uniform preparation failed, choose violating zeros z_j and use the translated and rotated solutions $y \mapsto u_j(z_j + R_{j,z_j} y)$. Local $C^{1,\alpha}$ bounds and compactness of $\mathcal{P}$ give an entire limit with exponent p_*. The weak equation passes to this limit by continuity of the flux and reaction, also when $p_* = 2$. Its value at the origin is zero and its gradient is $a_{p_*} e_n$. The equality case of Proposition 2.1 therefore identifies it as $g_{p_*}(y_n)$. Its gradient has a positive lower bound on each fixed ball. Its values also stay a positive distance from the wells on that ball. On a slightly larger fixed ball, C^1 convergence keeps the approximating gradients bounded below and their values away from the wells. The eigenvalues of $A_{p_j}(\nabla u_j)$ then have common positive lower and upper bounds. The flux and reaction are smooth on these compact ranges, uniformly in p_j. Interior Schauder estimates first give uniform $C^{2,\alpha}$ bounds on a smaller ball. Differentiate the equation and repeat the estimates to obtain bounds of every fixed order, using a larger initial ball if needed. Compactness at one further order improves the convergence to C^k. The profiles g_{p_j} converge smoothly to g_{p_*}. This contradicts the chosen violations.

To extend preparation away from zeros, it suffices to treat $0 \le b < 1$; the set is empty if $b < 0$. Choose $b_1 \in (b, 1)$. Lemma 2.3 puts every point with $|u_j| \le b$ within distance R_{b_1} of a zero. For such a point x, choose a zero z at that distance and use preparation on $B_{R_{b_1}+L+1}(z)$. Put $\tau = e_n \cdot R_{j,z}^{-1}(x - z)$, so $|\tau| \le R_{b_1}$. Translation of the zero-centered estimate gives

$$u_j(x + R_{j,z} y) = g_{p_j}(y_n + \tau) + o_{C^k(B_L)}(1),$$

uniformly in x and the chosen zero. At $y = 0$, the inverse profile is uniformly Lipschitz on the bounded phase range under consideration. Hence $\tau - U_j(x) \to 0$ uniformly. We may replace τ by $U_j(x)$ in the last estimate. The limiting profile gradients have a uniform positive lower bound on these fixed balls, which proves noncriticality throughout the core. If the rotation is chosen from the normal at x instead, its normal direction differs from $R_{j,z} e_n$ by $o(1)$, and the same estimate holds. The argument uses no stable classification theorem. □

2.2. **Weighted Jacobi calculus.** Return to $\mathcal{P} \Subset (2, \infty)$ for the weak identities across critical gradients. In this range $A_p(0) = 0$. At $p = 2$, $A_2 = I$. For $1 < p < 2$, the linearized matrix is singular at a zero gradient. Consider a weak solution $u \in C^1(\Omega)$ of $-\Delta_p u = f(u)$. Assume $f \in C^1$ on a neighborhood of the range of u. Define

$$s = |\nabla u|, \qquad \rho = s^{p-2}, \qquad A = A_p(\nabla u), \qquad \mathcal{R} = \{s > 0\}.$$

The matrix bounds are $\rho I \le A \le (p-1)\rho I$. Both matrices vanish on $\Omega \setminus \mathcal{R}$. For a relatively compact open set O, define $H^1_{\rho,0}(O)$ as the completion of $C_c^\infty(O)$ under

$$\|v\|^2_{H^1_\rho(O)} = \int_O v^2 + \rho |\nabla v|^2.$$

This gradient form is closable in $L^2(O)$. Indeed, suppose $v_j \to 0$ in L^2 and $\rho^{1/2}\nabla v_j$ is Cauchy in L^2. On every compact subset of $\{\rho > 0\}$, division by $\rho^{1/2}$ identifies an ordinary L^2 limit of ∇v_j. Distributional differentiation of $v_j \to 0$ identifies that limit as zero. These compact subsets exhaust the positive-weight region. On its complement, every weighted gradient $\rho^{1/2}\nabla v_j$ is zero, so its L^2 limit is zero there as well. We interpret local membership in H^1_ρ by multiplication with smooth compact cutoffs. The resulting weighted derivative is the ordinary weak derivative wherever ρ has a positive lower bound. To make this interpretation explicit, write $\mathcal{D}_\rho v$ for the L^2 limit of $\sqrt{\rho}\nabla v_j$

in a defining approximation. Closability makes this limit unique. It is zero almost everywhere on $\{\rho = 0\}$. An ordinary weak gradient across that set is not required. Writing $A = \rho\widehat{A}$ on $\{\rho > 0\}$, we interpret the form pairing as

$$\int A\nabla v \cdot \nabla w = \int \widehat{A}\mathcal{D}_\rho v \cdot \mathcal{D}_\rho w.$$

Any bounded choice of $\widehat{A}$ on the zero-weight set gives the same pairing.

The product and composition rules follow in this completed space. For a compact Lipschitz multiplier η, approximation gives

$$\mathcal{D}_\rho(\eta v) = \eta\mathcal{D}_\rho v + \sqrt{\rho}\, v\nabla\eta.$$

For smooth η, both terms converge in L^2 by the defining norm. For Lipschitz η, first form ηv_j and then mollify it in ordinary H^1. Boundedness of ρ makes this convergence weighted as well. A diagonal choice gives the asserted product rule.

For a vector v of form-domain functions, let G be C^1 with bounded derivative and $G(0) = 0$. Choose approximations $v^{(j)}$ that also converge almost everywhere. The derivative difference splits as

$$DG(v^{(j)})(\mathcal{D}_\rho v^{(j)} - \mathcal{D}_\rho v) + (DG(v^{(j)}) - DG(v))\mathcal{D}_\rho v.$$

The first term tends to zero in L^2 by the derivative bound. Dominated convergence handles the second. The Lipschitz bound on G gives convergence of the functions in L^2. Thus the chain rule and form membership pass to the limit. The condition $G(0) = 0$ preserves the zero exterior value. This argument uses approximability in the form norm, rather than finite weighted energy alone.

Proposition 2.5. *Locally, one has*

$$u_i \in H^1_\rho, \qquad |\nabla u|^{(p-2)/2}\nabla u \in H^1, \qquad F := |\nabla u|^{p-2}\nabla u \in H^1.$$

In particular,

$$\int_{K\cap\mathcal{R}} s^{p-2}|D^2 u|^2 < \infty \qquad (K \Subset \Omega). \tag{2.4}$$

Every compactly supported weighted form-domain test ψ satisfies

$$\int A\nabla u_i \cdot \nabla\psi - f'(u)u_i\psi = 0. \tag{2.5}$$

These estimates are uniform for $p \in \mathcal{P}$ if the local bounds for u, ∇u, and f' are uniform.

Proof. The qualitative flux and weak Jacobi identities follow from Farina–Sciunzi–Valdinoci [17, proof of Theorem 1, (6), (7), (13), and (14)]. These steps use the weak equation and do not require stability. On each compact subdomain, extend f as a C^1 function outside the compact range of u to meet their global nonlinearity hypothesis. We retain the estimates needed for the completed form domain and for uniformity on $\mathcal{P}$. Take a unit vector e and a nonzero increment h. Set $u_h(x) = (u(x + he) - u(x))/h$. On a smaller domain its equation is

$$-\operatorname{div}(A_h\nabla u_h) = c_h u_h, \qquad A_h = \int_0^1 DF_0\big((1-t)\nabla u(x) + t\nabla u(x+he)\big)\, dt.$$

Here $F_0(\xi) = |\xi|^{p-2}\xi$, and

$$c_h(x) = \int_0^1 f'\big((1-t)u(x) + tu(x+he)\big)\, dt.$$

Thus $|c_h|$ is bounded by the local bound for f'. The translated equation is tested only where both x and $x + he$ belong to the original domain. The fundamental theorem of calculus in the gradient

variable gives the displayed matrix A_h. The elementary power inequalities give

$$c_p(|\xi|+|\eta|)^{p-2}I \le \int_0^1 DF_0((1-t)\xi+t\eta)\,dt \le C_p(|\xi|+|\eta|)^{p-2}I.$$

Thus $A_h \ge c_p\rho(x)I$. For the lower secant bound, put $S=|\xi|+|\eta|$. If $|\xi|\ge S/2$, then $|(1-t)\xi+t\eta|\ge S/4$ for $0\le t\le 1/4$. Integrating the lower eigenvalue on that interval gives the bound with constant 4^{1-p}. If $|\eta|\ge S/2$, use $3/4\le t\le 1$. The upper eigenvalue is at most $(p-1)S^{p-2}$ throughout the segment. These constants are uniform on the standing compact exponent interval.

Moreover, A_h is locally bounded independently of h. Test with $u_h\zeta^2$ and use the local bound on u_h. Absorbing the cross term yields

$$\int \zeta^2 A_h\nabla u_h\cdot\nabla u_h \le C\int u_h^2(A_h\nabla\zeta\cdot\nabla\zeta+\zeta^2)\le C_\zeta.$$

For fixed h the function $u_h\zeta$ is compactly supported and Lipschitz. It therefore belongs to $H^1_{\rho,0}$: ordinary mollification converges in H^1, hence also in the weighted norm because ρ is locally bounded. Weak compactness and local uniform convergence $u_h\to u_e$ imply $u_e\zeta\in H^1_{\rho,0}$. Here weak compactness is taken in the completed form space. Its continuous embedding in L^2 identifies the limit with $u_e\zeta$. On each compact subset of $\mathcal{R}$, apply the noncritical bootstrap at the start of this section. Under the present $f\in C^1$ hypothesis it gives $u\in C^{2,\eta}$ locally. Thus the weighted derivative of u_i agrees there with its continuous ordinary derivative, and the regular level-set curvature is well defined. This also proves (2.4).

For the nonlinear gradient field, set $V_0(\xi)=|\xi|^{(p-2)/2}\xi$. The power estimate

$$|V_0(\xi)-V_0(\eta)|^2 \le C_p(F_0(\xi)-F_0(\eta))\cdot(\xi-\eta)$$

and the preceding bound control the ordinary L^2 difference quotients of $V_0(\nabla u)$ on compact subsets. After division by h^2, the scalar product is $A_h\nabla u_h\cdot\nabla u_h$. Thus $V_0(\nabla u)\in H^1_{\mathrm{loc}}$. One can see the power estimate along the same segment $\gamma(t)=(1-t)\xi+t\eta$. The largest eigenvalue of $DV_0(\gamma(t))$ is $(p/2)|\gamma(t)|^{(p-2)/2}$. Cauchy–Schwarz in t bounds the squared difference of V_0 by

$$\frac{p^2}{4}|\xi-\eta|^2\int_0^1|\gamma(t)|^{p-2}\,dt.$$

The lower eigenvalue of DF_0 makes this no greater than $(p^2/4)(F_0(\xi)-F_0(\eta))\cdot(\xi-\eta)$.

For $F=F_0(\nabla u)$, the flux identities in [17, proof of Theorem 1, (6) and (7)] also follow directly from the field just constructed:

$$F=|V_0(\nabla u)|^{(p-2)/p}V_0(\nabla u).$$

The outer map is C^1 and has bounded derivative on bounded sets. Its Sobolev chain rule gives $F\in H^1_{\mathrm{loc}}$. The level-set property gives $DF=0$ almost everywhere on $\{F=0\}$, which is exactly the critical set. On its complement, the ordinary chain rule identifies $\partial_iF=A\nabla u_i$. The bound $|A\nabla u_i|^2\le C\rho|\nabla u_i|^2$ on $\mathcal{R}$ then gives the quantitative flux estimate, uniformly on $\mathcal{P}$; here the local upper bound for ρ is used. Differentiate $-\operatorname{div}F=f(u)$ distributionally. The Sobolev chain rule gives $\partial_i f(u)=f'(u)u_i$, so $\partial_iF=A\nabla u_i$ gives (2.5) for smooth tests. This is the weak Jacobi identity in the cited proof. To pass to the completed test space, use

$$\left|\int A\nabla u_i\cdot\nabla\psi\right| \le (p-1)\left(\int\rho|\nabla u_i|^2\right)^{1/2}\left(\int\rho|\nabla\psi|^2\right)^{1/2}$$

on a compact neighborhood of the support. The potential term is continuous in L^2 because $f'(u)u_i$ is bounded there. Density therefore gives the full test class in the statement. All power estimates have constants uniform on $\mathcal{P}$.

□

We will use cutoffs of the form $\chi(s/\epsilon)$, with χ smooth and constant near zero. For fixed ϵ, these cutoffs lie in H^1_{loc}. Indeed, $s = |F|^{1/(p-1)}$, so the cutoff is a Lipschitz function of the H^1 field F on each bounded set. Its derivatives are supported in $\mathcal{R}$, where the ordinary chain rule applies.

For scalar-stable u, continuity and density extend

$$Q(\psi) = \int A\nabla\psi \cdot \nabla\psi - f'(u)\psi^2 \geq 0$$

to every compactly supported weighted form-domain test. Testing the Jacobi equation also gives

$$Q(u_e\zeta) = \int u_e^2 A\nabla\zeta \cdot \nabla\zeta \tag{2.6}$$

for a constant vector e and a smooth compactly supported ζ. The test in (2.5) is $u_e\zeta^2$.

Lemma 2.6 (Gluing across regular faces)**.** *Take finitely many pieces whose interface faces are piecewise smooth and lie in $\mathcal{R}$. On each piece let the test have compact support and finite weighted energy. Assume local weighted form membership up to the noninterface part of its support. Assume also ordinary H^1 membership in a collar of each interface it meets. Require that the closures of the interface portions met by the support, including their edges, be compactly contained in $\mathcal{R}$. At a noninterface boundary, the assumed local form membership refers to the zero extension of the test. If the two traces agree at every interface, the joined test belongs to $H^1_{\rho,0}$. Thus an exact Jacobi core $u_e\zeta$ can be joined to a Lipschitz connector with the same traces. For a core $\mathcal{C}$ with the relevant boundary faces in $\mathcal{R}$,*

$$Q_{\mathcal{C}}(u_e\zeta) = \int_{\mathcal{C}} u_e^2 A\nabla\zeta \cdot \nabla\zeta + \int_{\partial\mathcal{C}} u_e\zeta^2 A\nabla u_e \cdot n.$$

Only faces carrying nonzero test traces appear in this boundary integral.

Proof. The weight ρ has a positive lower bound on collars of the compact interface portions. Matching traces give an ordinary H^1 function across each collar. Use a smooth partition of unity to separate these collars from the piece interiors. Approximate the interior terms in the weighted norm and the collar terms in ordinary H^1. The latter convergence also holds in the weighted norm, since ρ is bounded above. Adding the approximations proves the gluing assertion. For the identity, test (2.5) with $u_e\zeta^2$ and a smooth approximation of the core indicator. The derivative of that indicator is supported in the regular collars. Green's formula applies there. Weighted energy integrability allows passage to the limit in the other terms. If the core also meets the critical set, first insert a cutoff that vanishes where $s \leq \epsilon$ and equals one where $s \geq 2\epsilon$. The extra term is bounded by

$$C_\zeta \int_{\{\epsilon<s<2\epsilon\}\cap\operatorname{supp}\zeta} s^{p-2}|D^2u|^2.$$

Indeed, $|u_e| \leq s$, the cutoff derivative is bounded by $C|D^2u|/\epsilon$, and $|A\nabla u_e| \leq C_p s^{p-2}|D^2u|$ there. The displayed integral tends to zero by (2.4). The regular boundary faces are unchanged for small ϵ. Expansion gives the stated formula. □

Lemma 2.7 (Recovery at a normal turn)**.** *For the fixed one-dimensional model, let $p > 3/2$. Let the one-dimensional profile V have a nondegenerate turn at zero: for small $|t|$,*

$$|V'(t)| \asymp |t|^{1/(p-1)}, \qquad |V''(t)| \asymp |t|^{-\alpha}, \qquad A(t) \asymp |t|^{\alpha}, \qquad \alpha = \frac{p-2}{p-1} < 1.$$

Each linear combination of V and V' admits Lipschitz recovery in the weighted form norm, with its endpoint data unchanged. For V', linear interpolation on $[-\epsilon, \epsilon]$ has squared weighted gradient cost $O(\epsilon^{1/(p-1)})$.

Proof. The interpolant has slope $O(\epsilon^{-\alpha})$. For a turn satisfying $-(|V'|^{p-2}V')' = f(V)$ and $f(V(0)) \neq 0$, put $Y = |V'|^{p-2}V'$. Then $Y(t) = -f(V(0))t + o(t)$. Inversion gives the stated orders for V' and V'', and $(p-1)|V'|^{p-2} \asymp |t|^{\alpha}$. Since $-1 < \alpha < 1$, both $|t|^{-\alpha}$ and $|t|^{\alpha}$ are integrable near zero. In particular, V'' is integrable, and continuity of V' joins the two half-interval derivatives into its weak derivative. The relevant energy densities obey

$$A|V''|^2 = O(|t|^{-\alpha}), \qquad A|V'|^2 = O(|t|^{2-\alpha}), \qquad A = O(|t|^{\alpha}).$$

This also explains why the fixed model permits the singular weight when $3/2 < p < 2$.

The squared gradient error is therefore at most

$$C\int_0^{2\epsilon} t^{-\alpha}\,dt + C\epsilon^{-2\alpha}\int_0^{2\epsilon} t^{\alpha}\,dt \le C\epsilon^{1-\alpha}.$$

Since $1-\alpha = 1/(p-1)$, this is the stated weighted gradient rate. The profile error is uniformly $O(\epsilon^{1/(p-1)})$ and is supported on an interval of length 2ϵ. Its squared L^2 norm is therefore $O(\epsilon^{1+2/(p-1)})$. The zero-order norm and every bounded potential term converge too. The interpolation is unchanged near the endpoints. Further local mollification gives Lipschitz recovery with the same traces. The V summand is already Lipschitz near the turn and may be left unchanged. For limits of actual gaps, use the standing range $\mathcal{P} \Subset (2,\infty)$. Keep ϵ fixed while passing to the coefficient limit. Then let ϵ tend to zero. Each fixed recovery function has bounded derivative. Uniform coefficient convergence therefore gives energy convergence on the compact interval. A diagonal choice then gives a recovery sequence for the limiting form. Here $A_p(\xi)$ is jointly continuous at $\xi = 0$ as well as elsewhere, uniformly on bounded sets for $p \in \mathcal{P}$; no uniform bound for the unrecovered V'' is used. For clarity, if $A_j \to A$ and $b_j \to b$ uniformly on a compact interval, then for a fixed Lipschitz recovery w_ϵ,

$$\left|\int (A_j - A)|w_\epsilon'|^2 + (b_j - b)w_\epsilon^2\right| \le \|A_j - A\|_\infty \int |w_\epsilon'|^2 + \|b_j - b\|_\infty \int w_\epsilon^2 \longrightarrow 0.$$

Choose $\epsilon_m \downarrow 0$ first. Choose increasing N_m so that the displayed difference is at most $1/m$ for $j \ge N_m$. Using w_{ϵ_m} for $N_m \le j < N_{m+1}$ gives the required diagonal recovery and fixes the order of the two limits.

□

For the directional and geometric inequalities below, assume that u is scalar-stable on Ω. This means $Q(\psi) \ge 0$ for every $\psi \in C_c^1(\Omega)$. Retain this assumption through Lemma 2.8.

For a fixed unit vector e define $P_e = I - e \otimes e$ and $w_e = |P_e \nabla u|$. Choose an orthonormal basis $e_1, \dots, e_{n-1}$ of $e^{\perp}$. On $\mathcal{R}$ put

$$\Gamma_e = \sum_{\alpha=1}^{n-1} A\nabla u_{e_\alpha} \cdot \nabla u_{e_\alpha} - A\nabla w_e \cdot \nabla w_e \ge 0.$$

Extend this expression by zero over the critical set. Lipschitz composition in the form domain gives $w_e \in H^1_\rho$. For a direct approximation, smooth the norm so that it vanishes where $w_e \le \epsilon$, equal w_e where $w_e \ge 2\epsilon$, and have a uniformly bounded derivative. Equation (2.4) gives weighted derivative convergence. The Sobolev level-set property makes the derivatives vanish almost everywhere on $\{w_e = 0\} \cap \mathcal{R}$. On $\{w_e > 0\}$, Cauchy–Schwarz gives

$$|\nabla w_e|_A^2 = \frac{1}{w_e^2}\left|\sum_\alpha u_{e_\alpha} A^{1/2}\nabla u_{e_\alpha}\right|^2 \le \sum_\alpha |\nabla u_{e_\alpha}|_A^2.$$

Thus $\Gamma_e \ge 0$ also has its asserted meaning at the zero set. Subtract stability for $w_e\eta$ from the sum of (2.6) in the directions e_α. This yields

$$\int \Gamma_e \eta^2 \le \int w_e^2 A\nabla\eta \cdot \nabla\eta. \tag{2.7}$$

The cross terms cancel by $w_e \nabla w_e = \sum_\alpha u_{e_\alpha} \nabla u_{e_\alpha}$. Applying the same norm calculation to $(u_1, \dots, u_n)$, whose norm is s, yields

$$\int_{\mathcal{R}} \left[s^p |B|^2 + (p-1) s^{p-2} |\nabla_T s|^2 \right] \eta^2 \le \int s^2 A \nabla \eta \cdot \nabla \eta \le (p-1) \int s^p |\nabla \eta|^2. \tag{2.8}$$

Here B and ∇_T belong to the regular level hypersurface through the integration point. This is [17, Theorem 1], retaining the exact matrix in the first upper bound. To compute the integrand, choose $\nabla u = s e_n$ and write $D^2 u = \left(\begin{smallmatrix} T & b \\ b^{\mathsf{T}} & h \end{smallmatrix} \right)$. Then

$$\sum_i A \nabla u_i \cdot \nabla u_i - A \nabla s \cdot \nabla s = s^{p-2} \left(|T|^2 + (p-1)|b|^2 \right),$$

with $T = sB$ and $b = \nabla_T s$. We begin with smooth η. Ordinary H^1 approximation, boundedness of $s^2 A$, and Fatou's lemma extend the estimates to compactly supported H^1 amplitudes.

Lemma 2.8 (Localization to a regular component)**.** *Let $\mathcal{C}$ be a connected component of $\mathcal{R}$. Inequality (2.8) holds with both integrals restricted to $\mathcal{C}$. No term is added on $\partial \mathcal{C}$.*

Proof. Start with a smooth compactly supported η. In (2.8), use $\mathbf{1}_{\mathcal{C}} \chi(s/\epsilon)\eta$, where $\chi = 0$ on $[0,1]$, $\chi = 1$ on $[2, \infty)$, and $|\chi'| \le C$. The cutoff observation above gives local H^1 regularity for fixed ϵ. The cutoff test avoids the critical boundary of $\mathcal{C}$. Within the compact support of η, its support in $\mathcal{C}$ stays a positive distance from $\partial \mathcal{C}$. Its zero extension is therefore in H^1. The new cutoff term obeys

$$\int_{\mathcal{C}} s^p \eta^2 |\nabla \chi(s/\epsilon)|^2 \le C_\eta \int_{\{\epsilon < s < 2\epsilon\} \cap \operatorname{supp} \eta} s^{p-2} |D^2 u|^2 \longrightarrow 0.$$

The limit follows from (2.4). Apply Young's inequality to the cross term. First send ϵ to zero, and then send the Young parameter to zero. Fatou's lemma handles the left-hand side. Dominated convergence handles the remaining right-hand side. Ordinary H^1 approximation extends the conclusion to the stated amplitudes. □

For the classical equation, the use of $q = 1 - u^2$ in the stability inequality appears in [29, Lemma 2.1]. We record its weighted counterpart for the matched equation. Return to a weak solution of $-\Delta_p u = u(1-u^2)^{p-1}$ on an open set Ω, with $-1 < u < 1$. Use the common phase variables and put $A_U = A_p(Z)$. On each compact subset of Ω, q is bounded away from zero and r is bounded. If u is the restriction of an entire solution $\mathbb{R}^n \to (-1, 1)$, Proposition 2.1 also gives $0 \le r \le 1$. The local identities below do not require this last bound. Stability is assumed only where it is explicitly stated. Let $L_u = -\operatorname{div}(A\nabla) - f'(u)$ with $f(u) = u(1-u^2)^{p-1}$. The identity $A\nabla u = (p-1)|\nabla u|^{p-2} \nabla u$ allows the weak product rule without differentiating ∇u. Indeed, $\nabla q = -2u \nabla u$ and $\operatorname{div} F = -f(u)$ give

$$-\operatorname{div}(A \nabla q) = 2(p-1)|\nabla u|^p - 2(p-1) u f(u).$$

The reaction satisfies $2(p-1)uf(u) + q f'(u) = q^p$. Also, $2(p-1)a^p = 1$ and $|\nabla u| = aqr$. Subtracting $f'(u)q$ therefore gives

$$L_u q = 2(p-1)|\nabla u|^p - q^p = -q^p D. \tag{2.9}$$

When u is scalar-stable, testing its stability inequality with $q\eta$ gives

$$\int q^{p+1} D \eta^2 \le \int q^2 A \nabla \eta \cdot \nabla \eta. \tag{2.10}$$

Here one expands $Q(q\eta)$ and tests (2.9) with $q\eta^2$. The functions involved are C^1 on the compact test support, so this step requires no second derivatives at a critical point.

Proposition 2.9. *On $\{r > 0\}$ set $\mathcal{H} = D^2U$ and*

$$\mathcal{S} = \operatorname{tr}(A_U \mathcal{H} A_U \mathcal{H}).$$

Put $\mathcal{S} = 0$ on $\{r = 0\}$. Then $D \in H^1_{\mathrm{loc}}$ and $\mathcal{S} \in L^1_{\mathrm{loc}}$, and

$$-\operatorname{div}(q^p A_U \nabla D) + 2p(p-1)a^2 q^{p+1} r^p D = pq^p \mathcal{S} \tag{2.11}$$

holds distributionally throughout the domain.

Proof. On compact subsets, q is bounded away from zero. Weighted regularity and the chain rule give

$$r^{p-2}|\mathcal{H}|^2 \in L^1_{\mathrm{loc}}, \qquad |Z|^{p-2}Z = a^{1-p}q^{1-p}F \in H^1_{\mathrm{loc}}.$$

For the first assertion use $\mathcal{H} = (aq)^{-1}D^2u + 2auZ \otimes Z$ on the regular set, together with $|\nabla u| = aqr$ and the weighted Hessian estimate. The identity $r^p = \big||Z|^{p-2}Z\big|^{p/(p-1)}$ expresses the power through a C^1 outer map. Its derivative is bounded on bounded sets, and hence $D \in H^1_{\mathrm{loc}}$. Fix a compact set $K \Subset \Omega$ and put $R_K = \max\{1, \sup_K r\} < \infty$. Since $p > 2$, one has

$$\mathcal{S} \le C_p r^{2p-4}|\mathcal{H}|^2 \le C_p R_K^{p-2} r^{p-2}|\mathcal{H}|^2 \quad \text{on } K.$$

The last expression is integrable by the weighted Hessian estimate. This uses local boundedness of r, rather than an entire-solution gradient bound.

For the matched equation, the reaction is smooth on each compact subinterval of $(-1, 1)$. The smooth-reaction clause of the same noncritical bootstrap therefore makes u and U smooth on the regular set. This is a local assertion on Ω. Derive the identity there, where the phase equation reads

$$\operatorname{div}(|Z|^{p-2}Z) = -2(p-1)auD.$$

Differentiate it and multiply the ith equation by $pr^{p-2}Z_i$. Writing $m_i = pr^{p-2}Z_i$, the two product-rule terms are

$$\operatorname{div}(A_U \nabla r^p) = \sum_i m_i \operatorname{div}(A_U \nabla Z_i) + \sum_i \nabla m_i \cdot A_U \nabla Z_i.$$

Since $\partial_j m_i = p\sum_\ell (A_U)_{i\ell}\mathcal{H}_{\ell j}$, the second sum is exactly $p\mathcal{S}$. The differentiated phase equation gives the other two terms below.

Summation gives

$$\operatorname{div}(A_U \nabla r^p) = p\mathcal{S} - 2p(p-1)a^2 q r^p D - 2p(p-1)aur^{p-2}Z \cdot \nabla D.$$

Use $A_U Z = (p-1)r^{p-2}Z$ and $\nabla D = -pr^{p-2}\mathcal{H}Z$. Since $\nabla(q^p) = -2pauq^p Z$, the product rule gives

$$\begin{aligned} -\operatorname{div}(q^p A_U \nabla D) &= q^p \operatorname{div}(A_U \nabla r^p) - \nabla(q^p) \cdot A_U \nabla D \\ &= pq^p \mathcal{S} - 2p(p-1)a^2 q^{p+1} r^p D. \end{aligned}$$

The added product term cancels the final first-order term in the preceding display. This proves (2.11) on $\{r > 0\}$.

Insert $\zeta\chi(r/\epsilon)$ to extend the identity across the critical set. For fixed ϵ, this is an H^1 test supported in the regular set. Apply the same cutoff observation to the H^1 field $|Z|^{p-2}Z$. The extra flux has the bound

$$\left| \int q^p A_U \nabla D \cdot \nabla\chi(r/\epsilon)\, \zeta \right| \le C_\zeta \int_{\{\epsilon<r<2\epsilon\}} r^{2p-4}|\mathcal{H}|^2 \le C_\zeta \epsilon^{p-2} \int_{\{\epsilon<r<2\epsilon\}} r^{p-2}|\mathcal{H}|^2 \longrightarrow 0.$$

Local integrability gives convergence of every other term. The Sobolev level-set property makes the weak gradients of $|Z|^{p-2}Z$ and D vanish almost everywhere on their constant-value critical set. There is therefore no additional measure there. The limiting flux is exactly the one in the statement. □

For the positive-discrepancy estimates, now let $u : \mathbb{R}^n \to (-1, 1)$ be an entire weak solution of the matched equation. Proposition 2.1 gives $0 \le r \le 1$. For the negative powers of D, also assume that u is nonplanar. Here nonplanar means that u is not one of the profiles in (2.3). The equality case of Proposition 2.1 then gives $D > 0$ everywhere. Indeed, $D = 0$ at any point would be equality in (2.2), and $-1 < u < 1$ there. Thus D has a positive minimum on a compact test support. Use $D^{-\theta}\zeta^2$ in (2.11), where $0 < \theta < 1$. The test is continuous, bounded and in H^1 on its compact support. Mollification gives smooth tests converging both uniformly and in H^1; uniform convergence handles the L^1 source $\mathcal{S}$. Thus the weak equation admits this test and yields

$$p\int q^p \mathcal{S} D^{-\theta}\zeta^2 + \frac{\theta}{2}\int q^p D^{-\theta-1}|\nabla D|^2_{A_U}\zeta^2 \le 2p(p-1)a^2\int q^{p+1}D^{1-\theta}\zeta^2 + \frac{2}{\theta}\int q^p D^{1-\theta}|\nabla\zeta|^2_{A_U}.$$

We have removed $r^p \le 1$ from the nonnegative zero-order term. Before absorption the coefficient of the derivative integral is θ, and the mixed integrand is bounded by $2XY$, with

$$X = q^{p/2}D^{-(\theta+1)/2}|\zeta|\,|\nabla D|_{A_U}, \qquad Y = q^{p/2}D^{(1-\theta)/2}|\nabla\zeta|_{A_U}.$$

The inequality $2XY \le (\theta/2)X^2 + (2/\theta)Y^2$ gives exactly the two coefficients retained above.

For the normal-current calculation, retain the entire-solution assumption; nonplanarity is not needed. Take a smooth unit field e on a smooth tube and write $u_e = e_i u_i$. For a Fermi tube $\Psi(z,t) = z + t\nu(z)$, its signed distance d satisfies $e = \nabla d$, $T = De = D^2 d$, and $Te = 0$. For a tangent vector τ at the base,

$$d\Psi(\tau) = (I + tB)\tau, \qquad T(I + tB)\tau = B\tau.$$

Thus the tangential part of T is $B(I+tB)^{-1}$. All test supports remain inside this smooth coordinate tube.

Weighted multiplication gives $u_e \in H^1_\rho$. Its weak equation is exactly

$$L_u(u_e) = -A_{jk}u_{ki}\partial_j e_i - \partial_j(A_{jk}(\partial_k e_i)u_i).$$

Test (2.5) with $e_i\psi$ and sum in i. The weighted product rule gives the weak pairing

$$\langle L_u(u_e), \psi\rangle = -\sum_i \int \psi A\nabla u_i \cdot \nabla e_i + \sum_i \int u_i A\nabla e_i \cdot \nabla\psi.$$

Collecting the last integral as a divergence gives the asserted equation. No derivative of A is taken.

Define the first contraction through $A\nabla u_i = \partial_i F \in L^2_{\mathrm{loc}}$. The formula then remains meaningful at critical points. For the Fermi normal field, $T = De$ is symmetric. Substitute

$$D^2 u = aq\mathcal{H} - 2a^2 uqZ \otimes Z, \qquad \nabla q = -2auqZ$$

to obtain

$$\frac{q}{a}L_u(u_e) = -a^{p-2}\left[q^p \operatorname{tr}(A_U\mathcal{H}T) + \operatorname{div}(q^p A_U TZ)\right].$$

In this substitution the Hessian contraction produces $2a^{p-1}uq^p A_U Z \cdot TZ$. Moving the factor q inside the divergence produces its negative. Symmetry of A_U makes the two scalar terms equal, so they cancel.

The amplitude

$$h = \frac{q + a^{-1}u_e}{2}, \qquad \varphi = \frac{h}{q} = \frac{1 + Z\cdot e}{2}$$

is admissible in the weighted form domain and satisfies $0 \le h \le q$. It obeys

$$\langle L_u h, h\eta^2\rangle = -\frac{1}{2}\int q^{p+1} D\varphi \eta^2 - \frac{a^{p-2}}{2}\int q^p \varphi \,\mathrm{tr}(A_U \mathcal{H} T)\eta^2 + \frac{a^{p-2}}{2}\int q^p A_U T Z \cdot \nabla(\varphi\eta^2).$$

The absolute value of each curvature current is controlled by an integral of a constant times

$$q^p(|T|\sqrt{\mathcal{S}}\,\eta^2 + |T|^2\eta^2 + |T|\,|\eta|\,|\nabla\eta|).$$

Indeed, $r \le 1$, while the condition number of A_U is at most $p-1$ on $\{r > 0\}$. More explicitly, $\mathcal{S} = |A_U^{1/2}\mathcal{H}A_U^{1/2}|^2$ and $\nabla\varphi = (\mathcal{H}e + TZ)/2$ there. These estimates give local integrability and justify the pairings for smooth compact amplitudes. For example, set $N = A_U^{1/2}\mathcal{H}A_U^{1/2}$, so $|N| = \sqrt{\mathcal{S}}$. Then

$$\mathrm{tr}(A_U\mathcal{H}T) = \mathrm{tr}(NA_U^{-1/2}TA_U^{1/2}), \qquad A_U TZ\cdot\mathcal{H}e = (A_U^{1/2}TZ)\cdot N(A_U^{-1/2}e).$$

The eigenvalue ratio gives bounds $\sqrt{p-1}|T|\sqrt{\mathcal{S}}$ and $\sqrt{p-1}r|T|\sqrt{\mathcal{S}}$, respectively. The remaining factors satisfy $|A_U TZ\cdot TZ| \le (p-1)r^p|T|^2$ and $|A_U TZ| \le (p-1)r^{p-1}|T|$.

In particular the divergence term is paired in the weighted form sense: if $J = q^p A_U TZ$, then $|J|^2/r^{p-2} \le Cq^{2p}r^p|T|^2$ on $\{r > 0\}$ and $J = 0$ on its complement. This is locally integrable, and q^{-1} and its first derivatives are bounded on the test support. Thus $\varphi\eta^2$ is a valid weighted test. Ordinary H^1 regularity of Z across the critical set is not needed. Multiplication by compact Lipschitz caps and Lipschitz base amplitudes preserves form membership.

Lemma 2.10 (Finite branch partitions)**.** *Suppose $T_0, \dots, T_k$ are bounded and compactly supported in $H^1_{\rho,0}$. Then $\Psi = (\sum_i T_i^2)^{1/2}$ belongs to that space, and*

$$Q(\Psi) = \sum_i Q(T_i) - \int_{\{\sum_i T_i^2 > 0\}} \frac{\sum_{i<j}|T_i\nabla T_j - T_j\nabla T_i|_A^2}{\sum_i T_i^2}.$$

Consequently $Q(\Psi) \le \sum_i Q(T_i)$.

Proof. For $\delta > 0$ consider

$$G_\delta(T) = \sqrt{\delta^2 + \sum_i T_i^2} - \delta.$$

This smooth Lipschitz composition preserves zero exterior values. Approximate T_i in the form norm. The derivative of G_δ is continuous and bounded, so the chain rule and form membership pass to the limit. On the positive-weight region, the Sobolev level-set property gives $\nabla T_i = 0$ where all T_i vanish. Weighted dominated convergence now gives $G_\delta(T) \to \Psi$ strongly in the form norm as $\delta \downarrow 0$. Apply the pointwise norm identity to $A^{1/2}\nabla T_i$. Writing $S = \sum_i T_i^2$, that identity is

$$S\sum_i |\nabla T_i|_A^2 - \left|\sum_i T_i\nabla T_i\right|_A^2 = \sum_{i<j}|T_i\nabla T_j - T_j\nabla T_i|_A^2.$$

On $\{S > 0\}$, use $\nabla\Psi = S^{-1/2}\sum_i T_i\nabla T_i$ and divide by S.

Its subtracted term is nonnegative and bounded by $\sum_i |\nabla T_i|_A^2$. It is therefore integrable. Finally, $\Psi^2 = \sum_i T_i^2$ separates the potential term. □

To apply the branch-partition lemma through normal turns, we justify zero extension at cap edges. Ordinary H^1 regularity of u_e is not needed. The argument also permits an irregular cap boundary. Write a branch as $w\kappa$ on an open tube O, with w bounded and locally in the form domain, and with $\int_O \rho\kappa^2|\nabla w|^2 < \infty$. If a cap κ vanishes on ∂O, then $|\kappa(x)| \le \mathrm{Lip}(\kappa)\,\mathrm{dist}(x, \partial O)$. For $\chi_\epsilon = \chi(\mathrm{dist}(x, \mathbb{R}^n \setminus O)/\epsilon)$, this gives

$$|w\kappa\nabla\chi_\epsilon| \le C\|w\|_\infty \mathrm{Lip}(\kappa)$$

on its intermediate layer. Put $v = w\kappa$ on O. The product rule gives

$$\mathcal{D}_\rho(v - v\chi_\epsilon) = (1 - \chi_\epsilon)\mathcal{D}_\rho v - \sqrt{\rho}\, v\nabla\chi_\epsilon.$$

The first term tends to zero by absolute continuity of its squared integral. Its integrability follows from the assumed energy of $\kappa\nabla w$ and the boundedness of $w\nabla\kappa$. The second term is supported in $\{x \in O : 0 < \operatorname{dist}(x, \partial O) < 2\epsilon\}$. Within the fixed compact support, these sets decrease to the empty set. Their ρ integrals therefore tend to zero by the preceding pointwise bound. The functions also converge in L^2 by dominated convergence. Each $v\chi_\epsilon$ has support compactly contained in O. Its local form approximations extend by zero to the ambient domain. A diagonal approximation and closedness then give the ambient zero extension of v in the same form domain.

2.3. **Relative phase estimates.** Small discrepancy controls how far the phase gradient is from a constant. The inverse-phase normalization and relative estimates follow the strategy of [35, Sections 5.1–5.4]. We derive the weighted identities needed for the matched equation. Denote the solution by v and use the common phase notation. We first work locally on a noncritical region. Proposition 2.1 gives $0 \le r \le 1$ and hence $0 \le D \le 1$. On the regular set define

$$n = Z/r, \qquad A = r^{p-2}(I + (p-2)n \otimes n).$$

Lemma 2.11 (Phase identities)**.** *For every $p > 1$, the equation for the phase is*

$$\Delta_p U = f(U)D, \qquad f(U) = -2ca\tanh(aU). \tag{2.12}$$

On a noncritical open set, differentiation yields

$$\operatorname{div}(A\nabla Z_i) + pf(U)r^{p-2}Z \cdot \nabla Z_i = f'(U)DZ_i. \tag{2.13}$$

At a regular zero, write $B = D_\Sigma n$ and $H = \operatorname{tr} B$. Then

$$H = \frac{c}{p}\frac{\partial_n D}{1 - D}. \tag{2.14}$$

Proof. The relation $\nabla v = aqZ$ gives the flux identity

$$\Delta_p v = a^{p-1}q^{p-1}\left(\Delta_p U - 2cavr^p\right).$$

Insert $\Delta_p v = -vq^{p-1}$ and $a^{1-p} = 2ca$ to obtain (2.12). The same calculation is valid weakly. To see this, test the equation for v with $q^{1-p}\phi$, where ϕ is compactly supported. On its support, q has a positive lower bound. Thus this is an admissible test, by approximation, and the product rule gives the displayed flux relation in distributions. No derivative of the normal field is needed at a critical point.

On a noncritical region the equation is smooth. Differentiate it and use $\partial_i D = -pr^{p-2}Z \cdot \nabla Z_i$ to obtain (2.13). At a regular zero, $f(U) = 0$ and $\Delta_p U = r^{p-2}(cU_{nn} + rH)$. Hence

$$cU_{nn} + rH = 0, \qquad \partial_n D = -pr^{p-1}U_{nn}.$$

These two identities give (2.14). □

Lemma 2.12 (Relative regularity)**.** *Given $L < \infty$ and an integer $j \ge 0$, choose $d_L > 0$ and $C_{L,j} < \infty$ uniformly for $p \in \mathcal{P}$. Suppose $v(z) = 0$, $d = D(z) \le d_L$, and $e = Z(z)/r(z)$. Then $B_L(z)$ is noncritical, and*

$$\|D\|_{C^j(B_L(z))} \le C_{L,j}d, \tag{2.15}$$

$$\|1 - U_e\|_{C^j(B_L(z))} \le C_{L,j}d, \tag{2.16}$$

$$\|D^2U\|_{C^j(B_L(z))} \le C_{L,j}\sqrt{d}. \tag{2.17}$$

Along the straight ray $z + se$ with $|s| \le L$,

$$|Z - (Z \cdot e)e| + |n - (n \cdot e)e| \le C_L d. \tag{2.18}$$

On the zero set this implies

$$|B| + |\nabla_\Sigma B| + |\nabla_\Sigma^2 B| \le C\sqrt{D}, \tag{2.19}$$

$$|H| \le CD, \qquad |\nabla_\Sigma^j D| \le C_j D \quad (j \ge 1). \tag{2.20}$$

If D is zero at any zero of the solution, the solution is a planar layer.

Proof. If $d = 0$, Proposition 2.1 gives a planar layer. All the estimates then hold with zero right-hand side. Assume $d > 0$.

We first obtain estimates that are independent of the phase at the center. On a unit ball centered at any y, subtract $U(y)$ from U. The bound $|\nabla U| \le 1$ controls this translated phase. The right-hand side of (2.12) is uniformly bounded. Moreover, every fixed derivative of $f(t+U(y))$ is bounded uniformly in $U(y)$ and $p \in \mathcal{P}$. Local $C^{1,\alpha}$ estimates therefore give one gradient Hölder bound on all such balls. If a fixed unit vector e satisfies $1 - U_e(y) \le 1/4$, this bound keeps $r \ge 1/2$ on $B_{4\rho}(y)$, for one fixed $\rho > 0$. On smaller balls the phase equation is uniformly elliptic: the eigenvalues of A are r^{p-2} and cr^{p-2}, and $1/2 \le r \le 1$. Their upper and lower bounds are uniform on $\mathcal{P}$. Interior elliptic estimates can therefore be iterated. The coefficients are smooth functions of Z on this range, and every fixed derivative of the translated reaction is uniformly bounded. This gives absolute bounds for every fixed derivative of the phase, independently of its value at the center.

Choose coordinates with $e = e_n$ and define $w = 1 - U_n \ge 0$. The convexity inequality at e_n is $|Z|^p \ge 1 + p(Z_n - 1)$. It yields

$$0 \le D \le pw. \tag{2.21}$$

For $\mathcal{L} = \operatorname{div}(A\nabla) + pf(U)r^{p-2}Z \cdot \nabla$, equation (2.13) becomes

$$\mathcal{L}w = -f'(U)DU_n.$$

Where $w > 0$, divide the right-hand side by w. Equation (2.21) and $|U_n| \le 1$ bound this quotient uniformly. At a zero of w, both D and the right-hand side vanish, so set the quotient equal to zero there. Thus w solves a linear equation with a bounded measurable zero-order coefficient. On each of the noncritical balls above, the other coefficients are uniformly bounded and elliptic. Harnack's inequality gives

$$\sup_{B_{2\rho}(y)} w \le C_H w(y)$$

whenever $w(y) \le 1/4$. At the original center,

$$w(z) = 1 - (1-d)^{1/p} \le C_0 d.$$

We continue this estimate before taking derivatives. Join z to any point of $B_{L+2}(z)$ by a segment. Subdivide it into at most N_L steps of length at most ρ. Use the same direction e and the same function w at each step. Choose d_L so that $C_0 C_H^{N_L+1} d_L \le 1/4$. Harnack's inequality then applies successively at every center. It gives $w \le C_L d$ throughout $B_{L+2}(z)$, and keeps that ball noncritical. Equation (2.21) gives $D \le C_L d$. Also, $|Z - e_n|^2 \le 2w$, so $Z_T = O_L(\sqrt{d})$. This choice of d_L does not depend on the derivative order.

To retain these relative orders under differentiation, use the original equations. This avoids differentiating the measurable quotient D/w. On buffered subballs, $\mathcal{L}$ has uniformly bounded smooth coefficients by the noncritical estimates above. Interior $W^{2,\ell}$ estimates, with ℓ greater than the dimension, apply directly to the displayed equation for w and to (2.13) for the tangential components. Their sources have sizes $O_L(d)$ and $O_L(d^{3/2})$, respectively. After reducing the Hölder exponent, these estimates give $w = O_{C^{1,\alpha}}(d)$ and $Z_T = O_{C^{1,\alpha}}(\sqrt{d})$. The exact identity

$$D = 1 - \left((1-w)^2 + |Z_T|^2\right)^{p/2}$$

then gives $D = O_{C^{1,\alpha}}(d)$. Indeed, every term contains a factor of w or two factors of Z_T. To see that this persists at higher orders, put

$$F(s,t) = 1 - ((1-s)^2 + t)^{p/2}, \qquad F_i(s,t) = \int_0^1 \partial_i F(\lambda s, \lambda t)\, d\lambda.$$

The exact decomposition $F(s,t) = sF_1(s,t) + tF_2(s,t)$ gives

$$D = wF_1(w, |Z_T|^2) + |Z_T|^2 F_2(w, |Z_T|^2).$$

The functions F_i have uniform smooth bounds on the noncritical range in use. The product rule therefore preserves the orders d and $\sqrt{d}$ in every fixed $C^{k,\alpha}$ norm.

Here is the induction used for the higher derivatives. Suppose the orders $w = O(d)$, $Z_T = O(\sqrt{d})$, and $D = O(d)$ hold in $C^{k,\alpha}$ on a buffered ball. The sources $-f'(U)DU_n$ and $f'(U)DZ_T$ then have $C^{k,\alpha}$ sizes $O(d)$ and $O(d^{3/2})$. The principal matrix A has uniform absolute $C^{k+1,\alpha}$ bounds. The drift coefficients have uniform $C^{k,\alpha}$ bounds. Both follow from the preceding noncritical estimates. Interior Schauder estimates give the same relative orders for w and Z_T in $C^{k+2,\alpha}$ on a smaller ball. The displayed decomposition gives $D = O_{C^{k+2,\alpha}}(d)$ there as well. For any prescribed derivative order, choose finitely many nested balls inside $B_{L+2}(z)$. The constants may depend on that order; the initial smallness threshold remains fixed. This proves (2.15) and (2.16). Since the derivatives of $Z = (Z_T, 1-w)$ have size $O_L(\sqrt{d})$, it also proves (2.17).

Along the ray, $Z_T = 0$ at $s = 0$. Commutation of mixed derivatives gives

$$\partial_s Z_T = \nabla_T U_n = -\nabla_T w = O_L(d).$$

Integration gives the bound for Z_T in (2.18). The bound for the tangential part of n follows by dividing by $r \geq 1/2$. At a zero, the second fundamental form is the tangential restriction of D^2U/r. The ambient derivative estimates and the lower bound for r control its first two covariant derivatives. More explicitly, with $\Pi = I - n \otimes n$,

$$Dn = \frac{\Pi D^2 U}{r}, \qquad B = \frac{\Pi D^2 U \Pi}{r}.$$

Every derivative of B contains a derivative of D^2U as one factor. That factor has size $O(\sqrt{d})$; all other factors are uniformly bounded. Tangential projection and restriction to the sheet preserve these orders.

Apply these fixed-radius estimates with each small-defect zero as the center. This gives (2.19) in terms of the defect at that zero, rather than the defect at the original center. Covariant surface derivatives are sums of projected ambient derivatives and terms containing derivatives of the projection. The latter are bounded by the same estimates for n. This proves the asserted bounds for every fixed surface derivative of D. Finally, (2.14) and $|\nabla D| \leq CD$ give $|H| \leq CD$, since $1 - D = r^p$ is bounded below at these small-defect zeros. This proves (2.20). □

The proof also gives (2.15)–(2.17) at every phase center x, with no restriction on $v(x)$ beyond $|v(x)| < 1$. For each fixed R, they hold on $B_R(x)$ when $D(x) \leq d_R$, with $e = Z(x)/|Z(x)|$ and constants independent of $v(x)$. The case $D(x) = 0$ again follows from Proposition 2.1. Only the conclusions about zero-sheet geometry require $v(x) = 0$.

For fixed $P \geq 1$, write $G = G_p$ and define

$$I^{\mathrm{fr}}_{p,P}(z) = \frac{2ca^2}{J_p} \int_{-P}^{P} G(s)^{p+1} D(z + sn(z))\, ds.$$

This mass uses the frozen weight G^{p+1}. Its definition does not require an injective normal map.

Corollary 2.13 (A fixed normal mass)**.** *Choose P first. At each zero with sufficiently small defect,*

$$c_P D(z) \leq I^{\mathrm{fr}}_{p,P}(z) \leq C_P D(z). \tag{2.22}$$

Thus, for δ below this threshold, $D \le \delta$ on the zero set implies $I^{\mathrm{fr}}_{p,P} \le C_P\delta$.

Proof. Equation (2.15) on $B_P(z)$ gives the upper bound. For the lower bound, use $|\nabla D| \le CD(z)$ on $B_1(z)$. It gives $D(z+sn(z)) \ge D(z)/2$ whenever $|s| \le s_0 = \min\{1,(2C)^{-1}\}$. On this interval the frozen weight has a positive lower bound uniform in $p \in \mathcal{P}$. Integration gives the result. □

Remark 2.14. Write $\nu = n(z)$. On a fixed window the relative phase estimates imply $q(z+s\nu)/G(s) = 1+O_P(D(z))$. Indeed, $\partial_s U = 1+O_P(D(z))$ and $U(z) = 0$ give $U(z+s\nu) = s+O_P(D(z))$, while $|(\log \operatorname{sech}^2(at))'| \le 2a$. The prefactors agree exactly, since $2ca^2/J_p = a^{2-p}/J_p = \sigma_q^{-1}$. Raising the weight ratio to the power $p+1$ therefore shows that replacing the frozen-profile mass by the physical mass changes it by the factor $1+O_P(D(z))$. We fix the window before decreasing the zero-defect threshold. This makes the replacement error smaller than each prescribed strict coefficient tolerance.

For this noncritical discrepancy equation, take $p > 1$. Let v be an entire solution of $-\Delta_p v = v(1-v^2)^{p-1}$ with $|v| < 1$. Use the common phase variables and set $T = D^2U$. The gradient bound gives $0 \le r \le 1$. On $r > 0$, use the phase matrix $A = A_p(Z)$ and the reaction f in (2.12).

Lemma 2.15 (Exact positive-source discrepancy equation)**.** *On the noncritical set one has*

$$\begin{aligned} &-\operatorname{div}(A\nabla D) + 2pca\, v r^{p-2} Z\cdot\nabla D + 2pca^2 q r^p D \\ &\quad = pr^{2p-4}|T|^2 + 2p(p-2)r^{2p-6}|TZ|^2 + p(p-2)^2 r^{2p-8}(Z\cdot TZ)^2 \ge 0. \end{aligned} \tag{2.23}$$

The source is nonnegative for every $p > 1$: it equals $p\operatorname{tr}(ATAT) = p|A^{1/2}TA^{1/2}|^2$. Here A is positive definite and T is symmetric.

Proof. One calculation proves the identity for every $p > 1$. Work on a noncritical region, where U is smooth, and put $F_U = r^{p-2}Z$. Differentiation of (2.12) gives

$$\operatorname{div}(A\nabla Z_i) = f'(U)DZ_i + f(U)\partial_i D.$$

Also $\nabla r^p = p\sum_i (F_U)_i \nabla Z_i$. Applying the product rule to this identity gives

$$\begin{aligned} \operatorname{div}(A\nabla r^p) &= p\sum_i \nabla(F_U)_i \cdot A\nabla Z_i + pr^{p-2}\sum_i Z_i \operatorname{div}(A\nabla Z_i) \\ &= p\operatorname{tr}(ATAT) + pf'(U)r^pD + pf(U)r^{p-2}Z\cdot\nabla D. \end{aligned}$$

For the trace term, use $DF_U = AT$ and the symmetry of A, T. Now insert $D = 1-r^p$, $f(U) = -2cav$, and $f'(U) = -2ca^2q$. Moving the last two terms to the left gives (2.23). Expanding the two factors of A gives its stated source:

$$\operatorname{tr}(ATAT) = r^{2p-4}|T|^2 + 2(p-2)r^{2p-6}|TZ|^2 + (p-2)^2 r^{2p-8}(Z\cdot TZ)^2.$$

When $1 < p < 2$, the middle coefficient is negative, so positivity must be checked for the sum. In an orthonormal frame with last vector $n = Z/r$, the eigenvalues of A are r^{p-2} in the orthogonal directions and cr^{p-2} in direction n. Thus

$$\operatorname{tr}(ATAT) = r^{2p-4}(|T_{TT}|^2 + 2c|T_{Tn}|^2 + c^2T_{nn}^2) \ge 0.$$

Here T_{TT} is the block on $n^\perp$ and T_{Tn} is its mixed column. Since $c > 0$, the source vanishes exactly when $T = 0$. For $p = 2$ the formula reduces to

$$-\Delta D + 4avZ\cdot\nabla D + 4a^2qr^2D = 2|T|^2.$$

For $p > 2$ it also agrees with (2.11) after division by q^p, using

$$\nabla \log q = -2avZ, \qquad AZ = cr^{p-2}Z.$$

The direct derivation above uses no weak identity at a critical point, so it applies unchanged for $1 < p < 2$. □

2.4. **Conjugation and the normal gap.** The normal spectral estimate requires the natural weighted norm. Return to the standing exponent range and set $k(u) = aq$ and $M = G^{p/2}$ for local preparation. We again denote the solution by u and use the frozen mass $I^{\mathrm{fr}}_{p,P}$ defined above. The comparison with the physical mass follows from the remark after Corollary 2.13. First verify the profile equation for every $p > 1$. Since $g_p' = a_p(1-g_p^2) > 0$ and $2(p-1)a_p^p = 1$, direct differentiation gives

$$-\big(|g_p'|^{p-2}g_p'\big)' + W_p'(g_p) = [2(p-1)a_p^p - 1]g_p(1-g_p^2)^{p-1} = 0.$$

The calculation takes place at finite s, where $-1 < g_p < 1$ and $g_p' > 0$, so it also applies to $1 < p < 2$. The profile derivative $r_0 = g_p'$ is therefore the translation zero mode: differentiating $g_p(s+t)$ at $t = 0$ gives r_0. Its Jacobi energy vanishes. A positive normal gap therefore requires orthogonality to this direction in the scalar product $\int r_0^{p-2}\varphi\psi\, ds$.

Lemma 2.16 (Exact conjugation)**.** *Put $\alpha = (2-p)/2$. For $\psi \in C_c^1(\mathbb{R}^n)$, define $\phi = k(u)^\alpha\psi$. Then*

$$Q_u(\phi) = \int_{\mathbb{R}^n} \big(A_p(Z)\nabla\psi\cdot\nabla\psi + \widetilde{V}\psi^2\big)\, dx, \tag{2.24}$$

with

$$\widetilde{V} = ca^2\left\{p^2 - p(p+1)q + (p-2)[p-cq]D\right\}. \tag{2.25}$$

For every $p > 1$, the same identity holds when the test support is contained in the noncritical set. In the standing range $p > 2$, it holds weakly across critical points of u. For a planar profile it becomes

$$V_0(s) = ca^2[p^2 - p(p+1)G(s)], \qquad -cM'' + V_0M = 0. \tag{2.26}$$

Proof. From $A_u = k^{p-2}A_p(Z)$, multiplication by k^α removes the scalar gradient weight: $k^{p-2}(k^\alpha)^2 = 1$. Put $h = k^\alpha$. The term containing one derivative of ψ is

$$2k^{p-2}h\psi A_p(Z)\nabla\psi\cdot\nabla h = k^{p-2}hA_p(Z)\nabla h\cdot\nabla(\psi^2).$$

Integrate this mixed term by parts. Its derivative of h cancels the term $k^{p-2}|\nabla h|^2_{A_p(Z)}\psi^2$ from the square. The resulting zero-order term is

$$k^{2-p}W_p''(u) - k^\alpha \operatorname{div}(k^{p-2}A_p(Z)\nabla k^\alpha).$$

Now $\nabla h = (p-2)auhZ$, $A_p(Z)Z = cr^{p-2}Z$, and $k^{p-2}h = h^{-1}$. The product rule and (2.12) give

$$h\operatorname{div}(k^{p-2}A_p(Z)\nabla h) = c(p-2)a^2\left\{[q-(p-2)u^2](1-D) - 2cu^2D\right\}.$$

Also

$$k^{2-p}W_p''(u) = 2ca^2\big[(2p-1)u^2 - 1\big].$$

Subtracting and using $u^2 = 1-q$ gives (2.25). For the constant part, the scalar simplification is

$$2[(2p-1)u^2 - 1] - (p-2)[q-(p-2)u^2] = p^2 - p(p+1)q.$$

The coefficient of D is $(p-2)[q-(p-2)u^2 + 2cu^2] = (p-2)(p-cq)$. Thus the two pieces give exactly the recorded potential.

These product rules use only the continuous phase flux $r^{p-2}Z$ and its weak divergence $f(U)D$. They remain valid at $Z = 0$ without differentiating the unit normal. The weak phase equation also justifies integration by parts. Because $k > 0$ on the test support, the wells cause no singularity. For the planar profile, $D = 0$ and $q = G$. Since $G(s) = \operatorname{sech}^2(as)$, direct differentiation gives

$$\frac{M'}{M} = -pa\tanh(as), \qquad \frac{M''}{M} = a^2\{p^2 - p(p+1)G(s)\}.$$

This proves both identities in (2.26). □

Proposition 2.17 (Sharp weighted normal gap)**.** *Suppose $p > 1$, $g = g_p$, and $r_0 = g' > 0$. Let φ belong to the weighted form domain obtained by completing $C_c^\infty(\mathbb{R})$ in the squared norm $\int_{\mathbb{R}} r_0^{p-2}((\varphi')^2 + \varphi^2)\,ds$. The potential satisfies $|W_p''(g)| \le C_p r_0^{p-2}$, so the Jacobi form is continuous in this norm. It obeys*

$$Q_p(\varphi) := \int_{\mathbb{R}} (c r_0^{p-2}(\varphi')^2 + W_p''(g)\varphi^2)\,ds = c\int_{\mathbb{R}} r_0^p \left|\left(\frac{\varphi}{r_0}\right)'\right|^2 ds. \tag{2.27}$$

Under the orthogonality condition

$$\int_{\mathbb{R}} r_0^{p-1}\varphi\,ds = 0, \tag{2.28}$$

it satisfies

$$Q_p(\varphi) \ge ca^2(2p-1)\int_{\mathbb{R}} r_0^{p-2}\varphi^2\,ds. \tag{2.29}$$

The coefficient is sharp.

Proof. Differentiating the profile equation makes r_0 a positive zero mode. Substitute $\varphi = r_0\zeta$ and integrate the mixed term. This proves (2.27) for compact tests. The cancellation can be written as

$$c r_0^{p-2}(r_0')^2 + W_p''(g)r_0^2 - (c r_0^{p-1}r_0')' = r_0\left[-(c r_0^{p-2}r_0')' + W_p''(g)r_0\right] = 0.$$

The last equality is the differentiated profile equation. Compact support removes the boundary term when integrating the mixed term.

To pass to the form domain, note that

$$\frac{r_0'}{r_0} = -2a\tanh(as), \qquad r_0^p\left|\left(\frac{\varphi}{r_0}\right)'\right|^2 = r_0^{p-2}\left|\varphi' - \frac{r_0'}{r_0}\varphi\right|^2.$$

The bounded logarithmic derivative makes the right-hand side continuous in the stated form norm. Approximation therefore gives (2.27) in that domain. The zero mode r_0 also belongs to this domain: $\int r_0^p < \infty$, and its bounded logarithmic derivative controls the derivative term. Smooth cutoffs converge in the same norm. The mean in (2.28) is also continuous, since

$$\left|\int r_0^{p-1}\varphi\right| \le \left(\int r_0^{p-2}\varphi^2\right)^{1/2}\left(\int r_0^p\right)^{1/2}.$$

Use the variable $x = as$. After cancellation of constant factors, the required inequality is

$$\int_{\mathbb{R}} \operatorname{sech}^{2p} x\,|\zeta_x|^2\,dx \ge (2p-1)\int_{\mathbb{R}} \operatorname{sech}^{2p} x\,\zeta^2\,dx, \qquad \int_{\mathbb{R}} \operatorname{sech}^{2p} x\,\zeta\,dx = 0.$$

This is the sharp weighted Poincaré inequality of Bonnefont–Joulin–Ma [5, Theorem 3.1, p. 27]. Indeed, set $z = \sinh x$, $f(z) = \zeta(\operatorname{arcsinh} z)$, and $\vartheta = p + 1/2$. Then

$$\operatorname{sech}^{2p} x\,dx = (1+z^2)^{-\vartheta}\,dz, \qquad |\zeta_x|^2 = (1+z^2)|f'(z)|^2.$$

Their parameter satisfies $\vartheta > 3/2$. Their best constant is therefore $2(\vartheta - 1) = 2p - 1$. The zero mean removes the mean square from their variance. Their normalizing constant cancels from the inequality. Apply its variance form to compact smooth approximations φ_j. They need not satisfy the zero-mean condition exactly. With $\zeta_j = \varphi_j/r_0$, the variance term in the original variable is

$$\int r_0^p\zeta_j^2 - \frac{\left(\int r_0^p\zeta_j\right)^2}{\int r_0^p} = \int r_0^{p-2}\varphi_j^2 - \frac{\left(\int r_0^{p-1}\varphi_j\right)^2}{\int r_0^p}.$$

The second term tends to zero by the continuity of the mean and (2.28). The first term and the derivative integral converge in the form norm. Finally, $x = as$ supplies the factor a^2 in the derivative term. This gives (2.29).

For sharpness, take $\zeta(x) = \sinh x$, which corresponds to their eigenfunction $f(z) = z$. Thus $\varphi(s) = r_0(s)\sinh(as)$ gives equality. It belongs to the stated form domain, since its value and derivative have finite weighted squared integrals and smooth cutoffs converge in that norm. Their weighted squares have tails $O(e^{-2a(p-1)|s|})$, which are integrable for every $p > 1$. Its weighted mean vanishes by oddness. For $p \in \mathcal{P} \Subset (1,\infty)$, a_p is bounded above and below by positive constants, and r_0'/r_0 is uniformly bounded. Thus the transformations and form-domain estimates are uniform. The exact coefficient $ca^2(2p-1)$ has a positive minimum on $\mathcal{P}$. □

Remark 2.18. The norm on the right-hand side of (2.29) is the weighted integral $\int r_0^{p-2}\varphi^2$. It is not the unweighted L^2 norm. For $p > 2$, translating a unit L^2 bump into a profile tail sends its unweighted Jacobi energy to zero. Also, q^{p-2} does not approach one uniformly throughout the tails as $p \downarrow 2$. Consequently the full preparation argument cannot be reduced to an immediate uniformly elliptic perturbation in those tails.

The tail obstruction in the preceding remark also persists after removing the translation mode. Choose a nonzero $\psi \in C_c^\infty(\mathbb{R})$ and set

$$\psi_R(s) = \psi(s-R), \qquad \alpha_R = \frac{\int r_0^{p-1}\psi_R\,ds}{\int r_0^p\,ds}, \qquad \varphi_R = \psi_R - \alpha_R r_0.$$

Then $\int r_0^{p-1}\varphi_R = 0$ and $\alpha_R \to 0$. The polarized ground-state identity gives $Q_p(r_0,\psi_R) = 0$, so

$$Q_p(\varphi_R) = Q_p(\psi_R) \longrightarrow 0, \qquad \|\varphi_R\|_{L^2} \longrightarrow \|\psi\|_{L^2} > 0.$$

Both Jacobi coefficients tend uniformly to zero on the translated support when $p > 2$. Hence the gap cannot be changed to an unweighted one even on the orthogonal subspace.

2.5. **Normal-ray propagation.** The local phase estimates let us propagate discrepancy along the initial phase normal. For propagation, return to the standing range $\mathcal{P} \Subset (2,\infty)$. Let u be an entire solution with $-1 < u < 1$. Thus $0 \le r \le 1$ by Proposition 2.1. Write

$$A = A_p(Z) = r^{p-2}I + (p-2)r^{p-4}Z \otimes Z,$$

with its continuous value $A_p(0) = 0$. For a planar layer all the propagation conclusions hold directly, with $D \equiv 0$ and constant Z. We henceforth exclude that case. The rigidity assertion in Proposition 2.1, applied at any phase value, then gives $D > 0$ everywhere.

Apply the phase-center version of Lemma 2.12. For fixed R, if $D(x) \le d_R$, it gives $D(y) \le C_R D(x)$ whenever $|x-y| \le R$. If $D(x) > d_R$, the same conclusion holds after increasing C_R, since $D(y) \le 1$. Apply the same argument with x and y exchanged to get the reverse bound. No bound on $U(x)$ is needed. Thus, for each finite R there is C_R, uniform on $\mathcal{P}$, such that

$$C_R^{-1}D(x) \le D(y) \le C_R D(x) \qquad (|x-y| \le R). \tag{2.30}$$

On sufficiently small-discrepancy balls we also have

$$|\nabla D| \le CD, \qquad |D^2U| \le C\sqrt{D}. \tag{2.31}$$

If $D(x)$ is small enough for a fixed R, these estimates hold on all of $B_R(x)$. Their constants depend only on R, the dimension, and $\mathcal{P}$.

Define $M = q^{p/2}$, $F = MD$, and $\mathcal{H} = D^2U$, $\mathcal{S} = \operatorname{tr}(A\mathcal{H}A\mathcal{H})$. Use the exact weighted equation (2.11). Its extension across critical gradients was proved above in the standing range $p > 2$. With the present notation it is

$$-\operatorname{div}(q^pA\nabla D) + 2pca^2q^{p+1}r^pD = pq^p\mathcal{S}.$$

Conjugation on the noncritical set gives

$$-\operatorname{div}(A\nabla F) + V_D F = pM\mathcal{S} \ge 0, \tag{2.32}$$

with

$$V_D = pca^2\{p - cq + (p - 2 - cq)D\}. \tag{2.33}$$

To see this, use

$$\nabla M = -pauMZ, \qquad \Delta_p U = -2cauD,$$

and differentiate to obtain

$$\frac{\operatorname{div}(A\nabla M)}{M} = pca^2[p - (p+1)q](1 - D) + 2pc^2a^2(1 - q)D.$$

Adding $2pca^2q(1 - D)$ gives (2.33). Finally apply

$$\operatorname{div}(M^2A\nabla D) = M\operatorname{div}(A\nabla F) - F\operatorname{div}(A\nabla M)$$

to obtain (2.32). In the standing range $p > 2$, this equation also holds distributionally across critical gradients. On each compact test support, M and M^{-1} are C^1 and bounded. Use ζ/M, with $\zeta \in C_c^\infty$, in (2.11). Smooth approximation in C^1 justifies this compact C^1 test. Apply the displayed product identity. The weak phase equation gives the stated bounded expression for $\operatorname{div}(A\nabla M)/M$, while $M\mathcal{S}$ is locally integrable. Thus this passage introduces no measure at the critical set.

The potential satisfies

$$V_D \geq pca^2(1 - D) \geq 0, \qquad V_D \leq c\beta^2 + pca^2(p - 2)D. \tag{2.34}$$

For the first bound, minimize the braces in (2.33) over $0 \leq q \leq 1$. Their minimum is $1 - D$. For the second, remove the two negative terms involving q. Neither estimate requires a planar approximation to U.

The conjugated equation permits comparison on long cylinders with an arbitrarily small loss in the rate.

Lemma 2.19. *For this linear comparison, let $\mathcal{P} \Subset (1, \infty)$. Given $\varepsilon > 0$, there exist $R < \infty$, $\omega > 0$, and $C < \infty$ uniformly on $\mathcal{P}$ with the following property. Use coordinates $(y, t) \in \mathbb{R}^{n-1} \times \mathbb{R}$ and let $F > 0$ satisfy*

$$\mathcal{L}F = -\operatorname{div}(A\nabla F) + VF \geq 0 \quad \text{in } B_R^{n-1} \times (-1, s), \qquad s > 0.$$

Assume

$$V \geq 0, \quad V \leq c\beta^2 + \omega, \quad |A - \operatorname{diag}(I, c)| + |\operatorname{div} A| \leq \omega.$$

If $F(y, s) \geq c_0 F(0, s)$ on the top disk, then

$$F(0, s) \leq Cc_0^{-1}e^{(\beta+\varepsilon)s}F(0, 0). \tag{2.35}$$

Proof. Set $\kappa = \beta + \varepsilon$ and $d = c(\kappa^2 - \beta^2) > 0$. Let ϕ be the first positive Dirichlet eigenfunction of $-\Delta_y$ on B_R, normalized by $\phi(0) = \max \phi = 1$. Denote its eigenvalue by λ_R. Choose R to ensure $2\lambda_R \leq \frac{1}{4}\min_{\mathcal{P}} d$. For $n \geq 2$, the normalization at the center is justified by radial symmetry. Simplicity of the first positive eigenfunction makes it invariant under orthogonal transformations of the disk. With $m = n - 1$, its radial equation is

$$(\varrho^{m-1}\phi'(\varrho))' = -\lambda_R\varrho^{m-1}\phi(\varrho).$$

Integration from zero gives $\phi' < 0$ away from the center. Also $\lambda_R = R^{-2}\lambda_1(B_1^{n-1})$, so the required choice of R is finite and independent of the cylinder length.

Positivity and the boundary point lemma show that ϕ and its gradient never vanish together on the closed disk. Consequently

$$\phi|D^2\phi| + \phi|\nabla\phi| \leq C_R(\phi^2 + |\nabla\phi|^2).$$

For $w(y, t) = e^{\kappa(t-s)}\phi(y)^2$, the constant-coefficient operator with potential $c\beta^2$ gives

$$(-\Delta_y - c\partial_t^2 + c\beta^2)w = e^{\kappa(t-s)}\{(2\lambda_R - d)\phi^2 - 2|\nabla\phi|^2\}.$$

Differentiating w gives

$$|D^2 w| + |\nabla w| + w \le C_R e^{\kappa(t-s)}(\phi^2 + |\nabla\phi|^2).$$

Here the terms $\phi D^2\phi$ and $\phi\nabla\phi$ are controlled by the preceding bound; the other terms already have the required form. Thus the drift and matrix perturbations contribute at most $C_R\omega e^{\kappa(t-s)}(\phi^2 + |\nabla\phi|^2)$ in absolute value. Choose ω so small that $\omega < \frac{1}{2}\min\{1, \min_{p\in\mathcal{P}}(p-1)\}$. Then A is uniformly elliptic on the whole exponent interval. Decrease the tolerance further to obtain

$$\mathcal{L}w \le -e^{\kappa(t-s)}\{\tfrac{1}{2}d\phi^2 + |\nabla\phi|^2\} \le 0.$$

The square on ϕ supplies the negative gradient term. It absorbs the errors even at the lateral boundary.

Put

$$B(y,t) = e^{\kappa(t-s)}\phi(y)^2 - e^{-\kappa(s+1)}.$$

Since $V \ge 0$, we have $\mathcal{L}B \le 0$. On the bottom and sides $B \le 0$, while $B \le 1$ on the top. The comparison principle applies because A is uniformly elliptic and $V \ge 0$. It gives $F \ge c_0 F(0,s)B$. For the weak comparison, put $G = F - c_0F(0,s)B$. Its negative part G_- has zero boundary trace. Testing $\mathcal{L}G \ge 0$ with G_- gives

$$0 \le -\int A\nabla G_- \cdot \nabla G_- - \int V G_-^2 \le 0.$$

Uniform ellipticity and the zero trace force $G_- = 0$.

Evaluation at $(0,0)$ gives

$$F(0,0) \ge c_0 F(0,s)e^{-\kappa s}(1 - e^{-\kappa}).$$

The last factor has a positive lower bound on $\mathcal{P}$. When $n = 1$, use $\phi \equiv 1$. □

A continuation argument turns this cylinder comparison into a bound on the whole normal ray.

Theorem 2.20 (Normal-ray propagation)**.** *For every $\varepsilon > 0$ there exists $C_\varepsilon < \infty$ uniformly for $p \in \mathcal{P}$ as follows. Take a point z with $Z(z) \ne 0$ and define $e = Z(z)/|Z(z)|$. For every $s \in \mathbb{R}$,*

$$D(z + se) \le C_\varepsilon D(z) e^{(2\beta+\varepsilon)|s|}. \tag{2.36}$$

There is also $b_\varepsilon > 0$ such that the condition $D(z)e^{(2\beta+\varepsilon)|s|} \le b_\varepsilon$ implies

$$D(z + se) \le C_\varepsilon D(z) e^{(2\beta+\varepsilon/2)|s|}, \tag{2.37}$$

$$|Z(z + se) - e| + |U(z + se) - U(z) - s| \le C_\varepsilon D(z) e^{(2\beta+\varepsilon/2)|s|}.$$

Every intermediate point on this segment is noncritical and obeys the same bounds.

Proof. The planar case has already been settled, so assume $D > 0$. Apply Lemma 2.19 with loss $\varepsilon/2$. This fixes R and the tolerance ω. All constants below are uniform for $p \in \mathcal{P}$. Choose an orientation tolerance ω_1 and then a discrepancy threshold d_*. We require that, at any x, the conditions

$$D(x) \le d_*, \qquad |Z(x) - e| \le \omega_1,$$

imply on $B_{R+2}(x)$

$$|A - \mathrm{diag}(I, c)| + |\operatorname{div} A| \le \omega, \qquad V_D \le c\beta^2 + \omega. \tag{2.38}$$

The last coordinate is parallel to e. Indeed, (2.30) gives $D \le C_{R+2}d_*$ on the ball. Equation (2.31) gives $|D^2U| \le C_R\sqrt{d_*}$ and hence $|Z - e| \le \omega_1 + C_R\sqrt{d_*}$. On this uniformly noncritical region, smoothness of A_p gives

$$|A_p(Z) - A_p(e)| \le C|Z - e|, \qquad |\operatorname{div} A_p(Z)| \le C|D^2U|.$$

The constants are uniform because $p \in \mathcal{P}$ and r stays in a fixed compact subinterval of $(0,\infty)$. Also $A_p(e) = I + (p-2)e \otimes e$ is precisely the reference matrix. The potential is controlled by (2.34). The order is R, ω, then ω_1, then d_*.

Choose a sign $\sigma \in \{-1, 1\}$ and put $x(t) = z + \sigma te$, $t \geq 0$, and $D_0 = D(z)$. First suppose D_0 is small enough for both initial inequalities to be strict. Let T be the first time at which $D(x(t)) = d_*$ or $|Z(x(t)) - e| = \omega_1$. Set $T = \infty$ if this never happens. Before T, the discrepancy threshold keeps r away from zero. The solution and the coefficients are therefore smooth there. For $0 < s < T$, use the radius-R cylinder around $-1 < t < s$. Every point of this cylinder lies in one of the fixed balls used in (2.38). Thus the cylinder is uniformly noncritical and satisfies those bounds. The part below $t = 0$ is contained in the fixed initial ball about z. If $\sigma = -1$, reverse the normal coordinate. The reference matrix is unchanged.

At the top disk, combine (2.30) with

$$|\nabla \log M| = \beta |uZ| \leq \beta$$

to obtain

$$F(y, s) \geq C_R^{-1} e^{-\beta R} F(x(s)) \qquad (|y| \leq R).$$

The top-disk constant is independent of s and D_0. The same fixed-ball bounds hold across the closed top disk, so comparison may be applied up to that disk. Apply Lemma 2.19 to (2.32) and use $M(z)/M(x(s)) \leq e^{\beta s}$. We get

$$D(x(s)) \leq C D_0 e^{K' s}, \qquad K' = 2\beta + \varepsilon/2, \qquad s < T. \tag{2.39}$$

Within this stopped cylinder, no smallness condition on $D_0 e^{K' s}$ was needed.

For the direction along the stopped ray, write $E(t) = |Z(x(t)) - e|$. The identities

$$\mathcal{H} Z = -\frac{\nabla D}{p r^{p-2}}, \qquad \mathcal{H} e = \mathcal{H} Z + \mathcal{H}(e - Z),$$

and (2.31) give, on $[0, T)$,

$$E(t) \leq C D_0 + C \int_0^t D(x(v))\, dv + C \int_0^t \sqrt{D(x(v))}\, E(v)\, dv.$$

Put $K = 2\beta + \varepsilon$. For $t < T$ with $D_0 e^{Kt} \leq b$, equation (2.39) yields

$$\int_0^t \sqrt{D(x(v))}\, dv \leq C\sqrt{b}, \qquad \int_0^t D(x(v))\, dv \leq C D_0 e^{K' t}.$$

These integrals have no factor proportional to the segment length. Indeed, integrating the exponential in (2.39) gives $\int_0^t e^{K' v/2}\, dv \leq 2e^{K' t/2}/K'$, and K' has a positive uniform lower bound. Since $K' < K$, the factor $\sqrt{D_0} e^{K' t/2}$ is at most $\sqrt{b}$. The integral of D is handled in the same way. Since $E(0) = 1 - r(z) \leq D_0$, Gronwall's inequality gives

$$E(t) \leq C D_0 e^{K' t}, \qquad |U(x(t)) - U(z) - \sigma t| \leq C D_0 e^{K' t}. \tag{2.40}$$

For $0 < b \leq 1$ these constants do not depend on t, D_0, b. The phase estimate follows by integrating $|\frac{d}{dt} U(x(t)) - \sigma| \leq E(t)$. The positive lower bound for K' on $\mathcal{P}$ bounds this integral by the same exponential.

To close the stopping argument, choose $b = b_\varepsilon$ small enough that $Cb < \min(d_*, \omega_1)/2$. Require also that the initial conditions hold for $D_0 \leq b$. If $T \leq K^{-1} \log(b/D_0)$, extend (2.39) and (2.40) by continuity to T. They contradict the first equality defining the stopping time. The stopped interval therefore contains the whole segment with $D_0 e^{Kt} \leq b$. This proves (2.37). The same argument applies at every earlier time, since $D_0 e^{Kt}$ increases with t. The discrepancy stays below $d_* < 1$ there, so the whole segment is noncritical.

Outside that segment, or when $D_0 > b$, use $D \leq 1$ to obtain $D(x(t)) \leq b^{-1} D_0 e^{Kt}$. Combine this with (2.39) on the accurate segment. The result is (2.36). The argument works for either sign. □

Corollary 2.21 (A fixed cone about the initial normal)**.** *For each $\varepsilon > 0$ choose $\eta_\varepsilon, b_\varepsilon > 0$ and $C_\varepsilon < \infty$ uniformly on $\mathcal{P}$. Suppose $Z(z) \neq 0$, $e = Z(z)/|Z(z)|$, and the unit vector v satisfies $|v - e| \le \eta_\varepsilon$. Then*

$$D(z + sv) \le C_\varepsilon D(z) e^{(2\beta+\varepsilon)|s|} \qquad (s \in \mathbb{R}).$$

If $D(z)e^{(2\beta+\varepsilon)|s|} \le b_\varepsilon$, the estimate holds with the smaller exponent $2\beta + \varepsilon/2$. Under this same smallness condition, on the whole segment one also has

$$|Z(z + sv) - v| \le C\eta_\varepsilon + C_\varepsilon D(z) e^{(2\beta+\varepsilon/2)|s|}. \tag{2.41}$$

For any prescribed angle $\theta > 0$, the parameters can be chosen so that the right-hand side is less than θ. The same propagation holds for $|v + e| \le \eta_\varepsilon$ after reversing v.

Proof. Use the cylinder and stopping argument from Theorem 2.20, now with axis v and reference matrix $I + (p-2)v \otimes v$. Stop when either $D < d_*$ or $|Z - v| < \omega_1$ fails. The same fixed-ball estimates give the coefficient bounds on the entire stopped cylinder, with constants independent of its length. They give $D(z + \sigma tv) \le CD_0 e^{K't}$ for either sign σ, where $K' = 2\beta + \varepsilon/2$.

The initial angle now obeys $E(0) \le \eta_\varepsilon + CD_0$. The integral inequality and Gronwall estimate give, under $D_0 e^{Kt} \le b$,

$$E(t) \le (\eta_\varepsilon + CD_0 e^{K't}) e^{C\sqrt{b}}.$$

First choose η_ε so that $\eta_\varepsilon e^C < \omega_1/4$. Then decrease $b \le 1$ until the displayed bound is less than $\omega_1/2$ and the discrepancy is less than $d_*/2$. The preceding first-exit argument applies. The conjugating factors still satisfy $M(z)/M(z + sv) \le e^{\beta|s|}$. This bound is independent of the phase error on the tilted ray, even when that error grows with its length. Beyond the accurate segment, use $D \le 1$ again. To obtain a prescribed final angle, first decrease η_ε and then decrease b. □

Corollary 2.22 (Opposing near-normal chords)**.** *Fix $\varepsilon > 0$. Choose $\eta_\varepsilon, c_\varepsilon > 0$ uniformly on $\mathcal{P}$. For distinct regular zeros z_-, z_+ put $\ell = |z_+ - z_-|$ and $v = (z_+ - z_-)/\ell$. If their oriented normals satisfy*

$$|\nu(z_-) - v| + |\nu(z_+) + v| \le \eta_\varepsilon,$$

then

$$D(z_-) + D(z_+) \ge c_\varepsilon e^{-(\beta+\varepsilon)\ell}. \tag{2.42}$$

In particular, the bound applies to prepared closest-foot chords once their endpoint angles are small enough.

Proof. Use Corollary 2.21 with loss 2ε and accurate-segment angle less than $1/2$. Choose η_ε below that corollary's cone aperture. Suppose the left side of (2.42) is less than $b_{2\varepsilon} e^{-(\beta+\varepsilon)\ell}$. For each endpoint, multiplication of its defect by $e^{(2\beta+2\varepsilon)\ell/2}$ gives a number below $b_{2\varepsilon}$. Thus both rays are accurate as far as their common midpoint, using the directions v and $-v$, respectively. By (2.41), the gradient there is within $1/2$ of both v and $-v$. This is impossible. We may take $c_\varepsilon = b_{2\varepsilon}$. □

Combining propagation with clearing makes the normal tails integrable.

Corollary 2.23. *Let $s \mapsto z + s\nu(z)$ range over the normal interval of a prepared tube, and put $D_0 = D(z) > 0$. Assume that on this interval, for every sufficiently small fixed clearing loss $\xi > 0$, $q(z + s\nu(z)) \le C_\xi e^{-(2a-\xi)|s|}$, and that its normal Jacobian satisfies $0 \le J \le C$. All integrals below are over this interval, with the displayed restriction $|s| > P$ where applicable. Choose the propagation and clearing losses sufficiently small. Then there are constants $C < \infty$ and $\gamma_{\mathrm{tail}} > 0$, uniform for $p \in \mathcal{P}$, such that*

$$\int q^p D_0^{1/2} D^{1/2} J \, ds \le CD_0, \qquad \int_{|s|>P} q^p (D/D_0)^{1/2} J \, ds \le Ce^{-\gamma_{\mathrm{tail}} P},$$

and

$$\int q^{p+1} DJ\, ds \le C D_0.$$

For a planar layer the unnormalized integrals vanish; the normalized tail estimate is needed only when $D_0 > 0$.

Proof. Apply (2.36) on the exact ray from its base. Write the clearing loss in its stated form, so that $q^p \le C_\xi e^{-(2\beta - p\xi)|s|}$. For the first two integrals, propagation bounds their integrands by $CD_0 e^{-(\beta - p\xi - \varepsilon/2)|s|}$ and $Ce^{-(\beta - p\xi - \varepsilon/2)|s|}$, respectively. For the nonfractional integral the bound is $CD_0 e^{-(2a-(p+1)\xi-\varepsilon)|s|}$. Choose $\xi, \varepsilon > 0$ so that both decay rates $\beta - p\xi - \varepsilon/2$ and $2a - (p+1)\xi - \varepsilon$ have positive lower bounds on $\mathcal{P}$. Take γ_{tail} below the positive uniform lower bound for the first decay rate. The Jacobian bound changes only the constants. □

3. Physical surface stability

Fix $\mathcal{P} = [p_-, p_+] \Subset (2, \infty)$. The estimates in this section are uniform on this interval. Their constants need not stay bounded as its upper endpoint tends to infinity. Use the phase, surface, and mass notation of Section 2 for matched solutions in the strict range. Write $G = G_p$. The fixed-core arguments use the noncritical estimates from that section. The ambient tests may also cross a normal turn; there we use the completed weighted form domain and its weak identities.

3.1. **Linear mean estimates.** The mean-curvature bound must retain a favorable curvature term. We adapt the positive-source decomposition and extreme-ray argument of [35, Lemma 5.4]. The normal operator depends on the exponent, and the comparison below also retains a transverse Laplacian term. For the two linear results below, allow $\mathcal{P} = [p_-, p_+] \Subset (1, \infty)$. For $Y \in \mathbb{R}^d$, with $d \ge 1$, define

$$\mathscr{L}_p = -\Delta_Y - \partial_x^2 + p^2 - p(p-1)\operatorname{sech}^2 x, \qquad j_p = \frac{\sqrt{\pi}\Gamma(p)}{\Gamma(p+1/2)}.$$

For $L > 0$ the mass as a function of Y is

$$\mathcal{M}_L(F)(Y) = \int_{-L}^{L} \operatorname{sech}^{p+2} x\, F(Y, x)\, dx.$$

For $0 \le b < 1/2$ set

$$R_p(b) = \frac{\Gamma(p-b+1)\Gamma(p-b)\Gamma(1+b)\Gamma(1/2-b)}{p\Gamma(p)\Gamma(p-b+1/2)\Gamma(1-b)}, \quad c_p(b) = \frac{j_p}{2R_p(b)}.$$

Extend by $c_p(1/2) = 0$. Gamma recurrence gives $c_p(0) = 1/2$. The function $c_p(b)$ is jointly smooth at both ends of the b interval. At $b = 1/2$ this follows from the simple zero of $1/\Gamma(1/2 - b)$. Define

$$\lambda_p(b) = 4b(p-b), \qquad K_{\mathcal{P}} := \sup_{\substack{p \in \mathcal{P} \\ 0 < b \le 1/2}} \frac{(c_p(b) - 1/2)_+}{\lambda_p(b)}.$$

This constant is finite and nonnegative. Indeed, if p_- is the left endpoint of $\mathcal{P}$, the mean-value theorem gives

$$(c_p(b) - 1/2)_+ \le b \sup_{\mathcal{P} \times [0,1/2]} |\partial_b c_p|, \qquad \lambda_p(b) \ge 4b(p_- - 1/2).$$

Thus no uniform control as $p \to \infty$ is needed.

Theorem 3.1. *For every $\varepsilon > 0$ choose $L_0 < \infty$ uniformly for $p \in \mathcal{P}$. For every $L \ge L_0$, let $F \ge 0$ be a smooth entire solution of*

$$\mathscr{L}_p F = S(Y)\operatorname{sech}^p x, \qquad S \ge 0, \qquad \Delta_Y S \ge 0.$$

Then

$$\frac{j_p}{2}|\partial_x F(Y, 0)| \le (1/2 + \varepsilon)\mathcal{M}_L(F)(Y) + K_{\mathcal{P}} \Delta_Y \mathcal{M}_L(F)(Y). \tag{3.1}$$

Proof. Begin with $\mathscr{L}_p F = 0$ and $F > 0$. Normalize by $F(0,0) = 1$ and call the resulting convex set $\mathcal{K}_p$. To prove compactness of $\mathcal{K}_p$ under smooth convergence on compact sets, use Harnack chains from the origin. They give uniform bounds on each compact set. Interior elliptic estimates then bound every fixed derivative there. Diagonal extraction gives compactness in C^∞_{loc}, a locally convex space. A limit is nonnegative, solves the same equation, and is still one at the origin; the strong maximum principle makes it positive. Thus the normalized set is closed as well. The topology used here is metrizable, for example by taking a countable exhaustion by compact balls and the seminorms of derivatives on them. Thus the sequential compactness just proved is compactness in the topology needed for the convex-hull argument.

Let F be an extreme point and take $h \in \mathbb{R}^d$. A horizontal Harnack chain gives

$$F(Y + h, x) \le C_h F(Y, x) \qquad \text{for all } (Y, x).$$

The constant is independent of the base point. This uses boundedness of the potential and its independence of Y. Subtracting a sufficiently small multiple of the translated function from F leaves a nonnegative solution. Normalize the two summands at the origin. Extremality forces every horizontal translate to be proportional to F. Indeed, choose $0 < \alpha < (2C_h)^{-1}$ and split

$$F = \alpha F(\cdot + h, \cdot) + [F - \alpha F(\cdot + h, \cdot)].$$

After normalization at the origin, the two convex weights are $\alpha F(h, 0)$ and $1 - \alpha F(h, 0)$, both strictly between zero and one. Extremality therefore gives $F(Y + h, x) = F(h, 0)F(Y, x)$. The factors form a continuous positive character of $\mathbb{R}^d$. Taking logarithms gives a continuous additive function, hence a linear function of h. Thus

$$F(Y, x) = e^{\xi \cdot Y} f(x), \qquad -f'' + [\mu^2 - p(p-1)\operatorname{sech}^2 x]f = 0, \qquad \mu^2 = p^2 - |\xi|^2. \tag{3.2}$$

At first μ^2 is merely real. To find its range, put $\nu = p - 1$ and factor

$$-\partial_x^2 - \nu(\nu+1)\operatorname{sech}^2 x + \nu^2 = (-\partial_x + \nu \tanh x)(\partial_x + \nu \tanh x). \tag{3.3}$$

The positive null function $w = \operatorname{sech}^\nu x$ lies in $L^2(\mathbb{R})$. The positive solution f gives a nonnegative ground-state form on compact tests. Indeed, for $\varphi \in C_c^\infty(\mathbb{R})$, integration by parts gives

$$\int_{\mathbb{R}} \{|\varphi'|^2 + [\mu^2 - \nu(\nu+1)\operatorname{sech}^2 x]\varphi^2\}\, dx = \int_{\mathbb{R}} f^2 \left|\left(\frac{\varphi}{f}\right)'\right|^2 dx \ge 0.$$

Insert $\varphi = \chi_R w$, where χ_R is one on $[-R, R]$, vanishes outside $[-2R, 2R]$, and $|\chi_R'| \le C/R$. Factorization (3.3) makes the left side equal to

$$(\mu^2 - \nu^2) \int_{\mathbb{R}} \chi_R^2 w^2\, dx + \int_{\mathbb{R}} |\chi_R'|^2 w^2\, dx.$$

Let $R \to \infty$. We obtain $\mu^2 \ge \nu^2$. Together with $\mu^2 \le p^2$, this gives

$$p - 1 \le \mu \le p, \qquad b = (p - \mu)/2 \in [0, 1/2], \qquad |\xi|^2 = 4b(p - b) = \lambda_p(b).$$

At $\mu = \nu$, reduction of order writes every solution as $w(x)[C_1 + C_2 \int_0^x w(t)^{-2}\, dt]$. Positivity at both ends forces $C_2 = 0$.

For $\mu > p - 1$, the solution decaying at the left end is

$$f_\mu(x) = e^{\mu x}\, {}_2F_1\left(1 - p, p; 1 + \mu; \frac{1 + \tanh x}{2}\right). \tag{3.4}$$

The hypergeometric equation verifies (3.2). For a direct check, set $z = (1 + \tanh x)/2$ and $f = e^{\mu x} H(z)$. Then $z' = 2z(1 - z)$ and $z'' = 4z(1 - z)(1 - 2z)$. The normal equation becomes

$$z(1 - z)H'' + (1 + \mu - 2z)H' + p(p - 1)H = 0.$$

These are the Gauss parameters $1 - p, p, 1 + \mu$ in the displayed formula. The solution takes value one at $z = 0$.

It also gives $f_\mu(x) \sim e^{\mu x}$ as $x \to -\infty$. This solution is positive. Otherwise its first zero produces a nonzero half-line Dirichlet test with zero quadratic-form energy. But (3.3) bounds that energy below by $[\mu^2 - (p-1)^2] \int f_\mu^2 > 0$. The decaying left asymptotic justifies both integration and truncation. Gauss's formula gives

$$\lim_{x\to+\infty} e^{-\mu x} f_\mu(x) = \frac{\Gamma(1+\mu)\Gamma(\mu)}{\Gamma(p+\mu)\Gamma(1+\mu-p)} > 0.$$

Hence $f_\mu(x)$ and $f_\mu(-x)$ are independent. The growing coefficients at the two ends show that a positive solution has nonnegative coefficients in this basis. If both coefficients were positive, the separated solution would split into two independent positive homogeneous solutions. Extremality therefore leaves only one of these two normal factors. The endpoint function is $f_{p-1} = 2^{1-p}\operatorname{sech}^{p-1} x$.

We next compute the mass with its exact normalization. Put $A = p - b = (p+\mu)/2$ and define

$$M_p(\mu) = \int_{\mathbb{R}} \operatorname{sech}^{p+2} x \, f_\mu(x)\, dx, \qquad M_{0,p}(\mu) = \int_{\mathbb{R}} \operatorname{sech}^{p} x \, f_\mu(x)\, dx.$$

Both integrals are finite for $b > 0$: the worst tail of f_μ is $e^{\mu|x|}$, whereas $\operatorname{sech}^p x$ decays as $e^{-p|x|}$ and $p - \mu = 2b > 0$. Apply the normal operator in (3.2) to $\operatorname{sech}^p x$ and integrate by parts. This gives

$$M_p(\mu) = \frac{p^2-\mu^2}{2p} M_{0,p}(\mu) = \frac{2Ab}{p} M_{0,p}(\mu). \tag{3.5}$$

The condition $\mu < p$ makes the boundary terms vanish. The differentiated test factor satisfies

$$(\operatorname{sech}^p x)'' = [p^2 - p(p+1)\operatorname{sech}^2 x]\operatorname{sech}^p x.$$

Applying the normal operator to it therefore gives $(\mu^2 - p^2)\operatorname{sech}^p x + 2p\operatorname{sech}^{p+2} x$. Integration against f_μ transfers that operator to f_μ, where it vanishes, and gives the stated mass reduction.

With $z = (1+\tanh x)/2$ we obtain

$$M_{0,p}(\mu) = 2^{p-1} \int_0^1 z^{A-1}(1-z)^{b-1} {}_2F_1(1-p, p; 1+\mu; z)\, dz.$$

The hypergeometric series converges absolutely at one, with excess $\mu > 0$. The sum of absolute coefficients bounds the series uniformly for $0 \le z \le 1$. Since $A, b > 0$, this bound times the beta weight is integrable. Dominated convergence therefore permits termwise integration. For the kth term of the Gauss series, the beta integral is

$$\int_0^1 z^{A+k-1}(1-z)^{b-1}\, dz = \frac{\Gamma(A)\Gamma(b)}{\Gamma(p)} \frac{(A)_k}{(p)_k}.$$

Here $(A)_k = \Gamma(A+k)/\Gamma(A)$ is the rising factorial. This is the factor that cancels the numerator parameter p.

Because $A + b = p$, the parameter p cancels, leaving

$$2^{p-1} \frac{\Gamma(A)\Gamma(b)}{\Gamma(p)} {}_2F_1(1-p, A; 1+\mu; 1).$$

Gauss's formula has positive excess $A > 0$, and gives

$$M_{0,p}(\mu) = \frac{2^{p-1}\Gamma(A)^2\Gamma(b)\Gamma(1+\mu)}{\Gamma(p)\Gamma(2A)\Gamma(1-b)}. \tag{3.6}$$

To compute the derivative, use the Ferrers representation

$$f_\mu(x) = \Gamma(1+\mu)\mathsf{P}_{p-1}^{-\mu}(-\tanh x).$$

The factor $\Gamma(1+\mu)$ converts its regularized hypergeometric function to the ordinary ${}_2F_1$ in (3.4). The formula

$$\frac{d}{dt}\mathsf{P}_\nu^m(t)\Big|_{t=0} = -\frac{2^{m+1}\sqrt{\pi}}{\Gamma((\nu-m+1)/2)\Gamma((-\nu-m)/2)}$$

then yields

$$f'_\mu(0) = \frac{2^{1-\mu}\sqrt{\pi}\,\Gamma(1+\mu)}{\Gamma(A)\Gamma(1/2-b)}. \tag{3.7}$$

These identities are [34, 14.3.1, 14.3.3, 14.5.2, and 15.4.20]. Combine (3.5)–(3.7) with gamma duplication.

The duplication identity used at this point is

$$\Gamma(2A) = \frac{2^{2A-1}}{\sqrt{\pi}}\Gamma(A)\Gamma(A+1/2).$$

Also $b\Gamma(b) = \Gamma(1+b)$ and $2A = p+\mu$. These relations cancel every power of two in the mass-to-derivative ratio. The exact ratio is

$$\frac{M_p(\mu)}{f'_\mu(0)} = \frac{\Gamma(A+1)\Gamma(A)\Gamma(1+b)\Gamma(1/2-b)}{p\Gamma(p)\Gamma(A+1/2)\Gamma(1-b)} = R_p(b). \tag{3.8}$$

Continuity extends it to $b = 0$ and gives $R_p(0) = j_p$. Here one may normalize by $f_\mu(0) = 1$, since the ratio is unchanged. The normal derivative bound proved below then bounds the integrand by a constant times $e^{-2|x|}$, uniformly on $\mathcal{P}$. This justifies continuity of the integral at that endpoint. At $b = 1/2$ the mass is positive and the derivative vanishes. Reflection preserves the mass and reverses the derivative. Every extreme normal factor therefore satisfies

$$\frac{j_p}{2}|f'(0)| = c_p(b)\int_{\mathbb{R}} \operatorname{sech}^{p+2} x\, f(x)\,dx.$$

This includes the ground-state endpoint, where both the left side and $c_p(1/2)$ vanish. Consequently

$$\frac{j_p}{2}|f'(0)| \le (1/2 + K_{\mathcal{P}}\lambda_p(b))\int_{\mathbb{R}} \operatorname{sech}^{p+2} x\, f(x)\,dx.$$

To control the discarded tail, normalize by $f(0) = 1$. The positive normal equation gives $|f'| \le pf$ and hence $e^{-p|x|} \le f(x) \le e^{p|x|}$. Indeed, $f'' \le p^2 f$. If $f'(x_0) > pf(x_0)$, put $h = f' - pf$. The inequality $h' + ph \le 0$ gives $h(x) \ge h(x_0)e^{p(x_0-x)}$ for $x < x_0$. Integrating $(e^{-px}f)' = e^{-px}h$ backwards then makes f negative far to the left, a contradiction. Apply the same argument to $f(-x)$ to exclude $f'(x_0) < -pf(x_0)$. The mass on $[-1,1]$ has a fixed positive lower bound $c_* > 0$ on the compact exponent interval. The discarded tail satisfies

$$\int_{|x|>L} \operatorname{sech}^{p+2} x\, f(x)\,dx \le Ce^{-2L}.$$

Also $0 \le \lambda_p(b) \le 2p_+ - 1$. For example, the central lower bound may be taken as $c_* = 2e^{-p_+}\operatorname{sech}^{p_+^2+2}(1)$, and the tail constant as $C = 2^{p_+^2+2}$. These follow from the two exponential bounds for f and $\operatorname{sech} x \le 2e^{-|x|}$. In particular the window choice is uniform without any growth assumption on the full solution.

Choose $L_0 \ge 1$ so that

$$[1/2 + K_{\mathcal{P}}(2p_+ - 1)]Ce^{-2L_0} \le \varepsilon c_*.$$

For $L \ge L_0$ it follows that

$$\frac{j_p}{2}|f'(0)| \le [1/2 + \varepsilon + K_{\mathcal{P}}\lambda_p(b)]\int_{-L}^{L} \operatorname{sech}^{p+2} x\, f(x)\,dx.$$

The transverse Laplacian of an extreme mass equals $\lambda_p(b)$ times that mass. Thus this is (3.1) at the origin.

To extend the bound beyond the extreme points, express the absolute-value inequality as two signed affine inequalities. Normal differentiation, finite-window mass, and its transverse Laplacian are continuous affine functionals under smooth convergence on compact sets. The Krein–Milman theorem says that the compact convex set $\mathcal{K}_p$ is the closed convex hull of its extreme points. It

therefore extends both signed inequalities to every normalized positive homogeneous solution, and scaling gives the result for every such F. Horizontal translation then gives the result at every Y.

For the source term, construct its minimal nonnegative Green potential. The potential of $\mathscr{L}_p$ is at least $p > 0$. Solve the zero-boundary Dirichlet problem on a nested exhaustion by smooth domains symmetric in x. Positivity of the potential gives the maximum principle. The solutions are nonnegative, increase with the domains, and are bounded above by F, which is a supersolution with nonnegative boundary values. Interior estimates therefore give a locally smooth limit, the minimal nonnegative Green potential G, with

$$0 \le G \le F, \qquad \mathscr{L}_p G = S(Y)\operatorname{sech}^p x, \qquad G(Y,-x) = G(Y,x).$$

The bound by F makes this limit finite on every compact set. If $W \ge 0$ is any supersolution with the same source, comparison on each exhaustion domain gives $G_R \le W$. Passing to the limit gives $G \le W$. Applying this observation to the limits from two exhaustions proves independence of the exhaustion as well as minimality.

Write the full minimal Green kernel as $\Gamma_p(z,x,t)$. It exists because the potential is at least p: the increasing Dirichlet kernels are bounded by the Green kernel of $-\Delta + p$. Minimality makes this kernel invariant under horizontal translations and rotations, and monotone convergence in the Dirichlet representation gives

$$G(Y,x) = \int_{\mathbb{R}^d}\int_{\mathbb{R}} \Gamma_p(z,x,t) S(Y-z)\operatorname{sech}^p t\, dt\, dz.$$

The bound $G \le F$ proves finiteness of this integral even when S has unrestricted growth. Take spherical averages in Y. Tonelli's theorem and the submean inequality for S give, for every $r > 0$,

$$\frac{1}{|\partial B_r|}\int_{\partial B_r} G(Y+z,x)\, dS_z \ge G(Y,x).$$

Indeed, for each fixed z,t, the submean inequality gives $\fint_{\partial B_r} S(Y-z+\omega)\, dS_\omega \ge S(Y-z)$. Multiply by $\Gamma_p(z,x,t)\operatorname{sech}^p t$ and integrate in z,t. Tonelli's theorem interchanges these nonnegative integrals, so it yields the displayed submean inequality for G. The averages are finite by $G \le F$ and local boundedness. Thus $\Delta_Y G \ge 0$ distributionally, and then pointwise by regularity. In particular,

$$\mathcal{M}_L(G) \ge 0, \qquad \Delta_Y \mathcal{M}_L(G) \ge 0, \qquad \partial_x G(Y,0) = 0.$$

The function $F - G$ is nonnegative and homogeneous. Apply the homogeneous estimate to it, allowing also the zero solution. Add the two nonnegative contributions of G. This proves (3.1). □

Remark 3.2. The source must be subharmonic for this strengthened conclusion. Evenness in x suffices for the scalar coefficient $\chi_p = \max_b c_p(b)$. It does not imply that the transverse Laplacian of the Green mass is nonnegative. For the nonlinear application below, it is enough to obtain a source $S(Y) = \gamma|T(Y)|^2$, where $\gamma \ge 0$ is constant and each component of the finite-dimensional vector T is harmonic. Then $\Delta_Y S = 2\gamma|\nabla_Y T|^2 \ge 0$, which gives the needed sign. The nonlinear argument below constructs this source and proves harmonicity.

Lemma 3.3. *In Theorem 3.1, suppose $S(Y) = \gamma|T(Y)|^2$, with constant $\gamma \ge 0$ and a finite-dimensional vector T whose components are harmonic. Interpret the conclusion as $0 \ge 0$ when $F \equiv 0$. Otherwise the strong maximum principle gives $F > 0$. For every fixed $L > 0$, the mass $J = \mathcal{M}_L(F)$ obeys*

$$\Delta_Y J \ge \frac{|\nabla_Y J|^2}{2J}. \tag{3.9}$$

Equivalently, $\sqrt{J}$ is subharmonic.

Proof. Use the preceding decomposition $F = F_{\mathrm{hom}} + G$. For a finite positive sum of homogeneous extreme rays the mass is

$$J_{\mathrm{hom}}(Y) = \sum_i A_i e^{\xi_i \cdot Y}, \qquad A_i \geq 0.$$

Cauchy–Schwarz yields

$$|\nabla J_{\mathrm{hom}}|^2 \leq J_{\mathrm{hom}} \sum_i A_i |\xi_i|^2 e^{\xi_i \cdot Y} = J_{\mathrm{hom}} \Delta_Y J_{\mathrm{hom}}.$$

Krein–Milman approximates every normalized positive homogeneous solution by finite convex combinations, smoothly on compact sets. The finite-window masses converge in C^2. The same inequality therefore holds for every J_{hom}.

For the source mass, integration over the normal variables in the Green formula gives

$$J_G(Y) = \int_{\mathbb{R}^d} K_L(z) |T(Y - z)|^2 \, dz.$$

Here K_L is a nonnegative kernel independent of Y. Its normalization includes γ. More explicitly, Tonelli's theorem gives

$$K_L(z) = \gamma \int_{-L}^{L} \int_{\mathbb{R}} \operatorname{sech}^{p+2} x \, \Gamma_p(z, x, t) \operatorname{sech}^p t \, dt \, dx.$$

All factors are nonnegative, so the normal integrations can be performed before the horizontal one. Finiteness of the resulting mass follows from the pointwise bound $G \leq F$. The kernel measure itself has finite mass. For source $\operatorname{sech}^p x$, the constant $1/p$ is a supersolution, since the potential is at least p. Green comparison therefore gives

$$\int_{\mathbb{R}^d} K_L(z) \, dz \leq \frac{\gamma}{p} \int_{-L}^{L} \operatorname{sech}^{p+2} x \, dx < \infty.$$

Thus $\sqrt{J_G(Y)}$ is the norm of $T(Y - \cdot)$ in $L^2(K_L(z)\, dz; \mathbb{R}^k)$. First restrict this norm to $|z| \leq R$. The componentwise harmonic mean-value formula is an identity in that Hilbert space: the harmonic vector is bounded on the compact sets now used, and the kernel measure is finite. Minkowski's inequality then gives

$$\|T(Y - \cdot)\|_{L^2(K_L; |z| \leq R)} \leq \fint_{\partial B_r} \|T(Y + \omega - \cdot)\|_{L^2(K_L; |z| \leq R)} \, dS_\omega.$$

Letting $R \to \infty$ gives

$$\sqrt{J_G(Y)} \leq \frac{1}{|\partial B_r|} \int_{\partial B_r} \sqrt{J_G(Y + \omega)} \, dS_\omega.$$

Monotone convergence applies on the left, and the right-hand side is dominated by the locally bounded function $\sqrt{J_G(Y + \omega)}$. This proves subharmonicity without first differentiating the Green integral. The mass J_G is smooth by the equation and integration over the fixed normal window. Where $J_G > 0$, smoothness gives

$$\Delta_Y J_G \geq \frac{|\nabla_Y J_G|^2}{2 J_G}.$$

At a zero of the smooth nonnegative function J_G, its gradient vanishes. The claimed subharmonicity remains a distributional statement there. For an explicit regularization, append the constant component $\sqrt{\delta}$ to the Hilbert-space vector. Its norm is $\sqrt{J_G + \delta}$, so the same mean-value argument applies. Let $\delta \downarrow 0$ to recover $\sqrt{J_G}$.

Cauchy–Schwarz shows that the property is preserved by addition:

$$\frac{|\nabla(J_{\mathrm{hom}} + J_G)|^2}{J_{\mathrm{hom}} + J_G} \leq \frac{|\nabla J_{\mathrm{hom}}|^2}{J_{\mathrm{hom}}} + \frac{|\nabla J_G|^2}{J_G}.$$

Apply this to the two parts of the mass to prove (3.9). Terms that vanish are omitted. □

3.2. **Nonlinear mean estimates.** We adapt the normalized-limit and finite-window arguments in [35, Lemmas 5.3 and 5.6]. We use the exponent-dependent phase equations and linear estimate proved above.

Return to the standing range $\mathcal{P} \Subset (2,\infty)$. Let u be an entire solution of the matched equation with $|u| < 1$. The gradient bound is $D \geq 0$. Near a zero of small defect, relative regularity gives, for fixed R, k,

$$\|D\|_{C^k(B_R(z))} \leq C_{R,k} D(z), \qquad \|D^2 U\|_{C^k(B_R(z))} \leq C_{R,k}\sqrt{D(z)} \tag{3.10}$$

provided $D(z)$ is small enough. The whole ball is then noncritical. The limiting argument uses these relative estimates, the exact phase equation (2.12), the conjugated discrepancy equation (2.32), and the planar equality case. Use the physical normal mass I_P defined in (2.1). We compute its normalization explicitly. The changes $x = as$ and $t = \tanh x$, followed by $v = t^2$ on the positive half of the integral, give

$$\begin{aligned} aJ_p &= \int_{\mathbb{R}} \operatorname{sech}^{2p} x\, dx = \int_{-1}^{1} (1-t^2)^{p-1}\, dt \\ &= \int_0^1 v^{-1/2}(1-v)^{p-1}\, dv = B(1/2,p) = \frac{\sqrt{\pi}\,\Gamma(p)}{\Gamma(p+1/2)} = j_p. \end{aligned}$$

Here B is Euler's beta integral. Its parameters $1/2$ and p are positive for every $p > 1$, so the beta–gamma identity applies. Since $a > 0$, this proves $J_p = j_p/a$. For fixed P and sufficiently small zero defect, this mass is smooth near the zero because every normal segment in its definition stays noncritical. When $n = 1$, the regular zero set has dimension zero. Its curvature, surface gradient, and surface Laplacian vanish. The conclusions below then follow from nonnegativity and the previously proved bounds for I_P. In the remaining proof take $n \geq 2$ and $d = n-1$.

Theorem 3.4. *For each $\varepsilon > 0$ choose $P < \infty$ and $\delta > 0$ uniformly for $p \in \mathcal{P}$. At every zero satisfying $D(z) < \delta$,*

$$\boxed{\beta|H(z)| \leq (1/2+\varepsilon) I_P(z) - (1/2-\varepsilon) b(z)^2 + \frac{K_{\mathcal{P}}}{ca^2}\Delta_\Sigma I_P(z).} \tag{3.11}$$

Proof. In dimension one, the regular zero sheet is zero-dimensional. Thus $H = b = 0$ and $\Delta_\Sigma I_P = 0$, while $I_P \geq 0$ by the gradient bound. The estimate is immediate in that case. Assume the ambient dimension is at least two. Use Theorem 3.1 with $\varepsilon/4$. Choose P so that aP contains its window for every $p \in \mathcal{P}$. Enlarge P now until the source fraction θ_P computed below satisfies $(1/2+\varepsilon/4)\theta_P \geq 1/2 - \varepsilon/2$ uniformly on $\mathcal{P}$. This choice depends only on ε and the exponent interval and is made before choosing a solution or a contrary sequence. Assume no positive δ works. Translate violating zeros to the origin and rotate their normals into the last direction. Take $\epsilon_j = D_j(0) \to 0$ and $p_j \to p_\infty \in \mathcal{P}$. A zero-defect solution is planar and cannot violate the estimate. For each fixed ball and derivative order, relative regularity bounds D_j/ϵ_j and $D^2U_j/\sqrt{\epsilon_j}$ uniformly. At the origin, $\nabla U_j = (1-\epsilon_j)^{1/p_j} e_s$. On each fixed ball, integration of the Hessian bound along segments gives $\nabla U_j - e_s = O(\sqrt{\epsilon_j})$. A second integration, using $U_j(0) = 0$, gives $U_j - s = O(\sqrt{\epsilon_j})$ there. The fixed derivative bounds therefore give, after diagonal extraction, smooth convergence on fixed compact sets:

$$U_j \to s, \qquad D_j/\epsilon_j \to \rho, \qquad D^2U_j/\sqrt{\epsilon_j} \to T, \qquad \rho(0) = 1.$$

In particular, $\rho \geq 0$. The extraction is taken over arbitrarily large fixed balls. Each such ball is noncritical for all sufficiently large j.

We retain enough fixed derivatives to include two tangential derivatives of the normal mass. Denote the limiting exponent and constants by p, c, a. The identity $\nabla D = -p|\nabla U|^{p-2} D^2U\,\nabla U$ and (3.10) imply $Te_s = 0$. Symmetry of third derivatives gives $\partial_s T_{ij} = \partial_i T_{sj} = 0$. Thus T does not depend on s. On the noncritical set, $\Delta_p U = r^{p-2}[\operatorname{tr} D^2U + (p-2)D^2U[n,n]]$. Divide the phase

equation by $\sqrt{\epsilon_j}$. Its right-hand side tends to zero, so $\operatorname{tr} T + (p-2)T_{ss} = 0$. Since $Te_s = 0$, this gives $\operatorname{tr} T = 0$. The same compactness bounds also apply to $\Phi_j = (U_j - s)/\sqrt{\epsilon_j}$, because its value and gradient at the origin are bounded. After extraction, $\Phi_j \to \Phi$ smoothly on compact sets and $D^2\Phi = T$. Equation (2.16) gives $\partial_s \Phi_j = O(\sqrt{\epsilon_j})$ on each fixed ball. Thus Φ depends only on y, and $\Delta_y \Phi = \operatorname{tr} T = 0$. Hessian compatibility then gives

$$\Delta_y T_{ij} = \partial_i \partial_j \operatorname{tr} T = 0.$$

For tangential indices this uses the compatibility relation twice:

$$\sum_\alpha \partial_\alpha \partial_\alpha T_{ij} = \partial_i \sum_\alpha \partial_\alpha T_{\alpha j} = \partial_i \partial_j \sum_\alpha T_{\alpha\alpha}.$$

The components with a normal index already vanish. Smooth convergence allows these derivatives to be commuted. No division by a component of T is involved, so the argument also covers $T \equiv 0$.

The higher derivative bounds in (3.10) justify these identities smoothly. Set $G(s) = \operatorname{sech}^2(as)$ and divide the conjugated discrepancy equation by ϵ_j. To see the limiting source explicitly, let $A_j = A_{p_j}(\nabla U_j)$ and $\mathcal{H}_j = D^2 U_j$. Then $A_j \to A_0 = \operatorname{diag}(I, c)$ smoothly on every fixed core, and

$$\frac{\operatorname{tr}(A_j \mathcal{H}_j A_j \mathcal{H}_j)}{\epsilon_j} \longrightarrow \operatorname{tr}(A_0 T A_0 T) = |T|^2,$$

because $Te_s = 0$. The conjugating factor tends to $G^{p/2}$, and (2.33) tends to $ca^2[p^2 - p(p-1)G]$. All these limits hold smoothly on compact sets. Hence $F = G^{p/2}\rho \geq 0$ solves

$$[-\Delta_y - c\partial_s^2 + ca^2\{p^2 - p(p-1)G(s)\}]F = pG^{p/2}|T(y)|^2.$$

Change variables by $Y = a\sqrt{c}\, y$, $x = as$ and retain the notation F for the rescaled function. The resulting equation is that of Theorem 3.1, with

$$S(Y) = \frac{p}{ca^2}\left|T\left(\frac{Y}{a\sqrt{c}}\right)\right|^2.$$

The source is nonnegative and subharmonic, because the components of T are harmonic. Also, $F(0,0) = 1$, so F is not zero. The strong maximum principle gives $F > 0$. Thus both Theorem 3.1 and Lemma 3.3 apply.

We retain the curvature contribution in the source mass. Define

$$\ell_p = -c\partial_s^2 + ca^2[p^2 - p(p-1)G(s)], \qquad M = G^{p/2}, \qquad E_p = p\ell_p^{-1}M.$$

The positive potential of ℓ_p gives a bounded positive E_p, uniformly on the compact exponent interval. One construction is to solve $\ell_p E_{p,R} = pM$ on $(-R, R)$ with zero endpoint values. Since $pca^2 \leq ca^2[p^2 - p(p-1)G] \leq p^2 ca^2$, the constant $(ca^2)^{-1}$ is a supersolution. Comparison gives $0 \leq E_{p,R} \leq (ca^2)^{-1}$ and monotonicity in R. Interior estimates give a bounded nonnegative even limit E_p, which solves $\ell_p E_p = pM$. If $E_p(s_0) = 0$, then s_0 is a minimum and $E_p''(s_0) \geq 0$. The equation would give $pM(s_0) = -cE_p''(s_0) \leq 0$, contrary to $M > 0$. Hence $E_p > 0$ on the whole line.

Work in physical variables (y, s). Let $\Gamma_p(z, s, t)$ be the full Green kernel for $-\Delta_y + \ell_p$. The minimal-source construction in the linear theorem gives

$$F_{\mathrm{src}}(y, s) = p\int_{\mathbb{R}^d}\int_{\mathbb{R}} \Gamma_p(z, s, t)M(t)|T(y - z)|^2\, dt\, dz.$$

This kernel is nonnegative and radial in the horizontal displacement. The integral is finite because the source potential is bounded above by F, which is locally bounded. Apply subharmonicity of $|T|^2$ to spherical averages inside that kernel. More explicitly, for every $R > 0$ the harmonic mean-value formula gives $\fint_{|z|=R} T(y - z)\, dS_z = T(y)$. Expand the square about this mean to obtain the

exact identity

$$\fint_{|z|=R} |T(y-z)|^2\,dS_z = |T(y)|^2 + \fint_{|z|=R} |T(y-z)-T(y)|^2\,dS_z \geq |T(y)|^2.$$

The cross term vanishes by the mean-value formula. Thus the discarded error is a nonnegative square at every radius. When $d=1$, the spherical average is the average at the two points $z=R$ and $z=-R$. At each radius the kernel is constant in the horizontal angular variable. Use polar coordinates in the Green integral and apply this inequality at every radius. Tonelli's theorem gives

$$F_{\mathrm{src}}(y,s) \geq |T(y)|^2\, p \int_{\mathbb{R}^d}\int_{\mathbb{R}} \Gamma_p(z,s,t) M(t)\,dt\,dz.$$

To identify the last integral, set

$$\widehat{E}_p(s) := p\int_{\mathbb{R}^d}\int_{\mathbb{R}} \Gamma_p(z,s,t) M(t)\,dt\,dz.$$

This is the minimal source potential with source $pM(s)$, independent of the horizontal variable. The bounded function $E_p(s)$ is a supersolution for this full equation. Comparison on the exhaustion domains therefore gives $0 \leq \widehat{E}_p \leq E_p$. In particular the integral is finite. Horizontal translation invariance makes this potential independent of the horizontal variable, and its equation reduces to $\ell_p \widehat{E}_p = pM$. Thus $w = E_p - \widehat{E}_p$ is bounded and nonnegative, and

$$w'' = a^2[p^2 - p(p-1)G(s)]w \geq 0.$$

A bounded convex function on the real line is constant. The coefficient in this equation is strictly positive, so that constant must be zero. Hence $\widehat{E}_p = E_p$. Substituting this identity in the Green lower bound gives

$$F_{\mathrm{src}}(y,s) \geq |T(y)|^2 E_p(s).$$

Positivity and Tonelli's theorem justify this bound without a growth condition on T. Thus the coefficient of $|T(y)|^2$ is exactly $p\ell_p^{-1}M$, with no dimensional normalization factor. The uniform resolvent bound proved above and the equation also control E_p'. Indeed, E_p and E_p'' are bounded, and one-dimensional interpolation bounds the first derivative. For example, Taylor's formula on a unit interval gives

$$\|E_p'\|_\infty \leq 2\|E_p\|_\infty + \tfrac{1}{2}\|E_p''\|_\infty.$$

From $\ell_p M = 2pca^2 GM$ we obtain

$$\frac{2ca^2}{J_p}\int_{\mathbb{R}} GME_p\,ds = \frac{1}{pJ_p}\int_{\mathbb{R}} (\ell_p M)E_p\,ds = \frac{1}{pJ_p}\int_{\mathbb{R}} M(pM)\,ds = 1.$$

The exponential decay of M, M' and boundedness of E_p, E_p' justify integration by parts. It follows that

$$\theta_P := \frac{2ca^2}{J_p}\int_{-P}^{P} GME_p\,ds \uparrow 1.$$

The convergence is uniform on $\mathcal{P}$: the resolvent bound for E_p and the uniform exponential decay of GM bound the omitted tails. In particular,

$$0 \leq 1-\theta_P \leq C_{\mathcal{P}} e^{-\gamma_* P}, \qquad \gamma_* := \min_{p\in\mathcal{P}} a_p(p+2) > 0.$$

Thus the prescribed source fraction is reached by a finite window chosen uniformly before the contrary sequence.

Hence the normalized physical source mass is at least $\theta_P|T(y)|^2$.

Apply Theorem 3.1 to the homogeneous part alone. Its mass is the total mass less the source mass. Its mass Laplacian is no greater than the total mass Laplacian, since $\Delta_Y \mathcal{M}_{aP}(F_{\mathrm{src}}) \geq 0$. Its normal derivative at zero is that of F, since the source potential is even in the normal variable. Here the source potential is expressed in (Y,x). The finite-window theorem applies with loss $\varepsilon/4$.

Multiply its inequality by $2ca^2/j_p$. The derivative term becomes $ca^2|\partial_x F(0,0)|$, and the total mass becomes I_∞ in the physical normalization identified below. The homogeneous mass is this total less the normalized source mass. Since $\Delta_y = ca^2\Delta_Y$, the Laplacian coefficient becomes $K_{\mathcal{P}}/(ca^2)$. The source mass bound and the chosen inequality for θ_P then give

$$\beta|H_\infty| \le (1/2+\varepsilon/2)I_\infty(0) - (1/2-\varepsilon/2)|T(0)|^2 + \frac{K_{\mathcal{P}}}{ca^2}(\Delta I)_\infty. \tag{3.12}$$

We identify the limiting quantities below in the physical normalization.

At the zero level, the exact relation is

$$H = \frac{c}{p}\frac{\partial_\nu D}{1-D}.$$

Hence $H_j/\epsilon_j \to H_\infty$ and

$$\frac{\beta H_j(0)}{\epsilon_j} \longrightarrow ca^2\partial_x F(0,0).$$

The implicit function theorem represents each fixed part of the zero sheet as a graph over the tangent plane once j is large. The relative derivative bounds make its graph function $O(\sqrt{\epsilon_j})$ in every fixed derivative norm. The implicit-function differentiation has a denominator bounded away from zero. For instance, if the graph height is h_j, then

$$\partial_\alpha h_j = -\frac{\partial_\alpha U_j}{\partial_s U_j},$$

$$\partial_{\alpha\gamma} h_j = -\frac{\partial_{\alpha\gamma}U_j + \partial_{\alpha s}U_j\partial_\gamma h_j + \partial_{\gamma s}U_j\partial_\alpha h_j + \partial_{ss}U_j\partial_\alpha h_j\partial_\gamma h_j}{\partial_s U_j}.$$

All phase derivatives are evaluated on the graph, where $\partial_s U_j \ge 1/2$ for large j. Higher graph derivatives follow by repeating this differentiation with the relative bounds already established.

Thus the graph metrics converge smoothly to the Euclidean metric. Their Fermi maps $X_j(y) + s\nu_j(y)$ converge smoothly to (y,s). On each fixed core, their difference from this limiting map, in every fixed derivative norm, is $O(\sqrt{\epsilon_j})$. The bounds for D_j/ϵ_j hold in every fixed derivative norm as well. Consequently composing this normalized function with the Fermi maps preserves its C^2 limit $\rho(y,s)$. The factors $q_j^{p_j+1}$ composed with those maps converge in C^2 to $G(s)^{p+1}$. Multiplication by the bounded C^2 functions D_j/ϵ_j and integration over the fixed interval introduce no loss after normalization. Thus the mass converges in C^2 as a function of the physical base variable y:

$$\frac{I_{P,j}}{\epsilon_j} \longrightarrow I_\infty(y) := \frac{2ca^2}{j_p}\mathcal{M}_{aP}(F)(a\sqrt{c}\,y),$$

and therefore

$$\frac{\Delta_{\Sigma_j} I_{P,j}}{\epsilon_j} \longrightarrow (\Delta I)_\infty := \frac{2ca^2}{j_p}\, ca^2\Delta_Y\mathcal{M}_{aP}(F)(0).$$

The additional factor ca^2 comes from $\Delta_y = ca^2\Delta_Y$. There is no geometric error divided by ϵ_j in this passage. Indeed, for $X_j(y) = (y, h_j(y))$ the induced metric is $(g_j)_{\alpha\gamma} = \delta_{\alpha\gamma} + \partial_\alpha h_j \partial_\gamma h_j$. Writing $\mathcal{I}_j = (I_{P,j}\circ X_j)/\epsilon_j$, the exact identity is

$$\frac{(\Delta_{\Sigma_j} I_{P,j})\circ X_j}{\epsilon_j} = \Delta_{g_j}\mathcal{I}_j = \frac{1}{\sqrt{\det g_j}}\partial_\alpha\left(\sqrt{\det g_j}\, g_j^{\alpha\gamma}\partial_\gamma\mathcal{I}_j\right).$$

The coefficients converge in the required derivative norms, while $\mathcal{I}_j$ converges in C^2. The same reasoning gives convergence of the normalized surface gradients.

To identify the curvature term, let $\Pi_j = I - \nu_j\otimes\nu_j$ be the orthogonal projection onto the zero sheet. Differentiating $\nu_j = \nabla U_j/r_j$ gives

$$D\nu_j = r_j^{-1}\Pi_j D^2 U_j, \qquad b_j^2 = r_j^{-2}|\Pi_j D^2 U_j \Pi_j|^2.$$

The norm is Hilbert–Schmidt. Indeed, the matrix $r_j^{-1}\Pi_j D^2U_j\Pi_j$ agrees with B_j on tangent vectors and vanishes on the normal direction. At the origin, $\nu_j = e_s$ and $r_j = (1-\epsilon_j)^{1/p_j} \to 1$. Put $\Pi_0 = I - e_s \otimes e_s$. The normalized Hessian convergence therefore yields

$$\frac{b_j(0)^2}{\epsilon_j} = r_j(0)^{-2}\left|\Pi_0 \frac{D^2U_j(0)}{\sqrt{\epsilon_j}}\Pi_0\right|^2 \longrightarrow |\Pi_0 T(0)\Pi_0|^2 = |T(0)|^2.$$

The last equality follows from $Te_s = 0$ and symmetry of T. Here T is the physical Hessian limit introduced above; no horizontal coordinate factor enters this curvature identity. Failure of the desired inequality would pass to

$$\beta|H_\infty| \geq (1/2+\varepsilon)I_\infty(0) - (1/2-\varepsilon)|T(0)|^2 + \frac{K_{\mathcal{P}}}{ca^2}(\Delta I)_\infty.$$

The limiting mass is positive, since $\rho(0,0) = 1$ and the normal weight is positive on the fixed window. The last lower bound exceeds the upper bound in (3.12) by $\varepsilon[I_\infty(0) + |T(0)|^2]/2 > 0$. Compactness of $\mathcal{P}$ makes the threshold uniform. □

The window conditions in this proof are lower bounds: aP must contain the linear window, and θ_P must exceed its prescribed source fraction. Thus any larger fixed P also works, after decreasing the defect threshold for that window. This observation is needed when the mean form and a later ambient test use one common core. Monotonicity of I_P alone would not preserve an inequality containing $\Delta_\Sigma I_P$.

The curvature–mass comparison follows the Green-kernel strategy of [35, Lemma 8.1], with the weighted source computed above.

Corollary 3.5. *For each $\varepsilon_I > 0$ choose a finite window P and then a positive defect threshold. At every zero below this threshold,*

$$I_P(z) \geq (1-\varepsilon_I)b(z)^2 \tag{3.13}$$

uniformly on $\mathcal{P}$. The same window may be used in Theorem 3.4.

Proof. Enlarge the window to ensure $\theta_P \geq 1 - \varepsilon_I/2$. If no threshold works, normalize a contrary sequence by $\epsilon_j = D_j(0)$ as in the proof of Theorem 3.4. The zero-defect case is planar and cannot violate the claim. The source comparison gives $I_\infty(0) \geq \theta_P|T(0)|^2$. Failure would give $I_\infty(0) \leq (1-\varepsilon_I)|T(0)|^2$. These are incompatible if $T(0) \neq 0$. If $T(0) = 0$, failure contradicts $I_\infty(0) > 0$ instead. Choose both windows before decreasing their common discrepancy threshold. □

Lemma 3.6. *Fix $P > 0$ and $\sigma > 0$. For sufficiently small $D(z)$, uniformly on $\mathcal{P}$,*

$$\Delta_\Sigma I_P(z) \geq \frac{|\nabla_\Sigma I_P(z)|^2}{2I_P(z)} - \sigma I_P(z). \tag{3.14}$$

Interpret the planar zero-defect case by continuity. Moreover, $I_P(z) \leq C_P D(z)$. If $P \geq 1$, then $c_I D(z) \leq I_P(z)$. The constant $c_I > 0$ is unchanged by later enlargement of P.

Proof. The planar case has $I_P \equiv 0$, and the asserted convention gives the result. Otherwise $D > 0$ everywhere by Proposition 2.1, so $I_P > 0$. If the estimate failed at arbitrarily small zero defects, use the normalization and relative C^2 compactness from the proof of Theorem 3.4. The limit has $I_\infty(0) > 0$, and Lemma 3.3 gives $\Delta I_\infty \geq |\nabla I_\infty|^2/(2I_\infty)$. The change of horizontal scale and the positive mass normalization multiply both sides by the same factor. The failed inequalities would pass to

$$\Delta I_\infty(0) \leq \frac{|\nabla I_\infty(0)|^2}{2I_\infty(0)} - \sigma I_\infty(0),$$

which is a contradiction. The positivity of the limiting mass also justifies passage through the quotient. In graph coordinates, division by the constant ϵ_j is exact:

$$\frac{1}{\epsilon_j}\frac{|\nabla_{\Sigma_j} I_{P,j}|^2}{2I_{P,j}} = \frac{|\nabla_{g_j}\mathcal{I}_j|^2}{2\mathcal{I}_j}.$$

Both the denominator and the metric therefore have regular limits at the base. A positive rescaling of the horizontal variables or of the mass multiplies the two sides of the limiting square-root inequality by the same positive factor.

For the upper mass bound, apply Corollary 2.13 and its physical-weight comparison on the fixed window $\max\{P, 1\}$. Positivity bounds I_P by the mass on that larger window. For the lower bound when $P \geq 1$, keep only the unit window: $I_P \geq I_1 \geq c_I D(z)$. Choose its threshold and c_I on that unit window. Later enlargement of P leaves this lower bound unchanged. □

The pointwise bound yields a quadratic-form estimate with a weighted cutoff error.

Theorem 3.7. *Given $\varepsilon, \eta > 0$, there is a finite lower window $P_0 \geq 1$ with the following property. For every fixed $P \geq P_0$, one can choose $\delta > 0$ uniformly for $p \in \mathcal{P}$. On the part of a zero sheet with $D < \delta$, each compactly supported C^1 function ϕ satisfies*

$$\boxed{\beta \int_\Sigma |H|\phi^2 \leq (1/2+\varepsilon)\int_\Sigma I_P\phi^2 - (1/2-\varepsilon)\int_\Sigma b^2\phi^2 + \eta\int_\Sigma |\nabla_\Sigma\phi|^2.} \tag{3.15}$$

This also holds for compactly supported H^1 tests. Before the last decrease of δ, one has the stronger weighted estimate

$$\beta \int_\Sigma |H|\phi^2 \leq (1/2+\varepsilon)\int_\Sigma I_P\phi^2 - (1/2-\varepsilon)\int_\Sigma b^2\phi^2 + C_{\mathcal{P}}\int_\Sigma I_P|\nabla_\Sigma\phi|^2, \tag{3.16}$$

The same constant $C_{\mathcal{P}}$ works for every fixed $P \geq P_0$. It is independent of the defect threshold. Keeping I_P in the cutoff error will be useful near the edge of the prepared region.

Proof. For a planar layer, $H = b = I_P = 0$, so both claims are immediate. Otherwise $D > 0$ by Proposition 2.1, so $I_P > 0$. Place the compact support inside the open prepared part of the sheet. Choose a lower window $P_0 \geq 1$ for the pointwise estimate with loss $\varepsilon/4$. Fix any $P \geq P_0$. The larger-window conclusion after Theorem 3.4 gives that pointwise estimate at this same P, with the same constant $K_{\mathcal{P}}$, after decreasing its local defect threshold. Apply Lemma 3.6 at this fixed window as well. Its threshold may depend on P and on the positive parameter σ chosen below. After choosing σ, take the smaller of the two thresholds. The window and both estimates are then fixed before any test is chosen. Write $I = I_P$ and define

$$X = \int_\Sigma \frac{|\nabla I|^2}{I}\phi^2, \quad Y = \int_\Sigma I|\nabla\phi|^2, \quad Z = \int_\Sigma (\Delta_\Sigma I)\phi^2, \quad M_0 = \int_\Sigma I\phi^2.$$

By Lemma 3.6 and integration by parts,

$$\frac{X}{2} - \sigma M_0 \leq Z = -2\int_\Sigma \phi\,\nabla I\cdot\nabla\phi \leq 2\sqrt{XY} \leq \frac{X}{4} + 4Y.$$

The last Young inequality follows by expanding $(\sqrt{X}/2 - 2\sqrt{Y})^2 \geq 0$. All integrals are finite because I is smooth and positive on the compact support. There are no boundary terms, since the support lies inside the prepared region. The two inequalities give $X \leq 16Y + 4\sigma M_0$. Substitute this into $Z \leq X/4 + 4Y$ to obtain

$$Z \leq 8Y + \sigma M_0. \tag{3.17}$$

Set $k_* = \sup_{\mathcal{P}} K_{\mathcal{P}}/(ca^2) < \infty$ and choose σ with $k_*\sigma \le \varepsilon/4$. Integrate the pointwise estimate and use (3.17). We obtain

$$\beta \int |H|\phi^2 \le (1/2+\varepsilon/2)\int I\phi^2 - (1/2-\varepsilon/4)\int b^2\phi^2 + 8k_*\int I|\nabla\phi|^2.$$

Now $I \le C_P\delta$. Decrease δ while keeping P fixed, until $8k_*C_P\delta \le \eta$. The resulting bound is stronger than (3.15). If $K_{\mathcal{P}} = 0$, omit the Laplacian step. For a compactly supported H^1 test, approximate in H^1 by smooth tests with support in a fixed compact subset of the prepared region. All coefficients are bounded there, so every integral converges. This gives both form estimates for H^1 tests. □

3.3. **Clearing and turn pressure.** Fix the ambient dimension n. For each small fixed $\varepsilon > 0$, Lemma 2.3 supplies R_0. A zero-free radius-R_0 ball then has $1 - |u| \le \varepsilon$ at its center. The Picone bound uses only the matched equation. We sharpen it to the exact one-dimensional decay rate.

Lemma 3.8 (Exact exponential rate with a polynomial loss)**.** *For every sufficiently large phase ball $B_R(x_0)$,*

$$1 - |u(x_0)| \le CR^{(n-1)/(2(p-1))}e^{-2aR}.$$

It follows that

$$(1-u(x_0)^2)^p \le CR^{p(n-1)/(2(p-1))}e^{-2\beta R}.$$

The constants and the minimum radius are uniform on $\mathcal{P}$.

Proof. By reflection, consider a positive phase and put $w = 1 - u$. Its equation is

$$\Delta_p w = \mathcal{F}_p(w), \qquad \mathcal{F}_p(w) = (1-w)[w(2-w)]^{p-1}.$$

Choose $v_* > 0$ so that $F(v_*)$ lies in a common interval where $\mathcal{F}_p$ increases. Here $F(V) = 2V/(1+V)$. Require in addition $v_*/(1+v_*) \le (p-2)/(2p)$ for every $p \in \mathcal{P}$. A common explicit choice is to put

$$w_* := \frac{1}{2}\min\left\{\frac{p_- - 2}{p_-},\ 1 - \frac{1}{\sqrt{2p_- - 1}}\right\}, \qquad v_* := \frac{w_*}{2-w_*}.$$

Then $F(v_*) = w_* > 0$. The required monotonicity follows from

$$\frac{\mathcal{F}_p'(w)}{\mathcal{F}_p(w)} = -\frac{1}{1-w} + \frac{2(p-1)(1-w)}{w(2-w)} > 0 \quad\Longleftrightarrow\quad (2p-1)(1-w)^2 > 1.$$

Both restrictions therefore hold throughout the common interval $0 < w \le w_*$.

Ordinary clearing gives $w \le F(v_*)$ on $B_{R-R_0}(x_0)$ for a fixed R_0.

Set $k = 2a$ and solve for ϕ in

$$c\phi''(r) + \frac{n-1}{r}\phi'(r) = ck^2\phi(r), \qquad \phi(0) = 1, \quad \phi'(0) = 0.$$

For $n > 1$ the solution is represented by

$$\phi(r) = C_{n,p}\int_{-1}^{1} e^{krt}(1-t^2)^{(n-1)/(2c)-1}\,dt,$$

with $C_{n,p}$ fixing its value at zero. For $n = 1$ take $\phi(r) = \cosh(kr)$. Either the representation or the equation gives

$$0 \le \phi'/\phi \le k, \qquad \phi(r) \ge C^{-1}(1+r)^{-(n-1)/(2c)}e^{kr} \quad (r \ge 1).$$

The exponent in the integral exceeds -1. For the lower bound when $n > 1$, put $\gamma = (n-1)/(2c) > 0$ and take $r \ge 2$. On $1 - r^{-1} < t < 1$, one has $e^{krt} \ge e^{kr-k}$ and $(1-t^2)^{\gamma-1} \ge C^{-1}(1-t)^{\gamma-1}$. Consequently the integral on this interval is at least $C^{-1}e^{kr}r^{-\gamma}/\gamma$. Both γ and the normalizing integral range over compact subsets of $(0,\infty)$ as p varies. The constants are therefore uniform. For $n = 1$ the same bound follows from $\cosh(kr) \ge e^{kr}/2$. Increase the constant to cover $1 \le r \le 2$.

Put $\rho = R - R_0$ and $V(r) = v_*\phi(r)/\phi(\rho)$. With $t(r) = V'(r)/(kV(r)) \in [0,1]$, radial differentiation gives

$$\Delta_p V = (V')^{p-2}\left(cV'' + \frac{n-1}{r}V'\right) = ck^pV^{p-1}t^{p-2}.$$

For an increasing scalar function F, the chain rule is

$$\Delta_p F(V) = (F'(V))^{p-1}\Delta_p V + c(F'(V))^{p-2}F''(V)|\nabla V|^p.$$

Here $F' = 2/(1+V)^2$, $F'' = -4/(1+V)^3$, and $ck^p = 2^{p-1}$. Substitution gives

$$\Delta_p F(V) = \frac{4^{p-1}V^{p-1}}{(1+V)^{2(p-1)}}t^{p-2}\left(1 - \frac{2V}{1+V}t^2\right).$$

On the other hand,

$$\mathcal{F}_p(F(V)) = \frac{4^{p-1}V^{p-1}}{(1+V)^{2(p-1)}}\left(1 - \frac{2V}{1+V}\right).$$

The choice of v_* makes $t^{p-2}(1 - 2Vt^2/(1+V))$ increasing on $[0,1]$. Its value is therefore at most its value at one. If $d = 2V/(1+V)$, the derivative that fixes this direction of comparison is

$$\frac{d}{dt}\left[t^{p-2}(1-dt^2)\right] = t^{p-3}\left[(p-2) - pd\,t^2\right] \geq 0 \qquad (0 < t \leq 1).$$

Here $d \leq (p-2)/p$ by the preceding choice.

Thus $F(V)$ is a supersolution of $-\Delta_p w + \mathcal{F}_p(w) = 0$ in the small well interval. At the center its gradient vanishes. The flux is C^1 there because $p > 2$, so the same inequality holds weakly throughout the ball. On ∂B_ρ we have $F(V) = F(v_*) \geq w$. The reaction is increasing on the range of both functions. For $\zeta = (w - F(V))_+$, subtraction of the weak equations gives

$$\int_{\{w>F(V)\}}(|\nabla w|^{p-2}\nabla w - |\nabla F(V)|^{p-2}\nabla F(V))\cdot\nabla(w - F(V)) + \int(\mathcal{F}_p(w) - \mathcal{F}_p(F(V)))\zeta \leq 0.$$

Both integrands are nonnegative. The zero boundary trace of ζ and strict monotonicity force $\zeta = 0$.

The comparison yields

$$w(x_0) \leq F\big(v_*/\phi(\rho)\big) \leq C\rho^{(n-1)/(2c)}e^{-2a\rho}.$$

Absorb the fixed R_0 to prove the first estimate. For the second use $1 - u^2 \leq 2(1-u)$. Reflection treats the negative phase. □

The same decay rate can be used in barriers built from several affine faces. Retain $w = 1 - u$, its equation, and the common small well interval above. Take inward affine unit distances l_i, constants $C_i > 0$, and define

$$V = \sum_i C_i e^{-2al_i}, \qquad F(V) = \frac{2V}{1+V}.$$

Lemma 3.9 (Logistic supersolution)**.** *For $p > 2$, there exists $v_* > 0$, uniform on $\mathcal{P}$, such that*

$$\Delta_p F(V) \leq \mathcal{F}_p(F(V)) \qquad (0 < V \leq v_*). \tag{3.18}$$

For one face with $C_i = 1$, the identity $F(e^{-2al_i}) = 1 - g(l_i)$ is exact for every $p > 1$.

Proof. Write $|\nabla V| = 2aVt$ with $0 \leq t \leq 1$. The Hessian of V is positive semidefinite, with trace $(2a)^2V$. Its value in any unit direction is at most its trace. The formula for $\Delta_p V$ therefore gives

$$\Delta_p V \leq c(2a)^pV^{p-1}t^{p-2}.$$

The chain rule and $c(2a)^p = 2^{p-1}$ give

$$\Delta_p F(V) \leq \frac{4^{p-1}V^{p-1}}{(1+V)^{2(p-1)}}t^{p-2}\left(1 - \frac{2V}{1+V}t^2\right).$$

The reaction is this prefactor times $1 - 2V/(1+V)$. Apply the scalar monotonicity calculation in the proof of Lemma 3.8, with $d = 2V/(1+V)$. The choice $V/(1+V) \le (p-2)/(2p)$ verifies its hypothesis and gives (3.18). It also holds where $\nabla V = 0$, by continuity of the flux derivative for $p > 2$. □

Lemma 3.10 (Buffered graph geometry)**.** *There are dimensional constants $c, C > 0$ with the following property. Let M be a relatively closed smooth embedded hypersurface without boundary in $B_{20r}(z)$, with $z \in M$, $|B| \le \kappa$, and $\kappa r \le c$. The piece through z contains a graph g over the tangent disk B_{4r}, with $g(0) = Dg(0) = 0$ and*

$$|Dg(y)| \le C\kappa|y|, \qquad |g(y)| \le C\kappa|y|^2, \qquad |D^2 g| \le C\kappa.$$

It contains the intrinsic radius-r ball about z. For $w \in M \cap B_r(z)$, the unoriented angle θ between the normals at z, w satisfies

$$\theta \le C(|z-w|/r + \kappa r). \tag{3.19}$$

Finally, let $(y, f(y))$, $y \in B_{4r}$, be a smooth graph with $|Df| \le 1/4$ and $|D^2 f| \le C\kappa$. If $|x'| \le r$ and $|f(x') - x_n| \le r$, the distance from x to this graph has a unique interior minimizing foot. Its projection and signed distance are smooth in a neighborhood of x.

Proof. Let P be projection onto $T_z M$. For each $y \in B_{4r}$ lift $t \mapsto ty$, starting at z, by solving

$$\dot{X}_y(t) = \left(dP|_{T_{X_y(t)}M}\right)^{-1} y, \qquad X_y(0) = z.$$

While the normal angle is small, $|\dot{X}_y| \le 2|y|$. Its total normal variation is then at most $2\kappa|y|$. For c sufficiently small this improves the angle bound, and the lift stays in $B_{8r}(z)$. If it stopped before $t = 1$, its bounded speed would give an ambient limit in this ball. Relative closedness puts the limit on M, and a regular inverse chart continues the lift there. Thus every lift exists up to $t = 1$. Smooth dependence for this smooth ODE gives a smooth map $y \mapsto X_y(1)$, and $P(X_y(1) - z) = y$ makes it a single-valued graph. The normal-variation bound gives the slope estimate; integration gives the height estimate, and the graph formula for B gives the Hessian estimate. A path of intrinsic length less than r from z cannot leave this graph: at a first exit its projected point still lies in B_r, and continuity of $X_y(1)$ and a local inverse chart continue the path inside the graph.

For the angle estimate put $d = |z - w|$. If $d \ge r/10$, increase C and use $\theta \le \pi/2$. Otherwise choose the signs of the two normals to have nonnegative dot product and project both patches onto the plane with their bisector normal. Each initial projection has inverse operator norm at most $\sqrt{2}$; the same radial-lift proof makes both patches graphs over a common disk of radius comparable to r, with bounded slope and Hessian at most $C\kappa$. Their affine height difference has constant term $O(d)$ and slope comparable to θ, with error $O(\kappa r^2)$. If the patches are disjoint, this difference cannot have opposite signs at the two ends of a diameter, giving (3.19). If they meet, embeddedness identifies their tangent planes there. The two lifted paths to that point have length $O(r)$, so curvature control gives $\theta \le C\kappa r$ instead.

For the last assertion minimize

$$F_x(y) = \tfrac{1}{2}(|y - x'|^2 + |f(y) - x_n|^2).$$

Its value at $y = x'$ is at most $r^2/2$, whereas for $|y| \ge 3r$ it is at least $2r^2$. Minimize first on $\overline{B_{3r}} \Subset B_{4r}$. The minimum is interior and is also the minimum over the whole graph. On the whole disk $|f(y) - x_n| \le 9r/4$, and

$$D^2 F_x = I + Df \otimes Df + (f - x_n) D^2 f \ge (1 - C\kappa r) I \ge \tfrac{1}{2} I.$$

Strict convexity gives uniqueness. The implicit function theorem applied to $D_y F_x = 0$ gives smooth projection. Its joining segment is normal to the graph, and the corresponding signed distance is smooth as well. □

The chamber comparison requires only local preparation. It does not invoke clean-ball preparation or finite density. Suppose two adjacent zero graphs are defined over a buffered disk of radius R. Their gap satisfies $\ell \le R/A$, for a prescribed large fixed A. Assume zero defects at most δ_0 and curvature at most κ. The fixed-order graph errors satisfy

$$\epsilon_0 = C\delta_0 + C_N\kappa^{\vartheta}R^N + Ce^{-c_0R}, \qquad \kappa R^2 \longrightarrow 0, \qquad \epsilon_0\ell \longrightarrow 0. \tag{3.20}$$

We use only finitely many derivative orders. Take the recorded exponents with $N \ge 2$ and $0 < \vartheta \le 1$. Then $\kappa R^2 \le \kappa^{\vartheta}R^N$ when $R \ge 1$ and $\kappa \le 1$. Fix these exponents before increasing the preparation scale. For example, let $m \ge 2$ be the largest required graph derivative order, and measure a graph height h from its tangent plane. The curvature bound gives $\|D^2h\|_\infty \le C\kappa$. Relative regularity supplies a fixed C^{2m-2} bound on larger unit charts. For $m \ge 3$, interpolation of D^2h between orders zero and $2m-4$ gives

$$\|D^jh\|_\infty \le C_m\kappa^{1-(j-2)/(2m-4)} + C_m\kappa \le C_m\kappa^{1/2}, \qquad 2 \le j \le m.$$

For $m = 2$, the curvature estimate suffices directly. Integration from the tangent point bounds the height and slope by $C\kappa R^2$ and $C\kappa R$. Thus one may take $\vartheta = 1/2$ and $N = m+2$, since

$$\max_{0\le j\le m} R^j\|D^jh\|_\infty \le C_m\kappa^{1/2}R^{m+2} \qquad (R \ge 1,\ \kappa \le 1).$$

The fixed larger chart buffer is used before interpolation. The condition $\epsilon_0\ell \to 0$ means that, for each required tolerance, $\epsilon_0(1+\ell)$ is below a fixed small number.

The unoriented angle is bounded by a sufficiently small fixed constant. These assumptions hold for the initial choice $R = A_0\log(1/\delta)$ with $\delta_0 \le C\delta$ and $\kappa = C\sqrt{\delta}$. With the displayed choice of exponents, the two relevant limits are

$$\kappa R^2 \le C\delta^{1/2}\log^2(1/\delta) \longrightarrow 0,$$

$$\epsilon_0\ell \le C\delta\log(1/\delta) + C_m\delta^{1/4}\log^{m+3}(1/\delta) + C\delta^{c_0A_0}\log(1/\delta) \longrightarrow 0.$$

The constants may depend on the fixed scale ratios and buffers.

A variable-envelope bound $\delta_0 \le CR^{-\alpha}$ with $\alpha > 1$ also gives $\delta_0\ell \to 0$, since $\ell \le R/A$. It does not by itself control curvature. In that setting the conditions $\kappa R^2 \to 0$ and $\kappa^{\vartheta}R^N\ell \to 0$ remain separate preparation hypotheses.

Apply Lemma 3.9 using both chamber faces. Choose a fixed depth S so that clearing places the deficit in the small well interval. Relative core preparation controls the deficit at this depth with relative error $O(\delta_0 + \kappa^{\vartheta}R^N)$. Shift the affine faces outward by their graph error, at most $C\kappa R^2$. Include this shift in the recorded graph-error term. At a point of the depth-S face, write its deficit as w_i and its affine inward distance as $l_i = S + O(\kappa R^2)$. Its exact exponential coefficient is $e^{2al_i}w_i/(2-w_i)$. Take the supremum of these coefficients on each buffered face. They equal $1 + O(\delta_0 + \kappa^{\vartheta}R^N)$ and give constant-coefficient logistic barriers above the actual face data.

Add lateral exponentials from a larger rectangular box. Choose their coefficients so that $V \ge v_*$ on its lateral faces. Compare on the part of the deep chamber where $V < v_*$. On the added boundary $V = v_*$, clearing gives $w \le F(v_*)$. On the remaining face boundaries the chosen coefficients give the same ordering. Both functions stay in the interval where the reaction increases. The positive-part comparison used in Lemma 3.8 therefore applies. A sufficiently large buffer makes each lateral exponential, relative to the own-face tail, $O(e^{-c_0R})$ in the smaller box. The identity for a single face is exact, so this construction has no fixed error depending only on the chosen depth S.

Use coordinates from the lower face and write the upper affine distance as

$$t_2 = \ell - b\cdot y - ts, \qquad |b|^2 + t^2 = 1, \qquad t > 0.$$

Along the true normal, the gradient bound gives $1-u \geq 1-\tanh(as)$ in Fermi coordinates. Combine this exact lower bound with the upper barrier. Below a fixed cropped middle band it yields

$$\left|\frac{1-u}{1-\tanh(as)}-1\right| \leq C\left(\epsilon_0 + e^{-2a[\ell - b\cdot y-(1+t)s]}\right), \qquad \epsilon_0\ell \longrightarrow 0.$$

Include the affine-coordinate error and its required fixed-order interpolated derivatives in ϵ_0. There is no fixed $O(e^{-S})$ error that is later multiplied by the corridor length. To distinguish coordinates, let τ be true signed distance from the lower graph and keep s for its tangent-plane coordinate. The graph slab gives $|\tau - s| \leq C\kappa R^2$ on the smaller buffered window. Since $|\partial_s \log(1-\tanh(as))| \leq 2a$, transferring the exact own-normal lower bound to s changes its ratio by only $O(\kappa R^2)$. For the upper bound, write $V = V_1 + V_2 + V_{\mathrm{lat}}$ with $V_1 = (1+O(\epsilon_0))e^{-2as}$. The exact logistic ratio is

$$\frac{F(V)}{F(e^{-2as})} = \frac{V}{e^{-2as}}\frac{1+e^{-2as}}{1+V}.$$

The opposite-face term divided by e^{-2as} has the exponential size already displayed, and the lateral term has relative size $O(e^{-c_0 R})$. These estimates give both sides of the stated own-face comparison.

Use the adjacent graphs with small relative angle and errors satisfying (3.20). Their fixed-order errors vanish on disks whose radius is a large multiple of the gap. Their zero defects also tend to zero. The own-face barrier $F(V) = 2V/(1+V)$ has coefficients $1+o(1)$. We use these two barriers to identify the turn profile.

Lemma 3.11 (Uniform fixed-band turn preparation)**.** *Fix $\varepsilon > 0$ and $T < \infty$. Choose a sufficiently small angle bound $\vartheta_0 > 0$ and sufficiently large gap and preparation thresholds. Consider a chamber as above. At an affine-bisector point, divide its well deficit by $h_0 = 2e^{-2aL}$, where L is the common affine-face distance. On the radius-T ball, this normalized deficit and its first derivatives are within ε of*

$$V_*(s) = V_p(s)/C_p^\infty, \qquad s = \frac{1}{2a}\int_1^{V_p(s)} (t^p-1)^{-1/p}\,dt \quad (s \geq 0),$$

extended evenly, with $C_p^\infty = \lim_{s\to\infty} e^{-2as}V_p(s)$. Thus $V_p(0) = 1$, $V_p'(0) = 0$, and $V_p(s) > 1$ for $s > 0$; it solves $\Delta_p V_p = 2^{p-1}V_p^{p-1}$ in one dimension. The integral definition excludes a spurious interval on which $V_p = 1$. The normal variable is the affine bisector coordinate. The estimates are uniform on buffered parts of the chamber and for $p \in \mathcal{P}$.

Proof. A failure gives a sequence with angles tending to zero and gaps tending to infinity. The preparation errors tend to zero, and the chosen points stay in their buffer. Translate each bisector point to the origin. Divide $w = 1-u$ by h_0. Write $p_j \to p$ after taking a subsequence, and use a for the limiting constant. The two face distances, less their common value L_j, converge on compact sets to s and $-s$. The true distances differ from these affine distances by $o(1)$. The gradient bound along the two shortest true segments yields

$$\liminf w/h_0 \geq e^{2a|s|}.$$

The own-face supersolution yields

$$\limsup w/h_0 \leq e^{2as} + e^{-2as}.$$

An additive $o(1)$ error in a distance changes its exponential by a factor $1+o(1)$. Thus the distance errors become relative errors after normalization. Both face coefficients are $1+o(1)$. The lateral exponentials divided by h_0 vanish, because the buffer radius is a sufficiently large fixed multiple of the gap. The bounds therefore hold on every fixed compact set.

For the normalized functions $v_j = w_j/h_{0,j}$, homogeneity of the p_j-Laplacian gives

$$\Delta_{p_j} v_j = \frac{\mathcal{F}_{p_j}(h_{0,j}v_j)}{h_{0,j}^{p_j-1}} = (1-h_{0,j}v_j)[v_j(2-h_{0,j}v_j)]^{p_j-1}.$$

On each bounded positive range, these reaction functions converge uniformly to $t \mapsto 2^{p-1}t^{p-1}$ as $h_{0,j} \to 0$ and $p_j \to p$. On a slightly larger fixed ball the barriers bound v_j and its right-hand side uniformly. Interior p-Laplace $C^{1,\alpha}$ estimates, with p_j in the fixed compact interval, give a C^1_{loc} convergent subsequence. Its entire limit v satisfies

$$\Delta_p v = 2^{p-1}v^{p-1}, \qquad e^{2a|s|} \le v(y,s) \le e^{2as} + e^{-2as}.$$

We identify this limit by comparison on a slab.

For an even profile with $m = V(0)$, on an interval where V' has one sign, multiply the one-dimensional equation by V'. Integration gives

$$\frac{p-1}{p}|V'|^p = \frac{2^{p-1}}{p}(V^p - m^p).$$

The value at the turn fixes the integration constant. Using $[2^{p-1}/(p-1)]^{1/p} = 2a$ gives the following form. The one-dimensional first integral is

$$|V'| = 2a(V^p - m^p)^{1/p}, \qquad m = V(0).$$

For an even positive solution, the equation gives $(|V'|^{p-2}V')' > 0$. Thus $V' > 0$ on $(0, M)$, and no constant interval is possible. Separation of variables gives $V(s) = mV_p(s)$. Conversely, the integral defining V_p gives this equation on $s > 0$. Its derivative tends to zero at the origin. Its odd flux has derivative $2^{p-1}V_p^{p-1}$ across the origin, so its even extension is a weak solution on the whole line.

Fix $M > 0$ and define

$$V_M^-(s) = \frac{e^{2aM}}{V_p(M)}V_p(s), \qquad V_M^+(s) = \frac{e^{2aM} + e^{-2aM}}{V_p(M)}V_p(s).$$

They have the lower and upper boundary values of v at $s = \pm M$. Also $V_p'/V_p \le 2a$ on the positive half-line, so $V_p(M) \le e^{2aM}$. Both comparison functions, and v, are therefore at least one on this slab. We justify comparison despite its unbounded horizontal directions.

Write $\Phi_p(\xi) = |\xi|^{p-2}\xi$. All gradients in question are bounded on $|s| \le M$. For v, this follows from interior estimates on the larger slab $|s| < M + 1$ and its global upper bound there. On any bounded gradient set, monotonicity gives

$$|\Phi_p(\xi) - \Phi_p(\eta)|^2 \le C\,[\Phi_p(\xi) - \Phi_p(\eta)] \cdot (\xi - \eta).$$

Indeed, integrate the positive semidefinite matrix $D\Phi_p$ along the segment from η to ξ and use its bounded operator norm. The reaction $2^{p-1}t^{p-1}$ has derivative bounded below by a positive constant on $t \ge 1$. Subtract the equations and test with $\chi^2\zeta$, where $\zeta = (V_M^- - v)_+$ or $\zeta = (v - V_M^+)_+$ and χ is a horizontal cutoff. The test has zero trace on the two horizontal faces. The displayed flux bound and Young's inequality absorb the gradient cross term.

Writing the larger and smaller functions as a_1, a_2, put $\mathcal{A} = \Phi_p(\nabla a_1) - \Phi_p(\nabla a_2)$ on $\{a_1 > a_2\}$. The gradient part of the difference test is $\int \chi^2 \mathcal{A} \cdot \nabla\zeta$. For any $\eta > 0$, the cross term satisfies

$$2|\chi\zeta\mathcal{A} \cdot \nabla\chi| \le \eta\chi^2|\mathcal{A}|^2 + \eta^{-1}\zeta^2|\nabla\chi|^2.$$

The preceding flux inequality absorbs the first term when η is fixed small enough. The reaction difference is bounded below by a positive constant times ζ on the bounded range in the slab. The positive reaction term then gives

$$\int_{|s|<M} \chi^2\zeta^2 \le C_M \int_{|s|<M} \zeta^2|\nabla_y\chi|^2.$$

For $n > 1$, take $\chi = 1$ on B_R^{n-1}, supported in B_{R+1}^{n-1}, with uniformly bounded gradient. If $E(R)$ is the integral of ζ^2 on $B_R^{n-1} \times (-M, M)$, this gives $E(R) \le C_M(E(R+1) - E(R))$. With $\tau = C_M/(1 + C_M) < 1$, iteration gives

$$E(R) \le \tau^j E(R+j) \le C_M\tau^j(R+j)^{n-1} \longrightarrow 0.$$

The last bound uses the uniform slab bound for ζ. Thus $E(R) = 0$. For $n = 1$, ordinary interval comparison gives the same conclusion directly. Hence $V_M^- \le v \le V_M^+$. The constants may depend on the fixed slab width M, which is held fixed during this limit.

The integral defining V_p shows that $\log V_p(s) - 2as$ has a finite limit. Indeed, the difference $(t^p - 1)^{-1/p} - t^{-1}$ is integrable at infinity. The singularity at $t = 1$ is integrable because $p > 1$. More explicitly,

$$C_p^\infty = \exp\left\{-\int_1^\infty [(t^p - 1)^{-1/p} - t^{-1}]\, dt\right\}.$$

The integral is continuous in p on $\mathcal{P}$ by dominated convergence. Near one, a common integrable bound is $C(t-1)^{-1/p_-}$, where $p_- = \min \mathcal{P}$. At infinity a common bound is $Ct^{-p_- - 1}$. Thus C_p^∞ has positive uniform lower and upper bounds. Both prefactors tend to $1/C_p^\infty$ as $M \to \infty$. The limit v is therefore V_p/C_p^∞. The integral formula and its first integral also give continuous dependence of V_p on p in C^1 on every fixed interval. The compactness above is already in C^1, so this contradicts the chosen failure. This argument also proves uniformity on $\mathcal{P}$. □

The turn comparison must also allow a small fixed tilt. For the two limiting affine faces, take lower inward distance s and upper inward distance

$$t_2 = \ell - b \cdot y - ts, \qquad |b|^2 + t^2 = 1, \qquad t > 0.$$

Their bisector is $L(y) = (\ell - b \cdot y)/(1 + t)$. The pressure at the middle has scale $e^{-2\beta L(0)}$. Use the positive Jacobi operator

$$\mathcal{H}_p^+ = -\Delta_y - c\partial_s^2 + ca^2[p^2 - p(p-1)G(s)], \qquad G = \operatorname{sech}^2(as).$$

A tangential exponential mode grows normally with exponent μ, where $\mu^2 = \beta^2 - \alpha^2|b|^2/c$. Matching its tilted top trace gives $\alpha(1 + t) - \mu = \beta$. Eliminating μ and using $|b|^2 = 1 - t^2$ gives

$$\alpha^2[(1 + t)^2 + (1 - t^2)/c] - 2\alpha\beta(1 + t) = 0.$$

The nonzero root is the transmission exponent recorded below.

The positive solution is

$$\alpha = \frac{2c\beta}{c(1 + t) + 1 - t}, \qquad \mu = \beta\frac{c(1 + t) - (1 - t)}{c(1 + t) + (1 - t)}. \tag{3.21}$$

For a sufficiently small fixed tilt, $\mu > 0$ and

$$\beta \le \alpha \le \frac{2\beta}{1 + t}, \qquad \frac{2\beta}{1 + t} - \alpha = \frac{2\beta(1 - t)}{(1 + t)[c(1 + t) + 1 - t]} \ge 0. \tag{3.22}$$

Thus $e^{-\alpha\ell}$, the core-transmission scale, is at least the capacity scale $e^{-2\beta\ell/(1+t)}$. A uniform comparison with $e^{-\beta\ell}$ is not needed. The lower exponent inequality also follows directly from

$$\alpha - \beta = \frac{\beta(c - 1)(1 - t)}{c(1 + t) + 1 - t} \ge 0.$$

This is one place where the standing condition $p > 2$, and hence $c > 1$, is used in the tilted comparison.

For the directional branch we need a fixed upper bound on turn pressure in terms of core mass.

Lemma 3.12 (Local turn scale is bounded by core defect)**.** *In the prepared flat chambers above, let ℓ be the shortest true distance from the base zero to its adjacent graph. Write the tangent plane as $\ell - b \cdot y - ts = 0$ with $|b|^2 + t^2 = 1$. Choose a small fixed angle and a large fixed ratio of the prepared horizontal radius to ℓ. Then*

$$P_{\mathrm{loc}} := e^{-2\beta\ell/(1+t)} \le CD(z) \le C'I(z).$$

The constants are uniform on $\mathcal{P}$. They do not depend on the gap, the curvature envelope, or the reach weight. Turn preparation also gives a two-sided comparison with middle pressure. At

the affine bisector on the base normal, $q^p \asymp P_{\mathrm{loc}}$ and $D \geq c_* > 0$. Thus $q^p D \asymp P_{\mathrm{loc}}$ there, uniformly under the same preparation. For every fixed T, the radius-T ball about this bisector point satisfies $q^p \leq Ce^{2\beta T} P_{\mathrm{loc}}$ after choosing its preparation threshold. Consequently any fixed positive normalization of this middle pressure is comparable to P_{loc} in both directions; replacing one by the other changes only C.

Proof. The minimizing foot on the buffered upper graph is interior and unique by Lemma 3.10: its edge is at distance comparable to R, whereas $\ell \ll R$ and $\kappa R \to 0$. The joining segment is normal at that foot, so the tangent-plane constant term is exactly the true distance ℓ used in the statement. Fix a bottom depth $P_0 \geq 1$, independently of the gap. We will choose a large fixed crop T_0 below. Turn preparation, with a sufficiently small angle and accurate preparation, gives on the cropped inclined top $s = S(y) = (\ell - b \cdot y)/(1+t) - T_0$

$$F(y, S(y)) \geq c_{T_0} e^{-\beta L(y)}, \qquad F = q^{(p-2)/2}(q - u_e/a), \qquad e = \nu(z).$$

The constant $c_{T_0} > 0$ is fixed. To check its sign, measure the turn variable from the middle toward the lower face. At the crop its value is T_0, and its profile is $V_* = V_p / C_p^\infty$. The relevant factor is

$$V_*(T_0) - \frac{V_*'(T_0)}{2a} > 0.$$

This follows from the strict first-integral inequality $V_p' < 2aV_p$. In the parallel limit, direct substitution of $w = h_0 V_*$ gives

$$e^{\beta L} F \longrightarrow 2^p V_*(T_0)^{(p-2)/2} \left(V_*(T_0) - \frac{V_*'(T_0)}{2a} \right).$$

This expression has a positive minimum on $\mathcal{P}$ for fixed T_0. Choose the C^1 turn tolerance after T_0, below that minimum by a fixed factor. A sufficiently small angle then preserves the lower bound. At the middle itself, the same preparation gives $q/h_0 \to 2V_*(0) > 0$ and $\nabla U = -\nabla w/(aq) \to 0$, because $V_*'(0) = 0$. Thus $q^p \asymp h_0^p \asymp e^{-2\beta L}$ and $D = 1 - |\nabla U|^p$ is bounded below by a fixed positive constant. These limits are uniform on $\mathcal{P}$, so they prove the stated two-sided middle-pressure comparison and $D \geq c_*$. The global bound $V_p(s) \leq e^{2a|s|}$ gives the stated upper estimate on fixed middle bands. The cropped estimate holds uniformly on $|y| \leq A\ell$ if the window has a fixed extra buffer and $A|b|$ is small enough. Choose A below, then decrease the angle bound.

On $-P_0 < s < S(y)$ we have the exact inequality $\mathscr{H}_u^+ F \geq 0$, where

$$\mathscr{H}_u^+ = -\operatorname{div}(A_p(Z)\nabla) + V_u^+, \quad V_u^+ = ca^2\{p^2 - p(p-1)q + (p-2)[p - cq]D\}.$$

For constant e, the weak Jacobi identity gives $L_u(q - u_e/a) = -q^p D$. Set $g_e = q - u_e/a = q(1 - Z_e)$, $\psi = g_e/q$ and $w = q^p$. Then $F = w^{1/2}\psi = q^{(p-2)/2} g_e$. Since $0 \leq r \leq 1$, convexity gives $D = 1 - r^p \leq p(1-r) \leq p(1 - Z_e)$. Thus $g_e \geq 0$ and $q^p D \leq pq^{p-1} g_e$, so $(L_u + pq^{p-1}) g_e \geq 0$.

We record the conjugation explicitly; it is the operator form of Lemma 2.16. The product rule and (2.9) give

$$\frac{L_u(q\psi)}{a^{p-2} q^{p-1}} = -w^{-1} \operatorname{div}(w A_p(Z) \nabla \psi) - 2ca^2 q D \psi.$$

Here $a^{2-p} = 2ca^2$. Multiplication by $w^{1/2}$ and the substitution $\psi = w^{-1/2} F$ remove the first-order term. The resulting potential, before adding pq^{p-1}, is

$$V = \frac{1}{2} \operatorname{div}(A_p(Z)\nabla \log w) + \frac{1}{4} |\nabla \log w|^2_{A_p(Z)} - 2ca^2 q D.$$

To evaluate it, use $\nabla \log w = -2pauZ$, $A_p(Z)Z = cr^{p-2} Z$ and the matched phase equation. They give

$$\operatorname{div}(A_p(Z)\nabla \log w) = -2pca^2 q r^p + 4pc^2 a^2 u^2 D,$$

$$\frac{1}{4} |\nabla \log w|^2_{A_p(Z)} = p^2 ca^2 u^2 r^p.$$

Substitute $u^2 = 1 - q$ and $r^p = 1 - D$. The terms independent of D are $ca^2[p^2 - p(p+1)q]$. The coefficient of D, divided by ca^2, is

$$-[p^2 - p(p+1)q] + 2pc(1-q) - 2q = (p-2)(p-cq).$$

This proves the exact conjugated potential used here. All factors divided out are positive, since $q > 0$. The calculation is classical on the noncritical cropped strip; the weak conjugation identity gives the same conclusion wherever a weak interpretation is needed. The added potential is $pa^{2-p}q = 2pca^2q$, which changes $-p(p+1)q$ in the exact conjugated potential to $-p(p-1)q$. Its potential is at least ca^2p. Indeed, $q \le 1$ gives $p^2 - p(p-1)q \ge p$, while $(p-2)[p-cq]D \ge 0$ because $p > 2$, $p - cq \ge 1$, and $D \ge 0$. This strict lower bound is uniform on $\mathcal{P}$. Use Euclidean coordinates with base tangent plane $s = 0$. Compare its nondivergence coefficients with

$$H_p^+ = -\Delta_y - c\partial_s^2 + ca^2[p^2 - p(p-1)G(s)]$$

The difference is at most

$$C\epsilon_0 + Ce^{-k[\ell - b\cdot y - (1+t)s]}, \qquad \epsilon_0\ell \longrightarrow 0,$$

for a fixed $k > 0$. We check this bound on the whole strip, including its fixed lower part. Choose a fixed depth S_c so that clearing gives the common small well interval for $s \ge S_c$. Use the fixed extra chamber buffers when taking unit balls or normal segments below. For $-P_0 - 1 \le s \le S_c + 1$, let z_x be the nearest foot on the lower zero graph, write $e_x = \nu(z_x)$, and let τ be signed distance from that graph. Buffered graph geometry gives

$$|\tau - s| \le C\kappa R^2, \qquad |e_x - e_s| \le C\kappa R.$$

The signed depth is in a fixed interval, independent of the gap. Fix $L_{\mathrm{core}} = P_0 + S_c + 5$. First require the graph-coordinate error to be at most one. Then decrease the preparation threshold so that

$$\delta_0 \le d_{L_{\mathrm{core}}}, \qquad C_{L_{\mathrm{core}},0}\delta_0 \le \frac{1}{2}.$$

All these choices are made before increasing the gap. Apply Lemma 2.12 at z_x, where $D(z_x) \le \delta_0$, with a fixed window containing these segments. Along the normal segment, (2.16) and (2.18) give

$$U = \tau + O(\delta_0), \qquad Z = e_x + O(\delta_0).$$

For the Hessian, freeze e_x as a constant vector on this segment. The C^2 estimate in (2.16) gives

$$\partial_{e_x} D^2U = D^2U_{e_x} = O(\delta_0).$$

At z_x, the tangential block of D^2U is rB, with $|B| \le \kappa$. Its normal row and column are $O(\delta_0)$ by $D^2U\, Z = -\nabla D/(pr^{p-2})$ and (2.15). Integration along the fixed segment therefore gives $|D^2U| \le C(\kappa + \delta_0)$ on this lower core. Consequently

$$|U - s| + |Z - e_s| + |D^2U| \le C(\delta_0 + \kappa R^2) \le C\epsilon_0.$$

The last inequality uses the recorded choices $R \ge 1$, $\kappa \le 1$, $N \ge 2$ and $0 < \vartheta \le 1$ in (3.20). These estimates include the negative segment $[-P_0, 0]$. Their constants depend on the fixed depths and $\mathcal{P}$, and are independent of the gap and the solution.

On the remaining well part $s \ge S_c$, retain the face-error order by first combining the own-face upper barrier with Modica. This gives a relative C^0 phase error $O(\delta_0 + \kappa^\vartheta R^N)$ plus the opposite tail above. For the value estimate, write $w_0 = 1 - \tanh(as)$. The exact inverse-profile difference is

$$U - s = \frac{1}{2a}\left[\log\frac{1 - w/2}{1 - w_0/2} - \log\frac{w}{w_0}\right].$$

On the small well interval, the first logarithm is bounded by $C|w - w_0|$, and the second by $C|w/w_0 - 1|$ when the ratio is close to one. Thus the relative own-face error gives the stated phase value error. On a fixed-radius ball its opposite-tail term changes only by a bounded factor, so the same bound supplies the local Schauder estimate.

We justify the gradient comparison before using a linear elliptic estimate. Fix a center $x_c = (y_c, s_c)$ with $s_c \geq S_c + 1$, and take $B_1(x_c)$ in the larger buffered chamber. Choose the fixed crop T_0 large enough that this ball remains below the previously fixed cropped middle band, even when its center approaches the final cropped top. The lower core estimate already covers $s \leq S_c + 1$. Put

$$h_c = w_0(s_c), \qquad v = \frac{w}{h_c}, \qquad v_0 = \frac{w_0(s)}{h_c}, \qquad \eta_c = \epsilon_0 + e^{-k[\ell - b\cdot y_c - (1+t)s_c]}.$$

The bound $|\partial_s \log w_0| \leq 2a$ gives $C^{-1} \leq v_0 \leq C$ on this ball. The relative own-face comparison and the bounded variation of its exponential term give

$$\|v - v_0\|_{L^\infty(B_1(x_c))} \leq C\eta_c.$$

For small preparation error and large fixed T_0, this also gives $C^{-1} \leq v \leq C$. Homogeneity of the flux gives exactly

$$\Delta_p v = (1 - h_c v)[v(2 - h_c v)]^{p-1}.$$

The right side is uniformly bounded, independently of $h_c > 0$. The bounded-source interior estimate [28, Theorem 3.2, p. 303] therefore gives a common $C^{1,\alpha_{\mathrm{reg}}}$ bound for v on $B_{3/4}(x_c)$, with $\alpha_{\mathrm{reg}} > 0$ uniform on $\mathcal{P}$. The explicit profile v_0 has the same bound. For their difference, a directional difference quotient of step ρ_{int} has error at most $C\eta_c/\rho_{\mathrm{int}} + C\rho_{\mathrm{int}}^{\alpha_{\mathrm{reg}}}$. Taking $\rho_{\mathrm{int}} = \eta_c^{1/(1+\alpha_{\mathrm{reg}})}$, after reducing the common smallness threshold so that $\rho_{\mathrm{int}} \leq 1/8$, gives

$$\|\nabla v - \nabla v_0\|_{L^\infty(B_{1/2}(x_c))} \leq C\eta_c^{\alpha_{\mathrm{reg}}/(1+\alpha_{\mathrm{reg}})}.$$

The exact phase-gradient formula is

$$\nabla U = -\frac{\nabla v}{av(2 - h_c v)}, \qquad e_s = -\frac{\nabla v_0}{av_0(2 - h_c v_0)}.$$

The denominators stay uniformly positive. These formulas and the preceding estimates put ∇U in a fixed small neighborhood of e_s on $B_{1/2}(x_c)$. They also give its common $C^{0,\alpha_{\mathrm{reg}}}$ bound. The fractional power is used only to enter this elliptic neighborhood. The phase value error remains $O(\eta_c)$ and will be the input to the linear Schauder estimate below, so no final error exponent is weakened. On this fixed elliptic neighborhood, put $\psi = U - s$. The phase equation has the form $A_p(e_s + \nabla\psi) : D^2\psi = f(U)(1 - |e_s + \nabla\psi|^p)$, where $f(U) = -2ca\tanh(aU)$. The right-hand side is a bounded smooth coefficient times $\nabla\psi$. Thus ψ solves a homogeneous linear equation. Its drift can be written without dividing by a derivative of ψ:

$$A_p(e_s + \nabla\psi) : D^2\psi + \mathbf{c}(x)\cdot\nabla\psi = 0, \qquad \mathbf{c}(x) = pf(U)\int_0^1 |e_s + \tau\nabla\psi|^{p-2}(e_s + \tau\nabla\psi)\, d\tau.$$

This follows by integrating the derivative of $|e_s + \tau\nabla\psi|^p$ from zero to one. The drift inherits uniform Hölder bounds in the fixed elliptic neighborhood.

Normalized interior estimates first give uniform Hölder bounds for its coefficients. Noncritical bootstrapping gives the fixed higher bounds needed here. Interior Schauder estimates then bound its C^2 norm by its C^0 norm on a slightly larger unit ball. The coefficient error therefore keeps the face-error order, without replacing that error by a smaller positive power. Its product with the corridor length tends to zero by (3.20). Include the graph errors and their interpolated fixed-order derivatives in ϵ_0. The potential stays uniformly positive.

For completeness, put $\eta(y, s) = \epsilon_0 + e^{-k[\ell - b\cdot y - (1+t)s]}$. On each enlarged unit ball in the well part, the exponential changes by at most a fixed factor. The preceding value estimate and homogeneous linear equation therefore give $|U - s| + |Z - e_s| + |D^2U| \leq C\eta$ there. The fixed-core estimate gives the same bound on the lower part. After increasing one common constant, both estimates hold throughout $-P_0 < s < S(y)$. They keep r in a fixed neighborhood of one after choosing T_0 large and the preparation error small. The maps $Z \mapsto A_p(Z)$ and their first derivatives are uniformly

smooth on this noncritical range. Also, $q = \operatorname{sech}^2(aU)$ has uniformly bounded derivatives as a function of U. Hence

$$|A_p(Z) - \operatorname{diag}(I, c)| + |\operatorname{div} A_p(Z)| \le C\eta,$$
$$|q - G(s)| + |D| \le C\eta,$$
$$|V_u^+ - ca^2[p^2 - p(p-1)G(s)]| \le C\eta.$$

These are exactly the principal, first-order and zero-order coefficient errors in nondivergence form. The constants are uniform on the fixed compact exponent interval. No new error is introduced at the overlap of the core and well regions, since both estimates concern the same coefficients.

Take the exponents α, μ from (3.21). Let $h = h_{b,P_0}$ solve

$$[-c\partial_s^2 + ca^2\{p^2 - p(p-1)G(s)\} - \alpha^2|b|^2]h = 0, \qquad h(-P_0) = 0,$$

with growing coefficient one at infinity. For small angles the ODE potential has a fixed positive lower bound. Start with $h(-P_0) = 0$, $h'(-P_0) = 1$ and divide by its growing coefficient. That coefficient is positive, since h, h' stay positive and the potential approaches $c\mu^2$ exponentially. To quantify the normalization, denote the initial-value solution by $\widetilde{h}$ and write $\widetilde{h}'' = V_0(s)\widetilde{h}$, with $0 < m_0^2 \le V_0 \le M_0^2$ uniformly in the allowed parameters. For $h_\lambda(s) = \sinh(\lambda(s + P_0))/\lambda$,

$$(\widetilde{h}'h_\lambda - \widetilde{h}h_\lambda')' = (V_0 - \lambda^2)\widetilde{h}h_\lambda.$$

Its initial value is zero. Choosing $\lambda = m_0, M_0$ gives

$$m_0 \coth(m_0(s + P_0)) \le \widetilde{h}'/\widetilde{h} \le M_0 \coth(M_0(s + P_0)).$$

Both $\widetilde{h}$ and $\widetilde{h}'$ stay positive because $\widetilde{h}'' > 0$ as long as $\widetilde{h} > 0$.

For the asymptotic coefficient, set $g = \widetilde{h}'/\widetilde{h}$, $Q(s) = a^2p(p-1)G(s)$, and $v = g - \mu$. The Riccati equation is $v' + (g + \mu)v = -Q$. The preceding lower bound gives $g + \mu \ge \eta_0 > 0$, so

$$|v(s)| \le |v(0)|e^{-\eta_0 s} + \int_0^s e^{-\eta_0(s-t)}Q(t)\,dt \qquad (s \ge 0).$$

This bound has uniformly finite integral and uniform tails. Consequently the growing coefficient is

$$A_\infty = \widetilde{h}(0)\exp\left(\int_0^\infty v(s)\,ds\right) > 0.$$

Comparison with the same constant-potential solutions bounds $\widetilde{h}(0)$ between $\sinh(m_0P_0)/m_0$ and $\sinh(M_0P_0)/M_0$. Thus A_∞ has uniform positive upper and lower bounds, and $h = \widetilde{h}/A_\infty$ has growing coefficient one.

The normalization and Riccati bounds give $h, h' > 0$, $h'' > 0$, and

$$h(s) \asymp e^{\mu s} \quad (s \ge 0), \qquad c_1 \le h'/h \le C(1 + (s + P_0)^{-1}).$$

These bounds are uniform. The exact reference solution is

$$B_0(y, s) = e^{-\alpha\ell + \alpha b\cdot y}h(s).$$

At the cropped top it is at most $Ce^{-\beta L(y)}e^{-\mu T_0}$.

Define $\delta_*(y, s) = \ell - b\cdot y - (1 + t)s$ and $S_{\max} = \max_{|y|\le A\ell} S(y)$. For a fixed large K use

$$w = K\epsilon_0(s - S_{\max}) + Ke^{-k\delta_*} - Ke^{-k(1+t)T_0}.$$

Then $w \le 0$ and $w \ge -CK\epsilon_0\ell - Ke^{-k(1+t)T_0}$. The gradient is a positive combination of e_s and $(b, 1+t)$. The Hessian is a positive multiple of $(b, 1+t)\otimes(b, 1+t)$. It follows that $H_p^+(B_0e^w)/(B_0e^w)$ is bounded above by

$$-cK(\epsilon_0 + e^{-k\delta_*})(1 + h'/h).$$

Writing $A_0 = \operatorname{diag}(I, c)$, the exact reference expression is

$$-2\langle A_0\nabla \log B_0, \nabla w\rangle - \operatorname{tr}(A_0D^2w) - |\nabla w|_{A_0}^2.$$

Here $\nabla \log B_0 = (\alpha b, h'/h)$ and $\nabla w = K\epsilon_0 e_s + Kke^{-k\delta_*}(b, 1+t)$. Their scalar product is positive and controls the displayed bound because h'/h has a uniform positive lower bound. The other two terms are nonpositive. Put $\eta = \epsilon_0 + e^{-k\delta_*}$.

The derivatives used in this bound are

$$\frac{\nabla(B_0 e^w)}{B_0 e^w} = \nabla \log B_0 + \nabla w,$$

$$\frac{D^2(B_0 e^w)}{B_0 e^w} = \frac{D^2 B_0}{B_0} + \nabla \log B_0 \otimes \nabla w + \nabla w \otimes \nabla \log B_0 + D^2 w + \nabla w \otimes \nabla w.$$

The quotient h''/h is bounded by the ODE. The remaining lower-end singular terms are only linear in h'/h. These formulas give the following bound with the same error η. Direct differentiation bounds the coefficient perturbation, after division by $B_0 e^w$, by

$$C\eta(1 + h'/h)(1 + K\eta + K^2\eta^2).$$

First fix K large enough to dominate the constant C. Then enlarge T_0 and improve preparation until $K\eta$ is uniformly small. The negative reference term absorbs this whole perturbation, so $\mathscr{H}_u^+(B_0 e^w) \le 0$. Also $\epsilon_0 \ell \to 0$ bounds w below by a fixed constant. This correction therefore causes no gap-dependent loss at the base.

A sufficiently small fixed multiple $B = c_* B_0 e^w$ lies below F on the cropped top. It vanishes on the bottom. The factor c_* is independent of ℓ. On the side, B has a fixed upper bound: its top value is exponentially small, and h increases with s.

Choose a small fixed $\omega > 0$. The positive potential and bounded coefficients make

$$\Psi(y) = \exp\{\omega\sqrt{1 + |y|^2} - \omega\sqrt{1 + A^2\ell^2}\}$$

a strict supersolution after reducing ω if necessary. Indeed, its image divided by Ψ is at least $\min_{\mathcal{P}} ca^2 p - C(\omega + \omega^2) > 0$. Here the coefficient and drift bounds are uniform on the noncritical corridor. Subtract $C\Psi$ from B. This subsolution is nonpositive on the side and bottom, and remains below F on the top. Comparison gives $F \ge B - C\Psi$ in the corridor. The cropped corridor is bounded and Lipschitz. If $G = F - B + C\Psi$, then $\mathscr{H}_u^+ G \ge 0$ and G has nonnegative boundary trace. Its negative part is an admissible H_0^1 test. It gives

$$0 \le -\int A_p(Z)\nabla G_- \cdot \nabla G_- - \int V_u^+ G_-^2 \le 0.$$

Uniform ellipticity and the positive potential force $G_- = 0$.

At $y = 0, s = 0$ we obtain

$$F(z) \ge ce^{-\alpha\ell} - Ce^{-\omega A\ell}.$$

Choose A with $\omega A > \max_{\mathcal{P}, |b| \le b_0} \alpha + 1$. Then decrease the angle until $A|b| \ll 1$, and enlarge the prepared window to contain the corridor. For all sufficiently large gaps this gives $F(z) \ge ce^{-\alpha\ell}$.

At the base zero, $q = 1$ and $e = Z/|Z|$, whence exactly

$$F(z) = 1 - |Z(z)| = 1 - (1 - D(z))^{1/p} \asymp D(z).$$

The exponent inequality $\alpha \le 2\beta/(1+t)$ gives the pressure bound. Corollary 2.13 and the following physical-mass comparison give $I(z) \ge c_I D(z)$, with the window chosen first. This gives the mass form of the estimate. For $0 \le D \le 1/2$, the endpoint comparison is explicitly

$$\frac{D}{p} \le 1 - (1 - D)^{1/p} \le \frac{2^{1-1/p}}{p} D.$$

The inequalities follow from the mean-value theorem and are uniform on the fixed exponent interval.

Choose these parameters before the directional branch's post-turn cutoff offset. The pressure constant is therefore independent of that later offset. □

3.4. **Physical surface tests.** The mean-curvature and turn-pressure bounds now enter the physical surface test. We adapt the lifting of surface amplitudes and the Jacobian and profile-deficit calculations from [35, Lemma 8.3 and Proposition 8.4]. The calculation below uses the weighted Jacobi form and the form-domain recovery in Lemma 2.7. The normal profile of the surface test must account for the neighboring transition. Cutting off q at the chamber middle would spend too much of the retained mass. The directional amplitude used below becomes small after the turn.

We require small zero discrepancy only in a buffered ambient neighborhood. Use the physical mass (2.1), with its window fixed before the discrepancy threshold. Relative regularity gives

$$b^2 + |H| + |\nabla_\Sigma H| \le CD, \qquad |\nabla_\Sigma D| \le CD, \qquad c_I D \le I_P \le C_P D. \tag{3.23}$$

The mass lower bound already holds on a unit normal core. For the mean-gradient term, differentiate $H = (c/p)\partial_\nu D/(1-D)$ tangentially. The derivative of ν contributes B, and therefore

$$|\nabla_\Sigma H| \le C(|B|\,|\nabla D| + |D^2 D| + |\nabla D|^2) \le CD.$$

Here $1 - D \ge 1/2$ and the first two relative derivative estimates for D are used on the fixed core.

For any small fixed $\xi > 0$, propagation has rate $2\beta + \xi$ and clearing has rate $2\beta - \xi$. Their reciprocal square-root tail has rate $\beta - O(\xi) > 0$. The $q^{p+1}D$ tail has rate $2a - O(\xi) > 0$. Both positive margins are uniform on $\mathcal{P}$. The Fermi-coordinate formulas will be used on each local inverse chart. For $\mathcal{N}(z,s) = z + s\nu(z)$ and $|s|b(z) < 1/2$,

$$d\mathcal{N}(\tau) = (I + sB)\tau, \qquad J(z,s) = \det(I + sB) > 0, \qquad \nabla(g \circ \pi) = (I + sB)^{-1}\nabla_\Sigma g.$$

Here π is the local projection to the base. The distance Hessian is zero on the normal direction and equals $B(I + sB)^{-1}$ on tangent directions. These are local formulas; global injectivity of the normal map is not assumed.

Proposition 3.13 (Small-defect physical surface estimate)**.** *For every $\eta \in (0, 1/10)$, $Q > 2$, and prescribed finite $P_{\min} > 0$, choose $P \ge \max\{1, P_{\min}\}$ and $A_1, A_0, C, \delta_0 > 0$, with $A_0 \gg A_1$, uniformly on $\mathcal{P}$. The choices may depend on $P_{\min}$. Let u be a scalar-stable entire solution of the matched equation, with $-1 < u < 1$, and suppose*

$$0 < \delta \le \delta_0, \qquad \sup_\Sigma D \le \delta.$$

Then each compactly supported $\psi \in C^1(\Sigma)$ satisfies

$$(1-\eta)\int_\Sigma \frac{I_P + b^2}{2}\psi^2 \le (1+\eta)\int_\Sigma |\nabla_\Sigma \psi|^2 + C\delta^Q \int_\Sigma \psi^2. \tag{3.24}$$

The estimate also holds locally if all zeros in the ambient $10A_0\log(1/\delta)$-neighborhood of $\operatorname{supp}\psi$ have defect at most δ. Choose all constants before δ and ψ.

Proof. For a planar layer, $D = b = 0$ and there is nothing to prove. Otherwise the gradient equality case gives $D > 0$ everywhere. Negative discrepancy powers are therefore defined on compact sets. We construct compatible caps and estimate every current. The symbol g below denotes a surface amplitude, which will finally be chosen as the prescribed test ψ. We choose the fixed core width above $P_{\min}$ before choosing the discrepancy threshold. The constants for that core may depend on its chosen width.

Set

$$L = A_1\log(1/\delta), \qquad R = A_0\log(1/\delta), \qquad \kappa = C\sqrt{\delta}.$$

Equation (3.23) makes each relevant zero regular, with curvature bounded by κ. Lemma 3.10, with $r = R/3$, extends the relevant sheet pieces over tangent disks of radius $4R/3$, with slope $O(\kappa R)$ and height error $O(\kappa R^2)$. Choose A_0/A_1 so that their $O(L)$ observation neighborhood has radius less than R. Then every required $B_{20R/3}$ buffer stays inside the prescribed $10R$ good neighborhood of the test support. Only regularity and relative closedness in this buffer are needed. Under the

global assumption the construction is available up to radius c/κ; here we use only fixed multiples of R, with $\kappa R \to 0$.

For all pieces based in the $O(L)$ observation window, (3.19) bounds their relative unoriented angle by $C(L/R + \kappa R)$. Choose A_0/A_1 large and then δ_0 small. They are then graphs of small slope over the reference tangent plane on a common disk of radius cR. Distinct pieces are ordered there, by embeddedness and continuity. Shrink the common graph window by a fixed factor and choose $r_* = c_g R$, where $c_g > 0$ is fixed. Each relevant piece is a graph $y \mapsto (y, f(y))$ on the convex disk $B^{n-1}_{4r_*}$, with

$$\|Df\|_\infty \le \frac{1}{4}, \qquad \|D^2 f\|_\infty \le C\kappa.$$

The graph formula for its second fundamental form gives the second bound from the curvature estimate and the first bound. Choose A_0/A_1 so that all observation bases lie over $B^{n-1}_{r_*/2}$ and $L/r_* \le 1/8$. These are fixed choices, since the observation radius is a fixed multiple of L.

Consider a supported normal representation $x = (y_0, f(y_0)) + t\nu(y_0)$, with $|t| \le 2L$. Then

$$|x'| \le r_*/2 + 2L \le r_*, \qquad |f(x') - x_n| \le \|Df\|_\infty |x' - y_0| + 2L \le \tfrac{5}{2}L < r_*.$$

The nearest-foot assertion of Lemma 3.10 therefore gives an interior minimizing foot. To check injectivity as well, consider the squared-distance function on the whole graph disk:

$$\Phi_x(y) = \tfrac{1}{2}(|y - x'|^2 + |f(y) - x_n|^2).$$

The known normal representation gives $|f(y_0) - x_n| \le 2L$. For every $y \in B^{n-1}_{4r_*}$, both y and y_0 lie in that disk, so

$$|f(y) - x_n| \le 2L + 8r_*\|Df\|_\infty \le 2L + 2r_*.$$

Consequently, for every horizontal vector v,

$$\begin{aligned} D^2\Phi_x(y)[v,v] &= |v|^2 + (Df(y)\cdot v)^2 + (f(y) - x_n)D^2 f(y)[v,v] \\ &\ge [1 - C\kappa(2L + 2r_*)]|v|^2 \ge \tfrac{1}{2}|v|^2 \end{aligned}$$

when δ_0 is small. Indeed, $r_* = c_g R$ and both κR and κL tend to zero. This estimate holds throughout the convex disk, including the segment between any two candidate feet. Every normal representation from this graph is a critical point of Φ_x. Strict convexity gives at most one such point. The signed normal parameter is then unique as well. Thus the normal map is injective on the supported part of each prepared graph piece. This is a statement inside the buffered graph window. It uses neither curvature improvement nor clean-ball preparation.

The distance between distinct local sheets tends to infinity as $\delta \to 0$. Suppose otherwise. Choose $\delta_j \downarrow 0$, exponents $p_j \in \mathcal{P}$, and zeros z_j, w_j on two distinct ordered graph pieces, with $|z_j - w_j| \le M$ for a fixed $M < \infty$. Translate z_j to zero and rotate its phase normal into e_n. The transformed functions, still denoted by u_j, remain entire solutions of the same matched equation, and $D_j(0) \le \delta_j$. After a subsequence, $p_j \to p_* \in \mathcal{P}$.

The local $C^{1,\alpha}$ estimates are uniform on every fixed ball. Compactness on successively larger balls and a diagonal subsequence therefore give an entire limit u_*, with $u_j \to u_*$ in $C^1_{\mathrm{loc}}(\mathbb{R}^n)$. To pass the equation, fix a compactly supported smooth test. On its support, the gradients converge uniformly. Joint continuity of the flux and the reaction in the exponent and their arguments passes the weak equation to the limit. Thus u_* is an entire matched weak solution with $-1 \le u_* \le 1$. Moreover,

$$u_*(0) = 0, \qquad \nabla u_*(0) = \lim_j a_{p_j}(1 - D_j(0))^{1/p_j} e_n = a_{p_*} e_n.$$

This nonzero gradient makes the limit nonconstant. Well-contact rigidity gives $-1 < u_* < 1$ everywhere. Equality holds in the gradient bound at zero, so Proposition 2.1 gives $u_*(y, s) = g_{p_*}(s)$, with the chosen increasing orientation.

The common graph radii tend to infinity, whereas the transformed point w_j stays in a fixed ball. Write $R_j = R(\delta_j)$ and $\kappa_j = \kappa(\delta_j)$ for the scales already chosen. The angle estimate (3.19) gives

$$\theta_j \le C(M/R_j + \kappa_j R_j) \longrightarrow 0$$

between the unoriented normals at the two chosen zeros. Choose the upward normals for these graph representations. Curvature control keeps those normals close to e_n on every fixed graph disk. The buffered graph construction therefore puts both pieces over a common fixed horizontal disk, with slopes at most $1/4$ for large j. Take its radius $H = 4(M+1)$ and retain a slightly larger fixed cylinder for the convergence argument. Both graph heights on B_H^{n-1} lie in $(-H, H)$ for large j. For the first graph this follows from its value zero at the origin and the slope bound. For the second, use its point w_j, whose height and horizontal distance are at most M, and integrate the same slope bound. On this fixed cylinder,

$$\partial_s g_{p_*}(s) \ge a_{p_*} \operatorname{sech}^2(a_{p_*} H) > 0.$$

The C^1 convergence therefore gives one positive lower bound for $\partial_s u_j$ there, for all large j. Each vertical line has at most one zero. The two graph functions must coincide on B_H^{n-1}, contrary to their choice as distinct ordered pieces. This argument also covers distinct local pieces of the same global connected component.

We check the quantitative inputs to the face, turn, and pressure barriers. Use a fixed smaller multiple of the common graph window, so that every comparison retains its larger graph buffer. These fixed radius ratios only change the constants in the estimates below. Each prepared graph has defect at most δ and height error $O(\kappa R^2)$ on this window. Relative estimates and interpolation make each required higher graph norm small. The coefficient error is consequently

$$\epsilon_0 = C\delta + C_N \kappa^{\vartheta} R^N + Ce^{-cR}, \qquad \epsilon_0 R \longrightarrow 0, \tag{3.25}$$

for fixed $N, \vartheta > 0$. Choose the associated constants before decreasing δ. The logistic barrier uses both actual face coefficients and lateral exponentials from the larger window. It gives, on the own side,

$$\left| \frac{1-u}{1-\tanh(as)} - 1 \right| \le C\left(\epsilon_0 + e^{-2a[\ell - b_* \cdot y - (1+t_*)s]}\right), \qquad |b_*|^2 + t_*^2 = 1.$$

The exact gradient inequality on the own normal supplies the lower comparison. No fixed error is multiplied by the chamber length.

Apply Lemma 3.11 with any fixed post-turn band needed below. Its hypotheses follow from (3.25), the separation of the sheets, and the small relative angle just proved. For a prescribed C^1 accuracy, first choose its angle threshold and then increase A_0/A_1; finally decrease δ_0 to meet the gap and preparation thresholds. No clean-ball preparation is used.

Lemma 3.12 applies on the same chamber, after increasing the fixed buffered ratio if necessary. If ℓ denotes the adjacent-face distance and $|b_*|^2 + t_*^2 = 1$ describes its affine normal, use the capacity normalization

$$\mathcal{P}_{\text{turn}} := e^{-2\beta\ell/(1+t_*)}.$$

The distance and affine normal are taken at the current base z. Turn preparation makes the actual middle pressure comparable to this quantity, with fixed positive constants. The pressure lemma gives

$$\mathcal{P}_{\text{turn}} \le CD(z). \tag{3.26}$$

In particular there is no additive $C\delta$ error when $D(z) \ll \delta$. Fix the pressure crop, lateral ratio, and pressure constant before choosing the later post-turn cutoff offset. Reflect u when treating a negative chamber. The matched equation, defect, and pressure comparison are invariant under this reflection.

The caps must agree wherever normal charts overlap. Use the even far cap $\zeta_0(s/L)$. It equals one for $|s| \le L$ and vanishes for $|s| \ge 2L$. If the observation window of size $8L$ contains a first neighbor on a side, let r_i, r_j be the true distances to the own sheet and that neighbor. Define

$$t = (r_i - r_j)/2.$$

The middle cap equals one through $t = T$ and vanishes for $t \ge T + 1$. Multiply it by the far cap. Since the distance functions have unit gradient, $|\nabla t| \le 1$. Thus a middle-cap derivative is uniformly bounded in physical coordinates, while a far-cap derivative is bounded by C/L. When there is no relevant neighbor, take the middle cap to be one. Enlarge the fixed observation multiple if needed. A graph entering or leaving that window then has its middle band outside $|s| = 2L$. The local formulas consequently agree. If a graph is at distance at least $8L$ from the base, then for $|s| \le 2L$ its distance from the ray point is at least $6L$ by the triangle inequality. It cannot affect a supported middle band. Enlarging the observation window provides a margin around every change in the local neighbor list. A sheet here means its selected graph piece in that window. Distinct pieces of the same global connected component are treated as distinct sheets until they leave the window.

Ordered disjoint graphs have a unique first neighbor. Their true-distance projections are unique in these flat windows. The cap definitions therefore agree on overlaps of charts.

The condition $2L\kappa \to 0$ keeps the normal map nonsingular. Indeed, on the whole supported tube,

$$\|(I + sB)^{\pm 1} - I\| + |J - 1| \le CL\kappa = o(1).$$

The metric and Jacobian therefore converge uniformly to the planar ones. Along either side of the own ray, differentiate t with respect to outward arclength. Its derivative is $(1 + t_*)/2 + o(1)$. This is positive and as near one as required. Each side of a ray meets at most one supported middle band. The cap ends before the adjacent zero because all gaps diverge while T stays fixed.

On every supported tube we have

$$\operatorname{dist}(z + s\nu(z), \Sigma) \ge |s| - C_T. \tag{3.27}$$

The own sheet is nearest on the own side of the bisector, where $t \le 0$. On the rest of the supported chamber, $0 < t < T + 1$ and $r_j = r_i - 2t \ge |s| - 2T - 2$. Other local sheets are outside the chamber. A segment from a chamber point to such a sheet must first meet one of the two boundary graphs. That first intersection is no farther from the point. Thus no nonboundary sheet can reduce the lower distance bound.

Zeros beyond the buffered window are farther away as well. Sharp clearing gives, for any small fixed $\xi > 0$,

$$q^p(z + s\nu(z)) \le C_T e^{-(2\beta - \xi)|s|}. \tag{3.28}$$

Choose A_1 large to make the far-cap cost at most $C\delta^{Q'}$ per unit base area, with any prescribed Q'. To see the choice explicitly, put $\beta_- = \min_{p \in \mathcal{P}} \beta > 0$. The derivative of the far cap is $O(1/L)$. The branch amplitude used below is bounded by q. Thus its kinetic cost is bounded by

$$\frac{C}{L^2} \int_{|s| \ge L} q^p J \, ds \le \frac{C_T}{L^2} e^{-(2\beta_- - \xi)L} = \frac{C_T}{L^2} \delta^{A_1(2\beta_- - \xi)}.$$

Take $A_1(2\beta_- - \xi) > Q'$. The same choice controls a cap enlarged by any fixed distance. There is no base derivative of L.

For the capped branches based in $\operatorname{supp} g$, each physical point has finitely many relevant normal preimages. Their bases lie in this compact regular set, and $|s| \le 2L$. The normal map is locally invertible because $2L\kappa < 1$. An infinite fiber would accumulate in this compact parameter region, contradicting local invertibility. All preimages lie in a common buffered window. Since a cap ends before the next zero, only the two boundary graphs of its phase chamber can contribute, each at most once. There are thus at most two nonzero branches. They overlap only in a fixed turn band. This justifies the root-sum test without an area-growth bound.

To estimate the physical branch and all of its normal currents, write

$$A = A_p(Z), \quad \mathcal{H} = D^2U, \quad \mathcal{S} = \operatorname{tr}(A\mathcal{H}A\mathcal{H}), \quad e = \nabla s, \quad T = De, \quad \varphi = (1 + Z \cdot e)/2, \quad h = q\varphi.$$

Here T is the distance Hessian. The post-turn offset retains its fixed value. The physical stability matrix is $A_u = A_p(\nabla u) = a^{p-2}q^{p-2}A$. For a compactly supported amplitude v, the weighted product rule gives

$$Q(hv) = \int h^2 A_u \nabla v \cdot \nabla v + \langle L_u h, hv^2 \rangle.$$

Both terms are defined in the form domain from Section 2. In fact $h = (q + a^{-1}u_e)/2$, so this is exactly the variable-field amplitude treated there. Apply its branch identity and (2.9). For a compact tube amplitude v they give

$$\begin{aligned} \langle L_u h, hv^2 \rangle = & -\frac{1}{2}\int q^{p+1} D\varphi v^2 - \frac{a^{p-2}}{2}\int q^p \varphi \operatorname{tr}(A\mathcal{H}T)v^2 \\ & + \frac{a^{p-2}}{2}\int q^p ATZ \cdot \nabla(\varphi v^2). \end{aligned} \tag{3.29}$$

At a critical point these hold in the weighted weak sense, using the natural weighted Hessian estimate. An unweighted estimate for D^2u is unnecessary.

Put $D_0 = D(z)$. On a prepared tube, $|T(z,s)| \le Cb(z)$. The absolute current bounds are

$$Cq^p(|T|\sqrt{\mathcal{S}}v^2 + |T|^2v^2 + |T||v||\nabla v|).$$

Choose $\theta = 1/2$. Corollary 2.23, with (3.28), yields

$$\begin{aligned} \int q^p D_0^\theta D^{1-\theta} J\, ds &\le C_T D_0, \\ \int_{|s|>P} q^p (D/D_0)^\theta J\, ds &\le C_T e^{-\gamma_{\text{tail}} P}. \end{aligned} \tag{3.30}$$

The reciprocal decay rate $\gamma_{\text{tail}} > 0$ is uniform on $\mathcal{P}$ when the fixed losses are sufficiently small. The same ray estimate gives $\int q^{p+1} DJ\, ds \le CD_0$, with rate $2a - O(\xi) > 0$. These are normal estimates; no isotropic propagation bound enters.

To bound the source term, apply the negative-power test estimate following Proposition 2.9, which was obtained from (2.11). Take $\theta = 1/2$ and $\zeta = D_0^{\theta/2} g(z) \widehat{\Theta}$, and omit its nonnegative weighted gradient term. The cap $\widehat{\Theta}$ extends a fixed distance beyond the actual cap. Choose it so that $0 \le \Theta \le \widehat{\Theta} \le 1$. Its derivatives have the same bounds as those of the actual cap. The enlarged post-turn band still ends before the next zero, because the gaps tend to infinity. Equation (3.23) bounds the base derivative of $D_0^{\theta/2}$. Explicitly,

$$|\nabla_\Sigma D_0^{\theta/2}| = \frac{\theta}{2} D_0^{\theta/2-1} |\nabla_\Sigma D_0| \le C D_0^{\theta/2}.$$

The inverse Fermi metric is uniformly bounded. Expansion of the test derivative therefore has the three squared costs

$$CD_0^\theta g^2 \widehat{\Theta}^2, \qquad CD_0^\theta |\nabla_\Sigma g|^2 \widehat{\Theta}^2, \qquad CD_0^\theta g^2 |\nabla \widehat{\Theta}|^2.$$

The negative-power estimate multiplies these by $q^p D^{1-\theta}$. Its zero-order term has the additional bounded factor q. The first line of the reciprocal estimate controls the first two costs. The third is estimated on the two kinds of cap below.

By (3.30), a derivative of g costs $D_0|\nabla g|^2$. At a post-turn cap, use $D \le 1$, the fixed band width, and $q^p \le C_T \mathcal{P}_{\text{turn}}$ from turn preparation. Its squared derivative cost per unit base area is therefore

at most $C_T D_0^\theta \mathcal{P}_{\mathrm{turn}}$. By (3.26) this is bounded by $C_T D_0^{1+\theta} \le C_T D_0$. At a far cap, $D_0^\theta \le 1$ and the far-cap bound give $C\delta^{Q'}$. Therefore

$$\int q^p \mathcal{S}(D_0/D)^\theta g^2 \widehat{\Theta}^2 \le C_T \int_\Sigma D_0(g^2 + |\nabla g|^2) + C\delta^{Q'} \int_\Sigma g^2. \tag{3.31}$$

Lifted integrals use the pullback Euclidean measure. The estimate may be summed over tubes even when they overlap. Indeed, the normal map is a local diffeomorphism on the lifted tube. Pull back the weak discrepancy equation on each inverse-function neighborhood. The resulting equation is local on the lifted tube and admits the compactly supported test above. Its measure is exactly $J\,ds\,d\Sigma$.

For the first current outside $|s| \le P$, apply Cauchy–Schwarz with the reciprocal weights in (3.30) and (3.31). The two squared factors are

$$\int q^p \mathcal{S}(D_0/D)^\theta g^2 \widehat{\Theta}^2 J\,ds\,d\Sigma$$

and

$$\int_{|s|>P} q^p |T|^2 (D/D_0)^\theta g^2 \frac{\Theta^4}{\widehat{\Theta}^2} J\,ds\,d\Sigma.$$

Set the quotient to zero where $\widehat{\Theta} = 0$. It is at most one because $0 \le \Theta \le \widehat{\Theta} \le 1$. The source estimate bounds the first factor; the second is at most $C_T e^{-\gamma_{\mathrm{tail}} P} \int_\Sigma D_0 g^2$. Writing $M_0 = \int D_0 g^2$, $E_0 = \int D_0 |\nabla g|^2$, and $N_0 = \int g^2$, the product of square roots is bounded by

$$C_T e^{-\gamma_{\mathrm{tail}} P/2} \left[\sqrt{M_0(M_0 + E_0)} + \sqrt{\delta^{Q'} M_0 N_0} \right].$$

Use $\sqrt{M_0(M_0+E_0)} \le M_0 + E_0$ and $2\sqrt{\delta^{Q'} M_0 N_0} \le M_0 + \delta^{Q'} N_0$. Since $D_0 \le 1$, this keeps the far-cap power Q' and gives the small Dirichlet coefficient after P is large. The offset T is fixed before this tail limit, so the constants C_T are already fixed.

Then use Young's inequality and $b^2 \le C D_0$. This gives

$$\left| \int_{|s|>P} q^p |T| \sqrt{\mathcal{S}} g^2 \Theta^2 \right| \le o_P(1) \int D_0 g^2 + o_P(1) \int |\nabla g|^2 + C\delta^{Q'} \int g^2.$$

We estimate the other two currents explicitly. Write $\widetilde{g}(z + s\nu(z)) = g(z)$ for the Fermi lift of the base amplitude on each lifted tube. On the supported caps, $|s| \le 2L$ and $2L\kappa \to 0$. The Fermi formulas therefore give

$$|T(z,s)| \le C b(z), \qquad |\nabla \widetilde{g}(z,s)| \le C |\nabla_\Sigma g(z)|, \qquad 0 < J(z,s) \le C.$$

Fix the clearing loss so that $c_{\mathrm{cl}} = 2 \min_{p \in \mathcal{P}} \beta_p - \xi > 0$. The clearing estimate above implies

$$\int_{\{|s|>P,\ \Theta(z,s) \ne 0\}} q^p J\,ds \le C_T e^{-c_{\mathrm{cl}} P}.$$

Since $0 \le \Theta \le 1$, the current containing $|T|^2$ satisfies

$$\int_{|s|>P} q^p |T|^2 g^2 \Theta^2 J\,ds\,d\Sigma \le C_T e^{-c_{\mathrm{cl}} P} \int_\Sigma b^2 g^2 \le C_T e^{-c_{\mathrm{cl}} P} \int_\Sigma D_0 g^2.$$

For the base derivative in $v = g\Theta$, use $\nabla v = \Theta\nabla\widetilde{g} + g\nabla\Theta$. Its first part has the bound

$$\begin{aligned}\int_{|s|>P} & q^p|T||g|\Theta^2|\nabla\widetilde{g}|J\,ds\,d\Sigma \\ &\le C_T e^{-c_{\mathrm{cl}}P}\int_\Sigma b|g||\nabla_\Sigma g| \\ &\le C_T e^{-c_{\mathrm{cl}}P}\left(\int_\Sigma D_0 g^2 + \int_\Sigma |\nabla_\Sigma g|^2\right).\end{aligned}$$

The last line uses $2b|g||\nabla_\Sigma g| \le b^2 g^2 + |\nabla_\Sigma g|^2$ and $b^2 \le CD_0$. The Fermi constants and C_T are uniform on $\mathcal{P}$. The offset T is already fixed, so these coefficients tend to zero as $P \to \infty$. This adds no far-cap error and does not change its power Q'. The terms containing $\nabla\Theta$ are the cap currents estimated next. Post-turn cap derivatives cost at most $C_T b\mathcal{P}_{\mathrm{turn}} g^2 \le C_T\sqrt{\delta}D_0 g^2$. Far-cap derivatives cost $C\delta^{Q'}g^2$. These estimates cover every tail current in (3.29). We split the paired integrals at $|s| = P$, without integrating by parts there. Thus this split creates no boundary term, and the exterior cap traces vanish.

On the fixed core, relative regularity gives

$$\varphi = 1 - D/(2p) + O_P(D_0^2), \quad \operatorname{tr}(A\mathcal{H}T) = b^2 + O_P(D_0^{3/2}), \quad ATZ = O_P(bD_0), \qquad \nabla\varphi = O_P(D_0).$$

Here and below the cap is one on the fixed core. To ensure this, decrease δ_0 after fixing P and the post-turn offset. The diverging gaps and $L \to \infty$ then contain that core. Normal propagation gives $Z - e = O_P(D_0)$ and $U - s = O_P(D_0)$. For this transport, freeze $e_0 = \nu(z)$ at the base. Along its straight ray the Fermi field e equals e_0. The C^2 estimate in (2.16) gives directly

$$\partial_{e_0}\mathcal{H} = D^2 U_{e_0} = -D^2(1 - U_{e_0}) = O_P(D_0).$$

All derivatives here are Cartesian derivatives in the fixed direction. The derivatives of the varying Fermi field remain in the separate currents through $T = De$. At the base, the tangential block of $\mathcal{H}$ is rB. Its normal row and column are $O(D_0)$, by $\mathcal{H}Z = -\nabla D/(pr^{p-2})$ and $Z(z) = r(z)e_0$. Integration along the fixed ray therefore gives $\mathcal{H} = B + O_P(D_0)$, with B extended by zero in the normal direction. Also $T = B(I + sB)^{-1} = B + O_P(D_0)$ and $Te = 0$. These facts give the trace and current expansions above. In particular, $A = I + (p-2)e\otimes e + O_P(D_0)$. Because the extension of B vanishes on e, the leading contraction is exactly $\operatorname{tr}(B^2) = b^2$. The product of an $O_P(\sqrt{D_0})$ curvature factor with an $O_P(D_0)$ remainder is $O_P(D_0^{3/2})$.

For the expansion of φ, first separate the component of Z parallel to the unit normal from its orthogonal component. The exact identity is

$$r - Z\cdot e = \frac{r^2 - (Z\cdot e)^2}{r + Z\cdot e} = \frac{|Z - (Z\cdot e)e|^2}{r + Z\cdot e}.$$

The relative ray estimate gives $Z - e = O_P(D_0)$. Orthogonal projection therefore gives $|Z - (Z\cdot e)e| \le |Z - e| = O_P(D_0)$. On the fixed core, both r and $Z\cdot e$ tend uniformly to one as the base defect tends to zero. The denominator is thus bounded below by a positive constant. Squaring the projection bound gives $r - Z\cdot e = O_P(D_0^2)$. Also $r = (1 - D)^{1/p}$. Since $D = O_P(D_0)$ on this core, Taylor's formula gives

$$r = 1 - D/p + O_P(D_0^2).$$

Substitution into the defining amplitude now yields

$$\varphi = \frac{1 + r}{2} - \frac{r - Z\cdot e}{2} = 1 - D/(2p) + O_P(D_0^2).$$

The actual core mass contains the factors $J\varphi$. They do not change its fixed-window target. Indeed, $J = \det(I + sB) = 1 + sH + O_P(b^2) = 1 + O_P(D_0)$, and hence

$$\left| \sigma_q^{-1} \int_{-P}^{P} q^{p+1} D\varphi J \, ds - I_P(z) \right| \le C_P D_0^2.$$

Furthermore $q = G + O_P(D_0)$ on the core. The curvature term has normalization $a^{p-2}/\sigma_q = 1/J_p$. Its leading normal integral is therefore $b^2 \int_{-P}^{P} G^p \, ds/(2J_p)$. The favorable normalized mass in (3.29) is consequently

$$\frac{1}{2} I_P + \frac{J_{p,P}}{2J_p} b^2 + o_P(D_0), \qquad J_{p,P} = \int_{-P}^{P} G^p.$$

The discrepancy mass outside the core is nonpositive in the branch form, so it may be omitted from its upper bound. A fixed-core base-current cross costs a small multiple of $D_0 g^2$ plus $o_\delta(1)|\nabla g|^2$. For example, its pointwise size is $C_P b D_0 |g||\nabla_\Sigma g|$. For any fixed $\tau > 0$, Young's inequality gives

$$C_P b D_0 |g||\nabla_\Sigma g| \le \tau D_0 g^2 + C_{P,\tau} b^2 D_0 |\nabla_\Sigma g|^2.$$

Since $b^2 \le C D_0 \le C\delta$, the last coefficient tends to zero with δ after P and τ are fixed.

We also estimate the explicit cutoff energy. For $v = g\Theta$, separate its gradient by the matrix Young inequality

$$|\nabla(g\Theta)|_{A_u}^2 \le (1+\tau)\Theta^2 |\nabla g|_{A_u}^2 + (1+\tau^{-1}) g^2 |\nabla \Theta|_{A_u}^2.$$

The fixed choice of τ determines the multiplicative factor that the later cap accuracy must absorb.

The small cap cost comes from the first integral at the turn. For the model profile, on its increasing half-line,

$$\frac{V_p'}{2aV_p} = (1 - V_p^{-p})^{1/p}, \qquad 0 \le 1 - \frac{V_p'}{2aV_p} \le V_p^{-p} \le C e^{-2\beta s_{\mathrm{b}}}.$$

Here s_{b} is unit bisector distance from the turn. We verify the uniform constant directly. Put $p_- = \min \mathcal{P} > 1$ and $p_+ = \max \mathcal{P}$. The inverse integral defining V_p gives

$$C_p^\infty = e^{-\mathcal{I}(p)}, \qquad \mathcal{I}(p) = \int_1^\infty [(t^p - 1)^{-1/p} - t^{-1}] dt.$$

Indeed, subtract $\log V_p(s)$ from its inverse-integral identity and let $s \to \infty$. The integrand is nonnegative. Uniformly for $p \in \mathcal{P}$, it is at most

$$C(t-1)^{-1/p_-} \quad (1 < t \le 2), \qquad C t^{-p_- - 1} \quad (t \ge 2).$$

For the first bound, use $t^p - 1 \ge p_-(t-1)$. For the second, write the difference as $t^{-1}[(1-t^{-p})^{-1/p} - 1]$ and use the mean value theorem on $0 \le t^{-p} \le 2^{-p_-} < 1$. Both majorants are integrable. They give $0 \le \mathcal{I}(p) \le C_{\mathcal{P}}$, and dominated convergence also makes $\mathcal{I}$ continuous in p. Thus $c_* := e^{-C_{\mathcal{P}}} \le C_p^\infty \le 1$. The same inverse-integral identity shows that $e^{-2as_{\mathrm{b}}} V_p(s_{\mathrm{b}})$ decreases from one to this limit. Consequently $V_p(s_{\mathrm{b}})^{-p} \le c_*^{-p_+} e^{-2\beta s_{\mathrm{b}}}$, which proves the required uniform exponential bound. In the parallel model, write the well deficit as $w = h_0 V_*(s_{\mathrm{b}})$. On the post-turn side $u_e = -h_0 V_*'(s_{\mathrm{b}})$, so

$$\frac{h}{q} = \frac{1}{2}\left(1 - \frac{V_*'(s_{\mathrm{b}})}{aV_*(s_{\mathrm{b}})(2 - h_0 V_*(s_{\mathrm{b}}))}\right) = \frac{1}{2}\left(1 - \frac{V_p'(s_{\mathrm{b}})}{2aV_p(s_{\mathrm{b}})}\right) + o(1).$$

The error tends to zero for each fixed band as the gap tends to infinity. Also $q^p \le C h_0^p e^{2\beta s_{\mathrm{b}}}$ there and $h_0^p = 2^p \mathcal{P}_{\mathrm{turn}}$. For tilted faces, projection of the unit bisector normal onto the own normal changes $Z \cdot e$ by $O(\vartheta^2)$. On the cap band $T \le t \le T+1$, the unit bisector coordinate obeys $s_{\mathrm{b}} = T + O(1 + \vartheta^2(T+2)) + o(1)$. Thus the condition $\vartheta^2(T+2) \ll 1$ keeps the exponential conversion bounded uniformly. The C^1 turn approximation transfers these value and derivative bounds to the actual chamber; after T is fixed, increase the gap threshold and decrease the preparation tolerance

until its ratio error is at most ϵ_T. At a fixed large post-turn offset, the one-dimensional profile satisfies

$$q^p \le C\mathcal{P}_{\mathrm{turn}} e^{2\beta T}, \qquad h/q \le C(e^{-2\beta T} + \vartheta^2 + \epsilon_T),$$

Here ϑ is the relative angle, and ϵ_T is the fixed-band C^1 error. The cap energy is at most

$$C\mathcal{P}_{\mathrm{turn}} e^{2\beta T}(e^{-2\beta T} + \vartheta^2 + \epsilon_T)^2 g^2.$$

Choose T large first. Then make the angle and preparation error small. Require also $\vartheta^2(T+2)$ to be small. The bisector cap coordinate and outward ray arclength differ by the factor $(1+t_*)/2 = 1 + O(\vartheta^2)$. This choice bounds the exponential change of variables on the fixed post-turn band. Projection of the opposite unit normal onto the own normal has an error of order ϑ^2. With the already fixed pressure constant, the cost becomes an arbitrarily small multiple of $D_0 g^2$. Separate tangential-cap cross terms using a fixed Young factor chosen before this accuracy. The far cap costs $C\delta^{Q'} g^2$. The normal integral of the tangential coefficient, divided by σ_q, is at most $1+o(1)$. Indeed, $h^2 A_u = a^{p-2} q^p \varphi^2 A$. On the fixed core, $r \to 1$ and $Z/r \to e$, so $A \to I + (p-2) e \otimes e$. The lifted base gradient is orthogonal to e, so the normal part of this limiting matrix contributes zero. Also $\varphi \to 1$ and $q \to G$. Its unnormalized tangential coefficient therefore approaches $a^{p-2} G^p I$. Clearing makes the coefficient integral outside that core arbitrarily small. More precisely, the lifted base gradient is $(I + sB)^{-1} \nabla_\Sigma g$. On the core its coefficient before normalization is $a^{p-2} G^p (I + O_P(\sqrt{\delta}))$. On the remaining tube it is bounded by Cq^p, since $2L\kappa \to 0$. Its normal integral there is small after increasing P.

Write $\mathcal{N}(z,s) = z + s\nu(z)$. Define the assembled test intrinsically by

$$\Psi(x)^2 = \sum_{\substack{z \in \operatorname{supp}\psi,\ \mathcal{N}(z,s)=x \\ |s| \le 2L}} h(z,s)^2 \Theta(z,s)^2 \psi(z)^2.$$

All bases in this formula belong to the regular buffered neighborhood of the test support; no normal is assigned to a critical zero elsewhere. Zero branch values contribute zero. The preceding fiber bound makes the sum locally finite. On each inverse-function neighborhood the summands are the ordinary physical branches. The definition agrees on overlaps because it sums over the fiber rather than choosing branch labels. Apply the gradient inequality in Lemma 2.10 on each inverse-function neighborhood. Its root-variance term is nonnegative because the form matrix is nonnegative. The potential terms add exactly, since the square of the root-sum is the sum of the branch squares. The assembled form is therefore no larger than the sum of branch forms. Integrating this local inequality and using the area formula for $\mathcal{N}$ gives precisely the pulled-back measure $J\,ds\,d\Sigma$ used above. No partition of unity or derivative of a chart label occurs. The base support is compact and the cap length is the fixed number $2L$. Thus the assembled support is compact. To justify admissibility through turns, use weighted approximation. Choose the open lifted tube slightly larger than the cap support. Weighted multiplication gives form membership of each capped branch there. On a smaller inverse chart, multiply the branch by a smooth compact cutoff. It then extends by zero as a form-domain function. Lemma 2.10 gives local form membership of Ψ. A finite cover of its compact support gives global compactly supported form membership. No switch between ambient and branch amplitudes is made, so no inaccurate-ray partition term is created.

Choose the current and cap tolerances small relative to ηc_I. All parameters preceding the core choice are now fixed. The tail estimates above decrease exponentially as P increases, and $J_{p,P}/J_p$ increases to one uniformly on $\mathcal{P}$. Choose one finite $P \ge \max\{1, P_{\min}\}$ so that all tail coefficients meet their chosen tolerances and $J_{p,P}/J_p > 1 - \eta/10$ for every $p \in \mathcal{P}$. A larger prescribed lower bound only enlarges this fixed core choice. The constants in the core errors can grow with P, so choose the discrepancy threshold after P. Reduce it until the cap equals one on $[-P, P]$, the required fixed-core estimates hold, and the $o_P(D_0)$ errors are absorbed in the I_P reserve. The lower bound $I_P \ge I_1 \ge c_I D_0$ uses only the unit core, so c_I is independent of this enlargement. This

proves the stated existence of a window above each prescribed lower bound; it makes no uniform assertion over unbounded core widths. Use $Q' \geq Q$ for the far-cap construction. For any prescribed small $\tau > 0$, the estimates above give

$$0 \leq \sigma_q^{-1} Q(\Psi) \leq (1+\tau) \int_\Sigma |\nabla_\Sigma \psi|^2 - \frac{1}{2} \int_\Sigma \left(I_P + \frac{J_{p,P}}{J_p} b^2 \right) \psi^2 + \tau \int_\Sigma D_0 \psi^2 + C\delta^{Q'} \int_\Sigma \psi^2.$$

Here τ is the mass and kinetic tolerance fixed at the start, small relative to ηc_I and η. The lower bound $I_P \geq c_I D_0$ absorbs the third term in the retained mass. Since $\delta \leq 1$ and $Q' \geq Q$, this proves (3.24).

Begin with the required mass and kinetic tolerances. Choose the pressure crop, lateral ratio, and pressure constant. Next choose the tangential Young factors. Then choose the post-turn offset, its angle tolerance, and its turn accuracy. Fix the current Young factors and the core P. Choose A_1 for the far-cap power, followed by A_0/A_1 for the buffers and angles. Only then decrease δ_0. The choices do not use curvature improvement, clean-ball decay, finite density, or a completed reach weight.

For the local statement, perform the same construction in the buffered ambient neighborhood of the base support. Every required sheet and capped ray stays inside that neighborhood. Zeros outside the window are farther away than the retained distances. They affect neither clearing nor compatibility. □

3.5. **Curvature improvement.** A local logarithmic estimate will recover curvature from the limiting planar geometry.

Lemma 3.14 (Newtonian logarithmic estimate)**.** *Let h be a smooth graph over the fixed Euclidean ball B_{r_0} in $\mathbb{R}^m$, with $m = 6$ and*

$$h(0) = 0, \quad Dh(0) = 0, \quad \|D^2 h\|_\infty \leq C_0.$$

In graph coordinates, suppose its scalar mean curvature is bounded by A and has Lipschitz constant at most B, where $0 < A \leq 1$. Then for $0 < r < r_0/4$,

$$|D^2 h(0)| \leq C r^{-2} \|h\|_{L^\infty(B_{2r})} + Cr^2 + CA \left[1 + \log_+ \left(\frac{rB}{A} \right) \right]. \tag{3.32}$$

The constants depend only on m, C_0, r_0. Here $\log_+ t = \max\{0, \log t\}$ for $t > 0$, and $\log_+ 0 = 0$.

Proof. Write the graph equation as

$$\operatorname{div} \frac{Dh}{\sqrt{1+|Dh|^2}} = \mathcal{H}.$$

For the upward graph normal, the convention $B = d\nu$ gives $\mathcal{H} = -H$. Only the absolute value and Lipschitz bound enter the argument, so this sign does not change the estimate.

The Hessian bound and $Dh(0) = 0$ give $|Dh(x)| \leq C_0 |x|$. Hence

$$\Delta h = \mathcal{H} + E, \qquad E = (\sqrt{1+|Dh|^2} - 1)\mathcal{H} + \frac{D^2 h[Dh, Dh]}{1+|Dh|^2}, \qquad |E(x)| \leq C|x|^2.$$

Choose a smooth radial cutoff η_r that is one on B_r and vanishes outside B_{2r}. Let Γ be the fundamental solution of Δ and define the Newtonian potential

$$w(x) = \int_{B_{2r}} \Gamma(x-y) \eta_r(y) (\mathcal{H}(y) + E(y))\, dy.$$

The source is smooth and compactly supported. Thus the second-derivative formula in Gilbarg–Trudinger [24, Lemma 4.2, formula (4.9), p. 55] applies, and $\Delta w = \mathcal{H} + E = \Delta h$ on B_r. Since

$|\Gamma(x-y)| \le C|x-y|^{2-m}$ and $|E| \le Cr^2$ on B_{2r}, radial integration gives

$$\|w\|_{L^\infty(B_r)} \le C(A+r^2)\int_0^{3r} t^{2-m}t^{m-1}\,dt \le CAr^2 + Cr^4.$$

The function $h-w$ is harmonic on B_r. Its interior derivative estimate [24, Theorem 2.10, p. 23] therefore gives

$$|D^2(h-w)(0)| \le Cr^{-2}\|h-w\|_{L^\infty(B_r)} \le Cr^{-2}\|h\|_{L^\infty(B_{2r})} + CA + Cr^2.$$

It remains to estimate $D^2w(0)$.

In the potential formula, the error E contributes at most Cr^2, since

$$\int_0^{2r} \frac{Ct^2}{t}\,dt \le Cr^2, \qquad E(0)=0.$$

The spherical cancellation of the Hessian kernel and radiality of the cutoff allow subtraction of $\mathcal{H}(0)$ in the other source term. Away from zero, the Hessian kernel is

$$K_{ij}(y) = \frac{1}{|\mathbb{S}^{m-1}|\,|y|^m}\left(\delta_{ij} - m\frac{y_iy_j}{|y|^2}\right).$$

Its integral on every sphere is zero, which is why radiality of the cutoff preserves the cancellation. The remaining source difference is controlled by its minimum of the uniform and Lipschitz bounds.

The distributional point mass costs at most $C|\mathcal{H}(0)| \le CA$. Since

$$|\mathcal{H}(x) - \mathcal{H}(0)| \le \min\{2A, B|x|\},$$

the remaining term is at most

$$C\int_0^{2r} \frac{\min\{2A, Bt\}}{t}\,dt \le CA\left[1 + \log_+\left(\frac{rB}{A}\right)\right].$$

To verify the last bound, first take $B>0$ and split at $2A/B$ if this radius lies in $(0,2r)$. The inner part is at most $2A$, and the outer part is $2A\log(rB/A)$. If $2A/B \ge 2r$, the integral is $2rB \le 2A$. For $B=0$ it vanishes. Combining these bounds with the harmonic remainder estimate proves (3.32).

□

Proposition 3.15 (Uniform curvature improvement)**.** *There are $C<\infty$ and $\delta_0>0$, uniform on $\mathcal{P}$, such that every scalar-stable entire solution in $\mathbb{R}^7$ with $p\in\mathcal{P}$ and $0<\delta=\sup_\Sigma D\le\delta_0$ satisfies*

$$K := \sup_\Sigma |B| \le C\delta\log(C/\delta). \tag{3.33}$$

Proof. Equation (3.23) gives $K \le C\sqrt{\delta}$ and $|H| + |\nabla_\Sigma H| \le C\delta$. In particular, K is finite. If (3.33) fails, choose $p_j \to p_* \in \mathcal{P}$, $\delta_j = \sup_{\Sigma_j} D_j \to 0$, and $K_j = \sup_{\Sigma_j}|B_j| > 0$ with

$$\frac{K_j}{\delta_j \log(1/\delta_j)} \longrightarrow \infty.$$

Take $z_j \in \Sigma_j$ satisfying $|B_j(z_j)| \ge K_j/2$. Every zero is regular because $D_j \le \delta_j < 1$. The zero set is closed, and each of its connected components is closed in the ambient space. An intrinsic Cauchy sequence is therefore an ambient Cauchy sequence with a limit on the same sheet. A regular graph chart at that limit gives intrinsic convergence. Thus each connected sheet is complete in its induced metric. Dilate the connected sheet through z_j by K_j. Use hats for the rescaled quantities. Then

$$|\widehat{B}_j| \le 1, \qquad |\widehat{B}_j(0)| \ge 1/2,$$
$$\|\widehat{H}_j\|_\infty \le C\delta_j/K_j \longrightarrow 0, \qquad \|\widehat{\nabla}\widehat{H}_j\|_\infty \le C\delta_j/K_j^2.$$

Fix a small $\eta>0$ and $Q>2$. First apply Corollary 3.5 with loss $\eta/4$. It gives a fixed window P_I and a defect threshold for which $I_{P_I} \ge (1-\eta/4)b^2$. Next apply Proposition 3.13 with loss $\eta/4$ and

prescribed lower bound $P_{\min} = \max\{1, P_I\}$. It chooses one fixed physical window $P \geq P_{\min}$ and its core, cap, and threshold constants. The entire gradient bound in Proposition 2.1 gives $D \geq 0$ along every normal ray. Therefore

$$I_P(z) - I_{P_I}(z) = \sigma_q^{-1} \int_{P_I < |s| \leq P} q^{p+1} D(z + s\nu(z))\, ds \geq 0.$$

Decrease the working defect threshold below both values. Thus the mass comparison holds for this same physical window. Its constants are fixed before $j \to \infty$. In particular,

$$(1 - \eta/4)\frac{I_P + b^2}{2} \geq (1 - \eta/4)(1 - \eta/8)b^2 \geq (1 - \eta)b^2.$$

The physical surface form then gives

$$(1 - C\eta) \int |B_j|^2 \psi^2 \leq (1 + \eta) \int |\nabla \psi|^2 + C_{\eta,Q} \delta_j^Q \int \psi^2. \tag{3.34}$$

Dilation by K_j changes the final coefficient to $C_{\eta,Q}\delta_j^Q / K_j^2$. On a six-dimensional sheet the exact scale factors are $d\widehat{\Sigma} = K_j^6 d\Sigma$, $|\widehat{B}_j| = K_j^{-1}|B_j|$, and $|\widehat{\nabla}\widehat{\psi}| = K_j^{-1}|\nabla \psi|$. Equivalently, for the test $\psi(x) = \widehat{\psi}(K_j(x - z_j))$,

$$\int |B_j|^2 \psi^2 = K_j^{-4} \int |\widehat{B}_j|^2 \widehat{\psi}^2, \qquad \int |\nabla \psi|^2 = K_j^{-4} \int |\widehat{\nabla}\widehat{\psi}|^2, \qquad \int \psi^2 = K_j^{-6} \int \widehat{\psi}^2.$$

Multiplying the original form inequality by K_j^4 therefore gives exactly the coefficient $C_{\eta,Q}\delta_j^Q / K_j^2$ above.

The contrary ratio implies $K_j \geq \delta_j$ for large j, so this coefficient is at most $C_{\eta,Q}\delta_j^{Q-2} \to 0$. For each η, fix the core and cap constants before letting $j \to \infty$.

We use intrinsic compactness, so nearby ambient sheets cause no loss of multiplicity information. Lemma 3.10, on a fixed sufficiently small rescaled radius, gives tangent graphs containing fixed intrinsic balls. Their slopes and Hessians are uniformly bounded, and every smaller intrinsic ball has a uniform positive volume lower bound. Completeness supplies the ambient continuation required by the lemma on the entire rescaled embedded zero sheet.

For an upper volume bound, use the Gauss equation to obtain a uniform lower Ricci bound. For a unit tangent vector, the equation reads

$$\mathrm{Ric}(v, v) = \widehat{H}_j \widehat{B}_j(v, v) - |\widehat{B}_j v|^2.$$

The curvature bound also bounds $|\widehat{H}_j|$, so the lower Ricci bound is independent of j. Together with the graph-ball volume lower bound, it makes the covering numbers uniform on each fixed intrinsic radius.

Choose a fixed $\rho > 0$ small enough that the graph charts contain intrinsic balls of radius 4ρ. Their slope bounds give $\mathrm{Vol}(B_\rho^{\mathrm{int}}) \geq v_\rho > 0$ at every center. A maximal 2ρ-separated set in an intrinsic ball of radius R has disjoint radius-ρ balls inside the ball of radius $R + \rho$. Bishop comparison gives an upper bound V_R for the latter volume. The number of centers is therefore at most V_R / v_ρ. Maximality makes the radius-2ρ balls cover the original ball, so their larger graph charts give a finite cover with uniform size. We make the atlas and completeness step explicit. The same packing argument applies at every radius $0 < r \leq \rho$: the graph bounds give $\mathrm{Vol}(B_r^{\mathrm{int}}) \geq cr^6$, while Bishop comparison bounds each larger intrinsic ball. Thus every fixed intrinsic ball has a uniform finite covering number at each prescribed positive scale. Choose finite nets at radii $1, 2, \ldots$ and scales $\rho, \rho/2, \rho/4, \ldots$. Extract a diagonal subsequence on which all distances between the labelled net points converge. The resulting distance space, after identifying zero distances and taking its completion, is a pointed metric limit X. The finite net bounds make its closed bounded balls compact. One can see this directly: each such ball has a finite net at every scale, and a diagonal choice of successively closer net points gives a convergent subsequence of any sequence in the ball.

At each net point, use a tangent graph over a fixed disk, with a larger disk kept as a buffer. Translate the whole immersion so that its original base point is at the origin. The ambient positions of centers in a fixed intrinsic ball are then bounded. The tangent frames lie in a compact orthogonal group. After another diagonal extraction, these centers and frames converge, and the graph heights converge locally in $C^{1,\alpha}$ for every $\alpha < 1$, and weakly in $W^{2,2}$. Include successively finer net centers so that these graph limits cover X. Each graph parametrization is uniformly bi-Lipschitz for intrinsic distance on a smaller disk. The lower bound follows by projecting ambient displacement onto its tangent plane. The upper bound follows by lifting the straight segment in that disk and using the slope bound. The buffered graph also contains a fixed intrinsic ball about each point of the smaller disk. These facts pass to the distance limit. They give a one-to-one chart onto an open neighborhood in X. Every point of X, including a point introduced in the metric completion, lies in such a neighborhood by the net coverings and the uniform smaller-ball inclusion.

Overlaps are recorded by intrinsic distance, rather than by coincident ambient images. For a compact subset of a limiting overlap, retain a positive margin inside both buffered graph disks. Points identified in X have approximating intrinsic distance tending to zero. The smaller-ball inclusion therefore puts their approximating points in both original charts for all large j. On this common domain the transition map is exactly the tangent projection for the receiving chart, applied to the ambient graph of the first chart. Its centers, frames and graph functions converge, so the transition maps converge in $C^{1,\alpha}$ on each smaller compact subset. The inverse and cocycle identities pass to the limit on these domains, since they hold for every approximating atlas. The limit charts therefore form a compatible $C^{1,\alpha}$ atlas. Their ambient graph maps agree under these intrinsic transition maps and define an immersion $\iota : X \to \mathbb{R}^7$. Different intrinsic points are retained even if their ambient images coincide.

The metric from this immersion agrees locally with the distance used to construct X. Indeed, chart metric coefficients converge uniformly. Lengths of fixed chart curves therefore converge. Conversely, for two sufficiently close points in a smaller graph disk, the approximating minimizing geodesics remain in the buffered disk: leaving it would cost a fixed positive length. Their projected paths have a uniformly convergent subsequence, and lower semicontinuity of length gives the opposite distance inequality. Thus the two local distances agree. The pointed limit is a length space. To check this, take approximating minimizing geodesics between two labelled points, parametrized with constant speed. They stay in a fixed bounded intrinsic ball. The finite net compactness and their common Lipschitz bound give a limit curve of no greater length, first for net points and then by approximation for all points of X. Consequently the induced path metric is the distance on X. It is complete by construction. The same curve argument shows that X is connected. These arguments also show that no boundary appears, since each point has a full buffered graph chart. Finally, the oriented normals converge in the same charts and agree on overlaps. They give a global unit normal along ι.

Since $\widehat{H}_j \to 0$ uniformly, the graph equations converge weakly to the minimal graph equation. The limiting graphs have bounded slope and are $C^{1,\alpha}$. Interior elliptic regularity makes them smooth. For a compactly supported smooth test on the limit, use the finitely many chart identifications and a subordinate partition of unity to transfer it to the approximating sheets. The test functions and their first derivatives converge in these charts. One may choose chart cutoffs χ_a supported strictly inside a finite cover of the limit support, with $\sum_a \chi_a = 1$ there. If X_a and $X_{a,j}$ are the limit and approximating chart maps, transport each $\chi_a\phi$ by $X_a \circ X_{a,j}^{-1}$ and sum the transported terms. Convergence of the transition maps gives a globally defined test ϕ_j converging with its first derivatives. On compact chart supports, B_j is linear in D^2h_j with coefficients converging uniformly with Dh_j. Weak $W^{2,2}$ convergence therefore gives weak L^2 convergence of the second fundamental forms in these identifications.

Thus the metrics and Dirichlet integrals converge. The transported curvature fields multiplied by ϕ_j converge weakly in L^2 to the limiting field multiplied by ϕ. Lower semicontinuity bounds

its squared norm by the lower limit of the approximating curvature integrals. For fixed sufficiently small η, the coefficient $1-C\eta$ is positive, so this passes (3.34) to the limit in the needed direction. Its final zero-order term has already vanished. Then let $\eta \downarrow 0$. The limiting immersion has coefficient-one minimal stability. It is smooth, connected, complete, boundaryless, and two-sided. It therefore satisfies every hypothesis of Hong–Li–Wang's stable Bernstein theorem [27, Theorem 1.1]. Its image is a plane. After choosing the tangent chart at the base point, the height functions therefore tend to zero in C^1 on a fixed small disk.

We must still recover the base curvature. The compactness above does not assert C^2 convergence. Write h_j for the height on a fixed tangent chart. Its Hessian is bounded, $Dh_j(0) = 0$, and $h_j \to 0$ in C^1. The scalar graph mean curvature has the bounds

$$A_j = C\delta_j/K_j, \qquad B'_j = C\delta_j/K_j^2.$$

The slope bound converts its intrinsic gradient bound to the same Lipschitz bound in graph coordinates. For large j, $A_j \le 1$. Lemma 3.14 gives, for each fixed small r,

$$|D^2h_j(0)| \le Cr^{-2}\|h_j\|_\infty + Cr^2 + C\frac{\delta_j}{K_j}\log(C/K_j).$$

The contrary ratio gives $K_j \ge \delta_j$ for large j. Equation (3.23) gives $K_j \to 0$. Thus

$$\frac{\delta_j}{K_j}\log(C/K_j) \le \frac{\delta_j}{K_j}\log(C/\delta_j) \longrightarrow 0.$$

Choose r so that $Cr^2 < 1/4$, and then send $j \to \infty$. This contradicts $|D^2h_j(0)| = |\widehat{B}_j(0)| \ge 1/2$ and proves (3.33). □

Remark 3.16 (Full normal derivative on fixed cores). The same estimate controls the full ambient derivative of the phase normal on each fixed core, after choosing a threshold for that core. Fix $L \ge 0$. Choose $0 < \delta_L \le \delta_0$ so that

$$\delta_L \le d_{L+2}, \qquad C_{L+2,0}\delta_L \le \frac{1}{2},$$

where the constants are from Lemma 2.12. Assume $0 < \delta \le \delta_L$. This order is essential: the fixed window is chosen before its discrepancy threshold. Fix a base zero z, put $e = \nu(z)$, and write $\mathcal{H} = D^2U$. At the base, its tangential block is rB. The identity

$$\nabla D = -pr^{p-2}\mathcal{H}Z$$

controls its remaining entries, since $Z(z) = r(z)e$ and $|\nabla D(z)| \le CD(z)$. Thus $|\mathcal{H}(z)| \le C(K + D(z))$. The chosen threshold permits Lemma 2.12 on $B_{L+2}(z)$. It gives $\|1 - U_e\|_{C^2(B_{L+1}(z))} \le C_L D(z)$ and $D \le 1/2$ there. Commuting derivatives yields

$$\partial_e\mathcal{H} = D^2U_e = -D^2(1 - U_e) = O_L(D(z)).$$

Integration along $z + se$, $|s| \le L$, gives $|\mathcal{H}(z + se)| \le C_L(K + \delta)$. The same relative estimates keep $r \ge 1/2$ on this fixed core. For the phase normal $\nu_U = Z/r$, the exact formula

$$\nabla\nu_U = r^{-1}(I - \nu_U \otimes \nu_U)\mathcal{H}$$

therefore gives

$$|\nabla\nu_U(z + se)| \le C_L(K + \delta) \le C_L\delta\log(C/\delta), \qquad |s| \le L.$$

The constant may depend on L. This estimate does not extend across a whole logarithmic gap.

Lemma 3.17 (Ambient local logarithmic curvature estimate)**.** *There are $C < \infty$ and $\delta_0 > 0$, uniform on $\mathcal{P}$, with the following property. For a scalar-stable entire solution in $\mathbb{R}^7$ with $p \in \mathcal{P}$, suppose $L > 0$, $0 < \delta \le \delta_0$, and $D \le \delta$ at each zero in $B_L(x)$. Then*

$$\sup_{\Sigma\cap B_{L/2}(x)} b \le C(\delta\log(C/\delta) + L^{-1}). \tag{3.35}$$

Proof. Suppose the estimate fails along a sequence, and let $K_j = \sup_{\Sigma_j \cap B_{L_j/2}(x_j)} b_j$. The relative estimate gives $K_j \le C\sqrt{\delta_j}$. For δ_j bounded away from zero, this bounds the ratio of K_j to $\delta_j \log(C/\delta_j)$. A failing sequence must therefore have $\delta_j \to 0$. It also satisfies

$$K_j L_j \to \infty, \qquad K_j/[\delta_j \log(C/\delta_j)] \to \infty.$$

Over all zeros in the ambient ball, select a maximum or a fixed-factor near maximum of $b(z)\,\mathrm{dist}(z, \partial B_{L_j}(x_j))$. The curvature is bounded by $C\sqrt{\delta_j}$ inside the ball. The product therefore extends continuously by zero at its boundary, so a positive maximum is attained in the interior. Every zero involved in its positive part is regular. At the selected point define

$$k_j = b(z_j), \qquad d_j = \mathrm{dist}(z_j, \partial B_{L_j}(x_j)).$$

Then

$$k_j d_j \ge c K_j L_j \to \infty, \quad k_j \ge c K_j, \quad b \le C k_j \text{ at every zero in } B_{d_j/2}(z_j).$$

Indeed, a zero in $B_{L_j/2}(x_j)$ has boundary distance at least $L_j/2$, so the maximum product is at least $K_j L_j/2$, up to the fixed factor used to approximate the supremum defining K_j. Since $d_j \le L_j$, this also gives $k_j \ge cK_j$. For any zero $y \in B_{d_j/2}(z_j)$, its boundary distance is at least $d_j/2$. Comparing the two products gives $b_j(y) \le 2k_j$ for an exact maximum. A near maximum changes only this fixed factor. These statements concern every nearby ambient sheet, rather than one intrinsic component.

Dilate by k_j at z_j. The connected zero sheet has bounded rescaled curvature on exhausting intrinsic balls. Its intrinsic distance from the ambient boundary is at least its ambient distance. Only the connected regular piece inside $B_{d_j/2}(z_j)$ is used here. A finite-length curve in a smaller ambient ball is an ambient Cauchy curve. Its limiting zero is regular because $D \le \delta_j < 1$ there, so its graph chart continues it intrinsically. Thus the ambient boundary is the only possible obstruction to continuation. Every fixed rescaled intrinsic ball is consequently contained in a compact regular region for large j. For any fixed rescaled radius R, one has $3R < k_j d_j/2$ for large j. Every curve of length at most $3R$ from the base remains inside the controlled ambient ball. The continuation just proved supplies the minimizing geodesics and graph charts on these fixed intrinsic balls. Thus local Bishop comparison and the preceding chart-packing argument use only the Ricci bound in this buffered region. No control of the sheet outside it is needed.

On these balls its rescaled mean curvature and its intrinsic gradient are bounded by $C\delta_j/k_j$ and $C\delta_j/k_j^2$, respectively. The first bound tends to zero. A compact rescaled support stays within ambient distance C/k_j of z_j. For large j it therefore has a boundary buffer comparable to d_j. As $k_j \le C\sqrt{\delta_j}$,

$$\frac{d_j}{\log(1/\delta_j)} = \frac{k_j d_j}{k_j \log(1/\delta_j)} \longrightarrow \infty.$$

Every such support thus has the complete $10A_0 \log(1/\delta_j)$ buffer needed by the local constant-cap surface form. All zeros in that buffer satisfy $D \le \delta_j$.

After the curvature–mass comparison, this form reads

$$(1-\eta)\int b^2\psi^2 \le (1+\eta)\int |\nabla\psi|^2 + C_{\eta,Q}\delta_j^Q \int \psi^2, \qquad Q > 2.$$

Dilation by k_j changes the final coefficient to exactly

$$C_{\eta,Q}\frac{\delta_j^Q}{k_j^2} \to 0.$$

Here we used $k_j/[\delta_j \log(C/\delta_j)] \to \infty$. Keep η fixed while passing to the geometric limit, then let $\eta \downarrow 0$. The pointed compactness and stability arguments in the proof of Proposition 3.15 apply on these exhausting intrinsic balls. They give a complete, connected, two-sided minimal immersion with coefficient-one stability. It is planar by [27, Theorem 1.1]. On the base chart the heights tend

to zero in C^1, while the rescaled base curvature is one. Apply Lemma 3.14 with $A_j = C\delta_j/k_j$ and $B'_j = C\delta_j/k_j^2$, exactly as in that proof. The logarithmic error vanishes because

$$(\delta_j/k_j)\log(C/k_j) \le (\delta_j/k_j)\log(C/\delta_j) \to 0.$$

This contradiction proves the ambient bound. □

4. Layer interactions and quantitative preparation

Unless another range is stated, take $p \in \mathcal{P} \Subset (2,\infty)$, with constants uniform for $p \in \mathcal{P}$. Use the phase, surface, profile, and mass notation of Section 2. Write

$$g(t) = \tanh(at), \qquad f(t) = -2ca\tanh(at).$$

The solution and dimension assumptions are specified for each construction.

4.1. **Phase comparison and stress.** We construct finite-slab comparison layers to compare their stress flux with that of the solution. The comparison with artificial layers on the two sides of one graph follows [35, proof of Proposition 8.5]. Here the artificial phases solve a quasilinear equation on finite slabs, and we compute their physical stress in the matched normalization. Here the interface dimension is $m \ge 1$, and the wider range $\mathcal{P} \Subset (1,\infty)$ is allowed. All estimates in this construction are uniform for $p \in \mathcal{P}$. Fix $q_0 > m+1$. On the slab $\mathcal{S}_L = \mathbb{R}^m \times (0,L)$, with $L \ge 2$, use $L^{q_0}_{\mathrm{uloc}}$ and $W^{2,q_0}_{\mathrm{uloc}}$ for the largest local norms over unit balls. Intersect the balls with the slab. Impose zero Dirichlet trace on the horizontal faces.

More precisely, take the norm on $B_1(x)\cap\mathcal{S}_L$ and then the supremum over $x \in \overline{\mathcal{S}_L}$. Changing the fixed ball radius or using unit cubes gives equivalent norms with constants independent of $L \ge 2$. Since $q_0 > m+1$, we have the uniform embedding $W^{2,q_0}_{\mathrm{uloc}} \hookrightarrow C^{1,\alpha}_b$ for $\alpha = 1-(m+1)/q_0 > 0$. We use $f(U) = -2ca\tanh(aU)$ for the phase coefficient in $\Delta_p U = f(U)(1-|\nabla U|^p)$. It is distinct from the reaction $f_p(u)$ in the original equation.

Lemma 4.1 (A finite-slab phase inverse)**.** *On $\mathcal{S}_L$, the operator*

$$\mathcal{P}_p = -\Delta_y - c\partial_t^2 + 2cpa\tanh(at)\partial_t$$

has a unique inverse among bounded functions with zero trace. It satisfies

$$\|\mathcal{P}_p^{-1}F\|_{W^{2,q_0}_{\mathrm{uloc}}} \le CL\|F\|_{L^{q_0}_{\mathrm{uloc}}}. \tag{4.1}$$

Proof. Uniform ellipticity and all smooth coefficient bounds are independent of L. Consider the barrier

$$b(t) = t + 1 - e^{-t}.$$

It has $b(0) = 0$ and $0 \le b(t) \le 1+t$. Moreover,

$$\mathcal{P}_p b = ce^{-t} + 2cpa\tanh(at)(1+e^{-t}) \ge c_0 > 0 \qquad (t \ge 0),$$

The lower bound is uniform in the exponent interval. Exhaust bounded cylinders to construct the positive Green kernel $G_L(x,z)$. Compare bounded approximations of the constant source one with b/c_0. Monotone convergence gives

$$\int_{\mathcal{S}_L} G_L(x,z)\,dz \le C(1+t_x) \le CL. \tag{4.2}$$

For an explicit lower bound, put

$$c_* = \min_{p\in\mathcal{P}}(p-1), \qquad a_* = \min_{p\in\mathcal{P}} a_p, \qquad p_- = \min\mathcal{P}.$$

On the unit depth interval, the first term is at least $c_* e^{-1}$. Beyond it, the second is at least $2c_* p_- a_* \tanh(a_*)$. Their minimum is a permitted c_0.

To treat uniformly local L^{q_0} data, estimate the kernel as a function of its source variable z. Away from the pole it solves the homogeneous adjoint equation. All its coefficients, including the zeroth-order coefficient, are uniformly bounded. The local boundedness estimate requires no sign on that coefficient. It bounds the kernel on each unit block by its integral over a fixed enlargement. The same estimate holds at the flat Dirichlet faces.

Here is a uniform estimate for the blocks near the pole. Write $d = m + 1$ and put $D_x = B_1(x) \cap \mathcal{S}_L$. Let H_x be the positive Dirichlet Green kernel of $\mathcal{P}_p$ on D_x, constructed by smooth interior exhaustion if the spherical and flat boundaries meet. For smooth f with compact support in D_x, its zero-boundary solution is

$$v_f(\xi) = \int_{D_x} H_x(\xi, z) f(z)\, dz.$$

Apply the Aleksandrov maximum estimate to each sign of v_f, with operator $-\mathcal{P}_p$ [24, Theorem 9.1, p. 220]. Its principal matrix is $\operatorname{diag}(I, c)$ and its zeroth-order coefficient is zero. The domain diameter is at most two. The determinant is $c > 0$, and the drift is uniformly bounded. The theorem therefore gives, on every smooth exhaustion domain,

$$\|v_f\|_{L^\infty} \le C\|f\|_{L^d(D_x)} \le C\|f\|_{L^{q_0}(D_x)}.$$

The constants are independent of the exhaustion, L, x, and $p \in \mathcal{P}$. Monotone convergence for nonnegative f, followed by duality, gives

$$\|H_x(x, \cdot)\|_{L^{q_0'}(D_x)} \le C.$$

This uses $q_0 > d$, as assumed. It requires no lower bound on the distance from x to a horizontal face.

Domain monotonicity gives $0 \le H_x(x, z) \le G_L(x, z)$ on D_x. Their difference $R_x(z)$ satisfies the homogeneous adjoint equation there: the two point sources at $z = x$ cancel in distributions. It has zero trace on the portion of a horizontal face inside D_x. For each fixed x, pole cancellation and interior elliptic regularity make R_x smooth across x. Away from the pole, both kernels are smooth up to the flat face on the smaller balls used below. Thus R_x belongs to $W^{1,2}$ on those balls or half-balls and has the required zero trace. In divergence form this equation is

$$-\operatorname{div}_z(A\nabla R_x + \mathbf{b}R_x) = 0, \quad A = \operatorname{diag}(I, c), \quad \mathbf{b}(z) = 2cpa\tanh(at_z)e_t.$$

The local boundedness estimate [24, Theorem 8.17] gives

$$\|R_x\|_{L^\infty(B_{1/4}(x)\cap\mathcal{S}_L)} \le C \int_{B_{3/4}(x)\cap\mathcal{S}_L} R_x(z)\, dz.$$

For clarity, its L^1 version follows from the L^2 estimate by $\|R_x\|_2^2 \le \|R_x\|_\infty \|R_x\|_1$, Young's inequality, and iteration on nested radii. The Young coefficient is chosen small enough for the geometric-radius series to converge. The same argument holds at a horizontal face. Indeed, odd reflection of the zero-trace solution and of the normal component of $\mathbf{b}$ gives an interior divergence equation with the same constant principal matrix and a bounded reflected lower-order coefficient. Its local constants remain uniform. Since $L \ge 2$, a radius-one neighborhood meets at most one horizontal face. The integral in the last bound is at most $\int_{\mathcal{S}_L} G_L(x, z)\, dz$.

Choose fixed small blocks Q_j with enlargements Q_j^* of diameter at most $1/8$ and with bounded overlap. These blocks give the same uniformly local norms as the unit blocks, up to a fixed constant. If Q_j^* misses $B_{1/8}(x)$, the adjoint local estimate used above gives

$$\|G_L(x, \cdot)\|_{L^{q_0'}(Q_j)} \le C \int_{Q_j^*} G_L(x, z)\, dz.$$

Only uniformly finitely many remaining blocks occur, and each lies in $B_{1/4}(x)$. On these blocks the decomposition $G_L(x,\cdot) = H_x(x,\cdot) + R_x$ and the two preceding bounds give

$$\|G_L(x,\cdot)\|_{L^{q_0'}(Q_j)} \le C + C\int_{\mathcal{S}_L} G_L(x,z)\,dz.$$

Thus the pole contribution is uniformly controlled, including when the pole approaches a horizontal boundary. The bounded overlap and (4.2) now give

$$\sum_j \|G_L(x,\cdot)\|_{L^{q_0'}(Q_j)} \le C + C\int_{\mathcal{S}_L} G_L(x,z)\,dz \le CL.$$

Hölder's inequality on each block therefore yields

$$\sup_x \int G_L(x,z)|F(z)|\,dz \le CL\|F\|_{L^{q_0}_{\mathrm{uloc}}}.$$

No global integrability of F is needed.

The local constants are uniform: the principal matrix is $\mathrm{diag}(I,c)$, all drift derivatives are bounded, and the boundary charts are fixed flat half-balls. Interior and flat-boundary W^{2,q_0} estimates now give (4.1). Approximation of the source and exhaustion produce the inverse.

For general uniformly local data, truncate the source in value and in space, and then smooth it. The approximations have uniformly bounded $L^{q_0}_{\mathrm{uloc}}$ norms and converge to F in L^{q_0} on each compact set. The estimates give local weak W^{2,q_0} compactness and local C^1 compactness of the solutions. A diagonal limit solves the equation with the same bound. Only local approximation is needed; compactly supported functions need not be dense in the uniformly local norm.

It remains to prove uniqueness for bounded solutions. Use the barrier $v_L(t) = 1 + b(t)/(L+1)$. It is bounded above and away from zero, and it satisfies $\mathcal{P}_p v_L \ge c_0/(L+1)$. Choose $\rho_L > 0$ small enough that

$$v_L(t)\exp(\rho_L\sqrt{1+|y|^2})$$

is a positive supersolution. On a sufficiently large cylinder, any positive multiple of this supersolution dominates the bounded homogeneous solution on the lateral face. It is nonnegative on the horizontal faces.

Put $r(y) = \sqrt{1+|y|^2}$. Since $|\nabla r| \le 1$ and $0 \le \Delta r \le m$,

$$e^{-\rho_L r}\mathcal{P}_p(v_L e^{\rho_L r}) = \mathcal{P}_p v_L - v_L(\rho_L \Delta r + \rho_L^2|\nabla r|^2) \ge \frac{c_0}{L+1} - 2(m\rho_L + \rho_L^2).$$

Choose ρ_L so that the subtracted term is at most $c_0/(2(L+1))$. The supersolution is then positive on the whole slab.

Apply comparison to each sign. Exhaust the tangential variable y, then send the multiple to zero. □

Proposition 4.2 (Artificial two-side cancellation)**.** *There are constants $\eta_0, C > 0$ with the following property. Let $R \ge L \ge 2$, $0 < \kappa \le 1$, and $h \in C_c^\infty(\mathbb{R}^m)$ satisfy*

$$h(0) = 0, \qquad Dh(0) = 0, \qquad \|Dh\|_\infty \le \kappa R, \qquad \|D^2 h\|_\infty \le \kappa, \qquad \kappa R^2 \le \eta_0.$$

For each $\sigma \in \{+1,-1\}$, there is a smooth function $U_\sigma = t + \psi_\sigma$ on $\mathcal{S}_L$, with $\psi_\sigma = 0$ on $t = 0, L$, such that

$$v_\sigma^*(y, h(y) + \sigma t) = g(U_\sigma(y,t))$$

solves the matched p-Laplace equation in its artificial slab. It is positive inside, equals zero on the graph, and has nonvanishing gradient. Moreover

$$\|\psi_\sigma\|_{W^{2,q_0}_{\mathrm{uloc}}} \le C\kappa L, \tag{4.3}$$

$$\|\psi_+ + \psi_-\|_{W^{2,q_0}_{\mathrm{uloc}}} \le C\kappa^2 L R^2. \tag{4.4}$$

In particular, denoting the inward normal derivatives at the origin on the upper and lower graph sides by ∂_{ν_+} and ∂_{ν_-}, respectively,

$$\left|\partial_{\nu_+} v_+^*(0) + \partial_{\nu_-} v_-^*(0) - 2a\right| \le C\kappa^2 L R^2. \tag{4.5}$$

Proof. Use the graph coordinates $x = (y, h(y) + \sigma t)$ and define

$$B_\sigma = \begin{pmatrix} I & -\sigma Dh \\ -\sigma Dh^{\mathsf{T}} & 1 + |Dh|^2 \end{pmatrix}, \qquad R_\sigma(\psi)^2 = (e_t + \nabla\psi) \cdot B_\sigma (e_t + \nabla\psi).$$

The Jacobian has absolute value one. In these coordinates the exact equation reads $\mathcal{F}_\sigma(\psi) = 0$, with

$$\mathcal{F}_\sigma(\psi) = \operatorname{div}\Big(R_\sigma(\psi)^{p-2} B_\sigma (e_t + \nabla\psi)\Big) - f(t+\psi)\left(1 - R_\sigma(\psi)^p\right).$$

At $h = 0, \psi = 0$, differentiation in ψ gives $-\mathcal{P}_p$. Differentiation in h gives $-\sigma\Delta h$. Thus the equation has the form

$$\mathcal{P}_p \psi = -\sigma\Delta h + N_\sigma(\psi). \tag{4.6}$$

Work in the zero-trace space $X = W^{2,q_0}_{\mathrm{uloc}}(\mathcal{S}_L)$. Uniform local Sobolev embedding gives $\|\psi\|_{C^1_b} \le C\|\psi\|_X$. If $\|Dh\|_\infty + \|\psi\|_X \ll 1$, then $1/2 < R_\sigma(\psi) < 2$. Hence the flux derivatives used here are bounded uniformly for $p \in \mathcal{P}$. The expansion stays away from critical gradients. Write the divergence in nondivergence form and apply the product rule. The result is

$$\|N_\sigma(0)\|_{L^{q_0}_{\mathrm{uloc}}} \le C\kappa^2 R^2, \tag{4.7}$$

$$\|N_\sigma(\psi) - N_\sigma(\varphi)\|_{L^{q_0}_{\mathrm{uloc}}} \le C(\kappa R + \|\psi\|_X + \|\varphi\|_X)\|\psi - \varphi\|_X. \tag{4.8}$$

For the first estimate, the exact residual is

$$\mathcal{F}_\sigma(0) = -\sigma \operatorname{div}_y\big((1 + |Dh|^2)^{(p-2)/2} Dh\big) - f(t)\big(1 - (1 + |Dh|^2)^{p/2}\big).$$

Remove the linear term $-\sigma\Delta h$. What remains is bounded by $C|Dh|^2|D^2 h| + C|Dh|^2$. For the Lipschitz estimate, the highest derivatives of ψ occur linearly. Their coefficient differs from its flat value by $O(|Dh| + |\nabla\psi|)$. The derivatives of B_σ have size at most $C\kappa$. The remaining factors involve bounded derivatives of f, ψ, and $\nabla\psi$. The same product estimate proves (4.8). Derivatives of h of order three are absent.

With $w = e_t + \nabla\psi$ and $B = B_\sigma$, the coefficient of $D^2\psi$ is exactly

$$C(B, w) = \mathcal{R}^{p-2} B + (p-2)\mathcal{R}^{p-4}(Bw) \otimes (Bw), \qquad \mathcal{R}^2 = w \cdot Bw.$$

All other derivatives of the flux fall on $B(Dh)$ and contain only $D^2 h$. At $Dh = \nabla\psi = 0$, this matrix is $\operatorname{diag}(I, c)$. To compare two highest-order terms, write

$$\begin{aligned} &[C(\psi) - C(0)]D^2\psi - [C(\varphi) - C(0)]D^2\varphi \\ &\quad = [C(\psi) - C(0)]D^2(\psi - \varphi) + [C(\psi) - C(\varphi)]D^2\varphi. \end{aligned}$$

The first coefficient is $O(\kappa R + \|\psi\|_X)$ in L^∞, and the second is bounded by $C\|\psi - \varphi\|_X$ there. The second-derivative factors are in uniformly local L^{q_0}. This proves the product estimate; the lower-order terms obey the same bound.

Apply Lemma 4.1 to (4.6). Define

$$\mathcal{T}_\sigma(\psi) = \mathcal{P}_p^{-1}(-\sigma\Delta h + N_\sigma(\psi)).$$

On the ball $\|\psi\|_X \le C_0\kappa L$, its Lipschitz constant is at most $CL(\kappa R + 2C_0\kappa L) \le C_{C_0}\kappa L R$. The linear forcing has image of norm at most $C\kappa L$. The image of the nonlinear terms has norm at most $C_{C_0}\kappa^2 L R^2$, using $L \le R$ in (4.7)–(4.8). Choose C_0 large enough for the linear term. Then choose η_0 small enough that $\kappa L R \le \kappa R^2 \le \eta_0$ makes the Lipschitz constant less than $1/2$ and the nonlinear image fit in the remaining half of the ball. The contraction theorem gives existence and (4.3). Local elliptic regularity away from critical gradients gives smoothness. Smallness in C^1

implies $U_{\sigma,t} > 1/2$. The zero traces then give $0 < U_\sigma < L$. Positivity and the nonvanishing physical gradient follow.

Sum (4.6) over the two signs. The linear contributions of h disappear, leaving

$$\mathcal{P}_p(\psi_+ + \psi_-) = N_+(\psi_+) + N_-(\psi_-).$$

Bounds (4.7)–(4.8) and (4.3) control the right side by $C(\kappa^2R^2+\kappa^2LR+\kappa^2L^2)$ in $L^{q_0}_{\mathrm{uloc}}$. Since $L \le R$, this is at most $C\kappa^2R^2$. Apply the inverse once more to obtain (4.4). At the origin, $Dh(0) = 0$, so the inward derivatives equal $a(1 + \partial_t\psi_\sigma(0, 0))$. Boundary Sobolev embedding gives (4.5). □

Remark 4.3 (Cutting off a local graph). Start with a small-slope graph over a disk of radius $4R$. Assume curvature at most κ and normalization $h(0) = Dh(0) = 0$. A cutoff on a comparable disk preserves the gradient and Hessian bounds up to a fixed factor. In the regime $L = A\log(1/\delta)$ and $R = L^k$, with $k > 1$ fixed, assume $\kappa^2 \log^{2k+1}(1/\delta) = o(\delta)$. Then the artificial slope error is $o(\delta)$. The half-strip argument below compares this layer with the actual solution.

The interaction limit requires the nonlinear mean-curvature flux. Linearizing that flux creates a cubic error in the slope. Curvature alone does not control that error. We therefore build the reference profile around the tilted plane.

Choose $b \in \mathbb{R}^m$ with $|b| \le 1/4$. Decompose the zero graph as

$$f(y) = b \cdot y + h(y), \qquad h(0) = Dh(0) = 0, \qquad \|Dh\|_\infty \le KR, \quad \|D^2h\|_\infty \le K.$$

Cut off h on a disk comparable to R. This changes the bounds only by a fixed factor. With $\lambda = (1 + |b|^2)^{1/2}$, take coordinates

$$x = (y, f(y) + \lambda t), \qquad 0 < t < L, \qquad R \ge L \ge 2.$$

Require KLR^2 to be small.

For $h = 0$, the affine phase $U = t$ has unit physical gradient $n = (-b, 1)/\lambda$. Its linearized operator is

$$\mathcal{P}_{p,b} = -\operatorname{div}(C_{p,b}\nabla) + 2(p-1)pa\tanh(at)\left(\partial_t - \frac{b}{\lambda}\cdot\nabla_y\right),$$

The constant symmetric principal matrix is

$$C_{p,b} = \begin{pmatrix} I + (p-2)b\otimes b/\lambda^2 & -(p-1)b/\lambda \\ -(p-1)b^{\mathsf{T}}/\lambda & p-1 \end{pmatrix}.$$

The matrix is uniformly elliptic. Its coefficients and the drift have uniform smooth bounds. The barrier $B(t) = t + 1 - e^{-t}$ obeys the identity

$$\mathcal{P}_{p,b}B = (p-1)e^{-t} + 2(p-1)pa\tanh(at)(1 + e^{-t}) \ge c_0 > 0.$$

The barrier is independent of y, so the mixed principal coefficients and tangential drift add no term to this calculation. Uniform ellipticity and the bounded coefficient derivatives give the same local adjoint estimates as before. The Green-mass construction therefore applies in these tilted coordinates as well. Together with local estimates, it gives

$$\|\mathcal{P}^{-1}_{p,b}F\|_{W^{2,q}_{\mathrm{uloc}}} \le CL\|F\|_{L^q_{\mathrm{uloc}}}, \qquad q > m + 1,$$

The inverse has zero trace on the horizontal faces.

For the curved graph, set $U = t + \psi$. Write $R_f(\psi)$ for the physical gradient length. At $\psi = 0$, it satisfies

$$R_f(0)^2 = \frac{1 + |b + Dh|^2}{\lambda^2}.$$

The phase equation becomes $\mathcal{P}_{p,b}\psi = F_h + N_h(\psi)$. Its source obeys

$$\|F_h\|_{L^q_{\mathrm{uloc}}} \le C\{K + |b|KR + K^2R^2\} \le CKR,$$

In a small ball of $X = W^{2,q}_{\mathrm{uloc}}$, the nonlinear part satisfies

$$\|N_h(\psi) - N_h(\varphi)\|_{L^q_{\mathrm{uloc}}} \le C(KR + \|\psi\|_X + \|\varphi\|_X)\|\psi - \varphi\|_X.$$

Only one flux derivative is taken. Thus only Dh, D^2h appear. To see the source estimate, evaluate the residual at $\psi = 0$:

$$-\lambda^{-1}\,\mathrm{div}_y(R_f(0)^{p-2}(b + Dh)) + 2(p-1)a\tanh(at)(1 - R_f(0)^p).$$

The two contributions are bounded by CK and $C(|b|KR + K^2R^2)$, respectively.

The inverse defines a contraction on $\|\psi\|_X \le C_0KLR$. Its contraction constant is at most $CKL^2R \le CKLR^2$. It produces an artificial layer $v^* = g_p(t + \psi)$ with

$$\|\psi\|_{W^{2,q}_{\mathrm{uloc}}} \le CKLR, \qquad \psi = 0 \quad (t = 0, L). \tag{4.9}$$

The phase gradient stays uniformly away from zero. Reverse the phase sign for the opposite side. This first-order flux estimate does not need cancellation between the two sides.

Fix the ambient Cartesian coordinates $x = (y, s)$ throughout this flux calculation. For a physical function v, write

$$\mathscr{F}_p[v](y, s) = |\nabla_x v(y, s)|^{p-2}\partial_s v(y, s)\nabla_y v(y, s) \in \mathbb{R}^m.$$

Here ∇_y means differentiation with the ambient coordinate s fixed. It is the mixed physical stress component, whose stress tensor is defined below. It is not the derivative at fixed pulled-back depth. For $\sigma \in \{+1, -1\}$ put

$$\Phi_\sigma(y, t) = (y, f(y) + \sigma\lambda t), \qquad v^*_\sigma(\Phi_\sigma(y, t)) = \sigma g_p(t + \psi_\sigma(y, t)), \qquad 0 < t < L_\sigma.$$

The two corrections are the one-sided tilted constructions, with $L_+ = L_{+1}$ and $L_- = L_{-1}$. On the lower side, apply that construction after reflecting s to $-s$, with graph $-f$ and slope $-b$. The signs give the same increasing physical planar transition on both sides of the zero graph. Changing the overall orientation multiplies both fields by -1; $\mathscr{F}_p[-v] = \mathscr{F}_p[v]$, so the current is unchanged. Define the artificial current per unit horizontal base measure by

$$\begin{aligned} J^*(y) &= \int_{f(y)-\lambda L_-}^{f(y)} \mathscr{F}_p[v^*_-](y, s)\,ds + \int_{f(y)}^{f(y)+\lambda L_+} \mathscr{F}_p[v^*_+](y, s)\,ds \\ &= \lambda \sum_{\sigma=\pm1} \int_0^{L_\sigma} \mathscr{F}_p[v^*_\sigma](\Phi_\sigma(y, t))\,dt. \end{aligned}$$

Both summands have the positive measure $\lambda\,dt$. On the lower half, the negative Jacobian $ds = -\lambda\,dt$ is canceled by reversing the integration limits. These are vertical fiber integrals; there is no zero-sheet surface-area factor. The two artificial halves are integrated separately, and their normal derivatives at the zero graph may differ. In the next central-fiber estimate, J^* means $J^*(0)$. Along that fiber $h = Dh = 0$. Compare with the planar layer

$$v^0(y, s) = g_p((s - b \cdot y)/\lambda).$$

Use (4.9) and the bounded logarithmic derivative of g'_p. The chain rule gives

$$|v^* - v^0| \le CKLR\,g'_p(t), \qquad |\nabla v^* - \nabla v^0| \le CKLR\,g'_p(t).$$

The mixed gradient flux $\xi \mapsto |\xi|^{p-2}\xi_s\xi_y$ is homogeneous of degree p. Its derivative has size at most $C|\xi|^{p-1}$, also through $\xi = 0$ because $p > 1$. The mean-value inequality and the preceding gradient comparison bound its pointwise error by $CKLR|g'_p(t)|^p$. Integration along either half-fiber gives an error at most $CKLR\int_0^L |g'_p|^p \le CKLR$.

Set $\sigma_p = \int_{\mathbb{R}} |g'_p|^p$. Since $g'_p = a_pG_p$, the normalizations from Section 2 give $\sigma_p = a_p^pJ_p = \sigma_E = a_p^2\sigma_q$. This is the tension in the stress flux and the projected Toda form. The reference flux is exact:

$$(v^0)_s = g'_p(t)/\lambda, \quad \nabla_y v^0 = -b\,g'_p(t)/\lambda, \quad ds = \lambda\,dt,$$

Adding the two complete half-fibers yields

$$\int_{\mathbb{R}} |\nabla v^0|^{p-2}(v^0)_s \nabla_y v^0 \, ds = -\sigma_p \frac{b}{\sqrt{1+|b|^2}}. \tag{4.10}$$

A truncation at depths L_+, L_- costs at most $C(e^{-2paL_+} + e^{-2paL_-})$. The finite-scale bound is therefore

$$\left| J^* + \sigma_p \frac{b}{\sqrt{1+|b|^2}} \right| \le CKR(L_+ + L_-) + C(e^{-2paL_+} + e^{-2paL_-}).$$

Here $L_+, L_- \ge 2$ and $R \ge \max\{L_+, L_-\}$, and the two artificial constructions require $K \max\{L_+, L_-\}R^2$ to be sufficiently small. The constants are uniform on the fixed compact exponent interval. This estimate retains the actual comparison radii and depths.

For an error $o(\sqrt{\rho})$, require in addition

$$KR(L_+ + L_-) = o(\sqrt{\rho}).$$

A sufficient regime is $L_+ + L_- + R \le C \log^q(1/\rho)$ for one fixed q, together with $K \log^M(1/\rho) = o(\sqrt{\rho})$ for every fixed M. The finite-scale construction alone does not require this extra relation. Actual half-gaps with pressure $O(\rho)$ have truncation error $O(\rho) = o(\sqrt{\rho})$, and the inactive caps have smaller tails.

The same notation permits different depths and moving terminal faces. Let $c_-(y) < f(y) < c_+(y)$ be those faces and put

$$H_+(y) = \frac{c_+(y) - f(y)}{\lambda}, \qquad H_-(y) = \frac{f(y) - c_-(y)}{\lambda}.$$

On a packet centered at y_0, use $L_\pm = H_\pm(y_0) \ge 2$. The variable-cap hypotheses mean $H_\pm(y) = L_\pm + o(1)$ on its buffered comparison disk, with small slopes and common local C^α bounds for the second derivatives. Use the same globally cut-off graph with a construction radius R^* satisfying $\max\{R, L_+ + 3, L_- + 3\} \le R^* \le 3R$. This is a fixed enlargement since $R \ge L_\pm \ge 2$. The global derivative bounds on the cut-off graph still apply, and $K \max\{L_+ + 3, L_- + 3\}(R^*)^2 \le CK \max\{L_+, L_-\}R^2$. Reduce the fixed smallness threshold by this constant. Construct each artificial solution in the deeper constant slab $0 < t < L_\pm + 3$ and restrict it to $0 < t < H_\pm(y)$. Since $H_\pm = L_\pm + o(1)$, each moving terminal face and a fixed neighborhood of width one stay strictly below the artificial top face $t = L_\pm + 3$ and above $t = 0$. The common bottom $t = 0$ remains a Dirichlet face, where we use the boundary estimates. Define the restricted J^* by replacing $L_\pm$ by $H_\pm(y)$ in the two physical integrals just displayed. The actual current uses exactly the same faces and physical measure:

$$J(y) = \int_{c_-(y)}^{f(y)} \mathscr{F}_p[u](y,s)\, ds + \int_{f(y)}^{c_+(y)} \mathscr{F}_p[u](y,s)\, ds.$$

Thus J and J^* are compared on identical truncated physical fibers. At y_0 their tilted half-depths are exactly L_+, L_-. The fixed extension by three changes the artificial error only by a uniform factor, since $L_\pm \ge 2$; the displayed central-fiber bound and its truncation terms keep their stated orders. The terminal slopes enter the tractions when the cell current is differentiated, rather than the definition of the fiber measure.

For this actual current, impose the buffered relative-profile hypotheses (4.32) and the scale hypotheses of (4.35) on both one-sided truncated domains. Use the exact variable-cap pullback for their moving faces. If a lower chamber is reflected to apply a one-sided estimate, transform the full tensor back to the fixed ambient coordinates before taking its mixed component. The absolute transmission bound is unchanged. Apply that finite bound on both sides. If these two transmission bounds are $o(\sqrt{\rho})$, the triangle inequality and the preceding artificial-flux estimate give, uniformly

on the smaller prepared disk,

$$J = -\sigma_p \frac{\nabla f}{\sqrt{1+|\nabla f|^2}} + o(\sqrt{\rho}).$$

In the Toda applications, each retained half-depth is comparable to $\ell \asymp \log(1/\rho)$ and the comparison radius is a fixed power $R \asymp \ell^k$, $k > 1$. The bounds in (4.35) then are $o(\sqrt{\rho})$: their first term has an exponential in ℓ beyond $\sqrt{\rho}$, their lateral term decays faster than every power of ρ, and their last term is $O(\rho)$. These are the local comparison radii; a larger disk used to index sheets is only a buffer. The nonlinear slope flux remains exact.

At a general base point y_0, repeat the construction after translation with the frozen slope $b = \nabla f(y_0)$. The constants are independent of y_0, so the resulting estimate is uniform on the smaller prepared disk. There is no need to differentiate the family of artificial profiles with respect to its center. The force equation uses the divergence of the physical flux in distributions: after multiplication by a test function, the derivative falls on that test, and the uniform flux error suffices.

To compare these layers with the actual solution, we estimate the pressure in a prepared gap. For the matched-gap estimates, return to $\mathcal{P} \Subset (2,\infty)$. In a positive chamber, define $w = 1-u$ and

$$\mathcal{F}_p(w) = (1-w)(2-w)^{p-1}w^{p-1}.$$

The deficit equation is $\Delta_p w = \mathcal{F}_p(w)$. Fix $w_* > 0$ sufficiently small. For $p \in \mathcal{P}$ and $0 < w < 2w_*$, the reaction satisfies

$$C^{-1}w^{p-2} \le \mathcal{F}_p'(w) \le Cw^{p-2}. \tag{4.11}$$

The derivative of the deficit reaction can be written without an unspecified expansion. Since $q(w) = w(2-w)$ and $q'(w) = 2(1-w)$,

$$\mathcal{F}_p'(w) = q(w)^{p-2}\big[2(p-1)(1-w)^2 - q(w)\big] = q(w)^{p-2}\big[(2p-1)(1-w)^2 - 1\big].$$

Consequently, uniformly for $p \in \mathcal{P}$,

$$\mathcal{F}_p'(w) = (p-1)2^{p-1}w^{p-2}(1+O_{\mathcal{P}}(w)) \qquad (w \downarrow 0).$$

Choose w_* so that the error has absolute value at most $1/2$ on $0 < w < 2w_*$. This proves the two-sided bound in (4.11). The same formula also shows $|\mathcal{F}_p'(w)| \le Cw^{p-2}$ for $0 < w \le 1$. This latter bound is useful on the bottom core, where the secant potential need not be positive.

The Modica bound gives

$$|\nabla w| \le a(2w - w^2) \le Cw. \tag{4.12}$$

Reflection gives the same identities in a negative chamber.

Let $0 < w, h < 2w_*$ be deficits in this small well interval, with the same equation and relative gradient bounds. Define $z = w - h$, $\sigma = w + h$, $\mu = \sigma^{p-2}$, and

$$\mathbb{M} = \int_0^1 A_p((1-t)\nabla h + t\nabla w)\,dt, \qquad b = \int_0^1 \mathcal{F}_p'((1-t)h + tw)\,dt.$$

Subtracting the equations yields

$$-\operatorname{div}(\mathbb{M}\nabla z) + bz = 0. \tag{4.13}$$

The secant formula and the relative bounds imply

$$0 \le \mathbb{M} \le C\mu I, \qquad C^{-1}\mu \le b \le C\mu, \qquad |\nabla \log \mu| \le C. \tag{4.14}$$

Thus every compactly supported test f satisfies

$$\int |\mathbb{M}| f^2 \le C \int \mu f^2. \tag{4.15}$$

At a degenerate turn, no lower ellipticity bound for $\mathbb{M}/\mu$ is claimed.

The positive lower bound for b remains available. In fact, for $r_1, r_2 \geq 0$ and $\theta = p - 2 > 0$,

$$c_{\mathcal{P}}(r_1 + r_2)^\theta \leq \int_0^1 ((1-t)r_1 + tr_2)^\theta \, dt \leq (r_1 + r_2)^\theta.$$

For the lower bound, suppose $r_2 \geq r_1$ and integrate on $1/2 \leq t \leq 1$, where the affine combination is at least $(r_1 + r_2)/4$. The logarithmic derivative estimate follows from $|\nabla(w + h)| \leq C(w + h)$.

Work on $\mathcal{C}_R = B_R^{n-1} \times (-L, L)$ with $R, L \geq 2$. Assume the well hypotheses in a fixed neighborhood of the closed cylinder. Also assume $w \leq h$ on $s = \pm L$. Set $z_+ = (w - h)_+$ and test the difference equation with $\eta^2 z_+$, where η is a compact tangential cutoff. The only cross term is $-2 \int \eta z_+ \mathbb{M} \nabla z_+ \cdot \nabla \eta$. Cauchy–Schwarz for the nonnegative matrix $\mathbb{M}$ bounds its absolute value by

$$\frac{1}{2} \int \eta^2 \mathbb{M}[\nabla z_+, \nabla z_+] + 2 \int z_+^2 \mathbb{M}[\nabla \eta, \nabla \eta].$$

Absorb the first term and use $b \geq 0$ to obtain

$$\int \eta^2 (\mathbb{M}[\nabla z_+, \nabla z_+] + b z_+^2) \leq 4 \int z_+^2 \mathbb{M}[\nabla \eta, \nabla \eta]. \tag{4.16}$$

Its trace vanishes on the horizontal faces. Monotonicity and the relative gradient bound give

$$\mathbb{M}[\nabla(w - h), \nabla(w - h)] \leq C\sigma^p.$$

Thus its natural energy is finite. Define

$$\mathcal{A}(r) = \int_{\mathcal{C}_r} (\mathbb{M}[\nabla z_+, \nabla z_+] + \mu z_+^2).$$

A tangential cutoff of width Λ gives

$$\mathcal{A}(r) \leq C\Lambda^{-2} \mathcal{A}(r + \Lambda). \tag{4.17}$$

Fix Λ so that $C\Lambda^{-2} < 1/2$, then iterate. The bounds $|z_+| + |\nabla z_+| \leq C\sigma$ and $\sigma \leq C$ imply

$$\mathcal{A}(R/2) \leq C R^{n-1} L e^{-cR}. \tag{4.18}$$

Consequently

$$\sup_{B_{R/3}^{n-1} \times [-L+1, L-1]} (w - h)_+ \leq C(R^{n-1} L)^{1/(n+p)} e^{-cR/(n+p)}. \tag{4.19}$$

Reverse the two solutions for the other comparison. The constants are uniform on $\mathcal{P}$.

To see the exponent in (4.19), let $d = z_+(x_0)$. The uniform Lipschitz bound gives $z_+ \geq d/2$ on a ball of radius d/C. Since $\mu z_+^2 \geq z_+^p$, this ball contributes at least $C^{-1} d^{n+p}$ to $\mathcal{A}(R/2)$. Combining this lower bound with (4.18) gives the claim. Here $n = m + 1$ is the ambient dimension.

Let $p_j \in \mathcal{P}$, and let u_j be bounded weak solutions with the Modica bound. Assume two adjacent zero graphs

$$s = h_{j,-}(y), \qquad s = h_{j,+}(y) \qquad (y \in B_{2R_j}^{n-1}),$$

enclose a positive chamber $u_j > 0$. Require

$$\ell_j \longrightarrow \infty, \qquad R_j/\ell_j \longrightarrow \infty, \qquad \sup_{B_{2R_j}} (|h_{j,-} + \ell_j/2| + |h_{j,+} - \ell_j/2|) \longrightarrow 0. \tag{4.20}$$

Also impose the following uniform properties.

(1) On every fixed normal neighborhood, the solutions and normals approach the correctly oriented layer $\tanh(a_{p_j} s)$ in C^1 and the vertical normal, respectively.

(2) Given $b < 1$, there is a fixed L_b such that all chamber points farther than this distance from the zero boundary satisfy $u_j > b$.

Uniform preparation of all fixed cores supplies the second property whenever it is already available.

Introduce the vertical stress and scalar discrepancy:

$$P_e = W_{p_j}(u_j) + \frac{|\nabla u_j|^{p_j}}{p_j} - |\nabla u_j|^{p_j-2}(u_j)_s^2, \qquad P = W_{p_j}(u_j) - \frac{p_j-1}{p_j}|\nabla u_j|^{p_j}.$$

At the gap midpoint they satisfy

$$P_e(0,0) = P(0,0) + o(e^{-p_j a_{p_j} \ell_j}) = (\Pi_{p_j} + o(1))e^{-p_j a_{p_j} \ell_j}, \tag{4.21}$$

The coefficient is

$$\Pi_p = \frac{4^p}{2p} \exp(-\gamma_E - \psi(1-1/p)). \tag{4.22}$$

Here ψ denotes the digamma function. The error $o(1)$ is uniform on the fixed compact exponent interval. The same conclusion holds uniformly at other central horizontal points whenever the hypotheses are uniform there.

Proof. Fix $S > L_b + 3$ large enough that the two distance-S faces lie in $1 - w_* < u < 1$. On $s = \pm(\ell_j/2 - S)$, preparation and (4.20) imply

$$(1-\rho_j)b_{p_j,S} \le 1 - u_j \le (1+\rho_j)b_{p_j,S}, \qquad b_{p,S} = 1 - \tanh(a_p S), \quad \rho_j \longrightarrow 0.$$

Clearing places the entire cylinder between these faces and a fixed small neighborhood in the same well range. The disk B_{2R_j} supplies the lateral padding.

Consider the positive symmetric one-dimensional deficits $h_j^{\pm}$ solving

$$(|h'|^{p_j-2}h')' = \mathcal{F}_{p_j}(h), \qquad h(\pm(\ell_j/2 - S)) = (1 \pm \rho_j)b_{p_j,S}.$$

The primitive of $\mathcal{F}_{p_j}$ is strictly convex in the small well range. Minimizing the Dirichlet functional there gives a solution between zero and the larger boundary value. Positive boundary values and the strong maximum principle make it positive. Here $\mathcal{F}_p(h)/h^{p-1}$ is bounded on the small well interval, which is the reaction condition needed for that principle. Testing the difference of two solutions with its positive part gives uniqueness. Reflection then makes the solution even. Equivalently, one may use the strictly monotone shooting integral. These solutions are portions of periodic layers. Their pressure is positive, and they satisfy (4.12). Since S is large, both extend a fixed distance beyond the faces inside a slightly larger well range.

The one-dimensional comparison functions satisfy the same relative gradient bound as the actual deficit. Indeed, their first integral has positive pressure, and therefore their slope is no larger than the zero-pressure slope $a(2h - h^2)$. This verifies the remaining hypothesis of the well comparison.

Use (4.19) for the positive parts $(1 - u_j - h_j^+)_+$ and $(h_j^- - (1-u_j))_+$. On every fixed midpoint ball,

$$h_j^- - E_j \le 1 - u_j \le h_j^+ + E_j, \qquad E_j \le C(R_j^{n-1}\ell_j)^C e^{-cR_j}. \tag{4.23}$$

Since $R_j/\ell_j \to \infty$, we obtain $E_j = o(e^{-a_{p_j}\ell_j})$.

The one-dimensional coefficient is determined exactly. Let $P > 0$ be the gap pressure.

Along a positive one-dimensional layer, multiplication of the original equation by u' gives the conserved quantity $W_p(u) - c|u'|^p/p = P$. Hence, on its increasing half,

$$u' = a(q^p - q_m^p)^{1/p}, \qquad q = 1 - u^2, \qquad q_m^p = 2pP.$$

Since $du = -dq/(2\sqrt{1-q})$, integration from the zero to the turn and then doubling gives the following width formula.

At its turning point let $q_m = (2pP)^{1/p}$ be the value of $1 - u^2$. The full width between zeros obeys

$$a_p \ell(P) = \int_{q_m}^1 \frac{dq}{\sqrt{1-q}(q^p - q_m^p)^{1/p}}.$$

The reference integral is elementary. Setting $v = \sqrt{1-q}$ gives

$$\int_{q_m}^{1} \frac{dq}{q\sqrt{1-q}} = 2\,\operatorname{arctanh}\sqrt{1-q_m} = \log\frac{1+\sqrt{1-q_m}}{1-\sqrt{1-q_m}} = \log(4/q_m) + O(q_m).$$

Subtract $\int_{q_m}^{1} dq/[q\sqrt{1-q}]$ and put $q = q_m t$. Uniformly on compact subsets of $(1,\infty)$, this gives

$$a_p \ell(P) = \log(4/q_m) + C_p + o(1), \qquad C_p = \int_1^{\infty} \left[(t^p-1)^{-1/p} - t^{-1}\right] dt = -\frac{\gamma_E + \psi(1-1/p)}{p}.$$

The integral is the negative logarithm of C_p^{∞} from Lemma 3.11. The uniform endpoint bounds used there apply here: near $t = 1$ the integrand is bounded by $C(t-1)^{-1/p_{\min}}$, and at infinity by $Ct^{-p_{\min}-1}$, where $p_{\min} = \min \mathcal{P} > 2$. In the difference integral the extra factor $(1-q_m t)^{-1/2}$ does not spoil this bound. On $t \le (2q_m)^{-1}$ it is bounded; the remaining endpoint portion is $O(q_m^{p_{\min}})$ after setting $v = q_m t$. Finally, setting $z = t^{-p}$ in C_p gives

$$C_p = \frac{1}{p}\int_0^1 \frac{(1-z)^{-1/p} - 1}{z}\, dz,$$

whose value follows from the digamma integral [34, Eq. (5.9.16)] at the positive argument $1 - 1/p$. Exponentiation gives $q_m = (4e^{C_p} + o(1))e^{-a_p \ell(P)}$. Since $P = q_m^p/(2p)$, its coefficient is $4^p e^{pC_p}/(2p) = \Pi_p$. Hence $P = (\Pi_p + o(1))e^{-p a_p \ell(P)}$. We justify the omitted zero-to-face lengths by their integrals. Write $d_j^{\pm} = (1 \pm \rho_j) b_{p_j,S}$ for the face deficits, $m_j^{\pm} = h_j^{\pm}(0)$, and $Q_j^{\pm} = m_j^{\pm}(2 - m_j^{\pm})$ for their turning values of q. Since the face-to-turn lengths are $\ell_j/2 - S \to \infty$, $Q_j^{\pm} \to 0$. Indeed, if $Q_j^{\pm}$ stayed above a positive constant, their quadratures would be uniformly bounded. The only turn singularity is $C(q - Q_j^{\pm})^{-1/p_j}$, which is uniformly integrable on the fixed compact exponent interval.

The length $\Lambda_j^{\pm}$ from a zero to the corresponding face is

$$\Lambda_j^{\pm} = \frac{1}{a_{p_j}} \int_0^{1-d_j^{\pm}} \left[(1-v^2)^{p_j} - (Q_j^{\pm})^{p_j}\right]^{-1/p_j} dv.$$

Because S is fixed, the numbers $d_j^{\pm}$ stay in a compact subinterval of $(0,1)$. Thus $1 - v^2 \ge q_{\mathrm{f}} > 0$ throughout these integrals, with q_{f} independent of j and of the sign. For large j, the ratio of their integrand to $(1-v^2)^{-1}$ is

$$\left[1 - \left(\frac{Q_j^{\pm}}{1-v^2}\right)^{p_j}\right]^{-1/p_j} = 1 + O_{\mathcal{P},S}((Q_j^{\pm})^{p_-}), \qquad p_- := \min \mathcal{P}.$$

This follows from the mean value theorem on a fixed interval bounded away from one. The majorant is integrable on the common integration range. Consequently

$$\Lambda_j^{\pm} = a_{p_j}^{-1} \operatorname{arctanh}(1 - d_j^{\pm}) + o(1) = S + o(1),$$

uniformly on $\mathcal{P}$. In the last equality, the derivative of $\operatorname{arctanh}(1-d)$ is bounded on this fixed interval of face deficits, and $\rho_j \to 0$. Each comparison profile therefore has full width $(\ell_j - 2S) + 2\Lambda_j^{\pm} = \ell_j + o(1)$. The $o(1)$ width change multiplies its exponential pressure by $1 + o(1)$, uniformly on $\mathcal{P}$. Thus its coefficient is still (4.22).

With the one-dimensional coefficient identified, define $\delta_j = e^{-a_{p_j}\ell_j}$. Its turning-value asymptotics and $m_j^{\pm} = Q_j^{\pm}/(1 + \sqrt{1 - Q_j^{\pm}})$ give, along any subsequence with $p_j \to p$,

$$Q_j^{\pm}/\delta_j \to 4e^{C_p}, \qquad m_j^{\pm}/\delta_j \to 2e^{C_p}.$$

We also need the profile, rather than only its midpoint value. Normalize each comparison by its own midpoint deficit, and write $\widehat{h}_j^{\pm} = h_j^{\pm}/m_j^{\pm}$. Its increasing half satisfies the exact inverse relation

$$s = \mathscr{T}_{p_j,m_j^{\pm}}(\widehat{h}_j^{\pm}(s)), \qquad \mathscr{T}_{p,m}(z) := \frac{1}{a_p}\int_1^z \left([v(2-mv)]^p - (2-m)^p\right)^{-1/p} dv.$$

On every fixed $1 \le v \le Z$ and for sufficiently small m, the derivative of $[v(2-mv)]^p$ is bounded above and below by positive constants. Hence the integrand has the common integrable majorant $C_Z(v-1)^{-1/p_-}$ near $v=1$. Away from one it converges uniformly when $p_j \to p$ and $m \to 0$. It follows that $\mathscr{T}_{p_j,m_j^{\pm}}$ converges uniformly on $[1,Z]$ to

$$\mathscr{T}_{p,0}(z) = \frac{1}{2a_p}\int_1^z (v^p-1)^{-1/p}\, dv.$$

This is the inverse integral defining V_p. For any fixed midpoint radius T, choose Z so large that $\mathscr{T}_{p,0}(Z) > T+1$ uniformly on $\mathcal{P}$; the lower bound $\mathscr{T}_{p,0}(Z) \ge (2\max_{\mathcal{P}} a_p)^{-1}\log Z$ permits this choice. Uniform convergence and strict monotonicity then allow inversion on $[0,T]$. They give $\widehat{h}_j^{\pm} \to V_p$ uniformly there. The derivative formula

$$(\widehat{h}_j^{\pm})' = a_{p_j}\left([\widehat{h}_j^{\pm}(2-m_j^{\pm}\widehat{h}_j^{\pm})]^{p_j} - (2-m_j^{\pm})^{p_j}\right)^{1/p_j}$$

has a continuous right side on these compact parameter and value ranges, including the turn where it is zero. Thus the derivatives converge uniformly as well. Evenness extends this C^1 convergence to $[-T,T]$. We conclude that $h_j^{\pm}/\delta_j \to 2e^{C_p}V_p$ on every fixed midpoint interval. The sandwich (4.23) and $E_j = o(\delta_j)$ now give $V_j = (1-u_j)/\delta_j \to 2e^{C_p}V_p(s)$ uniformly on fixed midpoint balls. These bounds also hold on slightly larger fixed balls, so the normalized equation below has a uniformly bounded right side. The rescaled equation is

$$\Delta_{p_j} V_j = (1-\delta_j V_j)(2-\delta_j V_j)^{p_j-1} V_j^{p_j-1}.$$

Local $C^{1,\alpha}$ compactness improves this to C^1 convergence. The positive limit has zero derivative at the midpoint. Hence $\nabla u_j(0,0) = o(\delta_j)$. Using $W_{p_j}(u_j) = (2\delta_j V_j - \delta_j^2 V_j^2)^{p_j}/(2p_j)$, both stresses in (4.21) agree with the one-dimensional pressure up to relative error $o(1)$. The same argument applies to every subsequence, so the conclusion is uniform.

More explicitly, the difference between the two stresses is

$$P_e - P = |\nabla u_j|^{p_j-2}|\nabla_y u_j|^2 = o(\delta_j^{p_j}),$$

while both gradient terms in P are of the same smaller order. Only the potential contributes to their leading coefficient. On a fixed neighborhood of either matching face $s = \pm M$, $M > 0$, the derivative of the limiting profile is nonzero. The normalized C^1 convergence gives uniform ellipticity there, so interior estimates give convergence in every fixed smooth norm. At the turn itself only C^1 convergence is used. If b_j denotes the actual midpoint deficit, the same calculation also gives

$$P_e = (\Gamma_{p_j} + o(1))b_j^{p_j}, \qquad \Gamma_p = \frac{2^{p-1}}{p}.$$

Indeed, $W_p(1-b) = \Gamma_p b^p(1+O(b))$, and the midpoint normalization gives $|\nabla u_j| = o(b_j)$.

□

We record why the prepared-gap comparison controls a growing half-gap. For a one-dimensional positive chamber, let s be distance from its left zero and put $q = 1-u^2$. On the increasing branch define

$$t_P(q) = \frac{1}{2a}\int_q^1 \frac{dr}{\sqrt{1-r}(r^p - q_m^p)^{1/p}}, \qquad t_0(q) = a^{-1}\operatorname{arctanh}\sqrt{1-q}.$$

Here $q_m \le q \le 1$ and $s = t_P(q)$. The integrand in $t_P - t_0$ is nonnegative. Its largest integral occurs at $q = q_m$, so the pressure-width computation gives

$$0 \le t_P(q) - t_0(q) \le \frac{1}{2a} \int_{q_m}^{1} \frac{(r^p - q_m^p)^{-1/p} - r^{-1}}{\sqrt{1-r}}\, dr = \frac{C_p + o(1)}{2a} \le C.$$

This bound is uniform on $\mathcal{P}$. Thus $(s-C)_+ \le t_0(q) \le s$. Inverting the planar relation $q = \operatorname{sech}^2(at_0)$ gives

$$ce^{-2as} \le q(s) \le Ce^{-2as}, \qquad \frac{q(s)}{2} \le 1 - u(s) \le q(s), \qquad 0 \le s \le \ell(P)/2.$$

In particular, the envelope remains valid up to the turn.

The secant comparison (4.19) is uniform throughout the interior strip between the two fixed well faces; its proof does not restrict the vertical variable to a fixed midpoint window. Apply it there to the one-dimensional profiles used in (4.23). Their virtual left zeros differ from the actual left zero by $o(1)$, which changes the envelope only by a bounded factor. The lateral error is $E_j = o(e^{-a_{p_j}\ell_j})$. Since $e^{-2a_{p_j}s} \ge ce^{-a_{p_j}\ell_j}$ on the first half, this error is absorbed uniformly there. The fixed-width strip next to a well face is covered by the relative gradient bound and preparation. Consequently

$$1 - u_j(y, h_{j,-}(y) + s) \le Ce^{-2a_{p_j}s}$$

on the entire buffered first half-gap. Here s is vertical depth above its own zero graph. We state the coordinate bounds used to compare this envelope with an artificial tilted layer. If the graph is $s = f(y)$, has slope at most $\varepsilon \le 1/4$ and curvature at most K, let τ be the true distance from $(y, f(y) + s)$ to that graph. The buffered nearest foot lies within distance Cs of $(y, f(y))$. On this disk the graph differs from its tangent plane by at most CKs^2. Comparison with that plane in both directions gives

$$\left| \tau - \frac{s}{\sqrt{1 + |\nabla f(y)|^2}} \right| \le CKs^2, \qquad |\tau - s| \le C(\varepsilon^2 s + Ks^2).$$

Thus a bounded value of $\varepsilon^2\ell + K\ell^2$ preserves the exponential envelope up to a fixed factor. If this quantity tends to zero, conversion also preserves a leading pressure coefficient. Under (4.20), the first conclusion also follows directly without a curvature rate: the own graph is within $o(1)$ of its horizontal reference plane, so its distance differs from vertical depth by $o(1)$ on the buffered half-gap.

For a frozen slope b in the artificial construction, retain its exact coordinate $t = s/\lambda$, $\lambda = \sqrt{1 + |b|^2}$. On a packet of radius R, $|\nabla f - b| \le CKR$, so the preceding bound gives

$$|\tau - t| \le CKs(R + s) \le CKLR \qquad (0 \le s \le L \le R).$$

The harmless fixed factors caused by λ are included in C. This comparison uses the small finite-scale error KLR and does not replace a fixed tilted depth by an unscaled vertical depth. The artificial phase estimate gives $1 - v_j^* \asymp e^{-2a_{p_j}t}$ and $|\nabla v_j^*| \ge c(1 - v_j^*)$. Combining the envelopes in the same coordinate proves (4.32) whenever the stated packet geometry is used. Integrating $Ce^{-2p_j a_{p_j} t}$ gives (4.30).

We next control the transmission of boundary errors from a gap to its core. Keep the secant notation and fix the dimension. The argument uses the natural energy and only the upper bound $\mathbb{M} \le C\mu I$. Set

$$\mathcal{D}_{R,L} = B_R^{n-1} \times (0, L), \qquad w = 1 - u, \quad h = 1 - v, \quad z = w - h, \quad \sigma = w + h, \quad \mu = \sigma^{p-2}.$$

The functions u, v solve the matched equation. They vanish on the bottom and take values in $(0, 1)$ above it. Their deficits obey $|\nabla w| \le Cw$ and $|\nabla h| \le Ch$. Their difference satisfies

$$\mathscr{L} z = -\operatorname{div}(\mathbb{M}\nabla z) + bz = 0.$$

We require uniform preparation as follows.

(1) On every fixed bottom strip, u, v and their first derivatives tend to the common layer $g_p(s) = \tanh(a_p s)$. The coefficients have uniform local Hölder bounds there, and their principal matrices are uniformly elliptic.

(2) For a fixed large S_0, the values of u, v on $s \geq S_0$ lie in a fixed convex well range. In particular, $b \geq c_0\mu$ on that region.

(3) Throughout the strip and a fixed neighborhood of its faces, the relative bounds imply $\mathbb{M} \leq C\mu I$.

Modica and the artificial phase estimates give these properties for the layers used here. In particular, uniform $W^{2,q}_{\mathrm{uloc}}$ bounds with $q > n$ give a fixed positive Hölder exponent for the phase gradients. The fixed bottom core is noncritical, so its secant coefficients have the regularity needed for boundary gradient estimates.

The same estimates apply when the zero sheet is curved. For a variable cap, first solve the artificial problem in the deeper slab of depth $L + 3$. The comparison depth is $L + o(1)$, so the moving top face and a fixed neighborhood of width one remain strictly inside that slab. The common bottom is its Dirichlet face and uses the boundary estimates. The artificial estimates change by only a fixed factor. We do not extend a Dirichlet solution through its top face. On each window use the physical domain

$$\{(y, s) : y \in B_R,\ f(y) < s < f(y) + \ell(y)\}, \qquad \ell(y) = L + o(1),$$

The slopes of f and ℓ tend to zero. Their second derivatives have uniform local C^α bounds. These are the geometry hypotheses for this variable-cap estimate. The prescribed cap construction below verifies them by the chain rule. Pull back both solutions by

$$\Phi(y, t) = (y, f(y) + t\ell(y)/L), \qquad 0 < t < L.$$

Set $P = D\Phi$ and $\mathcal{J}_\Phi = \det P$. The exact transformed coefficients are

$$\widetilde{\mathbb{M}} = \mathcal{J}_\Phi P^{-1}(\mathbb{M} \circ \Phi)P^{-T}, \qquad \widetilde{b} = \mathcal{J}_\Phi(b \circ \Phi), \qquad \widetilde{\mu} = \mathcal{J}_\Phi(\mu \circ \Phi).$$

The difference equation remains homogeneous. Indeed, for a test ζ, $(\nabla_x \zeta) \circ \Phi = P^{-T}\nabla_{(y,t)}(\zeta \circ \Phi)$ and $dx = \mathcal{J}_\Phi\, dy\, dt$. Substitution in the weak equation gives exactly $\widetilde{\mathbb{M}}$ and $\widetilde{b}$ above. Moreover, $P = I + o(1)$ and $\mathcal{J}_\Phi = 1 + o(1)$, and their first derivatives have uniform local Hölder bounds. The pullback preserves the natural energy. Thus the well bound, principal upper bound, IMS estimate, and cutoff iteration keep uniform constants. In the fixed bottom strip, the transformed coefficients converge to their planar values. The boundary gradient estimate therefore applies. No source term is introduced. Before projecting the force, transform the physical stress back by the exact tensor rule.

We justify physical stress conservation before using this change of coordinates. For either matched weak solution v, define

$$T[v] = |\nabla v|^{p-2}\nabla v \otimes \nabla v - \left(\frac{|\nabla v|^p}{p} + W_p(v)\right) I.$$

The first term is zero at a critical gradient. This sign convention makes the vertical pressure $-T[v]_{nn}$. The local regularity recalled in Section 1 gives the locally C^1 representatives of these solutions. Since $p > 2$, the reaction $-W_p'$ is C^1 on $\mathbb{R}$. Thus Proposition 2.5 applies and gives

$$V_v := |\nabla v|^{(p-2)/2}\nabla v \in H^1_{\mathrm{loc}} \cap L^\infty_{\mathrm{loc}}.$$

The Sobolev product and chain rules therefore give

$$T[v] = V_v \otimes V_v - \left(\frac{|V_v|^2}{p} + W_p(v)\right) I \in H^1_{\mathrm{loc}}.$$

On $\{\nabla v \neq 0\}$, the shared noncritical bootstrap from Section 2.1 gives $v \in C^2$ locally. With $F = |\nabla v|^{p-2}\nabla v$, the classical product rule gives

$$\sum_b \partial_b T[v]_{ab} = v_a(\operatorname{div} F - W_p'(v)) = 0.$$

On the critical set, $V_v = 0$. Every derivative term in $V_v \otimes V_v$ and $|V_v|^2$ has a factor V_v, so these weak derivatives vanish there almost everywhere. Also $\nabla W_p(v) = W_p'(v)\nabla v = 0$ there. Thus the weak derivative of $T[v]$ has zero divergence on both sets. Since it is locally in L^2, this proves $\operatorname{div}_x T[v] = 0$ in distributions across the critical set as well. The identity extends to compactly supported C^1 tests by approximation. Write T for the stress of the physical solution being used. Its exact transformation is

$$\widehat{T} = \mathcal{J}_\Phi (T \circ \Phi) P^{-T}, \qquad \sum_\alpha \partial_{\xi_\alpha} \widehat{T}_{i\alpha} = 0.$$

The index i remains a fixed physical Cartesian component, while α is the coordinate flux index. This is the Piola change of variables in the weak divergence equation. It keeps the Jacobian and the normal area vector in the boundary flux. The vertical pressure is $-T_{nn}$, and the mixed normal–tangential component is the integrated flux used below.

For sufficiently accurate fixed-core preparation, there is $\lambda_0 > 0$ independent of R, L such that

$$Q_{\mathrm{sec}}(\zeta) := \int_{\mathcal{D}_{R,L}} (\mathbb{M}[\nabla\zeta, \nabla\zeta] + b\zeta^2) \geq \lambda_0 \int_{\mathcal{D}_{R,L}} (\mathbb{M}[\nabla\zeta, \nabla\zeta] + \mu\zeta^2) \tag{4.24}$$

The tests are compactly supported and have zero trace on every face, including the bottom.

The required spectral gap comes from the planar form. Conjugate the exact physical Jacobi form by $\phi = (aG)^{(2-p)/2}F$. Its normal operator is

$$(p-1)\left[-\partial_s^2 + p^2a^2 - p(p+1)a^2G(s)\right], \qquad G(s) = \operatorname{sech}^2(as).$$

Apply Proposition 2.17 to the odd extension of a zero-trace half-line test. Its weighted mean is zero because g_p' is even. Approximation in the weighted form domain justifies this extension. Dividing both sides by two gives the sharp half-line gap $(p-1)a^2(2p-1)$, uniformly on the exponent interval. In the conjugated variables, the positive eigenfunction is $G^{(p-1)/2}\tanh(as)$. In physical variables the measure is $(aG)^{p-2}\,ds$. On this half-line it is uniformly comparable with $(2(1-g_p))^{p-2}\,ds$.

For the planar pair $w = h = 1 - g_p$, one has $\mathbb{M} = A_p(g_p' e_s)$ and $b = \mathcal{F}_p'(1-g_p) = -f_p'(g_p)$. Thus the secant form is exactly the planar Jacobi form. The odd extension already removes its even translation mode. Tangential derivatives contribute nonnegatively. The comparability of the two measures is also explicit:

$$\frac{(aG)^{p-2}}{(2(1-g_p))^{p-2}} = \left(\frac{a(1+g_p)}{2}\right)^{p-2},$$

which stays between two positive constants for $s \geq 0$ and $p \in \mathcal{P}$.

Combine the bottom gap with well coercivity by an IMS partition $\chi_0^2 + \chi_1^2 = 1$. Take χ_0 supported in $s < 2S$ and χ_1 supported in $s > S$, with $|\chi_i'| \leq C/S$. We choose the width before the preparation accuracy. Let $\mu_0 = (2(1-g_p))^{p-2}$ and let Q_0 be the planar secant form. The half-line gap and the lower bound for its potential give

$$Q_0(\xi) \geq \lambda_* \int \mu_0 \xi^2, \qquad \int (\mathbb{M}_0[\nabla\xi, \nabla\xi] + \mu_0\xi^2) \leq C_* Q_0(\xi),$$

with constants independent of the core width. For the second bound, write the gradient term as Q_0 minus its potential term. The potential coefficient is bounded below by $-C\mu_0$. The first bound then controls that subtraction.

Choose $S \geq S_0$ so large that the IMS error is below one quarter of both the planar and well mass gaps. On the now fixed strip $0 \leq s \leq 2S$, the planar gradient has a positive minimum. Sufficiently accurate preparation therefore gives

$$|Q_{\mathrm{sec}}(\xi) - Q_0(\xi)| \leq \varepsilon \int (\mathbb{M}_0[\nabla\xi, \nabla\xi] + \mu_0 \xi^2).$$

Choose $\varepsilon < 1/(2C_*)$ afterward. The core form retains at least half its planar gap, and μ is comparable with μ_0 there. On $s \geq S$, the bound $b \geq c_0\mu$ gives the well gap $Q_{\mathrm{sec}}(\chi_1\zeta) \geq c_0 \int \mu\chi_1^2\zeta^2$. The localization identity is

$$Q_{\mathrm{sec}}(\zeta) = \sum_{i=0}^{1} Q_{\mathrm{sec}}(\chi_i\zeta) - \int \sum_{i=0}^{1} \mathbb{M}[\nabla\chi_i, \nabla\chi_i]\zeta^2.$$

It follows by expanding the two gradients and using $\sum_i \chi_i \nabla\chi_i = 0$. Since $\mathbb{M} \leq C\mu I$, the last term is at most $C/S^2 \int \mu\zeta^2$. Its constant is uniform in S, so the chosen width absorbs it. Thus $Q_{\mathrm{sec}}(\zeta) \geq c \int \mu\zeta^2$. Finally, $b \geq -C\mu$ throughout the strip gives

$$\int \mathbb{M}[\nabla\zeta, \nabla\zeta] \leq Q_{\mathrm{sec}}(\zeta) + C \int \mu\zeta^2 \leq (1 + C/c) Q_{\mathrm{sec}}(\zeta).$$

These two bounds prove (4.24), with constants independent of R, L.

The argument never requires a lower bound for $\mathbb{M}/\mu$. Such a bound can fail before the relative-profile assumption imposed below.

Separate the top contribution from the lateral boundary data. Let $L \to \infty$ and choose $R = L^k$ with $k > 1$ fixed. Suppose $0 < B \ll 1$ and

$$0 < w, h \leq CB \quad \text{on a fixed neighborhood of } s = L,$$

Define $\rho = B^p$. Prepared-gap comparison gives these bounds when $L = \ell/2 + O(1)$ and $B \asymp e^{-a\ell}$. A one-sided clearing upper bound also suffices. No lower top-deficit bound is needed. The relative gradient assumptions give $|z(y, L)| + |\nabla_y z(y, L)| \leq C\sigma(y, L)$. Choose $\chi(y)$ equal to one on $B_{R/2}$ and supported in $B_{3R/4}$. Take a fixed smooth η with $\eta(0) = 1$ and support in $[0, 1)$. Lift the top trace by

$$F_{\mathrm{top}}(y, s) = \chi(y)\eta(L - s)z(y, L).$$

On fixed balls meeting the support, σ is comparable to its top-face value. The coefficient bounds give

$$|Q_{\mathrm{sec}}(F_{\mathrm{top}})| \leq CB^p R^{n-1} = C\rho R^{n-1}. \tag{4.25}$$

Indeed, the lift and its derivatives are $O(\sigma(y, L))$. The matrix and the absolute value of the potential are bounded by $C\sigma(y, L)^{p-2}$ on its support. Their product has size $O(\sigma(y, L)^p) \leq CB^p$. The same calculation bounds the lift's full squared natural norm by $C\rho R^{n-1}$.

Use the natural energy space to split the boundary data. Complete the compact smooth zero-boundary tests in the norm whose square is $\int(\mathbb{M}[\nabla f, \nabla f] + \mu f^2)$. This form is closable. Indeed, $\mu > 0$, and $\mathbb{M}$ is locally uniformly elliptic wherever at least one of $\nabla w, \nabla h$ is nonzero. On their common critical set it is zero. If tests tend to zero in $L^2(\mu)$ and their weighted gradients converge, local ellipticity makes the gradient limit zero off that set. It is also zero on the common critical set. Thus the completion has a well-defined $L^2(\mu)$ representative. The bound $|b| \leq C\mu$ makes Q_{sec} continuous in this norm. Coercivity gives a unique minimizer z_{top} in the affine space F_{top} plus the zero-boundary completion. It satisfies $\mathscr{L} z_{\mathrm{top}} = 0$. Write $z_{\mathrm{top}} = F_{\mathrm{top}} + h_0$, where h_0 lies in the zero-boundary space. The weak equation gives $Q_{\mathrm{sec}}(h_0, h_0) = -Q_{\mathrm{sec}}(F_{\mathrm{top}}, h_0)$. Write $\|\cdot\|_{\mathrm{nat}}$ for the natural norm just defined. The bound $|b| \leq C\mu$ and matrix Cauchy–Schwarz give $|Q_{\mathrm{sec}}(F, H)| \leq C\|F\|_{\mathrm{nat}}\|H\|_{\mathrm{nat}}$. Coercivity therefore yields

$$\lambda_0 \|h_0\|_{\mathrm{nat}}^2 \leq C\|F_{\mathrm{top}}\|_{\mathrm{nat}}\|h_0\|_{\mathrm{nat}}.$$

If h_0 is nonzero, divide by its norm; the zero case is immediate. The top-lift bound and the triangle inequality now give

$$\int (\mathbb{M}[\nabla z_{\mathrm{top}}, \nabla z_{\mathrm{top}}] + \mu z_{\mathrm{top}}^2) \le C\rho R^{n-1}. \tag{4.26}$$

Now $z_{\mathrm{side}} = z - z_{\mathrm{top}}$ has zero bottom trace. Its top trace vanishes on $B_{R/2}$. Use the extension $F_{\mathrm{side}} = (1-\chi)z$. Monotonicity and the gradient bounds imply

$$\mathbb{M}[\nabla z, \nabla z] \le C\sigma^p, \qquad \mu z^2 \le C\sigma^p \le C.$$

The remaining cutoff contribution $z^2\mathbb{M}[\nabla\chi, \nabla\chi]$ is controlled by the secant form bound. Hence $|Q_{\mathrm{sec}}(F_{\mathrm{side}})| \le CR^{n-1}L$. Correcting F_{side} by a zero-boundary function as above gives

$$\int (\mathbb{M}[\nabla z_{\mathrm{side}}, \nabla z_{\mathrm{side}}] + \mu z_{\mathrm{side}}^2) \le CR^{n-1}L. \tag{4.27}$$

The original difference lies in the sum of these two affine form spaces. More precisely, $z - F_{\mathrm{top}} - F_{\mathrm{side}}$ has zero trace on every face and has a C^1 representative. At a zero-trace face, its value is bounded by a constant times the distance to that face. Boundary cutoffs therefore converge in the natural norm: their extra gradient term is bounded by a constant times the integral of $|\mathbb{M}|$ over a shrinking boundary collar. This tends to zero. Interior smoothing finishes the approximation. The original difference and the sum of the two variational solutions thus differ by a zero-boundary homogeneous solution. Coercivity makes that difference zero.

Let v solve the homogeneous secant equation. For any cutoff ϑ such that v times the cutoff has zero trace, the exact identity is

$$Q_{\mathrm{sec}}(\vartheta v) = \int v^2 \mathbb{M}[\nabla\vartheta, \nabla\vartheta].$$

Use coercivity and the form bound with $f = v|\nabla\vartheta|$. Across a cutoff layer of fixed width Λ, chosen sufficiently large, the natural H^1 mass drops by a factor at most $1/2$.

For example, let $E(d)$ be the natural energy on the part whose distance from the source boundary is at least d. Choose a cutoff equal to one on that part and zero at distances less than $d - \Lambda$, with gradient at most C/Λ. On the retained part the cutoff is one, so its natural norm controls $E(d)$. Coercivity and the identity above give

$$E(d) \le \lambda_0^{-1} Q_{\mathrm{sec}}(\vartheta v) \le C\Lambda^{-2} \int_{\operatorname{supp}\nabla\vartheta} \mu v^2 \le C\Lambda^{-2} E(d - \Lambda).$$

After fixing Λ with $C\Lambda^{-2} \le 1/2$, iteration for $\lfloor d/\Lambda \rfloor$ steps gives $E(d) \le Ce^{-c_{\mathrm{dec}} d} E(0)$. Here $c_{\mathrm{dec}} > 0$ is fixed by the cutoff width. The constants are independent of the strip length. The same construction works from the lateral boundary using a radial tangential cutoff.

The mass is $\int(\mathbb{M}[\nabla v, \nabla v] + \mu v^2)$, while the cutoff term is bounded directly by $C\Lambda^{-2}\int \mu v^2$.

Apply this iteration vertically to z_{top}, starting at L and moving in s toward the bottom. For z_{side}, iterate tangentially inside $B_{R/2}$. Both horizontal traces vanish there. In particular,

$$\int_{B_{R/4}\times(0,L)} (\mu z_{\mathrm{side}}^2 + \mathbb{M}[\nabla z_{\mathrm{side}}, \nabla z_{\mathrm{side}}]) \le CR^{n-1} L e^{-cR}.$$

For a fixed $S < \infty$, this yields

$$\int_{B_{R/4}\times(0,S)} (\mu z^2 + \mathbb{M}[\nabla z, \nabla z]) \le C_S \rho R^{n-1} e^{-cL} + C_S R^{n-1} L e^{-cR}, \tag{4.28}$$

$$|\partial_s z(0,0)| \le C\rho^{1/2} R^{(n-1)/2} e^{-cL/2} + CR^{(n-1)/2} L^{1/2} e^{-cR/2}. \tag{4.29}$$

The final slope estimate uses local boundary gradient estimates in the fixed bottom neighborhood. Its coefficients are uniformly Hölder and elliptic, and z vanishes on the bottom. This preserves the square-root energy bound. A global Lipschitz interpolation would give a weaker exponent.

In that fixed bottom neighborhood the weight μ is bounded above and below by positive constants. The usual local boundedness estimate first controls $\|z\|_\infty$ by its local L^2 norm. The zero-trace boundary $C^{1,\alpha}$ estimate then controls the normal derivative by the same norm. The smooth flat boundary, or the uniformly controlled flattened boundary, makes its constant independent of L and R.

For top data bounded by a fixed constant, the same argument gives (4.29) with ρ replaced by one. Set $L = A\log(1/\varepsilon)$ and $R = L^k$, with $k > 1$ fixed. After fixing the energy-decay rate, choose A so that $cA/2 > 1$ with a strict margin. The polynomial factors are then absorbed, and the bottom slope error is $o(\varepsilon)$.

Assume $\rho = e^{-\beta\ell}$ and $L = \ell/2 + O(1)$. Choose a uniform $\eta > 0$. For large ℓ, the first contribution in (4.29) is at most $C\rho^{1/2+\eta}$. Any choice $\eta < c/(4\sup\beta)$ with strict margin absorbs its polynomial factor. Since $R = L^k$ with $k > 1$, the lateral contribution is smaller than every fixed power of ρ.

To estimate the stress flux outside the core, also assume the tail-mass estimate

$$\sup_y \int_S^L \sigma(y,s)^p\,ds \le Ce^{-\kappa S}, \qquad \sup_y \int_0^L \sigma(y,s)^p\,ds \le C, \tag{4.30}$$

supplied by the gap sandwich and artificial phase bound. Here $\kappa > 0$ is uniform. Define

$$\mathcal{J}(u)(y) = \int_0^L |\nabla u|^{p-2} u_s \nabla_y u\,ds.$$

The degree-p stress map $\xi \mapsto |\xi|^{p-2}\xi_s\xi_T$ satisfies

$$|F(\xi) - F(\zeta)|^2 \le C(|\xi| + |\zeta|)^p\,\mathbb{M}_{\xi,\zeta}[\xi - \zeta, \xi - \zeta].$$

This follows from flux monotonicity. Its left side is bounded by $C(|\xi| + |\zeta|)^{2p-2}|\xi - \zeta|^2$, and the secant form is bounded below by $c(|\xi| + |\zeta|)^{p-2}|\xi - \zeta|^2$. It follows that

$$|\mathcal{J}(u) - \mathcal{J}(v)| \le C\int_0^L \sigma^{p/2}\big(\mathbb{M}[\nabla z, \nabla z]\big)^{1/2}\,ds.$$

For either interval $I = [0, S]$ or $I = [S, L]$, Cauchy–Schwarz gives

$$\left\|\int_I (F(\nabla u) - F(\nabla v))\,ds\right\|^2_{L^2(B_{R/4})} \le C\left(\sup_y \int_I \sigma^p\,ds\right)\int_{B_{R/4}\times I} \mathbb{M}[\nabla z, \nabla z].$$

For the core, (4.28) supplies the extra factor e^{-cL}. For the tail, (4.26), (4.27) and tangential decay bound the second factor by $C\rho R^{n-1} + CR^{n-1}Le^{-cR}$. The first factor is at most $Ce^{-\kappa S}$ by (4.30). Divide by ρR^{n-1} and first let $\ell \to \infty$ with S fixed. The lateral error vanishes because $R = L^k$, $k > 1$, while $\rho = e^{-\beta\ell}$ and $L = \ell/2 + O(1)$. The upper limit of the normalized flux is therefore at most $Ce^{-\kappa S/2}$. Then let $S \to \infty$ to obtain

$$\frac{\|\mathcal{J}(u) - \mathcal{J}(v)\|_{L^2(B_{R/4})}}{R^{(n-1)/2}\sqrt{\rho}} \longrightarrow 0. \tag{4.31}$$

The bounds are uniform whenever the hypotheses are uniform.

To estimate the stress flux pointwise, suppose the first half-gap also satisfies

$$w \le Ch, \qquad |\nabla h| \ge ch, \qquad \sigma(y,s) \le Ce^{-2as}. \tag{4.32}$$

If h is the one-sided artificial profile, the prepared-gap sandwich gives the first and third bounds. Its almost unit phase gradient gives the second. The empty-side argument below gives the same upper comparison when no opposite zero is nearby. Use well faces at S and $4L - S$, then restrict to $[S, L]$. The reference turn stays beyond the interval where the relative envelope is needed.

These assumptions make the secant operator uniformly elliptic in its natural scale:

$$C^{-1}\mu I \le \mathbb{M} \le C\mu I. \tag{4.33}$$

Indeed, $|\nabla h| \geq c\sigma$ and $|\nabla w| + |\nabla h| \leq C\sigma$. Even when the gradient segment crosses zero, the integral of $|(1-t)\nabla h + t\nabla w|^{p-2}$ is comparable with $(|\nabla h| + |\nabla w|)^{p-2}$.

On fixed small balls centered at x_0, the normalized coefficients $\mathbb{M}/\mu(x_0)$ and $b/\mu(x_0)$ have uniform local Hölder bounds. To verify this at a crossing segment, divide the gradients by $\sigma(x_0)$. One endpoint then stays away from zero, and both remain bounded. On this compact set, the secant matrix is uniformly Hölder of every fixed order below $\min\{1, p_{\min} - 1\}$. Consider two endpoint pairs at distance $d < 1$. Split off the parameters where either segment is within Cd of zero. If this occurs, the segment velocity has a positive lower bound because one endpoint does. The exceptional set has length $O(d)$ and contributes at most $Cd^{p_{\min}-1}$. Off that set, the derivative estimate for A_p gives

$$Cd \int_d^C r^{p_{\min}-3}\, dr \leq C_{\mathcal{P}} d,$$

since $p_{\min} > 2$. Thus the secant map is uniformly Lipschitz there. Segments that stay away from zero are handled directly by smoothness. Compose with the local $C^{1,\alpha}$ estimates to obtain a positive Hölder exponent. No derivative bound for A_p at a zero gradient is required. The nonlinear estimates are also uniform in the small deficit. Put $s_0 = \sigma(x_0)$ and $W = w/s_0$, $H = h/s_0$ on a fixed ball. The relative gradient bounds give $W, H \leq C$ and $c \leq W + H \leq C$. Their equations have right-hand sides

$$s_0^{1-p}\mathcal{F}_p(s_0 W) = (1 - s_0 W)(2 - s_0 W)^{p-1} W^{p-1},$$

and the analogous expression for H, uniformly bounded there. Interior nonlinear $C^{1,\alpha}$ estimates therefore control $\nabla W, \nabla H$ with constants independent of s_0. Composing with the secant estimate just proved yields the asserted Hölder bound for $\mathbb{M}/\mu(x_0)$. For $b/\mu(x_0)$ use the same averaging argument for the scalar function $\mathcal{F}_p'$. The endpoint values are nonnegative and their sum is bounded below; the possible singular derivative near zero is integrable since $p_{\min} > 2$. This also gives a uniform positive Hölder exponent for the potential coefficient.

On a fixed small ball about x_0, the bound for $|\nabla \log \mu|$ makes μ comparable with $\mu(x_0)$. After division by $\mu(x_0)$, the equation is uniformly elliptic with the Hölder coefficient bounds just proved. Interior estimates and fixed-core boundary estimates therefore give

$$|\nabla z(x_0)| \leq C\mu(x_0)^{-1/2} \left(\int_{B_1(x_0)} (\mathbb{M}[\nabla z, \nabla z] + \mu z^2) \right)^{1/2}, \tag{4.34}$$

Shrink the fixed radius 1 if necessary. At the bottom, use the zero trace. The same local estimate also bounds $|z(x_0)|$ by the right-hand side of (4.34). A fixed distance below the top, the normal iteration yields

$$|\nabla z(y,s)| \leq C\mu(y,s)^{-1/2} \left(\sqrt{\rho}\, R^{(n-1)/2} e^{-c(L-s)/2} + R^C L^C e^{-cR/2} \right) \qquad (y \in B_{R/8}).$$

Combine $\sigma^{p-1}\mu^{-1/2} = \sigma^{p/2}$ with (4.32). The pointwise stress difference is bounded by

$$C\sqrt{\rho}\, R^{(n-1)/2} e^{-pas} e^{-c(L-s)/2} + R^C L^C e^{-cR/2} e^{-pas}.$$

On the last fixed-width top interval, the original relative gradient bounds suffice. They bound the stress difference by $C\sigma^p \leq Ce^{-2pas}$ without a boundary gradient estimate. Integration therefore gives, for a uniform $\kappa_1 > 0$,

$$\|\mathcal{J}(u) - \mathcal{J}(v)\|_{L^\infty(B_{R/8})} \leq C\sqrt{\rho}\, R^C L^C e^{-\kappa_1 L} + CR^C L^C e^{-cR/2} + C\rho = o(\sqrt{\rho}). \tag{4.35}$$

Here $R = L^k$, $k > 1$, and $\rho = e^{-2paL + O(1)}$. Consequently the weak projected error is small in every local L^q space, including exponents above the interface dimension. This is stronger than an unscaled L^2 bound.

To check the exponential integration, for any $A, B > 0$,

$$\int_0^L e^{-As} e^{-B(L-s)}\, ds \le L e^{-\min\{A,B\}L}.$$

Use $A = pa$ and $B = c/2$, where $c > 0$ is the fixed energy iteration rate. Choose κ_1 strictly below the uniform minimum of these rates, so the polynomial factor is absorbed.

It prevents concentration at an isolated minimum of a rescaled gap. The same estimate applies to the normal integral of each full stress-tensor component. The gradient terms are degree-p maps with the same secant estimate. For the potential,

$$|W_p(u) - W_p(v)| \le C\sigma^{p-1}|z|.$$

The local estimate (4.34) controls both $|z|$ and $|\nabla z|$. The potential contribution is therefore bounded by $C\sigma^{p/2}$ times the square root of the local natural energy. The tensor estimate permits a rotation of the normal chart before force projection.

An empty side requires a separate verification of the relative-profile hypothesis. Consider a prepared positive chamber based at a zero graph. On this chamber side, suppose there is no opposing boundary zero within depth $4L$ over the buffered comparison disk. Zeros on the other side of the base sheet do not enter this condition. Assume the chamber geometry is flat to $o(1)$ on disks of radius $R = L^k$, with $k > 1$. The ordered boundary graphs and their $o(1)$ oscillations must give the stated clearance uniformly on this whole disk. Choose S beyond the fixed clearing distance for a small convex well range. Compare between normal heights S and $4L - S$. Both faces and a fixed neighborhood then remain in the well. The subtraction of S at the far face leaves uniform clearance from an opposing zero just beyond depth $4L$.

Preparation at the near face gives $w \le (1 + o(1))(1 - g_p(S))$. At the far face, only $w \le w_*$ is needed. Solve the one-dimensional problem $\Delta_p h = \mathcal{F}_p(h)$ with these constant face values. The well secant comparison removes lateral effects at rate e^{-cR}. This solution is part of a periodic gap of virtual full width $4L + O(1)$, since the omitted zero-to-face lengths approach fixed constants.

For unequal face values, the one-dimensional deficit has strictly increasing flux $Y = |h'|^{p-2}h'$, because $Y' = \mathcal{F}_p(h) > 0$. Its two boundary values stay in a compact subset of $(0, 1)$ as $L \to \infty$. For a sufficiently long interval, Y must change sign. Indeed, if h were increasing on the whole interval, its minimum would be the fixed positive left boundary value, so $Y' \ge c > 0$. Integrating first for Y and then for h' would make h exceed its bounded right boundary value in a uniformly bounded length. The decreasing case follows by reflection. Thus the solution has exactly one interior minimum $h_m > 0$. Its first integral is

$$W_p(1 - h) - \frac{p-1}{p}|h'|^p = P, \qquad P = W_p(1 - h_m) > 0.$$

The minimum tends to zero. Otherwise $h_m \ge d > 0$ along a subsequence, and the travel integral from the minimum to either fixed face value would remain bounded. To see this directly, $\mathcal{F}_p(h) \ge c_d > 0$ on the corresponding compact interval, so the travel integrand is at most $C_d(h - h_m)^{-1/p}$ near the minimum. That singularity is integrable, uniformly on $\mathcal{P}$. The resulting bounded length contradicts $L \to \infty$.

Hence $P \to 0$. Extend this profile from each face to its adjacent zero of $u = 1 - h$. The first integral permits the extension because $W_p(1 - h) > P$ for $h_m < h \le 1$. The two extra lengths converge to their zero-pressure values. On the left that value is $S + o(1)$; on the right it is the fixed number $a_p^{-1} \operatorname{arctanh}(1 - w_*)$. The interval between the faces has length $4L - 2S$. Therefore the full virtual zero-to-zero width is $4L + O(1)$, uniformly in p. The same first integral gives $|h'| \le a_p h(2 - h)$, so this comparison profile satisfies the relative gradient bound used in the secant comparison.

To control the first quarter of this virtual gap, compare its positive-pressure quadrature with the zero-pressure one. For $q \geq 2q_m$ in the chosen small well,

$$0 \leq \int_q^1 \frac{(r^p - q_m^p)^{-1/p} - r^{-1}}{\sqrt{1-r}}\, dr$$
$$\leq C q_m^p \int_q^1 \frac{r^{-p-1}}{\sqrt{1-r}}\, dr \leq C(q_m/q)^p.$$

The middle inequality follows from $(1-x)^{-1/p} - 1 \leq Cx$ for $0 \leq x \leq 2^{-p}$. Split the last integral at $1/2$ to obtain its final bound. The virtual width formula gives $q_m \asymp e^{-4aL}$, whereas the zero-pressure lower comparison gives $q \geq ce^{-2as}$. Thus $q/q_m \to \infty$ uniformly for $S \leq s \leq L$. The extra travel time is bounded. A bounded shift of the virtual left zero changes the exponential by a fixed factor. Inverting the planar relation therefore gives

$$h(s) \leq Ce^{-2as} \qquad (S \leq s \leq L),$$

uniformly on $\mathcal{P}$.

Therefore, on a smaller horizontal disk,

$$1 - g_p(s) \leq w(y,s) \leq Ce^{-2as} + Ce^{-cR} \leq C(1 - g_p(s)) \qquad (S \leq s \leq L)$$

The lower estimate follows from Modica along a normal ray. A displacement $o(1)$ changes only its factor. Thus (4.32) holds on the truncated half-strip without a genuine opposite zero.

The top lift costs at most $Ce^{-2paL}R^{n-1}$. If $\rho = e^{-pa\ell_{\min}}$ and $L = C_0\ell_{\min}$, its cost is $O(\rho^{2C_0}R^{n-1})$. Choose $C_0 > 1$. The core energy error is then $o(\rho)$, and the uniform stress error is $o(\sqrt{\rho})$. At the cut itself the vertical stress is bounded by $C(1-u^2)^p \leq Ce^{-2paL} = o(\rho)$.

The Jacobi-core test also admits a closure with zero interaction. Cut off u_s over a fixed-width band at height L. Per horizontal point, the extra form cost is at most

$$C\int_{\text{band}} (u_s)^2 |A_p(\nabla u)| \leq C\int_{\text{band}} |\nabla u|^p \leq Ce^{-2paL}$$

Proposition 2.5 gives the weak Jacobi identity needed for this cutoff. The tensor $(u_s)^2 A_p(\nabla u)$ stays continuous across critical gradients. Thus inactive gaps, whether finite or infinite, have negligible outer force and stability costs.

For a moving terminal graph $s = c(y)$, choose the cutoff as $\eta(s - c(y))$, with a fixed-width transition and $|\eta'| \leq C$. Then

$$|\nabla \eta(s - c(y))|^2 \leq C(1 + |\nabla c(y)|^2).$$

The terminal slopes are uniformly small, so the weak Jacobi identity bounds its additional normal closure cost by

$$C\int_{\text{band}} (u_s)^2 |A_p(\nabla u)| \leq C\int_{\text{band}} |\nabla u|^p.$$

If a tangential amplitude $\zeta(y)$ is present, expand the gradient of $\eta\zeta$. The elementary inequality $|\nabla(\eta\zeta)|^2 \leq 2\eta^2|\nabla\zeta|^2 + 2\zeta^2|\nabla\eta|^2$ handles its cross term in the terminal band. Its horizontal contribution is bounded by the same exponentially small tail mass times $|\nabla\zeta|^2$, while its closure contribution is bounded by that mass times ζ^2. The coefficient of the kinetic term on the unchanged interior core is unaffected. No derivative of u_s needs to be estimated in that band.

We now prescribe the smooth caps used with these estimates. Let $\ell_{\min} \to \infty$, fix $C_0 > 1$, and put $L_{\text{cap}} = C_0\ell_{\min}$ and $\rho = e^{-pa\ell_{\min}}$. Choose a fixed smooth nonincreasing function χ that is one on $(-\infty, 0]$, zero on $[1, \infty)$, and satisfies $\chi(t) + \chi(1-t) = 1$ on $[0,1]$. Define a smooth nondecreasing

function Θ on $[0,\infty)$ by

$$\Theta(r)=\begin{cases} r/2, & 0\le r\le 1,\\ \dfrac{1}{2}+\dfrac{1}{2}\displaystyle\int_1^r \chi((t-1)/2)\,dt, & 1<r<3,\\ 1, & r\ge 3. \end{cases}$$

The symmetry of χ makes its integral on $[0,1]$ equal to $1/2$. Thus the middle formula has value one at $r=3$. The flat cutoff ends make the formulas smooth at both joins. Set $\theta(d)=L_{\mathrm{cap}}\Theta(d/L_{\mathrm{cap}})$. Then

$$\theta(d)=d/2\ (d\le L_{\mathrm{cap}}),\qquad \theta(d)=L_{\mathrm{cap}}\ (d\ge 3L_{\mathrm{cap}}),\qquad 0\le\theta'\le 1/2,\qquad \theta(d)\le d/2.$$

For every fixed $k\ge 2$, $|\theta^{(k)}|\le C_k L_{\mathrm{cap}}^{1-k}$. In the whole switching range $L_{\mathrm{cap}}\le d\le 3L_{\mathrm{cap}}$, monotonicity also gives $\theta(d)\ge L_{\mathrm{cap}}/2$.

For adjacent graphs $s=f_-(y)$ and $s=f_+(y)$, put $d=f_+-f_->0$. Their prescribed terminal faces are $f_-+\theta(d)$ and $f_+-\theta(d)$. For a Jacobi closure, fix $M\ge 0$ and use $\eta(s-f_--\theta(d)+M)$ on the lower side, where $\eta=1$ on $(-\infty,-1]$, $\eta=0$ on $[0,\infty)$, and $|\eta'|\le C$. Reflect this cutoff on the upper side. Its lower transition band has own depth between $\theta(d)-M-1$ and $\theta(d)-M$. For a missing neighbor use the constant depth $\theta=L_{\mathrm{cap}}$ and the same cutoff. Thus the switching-range band has depth at least $L_{\mathrm{cap}}/2-M-1$. This records the fixed offset before the terminal face; it changes only the constant in an exponential bound.

Here are the local geometry checks. On a buffered comparison window, assume the prepared graph slopes tend to zero, their second derivatives have common local C^α bounds, and their height oscillations are $o(1)$ in the chosen tangent chart. Since $|\theta'|\le 1/2$, the cap depth differs from its value at the center by $o(1)$ and has slope tending to zero. Moreover,

$$D^2[\theta(d)]=\theta'(d)D^2d+\theta''(d)\nabla d\otimes\nabla d.$$

The derivative bounds just proved give the required common C^α bounds for this Hessian. The fixed offset M does not change them. More precisely, if K bounds the two graph Hessians and ϑ bounds their slopes, then

$$|D^2[\theta(d)]|\le CK+\frac{C\vartheta^2}{L_{\mathrm{cap}}}\mathbf{1}_{\{L_{\mathrm{cap}}<d<3L_{\mathrm{cap}}\}}.$$

The second term need not be $O(K)$. It vanishes on every active midpoint region $d\le L_{\mathrm{cap}}$, where $\theta(d)=d/2$. On the switch it supplies only a uniform coefficient bound for the exact homogeneous pullback. It introduces no source term. These are precisely the variable-cap geometry hypotheses used above. All artificial comparisons are made in the deeper slab before restriction to the prescribed terminal face. The force and cutoff costs on the switch will be estimated from its exponentially small terminal stress and energy, rather than charged to K.

For $d\le 4L_{\mathrm{cap}}$, use the full prepared gap sandwich. Every prescribed face has own depth at most $d/2$. This includes $3L_{\mathrm{cap}}\le d\le 4L_{\mathrm{cap}}$, where no empty-side assertion is needed. For larger gaps or a missing neighbor, use the empty-side comparison at scale L_{cap}. If the gap at the center is only $4L_{\mathrm{cap}}$ up to an $o(1)$ graph oscillation, place its far well face at $4L_{\mathrm{cap}}-S-1$ instead of $4L_{\mathrm{cap}}-S$. The preceding proof still has virtual width $4L_{\mathrm{cap}}+O(1)$ and the same exponential envelope through depth $L_{\mathrm{cap}}+O(1)$. The fixed displacement therefore preserves all constants and supplies the required clearance on the whole window.

On every switching-range or farther-out cutoff band, the full-gap or empty-side envelope and the relative gradient bound now give

$$|T|+|\nabla u|^p\le C_M e^{-2pa(L_{\mathrm{cap}}/2-M-1)}\le C_M\rho^{C_0}.$$

The same bound holds for the terminal stress itself. The bounded terminal slopes and the weak Jacobi cutoff identity above bound its positive closure cost per horizontal point by $C_M\rho^{C_0}$. It is $o(\rho)$

because $C_0 > 1$ has a fixed margin. There is one further inactive case: $d \le L_{\mathrm{cap}}$ but $e^{-pad} = o(\rho)$ uniformly on the window. Here $\theta(d) = d/2$. The band is a fixed distance from the midpoint. The normalized turn comparison gives $|\nabla u|^p \le C_M e^{-pad} = o(\rho)$ there, and hence the same negligible Jacobi closure cost. These conclusions concern exactly this prescribed family of caps under the stated preparation and clearance hypotheses.

A coarser top bound can be obtained without a virtual gap. First fix the convex-well threshold used in coercivity and transmission. This fixes the positive energy-decay rate c_0. Later choose a threshold closer to one for the absorption barrier alone. That second choice leaves c_0 unchanged.

Let $p > 1$ and assume $\Delta_p w \ge \lambda w^{p-1}$ on B_T, with $0 < w \le A$. Set $q_* = p/(p-1)$ and

$$F(r) = (r^{q_*} + R_0^{q_*})^{1/q_*}, \qquad Z(r) = A \exp(k(F(r) - F(T))).$$

The identity $|F'|^{p-2}F' = r/F$ gives

$$\Delta_p F = \frac{n - (r/F)^{q_*}}{F} \le \frac{n}{R_0}, \qquad |F'|^p \le 1.$$

Hence

$$\Delta_p Z \le k^{p-1} \left(\frac{n}{R_0} + (p-1)k \right) Z^{p-1}.$$

Take $k < (\lambda/(p-1))^{1/p}$, then choose R_0 sufficiently large. Comparison for $-\Delta_p + \lambda w^{p-1}$ implies $w(0) \le Ce^{-kT}$. At the center, the flux is a bounded C^1 scalar multiple of $x/F(|x|)$. Thus the calculation also holds weakly there for the exponents considered here.

For the matched deficit, the coefficient λ approaches 2^{p-1} uniformly on $\mathcal{P}$ by restricting to the range $u \ge b_1$ with b_1 close to one. At a point at distance at least $L - O(1)$ from all zeros, clearing gives this closer well range on a ball of radius $L - L_{b_1} - O(1)$. The fixed loss in radius changes only the exponential factor. Choose b_1 after c_0. Then one can take $k = k_p$ satisfying

$$pk_p + c_0 > 2pa$$

with a uniform strict margin.

Put $a_* = \min_{p \in \mathcal{P}} a_p$ and $p_+ = \max \mathcal{P}$. Choose $0 < \varepsilon_* < \min\{a_*, c_0/(2p_+)\}$ and then take the closer well so that $k_p = 2a_p - \varepsilon_*$ is allowed. It follows that $pk_p + c_0 \ge 2pa_p + c_0/2$ uniformly.

The lift has energy $O(R^{n-1} e^{-pk_p L})$. Transmission gives the slope error

$$CR^C L^C e^{-(pk_p + c_0)L/2}.$$

For $L = \ell/2 - O(1)$ and $R = L^k$ with a fixed $k > 1$, this is bounded by $C\rho^\gamma$ for some uniform $\gamma > 1/2$, with $\rho = e^{-pa\ell}$. At this stage, the lateral radius must be a power of ℓ. A power of $\log(1/\delta)$ cannot be absorbed into ρ^γ until the relation between δ and ρ is known.

The same estimates hold in a prepared Fermi chart with quantitative metric bounds. Let B be the own sheet's shape operator and assume $|B| \le K$ and $K \max\{L_+, L_-\} \le 1/2$. Relative to the base metric, the tangential metric at depth t and its Jacobian are

$$g_t(v, v) = g_0((I + tB)v, (I + tB)v), \qquad J(t) = \det(I + tB).$$

The eigenvalues of tB have absolute value at most $1/2$. Hence

$$|J(t) - 1| + \|(I + tB)^{-1} - I\| \le CK|t|.$$

The lifted tangential gradient is $(I + tB)^{-1} \nabla_\Sigma$. These formulas give uniform equivalence of the physical natural energy and its coordinate expression, including all cutoff costs. They also show convergence to the product metric on every fixed core. Choose the metric accuracy below the fixed coercivity and Agmon margins. For the pointwise estimates, retain the fixed higher graph bounds already required by the prepared-chamber error budget. On buffered unit charts they give uniform Hölder bounds for the coordinate coefficients. More explicitly, their tangential metric error

is bounded by $C|t|\|B\|_{C^\eta}$; the condition $L\|B\|_{C^\eta} \le C$ supplies the uniform local bound, and the recorded stronger preparation makes this error small where needed.

In the physical stress integral, keep J and the inverse metric. Every difference from the corresponding product-metric factor is bounded by $CK|t|$. The relative-profile bounds give $|T(u)| + |T(v^*)| \le C\sigma^p$ and therefore bound the additional integrated stress error by

$$CK \int_{-L_-}^{L_+} |t|\sigma(t)^p \, dt \le CK.$$

The final bound uses the uniform exponential envelopes on both sides; with only a total mass bound the valid coarser bound is $CK(L_+ + L_-)$. All first moments here are finite with a common bound on $\mathcal{P}$. Thus metric errors are retained explicitly. In the Toda regime above, $K = o(\sqrt{\rho})$ makes this additional flux error $o(\sqrt{\rho})$. Energy transmission and the coarse fixed-angle comparisons use only the stated small metric error and the prepared-chamber bounds.

The well potential gives an explicit improvement of the transmission rate in the established secant framework. Take $w = 1 - u$, $h = 1 - v^*$, $z = w - h$, and

$$-\operatorname{div}(\mathbb{M}\nabla z) + bz = 0, \quad \mathbb{M} = \int_0^1 A_p(\nabla w_\theta)\, d\theta, \quad b = \int_0^1 \mathcal{F}_p'(w_\theta)\, d\theta, \quad w_\theta = (1-\theta)h + \theta w.$$

Here $\mathcal{F}_p(w) = (1-w)(2-w)^{p-1}w^{p-1}$ and $\mu = (w+h)^{p-2}$. The trace of z is zero at the bottom. The fixed bottom strips are prepared, and the upper region lies in a convex well. For the artificial deficit, it suffices to have $|\nabla h| \le 2a(1+o(1))h$. Modica gives $|\nabla w| \le 2aw$ for the actual deficit.

Lemma 4.4 (Pointwise well absorption and Agmon rate)**.** *Fix a compact $\mathcal{P} \Subset [2,\infty)$ and $\eta > 0$. By taking the well sufficiently close to one, and the artificial phase sufficiently prepared, throughout that well*

$$b \ge (1-\eta)4(p-1)a_p^2\lambda_{\max}(\mathbb{M}). \tag{4.36}$$

Consequently every rate $k_p < 2a_p\sqrt{p-1}$, uniformly separated from this endpoint on $\mathcal{P}$, is an admissible Agmon rate for the top-data solution in the prepared half strip.

Proof. At $p = 2$, use the convention $A_2 = I$ in the calculation below. For w_θ in the chosen well,

$$|\nabla w_\theta| \le 2a(1+o(1))w_\theta, \qquad \mathcal{F}_p'(w_\theta) \ge (1-o(1))(p-1)2^{p-1}w_\theta^{p-2}.$$

Since $\lambda_{\max}(A_p(\xi)) = (p-1)|\xi|^{p-2}$, we obtain

$$\lambda_{\max}(\mathbb{M}) \le (p-1)(2a)^{p-2}(1+o(1)) \int_0^1 w_\theta^{p-2}\, d\theta.$$

The ratio of the constants is $2^{p-1}/(2a)^{p-2} = 2a^{2-p} = 4(p-1)a^2$. This proves (4.36).

Fix S large. Choose the IMS core cutoff supported in $s < 2S$, with the well cutoff zero for $s < S$. Take a smooth exponent φ. It equals $k_p(L-s)$ above $2S+1$, is constant below $2S$, and obeys $|\varphi'| \le k_p$.

The constant value of φ on the bottom core is $k_pL + O_{S,\mathcal{P}}(1)$. To construct it, prescribe a smooth derivative between 0 and $-k_p$ on $[2S, 2S+1]$, and choose the constant to match $k_p(L-s)$ at $2S+1$. Thus $e^{2\varphi} \ge C_{S,\mathcal{P}}^{-1}e^{2k_pL}$ on $0 \le s \le S$. This explains the precise factor e^{-2k_pL} in the final weighted-energy estimate; the fixed loss in depth changes only its multiplicative constant.

Consider the modified form

$$Q_{\mathrm{sec}}(F) - \int \mathbb{M}[\nabla\varphi, \nabla\varphi]F^2$$

For the chosen rate, first take $0 < \eta_1 < \min(\eta, 1/2)$ so small that

$$k_p^2 \le (1-2\eta_1)4(p-1)a_p^2 \qquad (p \in \mathcal{P}).$$

Apply (4.36) with η_1. The modified well potential then satisfies

$$b - \mathbb{M}[\nabla\varphi, \nabla\varphi] \geq \frac{\eta_1}{1-\eta_1} b \geq c_{\eta_1}\mu.$$

Repeat the IMS argument proving (4.24). On the fixed core, $\varphi' = 0$, so its positive Dirichlet gap remains. The IMS error is at most $C/S^2 \int \mu F^2$. Choose S large enough to absorb it in the core and well margins. Then fix the required coefficient accuracy on that core. This gives coercivity in the full natural norm $\int(\mathbb{M}[\nabla F, \nabla F] + \mu F^2)$. The small metric errors from flattening are absorbed in the same strict margin.

To justify the lifting step, write $z_{\mathrm{top}} = F + H$, where F is the fixed-width top lift and H has zero trace on every face. Test the equation for H with $e^{2\varphi}H$. For each fixed strip, multiplication by this smooth bounded function preserves the zero-boundary form domain, since $\mathbb{M} \leq C\mu I$. The exact identity is

$$Q_{\mathrm{sec}}(e^{\varphi}H) - \int e^{2\varphi}\mathbb{M}[\nabla\varphi, \nabla\varphi]H^2 = -Q_{\mathrm{sec}}(F, e^{2\varphi}H).$$

On the support of F, both φ and $\nabla\varphi$ are bounded. The right side is consequently bounded by $C\mathcal{E}_{\mathrm{top}}^{1/2}$ times the natural norm of $e^{\varphi}H$. The modified coercivity and Young's inequality give a bound $C\mathcal{E}_{\mathrm{top}}$ for its squared natural norm. Adding back F, and expanding $\nabla(e^{\varphi}z_{\mathrm{top}})$, gives

$$\int e^{2\varphi}(\mathbb{M}[\nabla z_{\mathrm{top}}, \nabla z_{\mathrm{top}}] + \mu z_{\mathrm{top}}^2) \leq C\mathcal{E}_{\mathrm{top}}.$$

All lifting terms remain in the fixed top band. The energy in a fixed bottom strip is therefore at most $C\mathcal{E}_{\mathrm{top}}e^{-2k_pL}$. The fixed bottom core is uniformly elliptic and has the prepared coefficient bounds used in (4.29). The same boundary estimate gives the slope bound $C\mathcal{E}_{\mathrm{top}}^{1/2}e^{-k_pL}$. For the lateral piece, take the tangential radius to be a fixed power of L greater than one. Its exponential tangential decay then absorbs $e^{O(L)}$. □

Suppose the midpoint deficit satisfies $B \asymp e^{-2aL}$. The top lift then has energy $C\,\mathrm{poly}(L)B^p$. With $\rho = e^{-2paL}$, the bottom-slope error is bounded by $C\rho^{\gamma}$ for every exponent satisfying

$$\gamma < \frac{1}{2} + \frac{\sqrt{p-1}}{p}. \tag{4.37}$$

Indeed, if $\mathcal{E}_{\mathrm{top}} \leq CL^N e^{-2paL}$, then

$$\mathcal{E}_{\mathrm{top}}^{1/2}e^{-k_pL} \leq CL^{N/2}e^{-(pa+k_p)L} = CL^{N/2}\rho^{1/2+k_p/(2pa)}.$$

For any smaller exponent γ, the remaining positive power of ρ absorbs $L^{N/2}$ because $\rho = e^{-2paL}$. Letting the admissible rate approach $2a\sqrt{p-1}$ gives (4.37).

The strict margin absorbs polynomial factors. Rates and constants are uniform for p in a compact interval. Thus for any $\mathcal{P} \Subset (2, \infty)$, fix $1/2 < \gamma < \min_{p\in\mathcal{P}}(1/2 + \sqrt{p-1}/p)$. Choose the well and preparation thresholds after this margin. If radial clearing gives $2a - \varepsilon$ rather than $2a$, reduce γ slightly. The exact exponent one, needed later for the upper interaction comparison, still requires a separate argument.

4.2. **Toda stability projection.** The pressure profiles also provide weighted connectors for projecting stability onto the layers. Projecting ambient stability onto one amplitude per layer is part of the Wang–Wei reduction to a stable Toda system [47]. We compute the weighted connector and its core boundary flux explicitly for the matched equation.

The aim is to turn the ambient second variation into a quadratic form for one amplitude on each layer. The essential calculation is one-dimensional: the positive energy spent in a middle connector must be combined with the boundary flux coming from the two cores. After that flux is included, a negative multiple of the squared difference of the two amplitudes remains.

For this connector calculation, let $p > 1$ and set

$$c = p - 1, \qquad a = [2c]^{-1/p}, \qquad k = 2a, \qquad B = 2^{p-1}, \qquad \Gamma_p = \frac{B}{p}, \qquad \beta = pa. \tag{4.38}$$

Let $V = V_p$ be the even positive solution of

$$(|V'|^{p-2}V')' = BV^{p-1}, \qquad V(0) = 1, \quad V'(0) = 0. \tag{4.39}$$

Uniqueness follows in flux variables. The first integral is

$$c|V'|^p = B(V^p - 1), \qquad V'(t) = k(V(t)^p - 1)^{1/p} \quad (t > 0). \tag{4.40}$$

Equivalently,

$$t = k^{-1} \int_1^{V(t)} (z^p - 1)^{-1/p}\, dz \quad (t \geq 0).$$

The integrand is comparable to $(z-1)^{-1/p}$ at one and to z^{-1} at infinity. The first singularity is integrable because $p > 1$, and the second integral diverges. The strictly increasing integral therefore has one inverse for every $t \geq 0$. The resulting solution has $V'(0) = 0$. Its odd flux has derivative BV^{p-1} across zero, so even reflection gives a weak solution on the whole line. The same separation formula applies to every positive solution with these initial values, proving uniqueness.

Its logarithmic derivative satisfies

$$\frac{V'(t)}{V(t)} = k(1 - V(t)^{-p})^{1/p} \uparrow k \qquad (t \to +\infty). \tag{4.41}$$

The normalized normal Jacobi operator is

$$\mathcal{L}_p z = -(Az')' + Cz, \qquad A(t) = c|V'(t)|^{p-2}, \qquad C(t) = cBV(t)^{p-2}.$$

Both V and V' solve $\mathcal{L}_p z = 0$ on $\mathbb{R}$ in the weak sense. Indeed, the derivative of $AV' = c|V'|^{p-2}V'$ is $cBV^{p-1} = CV$, while

$$AV'' = BV^{p-1}, \qquad (AV'')' = CV'.$$

The flux limits agree across the turn. Hence there is no point source. We will use the exact Wronskian identity

$$A(VV'' - (V')^2) = BV^p - c|V'|^p = B. \tag{4.42}$$

We derive the turn bounds before using them in the weighted form. For $t \geq 0$, integrate (4.39) from zero and set

$$Y_p(t) = |V_p'(t)|^{p-2}V_p'(t) = B \int_0^t V_p(s)^{p-1}\, ds.$$

On a compact exponent interval in $(1, \infty)$, the separated formula gives a common $t_0 > 0$ such that $1 \leq V_p(t) \leq 2$ for $0 \leq t \leq t_0$. Indeed, the time at which $V_p = 2$ is positive and continuous in p. Near the lower integration endpoint its integrand has the common integrable bound $C(z-1)^{-1/p_-}$, where $p_- > 1$ is the lower endpoint of the exponent interval. It follows that

$$Bt \leq Y_p(t) \leq B2^{p-1}t, \qquad V_p'(t) = Y_p(t)^{1/(p-1)} \asymp t^{1/(p-1)}.$$

For $t > 0$, differentiation of the flux gives

$$V_p''(t) = \frac{B}{c} V_p(t)^{p-1} |V_p'(t)|^{2-p}.$$

Even reflection gives the corresponding bounds for $t < 0$. All comparison constants are uniform on the compact exponent interval. Thus the turning-point behavior is

$$|V'| \asymp |t|^{1/c}, \qquad V'' \asymp |t|^{-\alpha}, \qquad A \asymp |t|^{\alpha}, \qquad \alpha = \frac{p-2}{p-1} < 1. \tag{4.43}$$

Thus A^{-1} is locally integrable for every $p > 1$. The tangential weight $|V'|^{p-2}$ is locally integrable exactly when $p > 3/2$. For the one-dimensional calculation, use absolutely continuous functions

with finite normal energy $\int(A|z'|^2 + Cz^2)$ and the usual endpoint traces. The integrability of A^{-1} makes these traces compatible across the turn: the weighted Cauchy–Schwarz inequality controls the change of the function on any interval by its energy and the integral of A^{-1} there.

More precisely, weighted Cauchy–Schwarz gives

$$|z(t_2) - z(t_1)|^2 \le \left(\int_{t_1}^{t_2} A|z'|^2\,dt\right)\left(\int_{t_1}^{t_2} A^{-1}\,dt\right).$$

In particular, one cannot choose two unrelated traces on the two sides of the turn in this finite-energy class.

The products defining the Jacobi fluxes are interpreted by their continuous limits at zero. No value of the singular coefficient at that single point is needed when $p < 2$.

Choose $M > 0$ and set

$$V_M = V(M), \qquad d_M = V'(M) > 0, \qquad r_M = d_M/V_M.$$

Define

$$k_-(t) = \frac{1}{2}\left(\frac{V(t)}{V_M} - \frac{V'(t)}{d_M}\right), \qquad k_+(t) = \frac{1}{2}\left(\frac{V(t)}{V_M} + \frac{V'(t)}{d_M}\right). \tag{4.44}$$

These functions obey $k_-(-M) = k_+(M) = 1$ and $k_-(M) = k_+(-M) = 0$. They are nonnegative on $[-M, M]$ and solve the weighted Dirichlet problem for $\mathcal{L}_p$. Indeed, the Wronskian identity makes V'/V increase from $-r_M$ to r_M. The formulas in (4.44) are therefore nonnegative. The positive potential gives uniqueness in the finite-energy class with fixed endpoint values.

Lemma 4.5 (Exact interaction left after the core boundary flux)**.** *For arbitrary real numbers ζ_L, ζ_R, let*

$$\Phi(t) = d_M\zeta_L k_-(t) - d_M\zeta_R k_+(t).$$

Then

$$\int_{-M}^{M}(A|\Phi'|^2 + C\Phi^2)\,dt - A(M)d_M V''(M)(\zeta_L^2 + \zeta_R^2)$$
$$= -\frac{B}{2}r_M(\zeta_R - \zeta_L)^2 = -\lambda_p(M)\beta\Gamma_p(\zeta_R - \zeta_L)^2, \tag{4.45}$$

where

$$\lambda_p(M) = \frac{r_M}{k} = (1 - V_p(M)^{-p})^{1/p} \uparrow 1. \tag{4.46}$$

This identity holds for every $p > 1$ as a one-dimensional weighted identity. Tangentially varying connectors have finite energy when $p > 3/2$.

Proof. Put $\Phi = Vw$. The ground-state formula becomes

$$\int_{-M}^{M}(A|\Phi'|^2 + C\Phi^2) = \int_{-M}^{M} AV^2|w'|^2 + [AVV'w^2]_{-M}^{M}. \tag{4.47}$$

First integrate away from the turn, then pass to it. The boundary term disappears because $AVV' \to 0$, and (4.43) gives integrability. In particular, the derivative term near zero is bounded by $C|t|^{-\alpha}$, which is integrable since $\alpha < 1$. Thus this step uses the normal weighted energy for every $p > 1$. By (4.42),

$$\left(\frac{V'}{V}\right)' = \frac{B}{AV^2}, \qquad \int_{-M}^{M}\frac{dt}{AV^2} = \frac{2r_M}{B}.$$

Weighted Cauchy–Schwarz shows that, with fixed endpoint values, the least possible kinetic energy is

$$\frac{(w(M) - w(-M))^2}{\int_{-M}^{M}(AV^2)^{-1}\,dt}.$$

Equality holds exactly when w' is a constant multiple of $(AV^2)^{-1}$.

The minimizing w is therefore affine in V'/V. This is exactly the pair (4.44), with endpoint data $r_M\zeta_L$ and $-r_M\zeta_R$. Its kinetic contribution in (4.47) is

$$\frac{Br_M}{2}(\zeta_R+\zeta_L)^2,$$

while the boundary contribution is $A(M)d_M^3/V_M\,(\zeta_L^2+\zeta_R^2)$. Subtract the core boundary flux. Using $A(M)(V_MV''(M)-d_M^2)=B$ gives

$$\frac{Br_M}{2}(\zeta_R+\zeta_L)^2-Br_M(\zeta_R^2+\zeta_L^2),$$

which is (4.45), since $\beta\Gamma_p=(pa)(B/p)=Bk/2$. □

Given $0<\theta<1$, impose

$$M\ge\frac{1}{2a}\int_1^{(1-\theta^p)^{-1/p}}(z^p-1)^{-1/p}\,dz. \tag{4.48}$$

This ensures $\lambda_p(M)\ge\theta$.

By the formula for $\lambda_p(M)$, this condition is equivalent to $V_p(M)\ge(1-\theta^p)^{-1/p}$. The separated integral for V_p gives exactly the displayed window choice. On a compact exponent interval its integrable singularity at one has a common majorant. The endpoint and the factor $1/(2a_p)$ also vary continuously.

The required bound is finite and locally uniform in $p>1$. Thus a single window M works on a compact exponent interval once θ is fixed.

The one-dimensional formulas can be checked directly at $p=2$. In that case

$$a_2=\frac{1}{\sqrt{2}},\qquad V_2(t)=\cosh(\sqrt{2}\,t).$$

Its tail coefficient and logarithmic correction are

$$C_2^\infty:=\lim_{t\to\infty}e^{-\sqrt{2}t}V_2(t)=\frac{1}{2},\qquad C_2:=-\log C_2^\infty=\log 2.$$

For the pressure check, consider the positive one-dimensional arch of $u''=u^3-u$ between two consecutive zeros. Let $P=W_2(u)-|u'|^2/2\in(0,1/4)$ be its conserved stress. Let $\ell(P)$ be the distance between the zeros and let $q_m=1-u_{\max}^2=2\sqrt{P}$ be its value of $1-u^2$ at the turn. The first integral and the substitution $q=1-u^2$ give

$$\frac{\ell(P)}{\sqrt{2}}=\int_{q_m}^1\frac{dq}{\sqrt{1-q}\sqrt{q^2-q_m^2}}.$$

As $P\downarrow 0$, subtract the reference integral

$$\int_{q_m}^1\frac{dq}{q\sqrt{1-q}}=2\,\operatorname{arctanh}\sqrt{1-q_m}=\log(4/q_m)+O(q_m).$$

After setting $q=q_mt$, the difference tends to

$$\int_1^\infty\Big((t^2-1)^{-1/2}-t^{-1}\Big)\,dt=\lim_{T\to\infty}(\operatorname{arcosh}T-\log T)=\log 2.$$

To justify this limit, the difference integrand is bounded by $C(t-1)^{-1/2}$ near one and by Ct^{-3} at infinity. On $t\le(2q_m)^{-1}$ the remaining factor $(1-q_mt)^{-1/2}$ is bounded. The part with $(2q_m)^{-1}\le t\le q_m^{-1}$ is $O(q_m^2)$ after setting $v=q_mt$. Consequently

$$\frac{\ell(P)}{\sqrt{2}}=\log(8/q_m)+o(1),\qquad q_m=(8+o(1))e^{-\ell(P)/\sqrt{2}}.$$

Since $P = q_m^2/4$, the one-dimensional pressure coefficient is

$$\Pi_2 := \lim_{P\downarrow 0} Pe^{\sqrt{2}\ell(P)} = 16, \qquad P = (16 + o(1))e^{-\sqrt{2}\ell(P)}.$$

Also

$$\sigma_2 = \int_{\mathbb{R}} \frac{1}{2}\operatorname{sech}^4(s/\sqrt{2})\,ds = \frac{2\sqrt{2}}{3}, \qquad \lambda_2(M) = \tanh(\sqrt{2}M).$$

The integral for σ_2 follows by setting $z = \tanh(s/\sqrt{2})$ and integrating $1 - z^2$ on $[-1, 1]$. Since $\Gamma_2 = 1$ and $\beta_2 = \sqrt{2}$, the right side of (4.45) becomes $-\sqrt{2}\tanh(\sqrt{2}M)(\zeta_R - \zeta_L)^2$. This computes the pressure, kinetic, and connector normalizations from the same one-dimensional equation.

Lemma 4.6 (Recovery by regular connectors)**.** *For $p > 3/2$, the bases $k_\pm$ can be approximated by Lipschitz bases $k_{\pm,\varepsilon}$, equal to the original bases near $\pm M$, in the norm*

$$\int_{-M}^{M} \left(A|k'_{\pm,\varepsilon} - k'_\pm|^2 + (C + |V'|^{p-2})|k_{\pm,\varepsilon} - k_\pm|^2\right) dt.$$

The norm tends to zero as $\varepsilon \downarrow 0$.

Proof. Use the linear interpolant d_ε of V' from Lemma 2.7, and keep V unchanged. The turn bounds in (4.43) verify that lemma's hypotheses. It gives

$$\int A|d'_\varepsilon - V''|^2 \le C\varepsilon^{1-\alpha}, \qquad \|d_\varepsilon - V'\|_\infty \le C\varepsilon^{1-\alpha},$$

with the change supported in $[-\varepsilon, \varepsilon]$. The extra point here is convergence in the tangential weight, which can be singular when $3/2 < p < 2$. Since $C(t)$ is bounded near zero and $|V'|^{p-2} \asymp |t|^\alpha$,

$$\int C(t)|d_\varepsilon - V'|^2\,dt \le C\varepsilon^{2-2\alpha}\int_0^\varepsilon dt \le C\varepsilon^{3-2\alpha},$$
$$\int |V'|^{p-2}|d_\varepsilon - V'|^2\,dt \le C\varepsilon^{2-2\alpha}\int_0^\varepsilon t^\alpha\,dt \le C\varepsilon^{3-\alpha}.$$

Here $-1 < \alpha < 1$, so both exponents are positive and the last integral is finite. Substitute d_ε for V' in (4.44). The fixed factors V_M^{-1} and d_M^{-1} are finite, and the interpolant is unchanged near $\pm M$. This proves the claimed norm convergence with the same endpoint data. On a compact subset of $p > 3/2$, α stays in a compact subinterval of $(-1, 1)$. Thus the denominators $1 - \alpha$ and $1 + \alpha$ in the turn integrals have positive lower bounds. The flux bounds used to prove (4.43) give common comparison constants near zero. Away from zero, the separated formula and its nonzero derivative give continuous dependence of V_p and V'_p on p. In particular, $V_p(M)$ stays finite and positive, and $d_M = V'_p(M)$ has a positive minimum for fixed $M > 0$. All factors in (4.44) and in the preceding recovery estimates are therefore bounded uniformly. This proves uniform recovery on the compact exponent interval.

□

We first transfer the connector calculation to prepared gaps of the actual solution at a fixed $p > 2$. The argument also applies to $p_j \to p_*$ in $\mathcal{P} \Subset (2, \infty)$. In that case every power depending on j uses the actual exponent p_j. Thus the pressure normalization is $\Gamma_{p_j} b_j^{p_j}/\tau_j^2$, and the matrix normalization is $b_j^{2-p_j}\mathbb{A}_{p_j}(\nabla v_j)$. The limiting profile is V_{p_*}. We neither substitute $b_j^{p_*}$ nor assume $b_j^{p_j - p_*} \to 1$.

Take scalar-stable v_j on $\mathbb{R}^{m+1}$ and use coordinates (y, s). For each fixed j, the matched right-hand side has absolute value at most one. The local regularity theorem recalled in Section 1 therefore gives $v_j \in C^{1,\eta_j}_{\mathrm{loc}}$ throughout every core. For $p > 2$, the absolute potential extension W_p is $C^2(\mathbb{R})$. Thus $f = -W'_p$ is C^1 on a neighborhood of the range of v_j. These facts also hold for the actual exponents $p_j \in \mathcal{P}$. Proposition 2.5 now gives the exact weak Jacobi equation for $v_{j,s}$. It

also gives local membership in the weighted form domain. Ordinary H^1 membership is used only in collars of the noncritical matching faces, as in Lemma 2.6.

Choose $\tau_j \downarrow 0$ and put $Y = \tau_j y$. Over the compact horizontal support of the test, select finitely many consecutive cores. Write $s = f_{j,i}(y)$ for their ordered zero graphs. For gap i, between cores $i-1$ and i, let the middle graph be $s = m_{j,i}(y)$ and use the window $|s - m_{j,i}(y)| < M$. Choose a bounded open set Ω containing the rescaled test support and put $\Omega_j = \tau_j^{-1}\Omega$. The open midpoint windows and the remaining core domains are

$$\mathcal{W}_{j,i} = \{(y,s) : y \in \Omega_j,\ |s - m_{j,i}(y)| < M\}, \qquad \mathcal{C}_{j,i} = \{(y,s) : y \in \Omega_j,\ s \in \mathcal{C}_{j,i}(y)\}.$$

Here $\mathcal{C}_{j,i}(y)$ is the remaining vertical interval containing $f_{j,i}(y)$. At two paired faces it is exactly

$$\mathcal{C}_{j,i}(y) = (m_{j,i}(y) + M,\ m_{j,i+1}(y) - M).$$

At an unpaired exterior edge, replace the missing endpoint by the chosen terminal face, or by the corresponding infinite endpoint for an unbounded tail. The exterior cutoff below vanishes before that terminal face or at large depth in the tail. Thus the selected vertical region is partitioned into the core intervals and midpoint windows, apart from the terminal portions removed by the cutoff. The full core mass is always integrated over $\mathcal{C}_{j,i}(y)$, not over a midpoint window. The horizontal amplitudes vanish near $\partial\Omega$, so lateral boundary terms on $\partial\Omega_j$ are zero. All faces with a nonzero trace are the paired, prepared matching faces. Assume the following.

(1) The middle slopes tend uniformly to zero. Use deficit $w_j = 1 - v_j$ in positive chambers and $w_j = 1 + v_j$ in negative chambers. Write $b_{j,i}(y) = w_j(y, m_{j,i}(y)) \to 0$. After division by this midpoint value, the profiles converge uniformly with respect to base point to $V_p(t)$ in C^1 on fixed midpoint neighborhoods. Near the faces $t = \pm M$, convergence is in C^2. These are norms in all physical variables. On a neighborhood centered at $(y_0, m_{j,i}(y_0))$, divide by the constant $b_{j,i}(y_0)$. All these convergences are uniform for $Y = \tau_j y_0$ in a fixed compact set.

(2) On rescaled compact sets,

$$\frac{\Gamma_p b_{j,i}(Y/\tau_j)^p}{\tau_j^2} \longrightarrow \mathcal{P}_i(Y), \tag{4.49}$$

locally uniformly in Y, with locally bounded $\mathcal{P}_i \geq 0$. In particular, $b_{j,i}^p \leq C_K \tau_j^2$ on every rescaled compact set K. This is the actual normal-stress normalization. The midpoint gradient tends to zero after normalization, and $W_p(\pm(1-b))/b^p \to \Gamma_p$.

(3) Define

$$\mathbb{A}_p(\xi) = |\xi|^{p-2}(I + (p-2)\xi \otimes \xi/|\xi|^2), \qquad \sigma_p = \int_{\mathbb{R}} |g_p'|^p, \qquad g_p(s) = \tanh(as),$$

Require the full core masses to satisfy

$$\int_{\mathcal{C}_{j,i}(y)} v_{j,s}^2 [\mathbb{A}_p(\nabla v_j)]_{yy}\, ds \longrightarrow \sigma_p I_m. \tag{4.50}$$

Fixed-core convergence to g_p gives this once the tails of $\int |\nabla v_j|^p ds$ are uniformly tight. The convergence and tail tightness here are uniform for $Y = \tau_j y$ in a fixed compact set. The bound

$$v_{j,s}^2 \|[\mathbb{A}_p(\nabla v_j)]_{yy}\| \leq \sqrt{m}\,(p-1)|\nabla v_j|^p$$

controls the discarded tails. Here the norm is Hilbert–Schmidt. The tangential block is a positive semidefinite $m \times m$ matrix. Its eigenvalues are at most $(p-1)|\nabla v_j|^{p-2}$, so the square root of their squared sum is at most $\sqrt{m}\,(p-1)|\nabla v_j|^{p-2}$. Also $v_{j,s}^2 \leq |\nabla v_j|^2$. At a critical point, both sides of the displayed bound are zero because $p > 2$. The dimensional factor is used only in the tail majorant. The fixed-core integrand still converges exactly to $|g_p'|^p I_m$. Core convergence alone is insufficient.

More explicitly, first restrict to a fixed normal interval about the zero graph of core i. On that interval the integrand converges to $|g_p'|^p I_m$. The integral of the displayed majorant outside the interval tends to zero uniformly as the interval is enlarged. Taking the limits in this order proves (4.50).

(4) At every exterior edge with zero limiting interaction, a prepared window or tail closure must cost $o(\tau_j^{2-m})$ for the chosen test.

More precisely, the exterior closure can be measured by its positive Jacobi cutoff energy. Let $\chi_{j,i}$ equal one on every fixed normal neighborhood of its core and near each prepared matching face. Let it vanish near the unpaired exterior face. For each fixed compact horizontal test ζ, require

$$\tau_j^{m-2} \int_{\mathcal{C}_{j,i}} v_{j,s}^2 \zeta(\tau_j y)^2 \mathbb{A}_p(\nabla v_j)[\nabla \chi_{j,i}, \nabla \chi_{j,i}] \longrightarrow 0. \tag{4.51}$$

The empty-side Jacobi cutoff estimate above supplies this cost in the applications. Its nonnegativity is needed to control mixed cutoff terms; cancellation in the full quadratic form is not used.

The coefficients on fixed midpoint cylinders converge to

$$b_j^{2-p} \mathbb{A}_p(\nabla v_j) \longrightarrow |V'|^{p-2} \begin{pmatrix} I_m & 0 \\ 0 & c \end{pmatrix}, \qquad b_j^{2-p} W_p''(v_j) \longrightarrow cBV^{p-2}. \tag{4.52}$$

The potential limit is uniform because

$$W_p''(u) = (1-u^2)^{p-2}[(2p-1)u^2 - 1], \qquad (1 - v_j^2)/b_j \to 2V.$$

The matrix limit remains uniform for $p_j \to p_* \in (2, \infty)$. Indeed, $(p, \xi) \mapsto A_p(\xi)$ is jointly continuous on compact sets where $p > 2$, also at $\xi = 0$. Throughout this passage, the powers use p_j. No estimate on $b_j^{p_j - p_*}$ enters.

Let

$$Q_u(\phi) = \int \left(\mathbb{A}_p(\nabla u)[\nabla \phi, \nabla \phi] + W_p''(u)\phi^2 \right).$$

Proposition 2.5 says that $z = u_s$ solves the exact weak Jacobi equation, including at critical points. On a core $\mathcal{C}$ this gives

$$Q_{u,\mathcal{C}}(z\zeta) = \int_{\mathcal{C}} z^2 \mathbb{A}_p(\nabla u)[\nabla \zeta, \nabla \zeta] + \int_{\partial \mathcal{C}} z\zeta^2 \mathbb{A}_p(\nabla u) \nabla z \cdot n. \tag{4.53}$$

This is the core identity in Lemma 2.6, applied to $z = u_s$. The tangential test support is compact, and the matching faces used below are noncritical. Thus the required regular collars are available, even if the core contains critical points. The lemma retains exactly the outward flux in (4.53). No profile approximation enters this identity.

On core i use $v_{j,s}\zeta_i(\tau_j y)$. In a positive gap, impose the actual connector traces

$$\begin{aligned} d_j^-(y) &= v_{j,s}(y, m_j - M)\zeta_L(\tau_j y), \\ d_j^+(y) &= v_{j,s}(y, m_j + M)\zeta_R(\tau_j y), \\ \phi_{j,\varepsilon}(y, m_j + t) &= d_j^-(y) k_{-,\varepsilon}(t) + d_j^+(y) k_{+,\varepsilon}(t). \end{aligned} \tag{4.54}$$

The traces agree exactly at both faces. Reversing the phase changes both signs, so every quadratic expression stays unchanged. The limiting face data are

$$v_{j,s}(m_j - M)/b_j \to d_M, \qquad v_{j,s}(m_j + M)/b_j \to -d_M, \qquad \nabla v_{j,s}(m_j \pm M)/b_j \to -V''(M)e_s.$$

Thus the two outward core fluxes sum to

$$-b_j^p A(M) d_M V''(M)(\zeta_L^2 + \zeta_R^2) + o(b_j^p). \tag{4.55}$$

At a graph face, conormal times area equals $\pm(-\nabla m_j, 1)dy$. The error above includes its small slope correction. For a positive chamber, the left core has outward conormal in the positive s

direction. Its Jacobi value is $b_j d_M + o(b_j)$ and its outward flux is $-b_j^{p-1} A(M)V''(M) + o(b_j^{p-1})$. At the right core both of these signs reverse. Their products therefore have the same negative sign in (4.55).

Fix the regularization ε. The chain rule for each face trace contains $\nabla_y v_{j,s} + v_{j,ss}\nabla m_j$, evaluated at that face. After division by b_j, its first term tends to zero and its second does so because the middle slopes vanish. At fixed physical s, differentiation of the connector also gives the term $-\nabla m_j(d_j^- k'_{-,\varepsilon} + d_j^+ k'_{+,\varepsilon})$. The regularized basis derivatives are bounded for fixed ε. Thus

$$\frac{|\nabla_y \phi_{j,\varepsilon}|}{b_j} \le o_j(1)(|\zeta_L| + |\zeta_R|) + C_{M,\varepsilon}\tau_j(|\nabla\zeta_L| + |\nabla\zeta_R|). \tag{4.56}$$

All errors here are uniform on the chosen rescaled compact support. No derivative of b_j is required, since the traces are the actual Jacobi values. All constants are finite for fixed ε. Bounds (4.52) and (4.56) make the scaled tangential and mixed window terms vanish. The remaining normal and potential terms tend to the regularized model form of Lemma 4.5. Only after this sequence limit do we take $\varepsilon \downarrow 0$, using Lemma 4.6.

Here is an explicit estimate for the error on the growing physical support. Fix M, ε, and a rescaled compact set K. Let $\eta_j \to 0$ dominate the uniform errors in the normalized window coefficients, face traces and their first derivatives, and the middle slope on this set. The normalized matrix has a block-diagonal limit, so its mixed block is $O(\eta_j)$. Write

$$G_0 = \zeta_L^2 + \zeta_R^2, \qquad G_1 = |\nabla\zeta_L|^2 + |\nabla\zeta_R|^2.$$

Let $q_{j,\varepsilon}(y)$ be the physical energy density of this window, together with its two outward core fluxes, after integration in the normal variable. Let $q_{M,\varepsilon}(Y)$ be the corresponding model density obtained from $d_M\zeta_L(Y)k_{-,\varepsilon} - d_M\zeta_R(Y)k_{+,\varepsilon}$ and the model fluxes. The coefficient and trace estimates give

$$|q_{j,\varepsilon}(y) - b_j(y)^p q_{M,\varepsilon}(\tau_j y)| \le C_{M,\varepsilon} b_j(y)^p (\eta_j G_0(\tau_j y) + \tau_j^2 G_1(\tau_j y)).$$

For the tangential square, use (4.56). For the mixed term, use its extra coefficient $O(\eta_j)$ and $2ab \le a^2 + b^2$. The normal and potential terms, and the face fluxes, have error $C_{M,\varepsilon}\eta_j b_j^p G_0$ directly. Since $b_j^p/\tau_j^2 \le C_K$, multiplication by τ_j^{m-2} and the change of variables $Y = \tau_j y$ bound the integrated error by

$$C_{K,M,\varepsilon}\left(\eta_j \int_K G_0 + \tau_j^2 \int_K G_1\right) \longrightarrow 0.$$

Thus the errors do not accumulate as the physical support expands.

Theorem 4.7 (Fixed-exponent Toda stability projection)**.** *For $p > 2$, under the prepared hypotheses above, for every finite family $\zeta_i \in C_c^\infty(\mathbb{R}^m)$, with zero exterior amplitudes where its support ends,*

$$\sigma_p \sum_i \int |\nabla\zeta_i|^2 \ge \lambda_p(M)\beta \sum_i \int \mathcal{P}_i(\zeta_i - \zeta_{i-1})^2. \tag{4.57}$$

If preparation holds for every fixed M, then $M \to \infty$ gives the full coefficient $\lambda = 1$. One fixed M chosen by (4.48) retains any prescribed $\lambda < 1$.

Proof. Fix the finitely many amplitudes and then $\varepsilon > 0$. Include every gap incident to a node with nonzero amplitude. At a finite neighboring core outside the support, assign amplitude zero; the test can then terminate inside that core. An actual exterior end is closed by the assumed negligible tail or prepared-window construction. This adds only finitely many pieces. Use $v_{j,s}\zeta_i(\tau_j y)$ on each core and (4.54) on each midpoint window. Their traces match. Proposition 2.5 places the core pieces in the closed weighted domain. The regularized connectors are Lipschitz and have locally Lipschitz face data. For every sufficiently large j the matching faces are noncritical: their normalized normal derivatives converge to $\pm d_M \neq 0$. Thus the core pieces are ordinary H^1 in collars of these faces. Lemma 2.6 puts the assembled compact test in the weighted form domain and supplies smooth

approximation in that domain. Weighted density extends scalar stability to these tests. Apply it together with the core identity (4.53). On noncritical faces the fluxes are classical. This directly gives a nonnegative sum of the core and connector forms.

On a core ending at a tail cap, use $v_{j,s}\zeta_i(\tau_j y)\chi_{j,i}$ instead, with the closing cutoff from the exterior-closure hypothesis. Apply the exact Jacobi identity to $\zeta_i(\tau_j y)\chi_{j,i}$. Its kinetic term contains the factor $\chi_{j,i}^2$, while the pure cutoff term is exactly the expression in (4.51) after rescaling. Call these two nonnegative rescaled terms K_j and E_j. Cauchy–Schwarz for the positive matrix $\mathbb{A}_p$ bounds the absolute value of the rescaled mixed term by

$$2K_j^{1/2}E_j^{1/2} = o(1),$$

since the core mass bound gives $K_j = O(1)$ and (4.51) gives $E_j \to 0$. The cutoff-weighted core matrix still tends to $\sigma_p I_m$. Its integral is bounded above by the full core matrix. On every fixed normal interval the cutoff is eventually one and the integral converges to the planar value. Enlarge this fixed interval after taking the sequence limit. Positivity of the matrices and (4.50) give the same limit from above and below. Thus the later core formulas apply with $\chi_{j,i}^2$ understood on capped cores.

Rescale the inequality by τ_j^{m-2} and send $j \to \infty$. By (4.50), the core kinetic terms tend to $\sigma_p \sum_i \int |\nabla\zeta_i|^2$. Indeed, the core amplitude depends only on the physical horizontal coordinate, so its gradient is $(\tau_j \nabla_Y \zeta_i, 0)$. The change of variables $Y = \tau_j y$ then gives exactly the matrix integral in (4.50), with no derivative of a moving core boundary. For the windows use (4.55) and the coefficient and tangential bounds. Then take $\varepsilon \downarrow 0$. For each gap, the limiting model expression is $-\lambda_p(M)\beta\Gamma_p(\zeta_R - \zeta_L)^2$ by Lemma 4.5. Its physical multiplier is b_j^p. After the change $Y = \tau_j y$, the rescaled multiplier is $\Gamma_p b_j^p/\tau_j^2$, which tends to $\mathcal{P}_i$ by (4.49). Moving these negative gap terms to the other side of scalar stability gives exactly (4.57). All sums are finite because only finitely many nodes are tested. The assumed outer closure handles missing interactions. The limit order is $j \to \infty$, then $\varepsilon \downarrow 0$, and finally $M \to \infty$ if the full coefficient is desired. □

The same projection applies when $p_j \to p_* \in \mathcal{P} \Subset (2, \infty)$. Keep the actual exponent in every normalization, including $b_j^{2-p_j}\mathbb{A}_{p_j}(\nabla v_j)$ and $\Gamma_{p_j} b_j^{p_j}/\tau_j^2$. At fixed M and recovery accuracy ε, the coefficient, trace, and growing-support bounds above are uniform on $\mathcal{P}$. Pass to the sequence limit first and then send $\varepsilon \downarrow 0$. The recovery estimates are uniform on this compact interval as well. We justify continuity of the full normal integral. Since $g_p'(s) = a_p \operatorname{sech}^2(a_p s)$, the change $t = a_p s$ gives

$$\sigma_p = a_p^{p-1} \int_{\mathbb{R}} \operatorname{sech}^{2p} t \, dt.$$

For this continuity calculation, take any fixed compact subinterval of $(1, \infty)$ and denote its lower endpoint by $p_- > 1$. The prefactor a_p^{p-1} is continuous, and $\operatorname{sech}^{2p} t \le \operatorname{sech}^{2p_-} t$. The latter function is integrable, since it is at most $2^{2p_-} e^{-2p_-|t|}$. Dominated convergence therefore gives $\sigma_{p_j} \to \sigma_{p_*}$, including when $p_* = 2$. The quantities β_{p_j} and $\lambda_{p_j}(M)$ converge by their defining formulas and the preceding fixed-window continuity. Thus one fixed factor $0 < \lambda < \inf_{p\in\mathcal{P}} \lambda_p(M)$ is retained in the limit. The uniform window choice in (4.48) supplies such a common M for every prescribed $0 < \lambda < 1$.

4.3. **The Toda Liouville estimate.** The Liouville estimate must allow the stability factor retained by the projection. Let $\mathcal{I}$ be an interval of $\mathbb{Z}$. It may be finite or infinite in either direction. At node $i \in \mathcal{I}$ write the height as h_i. Edge i connects $i-1$ to i. Put $A_i = 0$ whenever one endpoint is missing. On each present edge take a smooth function $A_i > 0$ on $\mathbb{R}^m$ satisfying

$$-\Delta \log A_i = 2A_i - A_{i-1} - A_{i+1}. \tag{4.58}$$

For a retained factor $\lambda > 0$, assume

$$\lambda \sum_i \int_{\mathbb{R}^m} A_i(\phi_i - \phi_{i-1})^2 \le \sum_i \int_{\mathbb{R}^m} |\nabla \phi_i|^2 \tag{4.59}$$

for compact smooth node tests with finitely many nonzero entries and constraint $\sum_i \phi_i = 0$. Missing entries are zero. Hence every tested sum is finite.

For instance, the height equation

$$\Delta h_i = C\left(e^{-\beta(h_i - h_{i-1})} - e^{-\beta(h_{i+1} - h_i)}\right),$$

implies (4.58) for $A_i = \beta C e^{-\beta(h_i - h_{i-1})}$. In the physical normalization, let $\sigma > 0$ be the tension and $P_i = \Pi e^{-\beta(h_i - h_{i-1})}$ the pressure on edge i. The equations

$$\sigma \Delta h_i = P_i - P_{i+1}, \qquad \sigma \sum_i \int |\nabla \phi_i|^2 \ge \lambda\beta \sum_i \int P_i(\phi_i - \phi_{i-1})^2$$

become exactly (4.58)–(4.59) after setting $A_i = \beta P_i/\sigma$.

Theorem 4.8 (Robust Toda Liouville theorem)**.** *If $m < 8\lambda + 2$, there is no positive edge satisfying (4.58) and (4.59). This statement applies to a finite, one-sided infinite, or doubly infinite chain, and also to disjoint unions of such chains separated by identically zero edges. In particular, in $\mathbb{R}^6$ it suffices that $\lambda > 1/2$.*

At $\lambda = 1$, the threshold $m < 10$ is the one in Wang's infinite-chain theorem [45, Theorem 1.1]. The proof below adapts his adjacent-node and exponential tests [45, Section 2, (2.3)–(2.6)]. We retain the stability factor λ and the neighboring-edge terms. Testing two adjacent nodes is enough. Thus boundary nodes and infinite chains need no separate limit procedure.

Select a positive edge i and set

$$A = A_i, \qquad W = A_{i-1} + A_{i+1} \ge 0, \qquad v = \log A.$$

In the stability inequality (4.59) take $\phi_i = \eta$ and $\phi_{i-1} = -\eta$, with all other components zero. Both endpoints of the selected edge are present, even if it is the first or last edge of a finite chain. The amplitudes have zero sum at every horizontal point, so this pair is admissible under the constraint in (4.59). The selected edge contributes $4A\eta^2$, and its neighboring edges contribute $W\eta^2$. The kinetic term is $2|\nabla\eta|^2$. Divide by two to obtain

$$\lambda \int \left(2A + \frac{W}{2}\right)\eta^2 \le \int |\nabla \eta|^2. \tag{4.60}$$

The edge equation reads

$$-\Delta v = 2A - W. \tag{4.61}$$

Lemma 4.9 (Weighted estimate)**.** *For every $0 < q < 2\lambda$ and every smooth compactly supported real χ,*

$$\begin{aligned}(2\lambda - q)\int A^{2q+1}\chi^2 + \frac{\lambda + q}{2}\int W A^{2q}\chi^2 \\ \le \frac{1}{2}\int A^{2q}(|\nabla\chi|^2 - \chi\Delta\chi).\end{aligned} \tag{4.62}$$

Proof. Positivity $A > 0$ and compact supports justify all tests below. Set

$$I = \int A^{2q+1}\chi^2, \quad J = \int W A^{2q}\chi^2, \quad X = \int A^{2q}|\nabla v|^2\chi^2,$$

$$Y = \int A^{2q}\chi\nabla v\cdot\nabla\chi, \qquad Z = \int A^{2q}|\nabla\chi|^2.$$

Testing (4.61) with $A^{2q}\chi^2$ gives

$$2qX + 2Y = 2I - J. \tag{4.63}$$

Then insert $\eta = A^q\chi$ into (4.60) to get

$$2\lambda I + \frac{\lambda}{2}J \le q^2X + 2qY + Z. \tag{4.64}$$

Multiply (4.63) by $q/2$ to write $q^2X = qI - (q/2)J - qY$. Substitution into (4.64) gives

$$(2\lambda - q)I + \frac{\lambda + q}{2}J \le qY + Z.$$

The coefficient of I is positive because $q < 2\lambda$. Since $\nabla A^{2q} = 2qA^{2q}\nabla v$, integration by parts gives

$$qY = -\frac{1}{2}\int A^{2q}(|\nabla\chi|^2 + \chi\Delta\chi).$$

Substitution proves (4.62) with the stated coefficients. □

Corollary 4.10 (Scale-dependent interaction estimate)**.** *For $0 < q < 2\lambda$,*

$$\int_{B_R(x_0)} A_i^{2q+1} \le C(m, q, \lambda)R^{m-4q-2} \tag{4.65}$$

whenever the equations and stability hold on $B_{2R}(x_0)$. The constant does not depend on the chain or the solution.

Proof. Take $\zeta_R \in C_c^\infty(B_{2R}(x_0))$ with $0 \le \zeta_R \le 1$ and $\zeta_R = 1$ on $B_R(x_0)$. Require $|\nabla\zeta_R| \le C/R$ and $|\Delta\zeta_R| \le C/R^2$. Put $\chi = \zeta_R^k$, choosing the integer k at least $2q + 1$. The cutoff satisfies $|\nabla\chi|^2 - \chi\Delta\chi = k\zeta_R^{2k-2}|\nabla\zeta_R|^2 - k\zeta_R^{2k-1}\Delta\zeta_R \le CR^{-2}\zeta_R^{2k-2}$.

Here the precise factorization is

$$A^{2q}\zeta_R^{2k-2} = (A^{2q+1}\zeta_R^{2k})^{2q/(2q+1)}\zeta_R^{2k/(2q+1)-2}.$$

The last exponent is nonnegative, so that factor is at most one. This explains exactly why $k \ge 2q+1$ is required.

Discard the nonnegative J contribution in (4.62). This yields

$$I_R := \int A^{2q+1}\zeta_R^{2k} \le CR^{-2}\int A^{2q}\zeta_R^{2k-2}.$$

By Hölder's inequality and $k \ge 2q + 1$,

$$\int A^{2q}\zeta_R^{2k-2} \le I_R^{2q/(2q+1)}|B_{2R}|^{1/(2q+1)}.$$

If $I_R = 0$, the conclusion is immediate. Otherwise divide by $I_R^{2q/(2q+1)}$ and raise to the power $2q+1$. This gives $I_R \le CR^{m-2(2q+1)}$ and proves (4.65). □

Proof of Theorem 4.8. The dimension condition is $(m - 2)/4 < 2\lambda$. Since $\lambda > 0$, it permits the choice

$$\max\{0, (m - 2)/4\} < q < 2\lambda.$$

Then $m - 4q - 2 < 0$. For $R \ge 1$, bound the integral on $B_1(x_0)$ by that on $B_R(x_0)$ in (4.65), and take $R \to \infty$. This gives $\int_{B_1(x_0)} A_i^{2q+1} = 0$, contradicting positivity of the selected edge. The calculation involved only this edge and its two neighbors. An absent neighbor contributes zero to W. □

Remark 4.11 (The six-dimensional margin). If the projected factor tends to one, choose $q = 3/2$ once $\lambda > 3/4$. In interface dimension six this gives

$$\int_{B_R} A_i^4 \le CR^{-2}.$$

Thus the exact factor one need not be preserved. Any fixed factor above one half works after a suitable choice of q. Discarding the cross interaction between the two adjacent amplitudes would leave only the potential $A_i + (A_{i-1} + A_{i+1})/2$. Even with retained factor one, the corresponding calculation reaches only $m < 6$. To check this, set the retained factor to one and use that weaker potential in place of (4.60). With the notation of the weighted-estimate proof, the form test becomes

$$I + \frac{1}{2}J \le q^2 X + 2qY + Z.$$

The equation test is still $2qX + 2Y = 2I - J$. Substitution gives

$$(1-q)I + \frac{1+q}{2}J \le qY + Z = \frac{1}{2}\int A^{2q}(|\nabla\chi|^2 - \chi\Delta\chi).$$

A positive coefficient of I now requires $0 < q < 1$. The same cutoff calculation gives exponent $m - 4q - 2$. It is negative for some such q exactly when $m < 6$. The adjacent-node variation is therefore essential in dimension six.

4.4. **The uniform small-defect gap.** We first record a fixed-core fact that also applies to the local assembly below. Fix the ambient dimension and $\mathcal{P} \Subset (2, \infty)$. Given $0 < T < \infty$ and an integer $k \ge 1$, there is a nonnegative modulus $\omega_{T,k}$ with $\omega_{T,k}(0) = 0$ and $\omega_{T,k}(d) \to 0$ as $d \downarrow 0$ with the following property. Let $u : \mathbb{R}^n \to [-1, 1]$ be an entire matched weak solution with $p \in \mathcal{P}$, and let z be a zero with $D(z)$ sufficiently small. Choose an orthogonal map Q with $Qe_n = \nabla u(z)/|\nabla u(z)|$. The rotated solution is smooth on B_T, and

$$\|u(z + Qx) - g_p(x_n)\|_{C^k(B_T)} \le \omega_{T,k}(D(z)).$$

The modulus and threshold are independent of the solution and of the zero. Stability is not needed for this fact.

The claim follows directly from Lemma 2.12. For $d = D(z) < 1$, the zero is regular, so the solution is nonconstant. Remark 2.2 gives $-1 < u < 1$ everywhere. In the centered and rotated coordinates, $U(0) = 0$ and $\nabla U(0) = (1-d)^{1/p} e_n$. Fix T and k, and apply the relative estimates on B_{T+1}, using the Hessian estimate with derivative order $\max\{k-2, 0\}$. The relative estimates make B_{T+1} noncritical. The strict range and the smoothness of the matched reaction on $(-1, 1)$ permit the noncritical bootstrap in Section 2.1. Thus the rotated solution is smooth on B_T. The initial gradient error is $O(d)$, uniformly on $\mathcal{P}$. Integration of the Hessian along segments first gives $\|\nabla U - e_n\|_{C^0(B_T)} \le C_T\sqrt{d}$. A second integration gives $\|U - x_n\|_{C^0(B_T)} \le C_T\sqrt{d}$. For $k \ge 2$, the derivatives of orders 2 through k are bounded by the same relative Hessian estimate. Hence, also for $k = 0, 1$,

$$\|U - x_n\|_{C^k(B_T)} \le C_{T,k}\sqrt{d}.$$

The derivatives of g_p are uniformly bounded on the resulting fixed phase range. The chain rule applied to $u = g_p(U)$ gives the asserted estimate, for example with $\omega_{T,k}(d) = C_{T,k}\sqrt{d}$ below the chosen threshold. At $d = 0$, the equality case of Proposition 2.1 gives the profile exactly. The radius and derivative order are fixed before the threshold. For windows that grow with the sequence, we retain the separate relative and graph estimates below.

Theorem 4.12 (Uniform small-defect gap)**.** *For each compact $\mathcal{P} \Subset (2, \infty)$, there is $\delta_{\mathcal{P}} > 0$ such that every scalar-stable entire solution $u : \mathbb{R}^7 \to (-1, 1)$ satisfying*

$$\sup_{\{u=0\}} \big(1 - 2(p-1)|\nabla u|^p\big) \le \delta_{\mathcal{P}}, \qquad p \in \mathcal{P},$$

is a planar heteroclinic. Take $\delta_{\mathcal{P}} < 1$. The constant zero cannot satisfy the defect bound, and the preceding one-sign clearing result ensures that every nonconstant solution has a zero.

Proof. Argue by contradiction. Take nonplanar stable solutions v_j with $p_j \to p_* \in \mathcal{P}$ and $\delta_j = \sup_{\Sigma_j} D_j \downarrow 0$. Use $m = 6$, $n = 7$, $c_j = p_j - 1$, $a_j = [2c_j]^{-1/p_j}$, $\beta_j = p_j a_j$, and

$$\sigma_j = \int_{\mathbb{R}} |g_{p_j}'|^{p_j} = a_j^{p_j - 1} B(1/2, p_j), \qquad K_j = \sup_{\Sigma_j} |B_j|.$$

The fixed-core preparation above applies uniformly at every zero, since $D_j \le \delta_j$ there. Proposition 3.15 implies

$$K_j \le C\delta_j \log(C/\delta_j), \qquad \kappa_j := K_j + \delta_j \le C\delta_j \log(C/\delta_j). \tag{4.66}$$

On a fixed core, the full ambient derivative of the phase normal is bounded by $C_S \kappa_j$. Graph estimates may use K_j alone. Phase and stress estimates on whole cores use κ_j. The constants are uniform on $\mathcal{P}$.

Orient Σ_j by $\nu_j = \nabla v_j / |\nabla v_j|$. Define the nearest opposing-sheet distance

$$\ell_j = \inf\{|z - z'| : z, z' \in \Sigma_j, \nu_j(z) \cdot \nu_j(z') \le -1/2\},$$

Set it equal to $+\infty$ if no pair exists. Uniform preparation implies $\ell_j \to \infty$ whenever this distance is finite.

Set $r_j = \delta_j^{-1/4}$. All zeros are regular, so Σ_j is a closed smooth embedded hypersurface without boundary. Equation (4.66) gives $K_j r_j \to 0$. Lemma 3.10 therefore gives tangent graph disks of radius at least r_j, with its slope, height, and Hessian bounds. The entire zero set supplies every required ambient buffer. The same lemma gives, for curvature bound K, buffered radius r, and base separation d,

$$\theta \le C(d/r + Kr). \tag{4.67}$$

This estimate covers both meeting and disjoint local patches. Apply it with $r = r_j$ and $d = O(R+L)$. Choose $L = A \log(1/\delta_j)$ and $R = L^q$, with fixed $q > 1$.

Then the zero branches in the smaller cylinder form ordered graphs. Its oscillation over B_R is $o(1)$: the relative tilt is $O((R+L)/r_j + K_j r_j)$ and the graph oscillation is bounded by this quantity times R, plus $CK_j R^2$. Each term tends to zero because R and L are fixed powers of $\log(1/\delta_j)$, whereas $r_j = \delta_j^{-1/4}$ and (4.66) holds. Consecutive orientations alternate. Each branch reaches the central axis, and local regularity rules out accumulation at finite height. None of this requires a bound on the number of branches.

Choose a zero with $D_j \ge \delta_j/2$. Suppose neither adjacent graph lies within vertical distance $2L$. On either side of the base graph there is then a zero-free chamber of depth L and lateral radius R. Beyond a fixed core, clearing puts both chambers in the convex well. Compare with the two artificial finite-slab layers. Their bottom slopes sum to $2a_j + O(K_j^2 L R^2)$ by (4.5). This geometric error is $o(\delta_j)$, by (4.66) and the logarithmic choices of L, R. Half-strip transmission with generic top data bounds the difference between the actual and artificial bottom slopes by $CR^C L^C e^{-cL}$, as in (4.29). Its constants are fixed before A. Taking A large makes this error $o(\delta_j)$ as well. At the chosen zero, however, the actual sum is $2a_j(1 - D_j)^{1/p_j} = 2a_j - \Theta(\delta_j)$. This is a contradiction. Hence

$$\ell_j < \infty, \qquad \ell_j \le C \log(1/\delta_j). \tag{4.68}$$

Define

$$\rho_j = e^{-\beta_j \ell_j}.$$

We must show that curvature is negligible on this scale. The defect and ρ_j cannot simply be identified.

Use Lemma 3.8. Absorb its polynomial factor into a fixed small loss in the exponential rate. In a sufficiently deep well this gives $w(0) \le Ce^{-k_p R}$. The rate k_p can be chosen arbitrarily close

to $2a_p$ from below, uniformly on $\mathcal{P}$. First fix the convex-well threshold and its transmission gap. Only then choose the closer threshold that fixes k_p. Increasing k_p in this order does not change the transmission constant.

At any zero, translate its base to the origin and rotate its phase normal into the vertical direction. Take $L = \ell_j/2 - C_{\mathrm{top}}$, where $C_{\mathrm{top}} \geq 2$ is fixed, and $R = L^q$, with fixed $q > 1$. We check clearance on the whole slab top, including a fixed-width collar. Only graph pieces meeting the buffered comparison box are needed.

Use the earlier buffered radius $r_j = \delta_j^{-1/4}$. By (4.66) and (4.68), every zero in $B^m_{2R} \times (-4L, 4L)$ has a graph piece extending across B^m_{3R}. Its unoriented tilt from the horizontal plane is at most

$$s_j = C((R+L)/r_j + K_j r_j).$$

Indeed, its base is at distance $O(R+L)$ from the chosen zero, so (4.67) applies. The graph radius is much larger than R, so the radial-lift construction reaches the whole displayed disk before restriction. Its height differs from its value on the central axis by at most

$$\varepsilon_j^{\mathrm{osc}} = C(s_j R + K_j R^2) \longrightarrow 0.$$

All three terms tend to zero because $L \leq C\log(1/\delta_j)$, $R = L^q$, $r_j = \delta_j^{-1/4}$, and $K_j \leq C\delta_j \log(C/\delta_j)$. Thus this estimate does not use (4.69), which we are proving.

Identify graph pieces that coincide. The remaining pieces are ordered on their common disk. Every axis zero in $(-\ell_j, \ell_j)$ is represented, since $\ell_j < 3L$ for large j. These zeros are transverse and isolated. Order them on the axis. Consecutive phase orientations alternate. For large j, the small unoriented tilts make the dot product of consecutive oriented normals less than $-1/2$. Their axis points therefore form an opposing pair in the definition of ℓ_j, so their height difference is at least ℓ_j. Since the base height is zero, there can be no other axis zero in $(-\ell_j, \ell_j)$. Thus every other graph in the comparison box has axis height at least ℓ_j or at most $-\ell_j$. No bound on the number of pieces is used.

Write the base graph as $s = h_j(y)$. A point in either unit-width top collar has the form

$$x = (y, h_j(y) + \sigma L + t), \qquad |y| \leq R+1, \quad \sigma \in \{-1, 1\}, \quad |t| \leq 1.$$

If a zero w were at distance less than $L-3$ from x, then $|w'| < 2R$ and $|w_n| < 3L$ for large j. It therefore belongs to one of the graph pieces just described. On the base piece, its vertical distance from x is at least $L - 1 - 2\varepsilon_j^{\mathrm{osc}}$. On another piece, the axis-height separation gives the lower bound

$$\min\{L, \ell_j - L\} - 1 - 2\varepsilon_j^{\mathrm{osc}} \geq L - 1 - 2\varepsilon_j^{\mathrm{osc}}.$$

Euclidean distance is at least this vertical distance. Since $\varepsilon_j^{\mathrm{osc}} \to 0$, both bounds contradict the choice of w. Zeros outside the comparison box cannot be closer, by the same horizontal and vertical distance bounds. Hence

$$\mathrm{dist}(x, \Sigma_j) \geq L - 3$$

uniformly throughout both top collars. The slabs have the signs fixed by the base orientation, since their interiors contain no other zero. Let R_{well} be the clearing radius for the already chosen closer well range. The ball $B_{L-3-R_{\mathrm{well}}}(x)$ remains in that range. The radial absorption barrier therefore gives the actual deficit at x bounded by

$$Ce^{-k_{p_j}(L-3-R_{\mathrm{well}})} \leq C' e^{-k_{p_j} L}.$$

The radius loss is fixed and changes only the constant. On each artificial top face, the phase equals L, so its deficit is

$$1 - \tanh(a_j L) \leq 2e^{-2a_j L} \leq 2e^{-k_{p_j} L}.$$

Its bounded phase gradient gives the same bound on the inner unit-width top collar. Thus the lift energy is at most $CR^m e^{-p_j k L}$. With uniform transmission exponent $c_0 > 0$, the bottom slope error

is bounded by

$$CR^C e^{-(p_j k + c_0)L/2}.$$

Choose the clearing rates $k_p < 2a_p$ so close that $p(2a_p - k_p) < c_0/2$ for every $p \in \mathcal{P}$. This is possible uniformly on the compact interval, after c_0 is fixed. Then $pk_p + c_0 > 2pa_p + c_0/2$. Slightly reducing the resulting exponent absorbs the polynomial factors in ℓ_j. Here the comparison radius is a power of ℓ_j, as required before a relation between δ_j and ρ_j is known. Choose a fixed $1/2 < \gamma < 1$ strictly below $\inf_{p\in\mathcal{P}}(pk_p + c_0)/(4pa_p)$. This is possible because $pk_p + c_0 > 2pa_p$ with a uniform margin. The fixed loss in $L = \ell_j/2 - O(1)$ changes only the constant. The lateral error also lies below $C\rho_j^\gamma$, since R is a power of ℓ_j greater than one. Evaluation at a nearly maximal-defect zero therefore yields

$$\delta_j \le CK_j^2 \log^M(1/\delta_j) + C\rho_j^\gamma.$$

Use (4.66) to absorb the first term, since it is $O(\delta_j^2 \log^{M+2}(C/\delta_j)) = o(\delta_j)$. The function $t \mapsto t\log(C/t)$ is increasing for small t. Thus $\delta_j \le C\rho_j^\gamma$ also gives $\kappa_j \le C\rho_j^\gamma \log(C/\rho_j)$, proving the second assertion below with the same fixed exponent margin. We conclude

$$\delta_j \le C\rho_j^\gamma, \qquad \kappa_j \log^M(1/\rho_j) = o(\sqrt{\rho_j}) \quad \text{for every fixed } M. \tag{4.69}$$

Take an opposing pair whose distance is $\ell_j + o(1)$. Both points lie in the ordered graph region, since (4.68) is logarithmic. Replace the second point by its nearest foot on its local graph disk. Lemma 3.10 gives a unique interior foot, because the selected distance is logarithmic and the graph disk is much larger. The original normal constraint and the vanishing unoriented angle make the oriented scalar product $-1 + o(1)$. The new foot is within intrinsic distance $O(\ell_j)$ of the original second point, so its normal changes by $O(K_j\ell_j) = o(1)$. It remains in the same buffered region and satisfies the strict opposing-normal constraint. The new distance is therefore at most the old one and at least ℓ_j by definition.

The relative unoriented angle is at most $C\sqrt{K_j\ell_j}$. For $K_j > 0$, apply (4.67) at radius comparable to $\sqrt{\ell_j/K_j}$; it is admissible because $K_j\ell_j \to 0$. The case $K_j = 0$ consists of parallel flat pieces and gives angle zero. Move the first point to the origin and make its tangent plane horizontal. The vertical gap at that point differs from the normal-foot distance by $O(K_j\ell_j^2) = o(1)$. Every consecutive local vertical gap is at least ℓ_j, because adjacent orientations are opposite. The selected pair must therefore be adjacent; an intervening graph would give a vertical gap at least $2\ell_j$. Reflect the solution if needed so that this chamber is positive. The selected vertical gap is $\ell_j + o(1)$.

To index the graphs on all rescaled compact sets, enlarge the auxiliary scale. Set $\kappa_j = K_j + \delta_j + \rho_j$, $r_j = (\kappa_j\sqrt{\rho_j})^{-1/2}$, and $H_j = r_j/\ell_j^{M_0}$, with M_0 fixed after all polynomial window powers and sufficiently large. The added ρ_j keeps κ_j positive and preserves (4.69). Indeed, $r_j\sqrt{\rho_j} = (\sqrt{\rho_j}/\kappa_j)^{1/2}$ and $\kappa_j r_j = (\kappa_j/\sqrt{\rho_j})^{1/2}$. Equation (4.69), with arbitrarily large fixed logarithmic powers, shows that $r_j\sqrt{\rho_j}$ and $H_j\sqrt{\rho_j}$ diverge, whereas $\kappa_j r_j \to 0$. Fix

$$T_j = H_j\sqrt{\rho_j}/100 \longrightarrow \infty$$

and use the single horizontal disk $B_{T_j/\sqrt{\rho_j}} = B_{H_j/100}$. It contains every fixed rescaled ball for large j. Graph disks based in $|s| \le H_j$ and $|x| \le H_j/100$ have unoriented angle at most $C(H_j/r_j + \kappa_j r_j)$ from the reference normal, by (4.67). They extend across the whole chosen horizontal disk. Index the branches with axis heights in $[-H_j, H_j]$. Initially apply the equations to nodes in $[-H_j/2, H_j/2]$. Their height oscillation is $o(H_j)$, so the larger interval contains all logarithmic neighbors. Each fixed block joined to the selected edge by bounded normalized gaps has axis height $O_i(\ell_j) = o(H_j)$ and stays in the inner interval. If a fixed-index extension reaches the guard boundary, an earlier gap must be much larger than ℓ_j and is inactive. This construction works for every fixed T and finite index block. The size of the entire family is unrestricted. Every later extraction uses restrictions of this same indexing.

Write the resulting ordered graphs as $s = f_{i,j}(x)$. Their orientations alternate. In the coordinates $y = \sqrt{\rho_j}x$, the physical slopes $\nabla_x f_{i,j}(y/\sqrt{\rho_j})$ tend locally to zero. If $s_j = C(H_j/r_j + \kappa_j r_j)$ is the common slope bound, then $s_j^2\ell_j \to 0$ by (4.69) and the logarithmic margin in H_j. Thus normal and vertical gaps of size $O(\ell_j)$ differ by $o(1)$.

For example, the distance to an affine graph of slope b and vertical separation d is $d/\sqrt{1+|b|^2}$. Replacing the graph by its tangent plane changes this by at most CK_jd^2 on the relevant window. Therefore

$$|d - d^{\perp}| \le Cd\,s_j^2 + CK_jd^2 = o(1) \qquad (d \le C_0\ell_j).$$

It follows that $e^{-\beta_j d^{\perp}}/e^{-\beta_j d} = 1 + o(1)$. Thus the pressure constant is unchanged when the stress comparison is expressed in the common vertical coordinates.

Each consecutive vertical gap $d_{i,j} = f_{i+1,j} - f_{i,j}$ is at least ℓ_j: the normals on adjacent graphs are opposite up to $o(1)$, so their two vertically aligned points form an admissible pair in the definition of ℓ_j.

Consider $d_{i,j} \le C\ell_j$ and rotate at its lower foot. Apply (4.67) with radius comparable to $\sqrt{d_{i,j}/K_j}$ when $K_j > 0$. It bounds the relative tilt by $C\sqrt{K_jd_{i,j}}$. For $K_j = 0$ the pieces are parallel. On a logarithmic disk of radius $R = (C\ell_j)^3$, the upper graph oscillates by at most $C(\sqrt{K_jd_{i,j}}R + K_jR^2) = o(1)$. The lower oscillation is $O(K_jR^2) = o(1)$. Thus the pressure comparison applies and yields

$$P_{i,j} = (\Pi_{p_j} + o(1))e^{-\beta_j d_{i,j}}, \qquad \Pi_p = \frac{4^p}{2p}e^{-\gamma_E - \psi(1-1/p)}, \tag{4.70}$$

uniformly on rescaled compact sets. More explicitly, on each fixed compact set and finite index block there is $\eta_j \to 0$ such that

$$|P_{i,j} - \Pi_{p_j}e^{-\beta_j d_{i,j}}| \le \eta_j e^{-\beta_j d_{i,j}} \le C\eta_j\rho_j$$

whenever $d_{i,j} \le C_0\ell_j$. If uniformity failed, choose violating base points and indices. The same gap hypotheses persist: diverging widths, comparison radii much larger than those widths, vanishing graph oscillations, and uniform core preparation. The pressure lemma then contradicts the violation. Other sheets never enter the estimates. For larger gaps, cap each side at depth $C_0\ell_j$, taking C_0 fixed and large. Use the prescribed cap construction with $\ell_{\min} = \ell_j$, and denote its depth function by θ_j. It has $0 \le \theta_j' \le 1/2$, $\theta_j \le d/2$, and the stated inverse-power bounds on its higher derivatives. The lower top is $f_{i,j} + \theta_j(d_{i,j})$, and the upper bottom is $f_{i+1,j} - \theta_j(d_{i,j})$. Their slopes are $o(1)$. In the transition region, each terminal depth is at least $C_0\ell_j/2$. For $d_{i,j} \le 4C_0\ell_j$, use the full prepared gap and restrict its comparison to this terminal depth. For a larger gap, or a missing exterior neighbor, use the empty-side comparison. Its one-sided clearance holds on the whole buffered polynomial disk. Indeed, $s_j\ell_j^3 + K_j\ell_j^6 = o(1)$ by the choice of M_0 and (4.69). The ordered graphs therefore remain at their axis heights up to $o(1)$ on this window. No other zero enters the chosen chamber before depth $4C_0\ell_j - O(1)$. The guard interval also excludes unindexed graphs from that window. Thus the full-gap or empty-side comparison bounds every terminal stress and unit-band cutoff cost by $O(\rho_j^{cC_0}) = o(\rho_j)$ after increasing C_0. Neither force nor stability differentiates a discontinuous active-gap choice.

Use the stress $T = T[v_j]$ defined in the variable-cap argument, with the actual exponent p_j. Let $s = c_{i,j}^-(x)$ and $s = c_{i,j}^+(x)$ be the lower and upper faces of the cell of zero i, and set

$$J_{i,j}(x) = \int_{c_{i,j}^-}^{c_{i,j}^+} T_{sy}(x,s)\,ds, \qquad F_{i,j}^{\pm} = T_{ss}(x, c_{i,j}^{\pm}) - T_{sy}(x, c_{i,j}^{\pm}) \cdot \nabla c_{i,j}^{\pm}.$$

More explicitly, the identity $\operatorname{div} T = 0$ gives $\operatorname{div}_x T_{sy} = -\partial_s T_{ss}$. Leibniz' rule then gives

$$\operatorname{div}_x J_{i,j} = T_{ss}(c^-) - T_{ss}(c^+) + T_{sy}(c^+) \cdot \nabla c^+ - T_{sy}(c^-) \cdot \nabla c^-.$$

This is exactly the displayed difference of tractions. The calculation is distributional when a cell contains a critical point; it follows by testing the weak stress identity and approximating the two smooth graph faces.

By the definitions of the two tractions, this is

$$\operatorname{div}_x J_{i,j} = F^-_{i,j} - F^+_{i,j}.$$

Use gap midpoints at active boundaries and the caps above at inactive ones. At an active upper face $F^+_{i,j} = -P_{i,j} + o(\rho_j)$, and at an active lower face $F^-_{i,j} = -P_{i-1,j} + o(\rho_j)$. Indeed, the normalized tangential midpoint gradient tends to zero and the face slope is $o(1)$, so its additional traction is $o(e^{-\beta_j d_{i,j}})$. At inactive faces both the force and its error are $o(\rho_j)$. Consequently

$$\operatorname{div}_x J_{i,j} = P_{i,j} - P_{i-1,j} + o(\rho_j)$$

in distributions, with a uniformly bounded scalar error. The two half-strip pointwise flux estimates and artificial phase bounds give

$$J_{i,j} = -\sigma_j \frac{\nabla f_{i,j}}{\sqrt{1 + |\nabla f_{i,j}|^2}} + E_{i,j}, \qquad \|E_{i,j}\|_{L^\infty} = o(\sqrt{\rho_j}). \tag{4.71}$$

uniformly on rescaled compact sets. Along long tails, the gap sandwich or empty-side comparison bounds the actual deficit by a multiple of the artificial one. The reference gradient has a lower bound relative to this deficit. After division by the well weight, the secant matrix is uniformly elliptic with uniform local Hölder coefficients. Schauder estimates for the exponentially decaying natural energy give the integrated pointwise error $o(\sqrt{\rho_j})$. An L^2 bound alone would not suffice. Freeze the tilted planar layer at the foot. For graph slope b, its exact integrated flux is $-\sigma_j b/\sqrt{1 + |b|^2}$. Rotate back to the common vertical chart and integrate the complete stress tensor. Transmission and phase estimates leave the error $O(\kappa_j \log^M(1/\rho_j))$, which is small by (4.69). This retains the nonlinear slope flux and avoids an uncontrolled cubic slope error.

The retained half-depths satisfy $L_+, L_- \le C\ell_j$. Use the local comparison radius $R_{\mathrm{cmp},j} \asymp \ell_j^3$. The finite artificial-flux estimate therefore gives

$$K_j R_{\mathrm{cmp},j}(L_+ + L_-) \le C\kappa_j \ell_j^4 = o(\sqrt{\rho_j})$$

by (4.69). The contraction condition also holds, since $K_j \max\{L_+, L_-\} R^2_{\mathrm{cmp},j} \le C\kappa_j \ell_j^7 \to 0$. The larger radius H_j indexes the sheets and supplies the buffer; it is not the radius of this local comparison.

The errors have the following scales. Write $R_j = R_{\mathrm{cmp},j}$. There are fixed $C, c > 0$ and an integer M independent of j such that

$$\|E_{i,j}\|_\infty \le C\kappa_j \ell_j^M + C\sqrt{\rho_j}\, \ell_j^M e^{-c\ell_j} + C\rho_j + C\ell_j^M e^{-c\ell_j^3}.$$

The four contributions come from curvature of the artificial layer, top-data transmission, profile truncation, and lateral data. Here the curvature term uses the own zero graph and its artificial layer. It does not include the switched-cap term $\theta_j''(d_{i,j})|\nabla d_{i,j}|^2$. That term is zero on active midpoint regions. On its inactive support, the prescribed-cap estimate bounds terminal stress and positive Jacobi closure cost by $O_M(\rho_j^{C_0})$. The cap derivatives remain uniformly bounded, so they preserve this order in the physical traction and cutoff calculations. For the integrated flux, apply (4.35) with the terminal small parameter bounded by $C_M \rho_j^{C_0}$. Its square-root factor and fixed polynomial losses are absorbed by $C\rho_j$ after choosing the already large fixed $C_0 > 2$ with a strict margin. Thus this contribution is included in the truncation term, rather than in $C\kappa_j \ell_j^M$. Divide by $\sqrt{\rho_j}$. Each term tends to zero by (4.69) and $\rho_j = e^{-\beta_j \ell_j}$, because β_j stays bounded above and below. Once C_0 is large, inactive terminal cells have the further bound $C\rho_j^{(1+\theta)/2}$ for a fixed $\theta > 0$. Their stress is $O(\rho_j^{1+\theta})$. The pressure and slanted-boundary errors are $o(\rho_j)$ in physical variables. Hence

divergence errors vanish after division by $\sqrt{\rho_j}$, while scalar errors vanish after division by ρ_j. These are the distinct normalizations needed below.

Subtract the stress identities of consecutive cells and define

$$h_{i,j}(y) = \beta_j[d_{i,j}(y/\sqrt{\rho_j}) - \ell_j] \geq 0, \qquad C_j = \beta_j \Pi_{p_j}/\sigma_j, \qquad A_{i,j} = C_j e^{-h_{i,j}}.$$

On rescaled compact balls, the distributional equation is

$$\operatorname{div}(B_{i,j}\nabla h_{i,j}) = 2A_{i,j} - A_{i-1,j} - A_{i+1,j} + o(1) + \operatorname{div} e_{i,j}, \tag{4.72}$$

Here the matrices have the explicit form

$$B_{i,j} = \int_0^1 DF\big((1-t)\nabla f_{i,j} + t\nabla f_{i+1,j}\big)\, dt, \qquad F(b) = \frac{b}{\sqrt{1+|b|^2}},$$

evaluated at $x = y/\sqrt{\rho_j}$. The eigenvalues of $DF(b)$ lie between $(1+|b|^2)^{-3/2}$ and $(1+|b|^2)^{-1/2}$. Thus $B_{i,j} \to I$ uniformly and these matrices are uniformly elliptic. Before rescaling, subtraction of the two cell equations reads

$$\sigma_j \operatorname{div}_x(B_{i,j}\nabla_x d_{i,j}) = 2P_{i,j} - P_{i-1,j} - P_{i+1,j} + o(\rho_j) + \operatorname{div}_x(E_{i+1,j} - E_{i,j}).$$

This displays both the sign of the interaction and the different forms of the scalar and vector errors. To check the normalization, write $\nabla_x d_{i,j} = (\sqrt{\rho_j}/\beta_j)\nabla_y h_{i,j}$. Also $e^{-\beta_j d_{i,j}} = \rho_j e^{-h_{i,j}}$. Divide the consecutive stress identity by $\sigma_j\rho_j/\beta_j$. This gives exactly the interaction coefficient C_j above and the vector error $e_{i,j} = \beta_j(E_{i+1,j} - E_{i,j})/(\sigma_j\sqrt{\rho_j})$. Thus the vector error tends to zero in local L^∞. The scalar error $o(1)$ is uniform: it is the consecutive difference of the physical force errors, multiplied by $\beta_j/(\sigma_j\rho_j)$. The displayed errors include inactive terminal forces. Under the same local preparation estimates, the calculation also applies when $h_{i,j} \geq -C_{\text{low}}$ for a fixed constant C_{low}. On each active gap, $P_{i,j}/\rho_j = (\Pi_{p_j} + o(1))e^{-h_{i,j}}$. Since $e^{-h_{i,j}} \leq e^{C_{\text{low}}}$, the scalar remainder is still uniform. The flux normalization and ellipticity are unchanged. The Harnack and cutoff arguments below then apply to $h_{i,j} + C_{\text{low}} + 1$.

The factor $\sqrt{\rho_j}$ in the vector normalization is essential: one horizontal derivative transforms as $\operatorname{div}_x E = \sqrt{\rho_j}\operatorname{div}_y E$. A scalar traction error already occurs after taking that derivative, so its normalization is instead ρ_j.

The construction gives $0 \leq A_{i,j} \leq C$ and, on the selected gap, $h_{0,j}(0) \to 0$.

We need pointwise convergence of the selected interaction. On each fixed rescaled ball, (4.72) has a bounded scalar source and a divergence source tending to zero in L^∞. The ellipticity constant is uniform. Inhomogeneous Harnack therefore gives, for fixed T and each gap defined on B_{4T},

$$\sup_{B_T} h_{i,j} \leq C_T\big(\inf_{B_T} h_{i,j} + 1\big).$$

These constants do not depend on the index or the number of sheets. Since $h_{0,j}(0) \to 0$, this bounds $h_{0,j}$ on fixed balls. Interior De Giorgi estimates for bounded divergence data also give a common Hölder exponent and equicontinuity. Apply these bounds on a larger fixed ball before restricting to its interior. At a neighboring index, pass to a subsequence with bounded value at the origin or with value tending to infinity. The first case has the same compactness. In the second case, Harnack forces the infimum on every fixed ball to diverge. Hence $A_{i,j} \to 0$ locally uniformly.

The common guard band contains every fixed finite block linked to the selected edge by bounded normalized gaps. Its total axis height is $O_i(\ell_j) = o(H_j)$. On $B_{T/\sqrt{\rho_j}}$ its oscillation is $o(H_j)$ for fixed T. Thus an index can exit the guard band only beyond an inactive interaction. The explicit outer closure handles that edge. Diagonal extraction over integer radii T and finite blocks gives a connected positive chain containing edge zero. The chain is finite, one-sided infinite, or doubly infinite. No sum over all physical sheets is used.

For every bounded log gap, test (4.72) with $\eta^2 h_{i,j}$, where η is a compactly supported cutoff. Uniform ellipticity and Young's inequality give

$$\int \eta^2 |\nabla h_{i,j}|^2 \le C_\eta (1 + \|h_{i,j}\|^2_{L^\infty(\operatorname{supp}\eta)} + \|e_{i,j}\|^2_{L^\infty(\operatorname{supp}\eta)}).$$

For example, writing the equation as $\operatorname{div}(B\nabla h) = f + \operatorname{div} e$, the tested identity is

$$\int \eta^2 B\nabla h \cdot \nabla h = -\int f\eta^2 h + \int e \cdot \nabla(\eta^2 h) - 2\int \eta h B\nabla h \cdot \nabla \eta.$$

Apply $2ab \le \varepsilon a^2 + \varepsilon^{-1} b^2$ to every term containing $\eta|\nabla h|$, and choose ε smaller than a fixed fraction of the ellipticity constant. This absorbs those gradient terms into the left side and gives the stated bound.

Thus the gradients are bounded in local L^2. Use uniform convergence, weak H^1 convergence, and $B_{i,j} \to I$ to pass to the equation. Inactive neighboring interactions converge uniformly to zero. The limit satisfies

$$-\Delta \log A_i = 2A_i - A_{i-1} - A_{i+1} \quad \text{on } \mathbb{R}^6, \qquad A_i > 0, \tag{4.73}$$

with missing exterior edges interpreted as zero. Elliptic bootstrapping gives smoothness. Moreover, $A_0(0) = \lim C_j > 0$. The selected interaction has survived the limit.

There are two ways for an interaction to disappear. A gap may be larger than the cap scale $C_0\ell_j$. It may also satisfy $d_{i,j} = \ell_j + o(\ell_j)$ while $h_{i,j} = \beta_j(d_{i,j} - \ell_j) \to \infty$. The latter case still has a vanishing normalized pressure and needs its own closure estimate. We give one smooth cutoff that covers both cases.

Suppose edge i has $h_{i,j} \to \infty$ uniformly on the compact rescaled support under consideration. Fix $M > 2$ and put

$$c_{i,j}(x) = f_{i,j}(x) + \theta_j(d_{i,j}(x)) - M.$$

Choose $\chi \in C^\infty(\mathbb{R})$ with $\chi = 1$ on $(-\infty, -1]$, $\chi = 0$ on $[0, \infty)$, and $|\chi'| \le C$. On the lower core use the actual Jacobi field multiplied by

$$\chi_{i,j}(x, s) = \chi(s - c_{i,j}(x)).$$

The upper core is closed by the reflected cutoff. Since $0 \le \theta_j' \le 1/2$ and the graph slopes tend to zero, $|\nabla_{x,s}\chi_{i,j}| \le C$ uniformly.

If $d_{i,j} \le C_0\ell_j$, the cutoff band lies at the fixed turn coordinates $-M - 1 \le s - m_{i,j}(x) \le -M$. The normalized midpoint convergence and the pressure formula give

$$\sup_{\text{band}} |\nabla v_j|^{p_j} \le C_M b_{i,j}^{p_j} \le C_M \rho_j e^{-h_{i,j}} = o(\rho_j)$$

uniformly on the rescaled support. If $d_{i,j} > C_0\ell_j$, the depth of the cutoff band from its own zero is at least $C_0\ell_j/2 - M - 1$. The full-gap or empty-side envelope therefore gives $|\nabla v_j|^{p_j} \le C_M \rho_j^{C_0}$ there, up to a uniform $1 + o(1)$ factor from changing normal to vertical depth. Take $C_0 > 1$. These bounds prove that the integral of $|\nabla v_j|^{p_j}$ across every cutoff band is $o(\rho_j)$. The same function θ_j is used throughout; the two ranges are only used to estimate it and introduce no discontinuity in the test. A missing exterior neighbor is handled by the already proved empty-side cap.

Here is the boundary-flux accounting. Write $z = v_{j,s}$ and $\zeta_j(x) = \zeta(\sqrt{\rho_j}x)$. The exact weak Jacobi identity gives on the truncated core

$$Q_{v_j,\mathcal{C}}(z\chi_{i,j}\zeta_j) = \int_{\mathcal{C}} z^2 A_{p_j}(\nabla v_j)[\nabla(\chi_{i,j}\zeta_j), \nabla(\chi_{i,j}\zeta_j)] + \int_{\partial\mathcal{C}} z(\chi_{i,j}\zeta_j)^2 A_{p_j}(\nabla v_j)\nabla z \cdot n.$$

The terminal flux is zero because the cutoff vanishes there. At the other matching faces the cutoff is one, so their original fluxes are unchanged. All terms containing a derivative of the cutoff lie in its unit-width band. Using $z^2\|A_{p_j}(\nabla v_j)\| \le C|\nabla v_j|^{p_j}$, their absolute contribution per horizontal point is bounded by

$$o(\rho_j)(|\zeta|^2 + |\nabla\zeta|^2),$$

with the tests evaluated in rescaled coordinates. This includes the mixed term obtained by expanding the square. Integration over a physical tangential support of volume $O(\rho_j^{-m/2})$ gives $o(\rho_j^{1-m/2})$, which disappears in the projected stability inequality. The potential has been cancelled through the exact Jacobi identity; it is not being discarded by a sign argument. The removed tails do not change the limiting core mass, because the cutoff depth tends to infinity and the tails are uniformly tight.

It remains to pass stability to this limiting chain. Choose $M > 0$ so that $\inf_{p\in\mathcal{P}} \lambda_p(M) > 3/4$, using (4.48). The pressure and turn comparisons give the normalized profiles in C^1 on fixed midpoint neighborhoods. We also need their C^2 convergence near the two matching faces. Fix one of the finitely many prepared matching gaps and a physical base point x_0. Use its midpoint value $b = b_{j,i}(x_0) > 0$ as a constant normalization on the whole local neighborhood. In the notation of the projection statement, put

$$\widehat{w}_j(\xi, t) = \frac{w_j(x_0 + \xi, m_{j,i}(x_0) + t)}{b}.$$

Reflect the phase when necessary, so that w_j is the positive well deficit. Its exact normalized equation is

$$\Delta_{p_j}\widehat{w}_j = (1 - b\widehat{w}_j)[\widehat{w}_j(2 - b\widehat{w}_j)]^{p_j-1}.$$

The powers here use p_j before passing to a limit.

Choose a fixed small face-collar width $a_M > 0$, with $2a_M < M/2$. The inverse-integral profile and its first integral give

$$\min_{\substack{p\in\mathcal{P}\\ ||t|-M|\le 2a_M}} |V_p'(t)| > 0.$$

The uniform C^1 turn convergence therefore bounds $|\nabla\widehat{w}_j|$ above and away from zero on these enlarged collars, and bounds $\widehat{w}_j$ above and away from zero there. Uniform interior $C^{1,\alpha}$ estimates for the normalized equation apply on slightly larger fixed neighborhoods: its solutions and right-hand sides are bounded uniformly. On the noncritical collars, write the equation as

$$A_{p_j}(\nabla\widehat{w}_j) : D^2\widehat{w}_j = (1 - b\widehat{w}_j)[\widehat{w}_j(2 - b\widehat{w}_j)]^{p_j-1}.$$

Its principal eigenvalues have common positive lower and upper bounds. The principal coefficients have uniform $C^{0,\alpha}$ bounds, because the gradients have such bounds and stay away from zero. The reaction functions and their derivatives are uniformly bounded on the positive value range, since $p_j \in \mathcal{P}$ and $b \to 0$. Interior Schauder estimates now give uniform $C^{2,\alpha}$ bounds on smaller face collars.

Compactness at this higher order upgrades the known C^1 convergence to C^2 convergence there. The limit is still $V_p(t)$, since the C^1 limit has already been identified. The same argument works for a sequence of varying exponents, after taking $p_j \to p$; the model ODE gives continuous dependence of V_p'' on p away from the turn. If uniformity in the physical base point failed, choose failing base points with $\sqrt{\rho_j}x_0$ in one fixed compact set and repeat this local argument. The uniform turn comparison gives the same limit and a contradiction. Thus the C^2 convergence holds in all physical variables, uniformly at the matching faces. No second derivative convergence at the turn itself is used.

Fixed-core convergence and the well envelopes give (4.50). The smoothed inactive closures cost $o(\rho_j)$ per horizontal point. Hence Theorem 4.7 applies with $\tau_j = \sqrt{\rho_j}$. Retain the actual exponent p_j throughout normalization. In the limit it gives

$$\lambda \sum_i \int A_i(\zeta_i - \zeta_{i-1})^2 \le \sum_i \int |\nabla\zeta_i|^2$$

for every finite node test, with a fixed $\lambda > 3/4$. Indeed, the change $y = \sqrt{\rho_j}x$ multiplies each physical kinetic term by $\rho_j^{1-m/2}$. After division by $\sigma_j\rho_j^{1-m/2}$, the interaction coefficient is

$\beta_j P_{i,j}/(\sigma_j \rho_j) \to A_i$. All incident edges are included for each finite family of node tests. An inactive exterior edge contributes zero in the limit, with its closure cost $o(\rho_j)$ before rescaling. Thus this passage does not require a uniform bound on the number of sheets. In particular it applies to the zero-sum tests in (4.59). Proposition 2.5 supplies the Jacobi identity directly for v_j. The solution need not arise as a directional endpoint. By $6 < 8\lambda + 2$, Theorem 4.8 excludes $A_0(0) > 0$. The contradiction proves the uniform gap. □

Lemma 4.13 (Local graph and force assembly)**.** *Let $p_j \in \mathcal{P} \Subset (2, \infty)$ and let u_j be scalar-stable entire matched solutions. Suppose that all zeros in $B_{s_j}(0)$ are regular and satisfy $D_j \le \delta_j \to 0$ and $|B_j| \le K_j$. Here an opposing pair means two distinct zeros z, z' with $\nu_j(z) \cdot \nu_j(z') \le -1/2$, where $\nu_j = \nabla u_j/|\nabla u_j|$. Its intensity is $e^{-\beta_j|z-z'|}$, with $\beta_j = p_j a_{p_j}$. Choose an adjacent opposing pair in this ball with distance $\ell_j \to \infty$ and one foot at the origin. Require its open joining segment to contain no zero and its joining segment to be normal to the zero set at this base foot. The first condition specifies adjacency here. The second is the normal-chord selection condition. It holds if the other foot was fixed first and this base foot is an interior nearest opposing foot at which the normal constraint is strict. Put $\rho_j = e^{-\beta_j \ell_j}$ and $\kappa_j = K_j + \delta_j + \rho_j$. Assume that a fixed $C_* \ge 1$ bounds the intensity of every opposing pair contained in $B_{s_j}(0)$ by $C_* \rho_j$. Assume also that, for every fixed integer $N \ge 0$,*

$$\frac{\kappa_j \log^N(1/\rho_j)}{\sqrt{\rho_j}} \longrightarrow 0, \qquad \frac{s_j \sqrt{\rho_j}}{\log^N(1/\rho_j)} \longrightarrow \infty.$$

After rotation, the graph indexing and vertical guard construction above apply on every fixed compact set in $y = \sqrt{\rho_j} x$. The selected vertical gap is $\ell_j + o(1)$. For each fixed finite block, the physical slopes tend to zero and

$$h_{i,j} \ge -\log C_* + o(1), \qquad h_{0,j}(0) = o(1).$$

On gaps of width at most $C_0 \ell_j$, the pressure formula (4.70), normalized turn preparation, and core-mass condition (4.50) hold uniformly on rescaled compact sets. The cell currents satisfy (4.71), with vector error $o(\sqrt{\rho_j})$, and their scalar force errors are $o(\rho_j)$. Consequently (4.72) holds with uniform scalar errors and divergence errors tending to zero in local L^∞. Edges with vanishing normalized interaction admit the smooth Jacobi closures above. Their positive cutoff energy is $o(\rho_j)$ per horizontal point, as required by (4.51) with $\tau_j = \sqrt{\rho_j}$. The Harnack, Hölder, and local H^1 estimates following (4.72) apply to $h_{i,j} + \log C_* + 1$. All these conclusions use only the stated local bounds.

Proof. Each u_j has values in $[-1, 1]$ by the standing matched-solution assumption. The selected zero at the origin is regular, so $|\nabla u_j(0)| > 0$ and u_j is nonconstant. Remark 2.2 applies to this entire solution. Attaining either well would make u_j that constant well, which contradicts the selected zero. Hence $-1 < u_j < 1$ everywhere. This is the strict range required for the phase variables and the chamber comparisons below.

Choose

$$r_j = \min\{s_j/100, (\kappa_j \sqrt{\rho_j})^{-1/2}\}, \qquad H_j = r_j/\ell_j^{M_0}, \qquad T_j = H_j \sqrt{\rho_j}/100,$$

where M_0 exceeds the fixed polynomial window orders. The scale assumptions give $T_j \to \infty$ and $\kappa_j r_j \to 0$. Lemma 3.10 applies on the required buffered disks, all contained in $B_{s_j}(0)$. Its angle estimate gives the common slope bound $C(H_j/r_j + K_j r_j)$. Index the graphs with axis heights in $[-H_j, H_j]$ and initially use nodes in $[-H_j/2, H_j/2]$. Every fixed block connected to the selected edge by gaps of size $O(\ell_j)$ stays in the smaller guard interval. An exit through its boundary is preceded by a larger, inactive gap. Rotate the selected normal chord onto the vertical axis and label its adjacent graphs as the selected edge. At the axis their heights differ by exactly ℓ_j. Thus its normalized gap has $h_{0,j}(0) = 0$ in this chart; the stated $o(1)$ allows the equivalent normal-to-vertical descriptions used in the pressure comparison. For every same-base adjacent graph pair on a fixed

rescaled compact set, both vertically aligned zero points lie in $B_{s_j}(0)$. Their phase orientations alternate and their slopes tend to zero. Their normal product is therefore less than $-1/2$ for large j. They form an eligible opposing pair of Euclidean chord length $d_{i,j}$. The intensity assumption gives directly

$$e^{-\beta_j d_{i,j}} \le C_*\rho_j, \qquad h_{i,j} = \beta_j(d_{i,j} - \ell_j) \ge -\log C_*.$$

For a gap of width $O(\ell_j)$, the earlier geometric estimate also bounds the difference between its vertical and normal widths by $C\vartheta_j^2\ell_j + CK_j\ell_j^2 = o(1)$, with ϑ_j the common slope. The scale assumptions and the choice of M_0 give this convergence. Hence the same lower bound holds with an $o(1)$ error in normal-gap coordinates, and the pressure factor is unchanged.

The pressure and flux constructions above use polynomial disks of radius $O(\ell_j^q)$ and normal depths $O(\ell_j)$. These disks and all their fixed buffers lie in $B_{s_j}(0)$. The chamber errors tend to zero because $\delta_j \le \kappa_j$ and each fixed power of ℓ_j is a logarithmic power of $1/\rho_j$. Thus the local pressure comparison and the complete tilted-layer flux calculation apply with their actual exponents p_j. The four error terms displayed after (4.71), divided by $\sqrt{\rho_j}$, tend to zero under the scale assumptions. For the finite comparison windows, $R_{\mathrm{cmp}} = O(\ell_j^3)$ and $L_+ + L_- = O(\ell_j)$, so $K_j R_{\mathrm{cmp}}(L_+ + L_-) \le CK_j\ell_j^4 = o(\sqrt{\rho_j})$. Moreover, the common slope ϑ_j satisfies $\vartheta_j^2\ell_j \le C\ell_j^{1-2M_0} + C\kappa_j\ell_j/\sqrt{\rho_j} \to 0$. These are the finite-window flux and normal-gap accuracy conditions. The traction errors tend to zero after division by ρ_j. The preceding subtraction of consecutive cell identities now gives (4.72). Replacing the lower bound zero by $-\log C_* + o(1)$ changes only fixed constants, since $e^{-h_{i,j}}$ stays bounded.

The fixed-core limits above apply at every controlled zero. Their hypotheses hold because the u_j are entire bounded matched solutions, $p_j \in \mathcal{P}$, and $D_j \le \delta_j \to 0$ at these zeros. Together with the well envelopes and regular turn recovery, they are the local inputs used for (4.50) and the smooth inactive closures above. Use the prescribed cap construction with $\ell_{\min} = \ell_j$. Its graph and depth bounds follow from the polynomial-window geometry already checked. A switching-range band has depth at least $C_0\ell_j/2 - M - 1$ and costs $O_M(\rho_j^{C_0}) = o(\rho_j)$. For a smaller gap whose normalized interaction vanishes, its fixed turn-band cost is $O_M(e^{-\beta_j d_{i,j}}) = o(\rho_j)$ instead. All these bands stay in the controlled buffer. The same prescribed construction therefore gives positive Jacobi costs $o(\rho_j)$ per horizontal point. Finally, the Harnack and cutoff calculations use only uniform ellipticity, bounded scalar sources, vanishing local L^∞ divergence sources, and the fixed lower bound for $h_{i,j}$. They therefore give the asserted local estimates with the indicated shift. No information on zeros outside $B_{s_j}(0)$ enters these steps. □

4.5. **Local clean-ball preparation.** The uniform gap also gives quantitative preparation on large balls without bad zeros. Work in $\mathbb{R}^7$ with scalar-stable entire solutions of the matched equation, with $-1 < u < 1$ and $p \in \mathcal{P}$. At every zero, including a critical one, $D(z) = 1 - (|\nabla u(z)|/a_p)^p$ on Σ. Use Theorem 4.12 and the local graph and force assembly in Lemma 4.13. The physical surface form is Proposition 3.13, and its lower mass comparison is Corollary 3.5. The local stress and connector estimates used below are those of Section 4.1 and Theorem 4.7. Proposition 2.5 and Lemma 2.7 supply the weighted Jacobi identity and the turn recovery for these scalar-stable solutions.

Fix the exponents by

$$\frac{1}{2} < \gamma < \min_{p\in\mathcal{P}} \left(\frac{1}{2} + \frac{\sqrt{p-1}}{p}\right), \qquad 1 < \alpha < 2\gamma < 2. \tag{4.74}$$

Only the strict inequalities in (4.74) are needed for this preparation argument.

Theorem 4.14 (Local clean-ball preparation)**.** *There are constants*

$$0 < \delta_0 < 1, \qquad 1 < \alpha < 2\gamma, \qquad C < \infty, \qquad R_0 < \infty$$

depending only on $\mathcal{P}$ and the fixed dimension, with the following property. Define the closed bad-zero set

$$\mathcal{B}(u) = \{z \in \Sigma : D(z) \geq \delta_0\}.$$

If $R \geq R_0$ and $B_R(x_0) \cap \mathcal{B}(u) = \varnothing$, then

$$\sup_{\Sigma \cap B_{R/2}(x_0)} D \leq CR^{-\alpha}, \qquad \sup_{\Sigma \cap B_{R/2}(x_0)} |B| \leq C/R. \tag{4.75}$$

For each fixed $b \in (0,1)$ there are additionally $C_b, R_b < \infty$ such that, if $R \geq R_b$, then on $B_{R/4}(x_0) \cap \{|u| \leq b\}$,

$$|\nabla u| \geq C_b^{-1}, \qquad \left| D\left(\frac{\nabla u}{|\nabla u|}\right)\right| \leq C_b/R. \tag{4.76}$$

At zeros in $B_{R/4}(x_0)$, $|H| + |\nabla_\Sigma H| \leq CR^{-\alpha}$ as well.

When $\mathcal{B}(u)$ is empty, the small-defect theorem already makes u a planar layer. All the estimates then hold with $D = |B| = H = 0$.

Proof. The translated compactness and stability arguments established in Section 2 apply uniformly on $\mathcal{P}$. They apply to every translated sequence used below, including sequences with varying exponents. The zero constant is not a stable limit: for $p > 2$, its diffusion matrix is zero and its form on a nonzero compact test is $-\int \phi^2$. A zero-centered limit cannot be either constant well. Remark 2.2 therefore gives its strict range.

We retain the quantitative fixed-core estimates in addition to the uniform local convergence proved above. Fix $T < \infty$ and let z be a zero of one of these solutions. Put $d = D(z)$ and $e = \nu(z)$, with d below the threshold in Lemma 2.12 for radius $T + 1$. That lemma makes the whole ball noncritical and gives $\|D^2 U\|_{C^k(B_T(z))} \leq C_{T,k}\sqrt{d}$ for every fixed k. At the center,

$$U(z) = 0, \qquad \nabla U(z) = (1-d)^{1/p} e.$$

Since $|(1-d)^{1/p} - 1| \leq Cd$, integration along the segments from z to points of $B_T(z)$ gives

$$\sup_{B_T(z)} \left(|U - e \cdot (x - z)| + |\nabla U - e|\right) \leq C_T\sqrt{d}.$$

The higher Hessian bounds give the same convergence in each fixed C^k norm as $d \to 0$. For any zero z' in the ball, normalizing the phase gradient also gives $|\nu(z') - e| \leq C_T\sqrt{d}$. In particular, no pair with $D(z) \to 0$ and bounded separation can have $\nu(z) \cdot \nu(z') \leq -1/2$. Only the defect at the first zero enters this argument. For the quantitative transmission bounds we retain the sharper order-d estimates (2.15) and (2.16) themselves.

Choose δ_0 strictly below the threshold in Theorem 4.12. We claim that, given $\eta > 0$, some $L_\eta < \infty$ ensures

$$B_L(x) \cap \mathcal{B}(u) = \varnothing, \quad L \geq L_\eta \quad \Longrightarrow \quad \sup_{\Sigma \cap B_{L/2}(x)} D \leq \eta. \tag{4.77}$$

Otherwise there are clean balls with radii tending to infinity and zeros in their middle halves where $D \geq \eta$. Center these zeros and extract an entire stable limit u_*, with $u_*(0) = 0$ and $D_*(0) \geq \eta$. The boundaries of the clean balls recede to infinity after this translation. We show that every zero of u_* is approached by zeros of the sequence. If a limit zero had a one-sign neighborhood, reflect u_* if necessary and shrink the ball so that $0 \leq u_* < 1/2$ there. Then $-\Delta_{p_*} u_* = f_{p_*}(u_*) \geq 0$. The strong minimum principle forces $u_* = 0$ on that ball. The compactly supported stability test just given applies on that ball and gives a contradiction. Thus every neighborhood of a limit zero contains a positive and a negative point. Local uniform convergence preserves their signs, and a connecting segment supplies a zero of each approximating solution. Taking successively smaller neighborhoods gives approximating zeros. Their defects are at most δ_0 because they lie in the clean balls. Local C^1 convergence gives $D_* \leq \delta_0$ at every limit zero. Theorem 4.12 now makes u_* planar. This contradicts $D_*(0) \geq \eta$ and proves (4.77).

Use the choices in (4.74), so

$$1 < \alpha < 2\gamma < 2. \tag{4.78}$$

Assume the defect bound in (4.75) fails. There are clean balls $B_{R_j}(x_j)$ with zeros in $B_{R_j/2}(x_j)$ satisfying $DR_j^\alpha \to \infty$. Maximize $D(z)\,\mathrm{dist}(z, \partial B_{3R_j/4}(x_j))^\alpha$ over the zeros in $\overline{B_{3R_j/4}(x_j)}$. The product is continuous and vanishes on the boundary. Its maximum tends to infinity. After translation, this gives a zero at the origin, a radius $L_j \to \infty$, and defects $\varepsilon_j \to 0$ with

$$D_j(0) \geq c\varepsilon_j, \qquad D_j \leq \varepsilon_j \quad \text{on } \Sigma_j \cap B_{L_j}, \qquad \varepsilon_j L_j^\alpha \to \infty. \tag{4.79}$$

To verify these choices, let B be the maximization ball and let y_j attain its maximum. Set $s_j = \mathrm{dist}(y_j, \partial B)$, $L_j = s_j/2$, and $\varepsilon_j = 2^\alpha D_j(y_j)$. For a zero within $s_j/2$ of y_j, its distance to the boundary of the maximization ball is at least $s_j/2$. Maximality gives $D_j \leq 2^\alpha D_j(y_j) = \varepsilon_j$ there. Also $D_j(y_j) = 2^{-\alpha}\varepsilon_j$ and $\varepsilon_j L_j^\alpha = D_j(y_j) s_j^\alpha \to \infty$. Since $D_j \leq 1$, this forces $s_j \to \infty$. The ball $B_{s_j}(y_j)$ is clean, so (4.77) gives $D_j(y_j) \to 0$. This proves all assertions in (4.79).

For each zero in $B_{3L_j/4}$, the ball of radius $L_j/4$ about it lies in B_{L_j}. Apply Lemma 3.17 on these balls, with defect threshold ε_j. It gives, on all of $\Sigma_j \cap B_{3L_j/4}$,

$$|B_j| \leq \overline{K}_j := C\big(\varepsilon_j \log(C/\varepsilon_j) + L_j^{-1}\big). \tag{4.80}$$

For any fixed N,

$$\overline{K}_j^2 \log^N(C/\varepsilon_j) = o(\varepsilon_j). \tag{4.81}$$

Indeed, the square of the first term in $\overline{K}_j$, divided by ε_j, is $O(\varepsilon_j \log^{N+2}(C/\varepsilon_j)) = o(1)$. For the second term use $\log(C/\varepsilon_j) \leq C + \alpha \log L_j$ and write

$$\frac{\log^N(C/\varepsilon_j)}{\varepsilon_j L_j^2} \leq C \frac{L_j^{\alpha-2}(1+\log L_j)^N}{\varepsilon_j L_j^\alpha} \longrightarrow 0.$$

The mixed term is controlled by the sum of these squares.

The fixed-core hypotheses of the following slab and transmission comparisons follow from the local observation above, using Lemma 2.12 at their own zero bases. For each fixed T and every zero in $B_{3L_j/4}$, its radius-T ball lies in B_{L_j} for all large j, and its base defect is at most ε_j. These are local applications; no condition on the zeros outside B_{L_j} is needed. At the zero in (4.79), the two-sided slab argument finds an opposite zero with

$$\ell_j^0 \leq C \log(C/\varepsilon_j). \tag{4.82}$$

The construction stays inside $B_{L_j/4}$ for large j. We check its two comparison strips before applying transmission. Write

$$d_j = A \log(C/\varepsilon_j), \qquad R_j^{\mathrm{cmp}} = d_j^q, \qquad r_j^{\mathrm{aux}} = \varepsilon_j^{-1/4},$$

where $q > 1$ is fixed. The constant A will be fixed after the transmission rate. The relation $\varepsilon_j L_j^\alpha \to \infty$ gives $L_j^{-1} = o(\varepsilon_j^{1/\alpha})$. Since $\alpha < 2$,

$$r_j^{\mathrm{aux}} = o(L_j), \qquad R_j^{\mathrm{cmp}} = o(r_j^{\mathrm{aux}}), \qquad \overline{K}_j r_j^{\mathrm{aux}} R_j^{\mathrm{cmp}} \longrightarrow 0.$$

All fixed multiples of these graph buffers therefore lie in the region where every zero is regular and (4.80) holds. Rotate the normal at the origin to the vertical direction. Lemma 3.10 and (3.19) give the unoriented angle bound

$$\vartheta_j = C\left(R_j^{\mathrm{cmp}}/r_j^{\mathrm{aux}} + \overline{K}_j r_j^{\mathrm{aux}}\right).$$

This holds relative to the fixed plane for every zero in a fixed large multiple of $B_{R_j^{\mathrm{cmp}}}$. In particular, $\vartheta_j \to 0$, and

$$\omega_j = C\left(\vartheta_j R_j^{\mathrm{cmp}} + \overline{K}_j (R_j^{\mathrm{cmp}})^2\right) \longrightarrow 0.$$

Every zero in $B^6_{4R_j^{\mathrm{cmp}}} \times (-4d_j, 4d_j)$ lies on a graph over the common disk $B^6_{8R_j^{\mathrm{cmp}}}$. Here is an explicit reparametrization of the tangent graph from Lemma 3.10. For such a zero w, write it as

$$X_w(\xi) = w + \xi + g_w(\xi)n_w, \qquad \xi \in T_w\Sigma_j, \qquad |\xi| < cr_j^{\mathrm{aux}}.$$

The tangent normalization and curvature bound give $|g_w(\xi)| \le C\overline{K}_j|\xi|^2$ and $|Dg_w(\xi)| \le C\overline{K}_j|\xi|$. Let P be horizontal projection, and set

$$L_w = P|_{T_w\Sigma_j}, \qquad N_w(\xi) = g_w(\xi)Pn_w.$$

The angle bound gives $\|L_w^{-1}\|_{\mathrm{op}} \le 2$. For $y \in B^6_{8R_j^{\mathrm{cmp}}}$, solve

$$\xi = L_w^{-1}(y - Pw - N_w(\xi)).$$

Since $|Pw| \le 4R_j^{\mathrm{cmp}}$, its right side maps the tangent ball $|\xi| \le 32R_j^{\mathrm{cmp}}$ into itself for all large j. Indeed, the linear part has size at most $24R_j^{\mathrm{cmp}}$, and the remaining part is $O(\overline{K}_j(R_j^{\mathrm{cmp}})^2) = o(R_j^{\mathrm{cmp}})$. Its Lipschitz constant is at most $C\overline{K}_jR_j^{\mathrm{cmp}} = o(1)$, hence is less than $1/2$. The ball lies inside the tangent graph because $R_j^{\mathrm{cmp}}/r_j^{\mathrm{aux}} \to 0$. The contraction theorem gives a unique solution for every such y. The derivative $L_w + DN_w$ is invertible there, so the local inverse function theorem makes this solution smooth in y. Also $PX_w(\xi(y)) = y$, and at $y = Pw$ the solution is $\xi = 0$. Thus this is a graph on the full common disk through the original zero w. It stays in a fixed multiple of $B_{R_j^{\mathrm{cmp}}}$, where the angle bound above remains valid. Each graph differs from its height on the central axis by at most ω_j on this disk. Identify coincident graphs. The remaining graphs are disjoint and ordered. There are only finitely many meeting the displayed smaller box: their axis intersections lie in a compact interval, and these intersections are isolated because the vertical direction is transverse to the regular zero set. This argument uses no uniform bound on their number. Write the own graph as $s = f_{0,j}(y)$, with $f_{0,j}(0) = 0$.

Suppose that neither first axis neighbor has height within $2d_j$ of zero. The ordered family then has no other axis height in $(-2d_j, 2d_j)$. Since the graph oscillations tend to zero, both regions

$$|y| < 2R_j^{\mathrm{cmp}}, \qquad 0 < \sigma(s - f_{0,j}(y)) < d_j + 2, \qquad \sigma \in \{-1, 1\},$$

contain no zero. Each region is connected and adjoins the regular own graph. With the chosen normal, $u_j > 0$ in the upper region and $u_j < 0$ in the lower one. Thus clearing selects the positive well for u_j above and for $-u_j$ below. Consecutive graphs, if present, have opposite phase orientations. Thus a first axis neighbor within $2d_j$ would be an opposing zero and would prove (4.82).

Fix the small convex well required by the half-strip estimate, and let R_{well} be its clearing radius from Lemma 2.3. Choose a fixed $S > R_{\mathrm{well}} + 4$. For a point

$$x = (y, f_{0,j}(y) + \sigma t), \qquad |y| \le 2R_j^{\mathrm{cmp}}, \quad S \le t \le d_j + 1,$$

any zero within distance R_{well} would lie in the smaller box covered by the common graphs, for all large j. On the own graph its vertical distance from x is at least $t - 2\omega_j$. On any other graph it is at least $2d_j - t - 2\omega_j$, because its axis height has absolute value at least $2d_j$. These bounds also apply when the horizontal coordinates differ. Their minimum is at least $S - 1$ for large j, which is greater than R_{well}. This excludes such a zero. Zeros outside the common box cannot be closer, by the same horizontal and vertical localization. Hence clearing puts both actual strips in the chosen well, uniformly in y, from depth S through depth $d_j + 1$. For the lower strip reflect both the vertical coordinate and the solution. Its graph becomes $-f_{0,j}$, and the matched equation and all estimates are preserved.

Cut off $f_{0,j}$ outside a disk comparable to $B^6_{8R_j^{\mathrm{cmp}}}$, as in the remark after Proposition 4.2. Its gradient and Hessian have the required bounds with $\kappa = C\overline{K}_j$, and $\overline{K}_j(R_j^{\mathrm{cmp}})^2 \to 0$. Construct both artificial layers in depth $d_j + 3$ and compare only up to depth d_j. A fixed neighborhood of

the comparison top is then inside the artificial slab. Equation (4.3) gives $U_\sigma = t + O(\overline{K}_j(d_j+3))$ in C^1. Thus these artificial layers also lie in the same fixed well for $t \geq S$, after increasing S once if needed. Modica's estimate gives the actual relative gradient upper bound, and the bounded artificial phase gradients give its artificial counterpart. These bounds verify the principal upper bound in the natural-energy comparison throughout both enlarged strips. The well condition gives its positive well potential. The local fixed-core observation applies uniformly at every own zero on the comparison disk, since its defect is at most ε_j and its fixed ball remains in B_{L_j}. It gives the prepared, uniformly elliptic bottom coefficients and their fixed Hölder bounds. No lower ellipticity bound in the growing tails is used.

The hypotheses of (4.29) are now verified on both strips. Use its generic bounded-top-data version. The resulting difference of actual and artificial bottom slopes is at most $C(R_j^{\mathrm{cmp}})^C d_j^C e^{-cd_j}$, after reducing the fixed positive rate c to include both top and lateral errors. For the lateral term this follows from $R_j^{\mathrm{cmp}}/d_j = d_j^{q-1} \to \infty$; hence its exponential decay is faster than e^{-cd_j}. Its constants are fixed before A. Choose A so large that this error is $o(\varepsilon_j)$. The artificial slope sum is $2a_{p_j} + O(\overline{K}_j^2 d_j (R_j^{\mathrm{cmp}})^2)$ by (4.5); the change from d_j to d_j+3 changes only its fixed bound. Its geometric error is $o(\varepsilon_j)$ by (4.81). The actual sum is $2a_{p_j}(1-D_j(0))^{1/p_j}$. Since $D_j(0) \geq c\varepsilon_j$ in (4.79), this contradicts the two preceding $o(\varepsilon_j)$ estimates. An adjacent opposing zero must therefore occur within $2d_j$, which proves (4.82).

Choose ℓ_j^0 as the nearest opposing-zero distance from the defect base point and define $\rho_j^0 = e^{-p_j a_{p_j} \ell_j^0}$. The local separation consequence above gives $\ell_j^0 \to \infty$, since $D_j(0) \leq \varepsilon_j \to 0$ and the second foot has an opposing normal. For the top estimates, use tangent disks with $r \asymp \sqrt{\ell_j^0/\overline{K}_j}$. Their $20r$ buffers remain in the working ball because $\overline{K}_j \geq c/L_j$ and $r/L_j \leq C\sqrt{\ell_j^0/L_j} \to 0$. Also $\overline{K}_j r \to 0$. Equation (3.19) bounds their relative unoriented tilt by $C(\ell_j^0/r + \overline{K}_j r) = C\sqrt{\overline{K}_j \ell_j^0}$. The unoriented tilt bound and the opposing-normal constraint imply that the oriented normal product of the selected pair tends to -1. The constraint $\nu(z)\cdot\nu(z') \leq -1/2$ is strict at its nearest foot. The distance minimum is thus unconstrained on the second graph, so the joining segment is normal to that graph. It is also close to the own normal at the base. An earlier zero on this segment would be an opposing graph and would give a shorter eligible pair. Thus the selected pair is adjacent. For this transmission comparison use a buffered graph disk of radius $2R$, where $R = (C\ell_j^0)^q$ and $q > 1$ is fixed. Apply the estimates on its central radius-R disk. Since $\ell_j^0 \leq C\log(C/\varepsilon_j)$, the rotated graph oscillations are bounded by $C\sqrt{\overline{K}_j \ell_j^0} R + C\overline{K}_j R^2 = o(1)$. Tops at depth $\ell_j^0/2 - O(1)$ therefore stay at least $\ell_j^0/2 - O(1)$ from all zeros. Ordered nearby graphs and nearest-foot selection give this inside the disk. Zeros outside the larger buffered graph disk are farther away from this central comparison disk, since $R/\ell_j^0 \to \infty$. The same ordering gives a zero-free strip over the full comparison disk. Above a fixed sufficiently large own depth S, its distance to every zero is at least $S - O(1)$, uniformly in the horizontal point. Thus clearing supplies the required small well range uniformly over the whole buffered strip. Fix

$$\gamma < \gamma' < \min_{p\in\mathcal{P}} \left(\frac{1}{2} + \frac{\sqrt{p-1}}{p} \right).$$

Fix this γ' first. Then choose the Agmon rate and clearing loss in (4.37) so that their combined exponent is still strictly larger than γ'. Finally choose the well and preparation thresholds required by those estimates. At terminal depth $\ell_j^0/2 - O(1)$, radial clearing and improved Agmon transmission give a bottom-slope error bounded by $C(\ell_j^0)^M(\rho_j^0)^{\gamma'} \leq C(\rho_j^0)^\gamma$.

The last absorption uses the strict margin $\gamma' - \gamma > 0$:

$$(\ell_j^0)^M(\rho_j^0)^{\gamma'-\gamma} = (\ell_j^0)^M e^{-\beta_j(\gamma'-\gamma)\ell_j^0} \longrightarrow 0,$$

uniformly because $\inf_{p\in\mathcal{P}} \beta_p > 0$. The choice of a radius polynomial in ℓ_j^0 makes this absorption valid before any relation between ε_j and ρ_j^0 is available. The geometric slope error is bounded by $C\overline{K}_j^2 \operatorname{polylog}(1/\varepsilon_j)$. Consequently,

$$\varepsilon_j \le C\overline{K}_j^2 \operatorname{polylog}(1/\varepsilon_j) + C(\rho_j^0)^\gamma.$$

With (4.81), this implies

$$\rho_j^0 \ge c\varepsilon_j^{1/\gamma}, \qquad \rho_j^0 L_j^2 \ge c(\varepsilon_j L_j^\alpha)^{1/\gamma} L_j^{2-\alpha/\gamma} \longrightarrow \infty. \tag{4.83}$$

Its positive power of L_j dominates every fixed logarithmic loss. Here $\log(1/\rho_j^0) = O(\log(1/\varepsilon_j)) = O(\log L_j)$ by (4.82) and (4.79).

A smallest gap in a ball need not lie far from its boundary. Instead, for $z \in \Sigma_j \cap \overline{B_{L_j/2}}$ define

$$d_j(z) = L_j/2 - |z|.$$

Let $\ell_j(z)$ be the nearest opposite-zero distance, restricting the second point to $\overline{B_{3L_j/4}}$ and imposing $\nu(z)\cdot\nu(z') \le -1/2$. An empty infimum has intensity zero. All zeros in these compact balls are regular. The resulting intensity is upper semicontinuous. Indeed, for a convergent sequence of first points with positive limiting intensity, choose minimizing second points. A subsequence converges in the compact second-point set. Continuity of the normals preserves the closed normal constraint. The two limiting points cannot coincide, since their normal scalar product would then be one. Continuity of distance therefore gives an eligible distinct limit pair. Its intensity bounds the upper limit from above. If the limit intensity were positive at a point with an empty feasible set, the same argument would give a contradiction. Since the intensity is bounded by one, its product with d_j^2 extends by zero to the boundary of $B_{L_j/2}$. Upper semicontinuity on this compact zero set therefore gives a maximum of

$$e^{-\beta_j \ell_j(z)} d_j(z)^2, \qquad \beta_j = p_j a_{p_j}.$$

Call the first point z_j, its gap ℓ_j, its boundary distance s_j, and its intensity $\rho_j = e^{-\beta_j \ell_j}$. The original defect base at the origin is an admissible competitor: its nearest opposing point has distance $\ell_j^0 = o(L_j)$ and lies in the allowed second-point ball. Comparing the maximal product with this competitor, and using $s_j \le L_j/2$, gives

$$\rho_j s_j^2 \ge \rho_j^0 (L_j/2)^2, \qquad \rho_j \ge \rho_j^0, \qquad \ell_j \le \ell_j^0. \tag{4.84}$$

In particular,

$$s_j\sqrt{\rho_j} \ge \tfrac{1}{2} L_j \sqrt{\rho_j^0} \ge c(\varepsilon_j L_j^\alpha)^{1/(2\gamma)} L_j^{1-\alpha/(2\gamma)}.$$

The last power of L_j is positive, since $\alpha < 2\gamma$. Also $\log(1/\rho_j) \le \log(1/\rho_j^0) = O(\log L_j)$. Thus $s_j\sqrt{\rho_j}$ grows faster than every fixed power of $\log(1/\rho_j)$. The same local separation consequence gives $\ell_j \to \infty$: the selected base lies in $B_{L_j/2}$ and has $D_j(z_j) \le \varepsilon_j \to 0$, so a bounded opposing gap is excluded by the fixed-ball normal estimate above.

For zeros in $B_{s_j/2}(z_j)$, the boundary distance is at least $s_j/2$. Maximality yields

$$e^{-\beta_j \ell_j(z)} \le 4\rho_j. \tag{4.85}$$

This includes every neighboring graph needed at the interaction scale. Its relevant gap is logarithmic, whereas the boundary buffer is much larger. The geometric nearest-foot argument above applies again. For example, use tangent disks with

$$r = \frac{1}{100} \min\{s_j, \sqrt{\ell_j/\overline{K}_j}\}.$$

Since $\ell_j = o(s_j)$, the radius-$20r$ buffers about both selected feet lie in $B_{s_j/2}(z_j)$ for large j. All their zeros are regular and satisfy the curvature bound $\overline{K}_j$. Lemma 3.10 therefore applies. The tilt bound is at most $C(\ell_j/s_j + \sqrt{\overline{K}_j \ell_j}) = o(1)$. The second point is an interior point of its allowed ball,

since $\ell_j \le \ell_j^0 = o(L_j)$. Thus the normal constraint is inactive, the pair is adjacent, and the joining segment is normal at the second foot. Use this foot as the base and rotate its normal vertically. This moves the base point by $O(\ell_j) = o(s_j)$ and preserves a fixed fraction of the buffer. All later graph windows remain inside $B_{s_j/2}(z_j)$ for large j, so they retain the intensity bound (4.85).

For each fixed N, the curvature bound and (4.83)–(4.84) imply

$$\frac{\overline{K}_j \log^N(1/\rho_j)}{\sqrt{\rho_j}} \to 0, \qquad \frac{s_j\sqrt{\rho_j}}{\log^N(1/\rho_j)} \to \infty. \tag{4.86}$$

For the L_j^{-1} term in $\overline{K}_j$ use $L_j\sqrt{\rho_j} \ge cL_j\varepsilon_j^{1/(2\gamma)}$ together with $\alpha < 2\gamma$. For the remaining term use $\varepsilon_j \le C\rho_j^\gamma$ and the monotonicity of $t\log^N(C/t)$ for small t. This proves the scale separation without a reverse defect-pressure estimate.

Set $\kappa_j = \overline{K}_j + \rho_j$ and choose

$$r_j = \min\left\{s_j/100, (\kappa_j\sqrt{\rho_j})^{-1/2}\right\}, \qquad H_j = r_j/\ell_j^{M_0},$$

where M_0 is fixed and sufficiently large. Then

$$H_j \le L_j, \qquad H_j\sqrt{\rho_j} \to \infty, \qquad \kappa_j r_j \to 0.$$

The corresponding logarithmic versions follow from (4.86). To verify the scale claims, note that

$$r_j\sqrt{\rho_j} = \min\left\{\frac{s_j\sqrt{\rho_j}}{100}, \left(\frac{\sqrt{\rho_j}}{\kappa_j}\right)^{1/2}\right\}.$$

Both terms diverge faster than every fixed logarithmic loss by (4.86), while $\kappa_j r_j \le (\kappa_j/\sqrt{\rho_j})^{1/2} \to 0$. Apply Lemma 4.13 after the selected translation and rotation, with a fixed fraction of s_j as its working radius, defect bound ε_j, and curvature bound $\overline{K}_j$. The new base is the minimizing second foot. The normal constraint there is strict, so its joining segment is normal at this base. This verifies the lemma's normal-chord selection condition. Both selected feet and the whole reduced working ball lie in $B_{s_j/2}(z_j)$ for large j, since their base displacement is $O(\ell_j) = o(s_j)$. For any eligible pair z, z' in this working ball, the second point is still in the original allowed ball $\overline{B_{3L_j/4}}$. By the definition of $\ell_j(z)$,

$$e^{-\beta_j|z-z'|} \le e^{-\beta_j\ell_j(z)} \le 4\rho_j.$$

The last inequality is (4.85). Thus the required intensity bound holds with $C_* = 4$, before any normal-to-vertical transfer. Since $\overline{K}_j \ge c\varepsilon_j$ for large j, its parameter $\overline{K}_j + \varepsilon_j + \rho_j$ is bounded by a fixed multiple of the κ_j just chosen. Equation (4.86) therefore verifies both of its scale hypotheses. The lemma supplies one common graph indexing and vertical guard. Each fixed compact set in $y = \sqrt{\rho_j}x$ and every fixed finite block of active nodes lie inside these charts for large j. Every chart stays within the original clean ball. For the finite stress windows, $R_{\mathrm{cmp}} = O(\ell_j^3)$ and $L_+ + L_- = O(\ell_j)$. Hence

$$\overline{K}_j R_{\mathrm{cmp}}(L_+ + L_-) \le C\overline{K}_j\ell_j^4 = o(\sqrt{\rho_j})$$

by (4.86). The common slope is at most $\vartheta_j = C(\ell_j^{-M_0} + \kappa_j r_j)$, and

$$\vartheta_j^2\ell_j \le C\ell_j^{1-2M_0} + C\frac{\kappa_j\ell_j}{\sqrt{\rho_j}} \longrightarrow 0.$$

These bounds verify the finite-window flux accuracy and the normal-to-vertical conversion used by the local assembly.

The buffer estimate is uniform on every fixed rescaled ball: for fixed T, its physical radius is $T/\sqrt{\rho_j} = o(s_j)$. It is therefore contained in the selected intensity-control region for all sufficiently

large j. The additional polynomial windows of radius $C\ell_j^3$ and their vertical caps also fit there, because $s_j\sqrt{\rho_j}$ dominates every logarithmic power.

Every finite gap bounded by a fixed multiple of ℓ_j satisfies the prepared-gap assumptions on logarithmic disks. Close larger gaps by the smoothed empty-side caps. By (4.86), force and uniform stress errors vanish on rescaled compact sets. Define the normalized log gaps and interactions, in the coordinates $y = \sqrt{\rho_j}x$, by

$$h_{i,j}(y) = \beta_j(d_{i,j}(y/\sqrt{\rho_j}) - \ell_j), \qquad A_{i,j} = \frac{\beta_j \Pi_{p_j}}{\sigma_{p_j}} e^{-h_{i,j}}, \qquad \sigma_p = \int_{\mathbb{R}} |g_p'|^p.$$

Unlike the global construction, their lower bound is $h_{i,j} \geq -\log 4 + o(1)$ by (4.85), rather than $h_{i,j} \geq 0$. Thus $A_{i,j}$ is uniformly bounded. The local stress identity has exactly the normalization in (4.72). Scalar stress errors are divided by ρ_j, and integrated mixed-flux errors by $\sqrt{\rho_j}$. Lemma 4.13 supplies precisely these uniform scalar and vector error bounds on the buffered charts. Its local Harnack estimate applies to $h_{i,j} + \log 4 + 1$, using the lower bound for $h_{i,j}$ and the uniform upper bound for $A_{i,j}$. Its scalar source is bounded and its divergence source tends to zero in local L^∞. The selected gap is bounded on each fixed rescaled compact set. Neighboring gaps either have the same compactness or tend uniformly to infinity, in which case their interactions tend uniformly to zero. The guard interval and diagonal extraction therefore give a positive chain containing the selected edge. For bounded gaps, the local cutoff estimate in Lemma 4.13 bounds their gradients in local L^2. Uniform convergence, weak H^1 convergence, and convergence of the coefficient matrices to I pass the equation to the limit. Thus $-\Delta \log A_i = 2A_i - A_{i-1} - A_{i+1}$, with absent neighbors zero. Each present edge has $A_i > 0$. Local elliptic regularity for $\log A_i$, followed by bootstrapping on finite neighboring blocks, makes every present A_i smooth. The selected edge satisfies $h_{0,j}(0) = o(1)$, so $A_0(0) = \lim_j \beta_j \Pi_{p_j}/\sigma_{p_j} > 0$.

The stability test uses finitely many cores above a compact tangential support. The buffer keeps that support and all normal closures inside the clean ball. Lemma 4.13 supplies normalized turn preparation, the full core masses in (4.50), and the positive closure costs in (4.51). Choose a fixed window by (4.48) so that the connector retains $\lambda > 3/4$ uniformly on $\mathcal{P}$. We also verify the required regularity at the matching faces. Use the constant midpoint normalization from the proof of Theorem 4.12. For each physical base point, keep its midpoint deficit b_j constant on the local neighborhood. The normalized positive deficit $\widehat{w}_j$ satisfies

$$\Delta_{p_j}\widehat{w}_j = (1 - b_j\widehat{w}_j)[\widehat{w}_j(2 - b_j\widehat{w}_j)]^{p_j - 1}.$$

Write $M > 0$ for the fixed matching half-window just chosen. On fixed enlarged collars of $t = \pm M$ contained in $|t| \geq M/2$, the C^1 turn convergence and $\inf_{p\in\mathcal{P}} |V_p'(\pm M)| > 0$ bound the normalized gradients above and away from zero. The normalized values also stay in one compact subinterval of $(0, \infty)$. The pressure comparison (4.21) and (4.85) give $b_j^{p_j} \leq C\rho_j$ uniformly on these prepared gaps. Compactness of $\mathcal{P}$ and $\rho_j \to 0$ therefore give $b_j \to 0$ uniformly. The reaction and its fixed derivatives are therefore uniformly bounded. The normalized $C^{1,\alpha}$ estimates and noncritical Schauder bootstrap used in the earlier matching-face argument give uniform $C^{2,\alpha}$ bounds on smaller collars, for some common $\alpha > 0$. Compactness upgrades the known C^1 convergence to C^2 convergence there. In particular, the normalized Jacobi traces and their first derivatives converge. If uniformity over the rescaled compact support failed, choose violating base points and repeat this local argument. The common chart buffers and turn preparation give the same limit, a contradiction. No second derivative convergence at the turn is needed. Theorem 4.7, in its varying-exponent form, now applies with $\tau_j = \sqrt{\rho_j}$. After division by σ_{p_j}, its limiting interaction is $\beta_j P_{i,j}/(\sigma_{p_j}\rho_j) \to A_i$. The limiting stability inequality is

$$\lambda \sum_i \int A_i(\zeta_i - \zeta_{i-1})^2 \leq \sum_i \int |\nabla \zeta_i|^2$$

for every finitely supported node test with compact smooth tangential support. This includes the zero-sum node tests in (4.59). The chain is finite, one-sided infinite, or doubly infinite, and its missing exterior edges are zero. Theorem 4.8 therefore applies in dimension six. It contradicts $A_0(0) > 0$, since $6 < 8\lambda + 2$. This proves the defect estimate in (4.75).

Apply the defect estimate on slightly larger interior balls. Then use Lemma 3.17 with $\delta = CR^{-\alpha}$. Because $\alpha > 1$,

$$\delta \log(C/\delta) + R^{-1} \le C/R.$$

This proves the curvature bound in (4.75). Relative mean-curvature estimates give decay $R^{-\alpha}$ for H and its tangential derivative.

To extend the estimates from zeros to the fixed core, fix $b < 1$ and choose $b_1 \in (b, 1)$. For $x \in B_{R/4}(x_0)$ with $|u(x)| \le b$, Lemma 2.3 gives a zero at distance at most R_{b_1}. Choose a nearest zero z; it exists because Σ is closed. For large R, it lies in $B_{R/2}(x_0)$. It is regular, and the minimizing segment is normal there, so $x = z + s\nu(z)$ with $|s| \le R_{b_1}$. The relative phase estimates keep $r \ge 1/2$ on this fixed segment. The fixed-core transport argument following Proposition 3.15 uses only the base bound $|D^2U(z)| \le C(|B(z)| + D(z))$ and $\partial_e D^2 U = O_b(D(z))$ on the segment, where $e = \nu(z)$ is fixed. The latter is the C^2 estimate for $1 - U_e$ in Lemma 2.12. Its local hypotheses hold here because $D(z) \to 0$ and the segment length is fixed. Thus it gives

$$\left| \nabla \left(\frac{\nabla u}{|\nabla u|} \right)(x) \right| \le C_b(|B(z)| + D(z)) \le C_b/R.$$

Also $|\nabla u(x)| = a_p(1 - u(x)^2)r(x) \ge \frac{1}{2}(\min_{p\in\mathcal{P}} a_p)(1 - b^2) > 0$. This proves (4.76) and the theorem. □

The same constants apply if a later working bad threshold is $0 < \delta \le \delta_0$. Indeed, $\{u = 0, D \ge \delta_0\} \subset \{u = 0, D \ge \delta\}$. A ball clean for the latter set is therefore clean for the original set in Theorem 4.14. We apply that theorem with its original constants and exponents. They remain fixed when the later threshold decreases.

In particular, the zero-set defect obeys $D(z) \le C(1 + \operatorname{dist}(z, \mathcal{B}(u)))^{-\alpha}$. For finite distance d, apply the theorem on a clean ball of radius $d/2$ once d is large; the bound $D \le 1$ covers smaller d. When the bad set is empty, use the planar case and the convention $\operatorname{dist}(z, \varnothing) = \infty$. For each fixed $b \in (0, 1)$, choose d_b larger than twice the radius threshold for (4.76). If $|u(x)| \le b$ and $d = \operatorname{dist}(x, \mathcal{B}(u)) \ge d_b$, apply that estimate on the clean ball $B_{d/2}(x)$. It gives

$$|\nabla u(x)| \ge c_b > 0, \qquad \left| D \left(\frac{\nabla u}{|\nabla u|} \right)(x) \right| \le C_b/d.$$

The gradient lower bound makes this normal well defined. If $\mathcal{B}(u)$ is empty, the planar case gives the same conclusion with zero normal derivative and $d = \infty$.

We specify the limiting stability statement and its test class. Let $p_j \in \mathcal{P}$ and let u_j be scalar-stable entire matched solutions. Let $R_j \to \infty$, and put $\widetilde{\Sigma}_j = R_j^{-1}(\Sigma_j - x_j)$. Consider a smooth two-sided limiting hypersurface S in an open rescaled domain. For each relatively compact smooth domain $\Omega \Subset S$, assume that the relevant prelimit region S_j^Ω has a proper finite covering map $\pi_j : S_j^\Omega \to \Omega$, with uniformly bounded positive degree d_j. Assume that its local graph lifts converge to Ω in C^1 and weakly in $W^{2,2}$, with common local $W^{2,2}$ bounds over these lifts. For each compact $K \subset \Omega$, also assume that the physical set $x_j + R_j\pi_j^{-1}(K)$ has an ambient clean neighborhood of width $c_K R_j$, for some fixed $c_K > 0$. These are the local convergence and buffer conditions used in the clean-limit assertion below.

For $\zeta \in C_c^1(\Omega)$, lift it as $\widetilde{\psi}_j = \zeta \circ \pi_j$ on S_j^Ω and extend it by zero on the rest of $\widetilde{\Sigma}_j$. Properness and the gap between $\operatorname{supp}\zeta$ and $\partial\Omega$ make this a legitimate compact surface test. Its physical version is $\psi_j(x) = \widetilde{\psi}_j((x - x_j)/R_j)$. The clean-ball theorem gives, on the required physical neighborhood,

$$D \le CR_j^{-\alpha}, \qquad |B| \le C/R_j, \qquad |H| + |\nabla_\Sigma H| \le CR_j^{-\alpha}.$$

In particular, the rescaled mean curvature is $O(R_j^{1-\alpha}) \to 0$. The stated convergence therefore makes S minimal.

Fix $\eta > 0$ and $Q > 2$ first. Choose one normal window for Proposition 3.13 and Corollary 3.5, with $I_P \geq (1-\eta)|B|^2$, before decreasing the defect threshold. Here the constant-cap physical form means exactly the local inequality (3.24) for these compact surface tests. Its comparison radii and retained normal depths are bounded by fixed powers of $\log R_j$. Every two-face comparison used there has a gap ℓ within this range. Longer zero-free sides are closed by own-face caps at these depths. In (3.20), take the zero-defect bound to be $\delta_j = CR_j^{-\alpha}$. Then all three required errors tend to zero: $\delta_j \ell \leq CR_j^{-\alpha}(\log R_j)^C$, $\kappa R_{\mathrm{ch}}^2 \leq CR_j^{-1}(\log R_j)^C$, and $\kappa^\vartheta R_{\mathrm{ch}}^N \ell \leq CR_j^{-\vartheta}(\log R_j)^C$. Here R_{ch} denotes the chamber radius, $\kappa = C/R_j$, and the derivative orders and $\vartheta > 0$ are fixed first. The physical form requires an ambient neighborhood of width $10A_0 \log(1/\delta_j) = O(\log R_j)$. It fits inside the prescribed clean buffer for large j. Thus Proposition 3.13 applies to ψ_j.

Write $\widetilde{B}_j$ for the rescaled second fundamental form. Combining the physical form with the lower mass comparison and scaling gives

$$\begin{aligned}(1-\eta)(1-\eta/2) \int_{S_j^\Omega} |\widetilde{B}_j|^2 \widetilde{\psi}_j^2 \\ \leq (1+\eta) \int_{S_j^\Omega} |\nabla \widetilde{\psi}_j|^2 + C_{\eta,Q} R_j^{2-\alpha Q} \int_{S_j^\Omega} \widetilde{\psi}_j^2.\end{aligned}$$

The factor R_j^2 in the last coefficient is the ratio between surface mass and Dirichlet scaling. Since $\alpha > 1$ and $Q > 2$, $2 - \alpha Q < 0$. Divide by d_j and pass to the limit. Pass to a subsequence on which the bounded integer degree d_j is constant. Uniform metric convergence passes the Dirichlet integral to its limit. Weak $W^{2,2}$ convergence and lower semicontinuity give

$$\int_\Omega |B_S|^2 \zeta^2 \leq \liminf_j d_j^{-1} \int_{S_j^\Omega} |\widetilde{B}_j|^2 \widetilde{\psi}_j^2.$$

This is computed on local graph lifts and then summed over a finite chart cover of the compact test support. The normalized area on that support is bounded, so the cap term tends to zero. Finally let $\eta \downarrow 0$. We obtain

$$\int_\Omega |B_S|^2 \zeta^2 \leq \int_\Omega |\nabla_S \zeta|^2, \qquad \zeta \in C_c^1(\Omega).$$

Since the same argument applies on every such compact domain, this is coefficient-one stability for the clean minimal limit.

5. Blowdowns and finite-density classification

Keep the phase, surface, and profile notation of Section 2. Uniform estimates use a fixed compact interval $\mathcal{P} \Subset (2, \infty)$. Write $G = G_p$. For the energy and density, write

$$W_p(t) = \frac{(1-t^2)^p}{2p}, \qquad e_p(u) = \frac{|\nabla u|^p}{p} + W_p(u), \qquad \sigma_p = \int_{\mathbb{R}} |g_p'|^p, \qquad \omega_{n-1} = |B_1^{n-1}|,$$

$$M_R(u; y) = \frac{E_p(u; B_R(y))}{\sigma_p \omega_{n-1} R^{n-1}}.$$

Thus $\sigma_E = \sigma_p$. The dimension and the assumptions on the solution are specified below.

5.1. **Discrepancy and geometric stability.** We first consider a nonconstant stable entire solution. The well-rigidity argument in Remark 2.2 gives $-1 < u < 1$. Here the dimension is arbitrary. No energy-growth hypothesis is used. The constant wells have zero physical discrepancy and energy. When they occur below, interpret the products $q^p D$ and $q^{p+1} D$ as zero.

Define

$$A_u = A_p(\nabla u), \qquad L_u = -\operatorname{div}(A_u \nabla) + W_p''(u).$$

At critical gradients put $A_u = 0$. The weak identity (2.9) and the scalar stability test (2.10) give

$$\int q^{p+1} D\eta^2 \le \int q^2 A_u \nabla\eta \cdot \nabla\eta \le c a^{p-2} \int q^p |\nabla\eta|^2. \tag{5.1}$$

For the final inequality use $|\nabla u| = aq(1-D)^{1/p}$ and $p \ge 2$. This estimate also holds at $p = 2$, with $A_2 = I$. The ordinary Jacobi calculation gives $L_u q = -q^2 D$ there, and testing stability with $q\eta$ gives the same bound. The stable blowdown and unit-density arguments below retain the standing range $p > 2$. Take a cutoff supported in B_{2R} and equal to one on B_R. Then (5.1) gives

$$\int_{B_R} q^{p+1} D \le C R^{n-2}.$$

The pointwise factorization is

$$q^p D = (q^{p+1} D)^{p/(p+1)} D^{1/(p+1)}.$$

Apply Hölder with exponents $(p+1)/p$ and $p+1$, and bound the last power of D by one.

$$\int_{B_R} q^p D \le \left(\int_{B_R} q^{p+1} D \right)^{p/(p+1)} |B_R|^{1/(p+1)} \le C R^{n-2p/(p+1)}. \tag{5.2}$$

Since $W_p(u) - (p-1)|\nabla u|^p/p = q^p D/(2p)$, the nonnegative physical discrepancy satisfies

$$R^{1-n} \int_{B_R} \left(W_p(u) - \frac{p-1}{p} |\nabla u|^p \right) \le C R^{-(p-1)/(p+1)}. \tag{5.3}$$

No surface-energy density bound was needed.

Lemma 2.3, applied with a threshold strictly above $3/4$, bounds the distance of every point in the core specified next from the zero set. Its radius is uniform on the fixed exponent interval. With $b_* = 3/4$, this gives a finite $L < \infty$, uniform for $p \in \mathcal{P}$, for which every stable entire solution satisfies

$$\{|u| \le b_*\} \subset \{\operatorname{dist}(x, \{u = 0\}) \le L\}. \tag{5.4}$$

There are uniform constants $\rho > 0$ and $c_0 > 0$ for $p \in \mathcal{P}$ such that, at each zero z,

$$q(x)^{p+1} D(x) \ge c_0 D(z) \quad \text{on } B_\rho(z). \tag{5.5}$$

If $D(z) \le \delta_*$, apply Lemma 2.12 with its fixed-ball, single-zero hypothesis. It gives $|\nabla D| \le C D(z)$ throughout that ball, uniformly on $\mathcal{P}$. Choose its radius no larger than $1/(2C)$. Then

$$D(x) \ge D(z) - C D(z)|x - z| \ge \tfrac{1}{2} D(z).$$

If $D(z) = 0$, the desired source bound is automatic. Also $U(z) = 0$ and $|\nabla U| \le 1$, so on a fixed radius ball

$$q(x) = \operatorname{sech}^2(aU(x)) \ge \operatorname{sech}^2(a\rho) > 0.$$

The lower bound is uniform on the compact exponent interval.

If $D(z) > \delta_*$, the gradient bound first keeps q uniformly positive on a fixed ball about the zero. On that ball, $D = 1 - (|\nabla u|/(aq))^p$ has a uniform modulus of continuity by the global $C^{1,\alpha}$ bound for u. A smaller fixed radius therefore gives $D \ge \delta_*/2$. Since $D(z) \le 1$, this lower defect bound

is at least $(\delta_*/2)D(z)$. Take the smaller of the two radii. This proves (5.5). The radius ρ remains fixed as the later threshold δ tends to zero.

An exponentially weighted comparison relates the core mass to the discrepancy. For $t > 0$ define

$$\zeta_t(x) = e^{-t\sqrt{1+|x|^2}}, \quad M_t = \int q^p \zeta_t^2, \quad S_t = \int q^{p+1} D\zeta_t^2.$$

Both integrals are finite and $|\nabla \zeta_t| \le t\zeta_t$. Insert compact cutoffs to justify the PDE and stability tests below. The gradients and the reactions are uniformly bounded. On a cutoff annulus, the exponential factor dominates its polynomial volume, so every extra cutoff term tends to zero. Dominated convergence gives the stated noncompact tests.

Test $-\Delta_p u = uq^{p-1}$ against $u\zeta_t^2$ and apply Modica:

$$\begin{aligned}\int u^2 q^{p-1}\zeta_t^2 &= \int |\nabla u|^p \zeta_t^2 + 2\int u\zeta_t |\nabla u|^{p-2}\nabla u \cdot \nabla \zeta_t \\ &\le a^p \int q^p \zeta_t^2 + 2a^{p-1} t \int q^{p-1}\zeta_t^2.\end{aligned}$$

It follows that

$$\int [(1+a^p)u^2 - a^p - 2a^{p-1}t] q^{p-1}\zeta_t^2 \le 0. \tag{5.6}$$

There is a useful numerical margin here. Since $a^p \le 1/2$, on $|u| \ge 3/4$ one has

$$(1+a^p)u^2 - a^p \ge \frac{9}{16} - \frac{7}{16}a^p \ge \frac{11}{32}.$$

Choose $t_0 > 0$ so that $2a^{p-1}t_0 \le 11/64$ for every $p \in \mathcal{P}$. The bracket in (5.6) is then at least $11/64$ on this exterior part of the range, whereas its negative part on $|u| \le 3/4$ is bounded uniformly. Splitting the integral into these two sets gives

$$\int_{\{|u|>3/4\}} q^{p-1}\zeta_t^2 \le C \int_{\{|u|\le 3/4\}} q^{p-1}\zeta_t^2 \le C \int_{\{|u|\le 3/4\}} \zeta_t^2.$$

Consequently, for $0 < t \le t_0$,

$$M_t \le \int q^{p-1}\zeta_t^2 \le C \int_{\{|u|\le 3/4\}} \zeta_t^2. \tag{5.7}$$

The stability estimate (5.1) also gives

$$S_t \le Ct^2 M_t. \tag{5.8}$$

Suppose $u(0) = 0$ and assume

$$D(z) \ge \delta \quad \text{for every zero } z \in B_R. \tag{5.9}$$

Choose a maximal ρ-separated family of zeros in B_R. Use the radius ρ from (5.5). The $\rho/2$-balls are disjoint, while the ρ-balls cover the zeros in B_R. By (5.4), the core $\{|u| \le 3/4\}$ inside $B_{R-L-\rho}$ is covered by concentric balls of radius $L + \rho$.

The logarithm of ζ_t^2 has Lipschitz constant $2t$. For $t \le t_0$, the weighted volumes of these larger and smaller concentric balls are therefore uniformly comparable.

Write the selected centers as $z_1, \dots, z_N$. For any fixed radius $b > 0$ and $t \le t_0$,

$$e^{-2t_0 b}|B_b|\zeta_t(z_i)^2 \le \int_{B_b(z_i)} \zeta_t^2 \le e^{2t_0 b}|B_b|\zeta_t(z_i)^2.$$

This follows by integrating $|\log \zeta_t(x)^2 - \log \zeta_t(z_i)^2| \le 2t|x - z_i|$. Consequently the weighted volume of $B_{L+\rho}(z_i)$ is at most a uniform multiple of that of $B_{\rho/2}(z_i)$. On each small ball, (5.5) and (5.9) imply $q^{p+1}D \ge c_0\delta$. The large balls cover the indicated core, whereas the small balls are disjoint.

Summing over the disjoint small balls gives

$$\int_{\{|u|\le 3/4\}\cap B_{R-L-\rho}} \zeta_t^2 \le \frac{C}{\delta} S_t. \tag{5.10}$$

When $R \ge 2(L+\rho)$, ambient volume alone bounds the remaining core:

$$\int_{|x|>R-L-\rho} \zeta_t^2 \le Ct^{-n}e^{-tR/2}.$$

Indeed, use one factor $e^{-t|x|}$ for the tail bound $e^{-tR/2}$. Integrate the other over $\mathbb{R}^n$. Combining this with (5.7) gives

$$M_t \le \frac{C_1}{\delta} S_t + C_2 t^{-n} e^{-tR/2}. \tag{5.11}$$

Set $t = \kappa\sqrt{\delta}$, with $\kappa \le t_0$. By (5.8), the source term in (5.11) is at most $C_1 C\kappa^2 M_t$. Choose $\kappa > 0$ once so that $C_1 C\kappa^2 \le 1/2$. Absorption gives

$$M_t \le 2C_2 t^{-n} e^{-tR/2}.$$

Conversely, $u(0) = 0$ and the gradient bound give $q \ge 3/4$ on a fixed ball. Thus $M_t \ge c_1 > 0$ for $t \le t_0$. The two mass bounds would require

$$\frac{tR}{2} \le \log(2C_2/c_1) + n\log(1/t).$$

With $t = \kappa\sqrt{\delta}$, this is impossible if

$$R > C\delta^{-1/2}\log(C/\delta).$$

In every dimension, fix a compact interval $\mathcal{P} \Subset (2,\infty)$. There is $C = C(\mathcal{P}, n)$ with the following property. For every nonconstant stable entire solution, each zero z, and each sufficiently small $\delta > 0$, one can find a zero z' with

$$\boxed{|z' - z| \le C\delta^{-1/2}\log(C/\delta), \qquad D(z') < \delta.} \tag{5.12}$$

For the equivalent large-radius formulation, take $\delta = C_*(\log R)^2/R^2$. At large R,

$$C\delta^{-1/2}\log(C/\delta) = \frac{CR}{\sqrt{C_*}\log R}\log\left(\frac{CR^2}{C_*(\log R)^2}\right) \le \frac{3C}{\sqrt{C_*}}R.$$

Choose $\sqrt{C_*} > 3C$ and then take R large enough that δ is within the allowed small range. The guaranteed search radius is at most R.

Consequently,

$$\inf_{\{u=0\}} D = 0, \qquad \inf_{\{u=0\}\cap B_R(z)} D \le C\frac{(\log R)^2}{R^2} \quad (R \text{ sufficiently large}). \tag{5.13}$$

Fix the solution and its exponent p. Choose zeros z_j with $D(z_j) \to 0$. For large j their gradients are nonzero. Choose orthogonal maps O_j with $O_j e_n = \nabla u(z_j)/|\nabla u(z_j)|$, and set $v_j(x) = u(z_j + O_j x)$. The local $C^{1,\alpha}$ estimates give, after a subsequence, $v_j \to v$ in C^1_{loc}. The fluxes and reactions converge uniformly on compact sets, so the weak equation passes to v. At the centered zero,

$$v(0) = 0, \qquad \nabla v(0) = \lim_j a_p(1 - D(z_j))^{1/p} e_n = a_p e_n.$$

Thus the limit is nonconstant. Remark 2.2 gives $-1 < v < 1$. The gradient bound is an equality at the origin. Proposition 2.1 therefore gives $v(x) = g_p(x_n)$. This proves the planar translated-limit assertion in every dimension. We compute the density of this layer. The matched profile satisfies $W_p(g_p) = (p-1)|g_p'|^p/p$, so $e_p(g_p) = |g_p'|^p$ and $\int_{\mathbb{R}} e_p(g_p)\,ds = \sigma_p$. More generally, write a layer as $v(x) = g_p(\nu\cdot x - t_0)$, fix a center y, and put $m = n-1$ and $\tau = t_0 - \nu\cdot y$. Slicing $B_R(y)$ orthogonally to ν gives

$$M_R(v;y) = \frac{1}{\sigma_p}\int_{-R}^{R}\left(1-\frac{s^2}{R^2}\right)^{m/2}|g_p'(s-\tau)|^p\,ds \longrightarrow 1.$$

Indeed, extend the integrand by zero outside $(-R,R)$. Its radial factor is between zero and one and tends to one at every fixed s. The integrable majorant $|g_p'(s-\tau)|^p$ has integral σ_p, so dominated convergence proves the limit. For $n = 1$, use $m = 0$ and $\omega_0 = 1$; the radial factor is one inside the interval and is defined as zero outside it. Thus this calculation also includes dimension one. The same formula gives

$$0 \le M_R(v;y) \le 1 \qquad (R > 0,\ y \in \mathbb{R}^n),$$

since its radial factor is at most one and $\int_{\mathbb{R}} |g_p'(s-\tau)|^p\,ds = \sigma_p$. Thus every fixed rotation and translation of the layer has limiting density one, and $1 + \sup_y M_R(v;y) \le 2$ at every radius. When the original solution has finite limiting density, Proposition 5.6 below bounds its density at every translated center by $M_\infty(u)$. For every fixed r, local energy convergence at the almost-flat zeros z_j therefore gives

$$M_r(g_p(x_n);0) = \lim_j M_r(u;z_j) \le M_\infty(u).$$

Send $r \to \infty$ to obtain $M_\infty(u) \ge 1$. The same lower bound is automatic when $M_\infty(u) = +\infty$. The existence of a planar translated limit does not classify the original solution in high dimension.

Return to ambient dimensions $2 \le n \le 7$ and scalar-stable solutions. Write $s = |\nabla u|$ and

$$\mathsf{A} = A_p(\nabla u) = s^{p-2}(I + (p-2)\nu\otimes\nu), \qquad \nu = \nabla u/s \quad (s>0), \qquad \mathsf{A} = 0 \quad (s=0).$$

The earlier equation and Jacobi identities give

$$-\Delta_p u = u(1-u^2)^{p-1}, \qquad u_i \in H^1_{s^{p-2},\mathrm{loc}}, \qquad -\operatorname{div}(\mathsf{A}\nabla u_i) + W_p''(u)u_i = 0. \tag{5.14}$$

The scalar form is $Q_u(\phi) = \int(\mathsf{A}\nabla\phi\cdot\nabla\phi + W_p''(u)\phi^2)\,dx$. Local boundedness of its coefficients extends it to compactly supported H^1 tests. We now prove the exponential clearing estimate. This argument uses only the equation, the gradient bound, and Lemma 2.3. It works in every dimension and uniformly on compact exponent intervals in $[2,\infty)$. Fix $\varepsilon \in (0,1/2)$. The clearing lemma gives R_ε such that $|u| \ge 1-\varepsilon$ whenever $\operatorname{dist}(x,\Sigma) \ge R_\varepsilon$. Increase this radius slightly so that the same bound holds on the closed comparison balls below.

Put $d = \operatorname{dist}(x,\Sigma)$ and first take $R = d - R_\varepsilon > 0$. The ball $B_R(x)$ has one sign. After reflection, assume this sign is positive and write $w = 1-u$. Throughout the ball, $0 \le w \le \varepsilon$, and

$$\Delta_p w = (1-w)(2-w)^{p-1}w^{p-1} \ge c_* w^{p-1}, \qquad c_* := \tfrac{1}{2}.$$

Choose $\kappa_0 \in (0,1]$ uniformly on $[p_-,p_+] \subset [2,\infty)$ so that $(p_+ + n - 2)\kappa_0^{p_-} \le c_*$. For $r = |y-x|$ define

$$\mathfrak{b}(r) = \varepsilon\frac{\cosh(\kappa_0 r)}{\cosh(\kappa_0 R)}.$$

If $t = \kappa_0 r > 0$ and $T = \tanh t$, then $\mathfrak{b}' = \kappa_0\mathfrak{b}T$ and $\mathfrak{b}'' = \kappa_0^2\mathfrak{b}$. The radial p-Laplacian is therefore

$$\Delta_p\mathfrak{b} = \kappa_0^p\mathfrak{b}^{p-1}\left((p-1)T^{p-2} + (n-1)\frac{T^{p-1}}{t}\right) \le (p+n-2)\kappa_0^p\mathfrak{b}^{p-1} \le c_*\mathfrak{b}^{p-1}.$$

Here $T^{p-2} \le 1$ and $T^{p-1}/t \le 1$ for $p \ge 2$. At the center the inequality holds by its limiting value; the radial flux has no point source there. Since $\mathfrak{b} = \varepsilon \ge w$ on the boundary, weak comparison for $-\Delta_p + c_* t^{p-1}$ gives

$$w(x) \le \mathfrak{b}(0) = \frac{\varepsilon}{\cosh(\kappa_0 R)} \le 2\varepsilon e^{-\kappa_0 R}.$$

Indeed, test the difference with $(w - \mathfrak{b})_+$. Monotonicity of the p-Laplace flux and of t^{p-1} makes both resulting terms nonnegative, so this positive part is zero. Modica's inequality and $1 - u^2 = w(2-w) \le 2w$ give $e_p(u) \le pW_p(u)/(p-1) \le Cw^p$. Reflect back for a negative phase. For bounded d, the uniform pointwise energy bound enlarges only the prefactor. Thus

$$e_p(u)(x) \le C \exp[-c_0 \operatorname{dist}(x, \{u = 0\})]. \tag{5.15}$$

The constant wells satisfy this with zero energy and distance $+\infty$ from their empty zero sets. All constants are uniform on the fixed exponent interval. Neither the equality argument nor the detector below assumes energy growth.

Choose $e \in S^{n-1}$. Put $P_e = I - e \otimes e$ and $w_e = |P_e \nabla u|$, and choose an orthonormal basis $e_1, \dots, e_{n-1}$ for $e^\perp$. Define

$$\Gamma_e(u) = \sum_{\alpha=1}^{n-1} \mathsf{A} \nabla u_{e_\alpha} \cdot \nabla u_{e_\alpha} - \mathsf{A} \nabla w_e \cdot \nabla w_e. \tag{5.16}$$

The expression is independent of the basis, is nonnegative, and lies in L^1_{loc}. All derivatives in (5.16) are understood in the weighted sense of Proposition 2.5. On the regular set with $w_e = 0$, Sobolev level-set calculus makes every term zero almost everywhere. Define the weighted terms as zero on the critical set. On $\{w_e > 0\}$, we have

$$\Gamma_e = w_e^2 \sum_{\alpha=1}^{n-1} \mathsf{A} \nabla \theta_\alpha \cdot \nabla \theta_\alpha, \qquad \theta = P_e \nabla u / w_e. \tag{5.17}$$

Lemma 5.1 (Tangential geometric stability)**.** *Every compactly supported H^1 test η satisfies*

$$\int \Gamma_e \eta^2 \, dx \le \int w_e^2 \mathsf{A} \nabla \eta \cdot \nabla \eta \, dx \le (p-1) \int s^p (1 - (\nu \cdot e)^2) |\nabla \eta|^2 \, dx. \tag{5.18}$$

Set the final integrand equal to zero at critical points.

Proof. For smooth compact tests, the first inequality is exactly (2.7). Its proof already treats weighted composition and the critical set. The second follows from $\mathsf{A} \le (p-1) s^{p-2} I$ and $w_e^2 = s^2(1 - (\nu \cdot e)^2)$ on the regular set. For a compactly supported H^1 test, approximate smoothly in H^1 on one fixed compact support. Pass to a subsequence converging almost everywhere. Fatou applies to the nonnegative left side. Boundedness of $w_e^2 \mathsf{A}$ gives convergence of the middle form. This proves the stated test class. □

The full geometric inequality (2.8), with the same H^1 approximation, gives

$$\int_{\{s>0\}} \mathcal{G}_u \eta^2 \, dx \le \int s^2 \mathsf{A} \nabla \eta \cdot \nabla \eta \, dx \le (p-1) \int s^p |\nabla \eta|^2 \, dx, \tag{5.19}$$

On the regular set,

$$\mathcal{G}_u = s^p (|B|^2 + (p-1) |\nabla_T \log s|^2) \ge s^p |D\nu|^2. \tag{5.20}$$

Here B is the second fundamental form of the regular level hypersurface through the integration point, and ∇_T is its tangential gradient. To check the identity, choose an orthonormal frame with last vector ν, and put $b = \nabla_T s$ and $d = \partial_\nu s$. Then

$$D^2 u = \begin{pmatrix} sB & b \\ b^{\mathsf{T}} & d \end{pmatrix}, \qquad A_u = s^{p-2} \operatorname{diag}(1, \dots, 1, p-1).$$

Consequently

$$\sum_i A_u \nabla u_i \cdot \nabla u_i = s^p |B|^2 + p\, s^{p-2} |b|^2 + (p-1) s^{p-2} d^2,$$

$$A_u \nabla s \cdot \nabla s = s^{p-2} |b|^2 + (p-1) s^{p-2} d^2.$$

Subtracting yields (5.20). Also

$$\partial_\nu \nu = \frac{b}{s} = \nabla_T \log s, \qquad |D\nu|^2 = |B|^2 + |\nabla_T \log s|^2.$$

The inequality in (5.20) follows because $p - 1 \geq 1$.

Theorem 5.2. *If $\Gamma_e(u) = 0$ almost everywhere in $\mathbb{R}^n$ for one $e \in S^{n-1}$, then u is a planar heteroclinic. More generally, allowing the closed range $[-1, 1]$, the alternatives are a planar heteroclinic and the constant wells.*

Proof. If u is constant, the equation permits only $-1, 0, 1$, and scalar stability excludes 0. We may therefore assume that u is nonconstant. Remark 2.2 gives $-1 < u < 1$ everywhere. Set $\mathcal{R} = \{s > 0\}$, which is now nonempty, and choose a connected component $\mathcal{C}$ of $\mathcal{R}$.

On compact subsets of $\mathcal{R}$, the gradient is bounded away from zero. Noncritical elliptic regularity makes u smooth there, and the equation is uniformly elliptic. Its nonlinear coefficients are analytic: $\xi \mapsto |\xi|^{p-2}\xi$ is analytic when $\xi \neq 0$, and $t \mapsto t(1-t^2)^{p-1}$ is analytic for $-1 < t < 1$. Morrey proved interior analytic regularity in [32]. We apply its scalar form in [33, Theorem 5.8.6, p. 180]. In nondivergence form, the principal matrix is

$$A_u = |\nabla u|^{p-2} \left(I + (p-2) \frac{\nabla u \otimes \nabla u}{|\nabla u|^2} \right).$$

Its eigenvalues are $|\nabla u|^{p-2}$ and $(p-1)|\nabla u|^{p-2}$. They are bounded above and away from zero on each compact noncritical patch. The equation is analytic in the second-order jet near every such point, and u is smooth there. Subtracting its quadratic Taylor polynomial and making a linear change of coordinates gives the normalized equation used in that theorem. Hence u is analytic on $\mathcal{C}$. No analyticity at critical points is asserted.

If w_e vanishes identically on $\mathcal{C}$, choose any unit $a \in e^\perp$. Then $u_b = 0$ on $\mathcal{C}$ for every $b \perp \operatorname{span}\{e, a\}$. Otherwise take a nonempty connected patch of $\{w_e > 0\} \cap \mathcal{C}$. The integrand in (5.17) is continuous there, so its almost everywhere vanishing is pointwise. Positive definiteness of A forces every $\nabla\theta_\alpha$ to vanish. Connectedness then makes θ a fixed unit vector $a \in e^\perp$. On that patch, $u_b = 0$ for every $b \perp \operatorname{span}\{e, a\}$. Analytic continuation extends these identities over the connected set $\mathcal{C}$.

Set $E = \operatorname{span}\{e, a\}$. Throughout $\mathcal{C}$ we have $\partial_b \nabla u = \nabla u_b = 0$ for each $b \in E^\perp$. For $x \in \mathcal{C}$, continue the line $x + tb$ as long as it stays in $\mathcal{C}$. The gradient remains the same nonzero vector along it. At a finite endpoint, continuity would keep that gradient nonzero. The endpoint would lie in $\mathcal{R}$, and a small regular ball would place it in the same component. This is impossible. Hence these lines are complete, and

$$\mathcal{C} = G \times E^\perp \quad \text{after an orthogonal identification } E \simeq \mathbb{R}^2, \qquad u(x, y) = v(x), \tag{5.21}$$

for a connected open set $G \subset \mathbb{R}^2$. We do not assume that G fills $\mathbb{R}^2$ or impose boundary regularity on ∂G.

Apply Lemma 2.8 to $\mathcal{C}$. The integrand $\mathcal{G}_u$ is the one in (2.8). That lemma includes the critical-boundary cutoff and gives

$$\int_{\mathcal{C}} \mathcal{G}_u \eta^2 \, dx \leq (p-1) \int_{\mathcal{C}} s^p |\nabla \eta|^2 \, dx \tag{5.22}$$

for every compactly supported H^1 test η.

The cylindrical structure reduces the inequality to a planar capacity estimate. In the cylinder (5.21), choose $\psi \in C_c^1(\mathbb{R}^2)$ and set $\eta(x, y) = \psi(x)\zeta(y/R)$ with nonzero $\zeta \in C_c^1(E^\perp)$. Divide (5.22)

by $R^{n-2}\int_{E^\perp}\zeta^2$, then take $R\to\infty$. The transverse term is $O(R^{-2})$, since s is bounded and ψ has compact support. This step is unnecessary for $n=2$. We obtain

$$\int_G \mathcal{G}_v\psi^2\,dx \le (p-1)\int_G |\nabla v|^p|\nabla\psi|^2\,dx. \tag{5.23}$$

Here ψ is supported in the whole plane and need not vanish on ∂G.

For the logarithmic cutoff, fix $R_0>1$. Take ψ_R equal to one on $B^2_{R_0}$. Let it be linear in $\log|x|$ between R_0 and R, and zero outside B^2_R. Smooth approximation makes it admissible.

For the piecewise logarithmic cutoff, polar coordinates give

$$\int_{\mathbb{R}^2}|\nabla\psi_R|^2 = \frac{2\pi}{\log(R/R_0)^2}\int_{R_0}^R\frac{dr}{r} = \frac{2\pi}{\log(R/R_0)}\longrightarrow 0.$$

Since $s\le a$,

$$\int_G s^p|\nabla\psi_R|^2 \le \frac{2\pi a^p}{\log(R/R_0)}\longrightarrow 0.$$

Therefore (5.23) gives $\mathcal{G}_v=0$ almost everywhere on $G\cap B^2_{R_0}$. The integrand is continuous on G, so the vanishing is pointwise. As R_0 is arbitrary, $D\nu=0$ everywhere on $\mathcal{C}$.

Apply the argument to every component of $\mathcal{R}$. Each regular level component has constant normal and zero principal curvatures. Symmetry of D^2u also gives $\nabla_T|\nabla u|=0$ there. Moreover, $u\in C^1(\mathbb{R}^n)\cap C^2(\mathcal{R})$. Thus [16, Lemma 2.11, p. 757] applies and makes u globally one-dimensional.

Write $u(x)=h(x\cdot\omega)$. The one-dimensional Hamiltonian

$$\mathcal{H}=\frac{p-1}{p}|h'|^p - W_p(h)$$

is constant, including at a turn. Indeed, the flux $F=|h'|^{p-2}h'$ is C^1 and satisfies $F'=W_p'(h)$. The function $|F|^{p/(p-1)}$ is C^1 even at $F=0$, and differentiation gives $\mathcal{H}'=h'(F'-W_p'(h))=0$ everywhere. The gradient bound gives $\mathcal{H}\le 0$. Its negative is the physical discrepancy, so (5.3) gives

$$(-\mathcal{H})|B_1|R\le CR^{-(p-1)/(p+1)}.$$

Letting $R\to\infty$ forces $\mathcal{H}=0$. Equality therefore holds in the gradient bound at every point. Since the nonconstant solution has strict range, the equality case of Proposition 2.1 gives the planar heteroclinic. □

Remark 5.3. The argument never continues the equation across $\{\nabla u=0\}$. Analytic continuation is used only inside a noncritical component. The other regularity input is weighted Hessian integrability used in Lemma 2.8. Both inputs apply to every fixed finite $p>2$.

5.2. **Detectors and stationary blowdowns.** The detector and bad-center packing adapt the compactness and tangential-stability arguments of Florit-Simon–Serra [20, Section 4.3 and Proposition 5.2]. The weighted identities above give the corresponding estimates for our equation. The equality case has a quantitative consequence at each bad zero.

Theorem 5.4 (Large-window detector)**.** *Given $\mathcal{P}\Subset(2,\infty)$ and $\delta\in(0,1)$, there are $L<\infty$ and $b>0$ depending only on $\mathcal{P},n,\delta$ such that every stable solution with $p\in\mathcal{P}$, each zero z satisfying $D(z)\ge\delta$, and each $e\in S^{n-1}$ obey*

$$\int_{B_L(z)}\Gamma_e(u)\,dx\ge b. \tag{5.24}$$

Proof. If these constants L,b did not exist, center bad zeros and choose u_j,p_j,e_j with $\int_{B_j}\Gamma_{e_j}(u_j)<1/j$. Pass to $p_j\to p_*$, $e_j\to e_*$, and local C^1 convergence $u_j\to u_*$. The nonlinear fluxes and reactions converge locally uniformly, so u_* solves the matched equation at p_*. The diffusion matrices and potentials also converge locally uniformly, including at critical gradients because $\min\mathcal{P}>2$.

Passing the stability inequality to the limit for each smooth compact test therefore preserves scalar stability. The bad-zero condition persists under C^1 convergence, giving $u_*(0) = 0$ and $D_*(0) \geq \delta$. This excludes constant wells. The zero constant is excluded by its scalar form. Finally, the matched gradient bound prevents a nonconstant limit from attaining a well.

On compact sets where $|\nabla u_*|$ and $|P_{e_*}\nabla u_*|$ are positive, uniform ellipticity gives C^2 convergence. The direction formula (5.17), nonnegativity, and the vanishing integrals show that $\Gamma_{e_*}(u_*) = 0$ on those compact sets, and hence the limiting tangential unit direction is locally constant there. On the regular set where $|P_{e_*}\nabla u_*| = 0$, the Sobolev level-set rule gives $\Gamma_{e_*}(u_*) = 0$ almost everywhere; on the critical set it is zero by definition. Thus $\Gamma_{e_*}(u_*) = 0$ almost everywhere globally, without subtracting weak Hessian limits. Theorem 5.2 makes u_* planar, contradicting $D_*(0) \geq \delta$. □

Write $\mathcal{B}_\delta = \{u = 0, D \geq \delta\}$. Uniform C^1 continuity gives fixed source balls: there are $r, b_0 > 0$ such that

$$\int_{B_r(z)} q^{p+1} D\, dx \geq b_0 \qquad (z \in \mathcal{B}_\delta). \tag{5.25}$$

Also define the tilt mass in direction e by

$$T_e(R) = \int_{B_R} s^p (1 - (\nu \cdot e)^2)\, dx.$$

Corollary 5.5. *For each $h > 0$ and sufficiently large R,*

$$|B_h(\mathcal{B}_\delta) \cap B_R| \leq C_{h,\delta} R^{-2} \int_{B_{3R}} e_p(u)\, dx, \tag{5.26}$$

$$|B_h(\mathcal{B}_\delta) \cap B_R| \leq C_{h,\delta} R^{-2} T_e(3R) \quad \text{for every } e \in S^{n-1}. \tag{5.27}$$

In particular, under $E_p(u; B_R) \leq \Lambda R^{n-1}$ the left side is $O(R^{n-3})$. If a sequence satisfies $R^{1-n} T_e(3R) \to 0$, it improves to $o(R^{n-3})$.

Proof. To prove (5.26), choose a maximal disjoint family of the source balls centered at bad zeros in B_{R+h}. A fixed enlargement covers $B_h(\mathcal{B}_\delta) \cap B_R$. The enlargement factor depends only on h, δ and the fixed source radius. For sufficiently large R, every selected ball lies in B_{2R}. Their disjointness and (5.25) bound their number by $b_0^{-1} \int_{B_{2R}} q^{p+1} D$. Now use (5.1) with a cutoff supported in B_{3R} and the inequality $q^p \leq 2p e_p$.

For (5.27), choose a maximal disjoint family of detector balls $B_L(z)$ with centers in the same bad set. Their fixed enlargements cover the required neighborhood. For large R their union lies in B_{2R}, and (5.24) bounds their number by $b^{-1} \int_{B_{2R}} \Gamma_e$. Apply (5.18) with a cutoff equal to one on B_{2R}, supported in B_{3R}, and with gradient bounded by C/R. This gives the claimed uniform estimate for every e.

For the point counts, let $z_1, \ldots, z_N$ be d-separated bad zeros in B_R. The balls $B_r(z_i)$ have overlap at most $C(n)(1 + r/d)^n$: indeed, the disjoint balls $B_{d/2}(z_i)$ whose centers lie within r of a given point all fit in a ball of radius $r + d/2$. For large R, all source balls lie in B_{2R}. Summing (5.25) and using this overlap bound gives

$$N b_0 \leq C_{r,d} \int_{B_{2R}} q^{p+1} D \leq C_{r,d} R^{-2} E_p(u; B_{3R}).$$

The same argument with $B_L(z_i)$, (5.24), and the directional inequality gives $N b \leq C_{L,d} R^{-2} T_e(3R)$. This proves the same two counting bounds; no maximality of the collection is required.

□

Set

$$\mathcal{P}_u = W_p(u) - \frac{p-1}{p} s^p = \frac{q^p D}{2p} \geq 0.$$

We repeatedly use $e_p = s^p + \mathcal{P}_u$.

Proposition 5.6. *Let $n \geq 1$ be arbitrary and $p > 1$, and let $u \in C^1(\mathbb{R}^n)$ be an entire matched weak solution with $|u| \leq 1$. The stress tensor*

$$\mathbb{S} = e_p(u)I - |\nabla u|^{p-2}\nabla u \otimes \nabla u$$

satisfies, for every $X \in C_c^1(\mathbb{R}^n;\mathbb{R}^n)$,

$$\int_{\mathbb{R}^n} \Big(e_p(u)\,\mathrm{div}\, X - |\nabla u|^{p-2}\nabla u \cdot DX\nabla u \Big)\, dx = 0.$$

The second tensor term is defined as zero at critical gradients. Thus $\mathrm{div}\,\mathbb{S} = 0$ in distributions, including across the critical set. Interpret $s^{p-2}(\partial_r u)^2$ as zero where $s = 0$. For almost all $R > 0$,

$$\begin{aligned} &\frac{d}{dR}\Big(R^{1-n}E_p(u;B_R(y))\Big) \\ &\quad = R^{-n}\int_{B_R(y)} \mathcal{P}_u\, dx + R^{1-n}\int_{\partial B_R(y)} s^{p-2}(\partial_r u)^2\, d\mathcal{H}^{n-1}. \end{aligned} \tag{5.28}$$

Hence M_R is nondecreasing. If the limiting density is $\Lambda < \infty$, it is independent of the fixed center, and $M_R(u;y) \leq \Lambda$ for every y, R.

Proof. Use the C^1 extension $\widehat{W}_p(t) = |1-t^2|^p/(2p)$ and the integrand $\mathcal{L}(t,\xi) = |\xi|^p/p + \widehat{W}_p(t)$. It is C^1 and strictly convex in ξ. The solution is C^1, hence locally Lipschitz. These integrand and convexity properties hold for every $p > 1$. Apply the local Pucci–Serrin identity of Degiovanni–Musesti–Squassina [12, Lemma 1, identity (3), p. 319] with zero forcing. Since $\mathcal{L}$ has no explicit spatial dependence, it gives the stationary stress identity

$$\int \Big(e_p\,\mathrm{div}\, X - s^{p-2}\nabla u \cdot DX\nabla u \Big)\, dx = 0 \qquad (X \in C_c^1(\mathbb{R}^n;\mathbb{R}^n)).$$

Follow the radial computation in [8, proof of Theorem 4.4, Eq. (4.9), pp. 1467–1469]. Take $X(x) = \eta(r)(x-y)$, with $r = |x-y|$ and a smooth compactly supported radial cutoff η that is constant near zero. Then $\mathrm{div}\, X = n\eta + r\eta'$ and

$$s^{p-2}\nabla u \cdot DX\nabla u = s^p\eta + r\eta' s^{p-2}(\partial_r u)^2.$$

Thus stationarity reads

$$\int (ne_p - s^p)\eta + \int r\eta'(e_p - s^{p-2}(\partial_r u)^2) = 0.$$

Approximate the indicator of $(0,R)$ by these cutoffs. Coarea gives, for almost every R, $R\int_{\partial B_R}(e_p - s^{p-2}(\partial_r u)^2) = \int_{B_R}(ne_p - s^p)$. Since $E_p'(R) = \int_{\partial B_R} e_p$ at these radii, rearrangement gives

$$RE_p'(R) - (n-1)E_p(R) = \int_{B_R(y)} (e_p - s^p) + R\int_{\partial B_R(y)} s^{p-2}(\partial_r u)^2.$$

Here their normalization is $\Phi(t) = 2t^{p/2}/p$ and $F = \widehat{W}_p$, so their energy is twice ours. We retain the boundary term in their identity. Insert $e_p - s^p = \mathcal{P}_u$ to obtain (5.28). Its right side is nonnegative. Finally, the inclusions $B_{R-|y|}(0) \subset B_R(y) \subset B_{R+|y|}(0)$ prove independence of the asymptotic center. □

Proposition 5.7 (Finite-density discrepancy decay)**.** *Let $n \geq 2$ be arbitrary and $p > 2$. Let u be a scalar-stable entire matched solution with $|u| \leq 1$ and $M_\infty(u) \leq \Lambda$. For $R \geq 1$,*

$$R^{1-n}\int_{B_R(y)} \mathcal{P}_u\, dx \leq C\Lambda^{p/(p+1)}R^{-(2p-1)/(p+1)}. \tag{5.29}$$

The constant is uniform on compact exponent ranges.

Proof. For a constant well the discrepancy is zero, so the claim is immediate. Otherwise use (5.1) with a cutoff equal to one on $B_R(y)$ and supported in $B_{2R}(y)$. Since $q^p \le 2pe_p$ and monotonicity bounds the energy at every center, this gives $\int_{B_R(y)} q^{p+1}D \le C\Lambda R^{n-3}$. Hölder's inequality, $D \le 1$, and the volume of B_R give

$$\int_{B_R} q^p D \le \left(\int_{B_R} q^{p+1} D\right)^{p/(p+1)} |B_R|^{1/(p+1)} \le C\Lambda^{p/(p+1)} R^{n-3p/(p+1)}.$$

Divide by $2pR^{n-1}$ to finish. □

A varifold records both an interface measure and its tangent plane. For $m = n-1$, write $G(n,m)$ for the compact space of unoriented m-planes, identified with their orthogonal projection matrices. A varifold on $\mathbb{R}^n$ is a Radon measure on $\mathbb{R}^n \times G(n,m)$. Its weight $\|V\|$ is the marginal measure in x, and its first variation is

$$\delta V(X) = \int \operatorname{tr}(SDX(x))\, dV(x,S) \qquad (X \in C_c^1(\mathbb{R}^n;\mathbb{R}^n)).$$

It is stationary when this expression vanishes for every such X. It is rectifiable when, outside a set of weight zero, its planes are the tangent planes of a countably m-rectifiable set and its weight is a positive measurable multiple of $\mathcal{H}^m$ there. Rectifiability does not require that multiple to be an integer.

Theorem 5.8 (Stationary rectifiable blowdown)**.** *Let $n \ge 2$ be arbitrary. Let u_j be stable entire solutions in $\mathbb{R}^n$ with fixed $p > 2$ and a uniform finite-density bound. Take $R_j \to \infty$, arbitrary centers y_j, and define $\epsilon_j = R_j^{-1}$ and $v_j(x) = u_j(y_j + R_j x)$. Along a subsequence, the diffuse varifolds*

$$V_j(\Phi) = \int \Phi(x, I - \nu_j \otimes \nu_j)\epsilon_j^{p-1}|\nabla v_j|^p\, dx$$

(the plane is arbitrary where $\nabla v_j = 0$, since the weight there is zero) converge to a stationary rectifiable $(n-1)$-varifold V. Its weight also equals the limit of

$$\mu_j = \left(\frac{\epsilon_j^{p-1}}{p}|\nabla v_j|^p + \frac{1}{\epsilon_j}W_p(v_j)\right) dx.$$

The support of V is the local Hausdorff limit of the zero sets. On nonempty support its density has local positive lower and finite upper bounds. No integrality, stable-varifold regularity, or diffuse sheeting is asserted here.

Proof. Finite density bounds $\mu_j(B_r(x))$ by $C\Lambda r^{n-1}$ for all centers and radii. These bounds give local tightness as measures on the product of space and the compact Grassmannian. A diagonal subsequence on exhausting balls therefore gives a weak varifold limit. Energy and kinetic weights differ by

$$\mathcal{P}_j = \epsilon_j^{-1}W_p(v_j) - \frac{p-1}{p}\epsilon_j^{p-1}|\nabla v_j|^p \ge 0.$$

For each fixed $r > 0$, Proposition 5.7 gives

$$\int_{B_r(x)} \mathcal{P}_j \le C\Lambda^{p/(p+1)} r^{n-1}(R_j r)^{-(2p-1)/(p+1)} \longrightarrow 0.$$

Here $R_j r \ge 1$ once j is large. Thus the discrepancy tends to zero in local L^1. The two weights therefore share a limiting Radon measure μ.

For a vector field $X \in C_c^1(\mathbb{R}^n;\mathbb{R}^n)$, rescale the stress identity. If $k_j = \epsilon_j^{p-1}|\nabla v_j|^p$, then

$$\int \left((k_j + \mathcal{P}_j)\operatorname{div} X - k_j \nu_j \cdot DX\nu_j\right) = 0.$$

Since $\operatorname{tr}((I-\nu_j\otimes\nu_j)DX)=\operatorname{div}X-\nu_j\cdot DX\nu_j$, it follows that

$$\delta V_j(X)=-\int \mathcal{P}_j \operatorname{div} X.$$

The right side tends to zero. The integrand defining first variation is continuous and compactly supported on the space-plane product, so weak varifold convergence gives $\delta V(X)=0$.

For the lower density bound, start at an unscaled core point $|u|\le b<1$. The gradient bound gives a fixed ball with a positive lower bound for $W_p(u)$. Hence $E_p(u;B_{r_0})\ge c_1>0$. Monotonicity (5.28) then yields $E_p(u;B_r)\ge c_2 r^{n-1}$ for $r\ge r_0$. After scaling, each zero of v_j has this lower density at all radii at least $r_0\epsilon_j$.

Conversely, clearing (5.15) makes energy vanish on compact regions a fixed distance from the zero sets. The rescaled pointwise bound for the energy density is

$$C\epsilon_j^{-1}\exp[-c\operatorname{dist}(x,\{v_j=0\})/\epsilon_j].$$

If $z_j\in\{v_j=0\}$ and $z_j\to z$, the preceding lower bound at the zeros, applied on $B_{r/4}(z_j)$, gives $\mu(B_r(z))\ge cr^{n-1}$ for every fixed $r>0$. One passes to the weak limit using a continuous cutoff equal to one on $B_{r/2}(z)$ and supported in $B_r(z)$. Thus $z\in\operatorname{supp}\mu$. If, on the other hand, a point $z\in\operatorname{supp}\mu$ had distance at least $d>0$ from the zero sets along a subsequence, the exponential estimate would force $\mu(B_{d/4}(z))=0$, a contradiction. The limit and its support may both be empty.

These two sequential statements give convergence uniformly on compact subsets. On each compact set, any sequence of zeros with a convergent subsequence has its limit in $\operatorname{supp}\mu$; and every point of $\operatorname{supp}\mu$ is approached by zeros. A contradiction and compactness of a slightly larger ball make both distance statements uniform on compact subsets. This formulation avoids an artificial boundary issue caused by intersecting a closed set with the boundary of a chosen ball. If $V=0$, it says that zeros eventually leave every compact set.

In particular, every support point satisfies $\mu(B_r(z))\ge cr^{n-1}$ at every radius. For the upper bound, choose a continuous cutoff equal to one on $B_r(z)$ and supported in $B_{r+\tau}(z)$, with values in $[0,1]$. Weak convergence and the common energy cap give $\mu(B_r(z))\le C\Lambda(r+\tau)^{n-1}$. Send $\tau\downarrow 0$ to obtain $\mu(B_r(z))\le C\Lambda r^{n-1}$. Stationary-varifold monotonicity makes $\mu(B_r(z))/(\omega_{n-1}r^{n-1})$ nondecreasing in r. The two bounds therefore give a positive finite density limit as $r\downarrow 0$ at every support point. Allard's rectifiability criterion requires locally bounded first variation and positive upper density at μ-almost every point [1, Theorem 5.5(1), pp. 450–451]. Here the first variation is zero, and the density bounds hold everywhere on the support. The criterion therefore gives rectifiability.

All steps above are independent of a dimension upper bound. The discrepancy estimate, stress identity, clearing argument, and positive density bounds hold for every $n\ge 2$. Allard's rectifiability criterion applies in every interface dimension $m=n-1\ge 1$. Thus this compactness theorem requires no dimension restriction from stable minimal hypersurface classification. It gives no integrality conclusion. □

For later uses of equality, consider a stationary rectifiable m-varifold whose ball mass about z equals ϑr^m for every $r>0$. The stationary monotonicity formula then has zero nonnegative annular term, so

$$S^{\perp}(x-z)=0\quad\text{for } V\text{-almost every } (x,S).$$

There is no atom at z when $m\ge 1$. Testing stationarity with $X(x)=\psi(x)(x-z)$ gives

$$\int (m\psi+(x-z)\cdot\nabla\psi)\,d\mu=0. \tag{5.30}$$

To see the scaling of the measure, put

$$F(t) = t^{-m} \int \psi\left(z + \frac{x-z}{t}\right)\, d\mu(x).$$

For t in a compact interval of positive values, the integrands and their derivatives have support in one compact set. Differentiation under the integral and the preceding identity give

$$F'(t) = -t^{-m-1} \int \left[m\psi\left(z + \frac{x-z}{t}\right) + \frac{x-z}{t} \cdot \nabla\psi\left(z + \frac{x-z}{t}\right)\right] d\mu(x) = 0.$$

Thus the weight has degree m under dilation about z, with the normalization t^{-m} used above. Rectifiability determines the plane almost everywhere from that weight, so the entire varifold is invariant under these varifold dilations. Every blowup centered at z is consequently the same translated varifold. If one tangent varifold at z is a weighted plane, the original varifold is that affine plane.

5.3. **Unit-density rigidity.** The unit-density argument uses the geometric PDE method of [43] and the weak-solution paraboloid lemma [40]. We need neither a p version of Hutchinson–Tonegawa integrality nor a degenerate version of Wang's deep-well semiconcavity. In the notation of these sources, take $h_0 = W_p$. Then $W_p(t) \asymp (1 - |t|)^p$ near the wells and $W_p''(t) = (1-t^2)^{p-2}((2p-1)t^2 - 1) > 0$ sufficiently near either well. Thus the potential satisfies their order-p growth and well-convexity hypotheses, with constants uniform on compact exponent intervals. Their one-dimensional inverse profile is $H_{0,p} = a_p^{-1}\,\mathrm{arctanh}$ in our normalization. The final flatness results in [43] require minimization. We prove the PDE and energy replacements needed to apply their geometric argument here.

Write $n = m + 1 \geq 2$ and set $k_p(t) = a_p(1 - t^2)$. Then

$$\sigma_p = \int_{-1}^{1} k_p(t)^{p-1}\, dt = \int_{\mathbb{R}} |g_p'(s)|^p\, ds.$$

Theorem 5.9 (Unit ball-energy cap and asymptotic flatness)**.** *Let $p > 2$. Suppose $u \in C^1(\mathbb{R}^n)$ is a weak solution of*

$$-\Delta_p u = u(1-u^2)^{p-1}, \qquad -1 < u < 1, \qquad u(0) = 0,$$

and satisfies both Modica's inequality

$$|\nabla u| \leq k_p(u)$$

and the global ball-energy bound

$$\int_{B_R(y)} e_p(u) \leq \sigma_p \omega_m R^m \qquad (y \in \mathbb{R}^n,\ R > 0). \tag{5.31}$$

Suppose there are $L_j \to \infty$, $\Theta_j/L_j \to 0$, and unit vectors ν_j for which

$$\{u = 0\} \cap \{\, |x - (x\cdot\nu_j)\nu_j| < L_j,\ |x\cdot\nu_j| < L_j \,\} \subset \{|x\cdot\nu_j| \leq \Theta_j\}. \tag{5.32}$$

Then $u(x) = g_p(x\cdot\nu)$ for some unit vector ν.

We prove this statement through contact projection and flatness improvement. Every cylinder has a fixed outer buffer. Increasing that buffer changes only the universal interior shrink factors.

Lemma 2.3 gives one-sided clearing in zero-free balls. Combine it with Lemma 3.8 and the gradient bound to obtain

$$e_p(u)(x) \leq C \exp[-c\ \mathrm{dist}(x, \{u = 0\})]. \tag{5.33}$$

Only the equation and matched potential are used. These estimates therefore apply in the one-sided contact comparison.

Lemma 5.10 (Unit-density cylinder upper bound)**.** *Let u_j be entire solutions of the equation in Theorem 5.9, with $-1 < u_j < 1$, satisfying Modica's inequality and (5.31). Suppose their zeros in a buffered cylinder of size $C\ell_j$ lie in $|s| \le \theta_j$, with*

$$\ell_j \longrightarrow \infty, \qquad \theta_j/\ell_j \longrightarrow 0.$$

For each fixed $c < C$, choose the outer buffer greater than $c+1$. Then

$$\int_{B^m_{\ell_j}\times(-c\ell_j,c\ell_j)} e_p(u_j) \le (1+o(1))\sigma_p\omega_m\ell_j^m. \tag{5.34}$$

The conclusion is invariant under translations and rotations.

Proof. Put $H_j = \theta_j + A\log\ell_j$ and fix A large. Since $\theta_j/\ell_j \to 0$ and $\log\ell_j/\ell_j \to 0$, we have $H_j/\ell_j \to 0$. The central region $|s| \le H_j$, $|x'| < \ell_j$ fits inside a ball of radius $(\ell_j^2+H_j^2)^{1/2}$. Its energy is at most

$$\sigma_p\omega_m(\ell_j^2+H_j^2)^{m/2} = (1+o(1))\sigma_p\omega_m\ell_j^m.$$

On the rest, distance from every zero is bounded below by the smaller of $|s|-\theta_j$ and a fixed multiple of ℓ_j. The outer buffer ensures this also for uncontrolled zeros outside the cylinder. Indeed, their horizontal or vertical distance from the retained cylinder is at least a fixed multiple of ℓ_j. Clearing (5.33) bounds the remaining energy by $C\ell_j^{m+1}e^{-cA\log\ell_j} = o(\ell_j^m)$ once A is large. This uses no compactness theorem for minimizers. □

Two barrier families are used below. We recall their definitions to separate the radial trapping step from the contact projection step. Both constructions are due to Sciunzi–Valdinoci and Valdinoci–Sciunzi–Savin [40, Lemma 5.1, pp. 333–338] [43, Chapter 2, pp. 7–25]. Their potential h_0 is W_p here. Their N is our n.

For the radial trapping family, choose the constants $c_1, C_1, C_2 > 0$ in [43, Lemma 2.20, pp. 24–25], and let l be large. Put $d_l = e^{-c_1 l}$ and $m_l = -1 + d_l$. Define

$$h_l(s) = \begin{cases} W_p(s) - W_p(m_l) - \dfrac{C_2}{l}((1+s)^p - d_l^p), & m_l < s < 0, \\ W_p(s) + W_p(1-d_l) + \dfrac{C_2}{l}((1-s)^p + d_l^{p-1}(1-s)), & 0 \le s < 1, \end{cases}$$

$$H_l(s) = \int_0^s \left(\frac{p-1}{ph_l(\zeta)}\right)^{1/p} d\zeta.$$

The construction gives $h_l > 0$ on $(m_l, 1)$. Thus H_l is strictly increasing. Set $T_l = H_l(1)$. The profile g_l is H_l^{-1} on $(H_l(m_l), T_l]$, and it is m_l on $(-\infty, H_l(m_l)]$. There is a constant $c_b > 0$ such that

$$c_b l \le T_l \le l/2, \qquad -l/2 \le H_l(m_l) \le -c_b l, \qquad g_l(0) = 0.$$

The upper radial barrier is

$$\Psi_{y,l}(x) = g_l(|x-y| - l), \qquad |x-y| < l + T_l.$$

It equals one on the outer boundary. Its zero set is the sphere $\partial B_l(y)$. Its negative constant region is a ball. On every smooth nonconstant branch outside this zero sphere,

$$\Delta_p\Psi_{y,l} < W_p'(\Psi_{y,l}).$$

The two branches need not have matching flux at the zero sphere. We will not compare across that sphere by a global weak supersolution inequality. The inverse-profile estimate needed for trapping is

$$H_{0,p}(s) \le H_l(s) - \frac{C_1}{l}\log(1-|s|) \quad \text{if } |s| < 1 - e^{-c_1 l/2}.$$

These are exactly the properties in [43, Lemma 2.20, (2.58)–(2.63)]. On each compact subinterval of $(-1,1)$, $h_l \to W_p$ and $H_l \to H_{0,p}$. Hence $g_l \to g_p$ on bounded intervals.

The contact projection uses a different family. Fix $s_0 \in [-1/4, 1/4]$ and a large R. Choose the constants $\overline{C}_0 \gg \widehat{C}_0 \gg 1$ from [43, Definitions 2.1 and 2.5, pp. 7–10]. Let $s_R \in (-1, -1/2)$ be the negative solution of $W_p(s_R) = R^{-1}$, and set

$$\varphi_{s_0,R}(s) = \frac{W_p(s)R^p}{[R - \overline{C}_0(s - s_0)k_p(s)]^p},$$

$$h_{s_0,R}(s) = \begin{cases} W_p(s) - W_p(s_R) - \dfrac{\widehat{C}_0}{R}(s - s_R), & s_R \le s \le -1/2, \\ \varphi_{s_0,R}(s), & -1/2 < s < 1/2, \\ W_p(s) + W_p(s_R) + \dfrac{\widehat{C}_0}{R}(1 - s), & 1/2 \le s \le 1. \end{cases}$$

This potential is positive on $(s_R, 1]$. It may jump at $s = \pm 1/2$. Define

$$H_{s_0,R}(s) = H_{0,p}(s_0) + \int_{s_0}^{s} \left(\frac{p-1}{p h_{s_0,R}(\zeta)} \right)^{1/p} d\zeta.$$

Its inverse $g_{s_0,R}$ is extended by the constant s_R to the left of $H_{s_0,R}(s_R)$. For $Y = (y, s_0) \in \mathbb{R}^n \times [-1/4, 1/4]$, write

$$B_{Y,R}(x) = g_{s_0,R}(H_{0,p}(s_0) + |x - y| - R), \qquad \mathcal{S}(Y, R) = \operatorname{graph} B_{Y,R}.$$

The domain ends where $B_{Y,R} = 1$. The level s_0 has radius R. On the core, $H'_{0,p} k_p = 1$. Taking the inverse pth root of $\varphi_{s_0,R}$ and integrating therefore gives the exact identity

$$H_{s_0,R}(s) = H_{0,p}(s) - \frac{\overline{C}_0}{2R}(s - s_0)^2 \qquad (|s| < 1/2).$$

The endpoint heights have size at most $C \log R$. The barrier is a strict supersolution where its value lies outside $(-1/2, 1/2)$. At the two intermediate levels its radial derivative jumps down. A C^1 function cannot touch it from below there. The supersolution assertion is not needed on the core. These facts are proved in [43, Definitions 2.8, 2.11–2.12 and Proposition 2.13, pp. 13–16].

For completeness, the strict differential inequality on each smooth branch follows from a scalar calculation. If $(p-1)|b'|^p/p = h(b)$ and $b' > 0$, then

$$(p-1)|b'|^{p-2}b'' = h'(b), \qquad \Delta_p b - W'_p(b) = h'(b) - W'_p(b) + \frac{n-1}{r}(b')^{p-1}.$$

The modifications above make $h' - W'_p$ negative enough to dominate the last term. On a nonconstant branch, $r \ge l/2$ for $\Psi_{y,l}$ and $r \ge R/2$ for $B_{Y,R}$, once the radius is large. For $\Psi_{y,l}$, this holds on both sides of its zero sphere. For $B_{Y,R}$, it is asserted only outside the core.

We also give the contact argument at the flat part. It uses only weak comparison and the C^1 regularity of u. For either family, its negative plateau is a value $m \in (-1, 0)$. Suppose the barrier is above u and touches it at an interior point of that plateau. Since $u \le m$ and $u(x_0) = m$, $\Delta_p u = W'_p(u) \ge c_0 > 0$ on a small ball $B_\rho(x_0)$. Put $p' = p/(p-1)$ and choose $k_* > 0$ by $n(k_* p')^{p-1} = c_0/2$. The function

$$\psi(x) = m - k_*(\rho^{p'} - |x - x_0|^{p'})$$

equals m on the boundary. Its flux is $(k_* p')^{p-1}(x - x_0)$, so $\Delta_p \psi = n(k_* p')^{p-1} = c_0/2$ weakly in the ball. Weak comparison gives $u \le \psi$, contradicting $u(x_0) = m > \psi(x_0)$. The number c_0 is positive for this fixed barrier. No bound independent of its radius is needed here.

At the spherical junction of the plateau and the increasing branch, the profile is C^1 and its derivative is zero. Choose a ball $B_\rho(z)$ in the plateau which is internally tangent at a proposed

contact point x_0. The preceding argument gives $v = m-u > 0$ in that ball, and $\Delta_p v = -W_p'(u) \le 0$ there. Let $\gamma = \min_{\partial B_{\rho/2}(z)} v > 0$. On the annulus $\rho/2 < |x-z| < \rho$, compare v with

$$w(x) = \gamma \frac{\displaystyle\int_{|x-z|}^{\rho} t^{-(n-1)/(p-1)}\,dt}{\displaystyle\int_{\rho/2}^{\rho} t^{-(n-1)/(p-1)}\,dt}.$$

This function is p-harmonic, equals γ on the inner sphere, and vanishes on the outer sphere. Weak comparison gives $v \ge w$. Along the inward normal at x_0 it follows that $v(x_0 - t\nu) \ge ct$ for small $t > 0$. First contact between the C^1 functions instead gives $\nabla v(x_0) = 0$, a contradiction. The local behavior of the profile at this junction is $b - m \asymp (t - t_0)_+^{p/(p-1)}$. Its flux vanishes continuously there, so the branch inequalities also pass weakly across the flat junction. No uniform ellipticity or $C^{1,1}$ bound is used at that junction.

At a smooth nonconstant contact, $\nabla u = \nabla b \ne 0$. The equation for u is then locally uniformly elliptic and u is C^2. The Hessian comparison at a minimum of $b-u$ contradicts $\Delta_p b < W_p'(b) = \Delta_p u$. At a derivative jump of $B_{Y,R}$, the one-sided radial derivatives already exclude a C^1 lower contact. The outer value one cannot touch a strict-range solution. We have therefore justified the weak-solution contact rules of [43, Corollary 2.14 and Lemma 2.21, pp. 16–17 and 25]: interior contact with $\Psi_{y,l}$ can occur only on its zero sphere; interior contact with $\mathcal{S}(Y,R)$ can occur only at $|u| < 1/2$. At those latter contacts the barrier slope is bounded above and away from zero.

We use the contact geometry of Valdinoci–Sciunzi–Savin. The result in [43, Proposition 3.14, pp. 40–42] concerns a C^1 weak subsolution on a buffered cube, with $1/c < l < cR$. Its full center-family conditions (3.2)–(3.3) are part of the hypotheses. In particular, the boundary condition concerns every ordered center with a retained parameter projection. The projected parameter domain is the closure of an open set, and every parameter must reach a crossing. For the seed below, we use the local center-map calculation in [43, Lemma 3.6 and proof of Proposition 3.14]. We construct a selected family of negative centers, check its boundary exclusions, and derive its projected measure bound directly. The full construction and area-formula argument are given in the proof of Lemma 5.11. The two local extensions of a contact are [43, Lemmas 4.1–4.2, pp. 43–44]. The first requires a weak solution and a contact normal making angle at most $\pi/8$ with e_n. It recovers actual graph points above every projected point in a fixed neighborhood, with the inverse-profile height estimate used below. The second requires only a C^1 weak subsolution. Its contact has $|u(x_0)| < 1/2$, $|x_{0,n}| < l/4$ and $q = |x_0'| < l/4$, after translation of the horizontal center. For fixed constants $C_c, A > 1$, it also requires $q \ge C_c$ and $4R^{1/3} \le l \le cR$. It produces a fraction cq^m of new projected contacts at radius R/A. Their distance from x_0 is at most $2q$; their normal changes by at most $C_c q/R$; and their normal height increases by at most $C_c q^2/R + H_{0,p}(u) - H_{0,p}(u(x_0))$. The geometric covering result is [43, Lemma 4.3, pp. 44–45]. Its contact sets must be nested. We use cumulative contact sets below to satisfy that requirement. None of these three results requires minimization or scalar stability.

The minimizing hypothesis in [43, Lemma 5.1 and Proposition 5.2, pp. 47–59] is used to obtain their profile bound (5.1) from Lemma 2.23. The proof of that bound starts with lower-side clearing (2.64). We establish the same ordering directly below, using the radial family and our clearing estimate. The rest of the seed and contact enlargement argument uses exactly the preceding weak-solution estimates. In Chapter 6, the other minimizing inputs are the initial phase signs, well attainment at the buffered ends, and the final energy limit (6.16). We prove the signs by projection, the end values by clearing, and replace the final energy limit by Lemma 5.10. For harmonic approximation we use the weak-solution paraboloid lemma [40, Lemma 6.6, p. 346], with its initial well ordering checked below. Its minimizer corollary, Lemma 9.3, is not used.

The following one-sided projection estimate determines the exterior phase signs without assuming them.

Lemma 5.11 (One-sided contact projection)**.** *There are universal constants $C > 1$ and $\vartheta \in (0,1)$ with the following consequence. Let u_j be entire weak solutions as above, with $u_j(0) = 0$, and suppose*

$$u_j(x', s) < 0 \quad \text{for } s < -\theta_j$$

in a buffered cylinder of size $C\ell_j$. Fix $\theta_0 > 0$ and suppose $\theta_j \geq \theta_0$ and $\theta_j/\ell_j \to 0$. Given an integer $k \geq 1$, for all sufficiently large j there is a subset of the graph of u_j, consisting of points with $|x'| < \ell_j$, $|u_j| < 1/2$ and $|s| < C\ell_j$, whose projection onto $(x', t) = (x', u_j)$ covers at least

$$[1 - (1-\vartheta)^k]\,|B^m_{\ell_j} \times (-1/2, 1/2)|. \tag{5.35}$$

The contact set can be chosen with the more precise height bounds

$$-\theta_j - C_0 \leq s \leq C_0 C^k \theta_j + H_{0,p}(t), \qquad H_{0,p}(t) = a_p^{-1} \operatorname{arctanh} t. \tag{5.36}$$

Here C_0 and C are universal, while k is fixed before letting j tend to infinity. In particular its entire height range is $o(\ell_j)$.

Proof. We adapt the proof of Valdinoci–Sciunzi–Savin [43, Proposition 5.2]. The clearing estimate replaces their minimizing input, as checked below.

We first establish the upper profile bound used as (5.1) in [43]. This is the upper-barrier part of its Lemma 2.23, with its minimizing input replaced by clearing and weak comparison. Put $l = \ell_j$. All comparison balls below have radius at most $3l/2$. Enlarge the fixed outer buffer to contain their full paths. Their zero spheres lie below $s = -\theta_j$. Their upper profile domains may extend a distance $l/2$ above that plane; the equation and strict range hold on those domains.

Place the initial center at height $-\theta_j - 4\ell_j$. Its whole profile domain then lies a fixed multiple of ℓ_j from every zero. One-sided clearing gives $u_j \leq -1 + t_*/2$ there for large j, where $t_* > 0$ is fixed so that W_p' is increasing on $[-1, -1 + t_*]$. This gives the ordering with the radial barrier even on its plateau. Indeed, in the ball

$$\Omega_- = \{\Psi_{y,l} < -1 + t_*\}$$

the radial barrier is a weak supersolution, including its flat junction. Its boundary value is $-1 + t_*$. The function $\phi = (u_j - \Psi_{y,l})_+$ therefore has zero trace. Subtract the supersolution inequality from the equation and test with ϕ. This gives

$$\int_{\{u_j > \Psi_{y,l}\}} (|\nabla u_j|^{p-2}\nabla u_j - |\nabla \Psi_{y,l}|^{p-2}\nabla \Psi_{y,l}) \cdot \nabla \phi + \int_{\Omega_-} (W_p'(u_j) - W_p'(\Psi_{y,l}))\phi \leq 0.$$

Both terms are nonnegative. Flux monotonicity and the zero trace force $\phi = 0$. Outside Ω_- the clearing bound already gives $u_j < \Psi_{y,l}$. The first-contact rule proved above excludes equality anywhere inside the initial profile domain, since its zero sphere is still in the negative phase.

Fix a horizontal point x' in the retained base. Move this center vertically upward, with horizontal coordinate x', until it reaches

$$y = (x', -\theta_j - l).$$

At every earlier position the whole zero sphere lies strictly below $s = -\theta_j$. A first interior contact is therefore impossible. A contact on the moving outer boundary is also impossible, since that boundary value is one and $u_j < 1$. Compactness of the fixed buffered path and continuity justify the usual first-contact argument. The ordering holds at the final position by continuity. At a point (x', s) with $s \geq -\theta_j - l$ and $s + \theta_j < T_l$, its radial argument is

$$|(x', s) - y| - l = s + \theta_j.$$

We obtain $u_j(x', s) \le g_l(s + \theta_j)$ there.

For the remaining points $s < -\theta_j - l$ in the retained cylinder, the desired right side is the constant plateau m_l. The ball buffer and the lower exterior sign give distance at least l from every zero. Since $u_j < 0$ there, exponential clearing and the lower bound $W_p(u_j) \ge c(1 + u_j)^p$ yield

$$1 + u_j \le Ce^{-c_* l}.$$

In the radial construction choose $c_1 < c_*$, as permitted by the smallness choice of c_1 in [40, proof of Lemma 5.1, pp. 333–338]. For large j, this is at most $e^{-c_1 l} = 1 + m_l$. The upper profile bound follows on the whole retained cylinder where the profile is defined. For $s + \theta_j \ge T_l$, its pointwise extension by one gives the same bound from the strict range alone. This extension is used only for ordering, never as a weak barrier. Thus

$$u_j(x', s) \le g_{\ell_j}(s + \theta_j)$$

with that pointwise extension understood. This is precisely the upper profile input (5.1) of [43]. It uses no upper exterior sign. Apply the same argument to $-u_j(x', -s)$ to obtain the lower bound when the upper exterior phase is positive.

We now construct the seed directly for these weak solutions. This follows the geometric argument of [43, proof of Lemma 5.1, (5.3)–(5.31)]. We do not apply that lemma's minimizing hypothesis. Write $l = \ell_j$ and $\theta = \theta_j$ during this construction. Choose a large fixed C and then C'' with $(C'')^2 \gg C$. Put

$$R_0 = \frac{l^2}{C\theta}, \qquad \mathcal{D}_j = \overline{B}^m_{l/C''} \times [-1/4, 1/4].$$

The choices of C, C'' are independent of θ_0. The threshold in j may depend on θ_0. Keep a comparison cube $K_j = \{|x'| < C'l,\ |s| < C'l\}$, with fixed $C' > 1$, inside the larger buffer used for the profile ordering. Enlarge that fixed outer buffer if necessary.

For $(\zeta, s_0) \in \mathcal{D}_j$, use the contact barrier with center $Y = (\zeta, y_n, s_0)$ and radius R_0. Its zero sphere has exact radius

$$r_{s_0} = R_0 + H_{s_0,R_0}(0) - H_{0,p}(s_0) = R_0 - H_{0,p}(s_0) - \frac{\overline{C}_0 s_0^2}{2R_0} = R_0 + O(1).$$

Its outer radius is $r_{s_0} + H_{s_0,R_0}(1) - H_{s_0,R_0}(0)$. The added term is bounded by $D_H(1 + \log R_0)$, uniformly in s_0 and $p \in \mathcal{P}$. Here D_H is fixed. Also $R_0/l \to \infty$ and $\log R_0 = O_{\theta_0}(1 + \log l) = o(l)$. Extend the barrier by the value one outside this outer domain only for pointwise ordering. This extension is continuous. It is never used as a weak supersolution across that boundary.

We first compare the two inverse profiles on their whole common range. Uniformly for $s_0 \in [-1/4, 1/4]$ and $t \in [s_{R_0}, 1]$,

$$H_{s_0,R_0}(t) - H_{s_0,R_0}(0) \le H_l(t) + \theta/2$$

for all large j. Here are the endpoint checks in this scalar comparison. The exact core identity gives $H_{s_0,R_0}(t) - H_{s_0,R_0}(0) \le H_{0,p}(t) + C/R_0$ on the core. On the negative outer branch $h_{s_0,R_0} \le W_p$; on the positive outer branch $h_{s_0,R_0} \ge W_p$. Integrating in the appropriate direction preserves this upper bound on the whole common range. Choose $c_* > 0$ small in terms of θ_0. The displayed definitions of the potentials give

$$h_{s_0,R_0}(t) - h_l(t) \le -R_0^{-1} + C_2 c_*^p l^{-2} + Ce^{-cl} < 0 \quad (1 + t \le c_* l^{-1/p}),$$
$$h_{s_0,R_0}(t) - h_l(t) \ge R_0^{-1} - C_2 c_*^p l^{-2} - Ce^{-cl} > 0 \quad (1 - t \le c_* l^{-1/p}).$$

Indeed, $R_0^{-1} = C\theta/l^2 \ge C\theta_0/l^2$. The negative contact plateau lies above the radial plateau: $s_{R_0} > m_l$, since $1 + s_{R_0} \asymp R_0^{-1/p}$ whereas $1 + m_l = e^{-c_1 l}$. Moreover $1 + s_{R_0} = o(l^{-1/p})$. Thus the difference of the inverse profiles is nondecreasing on its negative endpoint strip and nonincreasing

on its positive endpoint strip. Its maximum occurs in $|t| \le 1 - c_* l^{-1/p}$. On this interval the earlier radial inverse bound gives

$$H_{s_0,R_0}(t) - H_{s_0,R_0}(0) - H_l(t) \le C/R_0 + C_{\theta_0} \frac{1 + \log l}{l} = o(\theta).$$

This proves the required $\theta/2$ bound, including the endpoints by continuity. These calculations use only the explicit profiles.

For each parameter, slide y_n upward from

$$y_n^{\text{start}} = -r_{s_0} - C'l - D_H(1 + \log R_0) - 1 \quad \text{to} \quad y_n^{\text{end}} = -\sqrt{r_{s_0}^2 - |\zeta|^2} + \theta/32.$$

At the initial position the entire outer domain is below K_j, so its extension is one and is strictly above u_j there. At the final position the origin is strictly inside the zero sphere. Hence the barrier is negative at the origin, where $u_j(0) = 0$. Every parameter therefore reaches a first contact. Both centers are negative for large j. Choose C'' after C so that

$$r_{s_0} - \sqrt{r_{s_0}^2 - |\zeta|^2} \le |\zeta|^2 / r_{s_0} \le \theta/32.$$

Throughout every sliding path, $y_n + r_{s_0} \le \theta/16$. In particular it is at most $\theta/8$.

We check every boundary before retaining a contact. Let $d_1 = |x - (\zeta, y_n)| - r_{s_0}$ at a point in the physical barrier domain. If $l \le |x'| \le C'l$, then $|x' - \zeta| \ge (1 - 1/C'')l$. The outer-radius bound gives $|x - (\zeta, y_n)| \le 2r_{s_0}$ for large j. The vertical coordinate is above the center, since $|s| \le C'l = o(R_0)$. Subtracting the vertical leg from the hypotenuse gives

$$d_1 \ge s - (y_n + r_{s_0}) + c\frac{l^2}{r_{s_0}} \ge s + 2\theta,$$

once C is fixed large enough. The scalar inverse comparison then gives $B_{Y,R_0}(x) > g_l(s+\theta) \ge u_j(x)$ in this side annulus. On a plateau, use $d_1 \le H_{s_0,R_0}(s_{R_0}) - H_{s_0,R_0}(0)$ in the same inequalities. They give $g_l(s+\theta) < s_{R_0}$ there too. If the radial profile is extended by one, the inequalities instead show that this point is outside the physical contact-barrier domain; its value one is again strictly above u_j. Thus all contacts have $|x'| < l$.

On the bottom $s = -C'l$, lower-side clearing gives $1 + u_j \le Ce^{-cl}$, whereas the barrier is at least s_{R_0} and $1 + s_{R_0} \asymp R_0^{-1/p} \ge c_{\theta_0} l^{-2/p}$. The bottom ordering is therefore strict. The top of the physical barrier domain is at most $\theta/8 + D_H(1 + \log R_0) = o(l)$, so it cannot meet the top of K_j. On its own outer boundary the barrier has value one and cannot touch the strict-range solution. All first contacts are therefore interior. The weak contact rules proved above give $|u_j| < 1/2$ and a nonzero barrier slope at each of them.

Let $\Xi_{0,j}$ collect the ordered contacts in these bounded negative-center sliding families. Continuity makes this set compact; every parameter occurs at a first contact. At such a contact put $t = u_j(x)$. The exact core identity and the triangle inequality give

$$\begin{aligned} d_1 &= H_{s_0,R_0}(t) - H_{s_0,R_0}(0) \le H_{0,p}(t) + C/R_0, \\ s &\le y_n + |x - (\zeta, y_n)| \le \theta/8 + H_{0,p}(t) + C/R_0 \le H_{0,p}(t) + \theta/4. \end{aligned}$$

The last step uses $(R_0\theta)^{-1} = C/l^2 \to 0$. Also $|x' - \zeta| \le 2l$ and $|x - (\zeta, y_n)| = R_0 + O(1)$. Thus the phase normal at the contact makes angle at most Cl/R_0 with e_n and has positive vertical component. The profile ordering also gives $s \ge -\theta - C$ at these core contacts.

For the projected measure, we use the local center-map calculation in [43, Lemma 3.6 and proof of Proposition 3.14, pp. 36–42]. We give the localization, since no condition on unrelated positive-center barriers is needed here. The shared noncritical bootstrap at the start of Section 2 makes u_j smooth near each retained contact. If $B = B_{Y,R_0}$ touches from above, then $D^2B - D^2u_j \ge 0$.

Differentiating the exact core inverse profile gives

$$|\nabla u_j| = \partial_r B = \frac{k_p(t)}{1-(\overline{C}_0/R_0)(t-s_0)k_p(t)}, \qquad c \le |\nabla u_j| \le C.$$

The bounds are uniform because $|t| < 1/2$, $|s_0| \le 1/4$, $k_p(t)$ has common positive bounds there, and $R_0 \to \infty$. The core potential is W_p times $[1-\overline{C}_0(t-s_0)k_p(t)/R_0]^{-p}$. Its bounded core derivatives therefore give $h'_{s_0,R_0} - W'_p = O(R_0^{-1})$ uniformly. The radial slope $\partial_r B$ is bounded above and away from zero, and $|x-y| = R_0 + O(1)$ at these core contacts. The radial identity proved above consequently gives

$$\Delta_p B - W'_p(B) = h'_{s_0,R_0}(B) - W'_p(B) + \frac{n-1}{|x-y|}(\partial_r B)^{p-1} = O(R_0^{-1}).$$

Combining this bound with the contact Hessian inequality gives

$$0 \le \operatorname{tr}(A_p(\nabla u_j)(D^2 B - D^2 u_j)) = \Delta_p B - W'_p(B) \le C/R_0.$$

The common core gradient bounds make $A_p(\nabla u_j)$ uniformly positive. Hence $|D^2 B - D^2 u_j| \le C/R_0$. The center of a touching barrier is recovered from $x, t = u_j(x)$, and $\xi = \nabla u_j(x)$ by

$$s_0 = t + \frac{R_0}{\overline{C}_0}\left(\frac{1}{|\xi|} - \frac{1}{k_p(t)}\right), \qquad y = x - \left[R_0 + H_{s_0,R_0}(t) - H_{0,p}(s_0)\right]\frac{\xi}{|\xi|}.$$

Call this map $\mathcal{Y}(x) = (y, s_0)$. Its derivative with respect to ξ has norm at most CR_0 on the retained core range. For a fixed barrier the same formula returns its constant center. Differentiate that constant-center identity and subtract it at a contact, where $u_j = B$ and $\nabla u_j = \nabla B$. The derivatives in x, t cancel, leaving

$$|D_x \mathcal{Y}| \le CR_0 |D^2 u_j - D^2 B| \le C \quad \text{on } \Xi_{0,j}.$$

This is the local estimate used in the cited contact argument.

The projection $\pi(x', s, u_j) = (x', u_j)$ is injective on $\Xi_{0,j}$. Indeed, suppose two contacts had the same (x', t) and heights $s_1 < s_2$. The barrier at the upper contact is ordered above u_j. Both points are above its far-below center, and the lower point is closer to that center. It is therefore inside the barrier domain and has barrier value strictly less than t, contradicting $u_j(x', s_1) = t$. At contacts $\partial_s u_j$ is bounded away from zero, so this projection has local inverse charts with uniformly bounded first derivatives. The map from (x', t) to the projected center (ζ, s_0) consequently has Jacobian bounded by a common constant. Partition $\pi\Xi_{0,j}$ into disjoint measurable pieces subordinate to finitely many such inverse charts. Apply the area formula on each piece and then sum. Every parameter in $\mathcal{D}_j$ occurs, so

$$|\mathcal{D}_j| \le C|\pi\Xi_{0,j}|.$$

No chart-multiplicity factor is introduced. Since $|\mathcal{D}_j| = \frac{1}{2}\omega_m(l/C'')^m$, this proves, with $Q_j = B^m_{\ell_j} \times (-1/2, 1/2)$,

$$|\pi\Xi_{0,j}| \ge \vartheta_* |Q_j|, \qquad s \le H_{0,p}(u_j) + \theta_j/4 \quad \text{on } \Xi_{0,j}, \qquad \vartheta_* > 0.$$

The fraction is fixed independently of j and θ_0. This is the quantitative seed used below. It has been proved from the weak equation, profile ordering, and contact geometry.

The scales in the contact iteration are explicit. Let $A > 1$ be the fixed opening factor in the local contact lemma recalled above. Keep the seed radius R_0 chosen above, and use $R_h = R_0 A^{-h}$ for $0 \le h \le k$. The required dimensionless ratios tend to zero with θ_j/ℓ_j. The cubic-root condition also follows from

$$\frac{R_h^{1/3}}{\ell_j} = \frac{A^{-h/3}}{C^{1/3}(\theta_j \ell_j)^{1/3}} \longrightarrow 0.$$

Here k is fixed first, $\theta_j \ge \theta_0 > 0$, and $\ell_j \to \infty$. The number of contact enlargements is never allowed to grow with j.

For the enlargement, let $\mathscr{C}_{\nu,j}$ be the projected contacts obtained with curvature radius R_ν in the construction of [43, proof of Proposition 5.2, pp. 56–59]. Keep its common buffered comparison cylinder. Choose a fixed angular constant C_n first and a height constant C_h second, as specified below. A retained witness x_ν has $|u_j(x_\nu)| < 1/2$ and satisfies

$$\angle(\nu_\nu, e_n) \le C_n \ell_j / R_\nu, \qquad s_\nu \le H_{0,p}(u_j(x_\nu)) + C_h \ell_j^2 / R_\nu, \qquad \nu_\nu = \frac{\nabla u_j(x_\nu)}{|\nabla u_j(x_\nu)|}.$$

Define the sets for the covering lemma cumulatively:

$$\mathscr{D}_{h,j} = \bigcup_{\nu=0}^{h} \mathscr{C}_{\nu,j}.$$

These sets are nested. The seed just constructed makes $\mathscr{D}_{0,j} \cap Q_j$ nonempty.

We check the local enlargement condition for this union. Let $Z_0 \in \mathscr{D}_{h,j} \cap (B^m_{2\ell_j} \times (-1/2, 1/2))$ and let $Z = (z', t)$ be a projected point with $a_0 \le q = |Z - Z_0| \le 2\ell_j$. Choose a witness of Z_0 with some index $\nu \le h$. Since both value coordinates are in $[-1/2, 1/2]$, its horizontal distance $q' = |x'_\nu - z'|$ obeys

$$\sqrt{q^2 - 1} \le q' \le q.$$

Choose the fixed a_0 large enough that $q' \ge q/2 \ge C_c$. Apply [43, Lemma 4.2] with the horizontal center translated to z', the old radius R_ν, and a comparison size which is a fixed large multiple of ℓ_j. Its contact satisfies $|x_{\nu,n}| = o(\ell_j)$ and $q' \le 2\ell_j$, so both are in the required inner quarter of that comparison cylinder. The normal angle is at most $\pi/8$ for large j. Every radius in this finite list obeys

$$R_\nu^{1/3} / \ell_j \longrightarrow 0, \qquad \ell_j / R_\nu \longrightarrow 0.$$

Thus $4R_\nu^{1/3}$ is below the local comparison size and that size is below cR_ν, as required by the lemma. The largest R_ν, namely R_0, gives the first restriction; the smallest, R_k, gives the second. Both were checked above.

Use the fixed opening factor A of that lemma. Its new radius is exactly $R_\nu/A = R_{\nu+1}$. The new projected contacts lie in $B_{q/10}(Z)$ after increasing a_0 once. Their measure is at least $c(q')^m \ge c' q^m$. On the projected strip $|t| < 1/2$, $|B_q(Z) \cap \{|t| < 1/2\}| \le C q^m$ for $q \ge a_0$. Hence they occupy a fixed fraction of the latter set.

Their normal changes by at most $C_c q'/R_\nu$. Thus the new angle is at most $(C_n + 2C_c)\ell_j/R_\nu$. Their displacement has size at most $2q'$, and its component along ν_ν is at most

$$H_{0,p}(u_j) - H_{0,p}(u_j(x_\nu)) + C_c (q')^2 / R_\nu.$$

Write this scalar inequality as

$$\nu_{\nu,n}(s - s_\nu) + \nu'_\nu \cdot (x' - x'_\nu) \le H_{0,p}(u_j) - H_{0,p}(u_j(x_\nu)) + C_c (q')^2 / R_\nu.$$

For large j, $\nu_{\nu,n} \ge 1/2$. Solving for $s - s_\nu$ introduces a horizontal error at most $C C_n \ell_j^2 / R_\nu$. It also multiplies the inverse-profile difference by $1/\nu_{\nu,n}$. That difference is bounded on the core, and

$$1/\nu_{\nu,n} - 1 = O(C_n^2 \ell_j^2 / R_\nu^2).$$

The extra term is $o(\ell_j^2/R_\nu)$, uniformly over this finite list of levels. The old height s_ν is added after this division. Its bound $C_h \ell_j^2 / R_\nu$ is therefore not multiplied. We obtain

$$s - H_{0,p}(u_j) \le [C_h + C_{\mathrm{loc}} + C_{\mathrm{geom}} C_n + o(1)] \ell_j^2 / R_\nu,$$

with fixed local and geometric constants. Choose C_n large enough to include the seed estimate and satisfy $C_n(A-1) \ge 2C_c$. Then choose C_h large enough to include the seed height bound and satisfy

$$C_h(A - 1) \ge C_{\mathrm{loc}} + C_{\mathrm{geom}} C_n + 1.$$

These choices are possible for the actual fixed $A > 1$. They put the new contact at generation $\nu+1$, since $R_{\nu+1} = R_\nu/A$. With $C_0 = \max\{C_n, C_h\}$, both advertised bounds follow. Since $\nu + 1 \le h + 1$, the new projection belongs to $\mathscr{D}_{h+1,j}$. Its bounds are in particular $C_0\ell_j/R_{h+1}$ and $C_0\ell_j^2/R_{h+1}$. The lower height bound follows from clearing, since a core value cannot lie a fixed distance below the negative strip. All these contacts remain in the common buffered cylinder. The auxiliary comparison agrees with the old one outside a region of horizontal radius Cq; see [43, Appendix A.2, (A.38) and (A.54)]. For the retained smaller spherical barrier, the stronger ordering in (A.138)–(A.139) there applies: outside a fixed multiple of that region it is above the old barrier, and hence above u_j. Its slide and first-contact construction therefore use the same comparison domain. Since $|z'| \le 4\ell_j$ and $q \le 2\ell_j$, all changed regions lie in one fixed larger base. Thus this step does not consume a new outer buffer at each generation.

This proves the local measure condition (P2) of [43, Lemma 4.3] for the nested sets $\mathscr{D}_{h,j}$. The seed gives its condition (P1). This argument applies also when Z_0 has an older witness $\nu < h$. It does not replace its radius by R_h. The same finite list of radius inequalities checks that application, and its new contacts have index $\nu + 1 \le h + 1$. Thus the nested-union definition retains both the local fraction and the height bound $CA^h\theta_j$ needed below.

Fix the large constant a_0 used for the neighborhoods in that covering lemma. Its enlarged projected sets are

$$\mathscr{E}_{h,j} = \{Y : |Y_{m+1}| < 1/2, \operatorname{dist}(Y, \mathscr{D}_{h,j}) < a_0\}.$$

Here $Y = (x', t) \in \mathbb{R}^m \times \mathbb{R}$ is a projected point. The lemma gives

$$|Q_j \setminus \mathscr{E}_{h,j}| \le (1-\vartheta)^h |Q_j| \qquad (1 \le h \le k),$$

with a fixed $\vartheta \in (0, 1)$. Thus the uncovered fraction decays geometrically. These are enlarged projected sets; their points need not themselves be contact points. Lemma 4.1 supplies the last link to the asserted graph set.

Indeed, for $Y = (x', t) \in \mathscr{E}_{h,j}$, that lemma finds an actual graph point x with $\pi(x, u_j(x)) = Y$, near a contact point x_h. Write s_h for its vertical coordinate. Put $\nu_h = \nabla u_j(x_h)/|\nabla u_j(x_h)|$. The distance $|x - x_h|$ is bounded by a constant depending only on the fixed a_0, and

$$(x - x_h) \cdot \nu_h \le H_{0,p}(t) - H_{0,p}(u_j(x_h)) + C/R_h.$$

Since $|\nu_h - e_n| \le C\ell_j/R_h$, its vertical coordinate obeys

$$\begin{aligned} s &\le s_h + H_{0,p}(t) - H_{0,p}(u_j(x_h)) + C(1 + \ell_j)/R_h \\ &\le H_{0,p}(t) + C_0 A^h \theta_j. \end{aligned}$$

The last step uses $\ell_j^2/R_h = CA^h\theta_j$ and $\ell_j \to \infty$, with the fixed constants enlarged once. This is the passage from the covering sets to actual graph points in (5.40)–(5.41). At $h = k$, the geometric measure bound therefore applies to graph points satisfying the required height bound. The universal constant C in the statement is taken large enough to cover this fixed enlargement and the choice of A.

The contact lemmas also require ℓ_j/R_h to be small. Here

$$\frac{\ell_j}{R_h} = CA^h \frac{\theta_j}{\ell_j} \le CA^k \frac{\theta_j}{\ell_j} \longrightarrow 0.$$

Together with the cubic-root calculation above, this checks the scale conditions at every one of the finitely many levels. If a later argument prescribes an uncovered fraction $\eta > 0$, choose k so that $(1-\vartheta)^k < \eta$, and only then choose j large. This uses a finite iteration for each prescribed fraction; no limit with $k = k(j)$ is needed.

With these scale conditions checked, estimates (5.41)–(5.42) in the cited proof give (5.35) and the upper bound in (5.36).

For the lower height bound, a point more than C_0 below $s = -\theta_j$ has a fixed ball contained in the negative phase. Choose C_0 large enough that clearing gives $u_j < -1/2$ there. It cannot be a retained contact point. This gives the lower bound in (5.36). The whole contact set lies in any fixed smaller vertical window for large j, since k is fixed and $\theta_j/\ell_j \to 0$.

The fixed level range also makes this explicit:

$$|H_{0,p}(t)| \le a_p^{-1} \operatorname{arctanh}(1/2) \quad (|t| \le 1/2).$$

Combining the upper and lower height bounds gives

$$\sup \frac{|s|}{\ell_j} \le C_k \frac{\theta_j + 1}{\ell_j} \longrightarrow 0$$

over the retained graph set. Thus its fixed vertical cap and the outer buffer can be chosen before j. The contact-level restriction and the height cap are used for different purposes: the first keeps contacts on nonflat profile cores, and the second keeps them inside the comparison cylinder.

The source constants can be chosen uniformly for $p \in \mathcal{P}$. Indeed, for $0 \le t \le 1$,

$$W_p(1-t) = W_p(-1+t) = \frac{t^p(2-t)^p}{2p}.$$

For $0 \le t \le t_* < 1/4$,

$$|W_p'(1-t)| = |W_p'(-1+t)| = (1-t)t^{p-1}(2-t)^{p-1}.$$

Thus the growth constants in (1.1)–(1.2) of [43] are uniform. The well neighborhood can also be fixed: $W_p''(t) > 0$ when $0 < 1 - |t| < t_*$, with one $t_* > 0$ for all $p \in \mathcal{P}$. On $[-3/4, 3/4]$, the potential is uniformly positive and its derivatives have uniform bounds. The modified profiles in Chapter 2 therefore have common core bounds. Their contact slopes are bounded above and away from zero, as in Lemma 3.5 there. These slope bounds make the contact equations uniformly elliptic, as used in the proof of Lemma 3.6, since p and $(p-1)^{-1}$ are bounded. The barrier, contact, and covering arguments use only these common data. Their smallness thresholds may depend on θ_0 and the fixed integer k. □

Remark 5.12. Every retained contact point lies on a nonflat part of its barrier, with $|u| < 1/2$. The barrier derivative is nonzero there. The solution is therefore noncritical at that contact, and its equation is locally uniformly elliptic.

Lemma 5.13 (Phase orientation)**.** *In Lemma 5.10, also assume $u_j(0) = 0$ and $\theta_j \ge \theta_0 > 0$. For large j, the signs of u_j above and below the smaller buffered zero strip differ. Reflect u_j or the normal coordinate to make the upper sign positive.*

Proof. Both exterior zero-free regions are connected and have constant sign. Suppose the signs agree along a subsequence. If necessary replace u_j by $-u_j$ to make both negative. At $s = \pm c\ell_j$ over $B^m_{\ell_j}$, clearing gives uniform convergence to -1. Here c is fixed strictly inside the buffer.

Given $\eta > 0$, choose k with $(1-\vartheta)^k < \eta$ and apply Lemma 5.11. Define

$$F_j = \{x' \in B^m_{\ell_j} : \max_{|s| \le c\ell_j} u_j(x', s) < 1/4\}.$$

After one fixed enlargement of the buffer, the contact set lies in this height interval. A fiber $x' \in F_j$ misses all projected levels $(1/4, 1/2)$. Fubini and (5.35) imply

$$|F_j| \le 4\eta |B^m_{\ell_j}|.$$

Every other fiber starts near -1, reaches at least $1/4$, and returns near -1. Young's inequality gives pointwise

$$\frac{|\partial_s u_j|^p}{p} + W_p(u_j) \ge k_p (u_j)^{p-1} |\partial_s u_j|. \tag{5.37}$$

For (5.37), note that $W_p(t) = (p-1)k_p(t)^p/p$. Apply Young's inequality $X^p/p + (p-1)Y^{p/(p-1)}/p \geq XY$ with $X = |\partial_s u_j|$ and $Y = k_p(u_j)^{p-1}$. A vertical fiber need not be monotone. Define $\mathscr{J}_p(t) = \int_{-1}^{t} k_p(\tau)^{p-1}\,d\tau$. Then

$$|\partial_s(\mathscr{J}_p \circ u_j)| = k_p(u_j)^{p-1}|\partial_s u_j|.$$

Split the fiber at a point where $u_j \geq 1/4$. The fundamental theorem of calculus on its two parts bounds this total variation below by twice the integral from the lower endpoint value to $1/4$. The endpoint errors vanish uniformly by clearing. The full energy bounds this one-dimensional quantity because $|\nabla u_j|^p \geq |\partial_s u_j|^p$.

Therefore

$$\int_{-c\ell_j}^{c\ell_j} e_p(u_j(x',s))\,ds \geq 2\int_{-1+o(1)}^{1/4} k_p(t)^{p-1}\,dt$$

Since the potential is even,

$$J_{p,+} := 2\int_{-1}^{1/4} k_p(t)^{p-1}\,dt > \sigma_p.$$

Fix η so small that $(1-4\eta)J_{p,+} > \sigma_p$. The endpoint errors tend to zero uniformly in x', because clearing is uniform on the two buffered horizontal faces. Integrating the fiber lower bound contradicts (5.34). Therefore the signs are opposite. □

Lemma 5.14 (Flatness improvement under the unit cap)**.** *There are $0 < \eta_1 < \eta_2 < 1$ such that, for every $\theta_0 > 0$, there exists $\varepsilon_0(\theta_0) > 0$ with the following property. Let u satisfy (5.31) and the equation in Theorem 5.9, with $u(0) = 0$. If its zero set in the cylinder of radius and height ℓ is contained in $|s| \leq \theta$, where*

$$\theta \geq \theta_0, \qquad \theta/\ell < \varepsilon_0(\theta_0),$$

then, after a rotation through an angle at most $C\theta/\ell$, its zero set in a cylinder of radius and height $\eta_2\ell$ is contained in a strip of height $\eta_1\theta$.

Proof. We first prove a Harnack contraction, then use compactness and harmonic approximation.

The two contact projections give an energy contradiction. Work in a smaller buffered cylinder with base radius $r = c\ell$, and write

$$Q_r = B_r^m \times (-1/2, 1/2), \qquad \pi(x', s, u(x', s)) = (x', u(x', s)).$$

Lemma 5.13 supplies the opposite exterior signs. Reflect the seed construction proved in Lemma 5.11. Its verified weak-solution inputs hold with the upper exterior sign supplied above. It gives a graph set Ξ_+ such that

$$s \geq H_{0,p}(u) - \theta/4, \qquad |\pi\Xi_+| \geq \vartheta_*|Q_r|, \qquad \vartheta_* > 0.$$

This seed estimate is distinct from the iterated near-full projection: its first-contact barriers have radius comparable to r^2/θ, and Proposition 3.14 of [43] gives the fixed positive fraction. All its comparison hypotheses were checked in the proof of Lemma 5.11.

Choose $A > 1$ large enough to absorb the constants in (5.36), and then fix k with $(1-\vartheta)^k < \vartheta_*/8$. Suppose a zero $x^* = (x^{*\prime}, s^*)$ has $|x^{*\prime}| < \delta r$ and $s^* < -(1-\delta)\theta$. Take $\delta \leq 1/(4A^k)$ and put $\theta^* = \theta/(4A^k)$. In coordinates centered at x^* the negative phase lies below $-\theta^*$, because $s^* - \theta^* \leq -\theta$. Lemma 5.11 therefore supplies a second graph set Ξ_-, whose projection misses at most $(1-\vartheta)^k|Q_r|$ in the translated base cylinder, and with

$$s \leq H_{0,p}(u) + s^* + A^k\theta^* \leq H_{0,p}(u) - \theta/2.$$

Here we also require $\delta \leq 1/4$. Choose δ still smaller so that the symmetric difference of B_r^m and $B_r^m + x^{*\prime}$ has measure at most $\vartheta_*|B_r^m|/8$. It follows that

$$|\pi\Xi_- \cap \pi\Xi_+| \geq (\vartheta_*/2)|Q_r|.$$

For the projection overlap, write $A_+ = \pi\Xi_+ \subset Q_r$ and $A_- = \pi\Xi_- \subset Q_r + (x^{*\prime}, 0)$. Then

$$|A_+ \setminus A_-| \le |Q_r \setminus (Q_r + (x^{*\prime}, 0))| + |(Q_r + (x^{*\prime}, 0)) \setminus A_-|$$
$$\le \frac{\vartheta_*}{8}|Q_r| + (1-\vartheta)^k|Q_r| \le \frac{\vartheta_*}{4}|Q_r|.$$

Since $|A_+| \ge \vartheta_*|Q_r|$, the overlap has measure at least $3\vartheta_*|Q_r|/4$, and in particular at least $\vartheta_*|Q_r|/2$ as claimed. This calculation also places the overlap inside the original base B_r^m, where the cylinder energy will be evaluated.

The graph sets themselves are disjoint, since their height bounds are separated by $\theta/4$. Thus every point in this overlap has at least two distinct preimages on its vertical fiber.

We now convert the overlap into an energy lower bound. Let $N(x', t)$ count the preimages in a fixed buffered vertical window. Its endpoints tend uniformly to opposite wells by clearing, so $N(x', t) \ge 1$ for $-1+o(1) < t < 1-o(1)$ and all $x' \in B_r^m$. Both contact sets lie in that window: their core values and clearing keep them within a fixed distance of the zero strip. The one-dimensional area formula and (5.37) now give

$$\int_{B_r^m \times (-c'\ell, c'\ell)} e_p(u) \ge \int_{B_r^m} \int_{-1}^{1} k_p(t)^{p-1} N(x', t)\, dt\, dx' \ge \sigma_p \omega_m r^m + c_* r^m - o(r^m),$$

where $c_* > 0$ follows from the overlap and $\min_{[-1/2,1/2]} k_p^{p-1} > 0$. The baseline term uses one preimage between the endpoint values. The overlap supplies at least one additional preimage on a set of positive (x', t) measure. Endpoint levels omitted by the finite window contribute $o(r^m)$, since their ranges shrink uniformly and k_p^{p-1} is bounded. Lemma 5.10 contradicts this bound as $\theta/\ell \to 0$. Reflection excludes a zero near the upper strip face as well, proving the Harnack contraction on the smaller base. The constants A, k, δ precede this limit; the translated projection has the fixed lower height bound $\theta_0/(4A^k) > 0$. This is the projection and energy argument in Chapter 6 of [43]; the cylinder cap replaces its minimizing energy limit (6.16).

The contraction gives the compactness needed for harmonic approximation. Suppose improvement fails. Choose a sequence with $\theta_j \ge \theta_0$ and $\theta_j/\ell_j \to 0$, and use coordinates $y' = x'/\ell_j$, $y_n = s/\theta_j$.

To track the oscillation, center the iteration at one zero and start with half-height $2\theta_j$. Use that same center at each step. The fixed contraction constants $c_*, \kappa \in (0, 1)$ give, for every fixed integer h and all sufficiently large j,

$$|x' - z_j'| \le c c_*^h \ell_j \quad \Longrightarrow \quad |s - z_{j,n}| \le 2(1-\kappa)^h \theta_j$$

for the retained zeros. The initial factor two is used only once. At each fixed h, the physical height is bounded below by $2(1-\kappa)^h\theta_0$, so the contraction applies for large j. Take the sequence limit before increasing h. The resulting Holder exponent can be chosen as

$$\gamma_H = \min\{1, \log(1-\kappa)/\log c_*\} > 0.$$

The oscillation at horizontal scale c_*^h is at most $C(1-\kappa)^h \le C c_*^{h\gamma_H}$. Thus convergence is uniform on compact subballs, even though the finite-scale zero sets need not already be graphs.

Take a local Hausdorff subsequence of the rescaled zero sets. The contraction can be centered at any zero in a smaller cylinder. After each fixed number of steps it bounds the vertical oscillation on the corresponding smaller base disk by a fixed geometric factor. Thus two limit points with the same horizontal coordinate have the same height. Every interior vertical fiber has a zero, by the opposite signs at its two buffered ends and continuity. The limit is therefore the graph of a function h on the whole smaller base, with $h(0) = 0$. The same oscillation bounds give its uniform Hölder modulus and uniform convergence of the zero heights.

The limit h is viscosity harmonic. We use the local trace test in [40, Lemma 6.6, p. 346]. Its exact parameters are a weak solution on $[-l, l]^n$, $|u| \le 1$, $u(0) = 0$, and negative phase below the

paraboloid q_1 defined next. We may replace M by its symmetric part. For $0 < \delta \le \delta_0$, it requires $\delta \le \theta$, $\theta/l \le \sigma(\delta)$, $\|M\| \le 1/\delta$ and $|\xi| \le 1/\delta$. Its conclusion is $\operatorname{tr} M \le \delta$. The published statement also assumes the decay condition (2.1) for the vertical direction. In its proof, the lower half of that condition starts the radial comparison (6.23); the upper half is not used. We justify that ordering by local clearing and then give the local strict-barrier comparison below. The minimizer version, Lemma 9.3, obtains the decay condition from Corollary 9.2 and is not used here.

To check the initial radial ordering, the initial comparison ball in (6.23) is centered at $(0, -l/2)$ and has radius $l/4 + T_{l/4}$. In the construction, $T_{l/4} \le l/8$, so this ball stays a fixed multiple of l below the touching paraboloid after taking the aspect ratio small. Interior sign-ball clearing therefore puts u in the required negative well throughout that initial ball. The small fixed buffer accommodates its boundary as well.

The signed-distance comparison also uses a smaller domain than the whole initial cylinder. Write its touching paraboloid as

$$q_1(x') = \frac{\theta}{2l^2} x' \cdot Mx' + \frac{\theta}{l} \xi \cdot x'.$$

Take the tangent-sphere radius $R = l/4$. Fix $\bar{c} \in (0, 1/10)$ so that the radial transition length satisfies $T_R \ge \bar{c}l$. For small aspect ratio, $|q_1| \le \bar{c}l/64$ and $|\nabla q_1| \le 1/16$ on $|x'| \le l/2$. Use

$$\mathcal{K} = \{|x'| \le l/4,\ -3l/16 \le s \le \bar{c}l/8\}.$$

A vertical competitor bounds the distance from each $x \in \mathcal{K}$ to the paraboloid by $3l/16 + \bar{c}l/64 < l/4$. Any minimizing foot consequently lies in $|x'| < l/2$, rather than on the edge of the comparison graph. It is unique: the horizontal Hessian of its squared-distance function is

$$2(I + Dq_1 \otimes Dq_1 + (q_1 - s)D^2 q_1),$$

which is positive definite on this disk, since $l\|D^2 q_1\| = O_\delta(\theta/l) \to 0$. The signed distance d_1, positive above the paraboloid, therefore has its usual smooth normal coordinates there and satisfies

$$-3l/16 - \bar{c}l/64 \le d_1 \le 9\bar{c}l/64.$$

If X is this foot and N_X the upward unit normal, then $x = X + d_1 N_X$ and the lower tangent-sphere center is $Y = X - RN_X$. Thus $R + d_1 > 0$, $d_1 < T_R$, and $|x - Y| - R = |R + d_1| - R = d_1$. Both the radial sign and the profile domain used in the comparison are now fixed. The subsequent transition has width $O_\delta(\log(1/\epsilon)) = o(l)$, where $\epsilon = \theta/l^2$, and fits inside this smaller cylinder.

We first complete the radial comparison on $\mathcal{K}$. Let $Y_0 = (0, -l/2)$, and let $Y_X = X - RN_X$ be the lower sphere center associated with a foot X above. Since $R\|D^2 q_1\| = o(1)$, the ball $B_R(Y_X)$ lies below q_1. One can check this on its upper spherical graph: its negative Hessian dominates $D^2 q_1$, and the two graphs are tangent at X. The initial ball $B_R(Y_0)$ lies below q_1 by at least cl. The whole outer profile domains along the segment from Y_0 to Y_X stay inside $[-l, l]^n$, since $R + T_R \le 3l/8$, $|X'| < l/2$, and the slopes tend to zero.

To check the sign along that segment, fix $|\eta| \le 1$ and put $z_0 = Y_0 + R\eta$, $z_1 = Y_X + R\eta$ and $z_\tau = (1 - \tau)z_0 + \tau z_1$. The exact quadratic interpolation formula gives

$$\begin{aligned} q_1(z'_\tau) - z_{\tau,n} &= (1 - \tau)(q_1(z'_0) - z_{0,n}) + \tau(q_1(z'_1) - z_{1,n}) \\ &\quad - \frac{\tau(1 - \tau)}{2}(Y'_X - Y'_0) \cdot D^2 q_1 (Y'_X - Y'_0) \\ &\ge (1 - \tau)(cl - C_\delta \theta) > 0 \qquad (\tau < 1). \end{aligned}$$

Thus every earlier zero sphere lies in the negative phase. The initial negative-well ordering and the weak contact rule proved above allow the radial barrier to move along this segment without contact. Its outer boundary has value one. At the final position it is still above u. For $x \in \mathcal{K}$, the already checked inequalities $R + d_1 > 0$ and $d_1 < T_R$ give its radial argument exactly as d_1. We conclude $u(x) \le g_R(d_1(x))$ throughout $\mathcal{K}$. This is the local form of (6.23) needed below.

We spell out the strict profile used in this last comparison. This is the construction in [40, Lemma 5.2, pp. 339–342]. The initial radial ordering in (6.23) follows from the negative-well comparison already proved, which is [40, Corollary 6.4, p. 345]. Suppose, toward a contradiction, that $\operatorname{tr} M > \delta$. Put

$$\sigma = \theta/l, \qquad d = \delta/2, \qquad M_2 = M - \frac{\delta}{4m} I, \qquad q_2(x') = q_1(x') - \frac{\epsilon\delta}{8m}|x'|^2.$$

Then $q_2 = \epsilon x' \cdot M_2 x'/2 + \sigma\xi \cdot x'$, $\operatorname{tr} M_2 > 3\delta/4 > d$, $\|M_2\| \le 2/d$, and $|\xi| \le 1/d$. For large l, $0 < \epsilon \le \sigma \le d$. The required further bound $\sigma \le \sigma_0(d)$ follows by choosing the aspect ratio small after δ is fixed. These are the parameter hypotheses of that lemma.

Choose a fixed nondecreasing $\rho \in C^1(\mathbb{R})$, equal to -1 on $(-\infty, -1/2]$, equal to one on $[1/2, \infty)$, and with $\rho(0) = 0$. For a small constant $c_\Gamma > 0$, set

$$h_\Gamma(s) = \max\{0, W_p(s) + c_\Gamma d\epsilon\rho(s)\}.$$

Let $m_\Gamma \in (-1, -1/2)$ solve $W_p(m_\Gamma) = c_\Gamma d\epsilon$. The function h_Γ is positive on $(m_\Gamma, 1]$. Define

$$H_\Gamma(s) = \int_0^s \left(\frac{p-1}{p h_\Gamma(\zeta)}\right)^{1/p} d\zeta, \qquad T_\Gamma = H_\Gamma(1),$$

and let $g_\Gamma = H_\Gamma^{-1}$ on $[H_\Gamma(m_\Gamma), T_\Gamma]$, extended by m_Γ to the left. The well growth gives

$$1 + m_\Gamma \asymp (d\epsilon)^{1/p}, \qquad |H_\Gamma(m_\Gamma)| + T_\Gamma \le C_\delta \log(1/\epsilon).$$

These are the endpoint estimates (5.23), (5.27)–(5.28) of the cited construction. All its nonconstant part lies in this narrow normal strip. The same squared-distance argument used for q_1 gives smooth normal coordinates for q_2 there. It also does so after vertical translations of size $O_\delta(\log(1/\epsilon))$. This size is $o(l)$, since $\theta \ge \delta$.

Write d_2 for the signed distance above q_2. On the nonconstant strip, its principal-curvature formula gives

$$\Delta d_2 \le -cd\epsilon.$$

Indeed, the positive trace is at least $d\epsilon$, the slopes tend to zero, and the normal-offset error is $O_\delta(\epsilon^2 \log(1/\epsilon)) = o_\delta(\epsilon)$. The scalar first integral therefore gives

$$\Delta_p(g_\Gamma(d_2)) - W_p'(g_\Gamma(d_2)) \le c_\Gamma d\epsilon\rho'(g_\Gamma) - cd\epsilon(g_\Gamma')^{p-1} < 0.$$

Where ρ' is nonzero, the value lies in $[-1/2, 1/2]$ and g_Γ' has a positive lower bound. Choose c_Γ small there. Elsewhere $\rho' = 0$ and the second term is strictly negative. This proves the strict inequality on every increasing branch. It is the calculation (5.29)–(5.32) in [40].

We check the boundary ordering used to slide this profile. Write $m_R = -1 + e^{-c_1 R}$ for the plateau value of the radial profile g_R. The polynomial lower bound on $1 + m_\Gamma$ and $\theta \ge \delta$ give $m_\Gamma > m_R$ for large l. On the bottom of $\mathcal{K}$, $d_1 < -l/8$, so the radial comparison already established gives $u \le g_R(d_1) = m_R < m_\Gamma$. This ordering persists under the translations just described.

On the lateral face, compare the inverse profiles on $[m_\Gamma, 1]$. Near the negative well, $h_\Gamma - h_R \le -c_\Gamma d\epsilon + Cd\epsilon/l + Ce^{-cl} < 0$. Near the positive well the reverse inequality holds. Thus the maximum of $H_\Gamma - H_R$ occurs at levels whose distance from either well is at least a fixed multiple of $(d\epsilon)^{1/p}$. At those levels $H_\Gamma \le H_{0,p}$, because ρ has the sign of its argument. The radial inverse-profile bound then gives

$$\sup_{[m_\Gamma, 1]} (H_\Gamma - H_R) \le \frac{C}{l} \log \frac{1}{d\epsilon} = o(1).$$

This also follows from (6.26)–(6.31) in [40]. The estimate is applicable to the radial profile because $(d\epsilon)^{1/p}$ is much larger than $e^{-c_1 R/2}$.

For $|x'| = l/4$, the two comparison graphs have the vertical gap $\delta\theta/(128m)$. Consider a point where $g_R(d_1) \ge m_\Gamma$ and $d_2 \le T_\Gamma$. Put $L_\epsilon = \log(1/\epsilon)$ and $t_i = s - q_i(x')$. The small-slope distance comparison and $H_R(m_\Gamma) \ge -C_\delta L_\epsilon$ give $t_1 \ge -C_\delta L_\epsilon$. The bound $d_2 \le T_\Gamma$ gives $t_2 \le C_\delta L_\epsilon$ in the

same way. Since $q_2 \le q_1$, we have $t_1 \le t_2$. Both vertical heights therefore have size $O_\delta(L_\epsilon)$. Vertical competitors then give $|d_i| \le |t_i|$. Their normal feet stay in $l/5 < |y'| < l/3$.

On this annulus, $|Dq_i| = O_\delta(\sigma)$ and $|D^2 q_i| = O_\delta(\epsilon)$. If $Y_i = (y_i', q_i(y_i'))$ is the normal foot, then

$$x' - y_i' = -\frac{d_i Dq_i(y_i')}{\sqrt{1 + |Dq_i(y_i')|^2}}.$$

Taylor expansion of $q_i(x')$ at y_i' consequently gives

$$d_i = t_i + O_\delta(\sigma^2 L_\epsilon + \epsilon L_\epsilon^2).$$

Subtract the two formulas. On the lateral face,

$$d_2 - d_1 = \frac{\delta\theta}{128m} + O_\delta(\sigma^2 L_\epsilon + \epsilon L_\epsilon^2).$$

The error divided by $\delta\theta$ is at most

$$C_\delta \left(\frac{\theta L_\epsilon}{l^2} + \frac{L_\epsilon^2}{l^2} \right) = o(1).$$

Here $L_\epsilon \le C_\delta \log l$, because $\theta \ge \delta$, and $\theta/l \to 0$. Thus $d_2 - d_1 \ge c\delta\theta \ge c\delta^2$. This exceeds the inverse-profile error for large l. It follows that $g_\Gamma(d_2) > g_R(d_1) \ge u$ on that face. If $g_R(d_1) < m_\Gamma$, the same strict inequality follows from $g_\Gamma \ge m_\Gamma$ directly. This proves the lateral ordering, with no assertion about distance to a truncated graph outside its normal-coordinate region.

Start with the graph of q_2 translated downward by $A_\delta \log(1/\epsilon)$. Choose the fixed A_δ large. The whole translated profile domain in $\mathcal{K}$ then lies a large multiple of $\log(1/\epsilon)$ below q_1. The lower phase is negative there. Clearing makes $1 + u$ smaller than $1 + m_\Gamma$, so this initial profile is strictly above u. Move it upward to its unshifted position. The lateral ordering holds throughout: a downward translation increases the signed distance at a fixed point, and its profile is increasing. The bottom ordering was checked above. Its moving top has value one and cannot touch u. The fixed top of $\mathcal{K}$ is outside the translated profile domains for large l. All transition strips remain inside the normal-coordinate region. At the final position the profile and u both vanish at the origin, so a first interior contact must occur.

Such a contact is impossible. At a regular branch the strict differential inequality and the classical Hessian comparison apply. On the plateau and at its junction, use the explicit ball and p-harmonic annulus comparisons proved above. The plateau level m_Γ belongs to $(-1, 0)$, and its level surface has an interior ball on the plateau side in these smooth normal coordinates. Here $h_\Gamma'(m_\Gamma+) = W_p'(m_\Gamma) > 0$, so the same flat-junction calculation applies. The flux vanishes continuously at the junction; no measure appears there in weak integration by parts. Only C^1 regularity of the profile is required. The positive comparison constants here belong to each fixed barrier and need not have radius-independent lower bounds. We have justified the local version of the trace conclusion of [40, Lemma 6.6], using clearing in place of its initial decay hypothesis.

Below the touching paraboloid there are no zeros. Every point in that region can be joined vertically to the negative bottom phase without crossing a zero. Thus $u < 0$ throughout the required lower region. In rescaled coordinates, the trace bound excludes a strictly subharmonic quadratic touching h from below. More explicitly, suppose a quadratic P with $\operatorname{tr} D^2 P > 0$ touches h. On a fixed ball of radius r, subtract a smaller positive multiple of $|y'|^2$ to make contact strict. Uniform convergence then gives translated contact paraboloids for the original zero sets. Their base scale is $l_j = r\ell_j$ and height parameter $\widetilde{\theta}_j = r^2\theta_j$. At fixed r, the matrices and normalized slopes in Lemma 6.6 remain bounded. Choose $\delta > 0$ below $r^2\theta_0$, half the positive trace, the universal threshold, and the reciprocal matrix and slope bounds. Since $\widetilde{\theta}_j/l_j = r\theta_j/\ell_j \to 0$, all hypotheses hold for large j. The trace conclusion at most δ is impossible. Reflection supplies the reverse inequality. Hence $\Delta h = 0$.

The equation for h is initially in the viscosity sense. The touching inequalities also hold for C^2 tests: use their second-order Taylor polynomials with an arbitrarily small quadratic perturbation. To see that h is classically harmonic, take any ball $B_r(y_0')$ compactly contained in its domain. Let H be the Poisson extension of the continuous boundary values of h. For $\varepsilon > 0$, put $H_\pm(y') = H(y') \pm \varepsilon(r^2 - |y' - y_0'|^2)$. Both functions equal h on the boundary, and $\Delta H_\pm = \mp 2m\varepsilon$. If $h - H_+$ had a positive maximum, it would occur inside the ball. Adding that maximum to H_+ would give an upper touching test with negative Laplacian, contrary to the viscosity inequality. A negative minimum of $h - H_-$ similarly gives a lower touching test with positive Laplacian. Hence $H_- \le h \le H_+$. Letting $\varepsilon \downarrow 0$ gives $h = H$ on the ball. Thus h is a smooth harmonic function, and interior harmonic estimates give $|h(y') - \nabla h(0) \cdot y'| \le C|y'|^2$. Choose η_2 small, then η_1 satisfying $C\eta_2^2 < \eta_1 < \eta_2$. Rotate the original coordinates so that $s = (\theta_j/\ell_j)\nabla h(0) \cdot x'$ is the new horizontal plane. This rotation has angle at most $C\theta_j/\ell_j$. For large j the zero set lies in the improved cylinder. This contradicts the defining failure.

The contradiction argument stays within the class (5.31). That class is unchanged by rotations, translations, and sign reflection of u. Thus its thresholds are uniform and can be iterated. Enlarging outer buffers only decreases the universal inner-cylinder factors. □

Proof of Theorem 5.9. Fix $\theta_0 > 0$ and $\varepsilon < \varepsilon_0(\theta_0)$. Choose a cylinder in (5.32) satisfying $\Theta_j/L_j < \varepsilon$ and $L_j > \theta_0/\varepsilon$. If $\Theta_j < \theta_0$, this cylinder already has the desired height and a radius tending to infinity as $\varepsilon \downarrow 0$. Otherwise apply Lemma 5.14 until the height falls below θ_0. The successive scales are

$$\ell_h = \eta_2^h L_j, \qquad \theta_h = \eta_1^h \Theta_j.$$

Since $\eta_1 < \eta_2$, every step lowers the aspect ratio. Thus the assumptions remain valid. In this second case, at the final step,

$$\eta_1\theta_0 \le \theta_h < \theta_0, \qquad \ell_h > \eta_1\theta_0/\varepsilon.$$

Take $\varepsilon \downarrow 0$ and a convergent sequence of normals. The entire zero set lies in a strip of height θ_0 around a plane through zero. Next take $\theta_0 \downarrow 0$ and again extract normals. The zero set is contained in one hyperplane.

To recover the profile, apply Lemma 5.13 in arbitrarily large cylinders. The two half-spaces have opposite signs. By continuity, their common hyperplane consists entirely of zeros. Use the two one-sided trapping comparisons established in the proof of Lemma 5.11, with the radial profiles from [43, Lemmas 2.20–2.22]. Let the cylinder radius diverge while the positive strip height θ stays fixed. The modified profile barriers converge locally to the heteroclinic, giving

$$g_p(s - \theta) \le u(x', s) \le g_p(s + \theta).$$

after orientation. For fixed θ and a fixed compact set, the barrier domains contain that set once the radius is large. Their inverse profiles converge to $H_{0,p}$ on compact subintervals of $(-1, 1)$, so their profiles converge to g_p on bounded height intervals. Send $\theta \downarrow 0$ to obtain $u = g_p(s)$. The first integral $(p-1)|g'|^p/p = W_p(g)$ gives precisely $g_p(s) = \tanh(a_p s)$. □

Consider now arbitrary stable entire solutions with $p > 2$. The preceding results give four inputs.

(1) Zero-centered stable solutions are compact in C^1_{loc}, and stability passes to their limits.
(2) The density $M_R(u; y)$ is nondecreasing in R.
(3) Every nonconstant solution has $M_\infty \ge 1$, with $M_\infty = +\infty$ allowed.
(4) Under finite density, energy measures and kinetic varifolds have the same stationary rectifiable limit. Its support is the local Hausdorff limit of the zeros.

The third input is the flat-point density consequence following (5.13). None of these inputs assumes integer multiplicity.

Lemma 5.15 (Uniform lower density at every zero)**.** *Given $\eta > 0$, there is $R_\eta < \infty$ such that*

$$M_R(v; z) \geq 1 - \eta$$

for every stable entire solution at fixed $p > 2$, every zero z of v, and every $R \geq R_\eta$.

Proof. If this failed, translate solutions v_j to their chosen zeros and take $R_j \to \infty$ with $M_{R_j}(v_j; 0) \leq 1 - \eta$. Compactness gives a stable limit v with $v(0) = 0$. It cannot be zero because that constant has form $-\int \phi^2$. Thus v is nonconstant. At any fixed L, monotonicity and energy convergence give $M_L(v; 0) \leq 1 - \eta$. Hence $M_\infty(v) \leq 1 - \eta$, contrary to the third input. □

Theorem 5.16 (Stable unit-density rigidity)**.** *For $p > 2$, every nonconstant scalar-stable entire solution with $M_\infty \leq 1$ is a planar heteroclinic. Thus no nonplanar stable solution has asymptotic density one.*

Proof. The local regularity and gradient bound in Section 2.1 give a C^1 representative and $|\nabla u| \leq k_p(u)$. Since u is nonconstant, Remark 2.2 also gives $-1 < u < 1$. Its zero set is nonempty. Otherwise continuity on the connected space $\mathbb{R}^n$ would give one strict sign. For every $b \in (0, 1)$, apply Lemma 2.3 at each point in a zero-free ball of its required radius R_b. This gives $|u(x)| \geq b$ for every x. Letting $b \uparrow 1$ would make u the corresponding constant well, contrary to the hypothesis. Translate a zero to the origin. Monotonicity and independence of the limiting center give (5.31). Take $R_j \to \infty$ and a stationary rectifiable limit V of weight μ. The cap implies

$$\mu(B_r(x)) \leq \sigma_p \omega_m r^m \quad \text{for all } x, r.$$

At a support point $x \in \operatorname{spt} \mu$, choose rescaled zeros x_j tending to x. For fixed $r > 0$ and small $\tau > 0$, apply Lemma 5.15 at the physical zero $R_j x_j$ with radius $(r - \tau)R_j$. For every fixed $\eta > 0$, this radius exceeds R_η for large j. Hence the rescaled mass of $B_{r-\tau}(x_j)$ is at least $(1 - \eta)\sigma_p \omega_m (r - \tau)^m$. For large j this ball lies in $B_r(x)$. At a continuity radius of μ about x, the mass of the latter ball converges. First pass to that limit, then let $\eta \downarrow 0$ and $\tau \downarrow 0$. This gives the exact lower bound at that radius. For an arbitrary radius, use continuity radii increasing to it and monotonicity of the mass of balls. We conclude

$$\mu(B_r(x)) = \sigma_p \omega_m r^m \quad (x \in \operatorname{spt} \mu,\ r > 0). \tag{5.38}$$

The equality mechanism explained after the blowdown theorem applies here. Choose a point x where the rectifiable tangent varifold exists. It is a multiple of the approximate tangent plane. The mass identity (5.38) makes this multiple σ_p. Equality in stationary monotonicity makes V a cone about x. Every rescaling about x is therefore identical. Its tangent-plane limit is V itself. Hence V is a single affine plane with weight σ_p. The anchored zero puts the origin on that plane. This uses rectifiability and stationarity, without integrality or stable minimal-surface regularity.

The zero-set Hausdorff limit now gives (5.32) on each fixed interior cylinder. For example, take physical cylinders with radius and height $L_j = R_j/2$. They lie in a fixed buffered rescaled ball. Their zero heights are at most $o(R_j)$, so the corresponding strip heights satisfy $\Theta_j / L_j \to 0$. The C^1 regularity, strict range, centered zero, and Modica bound were checked at the start of the proof. Together with (5.31) and this flatness sequence, they verify every hypothesis of Theorem 5.9. That theorem gives the required planar profile. □

5.4. **Critical density and integer sheets.** We adapt the critical-density and blowdown scheme of Florit-Simon–Serra [20, Section 3.1 and Appendix B]. For the critical-density construction assume $3 \leq n \leq 7$ and fix $p \in (2, \infty)$. The small-defect theorem in dimension seven applies in these dimensions by constant extension. The stability calculation is given in the critical-cone argument below. Choose $\delta_0 > 0$ below the threshold of Theorem 4.12. Every nonplanar stable solution then contains a zero satisfying $D \geq \delta_0$.

Assume that nonplanar stable finite-density solutions exist. Set

$$K_* := \inf\{M_\infty(v) : v \text{ is stable, nonplanar, and has finite density}\}.$$

Proposition 5.17. *Some nonplanar stable solution u_* attains $M_\infty(u_*) = K_*$ and has $D_*(0) \geq \delta_0$. In addition, for each $\eta > 0$ there is $R_\eta < \infty$ such that every critical solution with $M_\infty = K_*$ satisfies*

$$K_* - \eta \leq M_R(u; z) \leq K_* \qquad (R \geq R_\eta,\ z \in \mathcal{B}_{\delta_0}(u)). \tag{5.39}$$

Almost-flat zeros imply $K_* \geq 1$. Unit-density rigidity then gives $K_* > 1$ at the attained nonplanar solution.

Proof. Take nonplanar stable solutions u_j with $M_\infty(u_j) \downarrow K_*$, and translate a δ_0-bad zero of each to the origin. Local C^1 compactness preserves stability, the value zero, and the badness bound. The limit cannot be the zero constant, by scalar stability. It cannot attain either well, since well-contact rigidity would make it constant. Thus it remains in the strict range $(-1, 1)$. For each fixed R, local energy convergence and monotonicity give $M_R(u_*; 0) \leq \lim_j M_\infty(u_j) = K_*$. The limit has finite asymptotic density at most K_*. It is nonplanar, since its anchored zero is bad and a layer has zero defect. The infimum therefore forces equality. The fixed core at the anchored zero also gives $K_* > 0$.

If (5.39) were not uniform, center violating bad zeros and take $R_j \to \infty$ with $M_{R_j} \leq K_* - \eta$. The same compactness argument would give a nonplanar stable limit with density at most $K_* - \eta$. Indeed, every fixed r is eventually below R_j, so $M_r(u_j; 0) \leq M_{R_j}(u_j; 0) \leq K_* - \eta$ before passing to the limit. Then let $r \to \infty$. This contradicts the definition of K_* and proves saturation.

Lemma 5.15 gives $K_* \geq 1$. If $K_* = 1$, Theorem 5.16 would make the attained solution planar, contrary to its anchored bad zero. Hence $K_* > 1$. □

To study blowdowns at the first nonplanar density, fix $p > 2$ and $3 \leq n \leq 7$, and let $m = n - 1$. All solutions here are stable and entire, with range $(-1, 1)$. The local analytic estimates are uniform on compact subsets of $(2, \infty)$. The critical-density moduli are taken at fixed p and fixed K. The local preparation results stated in dimension seven apply in every smaller ambient dimension used here. Indeed, if u is stable in $\mathbb{R}^n$, its constant extension $\widetilde{u}(x, y) = u(x)$ to $\mathbb{R}^7$ is a solution and is stable: for $\phi \in C^1_c(\mathbb{R}^7)$ its quadratic form is

$$Q_{\widetilde{u}}(\phi) = \int_{\mathbb{R}^{7-n}} Q_u(\phi(\cdot, y))\, dy + \int_{\mathbb{R}^7} |\nabla u|^{p-2} |\nabla_y \phi|^2 \geq 0.$$

A clean n-dimensional ball gives a clean seven-dimensional ball of the same radius, since the bad-zero set of the extension is $\mathcal{B}(u) \times \mathbb{R}^{7-n}$. Restricting the resulting estimates to $y = 0$ proves the asserted local bounds; no global energy bound for the extension is needed.

The following inputs were proved above, without a general phase-field varifold regularity theorem.

(1) Translates are compact in C^1_{loc}. Stability passes to the limit, and a zero-centered stable limit cannot be the zero constant. The weak Jacobi identities and full geometric inequality hold. With $s = |\nabla u|$, they imply

$$\int_{\{s>0\}} s^p |B|^2 \eta^2 \leq (p-1) \int s^p |\nabla \eta|^2. \tag{5.40}$$

Here B is the curvature of the level set through each regular point, not just the zero set.

(2) Choose the uniform threshold $\delta_0 > 0$ and define

$$\mathcal{B}(u) = \{z : u(z) = 0,\ D(z) \geq \delta_0\}.$$

If $\mathcal{B}(u) = \varnothing$, the small-defect theorem makes u planar. In a sufficiently large ball disjoint from $\mathcal{B}(u)$, preparation on a smaller ball gives

$$D \leq CR^{-\alpha}, \qquad |B| \leq C/R, \qquad |H| + |\nabla_\Sigma H| \leq CR^{-\alpha}, \qquad \alpha > 1. \tag{5.41}$$

The defect and mean bounds here are at zeros. For each fixed $b < 1$, the full core $\{|u| \le b\}$ in a smaller ball satisfies

$$|\nabla u| \ge c_b, \qquad \left| D\left(\frac{\nabla u}{|\nabla u|}\right)\right| \le C_b/R. \tag{5.42}$$

As $R \to \infty$, each fixed normal window at a clean zero tends uniformly to the oriented layer g_p.

(3) Rescaled clean minimal limits satisfy coefficient-one stability. If mean curvature vanishes and curvature stays locally bounded, the physical form and mass comparison give

$$\int_\Sigma |B_\Sigma|^2\zeta^2 \le \int_\Sigma |\nabla_\Sigma \zeta|^2. \tag{5.43}$$

The clean-limit passage after Theorem 4.14 uses Proposition 3.13 and the pointwise comparison in Corollary 3.5. For each $\varepsilon_I > 0$, choose its finite normal window and then its defect threshold, uniformly on $\mathcal{P}$. On zeros below this threshold, $I_P \ge (1-\varepsilon_I)|B|^2$. Use a common fixed core for this comparison and the physical form. Its mass $S = (I_P + |B|^2)/2$ therefore satisfies $S \ge (1-\varepsilon_I/2)|B|^2$ before taking a geometric limit. The tolerance tends to zero only after that limit; it is not a coefficient limit at an arbitrary fixed window. Below we apply the form to one lifted compact test on all sheets and check that the covering multiplicity cancels. The rough coefficient $p-1$ in (5.40) would not suffice.

(4) Finite density gives stationary rectifiable blowdown compactness and vanishing pressure discrepancy. The zero sets converge locally in Hausdorff distance to the support. Every fixed core $\{|u| \le b\}$ has the same limiting support. This input contains no integrality statement. We also have clearing

$$e_p(u)(x) \le C\exp[-c\ \mathrm{dist}(x, \{u = 0\})]. \tag{5.44}$$

(5) For each fixed width h, the bad-set bounds are

$$|B_h(\mathcal{B}(u)) \cap B_R(x)| \le C_h R^{-2} \int_{B_{3R}(x)} e_p(u), \tag{5.45}$$

$$|B_h(\mathcal{B}(u)) \cap B_R(x)| \le C_h R^{-2} \int_{B_{3R}(x)} |\nabla u|^p(1-(\nu\cdot e)^2) \qquad (e \in S^{n-1}). \tag{5.46}$$

A maximal fixed-radius bad-center net has the same bounds up to a constant. These estimates come from the tangential detector. No continuation through critical points is used.

(6) Every nonconstant stable solution has $M_\infty \ge 1$. By unit-density rigidity, equality $M_\infty = 1$ forces a layer. Moreover, $\inf_{\{u=0\}} D = 0$. A translated zero-centered limit with $D(0) = 0$ is planar.

The cone classification uses Simons' cone-instability result [41, Lemma 6.1.1, p. 102]. A smooth two-sided stable minimal hypersurface cone in ambient dimension between three and seven is a hyperplane, provided only its vertex can be singular.

Continue to assume that a nonplanar stable solution of finite density exists, and write $K = K_*$. Proposition 5.17 supplies a solution attaining this critical density and gives $1 < K < \infty$. Let $\mathscr{C}_K$ be the family attaining density K. Every member has a nonempty closed bad-zero set.

Proposition 5.17 gives uniform saturation:

$$\inf_{\substack{u\in\mathscr{C}_K \\ z\in\mathcal{B}(u)}} M_R(u;z) \longrightarrow K \qquad (R\to\infty). \tag{5.47}$$

Center independence also yields the global cap

$$M_R(u;x) \le K \qquad (u \in \mathscr{C}_K,\ x\in\mathbb{R}^n,\ R>0). \tag{5.48}$$

This is the center-independent cap in Proposition 5.6.

Theorem 5.18 (Critical planar blowdowns)**.** *Every blowdown of members of $\mathscr{C}_K$ about their bad zeros is the varifold of a hyperplane, with constant weight $K\sigma_p$. This assertion allows the solutions and the bad centers to vary with the blowdown sequence.*

Proof. Choose $u_j \in \mathscr{C}_K$, $z_j \in \mathcal{B}(u_j)$, and $R_j \to \infty$, and define

$$\epsilon_j = R_j^{-1}, \qquad v_j(x) = u_j(z_j + R_j x), \qquad Z_j = R_j^{-1}(\mathcal{B}(u_j) - z_j).$$

After diagonal extraction, the closed sets Z_j converge locally to a closed set Z_∞ containing zero. The kinetic varifolds converge to a stationary rectifiable V with weight μ. Their measures are

$$d\mu_j^{\mathrm{kin}} = \epsilon_j^{p-1} |\nabla v_j|^p \, dx.$$

The energy measures have the same limit. The cap gives

$$\mu(B_r(x)) \le K\sigma_p \omega_m r^m. \tag{5.49}$$

There is also the lower bound

$$\mu(B_r(x)) \ge \sigma_p \omega_m r^m \qquad (x \in \operatorname{spt}\mu,\ r > 0). \tag{5.50}$$

Indeed, approach x by rescaled zeros and apply Lemma 5.15 with physical radius $(r - \tau)R_j$. Pass at continuity radii and then send $\tau \downarrow 0$, exactly as in the proof of Theorem 5.16. In particular the limit is nonzero, since the origin is a rescaled zero. For $z \in Z_\infty$, take $z_j' \in Z_j$ converging to z. Apply (5.47) at the physical bad point represented by z_j', with radius rR_j. Use ball inclusions with radii $r - \tau$ and $r + \tau$ to account for the moving centers. Pass first at continuity radii and then send $\tau \downarrow 0$ to obtain

$$\mu(B_r(z)) = K\sigma_p \omega_m r^m \qquad (z \in Z_\infty,\ r > 0). \tag{5.51}$$

Equality in monotonicity makes V conical about each z.

Set $L = \operatorname{span} Z_\infty$ and $d = \dim L$. Write $D_{z,t}(x) = z + t(x - z)$ for dilation about z by $t > 0$. Then

$$D_{z,t} \circ D_{0,t^{-1}}(x) = x + (1 - t)z.$$

Both factors preserve the cone varifold. Thus translations by $(1 - t)z$ preserve it as well. Their inverses give the opposite translations, so every translation parallel to z preserves V. Composing these translations for $z \in Z_\infty$ gives invariance along all of L.

Rectifiability then implies that L lies in the approximate tangent m-plane at almost every point. One way to see this is to take a point where the rescaled weight measures converge to the tangent plane measure. Translation invariance is preserved in this blowup, so that tangent measure is invariant in each direction of L. A flat m-plane measure has that invariance only for vectors in the plane. Thus $d \le m$.

Suppose first that $d = m$. Translation invariance and the tangent-plane property give $\mu = \mathcal{H}^m \lfloor L \otimes \lambda$ for a Radon measure λ on the orthogonal line. Under dilation, both μ and the m-dimensional Lebesgue factor scale by t^m. Their factors cancel, so $\lambda(tA) = \lambda(A)$ for every $t > 0$. If an annulus away from zero had positive λ-mass, choose a fixed dilation ratio that makes its shrinking copies disjoint. All copies would lie in one compact interval and would have the same positive mass. This contradicts the Radon property. Thus λ is supported at zero. The mass identity (5.51) fixes its value as $K\sigma_p$.

We turn to the geometry away from L, assuming $d \le n - 2$. Each compact subset of $\mathbb{R}^n \setminus L$ eventually stays a fixed distance from Z_j. Its physical balls are therefore clean, with radii comparable to R_j. We construct the limiting sheets and transport stability to arbitrary compactly supported tests on them. Coincident limiting sheets are identified geometrically; their prelimit lifts will be counted separately.

Put $U = \mathbb{R}^n \setminus L$. Fix a compact subset of U and a slightly larger compact buffer. After rescaling, (5.41) gives

$$|B_j| \le C, \qquad |H_j| \le C R_j^{1-\alpha} \longrightarrow 0.$$

The zero surfaces are regular and relatively closed in this buffer. Choose a fixed rescaled radius r so that each B_{20r} lies inside this buffer and Cr is below the threshold in Lemma 3.10. That lemma gives a tangent graph disk through each zero, with uniformly bounded Hessian and slope as small as desired after decreasing r. Relative closedness and noncriticality are exactly its continuation hypotheses.

These graph disks can be put over a common plane near any limiting point. By (3.19), for two zeros z, z' with curvature bound K_j in a sufficiently large buffer, the angle between their unoriented normal lines obeys

$$\angle(\nu(z), \nu(z')) \le C(|z - z'|/r + K_j r).$$

Here both points have graph disks of radius r, and $K_j r$ is small. In rescaled coordinates K_j is bounded. First choose r small enough to make the second term small, and then restrict to a central ball of radius much smaller than r to make the first term small. All disks meeting that central ball are consequently transverse to one fixed direction e, with

$$|\nu \cdot e| \ge \gamma_0 > 0,$$

and their projections contain a common smaller disk in $e^\perp$. If two graph disks intersect, regularity of the level set makes them the same graph on their common base. Distinct graphs are disjoint and are ordered by height. At this stage we have not yet assumed that there are finitely many of them.

Write the graphs in physical coordinates as $x' + f_{i,j}(x')e$. On any smaller base disk their distinct heights have uniformly diverging physical separation:

$$\inf_{x'} \inf_{i \ne k} |f_{i,j}(x') - f_{k,j}(x')| \longrightarrow \infty.$$

To prove this statement without assuming a sheet count, suppose that a pair of graphs and a point x'_j violate it. Translate the first zero on the fiber to the origin. Its defect tends to zero, so local compactness and the equality case in the gradient bound give a planar-layer limit. After extracting the normal direction, this limit has the form $g_p(\nu_\infty \cdot x)$, up to reversal, with $|\nu_\infty \cdot e| \ge \gamma_0$. The two alleged zeros lie on one bounded segment parallel to e. On a slightly larger fixed segment the limiting derivative in that direction has one sign and has absolute value bounded below by a positive constant. Local C^1 convergence gives the same property for u_j, which permits at most one zero on this segment. This is a contradiction. The argument allows the two zeros to approach each other and is uniform over the graph indices and the smaller base disk.

We count the sheets with vertical windows, so no injectivity claim about their normal tubes is needed here. Fix a physical number $T > 0$ and put

$$W_{i,j} = \{x' + te : x' \in D_j,\ |t - f_{i,j}(x')| < T\},$$

where the common physical disk D_j has area at least cR_j^m and is buffered inside every graph domain. The separation just proved makes these sets pairwise disjoint for large j. Translated profile convergence gives a uniform lower bound for the energy on each vertical segment. Indeed, write a limiting planar profile as $u_\infty(x) = g_p(\nu_\infty \cdot x)$, up to reversal, and put $\gamma = |\nu_\infty \cdot e| \in [\gamma_0, 1]$. The Hamiltonian identity gives

$$\begin{aligned}\int_{-T}^{T} e_p(u_\infty)(te)\, dt &= \frac{1}{\gamma} \int_{-\gamma T}^{\gamma T} |g_p'(s)|^p\, ds \\ &\ge \int_{-\gamma_0 T}^{\gamma_0 T} |g_p'(s)|^p\, ds =: c_T > 0.\end{aligned}$$

In the first line, the energy density is that of the full planar solution, whose gradient has magnitude $|g_p'(\gamma t)|$. Uniformity follows by the same translated compactness argument: a sequence of points where the lower bound fails would have one of these planar profiles as its limit. Thus, for large j,

$$\int_{W_{i,j}} e_p(u_j) \geq \tfrac{1}{2} c_T |D_j| \geq c_T' R_j^m.$$

All these windows lie in one fixed rescaled compact buffer, whose physical energy is at most CR_j^m. Their disjointness therefore bounds the size of every finite collection of graphs by C/c_T'. Thus there are only finitely many graphs, with a bound independent of j.

The graph functions have uniformly bounded first and second derivatives on smaller disks. A subsequence converges in $C^{1,\theta}$ for each $\theta < 1$. Passing the graph mean-curvature equation to the limit gives a weak minimal graph. Interior elliptic regularity makes that graph smooth. Two ordered limiting minimal graphs either coincide or have strictly positive difference on the connected base disk. This follows from the strong maximum principle for their difference. After coincident graphs have been identified, each difference has a positive minimum on a compactly contained smaller disk. There are only finitely many pairs, so their minimum isolates every remaining geometric sheet there. Thus different limiting sheets cannot accumulate on another sheet in a compact subset of U. A diagonal extraction gives the locally finite smooth embedded support. This local isolation will also supply the whole-test covering argument.

We also need convergence of curvature. Write the graph equation in nondivergence form as

$$a^{ab}(Df_j)\,\partial_{ab} f_j = h_j, \qquad h_j \longrightarrow 0,$$

where the sign of h_j depends on the graph orientation and is irrelevant here. On a smaller disk the matrices are uniformly elliptic and uniformly Hölder continuous. If f is the limiting minimal graph, then $w_j = f_j - f$ satisfies

$$a^{ab}(Df_j)\partial_{ab} w_j = h_j + [a^{ab}(Df) - a^{ab}(Df_j)]\partial_{ab} f.$$

The right side converges to zero in every finite L^q, while $w_j \to 0$ uniformly. The matrices have common ellipticity constants and common Hölder bounds, since the graph Hessians are bounded. For disks $D' \Subset D$ in the chart, the interior estimate gives

$$\|w_j\|_{W^{2,q}(D')} \leq C_q\big(\|w_j\|_{L^q(D)} + \|h_j + [a^{ab}(Df) - a^{ab}(Df_j)]\partial_{ab} f\|_{L^q(D)}\big) \longrightarrow 0.$$

Thus $f_j \to f$ strongly in $W^{2,q}$ on smaller disks. In particular, the second fundamental forms converge strongly in L^2, and the metrics converge uniformly. This conclusion is uniform over the finitely many lifts in a fixed chart. It uses no derivative estimate for the rescaled mean curvature.

Let S be a connected limiting sheet and let $\zeta \in C_c^\infty(S)$. Write $\mathcal{S}$ for the limiting union of geometric sheets. Choose connected smooth domains

$$\operatorname{supp}\zeta \Subset \Omega \Subset \Omega_1 \Subset \Omega_2 \Subset S$$

inside the clean region. We first isolate the sheet over $\overline{\Omega}$. At every point of $\overline{\Omega}_1$, a graph chart contains only finitely many limiting graphs. Distinct ones have a positive separation on a smaller chart; coincident ones have already been identified. Moreover, the local part of S in a sufficiently small chart lies in Ω_2. A finite collection of these charts therefore gives an ambient neighborhood of $\overline{\Omega}_1$ in which every point of the limiting support lies on this local portion of S. In particular, there is an ambient neighborhood W of $\overline{\Omega}$, with $\overline{W} \Subset U$, such that

$$W \cap \mathcal{S} \subset \Omega_1.$$

This assertion also excludes any remote part of the same connected sheet: if such a part approached $\overline{\Omega}$, it would eventually enter one of the finitely many isolating graph charts.

The normal-bundle map

$$\mathcal{N} : NS \longrightarrow \mathbb{R}^n, \qquad \mathcal{N}(z, v) = z + v, \qquad v \perp T_z S,$$

is locally invertible at every $(z,0)$. Choose a tube radius $r_{\mathrm{tub}} > 0$. By compactness and after shrinking this radius, its derivative remains nonsingular and it is injective on the normal vectors of length less than $2r_{\mathrm{tub}}$ over Ω_1, and the closed normal tube of radius r_{tub} over $\overline{\Omega}$ lies in W. Here is the compactness argument for injectivity near the retained domain. Otherwise there would be two different normal vectors (z_k, v_k) and (z'_k, v'_k), both tending to the zero section over a fixed compact buffered part of S, with $z_k + v_k = z'_k + v'_k$. Then $|z_k - z'_k| \to 0$; after extraction both bases tend to one point of S. Local invertibility of $\mathcal{N}$ near that point contradicts the equality. The inverse supplies a smooth projection

$$\pi : \mathcal{T}_{2r_{\mathrm{tub}}}(\Omega_1) \longrightarrow \Omega_1.$$

The use of normal vectors, rather than a signed normal coordinate, allows S to be one-sided.

The compact fiber boundary

$$\{z + v : z \in \overline{\Omega},\ v \perp T_z S,\ |v| = r_{\mathrm{tub}}\}$$

is disjoint from the limiting support. Indeed, any support point there would lie in Ω_1 by the choice of W, and would have two normal representations, one with normal vector zero and one of length r_{tub}. Local Hausdorff convergence therefore excludes all prelimit zeros from this fiber boundary for large j. Every prelimit zero in the retained tube is close to S, and the C^1 graph convergence makes the restriction of π to its tangent space invertible.

Set

$$S_j^{\Omega} = \{x \in \Sigma_j \cap \mathcal{T}_{r_{\mathrm{tub}}}(\Omega_1) : \pi(x) \in \Omega\}.$$

For any compact $F \subset \Omega$, the set $(\pi|_{S_j^{\Omega}})^{-1}(F)$ is closed in the compact closed normal tube of radius r_{tub} over F. It cannot approach the fiber boundary, where there are no zeros. Relative closedness of the zero set inside U then makes this inverse image compact. Thus $\pi : S_j^{\Omega} \to \Omega$ is a proper local diffeomorphism. Its image is open by local invertibility, closed by properness, and nonempty by convergence to S. Since Ω is connected, the projection is onto.

Each fiber is compact and discrete, hence finite. To see the covering property directly, take the finitely many preimages of a point and inverse-function neighborhoods of them. If no smaller base neighborhood excluded all additional preimages, a sequence of additional preimages above points tending to the original base point would have a convergent subsequence by properness. Its limit would be one of the already listed preimages, a contradiction. Consequently a sufficiently small base neighborhood is evenly covered. The degree $d_j \geq 1$ is locally constant and therefore constant on Ω. It is also uniformly bounded in j, by the graph count in any chart meeting one fiber.

This reasoning explains why a prelimit sheet cannot disappear inside the domain: it has no critical point of the projection, cannot leave through the normal-fiber boundary, and cannot end at an interior point of the relatively closed regular zero set. No global numbering is needed; possible monodromy changes only the way the local lifts are connected. At this stage d_j counts zero sheets, and has not been identified with varifold multiplicity.

It suffices first to take $\zeta \in C_c^{\infty}(S)$; compactly supported C^1 tests follow by approximation on a fixed buffered domain. Lift this one test to $\zeta_j = \zeta \circ \pi$ on S_j^{Ω}, and extend it by zero on the rest of the zero surface. Properness makes the inverse image of $\operatorname{supp}\zeta$ compact. The gap between $\operatorname{supp}\zeta$ and $\partial\Omega$ ensures that the extension is smooth. It is therefore one legitimate compact surface test, independently of whether the covering admits a global numbering of its sheets.

Fix $0 < \eta < 1/10$ and $Q > 2$ before passing to the sequence limit. Apply Corollary 3.5 with $\varepsilon_I = \eta$. This chooses a finite window P_I and a positive defect threshold. Choose the physical form of Proposition 3.13 with loss η and a fixed core $P \geq P_I$, using the fixed-window choice in its construction. Decrease one common defect threshold to satisfy both results and the relative estimates on this fixed core. Since $D \geq 0$, the definition of the mass gives $I_P \geq I_{P_I} \geq (1-\eta)|B|^2$

at every zero below that threshold. Consequently

$$\frac{I_P + |B|^2}{2} \geq (1 - \eta/2)|B|^2.$$

All windows, thresholds and cap constants are uniform on $\mathcal{P}$ and are fixed independently of j and of the number of lifts of the test.

On a fixed slightly larger physical support, (5.41) gives $D \leq \delta_j = CR_j^{-\alpha}$ at all zeros. For large j, this is below the common threshold. The normal core has fixed length, and the required ambient neighborhood has width $10A_0 \log(1/\delta_j) = O(\log R_j) = o(R_j)$. Both fit inside the clean buffer, whose rescaled distance from L is positive. The same larger buffer controls every lifted component of the one test. The defect and curvature bounds also verify the separate chamber-error conditions, as checked after Theorem 4.14.

Use the displayed mass inequality in the physical surface form. Dilation by R_j^{-1} multiplies both the curvature and Dirichlet terms by the same factor R_j^{2-m}, while its zero-order error gains an additional factor R_j^2. Thus the prelimit coefficient is exactly $(1-\eta)(1-\eta/2)$, and the dilated form is

$$(1-\eta)(1-\eta/2)\int_{S_j^\Omega} |B_j|^2\zeta_j^2 \leq (1+\eta)\int_{S_j^\Omega} |\nabla\zeta_j|^2 + CR_j^{2-\alpha Q}\int_{S_j^\Omega} \zeta_j^2,$$

where $Q > 2$ is fixed. The exponent $2 - \alpha Q$ is negative. Use a finite partition of unity only to compute these integrals in charts. Strong $W^{2,2}$ convergence on each lift and uniform metric convergence give

$$d_j^{-1}\int_{S_j^\Omega} |B_j|^2\zeta_j^2 \longrightarrow \int_\Omega |B_S|^2\zeta^2,$$

$$d_j^{-1}\int_{S_j^\Omega} |\nabla\zeta_j|^2 \longrightarrow \int_\Omega |\nabla_S\zeta|^2,$$

$$d_j^{-1}\int_{S_j^\Omega} \zeta_j^2 \longrightarrow \int_\Omega \zeta^2.$$

Each chart has exactly d_j lifts counted with covering multiplicity, so the same factor occurs in all three lines. Divide the form by d_j, let $j \to \infty$, and then let $\eta \downarrow 0$. This proves (5.43) for the original whole test.

No mass remains concentrated on L. Cover a bounded part of its ρ-tube by $C_R\rho^{-d}$ balls of radius $C\rho$. The cap (5.49) gives

$$\mu(B_R \cap B_\rho(L)) \leq C_R\rho^{m-d} \longrightarrow 0. \tag{5.52}$$

The same estimate holds uniformly for prelimit energy and kinetic measures, independently of j.

Each fixed core level has this same limiting support, with the same unoriented normals. To check this directly, use (5.42). The gradient flow parametrized by the value of u has vector field $\nabla u/|\nabla u|^2$. Throughout $|u| \leq b$ this field is smooth and has speed at most c_b^{-1}, and the value of u changes at unit speed along a trajectory. Thus the path from a level $t \in [-b, b]$ to zero has length at most b/c_b. It exists until zero is reached: this length is independent of j, whereas the physical clean buffer has radius comparable to $R_j \to \infty$, so the path cannot leave the buffer first. Integrating the full-normal estimate along the path bounds its normal change by $C_b b/(c_b R_j)$. Conversely, the fixed profile window at a zero crosses every level $t \in [-b, b]$. After dilation, both the distances and the normal changes tend to zero uniformly in these levels.

Thus the support outside L is nonempty and determines the varifold.

Translation invariance and conicality give support $C \times L$. The transverse cone C is a smooth embedded minimal hypersurface in $\mathbb{R}^{n-d} \setminus \{0\}$. Local curvature and sheet-count bounds make it proper away from its vertex. In particular its link is a compact embedded smooth hypersurface of the unit sphere, with finitely many components. The minimal cone equation makes this link

minimal in the sphere. On each smooth component stationarity makes the weight constant. The support lower bound (5.50) shows that this weight is at least σ_p, without assuming that it is an integer multiple of σ_p. For a compactly supported test ζ on a component of C, apply (5.43) to $\zeta(\xi)\chi(y/T)$ on its product with L. Divide by $\int_L \chi(y/T)^2$ and send $T \to \infty$; the tangential-cutoff cost is $O(T^{-2})$. This passes stability to each component of C. When $d = 0$, there is no tangential factor and the same conclusion is immediate.

Suppose next that $d \le n-3$. Then $3 \le n-d \le 7$. Each component of the compact embedded link separates the sphere S^{n-d-1} and has a global unit normal. Extending that normal radially makes the cone component two-sided. The link has no boundary, and the cone is smooth away from its vertex. The sphere has dimension $n-d-1 \le 6$, which is precisely Simons' dimension range. These are the geometric hypotheses in his cone argument. Its negative variation for a nonflat link is supported on a truncated cone away from the vertex. The test vanishes on both radial boundary faces. Extend it by zero and approximate it in H^1 by smooth compact tests on the cone. The curvature potential is bounded on this compact annulus, so its quadratic form is continuous under this approximation. The negative sign persists, contradicting the stability just proved. Thus no test through the vertex, or integrability of its curvature potential there, is required. Simons' theorem makes each component a plane. Distinct planes would intersect away from the vertex, contradicting the smooth embedded lamination. Hence there is one plane. Stationarity makes its weight constant, and (5.51) identifies it as $K\sigma_p$.

It remains to consider $d = n-2$. The cone in $\mathbb{R}^2$ is a finite union of rays. Its annular mass is finite, and each local sheet has positive lower mass, so only finitely many rays occur. The next lemma excludes a genuine junction. The cone is then a line. Stationarity across L equates the weights of its two half-hyperplanes. The density identity gives $K\sigma_p$ times a hyperplane. □

Lemma 5.19 (Transverse normal-variation contradiction)**.** *In the two-dimensional transverse case of Theorem 5.18, the support C is a line.*

Proof. Use coordinates $x = (\xi, y) \in \mathbb{R}^2 \times \mathbb{R}^{n-2}$, so $L = \{0\} \times \mathbb{R}^{n-2}$. Suppose C is not a line. Work in a fixed buffered product inside B_3. Dimensional reductions of the radii do not affect the proof. Fix $b \in (0,1)$, for instance $b = 1/2$.

Choose a physical radius R_0 large enough that absence of bad zeros within a fixed large multiple of R_0 gives the core gradient bound (5.42). Remove projected bad collars from the transverse parameter ball:

$$U_j = B_1^{n-2} \setminus \pi_y(B_{CR_0\epsilon_j}(Z_j \cap B_4)).$$

By coarse packing, the relevant physical bad set is covered by at most $C_{R_0} R_j^{n-3}$ fixed-radius balls. After scaling and projection, their total $(n-2)$-dimensional volume is at most

$$C_{R_0}\epsilon_j^{-(n-3)}\epsilon_j^{n-2} = C_{R_0}\epsilon_j.$$

Thus

$$|U_j| \ge c_0 > 0 \tag{5.53}$$

for large j. For $y \in U_j$, every point in the sliced core $\{|v_j| \le b\}$ satisfies

$$\epsilon_j |\nabla v_j| \ge c_b > 0. \tag{5.54}$$

The buffers defining U_j exclude any effect from bad points outside B_4.

The full-gradient bound (5.54) does not itself make $\xi \mapsto v_j(\xi, y)$ a submersion. On a neighborhood of the core, however, v_j is smooth by noncritical elliptic regularity. For each y, Sard's theorem, applied on this open noncritical neighborhood, says that almost every $t \in (-b, b)$ is a regular value of the sliced function in the disk. The exceptional set of pairs is measurable: on compact subsets it is the projection of the closed set where $v_j = t$ and $\nabla_\xi v_j = 0$. A countable compact exhaustion

and Fubini's theorem therefore give regularity for almost every pair (y, t). For these pairs, the level slice

$$S_{j,y,t} = \{\xi : v_j(\xi, y) = t\}$$

is a smooth embedded curve. We use only these regular pairs in the coarea and slicing arguments below.

Let B_j be the full curvature of the regular level hypersurface of v_j. Scale (5.40) and use a cutoff on the buffered product. The rescaled curvature is $B_j = R_j B_{u_j}$, while the kinetic measure is the physical kinetic measure multiplied by R_j^{-m}. The physical inequality contributes R_j^{m-2}, so these factors cancel. This yields

$$\int |B_j|^2 \, d\mu_j^{\mathrm{kin}} \le C. \tag{5.55}$$

Since $d = n - 2 = m - 1$, the tube estimate gives

$$\mu_j^{\mathrm{kin}}(B_\rho^2 \times B_1^{n-2}) \le C\rho.$$

Cauchy–Schwarz bounds its curvature integral by $(\int |B_j|^2 \, d\mu_j^{\mathrm{kin}})^{1/2}$ times the square root of this tube mass. Hence

$$\int_{B_\rho^2 \times B_1^{n-2}} |B_j| \, d\mu_j^{\mathrm{kin}} \le C\sqrt{\rho}. \tag{5.56}$$

On the good core, (5.54) gives the conversion appropriate to the p-equation:

$$\epsilon_j^{p-1} |\nabla v_j|^p = (\epsilon_j |\nabla v_j|)^{p-1} |\nabla v_j| \ge c_b^{p-1} |\nabla v_j|.$$

Apply coarea first to v_j, then to π_y on each full hypersurface. The latter tangential Jacobian is $|\pi_\xi \nu_j| \le 1$.

Indeed, in an orthonormal basis of the parameter space, the squared Jacobian is

$$\det(I - \nu_{j,y} \otimes \nu_{j,y}) = 1 - |\nu_{j,y}|^2 = |\nu_{j,\xi}|^2.$$

This is an upper bound for coarea in the required direction; no positive lower bound for this projection is needed.

The two coarea formulas therefore read

$$\begin{aligned} &\int_{-b}^{b} \int_{U_j} \int_{S_{j,y,t} \cap B_\rho^2} |B_j| \, d\mathcal{H}^1 \, dy \, dt \\ &\quad = \int_{(B_\rho^2 \times U_j) \cap \{|v_j| < b\}} |B_j| \, |\nabla v_j| \, |\nu_{j,\xi}| \, dx \\ &\quad \le c_b^{1-p} \int_{B_\rho^2 \times B_1^{n-2}} |B_j| \, d\mu_j^{\mathrm{kin}}. \end{aligned}$$

Together with (5.56), this gives

$$\int_{-b}^{b} \int_{U_j} \int_{S_{j,y,t} \cap B_\rho^2} |B_j| \, d\mathcal{H}^1 \, dy \, dt \le C\sqrt{\rho}. \tag{5.57}$$

The integrand must use full hypersurface curvature, rather than the curvature of the planar slice.

We now obtain a lower bound contradicting (5.57). Fix $\rho > 0$ small before taking $j \to \infty$. On the annulus $\rho/2 \le |\xi| \le \rho$ and a buffered y-ball, clean compactness sends every fixed core level and its normals uniformly to $C \times L$. Each ray occurs: every clean zero sheet crosses every level in $[-b, b]$ inside a fixed profile window. The annular graph pieces cover the buffered y-ball, since their normals converge to normals orthogonal to L.

We explain why every ray occurs above every retained parameter. Choose disjoint narrow angular sectors around the finite rays, using the larger annulus $\rho/3 < |\xi| < 2\rho$ and a larger y-ball as buffers. Uniform core convergence excludes all core points from the angular faces of these sectors. In each

sector, the unoriented normal lines are close to its angular direction. Consequently, on every core level the map

$$\Pi(\xi, y) = (|\xi|, y)$$

is a local diffeomorphism: its one-dimensional ambient kernel is the angular direction, which is transverse to the level hypersurface. Restrict its base to $(\rho/2, 3\rho/2) \times B_1^{n-2}$. The map is proper over this base, because the inverse image of a compact base set stays away from the radial and y faces, cannot reach an angular face, and lies in a relatively closed regular level hypersurface. Its image is nonempty for every $t \in [-b, b]$: near one zero approaching the chosen ray, the fixed profile window crosses that level. The same open-and-closed argument as in the clean-covering argument above therefore makes it onto the connected base, as a finite covering. Since the base is contractible, each connected component of this covering is one graph. In particular, each ray contributes points above every retained y at radius ρ.

Thus every sliced level under consideration has at least one endpoint group at each ray. The energy bound and fixed-window separation bound the number of crossings. In this annulus the projected normals have length tending to one and are perpendicular to the corresponding rays. The sliced curves therefore meet the outer circle transversely, in endpoint groups near the rays. Any regular outer radius between $3\rho/4$ and ρ would work as well. Annular convergence makes all constants uniform in that choice.

Fix one regular pair (t, y). The set

$$S_{j,y,t} \cap \overline{B_\rho^2}$$

is a compact smooth one-dimensional manifold with boundary. Regularity of the pair gives $\nabla_\xi v_j \neq 0$ at every point of this set; annular graph convergence gives transversality at the outer circle. A compact one-dimensional manifold has only finitely many connected components, each a circle or an arc. Only the arcs have boundary points, and each pairs two of the endpoints on ∂B_ρ^2. Every limiting ray group has at least one endpoint, by the annular covering assertion above.

These pairings are noncrossing. More precisely, two disjoint embedded arcs in the closed disk cannot have alternating boundary endpoints. To verify this, join the first arc to either of the two boundary-circle arcs with the same endpoints. Their union is a simple closed curve. The two alternating endpoints of the second arc determine small interior portions of that arc lying on different sides of this curve: one endpoint lies on the closing boundary arc and enters its interior side, while the other lies on the complementary boundary arc. The Jordan curve theorem would force the second arc to meet the first one in the interior.

We now show that some arc has endpoints whose full oriented normals differ by a fixed positive amount. There are three cases. First, suppose an arc joins different ray groups whose directions $e_a, e_b \in S^1$ are not opposite. Annular convergence puts its endpoint normals close to choices of Je_a and Je_b, where J is rotation by $\pi/2$. Regardless of their signs, the limiting normal distance is at least

$$d_{ab} := \min_{\sigma,\tau\in\{-1,1\}} |\sigma Je_a - \tau Je_b| = \sqrt{2(1 - |e_a \cdot e_b|)} > 0.$$

Second, suppose an arc has both endpoints in the same ray group. Along this arc, its continuous unit tangent can be chosen as

$$\tau = J\nu_{j,\xi}/|\nu_{j,\xi}|.$$

This is well-defined because the regular-pair condition gives $\nu_{j,\xi} \neq 0$. Orient the arc by this tangent. At its initial endpoint the tangent points into the disk, whereas at its terminal endpoint it points out. Both endpoints are close to the same radial direction, and their tangents are nearly radial. Their projected normals therefore have opposite signs. Their full normals have y-components tending uniformly to zero in the annulus, so their full oriented normal distance tends to 2.

The third possibility would be that every arc joins opposite ray groups. If any ray has no opposite partner, this is already impossible, since that group has a nonempty set of endpoints. Otherwise,

because C is not a line, there are at least two different opposite pairs. Choose an arc joining one pair and an arc joining the other pair. Their four endpoints alternate on the circle, once the small endpoint groups have been chosen disjoint. This contradicts the noncrossing property.

We have proved that one of the first two cases must occur. The ray set is finite, so the positive distances d_{ab} in the first case, together with the number 2 in the second, have a positive minimum. Choose a smaller constant $c_1 > 0$. For fixed ρ and all sufficiently large j, annular convergence makes the endpoint errors small enough that the selected full normals differ by at least c_1, uniformly in the regular pair (t, y). The constant c_1 depends on the limiting ray directions, and is independent of ρ; only how large j must be depends on ρ.

This is the normal-variation argument of Florit-Simon–Serra [20, proof of Lemma B.5, p. 46], with the regular-pair restriction and the endpoint pairing checked above. Parametrize the selected arc by arclength. Its tangent is tangent to the full level hypersurface, so the derivative of the full oriented normal has norm at most $|B_j|$. The fundamental theorem of calculus bounds its endpoint difference by the curvature integral along that arc. The previously chosen $c_1 > 0$ depends only on the limiting ray angles. It is independent of ρ and of the regular pair. After ρ is fixed, all sufficiently large j satisfy

$$\int_{S_{j,y,t}\cap B^2_\rho} |B_j|\, d\mathcal{H}^1 \geq c_1.$$

Integrate over levels and good parameters. By (5.53), the left side of (5.57) is at least $2bc_0c_1 > 0$. Choose ρ small and then j large. This contradicts the upper estimate. Thus C is a line. □

Remark 5.20. This is the codimension-two topology and slicing argument from Florit-Simon–Serra, Appendix B [20]. For our equation, the additional checks are the weighted conversion $(\epsilon|\nabla v|)^{p-1}|\nabla v| \geq c|\nabla v|$ and the almost-everywhere regular-value restriction. A transverse gradient lower bound is not assumed. Nor is a general p-phase-field sheeting theorem used.

Corollary 5.21. *There is a modulus $\omega(\tau) \to 0$ as $\tau \downarrow 0$ with the following properties, uniform for $u \in \mathscr{C}_K$ and $z \in \mathcal{B}(u)$. For all sufficiently large R, there is a hyperplane $P_{z,R}$ through z such that*

$$\{|u| \leq b\} \cap B_R(z) \subset B_{\omega(1/R)R}(P_{z,R}), \tag{5.58}$$

$$M_{\omega(1/R)R}(u; y) \geq K - \omega(1/R) \qquad (y \in P_{z,R} \cap B_R(z)). \tag{5.59}$$

Here $b < 1$ is fixed, and changing b changes the modulus. For every fixed h,

$$\sup_{\substack{u\in\mathscr{C}_K\\ z\in\mathcal{B}(u)}} R^{3-n}|B_h(\mathcal{B}(u)) \cap B_R(z)| \longrightarrow 0. \tag{5.60}$$

Proof. Fix a tolerance and positive relative radius. Theorem 5.18, support convergence, and compactness give uniform slab containment. They also give uniform ball-mass convergence at centers on the limiting plane. For moving centers, extract a convergent scaled subsequence. Every sphere has zero plane measure, so convergence still holds. A sufficiently slow diagonal modulus gives (5.58) and (5.59) at once. The balls in (5.59) may need a larger fixed buffered blowdown ball. Entire solutions provide it. Enlarge the modulus, if necessary, so that $\omega(1/R) \geq R^{-1/2}$. Then its physical no-gap radius also tends to infinity. Enlarging it preserves (5.58); monotonicity preserves (5.59).

The kinetic limits are the same planar varifolds. Their tilt relative to $P_{z,R}$ therefore vanishes uniformly:

$$R^{-m} \int_{B_{3R}(z)} |\nabla u|^p (1 - (\nu \cdot e_{z,R})^2) \longrightarrow 0,$$

Here choose the approximating plane on a sufficiently large buffered ball, and let $e_{z,R}$ be its unit normal. Insert this bound into (5.46). As $m - 2 = n - 3$, we obtain (5.60). □

We can now determine the multiplicity of these planar blowdowns.

Theorem 5.22. *The first critical density is an integer $K \geq 2$.*

Proof. Fix $u \in \mathscr{C}_K$. Since $\inf_{\{u=0\}} D = 0$, choose zeros x_j with $D(x_j) \to 0$. Define

$$s_j = \operatorname{dist}(x_j, \mathcal{B}(u)).$$

Then $s_j \to \infty$. To see this, any C^1_{loc} limit of the translates $u(x_j + \cdot)$ has Modica equality at the origin and is a planar layer. If s_j stayed bounded, bad zeros would persist at bounded distance in that layer limit. Its defect is zero, a contradiction.

Take a nearest bad zero y_j with $|x_j - y_j| = s_j$. It exists because the bad set is closed and nonempty. Apply Theorem 5.18 at y_j and scale $4s_j$. After rotations and extraction, the limit is a plane weighted by $K\sigma_p$. The scaled images of x_j converge to a point on that plane because they are zeros. Choose a fixed small $\theta > 0$. The physical ball $B_{\theta s_j}(x_j)$ becomes a ball of radius $\theta/4$ whose center converges to that point. Its boundary has zero mass for the weighted plane. Energy convergence and the factor $(4s_j)^m$ therefore give

$$M_{\theta s_j}(u; x_j) \longrightarrow K. \tag{5.61}$$

The uniform flatness and no-gap estimates give the same conclusion. No macroscopic clean ball around the original bad center is assumed.

The ball $B_{s_j}(x_j)$ is clean. On a smaller buffered cylinder, (5.41) gives $|B| \le C/s_j$. At the same time (5.58) confines the zeros to a slab of width $o(s_j)$. These facts force all sheet tilts to vanish uniformly in a smaller cylinder. Indeed, suppose an interior tangent plane has vertical component ϑ. Follow a unit intrinsic geodesic in that direction. Over length ℓ, curvature changes its vertical velocity by at most $C\ell/s_j$. Following it for a fixed small multiple of ϑs_j gives displacement at least $c\vartheta^2 s_j$ while staying inside the buffer. Slab containment then implies $\vartheta \to 0$. The geodesic cannot end in the interior, since the zero set is closed and noncritical.

Each sheet now projects locally diffeomorphically to the limiting plane. In a smaller cylinder the projection is proper over the base disk. An interior end cannot leave through the top or bottom by slab containment, and cannot become critical by (5.42). The image of each component is open by local invertibility and closed by properness. It is therefore the whole connected disk. Each component is a covering of that disk and hence a single graph. We obtain finitely many ordered graphs

$$t = f_{1,j}(x') < \cdots < f_{N_j,j}(x'), \qquad \max_i (\|f_{i,j}\|_\infty / s_j + \|\nabla f_{i,j}\|_\infty) \longrightarrow 0.$$

The fixed-window counting argument in the proof of Theorem 5.18 applies on this clean buffer. Its hypotheses are the same uniform profile convergence, local noncritical graph uniqueness, and the energy cap (5.48). It gives $1 \le N_j \le C$ and uniform divergence of adjacent physical separations on smaller disks. Take a subsequence with fixed $N_j = N$.

We next quantify the energy of this clean packet. For fixed T, the disjoint windows $[-T, T]$ about its N sheets carry

$$(1 + o_j(1)) \int_{-T}^{T} |g_p'(t)|^p \, dt$$

per unit base area. Their Jacobians tend to one and their slopes vanish. Outside these windows, the energy per unit base area is at most

$$CNe^{-cT} + Cs_j e^{-cs_j}.$$

To justify the tail estimate, put $d_j(x', t) = \min_i |t - f_{i,j}(x')|$. The graphs have uniformly small slopes. Zeros within the larger buffered cylinder are therefore at distance at least $cd_j(x', t)$, while all other zeros are at distance at least cs_j. Clearing (5.44) therefore gives

$$e_p(u)(x', t) \le Ce^{-cd_j(x',t)} + Ce^{-cs_j}.$$

Outside the fixed normal windows, vertical distance is at least a fixed multiple of T, up to an error tending to zero for fixed T. Use $e^{-c\min_i a_i} \le \sum_i e^{-ca_i}$ and integrate vertically. This gives Ce^{-cT}

per graph and $Cs_j e^{-cs_j}$ for the buffer remainder. It avoids multiplying a fixed tail bound by the cylinder height. Thus $s_j \to \infty$ may precede $T \to \infty$.

The same quantization holds in $B_{\theta s_j}(x_j)$. Fix $0 < \eta < \theta/2$ within the clean cylinder fractions. The graph heights are $o(s_j)$. For fixed T and large j, the windows over the disk of radius $(\theta - \eta)s_j$ lie inside this ball, while its transition part is contained in the windows and tails over the disk of radius $(\theta + \eta)s_j$. Energy outside a slab of any fixed positive relative height is exponentially small. The packet estimates therefore give

$$N\left(\frac{\theta-\eta}{\theta}\right)^m \le \liminf_j M_{\theta s_j}(u; x_j) \le \limsup_j M_{\theta s_j}(u; x_j) \le N\left(\frac{\theta+\eta}{\theta}\right)^m$$

after taking $j \to \infty$ and then $T \to \infty$. Finally send $\eta \downarrow 0$. The result is

$$M_{\theta s_j}(u; x_j) \longrightarrow N.$$

Compare with (5.61) to obtain $K = N \in \mathbb{N}$. Unit-density rigidity excludes $K = 1$. □

The same argument gives local packets of K sheets.

Corollary 5.23. *There are fixed positive buffered-cylinder fractions such that, for every $u \in \mathscr{C}_K$ and every zero x with $s = \operatorname{dist}(x, \mathcal{B}(u))$ sufficiently large, a cylinder of radius comparable to s about x contains exactly K ordered zero graphs over a common base disk. Their heights are $o(s)$, their unoriented slopes are $o(1)$, and their physical separations tend to infinity as $s \to \infty$, uniformly in the critical family. They satisfy*

$$|B| \le C/s, \qquad |H| + |\nabla_\Sigma H| \le Cs^{-\alpha},$$

and every fixed normal window tends to the appropriate oriented matched profile.

Proof. Choose a nearest bad zero. Apply (5.58)–(5.59) at a fixed multiple of s. The ball of radius s around x is clean. The graph, separation, and quantization arguments of Theorem 5.22 are uniform on smaller fixed cylinders. The graph counts are bounded integers, so a violating sequence has a subsequence with fixed count N. The same window and tail calculation then gives normalized mass N. The planar bad-center limit has mass K, so $N = K$. The remaining estimates follow from clean-ball preparation with its established exponent $\alpha > 1$. This argument does not need a stronger numerical exponent. □

The packets need only be ordered locally; no global numbering is required.

5.5. **Finite-density classification.** These local packets lead to the finite-density contradiction. The nearest-sheet comparison adapts the tube, Hardy, and Green tests in [35, Sections 6.2–6.5]. We retain the weights and coefficients of the matched equation. Let $N = m + 1$ and $3 \le m \le 6$, and take p in $\mathcal{P}$. For a nonconstant stable entire solution, write $e_u = e_p(u)$ and $M_R(y) = M_R(u; y)$. The energy identity is

$$e_u = \frac{|\nabla u|^p}{p} + \frac{q^p}{2p} = a^p q^p (1 - D/p).$$

Below, r also denotes radial distance when written explicitly as $r = |x - y|$. In those formulas, write the phase speed as $r_{\mathrm{ph}} = |Z|$.

The critical-cone argument gives the following structure for a least-density nonplanar solution.

(1) The limiting density $M_\infty = K$ is a finite integer satisfying $K \ge 2$. There is a nonempty closed set $\mathcal{Z}$ with $M_R(y) \to K$ uniformly for $y \in \mathcal{Z}$. For each fixed $h < \infty$,

$$\sup_{y \in \mathcal{Z}} R^{2-m} |B_h(\mathcal{Z}) \cap B_R(y)| \longrightarrow 0.$$

(2) Put $s(x) = \operatorname{dist}(x, \mathcal{Z})$. If a zero z has large $s(z)$, then a buffered cylinder of radius $c_* s(z)$ contains K ordered regular graphs over a common comparable disk. Their heights are $o(s)$, their slopes are $o(1)$, and $|B| \le C/s$ uniformly. Their separations diverge. Every fixed normal window tends smoothly to the oriented matched layer.

(3) On these graphs the mean curvature satisfies

$$|H| \le C s^{-\alpha} \qquad \text{for some } 1 < \alpha < 2. \tag{5.62}$$

The argument below uses only this local smoothness and these bounds. It does not require mean-curvature decay s^{-2}.

Theorem 5.24 (Exclusion of a critical finite density)**.** *For fixed $p > 2$ and $3 \le m \le 6$, no stable solution has the critical structure above with $K \ge 2$.*

Proof. By Proposition 5.6, for almost every R one has

$$M_R'(y) = \frac{1}{\sigma_E \omega_m} \left[R^{-m} \int_{\partial B_R(y)} |\nabla u|^{p-2} u_R^2 + \frac{c a^p}{p} R^{-m-1} \int_{B_R(y)} q^p D \right]. \tag{5.63}$$

Both terms are nonnegative. Since the limiting density is center independent, $M_R(x) \le K$. Integrate (5.63). Coarea for the first term and Tonelli for the second give

$$\int_{|x-y|>R_0} \left(|\nabla u|^{p-2} u_R^2 + q^p D\right) |x-y|^{-m} \, dx \le C(K - M_{R_0}(y)). \tag{5.64}$$

The second calculation uses the exact identity

$$\int_{R_0}^{\infty} t^{-m-1} \int_{B_t(y)} F \, dx \, dt = \frac{1}{m} \left[R_0^{-m} \int_{B_{R_0}(y)} F + \int_{|x-y|>R_0} |x-y|^{-m} F \right].$$

Uniform saturation at $y \in \mathcal{Z}$ makes the right side tend to zero as $R_0 \to \infty$, uniformly in y. Thus both tails in (5.64) vanish uniformly. Integrating over $[R, 2R]$ also gives $R^{-m} \int_{B_R(y)} q^p D = o(1)$ uniformly: the second term in (5.63), integrated over $[R, 2R]$, is bounded below by a positive constant times $R^{-m} \int_{B_R(y)} q^p D$. Since $e_u = a^p q^p (1 - D/p)$ and $\sigma_E = a^p J_p$, this gives

$$R^{-m} \int_{B_R(y)} q^p \, dx \longrightarrow J_p K \omega_m. \tag{5.65}$$

Whenever a finite normal width P is fixed, increase S so that its tubes over Σ_S are jointly injective and have Jacobian $1 + o_S(1)$. They carry the fixed-window profile convergence of Corollary 5.23. Indeed, two colliding fibers have bases at distance at most $2P$. Those bases lie in a common buffered packet chart when S is large. The diverging separations exclude fibers from distinct graphs. On one graph, use the nearest-foot part of Lemma 3.10 on a buffered disk of a small fixed multiple of s. The candidate distance is at most $P = o(s)$, its minimizing foot stays in the interior, and the squared-distance Hessian is positive definite since $P|B| = o_S(1)$. Thus its normal representation in this fixed tube is unique as well. All uses of a finite tube below follow this order: fix its width, then increase the good-region threshold. A unit tube and the density cap give

$$\mathcal{H}^m(\Sigma_S \cap B_R(x)) \le C R^m, \qquad \Sigma_S = \Sigma \cap \{s > S\}, \quad R \ge 1.$$

Relative defect comparison on that tube and the exact q stability test imply

$$\int_{\Sigma_{S/2} \cap B_R(x)} D \, dS \le C \int_{B_{R+1}(x)} q^{p+1} D \, dx \le C R^{m-2}. \tag{5.66}$$

Here the first inequality follows by integrating the relative comparison $D(z) \le C D(z + t\nu(z))$ over an injective fixed unit tube. On that tube q and the tube Jacobian have uniform positive lower bounds. For the second, use the scalar q-test (5.1), $q^2 A_u \le c a^{p-2} q^p I$, and a cutoff equal to one on

$B_{R+1}(x)$, supported in $B_{2R+2}(x)$, with gradient $O(R^{-1})$. The density cap bounds its q^p integral by CR^m.

We transfer the radial kinetic term on each injective unit tube. There q and r_{ph} are bounded below, and the phase normal differs from its base normal by $O(\sqrt{D(z)})$. On a shell $|z-y| \asymp R$, the radial direction changes by $O(R^{-1})$ along the fiber. Since

$$|\nabla u|^{p-2} u_R^2 = a^p q^p r_{\mathrm{ph}}^p (n \cdot e_R)^2,$$

the elementary inequality $|v|^2 \le 2|v-w|^2 + 2|w|^2$ bounds the squared base radial-normal component by its unit-fiber kinetic integral plus $C(D(z) + R^{-2})$. Use the area bound and (5.66) on each dyadic shell. After multiplication by R^{-m}, these two errors are $O(R^{-2})$. Choose the fixed preparation threshold to be below the one in Lemma 2.12. The zero-level estimate (2.20) then gives $|H| \le C_{\mathcal{P}} D$. Its contribution on that shell is therefore $O(R^{-1})$ with weight R^{1-m}. Summing the shells and applying (5.64) gives

$$\int_{\Sigma_S \cap \{|z-y|>R_0\}} \Big((e_R \cdot \nu)^2 |z-y|^{-m} + |H||z-y|^{1-m} \Big)\, dS \le \eta(R_0), \qquad \eta(R_0) \longrightarrow 0, \tag{5.67}$$

uniformly for $y \in \mathcal{Z}$ and large S. The summable geometric errors tend to zero with R_0. The function η can be fixed at one unit-tube threshold. Increasing S only restricts the surface integrals, so it leaves this choice unchanged.

For example, after fixing a single unit-tube threshold and enlarging the constant, one can take, for $R_0 \ge 4$,

$$\eta(R_0) = C \left[\sup_{y \in \mathcal{Z}} (K - M_{R_0/2}(y)) + R_0^{-1} \right].$$

The first term controls the diffuse radial tail. The dyadic defect and area errors contribute $O(R_0^{-2})$, and the mean term contributes $O(R_0^{-1})$. This choice makes explicit that η is independent of every later fixed normal window and every subsequent increase of S.

The exact q stability inequality is

$$\int q^{p+1} D\eta^2 \le a^{p-2} \int q^p r_{\mathrm{ph}}^{p-2} \left(|\nabla \eta|^2 + (p-2)(n \cdot \nabla \eta)^2 \right).$$

For radial tests, $r_{\mathrm{ph}}^{p-2} \le 1$. Its extra normal term satisfies

$$r_{\mathrm{ph}}^{p-2} (n \cdot e_R)^2 \le r_{\mathrm{ph}}^p (n \cdot e_R)^2 + D. \tag{5.68}$$

Indeed, $1 - r_{\mathrm{ph}}^2 \le D$ and $r_{\mathrm{ph}}^{p-2} \le 1$. After multiplication by $p-2$ it extends continuously through a critical gradient. At $p = 2$ the term is absent. Also,

$$a^p q^p r_{\mathrm{ph}}^p (n \cdot e_R)^2 = |\nabla u|^{p-2} u_R^2.$$

Thus the extra radial-normal term, with weight $|x-y|^{-m}$, is controlled by the finite tails (5.64). It contributes $O(1)$, rather than another leading multiple of $\log R$.

Set $h = (m-2)/2$ and test with $\eta(x) = |x-y|^{-h}$ on $R_0/2 < |x-y| < 4R$. Use cutoffs on the adjacent dyadic shells. Their cost is $O(1)$ because $\int_{B_t(y)} q^p \le C t^m$. Write $Q_y(t) = \int_{B_t(y)} q^p$. Stieltjes integration gives

$$\int_{b<|x-y|<cR} q^p |x-y|^{-m}\, dx = [t^{-m} Q_y(t)]_b^{cR} + m \int_b^{cR} t^{-m-1} Q_y(t)\, dt.$$

The boundary term is bounded.

For the logarithmic average, write $t^{-m} Q_y(t) = J_p K \omega_m + \epsilon_y(t)$, where $\epsilon_y(t) \to 0$ uniformly in the bad center. For any fixed T_0, the integral up to T_0 is bounded independently of R. Above T_0, its absolute error is at most $\sup_{t \ge T_0, y} |\epsilon_y(t)| \log R$. Divide by $\log R$, first send R to infinity, and then increase T_0. This proves that the logarithmic remainder is $o(\log R)$ uniformly in the center.

Equation (5.65) and logarithmic averaging therefore give

$$\int_{b<|x-y|<cR} q^p|x-y|^{-m}\,dx = J_pKm\omega_m \log R + o(\log R)$$

for fixed b, c. Hence

$$\limsup_{R\to\infty} \frac{1}{\sigma_q \log R} \int_{R_0/2<|x-y|<4R} q^{p+1} D|x-y|^{2-m}\,dx \le Km\omega_m \frac{(m-2)^2}{4}. \tag{5.69}$$

The leading coefficient is independent of p. The crude matrix bound $(p-1)I$ would lose this coefficient and cannot be used here.

The diffuse mass also gives a lower bound for the weighted area of regular zero sheets. Choose a smooth ambient mask χ. It vanishes for $s \le S$ and equals one for $s \ge 2S$. Its derivatives are supported strictly inside the regular collar and bounded by C/S and C/S^2. To construct it, mollify the distance s at scale $S/8$, then compose with a fixed cutoff. Given nonnegative Lipschitz F supported in $[1/2, 2]$ and tolerance $\varepsilon > 0$, choose S large. Then

$$\liminf_{R\to\infty} R^{-m} \int_\Sigma \chi F(|z-y|/R)\,dS \ge Km\omega_m \int_0^\infty t^{m-1}F(t)\,dt - \varepsilon. \tag{5.70}$$

A fixed ball about every zero has uniformly positive energy, by the range bound, local C^1 control, and $W_p(0) > 0$. The density bound therefore limits the number of unit-separated zeros in $B_R(y)$ by $C(R+1)^m$. To make the clearing estimate uniform in $R \ge 1$, cover the zeros in $B_{2R+j+2}(y)$ by unit balls centered at a maximal separated family. The part of $B_{2R}(y)$ at zero-set distance between j and $j+1$ lies in the union of the concentric balls of radius $j+2$. Its volume is at most $C(R+j+2)^m(j+2)^{m+1}$. Clearing and $e_u \asymp q^p$ therefore give

$$\int_{B_{2R}(y)\cap\{\operatorname{dist}(x,\Sigma)>P\}} q^p\,dx \le CR^m \sum_{j\ge\lfloor P\rfloor} (j+2)^{2m+1} e^{-cj} = o_P(1)R^m.$$

For fixed P, points within P of zeros with $s \le 2S$ carry mass $o(R^{m-2})$ by sparse packing. All remaining mass lies in injective good tubes. Their normal integral is $J_{p,P} + o_S(1) \le J_p + o_S(1)$, where $J_{p,P} = \int_{-P}^{P} G^p$. Changing the radial argument in F costs $O(P/R)$ times an $O(R^m)$ tube mass. Equation (5.65) gives the limiting weighted mass $J_pKm\omega_m \int t^{m-1}F(t)\,dt$. First obtain this for step functions of the radius. Uniform approximation and the mass cap then give it for the continuous weight F. The area formula and the preceding decomposition yield the explicit inequality

$$\begin{aligned}\int q^p F(|x-y|/R)\,dx \le (J_p + o_S(1)) \int_\Sigma \chi F(|z-y|/R)\,dS \\ + o_P(1)R^m + o_{R\to\infty;P,S}(R^{m-2}) + C_P R^{m-1}.\end{aligned}$$

The first error is the mass farther than P from the zero set, the second is the fixed bad-set collar, and the last is the change of the radial test along a fiber. These estimates are uniform in $y \in \mathcal{Z}$. Divide by R^m, let $R \to \infty$ with P, S fixed, and use the already computed radial mass limit. For a prescribed final tolerance, first choose P so that the tail error is small, and then choose S so that the upper fiber-mass error is small. This proves the asserted lower surface-mass bound without an exchange of the limits in P, S, R.

The same construction gives the collar-area estimate. Injective unit tubes over zeros with $S < s < 2S$ remain in a fixed-width neighborhood of $\mathcal{Z}$. Their Jacobians and unit normal energy integrals are bounded below. The sparse-volume bound for that fixed neighborhood consequently gives, for fixed large S,

$$\sup_{y\in\mathcal{Z}} \mathcal{H}^m(\Sigma \cap \{S < s < 2S\} \cap B_R(y)) = o(R^{m-2}). \tag{5.71}$$

For $z \in \Sigma_S$ let $\ell(z)$ be the distance to the nearest other local graph in a buffered chart. Since $K \geq 2$, the packet contains another graph at vertical distance $o(s)$. Its vertical point is a competitor, so $\ell = o(s)$. Points outside the larger buffered chart are at distance comparable to s. Thus every closer competitor stays inside that chart, which makes the definition independent of the chart. The minimizing graph is adjacent: a chord to a farther graph crosses an intervening zero at a smaller distance. The packet separation also gives $\ell \to \infty$, uniformly as $s \to \infty$. Locally, ℓ is the minimum of one or two adjacent smooth distance branches ℓ_b, whose feet are π_b.

The target curvature is $O(s^{-1})$. Choose a disk radius r equal to a sufficiently small fixed multiple of s, so that its larger disk stays in the packet and $r|B|$ is below the threshold in Lemma 3.10. The graph slopes are small, and the vertical competitor is at distance $o(s) < r$ for large S. The nearest-foot part of that lemma therefore gives a unique interior foot, smooth in the source point. Denote it by $\pi_b(X)$ for the chosen adjacent target branch. Orient the target normal N toward the source, so

$$X = \pi_b(X) + \ell_b(X)N(\pi_b(X)).$$

Differentiating the minimized distance also gives $\nabla \ell_b(X) = N(\pi_b(X))$.

This also puts the foot inside a nonsingular normal tube. With the target normal N directed toward the source, the ambient distance Hessian has tangential eigenvalues $\kappa_i/(1 + \ell_b\kappa_i) \leq \kappa_i$ and zero normal eigenvalue. Its norm is $O(s^{-1})$. For a source normal ν, the exact restriction formula is

$$\Delta_\Sigma \ell_b = \operatorname{tr} D^2\ell_b - D^2\ell_b[\nu, \nu] - H\,\nu \cdot N.$$

Indeed, $\ell_b|B_N| = o(1)$ makes every denominator positive, and

$$\operatorname{tr} D^2\ell_b = \sum_i \kappa_i - \ell_b \sum_i \frac{\kappa_i^2}{1 + \ell_b\kappa_i} \leq \sum_i \kappa_i \leq |H(\pi_b(z))|.$$

Since $D^2\ell_b N = 0$, its value on ν depends only on the component orthogonal to N. That component has squared length $1 - (\nu \cdot N)^2 = |\nabla_\Sigma \ell_b|^2$. The Hessian norm bound therefore proves

$$\Delta_\Sigma \ell_b \leq |H(z)| + |H(\pi_b(z))| + \frac{C}{s}|\nabla_\Sigma \ell_b|^2. \tag{5.72}$$

The exact foot-map Jacobian is

$$J_{\pi_b} = \frac{|\nu(z) \cdot N(\pi_b(z))|}{|\det(I + \ell_b B_N(\pi_b(z)))|} = 1 + o_S(1).$$

To obtain this Jacobian, differentiate $X = \pi_b(X) + \ell_b(X)N(\pi_b(X))$ and project onto $N^\perp$:

$$(I + \ell_b B_N)d\pi_b = P_{N^\perp}|_{T_z\Sigma}.$$

The projection on the right has absolute determinant $|\nu \cdot N|$. Taking determinants gives the displayed formula. Its limit follows from $\ell_b|B_N| = o(1)$ and the common small-slope packet. An active measurable foot selection has inverse multiplicity at most two. Indeed, all sources for one target lie within $o(s)$ in one graph chart. Its normal line meets each graph at most once. Only the immediately adjacent graph on either side can be a nearest source.

The area formula can be used for this selected map even though the selection need not be continuous at a tie. To justify this precisely, cover the prepared surface by countably many charts. Partition it into disjoint measurable subsets, each lying in one chart and selecting one of its smooth branches. Apply the area formula on each subset and then sum. If $\varepsilon_S \downarrow 0$ is chosen so that $J_{\pi_b} \geq 1 - \varepsilon_S$ on every branch, then every nonnegative measurable g and measurable source set A satisfy

$$\begin{aligned} \int_A g(\pi(z))\, dS_z &\leq \frac{1}{1 - \varepsilon_S} \int_{\pi(A)} g(w) \#\{z \in A : \pi(z) = w\}\, dS_w \\ &\leq \frac{2}{1 - \varepsilon_S} \int_{\pi(A)} g(w)\, dS_w. \end{aligned} \tag{5.73}$$

This is the precise change-of-variables estimate used below.

Moreover, $s(\pi(z))/s(z) = 1 + o_S(1)$ and $|\pi(z) - y|/|z - y| = 1 + o_S(1)$ for $y \in \mathcal{Z}$.

To make the gradient term nonpositive, take an increasing concave ψ such that $-\psi''(\ell)/\psi'(\ell) \geq C/s$ throughout the region in question. The capped transform $w = \min\{T, \psi(\ell)\}$ satisfies

$$\Delta_\Sigma w \leq L_\psi(|H| + |H \circ \pi|), \qquad L_\psi = \sup \psi'. \tag{5.74}$$

Each distance branch is smooth with locally bounded Hessian on the regular chart. On an active branch, the chain rule gives

$$\Delta_\Sigma \psi(\ell_b) \leq \psi'(\ell_b)(|H| + |H \circ \pi_b|) + \left(\frac{C}{s}\psi'(\ell_b) + \psi''(\ell_b)\right)|\nabla_\Sigma \ell_b|^2.$$

The last term is nonpositive. The finite minimum of these smooth branches and the constant T is semiconcave. Its singular Hessian is negative semidefinite. Almost everywhere, the ordinary Hessian of w equals that of an active branch, or is zero on the capped set. On positive-measure tie sets the derivatives of the equal smooth branches agree almost everywhere, so any measurable active choice gives the same conclusion.

In local coordinates the surface Laplacian is $g^{ij}\partial_{ij} + b^j\partial_j$, where (g^{ij}) is positive definite and the coefficients are smooth. The first order term is locally integrable, since w is locally Lipschitz. The singular part of its second order term is nonpositive, because it is the trace of a negative semidefinite matrix-valued measure against (g^{ij}). The almost-everywhere active-branch bound therefore proves the same distributional inequality. This also explains why neither a global sheet numbering nor a continuous selection of π is required.

A radial Green kernel compares the transformed gap with the mean-curvature terms. Choose $F \in C_c^\infty((1/2, 2))$ nonnegative and positive near one, normalized by $\int t^{m-1}F(t)\,dt = m - 2$. Define

$$\phi_R(r) = \int_r^{2R} t^{1-m}\left(R^{-m}\int_t^{2R} v^{m-1}F(v/R)\,dv\right)dt \quad (0 < r < 2R),$$

and set it to zero for $r \geq 2R$. Put $A_R(t) = R^{-m}\int_t^{2R} v^{m-1}F(v/R)\,dv$. Then $0 \leq A_R \leq m - 2$ and $A_R' = -R^{-m}t^{m-1}F(t/R)$. Direct differentiation gives

$$\phi_R' = -r^{1-m}A_R(r), \qquad \phi_R'' = (m-1)r^{-m}A_R(r) + R^{-m}F(r/R).$$

Therefore

$$\phi_R'' + \frac{m-1}{r}\phi_R' = R^{-m}F(r/R), \qquad 0 \leq \phi_R \leq r^{2-m}, \qquad r|\phi_R'| + r^2|\phi_R''| \leq Cr^{2-m}. \tag{5.75}$$

The bound on ϕ_R uses $(m-2)\int_r^\infty t^{1-m}dt = r^{2-m}$. Because F vanishes near 2, the extension at $2R$ is smooth. For $r < R/2$ the exact singular part of ϕ_R is r^{2-m}, up to a constant depending on R.

Test (5.74) with this kernel, the mask χ, and an inner radial cutoff. We obtain

$$\begin{aligned} R^{-m}\int_\Sigma w\chi F(r/R)\,dS \leq L_\psi \int_{\Sigma_S \cap \{R_0 < r < 2R\}} & r^{2-m}(|H| + |H \circ \pi|)\,dS \\ & + C_{R_0,S} + o_{R\to\infty;R_0,S}(\log R) + T\eta(R_0). \end{aligned} \tag{5.76}$$

Both constants and the remainder $o(\log R)$ are independent of T.

Choose an inner radial cutoff τ, equal to zero for $r \leq R_0$ and to one for $r \geq 2R_0$, with its first two derivatives bounded by C/R_0 and C/R_0^2. Test (5.74) with the nonnegative smooth compact function $\chi\tau\phi_R$. Distributional integration by parts twice is legitimate for the locally Lipschitz semiconcave w and gives

$$\int_\Sigma w\Delta_\Sigma(\chi\tau\phi_R) \leq L_\psi \int_\Sigma (|H| + |H \circ \pi|)\chi\tau\phi_R.$$

There is no boundary term at the irregular part of Σ: the mask vanishes on a neighborhood of that part. For $R > 4R_0$, the support of $F(r/R)$ lies where $\tau = 1$.

The product rule on the surface is

$$\Delta_\Sigma(\chi\tau\phi_R) = \chi\tau\Delta_\Sigma\phi_R + \chi\phi_R\Delta_\Sigma\tau + \tau\phi_R\Delta_\Sigma\chi + 2\chi\nabla_\Sigma\tau\cdot\nabla_\Sigma\phi_R + 2\tau\nabla_\Sigma\chi\cdot\nabla_\Sigma\phi_R + 2\phi_R\nabla_\Sigma\chi\cdot\nabla_\Sigma\tau.$$

All derivatives fall on the smooth test. In particular, no estimate for $\nabla_\Sigma w$ is needed at a switching set. The terms containing a derivative of τ are supported in the fixed inner annulus. Every other cutoff term contains a derivative of χ and is supported in its fixed collar. On both sets $w \le \psi(\ell)$ is bounded independently of T. Constants in the cutoff estimates may depend on the fixed function ψ, as well as on R_0, S, but never on the cap.

On the inner transition $R_0 < r < 2R_0$ one has $s \le r \le 2R_0$, because $y \in \mathcal{Z}$. The packet bound $\ell \le Cs$ therefore bounds $w \le \psi(\ell)$ by a constant depending on R_0, not on T. Area and derivative bounds control all terms containing a derivative of τ by a constant $C_{R_0,S}$. On the collar supporting derivatives of χ, one has $s \asymp S$ and hence $\ell \le C_S$. Again $w \le \psi(C_S)$ is independent of T. Also

$$|\Delta_\Sigma\chi| \le C|D^2\chi| + |H||\nabla\chi| \le C_S.$$

On a dyadic shell $r \asymp t$, all terms containing derivatives of χ are bounded by

$$C_S t^{2-m}\mathcal{H}^m(\Sigma \cap \{cS < s < CS\} \cap \{r \asymp t\}) = o_t(1)$$

using (5.71). There are $O(\log R)$ shells. If a bounded sequence tends to zero, its sum over the first N terms is $o(N)$; applying this elementary Cesàro argument to the dyadic shell errors gives $o(\log R)$.

They are uniform for $y \in \mathcal{Z}$, by the uniform density cap, collar packing, and packet bounds. Finally, radial restriction gives

$$\Delta_\Sigma\phi_R = \phi_R'' + (m-1)\phi_R'/r + (\phi_R'/r - \phi_R'')(e_R\cdot\nu)^2 - H\phi_R'(e_R\cdot\nu).$$

The first two terms are $R^{-m}F(r/R)$ by (5.75). Since $0 \le w \le T$, the remaining geometric terms are bounded by $CT\eta(R_0)$ using (5.67); absorb the fixed constant into η. This proves (5.76), with all cutoff errors independent of T.

Begin with $\Psi(t) = \sqrt{t}$. The property $\ell = o(s)$ gives its required concavity, since $-\Psi''/\Psi' = 1/(2\ell)$ and $\ell/s \to 0$. Its derivative is at most one far from the bad set. Bounds (5.66) and (5.76) yield

$$\ell(z) \le C\{(\log s(z))^2 + s(z)^{2-\alpha}\}. \tag{5.77}$$

To verify this, fix z with $s(z) = R$ and choose $y \in \mathcal{Z}$ with $|z - y| = R$. The adjacent height difference $h > 0$ solves a uniformly elliptic divergence equation with forcing $O(R^{-\alpha})$.

Indeed, put $\mathscr{F}(\xi) = \xi/\sqrt{1+|\xi|^2}$ and

$$a^{ij}(x') = \int_0^1 \partial_{\xi_j}\mathscr{F}_i(Df_- + t(Df_+ - Df_-))\,dt.$$

Subtracting the graph mean-curvature equations gives

$$\partial_i(a^{ij}\partial_j h) = F_0, \qquad |F_0| \le CR^{-\alpha}.$$

Here the two graph normals are both chosen upward for this calculation; their signed means have the same absolute bounds as the original means. The eigenvalues of $D\mathscr{F}(\xi)$ are $(1+|\xi|^2)^{-1/2}$ and $(1+|\xi|^2)^{-3/2}$. Thus a^{ij} has fixed positive ellipticity constants. The inhomogeneous Harnack inequality on the buffered disk after rescaling to unit radius, gives

$$\sup_{B^m_{cR}} h \le C_H\left(\inf_{B^m_{cR}} h + R^2\|F_0\|_{L^\infty}\right).$$

The disk is centered at the horizontal projection x_z' of z, and its fixed relative radius $c > 0$ can be decreased below the packet buffer radius. In particular,

$$\inf_{B^m_{cR}} h \ge C_H^{-1}h(x_z') - CR^{2-\alpha}.$$

The exponent $2-\alpha$ is the result of multiplying the forcing bound $CR^{-\alpha}$ by the square of the radius.

If the source has two neighbors, apply this estimate to both gaps on the same smaller disk. Their central heights are at least $\ell(z)$. The following comparison transfers the lower bounds to Euclidean distance.

We make the comparison with Euclidean gaps explicit. If the target graph has Lipschitz constant at most L, its vertical gap $h_b(x')$ from the source at the same horizontal point and its distance branch ℓ_b satisfy

$$\frac{h_b(x')}{\sqrt{1+L^2}} \le \ell_b(x') \le h_b(x').$$

The upper bound uses the vertical candidate. For the lower bound, if $Y = (y', f(y'))$ is any point on the target and X is the source point, then

$$h_b(x') \le |X_n - f(y')| + L|x' - y'| \le \sqrt{1+L^2}\,|X - Y|.$$

Take the infimum in Y. Minimizing feet are in the interior of the common buffered chart, so this estimate applies to the distance branches under consideration.

Thus either $\ell(z) \le CR^{2-\alpha}$, or a source patch of area cR^m has $\ell \ge c\ell(z)$. Reduce the fixed patch radius so that $F(|\cdot - y|/R)$ is bounded below there. The patch lies in $\{s > 2S\}$ when R is large. In the latter case choose $T = \sqrt{\ell(z)}$ in (5.76). Its left side is at least $c'T$. By (5.66) and foot multiplicity two, the mean integral in each dyadic shell is bounded by a fixed constant. Its sum is therefore $O(1 + \log R)$. Equation (5.76) gives

$$c'T \le C_{R_0,S} + C\log R + o(\log R) + T\eta(R_0).$$

Fix R_0 so that $\eta(R_0) < c'/2$ and absorb the last term. Since $T = \sqrt{\ell(z)}$, this gives $\ell(z) \le C(\log R)^2$ in the second case. The first case gives $\ell(z) \le CR^{2-\alpha}$. Together they prove (5.77).

To obtain a transform whose derivative tends to one, use the already fixed exponent $1 < \alpha < 2$ from (5.62), and take the second transform

$$\Psi(t) = t + t^{2-\alpha}, \qquad t \ge 1. \tag{5.78}$$

It satisfies $\Psi(t) \ge t$ and $\Psi'(t) \to 1$. More explicitly,

$$-\frac{\Psi''(t)}{\Psi'(t)} = \frac{(\alpha-1)(2-\alpha)t^{-\alpha}}{1+(2-\alpha)t^{1-\alpha}} \asymp t^{-\alpha} > 0.$$

Thus it is increasing and concave in precisely the range needed here. By (5.77),

$$\frac{\ell^\alpha}{s} \le C\left\{\frac{(\log s)^{2\alpha}}{s} + s^{\alpha(2-\alpha)-1}\right\} \to 0, \qquad \alpha(2-\alpha) - 1 = -(\alpha-1)^2 < 0.$$

Thus (5.76) applies with $L_\Psi \le 1 + \varepsilon$ after increasing S by a fixed amount. Within the established range $1 < \alpha < 2$, no stronger decay exponent is needed.

The weighted mean form controls the leading upper coefficient. Transfer the receiving-foot integral first. The inverse multiplicity is at most two, and the radial Jacobian differs from one by $o_S(1)$. Adding the source integral gives at most

$$(3 + o_S(1)) \int_{\Sigma_{S/2} \cap \{R_0/2 < r < 4R\}} |H| r^{2-m}\, dS.$$

Choose a smooth μ_S, zero for $s \le S/4$ and one for $s \ge S/2$, with $|\nabla \mu_S| \le C/S$. Take $\zeta_{R_0,R}$ equal to one on the required annulus and zero outside $R_0/4 < r < 8R$, with standard dyadic derivative bounds. Fix the normal window P supplied by Theorem 3.7 for the chosen tolerance, and increase S to meet its defect threshold on $s > S/4$. Apply its weighted estimate before bounding I_P by a constant. Use the compact test

$$\varphi = \mu_S \zeta_{R_0,R} r^{-(m-2)/2}.$$

Discard the favorable negative curvature term. This gives

$$\beta \int |H|\varphi^2 \le (1/2+\varepsilon)\int I\varphi^2 + C_{\mathcal{P}}\int I|\nabla\varphi|^2. \tag{5.79}$$

Its derivative error is $O_{R_0,S}(1) + o_{R\to\infty;S}(\log R)$. On shells $r \asymp t$, the radial and annular terms are bounded by $Ct^{-m}\int I \le Ct^{-2}$ using (5.66). These bounds are summable. The mask derivative costs at most

$$C_S t^{2-m}\mathcal{H}^m(\Sigma \cap \{S/4 < s < S/2\} \cap \{r \asymp t\}) = o_t(1)$$

on each shell, by (5.71) and $I \le C$. At fixed S, Cesàro summation gives $o(\log R)$. Hence there is no leading Dirichlet contribution.

At distant bases, these fixed P-tubes are jointly injective, with Jacobian $1 + o_S(1)$. On each fiber, the radial coordinate changes by at most P. For $R_0 > 8P$, its factor r^{2-m} therefore differs from the base value by a relative error $O(P/r)$. The scalar q-test and density cap used above give $\int_{B_t(y)} q^{p+1}D \le Ct^{m-2}$. Thus the resulting weighted error on a dyadic shell $r \asymp t$ is at most

$$CPt^{1-m}\int_{B_{Ct}(y)} q^{p+1}D\,dx \le CP/t.$$

These errors are summable, so their total is bounded independently of R. The area formula and $0 \le \mu_S, \zeta_{R_0,R} \le 1$ now bound $\int I\varphi^2$ by $1 + o_S(1)$ times the normalized ambient integral in (5.69), on fixed enlargements of the annuli, plus this bounded error. The annular enlargements do not change the leading logarithmic coefficient. We obtain

$$\limsup_{R\to\infty} \frac{L_\Psi}{\log R}\int_{\Sigma_S\cap\{R_0<r<2R\}} r^{2-m}(|H| + |H\circ\pi|)\,dS \le \frac{3(1/2+O(\varepsilon))}{\beta} Km\omega_m \frac{(m-2)^2}{4}. \tag{5.80}$$

For the lower coefficient, fix $\epsilon > 0$. At adjacent feet the oriented normals point in opposite directions. Replace u by $-u$ if necessary so that the intervening phase is positive; this leaves D unchanged and gives the orientation required by Corollary 2.22. The minimizing chord is normal to the receiving graph, and the vanishing packet slopes place it in the fixed cone about the source normal. After increasing S, that corollary applies uniformly and gives

$$D(z) + D(\pi(z)) \ge c_\epsilon e^{-(\beta+\epsilon)\ell(z)}. \tag{5.81}$$

The opposing-chord estimate already gives this bound. No further propagation or Hessian estimate across a critical gradient is needed.

Choose $0 < \varepsilon < 2/\beta$, set $T = (2/\beta - \varepsilon)\log R$, and then choose $\epsilon > 0$ so that $b_* = (\beta+\epsilon)(2/\beta-\varepsilon) < 2$. On $R/2 < r < 2R$, the endpoint estimate gives $D(z) + D(\pi(z)) \ge c_\epsilon R^{-b_*}$ whenever $\ell(z) < T$. Use the area formula, the inverse multiplicity bound two, and (5.66) on an enlarged annulus. The short-gap set consequently has area at most

$$CR^{b_*}\int (D + D\circ\pi)\,dS \le CR^{m-2+b_*} = o(R^m).$$

The finite windows can be chosen together without changing the mean estimate. First choose one for the mean form and one for the profile tail, then take P at least as large as both. The quantity I_P is nondecreasing in P. The two terms multiplying it in the mean form have nonnegative coefficients, so that enlargement preserves the estimate. Choose S afterward for all fixed-window, Jacobian, chord, and mass tolerances. Fix $R_0 > 16P$ before taking the outer radius to infinity. Then increase R_0 with P, S fixed, and only finally decrease the tolerances. For each new tolerance, choose new finite P, S before the next outer-radius limit. No constant attached to a fixed window or collar must be uniform in that last step.

For S chosen at the prescribed mass tolerance, (5.70) shows that the retained surface mass has lower limit at least $Km\omega_m\int t^{m-1}F(t)\,dt - \varepsilon$ after division by R^m. Removing the short-gap set

changes this normalized integral by $o(1)$. Since (5.78) dominates the identity, the rest has $w = T$. Thus

$$\liminf_{R\to\infty} \frac{R^{-m}}{\log R} \int w\chi F(r/R)\, dS \geq (2/\beta - \varepsilon)[Km\omega_m(m-2) - \varepsilon].$$

In (5.76) first take $R \to \infty$, then $R_0 \to \infty$. Finally send all fixed tolerances to zero. Comparing with (5.80) gives

$$\frac{2}{\beta} Km\omega_m(m-2) \leq \frac{3}{2\beta} Km\omega_m \frac{(m-2)^2}{4}.$$

The factor $Km\omega_m(m-2)/\beta$ is positive, since $m \geq 3$. Dividing by it would imply

$$2 \leq \frac{3}{8}(m-2).$$

This is impossible for $3 \leq m \leq 6$. Equivalently, the effective mean coefficient $1/2$ lies strictly below the critical value $2/3$ in dimension six. The critical family is therefore excluded for every exponent in $\mathcal{P}$.

□

Return to $\mathbb{R}^7$ and take $p > 2$. At a critical gradient the linearized matrix is $A_p(0) = 0$. Call the constants $u \equiv -1, 1$ wells. A layer is $g_p(x \cdot e - t)$ with $|e| = 1$. We consider bounded entire weak solutions with $|u| \leq 1$. By Remark 2.2, each nonconstant solution has strict range $(-1, 1)$. Thus all preceding strict-range results apply.

Theorem 5.25 (Finite-density stable classification)**.** *Let $p > 2$. Every scalar-stable entire solution with $|u| \leq 1$ and*

$$M_\infty(u) := \lim_{R\to\infty} M_R(u; 0) < \infty$$

is a well or a layer.

Proof. The constant solutions are $-1, 0, 1$. The zero constant is unstable, with form $-\int \phi^2$. It remains to exclude a nonconstant nonplanar solution of finite density.

Fix p. Theorem 4.12 gives a threshold $\delta_0 > 0$ such that every nonplanar stable solution has a zero with $D \geq \delta_0$. Proposition 5.17 gives a stable solution anchored at such a zero that attains the least nonplanar density K_*. Almost-flat zeros give $K_* \geq 1$. Proposition 5.17 also gives uniform saturation at all bad zeros of the critical family.

By Theorem 5.16, $K_* = 1$ is impossible. Theorems 5.18 and 5.22 show that K_* is an integer at least two. Corollary 5.21 gives uniformly flat blowdowns and little-oh bad-collar packing. Corollary 5.23 supplies precisely K_* ordered local graphs on disks comparable to their distance from the bad set. Neither a global numbering nor one common plane is needed.

Clean-ball preparation provides $1 < \alpha < 2$, as in (4.74). These packets meet every hypothesis of Theorem 5.24. Its logarithmic comparison would require $2 \leq 3(m-2)/8 = 3/2$ at $m = 6$. This contradiction excludes K_* and proves the classification.

□

Proposition 5.26 (Uniform interior preparation from one density cap)**.** *Fix a compact exponent interval $\mathcal{P} \Subset (2, \infty)$, a finite number Λ, an interior fraction $0 < \theta < 1$, and a threshold $\delta > 0$. There exists $R_* = R_*(\mathcal{P}, \Lambda, \theta, \delta) < \infty$ such that every scalar-stable entire solution at an exponent $p \in \mathcal{P}$ satisfying*

$$R \geq R_*, \qquad M_R(u; x_0) \leq \Lambda$$

has

$$D(z) < \delta \quad \text{for every } z \in \{u = 0\} \cap B_{\theta R}(x_0).$$

In particular, for every smaller fraction $0 < \theta' < \theta$, the established clean-ball estimates give, for all sufficiently large R,

$$\sup_{\{u=0\}\cap B_{\theta' R}(x_0)} (D + |H| + |\nabla_\Sigma H|) \le C R^{-\alpha}, \qquad \sup_{\{u=0\}\cap B_{\theta' R}(x_0)} |B| \le C/R.$$

For every fixed $b < 1$ the corresponding fixed-core transversality and full-normal curvature estimates hold on a smaller fraction. Every fixed normal window at such zeros converges uniformly, through each fixed derivative order, to the appropriately oriented matched profile as $R \to \infty$. This proposition does not assert that the interior zero set has only one component.

Proof. If the first conclusion failed, choose $p_j \in \mathcal{P}$, $R_j \to \infty$, stable solutions u_j, centers x_j, and zeros $z_j \in B_{\theta R_j}(x_j)$ with $D_j(z_j) \ge \delta$ and $M_{R_j}(u_j; x_j) \le \Lambda$. Translate z_j to zero. Along a subsequence, $p_j \to p_* \in \mathcal{P}$ and $u_j \to v$ in C^1_{loc}. The limit has $v(0) = 0$. For fixed L, monotonicity centered at z_j and the inclusion $B_{(1-\theta)R_j}(z_j) \subset B_{R_j}(x_j)$ give

$$M_L(u_j; z_j) \le \frac{\Lambda}{(1-\theta)^6} \quad \text{for all sufficiently large } j.$$

Energy convergence and continuity of σ_{p_j} imply $M_\infty(v) \le \Lambda(1-\theta)^{-6}$.

The compactness and stability argument in Section 2 applies to this varying-exponent sequence. The zero-centered limit cannot be a well, and the zero constant is unstable. Continuity preserves the anchored defect: $D_v(0) \ge \delta$. Apply Theorem 5.25 to this one limit at exponent p_*. It makes v a layer, contradicting the positive anchored defect. Thus the compactness argument gives a radius uniform on $\mathcal{P}$, although the classification was stated at a fixed exponent.

For the quantitative bounds, first apply the established assertion using the fixed bad threshold of Theorem 4.14. Each point in $B_{\theta' R}(x_0)$ then has a clean ball of radius comparable to $(\theta - \theta')R$. Apply Theorem 4.14 on these balls, using its already fixed exponents. Its core conclusion gives transversality and full normal control. Relative phase estimates give convergence on every fixed normal window. These windows stay uniformly noncritical, so interior elliptic regularity improves convergence through every fixed derivative order. □

Proposition 5.26 supplies the finite-density preparation needed for reach geometry. Chord confinement and separation of scales will also isolate a single graph.

6. Reach geometry and global stability

Fix $\mathcal{P} \Subset (2, \infty)$ and work in $\mathbb{R}^7$, with $m = 6$. Keep the phase and surface notation of Section 2 and the energy and density notation of Section 5. Write $g = g_p$ and $M = G^{p/2}$. The global argument concerns a nonplanar stable entire matched solution with unbounded energy density. The local estimates retain their stated scopes. For an orientation N, write B_N, H_N. In matrix formulas, I is the identity on the relevant ambient or tangent space. The scalar normal mass introduced below is $I = I_P$ from (2.1).

6.1. Penalized reach and tangent balls.

Define $K(R) = 1 + \sup_x M_R(u; x)$. Exact monotonicity and a uniform pointwise energy bound imply

$$1 \le K(R) \le 1 + CR, \quad K(R) \longrightarrow \infty, \qquad K(t) \le C(t/R)K(R) \quad (t \ge R \ge 1).$$

To obtain the last inequality, cover $B_t(x)$ by at most $C(t/R)^7$ balls of radius R. Their energies are bounded by $CR^6K(R)$. Divide the sum by t^6 and then take the supremum over x.

We also need the good-surface area bound

$$\mathcal{H}^6(\Sigma_g \cap B_R(x)) \le CK(R)R^6 \qquad (R \ge 1). \tag{6.1}$$

This holds for $\Sigma_g = \{z \in \Sigma : D(z) < \delta\}$ with $0 < \delta \le \delta_* < 1$. Its constants depend only on the fixed upper threshold δ_* and $\mathcal{P}$. At a good zero z, the gradient estimate and its defect bound give

$|\nabla u(z)| \geq c > 0$ uniformly on $\mathcal{P}$. Uniform local $C^{1,\alpha}$ regularity gives one radius $r_* > 0$ on which a fixed component of the gradient stays positive. The zero set in that ball is therefore one graph over its orthogonal plane, with a uniform slope bound. In particular its area in a fixed smaller ball is at most Cr_*^6. Decrease r_* so that the global gradient bound also gives $|u| \leq 1/2$ on every $B_{r_*}(z)$ centered at a zero. The potential then has a uniform positive lower bound there, and

$$\int_{B_{r_*/16}(z)} e_p(u) \geq cr_*^7.$$

Take a maximal $r_*/8$-separated family of good zeros in $B_R(x)$. Its $r_*/16$-balls are disjoint and its $r_*/8$-balls cover that good zero set. The graph area estimate bounds its area by a fixed multiple of the number of centers. The preceding energy lower bound bounds that number by $C\int_{B_{R+r_*}(x)} e_p(u)$. The definition of K and its covering bound therefore give

$$\mathcal{H}^6(\Sigma_g \cap B_R(x)) \leq CE_p(u; B_{R+r_*}(x)) \leq CK(R+r_*)(R+r_*)^6 \leq CK(R)R^6.$$

All radii and constants are fixed by the good threshold and $\mathcal{P}$. This proof uses local graphs and disjoint energy balls. It requires no global normal parametrization.

Relative discrepancy comparison and phase clearing yield

$$D(x) \leq Ce^{C|x-y|}D(y), \tag{6.2}$$

$$e_p(u)(x) \leq Ce^{-c\,\mathrm{dist}(x,\Sigma)}. \tag{6.3}$$

For the first bound, apply the unit-ball comparison (2.30) along a chain of at most $1 + |x - y|$ unit steps. Its constants are independent of the phase value. We use this isotropic estimate only when the exact exponential rate is irrelevant.

Take distinct zeros z, y and assume that z is regular. Set

$$\ell = |y - z|, \qquad e = (y - z)/\ell, \qquad s = |\nu(z) \cdot e|.$$

If $s > 0$, then ℓ/s equals twice the radius of the sphere through the two zeros whose center lies on the base normal line. Along a principal direction, the diagonal limit is $2/|\lambda_i|$ whenever the curvature is nonzero.

Following the penalized-reach construction in [35, Section 7.1], we change the quotient only for nearly normal chords. Choose the angular factor $\Gamma(s)$ to agree with $1/s$ away from $s = 1$. It will remain uniformly close to $1/s$ and become constant near 1. On this constant part, the chord is normal at its second contact. This fact permits the signed transfer in Proposition 6.7. A small weighted second derivative keeps the differentiation error arbitrarily small.

Fix the accuracy $\varepsilon_1 > 0$. Choose $0 < \delta_1 < 1/10$, then $L > 1$, and set $\delta_2 = \delta_1 e^{-L}$. Take a smooth nonincreasing function $\chi : \mathbb{R} \to [0, 1]$. Require it to equal 1 on $(-\infty, 0]$ and 0 on $[1, \infty)$. For $s \geq 1 - \delta_1$ put

$$\Gamma(s) = \frac{1}{1-\delta_1} - \int_{1-\delta_1}^{s} u^{-2}\chi\left(\frac{\log(\delta_1/(1-u))}{L}\right) \mathrm{d}u,$$

and use $\Gamma(s) = s^{-1}$ below $1 - \delta_1$. Set the integrand to zero near $u = 1$. The resulting function is smooth on $(0, 1]$. Its value is the constant G_1 for $s \geq 1 - \delta_2$. Differentiation gives comparison bounds on $(0, 1]$ and derivative bounds on $[1 - \delta_1, 1]$:

$$\begin{gathered} s^{-1} \leq \Gamma(s) \leq (1 + C\delta_1)s^{-1}, \qquad 1 \leq G_1 \leq \Gamma(s), \\ -(1-\delta_1)^{-2} \leq \Gamma'(s) \leq 0, \qquad 0 \leq (1-s)\Gamma''(s) \leq \frac{2\delta_1}{(1-\delta_1)^3} + \frac{\|\chi'\|_\infty}{L(1-\delta_1)^2} \leq \varepsilon_1 \quad (s \geq 1 - \delta_1). \end{gathered} \tag{6.4}$$

First decrease δ_1 and then increase L to obtain the last inequality. Thus $G_1 - 1$ and ε_1 can both be made arbitrarily small. Fix these parameters before taking any large-reach or exhaustion limit.

Indeed, on the transition interval, if $\tau(s) = L^{-1}\log(\delta_1/(1-s))$, then

$$\Gamma'(s) = -s^{-2}\chi(\tau(s)), \qquad \Gamma''(s) = 2s^{-3}\chi(\tau(s)) - \frac{s^{-2}\chi'(\tau(s))}{L(1-s)}.$$

Both terms in the second formula are nonnegative. Also

$$\Gamma(s) - s^{-1} = \int_{1-\delta_1}^{s} u^{-2}(1-\chi(\tau(u)))\,du \in [0, C\delta_1].$$

These identities prove (6.4).

On Σ_{reg} define

$$d(z) = \inf_{y\in\Sigma\setminus\{z\}} \ell\Gamma(s), \qquad \ell = |y-z|, \quad e = \frac{y-z}{\ell}, \quad s = |\nu(z)\cdot e|. \tag{6.5}$$

A competitor with $s = 0$ is assigned value $+\infty$. An *active contact* is a competitor or a diagonal limit attaining this infimum. In the latter case, the second point tends to the base. Proposition 6.7 shows that one of these alternatives always occurs.

Lemma 6.1 (Elementary chord bounds)**.** *The function d is finite and positive on Σ_{reg}. At every such point, writing λ_i for the eigenvalues of B,*

$$d\max_i|\lambda_i| \le 2, \qquad |(y-z)\cdot\nu(z)| \le C_\Gamma|y-z|^2/d(z) \quad (y\in\Sigma). \tag{6.6}$$

If an active contact is diagonal, its limiting unit tangent direction v satisfies

$$|B_z(v,v)| = 2/d(z), \qquad b(z)d(z) \ge 2, \qquad b = |B|.$$

Such a direction belongs to an eigenspace with maximal absolute principal curvature. Fix $J < \infty$. There is a uniform $\delta(J) > 0$ for which $u(z) = 0$ and $D(z) < \delta(J)$ imply $d(z) > J$. At each regular base, the two normal directions give tangent open balls of radius $c_\Gamma d(z)$. Both balls are zero-free.

Proof. A local smooth graph bounds $|\nu(z)\cdot(y-z)|/|y-z|^2$ for nearby zeros. Outside a fixed small ball, the competitors have a positive lower bound since $\Gamma \ge 1$. Hence the reach is positive. If all zeros belonged to the tangent plane at z, (6.3) would bound the energy by $Ce^{-c\,\mathrm{dist}(x,T_z\Sigma)}$. The density would then be finite, contrary to the standing assumption.

Indeed, integrate the clearing bound first in the normal variable and then in the six tangential variables. The normal exponential is integrable. The tangential section of any radius-R ball has volume at most CR^6, uniformly in its center. Thus every such ball has energy at most CR^6 for $R \ge 1$.

Thus a finite competitor exists.

For a unit principal direction, let γ be the surface geodesic with that initial velocity. The convention $B = d\nu$ gives $\gamma''(0)\cdot\nu = -\lambda_i$. Taylor's formula yields

$$|\gamma(t) - z| = |t| + O(t^2), \qquad |\nu(z)\cdot(\gamma(t)-z)| = \tfrac{1}{2}|\lambda_i|t^2 + o(t^2).$$

For $\lambda_i \neq 0$, the incidence tends to zero. The penalty is then exactly the ordinary chord factor, so this limit is an admissible upper bound for the infimum.

This proves $d\max_i|\lambda_i| \le 2$. To treat every active diagonal direction, let $y_k \to z$ be its attaining chord sequence. In a smooth graph over $T_z\Sigma$, write its tangential displacement as $t_k v_k$, where $t_k > 0$ and $|v_k| = 1$. Pass to a subsequence with $v_k \to v$. Taylor's formula is uniform on the unit tangent sphere and gives

$$|y_k - z|^2 = t_k^2 + o(t_k^2), \qquad |\nu(z)\cdot(y_k - z)| = \tfrac{1}{2}|B_z(v_k,v_k)|t_k^2 + o(t_k^2).$$

Thus the incidence s_k tends to zero. For large k the penalty is exactly $\Gamma(s_k) = 1/s_k$, rather than its modified near-normal branch. Taking reciprocals of the chord costs gives

$$\frac{1}{|y_k - z|\Gamma(s_k)} = \frac{|\nu(z)\cdot(y_k-z)|}{|y_k-z|^2} \longrightarrow \tfrac{1}{2}|B_z(v,v)| = \frac{1}{d(z)}.$$

If $B_z(v,v)=0$, this limit is zero and the cost diverges, so that direction cannot attain the finite reach. Hence every active diagonal direction has $|B_z(v,v)|=2/d(z)$. The Frobenius norm satisfies $b(z)\geq|B_z(v,v)|$, proving $b(z)d(z)\geq 2$. Moreover

$$|B_z(v,v)|\leq\max_i|\lambda_i|\leq 2/d(z)=|B_z(v,v)|.$$

Equality in this Rayleigh-quotient bound places v in an eigenspace with maximal absolute principal curvature. After a choice of normal, it is the top positive eigenspace used below. For the chord bound, use the upper estimate on the penalty: $d\leq\ell\Gamma(s)\leq C_\Gamma\ell/s$.

Suppose, to the contrary, that $D(z_j)\to 0$ and $d(z_j)\leq J$ for a fixed $J>0$. Translate and rotate the phases U. Equation (6.2) and the phase equation give smooth convergence on every fixed ball to x_n. In particular, the zeros in B_{4J+4} form one graph whose slope and curvature tend to zero. Choose a genuine competitor with cost at most $J+1$; this uses only the definition of the infimum. Its length satisfies $\ell\leq J+1$. Taylor's formula for this graph, based at z_j, gives $s\leq o(1)\ell$, uniformly also when $\ell\to 0$. Thus its cost satisfies $\ell\Gamma(s)\geq\ell/s\to\infty$, a contradiction. No attainment of the infimum is needed here. Compactness in $p\in\mathcal{P}$ gives the uniform threshold. For the tangent balls, consider $B_{c_\Gamma d}(z\pm c_\Gamma d\nu)$ with c_Γ small enough. A point in either ball satisfies $|\nu\cdot(y-z)|>|y-z|^2/(2c_\Gamma d)$, contrary to (6.6). Hence the balls are zero-free. □

Remark 6.2 (Jointly injective ordinary tubes). For the final packing estimate, we use the ordinary tube construction of [35, Remark 7.2]. The chord bound gives its injectivity. If $d(z)>L$, choose $0<c<\min\{1/(2C_\Gamma),1/2\}$. For $0<|t|<cL$ and every other zero y,

$$|z+t\nu(z)-y|^2-t^2=|y-z|^2-2t\nu(z)\cdot(y-z)\geq(1-2cC_\Gamma)|y-z|^2>0.$$

Thus z is the only zero that minimizes distance from $z+t\nu(z)$. This proves joint injectivity on both normal sides for tubes based where $d>L$, throughout the stated range. The curvature bound $\max|\lambda_i|\leq 2/d$ implies $\det(I+tB)\geq(1-2c)^m>0$. Consider these bases in B_{2R}. Their area is bounded by CR^{m+1}/L if $L\leq R$: integrate the positive normal Jacobian in both directions and use the enclosing volume $|B_{3R}|$.

The next tube argument is geometric. It has no exponent restriction once the regular zero set and its reach are given.

For measurable bases $E\subset B_{2R}$ with $d>L$ and $L\leq R$, write the tube-volume calculation from Remark 6.2 as

$$2cL(1-2c)^m\mathcal{H}^m(E)\leq\int_E\int_{-cL}^{cL}\det(I+tB)\,dt\,dS\leq|B_{3R}|.$$

For general measurable sets of regular bases, exhaust by compact subsets of coordinate charts. This gives the same estimate without a compactness assumption on the set of bases.

Lemma 6.3 (Exact nearest-zero tube depth)**.** *For the angular penalty used in the reach construction one has*

$$s\Gamma(s)\leq G_1\qquad(0<s\leq 1).$$

Consequently, at every regular zero z, every point $x=z+t\nu(z)$ with $|t|<d(z)/(2G_1)$ has z as its unique nearest zero. These open normal tubes are jointly injective.

Proof. Outside the transition region we have $\Gamma(s)=1/s$. Within that region, $\Gamma'=-s^{-2}\chi_{\mathrm{tr}}(s)$ with $0\leq\chi_{\mathrm{tr}}\leq 1$ and $\Gamma\geq 1/s$. It follows that

$$(s\Gamma(s))'=\Gamma(s)-\chi_{\mathrm{tr}}(s)/s\geq 0.$$

At $s=1$ this product has value G_1. This proves the first claim.

Assume that another zero y is at least as close to x as z is. Write $\ell=|y-z|$ and $s=|\nu(z)\cdot(y-z)|/\ell$. Then $|x-y|^2\leq|x-z|^2=t^2$ implies $s>0$ and $\ell/s\leq 2|t|$. By the definition of reach,

$$d(z)\leq\ell\Gamma(s)=\frac{\ell}{s}\,s\Gamma(s)\leq 2G_1|t|,$$

This is impossible. Two distinct tubes could meet only at a point with two nearest zeros. The tubes are therefore jointly injective. Their normal maps are nonsingular as well. Indeed, $d|\lambda_i| \le 2$ and $G_1 \ge 1$ imply $|t\lambda_i| < 1/G_1 \le 1$. □

To control derivatives of the zero sheets, first find a scale with bounded density. For the density-window and isolated-sheet estimates, let u be a nonplanar scalar-stable entire matched solution with values in $(-1, 1)$, and set $T = D^2U$. We use the Modica bound, the relative phase estimates, exponential comparison for D, and phase clearing.

The density-window and isolated-sheet estimates use only the elementary properties of the reach d. They are

$$d(z)|B(z)| \le C, \qquad |(y-z)\cdot\nu(z)| \le C|y-z|^2/d(z) \quad (y \in \{u=0\}), \tag{6.7}$$

and the existence of empty tangent phase balls with radius comparable to $d(z)$. We will then differentiate the penalty and count receiving feet.

The stress identity links this local geometry to energy density. Set

$$e = e_p(u), \qquad \mathbb{S} = eI - |\nabla u|^{p-2}\nabla u \otimes \nabla u.$$

Proposition 5.6 gives $\operatorname{div}\mathbb{S} = 0$ distributionally, including at critical gradients. Its arbitrary compact vector-field identity is the one used below.

With $k = aq$, the energy and stress are

$$e = \frac{k^p}{p}(r^p + c), \qquad \mathbb{S} = k^p\left(\frac{r^p + c}{p}I - r^{p-2}Z \otimes Z\right).$$

When $Z \neq 0$, the eigenvalue of $\mathbb{S}$ in its direction is ck^pD/p, and its other eigenvalues are e. At $Z = 0$, one has $\mathbb{S} = eI$. Thus the Modica bound gives $0 \le \mathbb{S} \le eI$. Applying this eigenvalue bound to $\mathbb{S}^2$ gives $|\mathbb{S}v|^2 \le e\, v\cdot\mathbb{S}v$ for every vector v. Choose coordinates with last direction n. The stress then satisfies

$$|\mathbb{S}e_n|^2 \le e\,\mathbb{S}_{nn}, \qquad \mathbb{S}_{nn} = k^p\left(\frac{c}{p}D + r^{p-2}|Z_T|^2\right) \ge c_{\mathcal{P}}k^p(1 - Z_n^2). \tag{6.8}$$

For the final inequality, if $r \le 1/2$, then $D \ge 1 - 2^{-p_-} > 0$, whereas $1 - Z_n^2 \le 1$. If $r \ge 1/2$, compactness of $\mathcal{P}$ gives $1 - r^2 \le C(1 - r^p) = CD$ and $r^{p-2} \ge c_{\mathcal{P}} > 0$. Use $1 - Z_n^2 = 1 - r^2 + |Z_T|^2$ in this case. The formula for e also shows that it is uniformly comparable to k^p.

Write $M_R(z) = M_R(u; z)$. Proposition 5.6, with $n = 7$, gives

$$M_R'(z) = \frac{1}{\sigma_E\omega_6}\left[R^{-6}\int_{\partial B_R(z)}|\nabla u|^{p-2}u_R^2 + R^{-7}\int_{B_R(z)}\left(W_p(u) - \frac{c}{p}|\nabla u|^p\right)\right] \ge 0. \tag{6.9}$$

Set $\mathcal{K}(R) = 1 + \sup_z M_R(z)$. The pointwise energy bound implies $\mathcal{K}(R) \le 1 + CR$.

Lemma 6.4 (Delayed density window)**.** *There are fixed C, K_0 such that, for every regular zero with sufficiently large reach $d = d(z)$,*

$$h = d\exp\{-C\sqrt{\log(\mathcal{K}(d) + 1)}\}, \qquad M_h(z) \le K_0.$$

The constant C may be increased. In particular $h \ge d^{1-o(1)}$.

Proof. We adapt the stress estimate and delayed iteration in [35, proof of Lemma 7.3] to the matched stress tensor. Move z to the origin and arrange $\nu(z) = e_7$. By the chord bound, the zeros in B_{3R} lie in $|x_7| \le CR^2/d$. Test stress conservation with $x_7e_7\zeta^2$. Choose ζ equal to one on B_R and zero outside B_{2R}.

The stress test is explicit. Its compact support gives

$$\int \zeta^2\mathbb{S}_{77} = -2\int x_7\zeta\,\mathbb{S}e_7\cdot\nabla\zeta \le \frac{1}{2}\int\zeta^2\mathbb{S}_{77} + C\int e\,x_7^2|\nabla\zeta|^2.$$

Absorb the first term on the right and use (6.8) together with $e \asymp k^p$ to obtain

$$\int_{B_R} k^p(1 - Z_7^2) \le C \int_{B_{2R}} k^p (x_7/R)^2.$$

Outside $|x_7| \le CR^2/d + \sqrt{R}$, every zero is at distance at least $c\sqrt{R}$. Clearing bounds the energy there by $CR^7 e^{-c\sqrt{R}}$. Within the remaining slab, $(x_7/R)^2 \le C((R/d)^2 + R^{-1})$. In (6.9), the discrepancy is bounded by $Ck^p(1 - Z_7^2)$. On annuli with fixed radius ratio, the radial term is bounded by $Ck^p(1 - Z_7^2 + (x_7/R)^2)$. Integrating (6.9) from $R/8$ to R and using the height estimate at a fixed larger radius yields

$$0 \le M_R - M_{R/8} \le C\{(R/d)^2 + R^{-1}\} M_{8R} \qquad (10 \le R \le c_0 d).$$

More explicitly, for nonnegative integrands a_1, a_2,

$$\int_{R/8}^{R} t^{-6} \int_{\partial B_t} a_1 \, dS \, dt \le CR^{-6} \int_{B_R \setminus B_{R/8}} a_1 \, dx,$$

$$\int_{R/8}^{R} t^{-7} \int_{B_t} a_2 \, dx \, dt \le CR^{-6} \int_{B_R} a_2 \, dx.$$

In the second estimate, the factor R from integration in t cancels one of the seven inverse powers of R.

At the zero center, the uniform gradient bound keeps $|u| \le 1/2$ on a fixed ball. The potential contributes a uniform positive amount of energy there. Monotonicity therefore gives $M_R \ge c > 0$ for $R \ge 1$. This absorbs the exponential remainder.

The larger radius in the preceding bound must be retained. Set $R_j = d8^{-j}$, $k_j = M_{R_j}$, and $b_j = C(8^{-2j} + R_j^{-1})$. Choose a fixed starting index j_0 and a fixed terminal radius R_{end}. Take both large enough that $R_j \le c_0 d$ and $b_j < 1/8$ whenever $j \ge j_0$ and $R_j \ge R_{\text{end}}$. Let n be the last such index. Thus $R_{n+1} \in [R_{\text{end}}/8, R_{\text{end}})$, and

$$0 \le k_j - k_{j+1} \le b_j k_{j-1} \qquad (j_0 \le j \le n).$$

Choose $J = \lceil C_0 \sqrt{\log(\mathcal{K}(d) + 1)} \rceil$, with C_0 large enough that $J/2 \ge j_0 + 2$. For large d, the bound $\mathcal{K}(d) \le 1 + Cd$ gives $J < n/2$. Suppose $k_{j+1} < k_j/2$ for some $J - 2 \le j \le n$. Then $k_{j-1} > k_j/(2b_j) > 4k_j$, so the same strict halving propagates backward. On $\lfloor J/2 \rfloor \le i \le J - 3$, one has $b_i \le Ce^{-cJ}$: here $8^{-2i} \le Ce^{-cJ}$ and $d^{-1}8^i \le Ce^{-cJ}$, since $J = O(\sqrt{\log d})$. There are at least $J/3$ such indices for large J. At each one, the backward growth factor is at least $C^{-1}e^{cJ}$. Since $k_i \ge c > 0$ and every $k_i \le \mathcal{K}(d)$, multiplication gives

$$\mathcal{K}(d) \ge c(C^{-1}e^{cJ})^{J/3} \ge ce^{c'J^2}.$$

But $J \ge C_0 \sqrt{\log(\mathcal{K}(d) + 1)}$. Choosing C_0 sufficiently large makes these two inequalities incompatible. All constants here were fixed before d was sent to infinity. Hence $k_{j-1} \le 2k_j$ for $J \le j \le n$, and

$$k_{j+1} \ge (1 - 2b_j)k_j \qquad (J \le j \le n).$$

Moreover, $\sum_{j=J}^{n} b_j$ is bounded uniformly: the first summand is a convergent geometric series, and $\sum_{j=J}^{n} R_j^{-1} \le C/R_{\text{end}}$. The pointwise energy bound gives $k_{n+1} \le CR_{\text{end}}$.

The product has a quantitative lower bound. Since $2b_j < 1/4$, one has $\log(1 - 2b_j) \ge -4b_j$. Thus

$$\prod_{j=J}^{n} (1 - 2b_j) \ge \exp\left(-4 \sum_{j=J}^{n} b_j\right) > c > 0.$$

The constant is uniform because the sum was bounded before the terminal index was chosen.

It follows that $k_J \le K_0$. Choose C so that $h \le R_J$ and apply monotonicity. □

A zero-free tangent ball gives the next estimate without a separate isolation hypothesis.

Lemma 6.5 (One-sided tangent-ball estimates)**.** *Suppose a zero-free open ball of radius r is tangent at a zero x, and let N be its inward normal. For sufficiently large r, the zero is regular and*

$$D(x) \le C/r, \qquad |B(x)| \le C/r, \qquad H_N(x) \le Cr^{-1-\gamma}$$

for a uniform $\gamma > 0$. After orienting the positive phase into the ball, for each fixed $\epsilon < 1/2$ and large r,

$$s - C(1+s)/r \le U(x+sN) \le s \qquad (0 \le s \le r^{\epsilon}). \tag{6.10}$$

The improved mean estimate is one-sided.

Proof. The tangent-ball comparison follows the strategy of [35, proof of Lemma 7.6]. We construct the barrier and check its equation for the matched operator. Start with the normal operator $P_0 = -c\partial_s^2 + 2c\beta g(s)\partial_s$. Define $b_0(0) = 0$ by

$$b_0'(s) = \frac{1}{c} M(s)^{-2} \int_s^{\infty} M(t)^2 \, dt.$$

Since $M'/M = -\beta g$, differentiation gives $P_0 b_0 = 1$. Also, b_0, b_0' are nonnegative and $b_0 \le C(1+s)$. All positive-order derivatives used below are bounded uniformly. In the annulus with inward depth $0 < s < Q = \sqrt{r}$, set $f(s) = s - Kb_0(s)/r$. For $v_* = g(f(s))$, radial differentiation gives

$$\frac{\Delta_p v_* + v_*(1-v_*^2)^{p-1}}{g'(f)^{p-1}} = cf'^{p-2}f'' - \frac{6}{r-s} f'^{p-1} + 2cag(f)(1-f'^p).$$

Its denominator is $(a(1-g(f)^2))^{p-1} = g'(f)^{p-1}$. The right side is $K/r - 6/(r-s) + O(K^2(1+s)/r^2)$. First fix K large, then take r large. The function v_* is then a strict positive subsolution.

The phase in the ball has a fixed sign. Reflect u so that it is positive. Put $v = (1 - e^{-c_1 Q})v_*$, with a fixed small $c_1 > 0$. Every point of the inner sphere is at least Q from the zero set. Clearing therefore gives $u \ge 1 - Ce^{-cQ}$ there. Choose $c_1 < c$; then $v < u$ on that sphere for large r. On the outer sphere, $v = 0 \le u$. For every $0 < \tau \le 1$, the function τv is still a strict subsolution. Indeed, the quotient of the reaction by t^{p-1} is $t^{2-p}(1-t^2)^{p-1}$, which is strictly decreasing on $(0,1)$.

We justify comparison by amplitude sliding. First compare u with the radial p-harmonic function that equals zero on the outer sphere and a fixed positive constant below $\min u$ on the inner sphere. The equation gives $-\Delta_p u > 0$ in the annulus, so ordinary weak comparison applies. This barrier has a positive inward derivative on the outer sphere. Since v also vanishes linearly there, it follows that $u \ge \tau_0 v$ for some $\tau_0 > 0$ on the whole annulus. Let τ be the largest admissible amplitude not exceeding one. If $\tau < 1$, an obstruction can occur only in the interior or at an outer zero. The strict inner-boundary inequality excludes the inner sphere. At an interior contact the two gradients agree with $\tau\nabla v \ne 0$. The local secant equation is uniformly elliptic, and the strong maximum principle excludes that contact. At an outer zero the preliminary p-harmonic barrier already gives a positive normal derivative for u. The secant equation is again locally uniformly elliptic, and the boundary point principle makes the inward derivative of $u - \tau v$ strictly positive. Compactness of the boundary zero set, together with the positive interior and inner-boundary margins, then permits a larger amplitude. This contradiction proves $v \le u$.

For $s \le r^{\epsilon}$, the amplitude loss divided by $g'(s)$ is smaller than every fixed negative power of r, since $\epsilon < 1/2$. Also $f(s) \ge s - C(1+s)/r$. Invert g and combine with the Modica upper bound $U(x+sN) \le s$ to obtain (6.10). At the touching point, differentiation gives $u_N(x) \ge a - C/r > 0$. The tangential derivatives vanish there because the restriction of u to the outer sphere has a minimum at x. Thus the zero is regular with normal N, and $D(x) \le C/r$.

At a regular zero, the exact mean identity reads

$$H = \frac{c}{p} \frac{\partial_\nu D}{1 - D}. \tag{6.11}$$

This is (2.14), with the phase oriented so that $\nu = N$ at the touching zero.

The relative bounds imply $|H| \le CD \le C/r$. The tangent ball bounds each inward principal curvature below by $-1/r$. Their trace bound then bounds each above by C/r. Hence $|B| \le C/r$.

To improve the signed mean bound, set $x = 0$ and $N = e_7$. Work on a box of any fixed shape with logarithmic radius $P = \alpha_0 \log r$. Exponential comparison and relative derivative bounds give

$$\|D\|_{C^k} \le C_k r^{-1+C\alpha_0}, \qquad \|T\|_{C^k} + \|U - s\|_{C^k} \le C_k r^{-1/2+C\alpha_0}.$$

Fix all required derivative orders before choosing α_0. Differentiating $\nabla D = -p|Z|^{p-2}TZ$ yields

$$\nabla_Z T = -T^2 - \frac{|Z|^{2-p}}{p}\nabla^2 D - (p-2)|Z|^{-2}(TZ)\otimes(TZ).$$

Thus $\partial_s T = O(r^{-1+C\alpha_0})$. Replacing Z by e_7 costs $O(|Z - e_7||\nabla T|) = O(r^{-1+C\alpha_0})$.

Conjugate the exact discrepancy equation by $M(s)$. The result is

$$\left[-\Delta_y - c\partial_s^2 + ca^2\{p^2 - p(p-1)G(s)\}\right](MD) = pM(s)|T_T(y,0)|^2 + \mathcal{E}, \qquad \|\mathcal{E}\|_\infty \le Cr^{-3/2+C\alpha_0}. \tag{6.12}$$

Here T_T is the tangential block. Coefficient errors multiply D or its derivatives, of size $r^{-1+C\alpha_0}$, by $O(r^{-1/2+C\alpha_0})$. Further, $Te_7 = O(r^{-1+C\alpha_0})$. The normal derivative identity just proved shows that replacing $T_T(y,s)$ by $T_T(y,0)$ in its square costs $O(r^{-3/2+C\alpha_0})$. Absorb logarithmic factors into $C\alpha_0$.

Set $F = MD$. Write $\eta = s/\sqrt{c}$, and let $\lambda_-^2 = ca^2p$ and $\lambda_+^2 = ca^2p^2$. The operator in (6.12) becomes $-\Delta_{y,\eta} + V(\eta)$. Its potential is even and lies between λ_-^2 and λ_+^2. Choose

$$t = \sqrt{p}, \qquad A = t + \sqrt{1+t^2}, \qquad \Omega_P = \{(y,\eta) : |y|^2/P^2 + \eta^2/(A^2P^2) < 1\}, \qquad X_* = (0, tP).$$

This domain is symmetric in η. Its boundary is at least P from zero. Each point on its upper boundary is at most $P\sqrt{1+t^2}$ from X_*. To check this, put $r = \eta/P \in [0, A]$. The squared distance divided by P^2 is

$$1 + t^2 - 2tr + (1 - A^{-2})r^2.$$

This is convex in r and has equal endpoint values $1 + t^2$. The incidence angle also stays positive. Indeed, the dot product of the unnormalized outward normal $(y/P, \eta/(A^2P))$ with $(X - X_*)/P$ is $1 - rt/A^2 \ge 1 - t/A > 1/2$.

To estimate the normal derivative at zero, solve the even nonnegative source and the error separately with zero boundary values. The source solution is even, so its normal derivative at zero vanishes. Positivity of the potential and interior estimates bound the error solution in C^1 by $O(r^{-3/2+C_A\alpha_0})$ on fixed interior balls. Split the remaining homogeneous solution as $u_+ + u_-$, using the nonnegative data on the upper and lower boundary halves. Reflection and comparison imply $\partial_\eta u_-(0) \le 0$.

Denote the integral of the upper boundary data by $\mathcal{B}_\partial$. For each fixed $\zeta > 0$, constant-potential Poisson comparison gives

$$u_+(0) \le CP^C e^{-\lambda_- P}\mathcal{B}_\partial, \qquad u_+(X_*) \ge c_\zeta e^{-(\lambda_+ + \zeta)\sqrt{1+t^2}P}\mathcal{B}_\partial.$$

We prove both estimates next. Their shapes and masses range over compact sets for $p \in \mathcal{P}$, so the constants will be uniform.

Write out the kernel comparison as follows. Write K_λ for the Dirichlet Poisson kernel of the ellipsoid and f for nonnegative boundary data supported on its upper half. In this calculation, let $B = \sqrt{1+t^2}$ and let $\kappa > 0$ be a uniform lower bound for the scaled incidence and interior distance.

For a fixed nonnegative boundary datum, increasing a positive potential decreases the solution. Indeed, subtract the two equations and apply the maximum principle to the difference. It is therefore enough to prove the upper estimate for $-\Delta + \lambda_-^2$ and the lower estimate for $-\Delta + \lambda_+^2$. Equivalently, their Poisson kernels bound the kernel for V above and below. The kernel inequalities

follow first after integration against arbitrary nonnegative smooth data, and then pointwise by approximation of a boundary delta mass.

We give details of the two constant-potential estimates. The fundamental solution of $-\Delta + \lambda^2$ has the positive heat-kernel representation

$$\Phi_\lambda(X) = \int_0^\infty (4\pi\tau)^{-7/2} \exp\{-|X|^2/(4\tau) - \lambda^2\tau\}\, d\tau.$$

For $|X| \geq 1$, this is at most $C(1 + |X|)^C e^{-\lambda|X|}$, uniformly in the allowed masses. For example, substitute $\tau = |X|v$ and use $1/(4v) + \lambda^2 v \geq \lambda$; split the v integral into a fixed neighborhood of its minimizer and its two tails. The Dirichlet Green function is bounded above by Φ_λ. A unit-scale boundary gradient estimate, applied near a boundary point Q, thus bounds its inward normal derivative by $CP^C e^{-\lambda|Q|}$. The boundary curvatures are $O(P^{-1})$, so the constants in this estimate are uniform. Since $|Q| \geq P$, this proves the upper kernel bound.

The lower estimate can be obtained without an asymptotic formula for the Green function. Choose a fixed radius R_ζ so large that the first Dirichlet eigenvalue μ_1 of $-\Delta$ on a six-dimensional ball of this radius satisfies

$$\sqrt{\lambda_+^2 + \mu_1} \leq \lambda_+ + \zeta.$$

The radius can be chosen uniformly because λ_+ remains in a compact positive interval and $\mu_1 = C/R_\zeta^2$. The segment from X_* to Q stays inside the ellipsoid. Except for portions of length CR_ζ at its ends, a cylinder of radius R_ζ about this segment also stays inside. Here is an explicit convexity argument. The ball $B_{\kappa P}(X_*)$ lies in Ω_P. For $0 < \vartheta \leq 1$, convexity gives

$$B_{\vartheta\kappa P}((1-\vartheta)Q + \vartheta X_*) \subset \Omega_P.$$

The center is at distance $\vartheta|Q - X_*| \leq \vartheta BP$ from Q. Its distance from the boundary is therefore at least (κ/B) times that distance along the segment. After discarding length BR_ζ/κ at the boundary end, this contains the required fixed-width cylinder.

Let ϕ be the positive first eigenfunction on its transverse disk, normalized by $\max \phi = 1$, and let ℓ be its length. Along the cylinder coordinate $\rho \in [0, \ell]$, the function

$$\phi(Y)\, \frac{\sinh(\sqrt{\lambda_+^2 + \mu_1}\,(\ell - \rho))}{\sinh(\sqrt{\lambda_+^2 + \mu_1}\,\ell)}$$

solves the homogeneous massive equation, vanishes on the lateral boundary and the far cap, and equals ϕ at the near cap. A fixed ball around X_* lies in Ω_P. Comparison with its Green function gives a fixed positive lower bound for the Green function from X_* on the cylinder's near cap. Here the ball and the near cap are chosen at distances depending on R_ζ alone. The near cap stays away from the pole X_* and from that fixed ball's boundary. Positivity and compactness of the fixed-ball Green function give the claimed lower bound, uniformly in P. Comparison with the displayed function therefore gives a lower bound $c_\zeta e^{-(\lambda_+ + \zeta)\ell}$ on a fixed interior ball near the far end. Harnack comparison over the remaining distance $O(R_\zeta)$, followed by the boundary point estimate in a fixed interior ball tangent at Q, gives the same lower bound for the Poisson kernel at Q. Indeed, take the far interior ball a fixed positive distance before the cap where the displayed comparison function vanishes. The numerator of its hyperbolic-sine factor is then bounded below by a positive constant, while its denominator is at most $Ce^{\sqrt{\lambda_+^2 + \mu_1}\,\ell}$. The angle hypothesis at Q and the uniform boundary curvature give a chain of a bounded number of fixed-size interior balls joining this ball to the center of a ball tangent at Q. Harnack's inequality transfers the bound along this chain. In the final ball, compare on a concentric annulus with the radial massive solution having that fixed positive inner datum and zero outer datum. Its inward derivative on the outer sphere is a fixed positive number. This bounds the inward derivative of the Green function at Q below by the same constant multiple and proves the boundary point estimate used here. All these last distances and ball radii are fixed

after ζ is chosen. Their constants may depend on ζ but not on P or Q. Since $\ell \le |Q - X_*| + CR_\zeta$, the fixed additive length changes only that constant. Thus

$$K_{\lambda_+}(X_*, Q) \ge c_\zeta e^{-(\lambda_+ + \zeta)|Q - X_*|} \ge c_\zeta e^{-(\lambda_+ + \zeta)BP}.$$

Integration against f proves the lower kernel bound.

The radial phase bound at X_* yields

$$D(X_*) \le C(1+P)/r, \qquad F(X_*) \le C(1+P)r^{-1}e^{-\lambda_+ tP}.$$

For the first inequality, write $s_* = \sqrt{c}\,tP$ for the physical normal coordinate of X_*. On $[s_*-1, s_*+1]$, $0 \le D \le p(1-U_s)$. The phase bound (6.10) gives

$$\int_{s_*-1}^{s_*+1} (1-U_s)\,ds = 2 - U(s_*+1) + U(s_*-1) \le C(1+P)/r.$$

The interval lies in the allowed phase range for large r, since $P = \alpha_0 \log r$. By (2.30), $D(X_*)$ is at most a fixed multiple of the integral of D on this interval. This proves the pointwise bound. The second inequality includes the normal factor M. Positivity of the even source gives $u_+(X_*) \le F(X_*) + Cr^{-3/2+C_A\alpha_0}$. Combining the comparisons, the first contribution to $u_+(0)$ has exponent

$$-\lambda_- + \lambda_+(\sqrt{1+t^2} - t).$$

Since

$$p(\sqrt{p+1} - \sqrt{p}) = \frac{p}{\sqrt{p+1} + \sqrt{p}} < \frac{\sqrt{p}}{2},$$

this exponent is less than $-\lambda_-/2$. Choose ζ with loss below $\lambda_-/4$. Then choose $\alpha_0 > 0$ small enough that the error still decays faster than r^{-1}. Spend half of the positive gain to absorb powers of P. Thus $u_+(0) = O(r^{-1-\gamma_0})$ for some $\gamma_0 > 0$. Interior Harnack and gradient estimates for this positive homogeneous solution give $|\partial_\eta u_+(0)| \le Cu_+(0)$. Its normal derivative has the same bound. Combine this with the sign of u_- and the evenness of the source. The zero-level identity $H = c\partial_\nu D/[p(1-D)]$ gives

$$H_N(0) \le Cr^{-1-\gamma_0}. \tag{6.13}$$

The constants and γ_0 are uniform on $\mathcal{P}$.

□

Lemma 6.6 (Isolation and mean-curvature derivatives)**.** *Let h be the density-window scale from Lemma 6.4. After increasing its fixed constant, the zero set in a fixed inner fraction of $B_h(z)$ is a single graph with a fixed fractional buffer. On every smaller fixed fractional patch, for constants $0 < \theta < \alpha - 1$ and $\gamma_0 > 0$ depending on $\mathcal{P}$,*

$$|B| + h^\theta [B]_{C^\theta} \le Ch^{-1}, \quad |\nabla_\Sigma B| \le Ch^{-2}, \quad |H| \le Ch^{-2}, \quad |\nabla_\Sigma H| + |\nabla_\Sigma^2 H| \le Ch^{-3-\gamma_0}.$$

Proof. We follow the graph isolation and two parity projections in [35, proof of Lemma 7.5]. The normal operator and its weighted projections are derived below for the matched equation. Lemma 6.4 gives the single-ball bound $M_h(u; z) \le K_0$. Also $h \to \infty$ as $d \to \infty$, since $\mathcal{K}(d) \le 1 + Cd$. Apply Proposition 5.26 with this fixed cap and with a fixed interior fraction. Its threshold is the fixed one used for clean-ball preparation. The proposition applies to the present entire stable solution; it needs only the cap at scale h, rather than a bound for its limiting density. A smaller fixed fraction is therefore clean for large h. Absorb this fixed reduction into the constant defining h. The quantitative conclusion of that proposition gives

$$D + |H| \le Ch^{-\alpha}, \qquad b \le C/h. \tag{6.14}$$

In a smaller cylinder with a fixed buffer, chord confinement places every zero in a slab of height Ch^2/d, and the curvature is bounded by C/h. Following a unit tangent vector in both directions for length comparable to $h\sqrt{h/d}$, the slab bound and the geodesic Taylor estimate show that

its component in the confining normal direction is at most $C\sqrt{h/d}$. Thus tangent projection is nonsingular. The fixed buffer lets the local inverse branches extend over a common smaller disk. They therefore give graphs.

Write $A_h = Ch^2/d$ for the confining height.

To see that this gives simultaneous graphs, rather than just separate local graph charts, fix a cylinder

$$\mathcal{C} = \{\, |y| < \rho_0,\ |x_7| < h/8 \,\}, \qquad \rho_0 = h/8,$$

slightly reducing the fixed fractions if needed so that its closure lies inside the prepared ball. Every zero in $\mathcal{C}$ has $|x_7| \le A_h < h/16$. Let $E = \Sigma \cap \mathcal{C}$ and let $\pi : E \to B^6_{\rho_0}$ be horizontal projection. This map is a local diffeomorphism by the preceding tangent estimate. It is proper: if $K \Subset B^6_{\rho_0}$, then $\pi^{-1}(K)$ is a closed subset of the compact set $K \times [-A_h, A_h]$. It follows that π is a covering onto the disk. For clarity, its image is open by the local inverse theorem and closed by properness; it is nonempty since $0 \in E$. Each fiber is compact and discrete, hence finite. Each connected component is closed in E, so the restriction of π to it is likewise proper over the disk, and its image is both open and closed. Thus every component covers the whole disk. Since a disk is simply connected, each component is a single graph over this same disk. This also proves that no regular zero sheet in the cylinder has been omitted.

A second graph would meet the base normal line at distance $O(h)$ and provide a competitor with value at most CG_1h. Increase the fixed constant in the density-window scale so that $d/h > 2CG_1$. This contradicts the definition of d. Thus exactly one graph occurs. Its unit-scale jets are bounded, and every fixed normal window converges to the matched profile.

Choose

$$\theta = (\alpha - 1)/2, \qquad \sigma = (\alpha - 1)/4. \tag{6.15}$$

Then $0 < \sigma < \theta < \alpha - 1 < 1$. Fix the target derivative orders below, and later take the loss $\epsilon > 0$ smaller than all stated exponent margins. Interpolation with sufficiently high unit-scale derivative bounds gives, through these target orders, $|\partial_y^k H| \le C_\epsilon h^{-\alpha+\epsilon}$. The original-coordinate C^θ seminorm has the same bound. Split pairs at distance one and use the first derivative bound for closer pairs. The rescaled mean has bounded C^θ norm since $h^{1+\theta-\alpha+\epsilon}$ remains bounded for a small fixed $\epsilon > 0$. Elliptic graph estimates give

$$[B]_{C^\theta} \le Ch^{-1-\theta}.$$

Subtract the curvature at a unit-chart center. Interpolate its seminorm with sufficiently high uniformly bounded derivatives. Fixing that interpolation order gives

$$|\partial_y^k B| \le Ch^{-1-\sigma} \quad (1 \le k \le 30), \qquad |\partial_y^k H| \le C_\epsilon h^{-\alpha+\epsilon} \quad (0 \le k \le 30). \tag{6.16}$$

Frame changes contain an extra curvature factor and satisfy the same bounds. Any sufficiently large fixed target order would work in place of thirty.

For example, local interpolation bounds the derivative of order k by a constant times $\|H\|_\infty^{1-k/N}\|H\|_{C^N}^{k/N}$, with an additional lower-order supremum term. Taking $N > \alpha k/\epsilon$ gives the stated loss. For the curvature interpolation, choose N so that $(1+\theta)(1-30/N) > 1+\sigma$. These are auxiliary orders; the target orders and the powers of the expanding box remain fixed.

After the box exponent ϵ is chosen, fix the finite auxiliary interpolation orders needed for its loss bounds. All these orders are fixed before $h \to \infty$.

To estimate the phase on this graph, take Fermi coordinates $X(y) + s\nu(y)$ on nested boxes of size h^ϵ. Choose $\epsilon > 0$ small at the end. The single graph and $b \le C/h$ ensure that these boxes are injective. Each side has zero-free tangent balls of radius c_0h for a small fixed $c_0 > 0$. Indeed, the graph height above its tangent plane is at most $C|y|^2/h$. The tangent sphere has quadratic height at least $|y|^2/(2c_0h)$ on the corresponding side. Choose c_0 so that the sphere stays strictly on that side of the graph, and so that the whole ball lies in the fixed graph buffer. No other zero sheet

occurs there. Their radial phase bounds give, with $\psi = U - s$,

$$\psi(y,0) = 0, \qquad |\psi(y,s)| \le Ch^{-1}(1+|s|), \qquad \|\psi\|_{C^k} \le C_\epsilon h^{-1+C\epsilon}$$

This holds for every fixed derivative order required below. We justify the derivative estimate before dividing by the small profile derivative. On a unit ball centered at positive normal depth s_0, put $a_0 = 1 - g(s_0)$ and $v = (1-u)/a_0$. The phase bound and the bounded logarithmic derivatives of $1-g$ make v differ by $O(h^{-1+C\epsilon})$ in C^0 from the corresponding normalized ambient planar solution. Its exact equation is

$$\Delta_p v = (1 - a_0 v)[v(2 - a_0 v)]^{p-1}.$$

The normalized deficit stays in a fixed positive compact interval on the unit ball. The equation therefore gives uniform local $C^{1,\vartheta}$ bounds, independent of the small a_0. Compactness first shows that its gradient is close to that of the normalized planar deficit and hence is bounded away from zero on a smaller ball. In particular, $|\nabla U| \ge 1/2$ there for large h. The difference from the normalized planar solution now satisfies a uniformly elliptic secant equation with uniformly bounded coefficients of each fixed order. To see both assertions, write v_{pl} for the normalized ambient planar solution and $w = v - v_{\mathrm{pl}}$. Then

$$\operatorname{div}(B_v \nabla w) = c_v w,$$

$$B_v = \int_0^1 A_p(\nabla v_{\mathrm{pl}} + t(\nabla v - \nabla v_{\mathrm{pl}}))\, dt.$$

Here c_v is the secant quotient of the displayed reaction. The preceding C^1 closeness keeps every gradient in this integral in a fixed neighborhood of a nonzero planar gradient. Thus B_v has uniform positive upper and lower ellipticity bounds. The values of v, v_{pl} stay in a positive compact interval, so c_v and all the required reaction derivatives are bounded uniformly, also when a_0 tends to zero. Uniformly elliptic regularity first bootstraps the individual solutions to uniform bounds of each fixed order. Applied next to their linear difference equation, interior estimates give

$$\|w\|_{C^k(B_{1/4})} \le C_k \|w\|_{C^0(B_{1/2})} \le C_{k,\epsilon} h^{-1+C\epsilon}.$$

The fractions refer to the same unit ball and can be replaced by any two fixed nested fractions. Converting back to the phase involves only the normalized planar derivative, which is bounded away from zero on this unit ball. This proves the asserted quantitative phase derivative bounds. For negative depths use $(1+u)/(1-g(|s_0|))$ and reflect. The fixed central phase window is uniformly noncritical by the previous preparation and is treated directly. Coordinate changes to the Fermi chart have uniformly controlled derivatives by (6.16).

Arrange the exact phase equation as

$$\begin{aligned} P\psi &= H_s + (g_s^{ij} - \delta^{ij})\psi_{ij} + b_s^i \psi_i + cH_s\psi_s + \mathcal{N}(\psi), \\ P &= -\Delta_y - c\partial_s^2 + 2c\beta \tanh(as)\partial_s, \qquad H_s = \operatorname{tr}[B(I+sB)^{-1}]. \end{aligned} \tag{6.17}$$

For this formula, the parallel metric is $g_s(v,w) = g_0((I+sB)v, (I+sB)w)$ and its second fundamental form is denoted by II_s.

We give an exact formula to identify all nonlinear terms. Put $w = 1 + \psi_s$, $\xi^i = g_s^{ij}\psi_j$, $\mathfrak{r}^2 = w^2 + g_s^{ij}\psi_i\psi_j$, and let II_s be the second fundamental form of the parallel sheet with normal ∂_s. The Hessian formulas in Fermi coordinates give

$$\begin{aligned} \Delta_p U = {} & \mathfrak{r}^{p-2}(\psi_{ss} + \Delta_{g_s}\psi + H_s w) \\ & + (p-2)\mathfrak{r}^{p-4}\left[w^2\psi_{ss} + 2w\xi^i\psi_{si} + \xi^i\xi^j(\nabla_{g_s}^2\psi)_{ij} - w\,\mathrm{II}_s(\xi,\xi)\right]. \end{aligned}$$

For example, $\operatorname{Hess} U(\partial_s, \partial_i) = \psi_{si} - (S_s)_i^{\ j}\psi_j$ and $\operatorname{Hess} U(\partial_i, \partial_j) = (\nabla_{g_s}^2\psi)_{ij} + w(\mathrm{II}_s)_{ij}$. The two mixed contributions combine with the tangential one to give the displayed minus sign before $w\,\mathrm{II}_s(\xi,\xi)$.

The term $\mathcal{N}$ is a smooth sum of products containing at least two phase jets. Their orders are at most two, and at most one has order two. The terms arise from $O(|\nabla\psi|)D^2\psi$, $O(\psi\nabla\psi)$, and

$O(|\nabla\psi|^2)$. The normal metric divergence is $H_s[1 + c\psi_s + O(|\nabla\psi|^2)]$. These descriptions remain valid after all the finitely many derivatives used below. Expanding the metric and applying (6.16) gives

$$H_s = H - sb^2 + s^2 \operatorname{tr} B^3 + O(h^{-4+C\epsilon}), \qquad H_s = O(h^{-\alpha+C\epsilon}), \qquad g_s^{-1} - I,\ b_s = O(h^{-1+C\epsilon}).$$

There is a useful additional restriction on the geometric coefficients. Every geometrically dependent nonlinear term either contains a positive tangential derivative of ψ, or its geometric dependence is only through H_s. This follows directly from the exact formula: when all tangential phase jets vanish, the nonlinear remainder is

$$\begin{aligned} &c(w^{p-2} - 1)\psi_{ss} + H_s(w^{p-1} - 1 - c\psi_s) \\ &\qquad + 2ca\big[g(s + \psi)(1 - w^p) + p\, g(s)\psi_s\big]. \end{aligned}$$

Only the middle term contains geometry. This observation will be used in the second derivative improvement and avoids an unproved assumption about the structure of the remainder.

Put $F = M\psi$. Exact planar conjugation yields

$$(-\Delta_y + L_s)F = MP\psi, \qquad L_s = -c\partial_s^2 + ca^2[p^2 - p(p + 1)G(s)].$$

By Proposition 2.17 and the conjugation in Lemma 2.16, the even function M spans the kernel and the gap on its orthogonal complement is $ca^2(2p - 1) > 0$, uniformly for $\mathcal{P}$. Indeed, under that conjugation the original weighted orthogonality becomes a nonzero constant multiple of $\int FM\, ds = 0$. The Dirichlet half-line operator has this same positive lower bound, by odd reflection. Retain the actual quasilinear principal coefficient on the left of (6.17). Factor one phase jet from each remaining nonlinear term, using Taylor's formula along the known phase, and collect the coefficients. This gives an exact linear equation for this solution:

$$\mathcal{L}_{\mathrm{ph},h}\psi = H_s, \qquad \mathcal{L}_{\mathrm{ph},h} = -a_h^{ij}\partial_{ij} + b_h^i\partial_i + d_h.$$

The matrix a_h is symmetric. Use an outer half-box $\mathcal{Q}_r^+ = \{|y_i| < r,\ 0 < s < r\}$, where $r = 2h^\epsilon$. The earlier nested Fermi boxes permit this fixed enlargement for large h, since $h^\epsilon = o(h)$. With $A_0 = \operatorname{diag}(I_6, c)$ and $b_0 = 2c\beta \tanh(as)e_s$, the coarse phase bounds and (6.16) give

$$\|a_h - A_0\|_{C^{17}} + \|b_h - b_0\|_{C^{17}} + \|d_h\|_{C^{17}} \le C_\epsilon h^{-1+C\epsilon}, \qquad \|H_s\|_{C^{17}} \le C_\epsilon h^{-\alpha+C\epsilon}.$$

These are bounds in the Fermi coordinates on the outer half-box. They use only the already available coarse phase derivatives through a fixed order at most twenty and geometric derivatives through order thirty. In particular, normal derivatives of H_s contain powers of B; the quadratic terms are smaller than $h^{-\alpha}$ because $\alpha < 2$. All derivative orders are fixed before ϵ is chosen.

Put $\mu = \nabla \log M = -\beta \tanh(as)e_s$. For $F = M\psi$, direct differentiation gives

$$\begin{aligned} \mathcal{H}_h F &= MH_s, \\ \mathcal{H}_h &= -a_h^{ij}\partial_{ij} + (b_h^i + 2a_h^{ij}\mu_j)\partial_i \\ &\qquad + d_h - b_h \cdot \mu - a_h^{ij}(\mu_i\mu_j - \partial_i\mu_j). \end{aligned}$$

At the flat coefficients the first-order term is zero, and the zeroth-order term is $ca^2[p^2 - p(p+1)G(s)]$. Thus $\mathcal{H}_h$ differs from $\mathcal{H}_0 = -\Delta_y + L_s$ by $O(h^{-1+C\epsilon})$ in C^{17}. It remains uniformly elliptic. The source MH_s has C^{17} norm at most $C_\epsilon h^{-\alpha+C\epsilon}$. On the positive half-line use the bounded positive function

$$v_0(s) = M(s)\left[1 + \frac{1}{c}\int_0^s M(t)^{-2}\int_t^\infty M(r)\, dr\, dt\right] \quad (s \ge 0). \tag{6.18}$$

It satisfies $L_s v_0 = 1$. To verify this, write $v_0 = Mw$ and use $L_s(Mw) = -cM^{-1}(M^2w')'$. The integral defining w gives $(M^2w')' = -M/c$. For boundedness, $M(s) \sim 2^p e^{-\beta s}$ and $\int_s^\infty M(r)\, dr \sim M(s)/\beta$ imply $v_0(s) \to 1/(c\beta^2)$ as $s \to \infty$. These asymptotics are uniform on $\mathcal{P}$. Together with $v_0(0) = 1$ and positivity, they give uniform positive upper and lower bounds. Its ODE then bounds

every fixed derivative. This scalar construction uses only $p > 1$. Its bounds are uniform on each compact subset of $(1, \infty)$, as needed for the artificial-layer comparison below. Choose $\epsilon > 0$ so small that $h^{-1+C\epsilon} \to 0$. The bounds for v_0 and its derivatives give $\mathcal{H}_h v_0 = 1 + O(h^{-1+C\epsilon}) \geq 1/2$ for large h. Dividing by v_0 turns $\mathcal{H}_h$ into a uniformly elliptic operator whose zeroth-order coefficient is $(\mathcal{H}_h v_0)/v_0 \geq c_1 > 0$. The maximum principle therefore applies to this quotient operator, even though the untransformed potential may be negative.

Choose a fixed $\lambda > 0$ with $\lambda^2 \sup v_0 \leq 1/4$ and define, for $1 \leq i \leq 6$,

$$\mathfrak{B}_i(y, s) = v_0(s) \frac{\cosh(\lambda y_i)}{\cosh(\lambda r)}.$$

For the flat operator,

$$\mathcal{H}_0 \mathfrak{B}_i = \frac{\cosh(\lambda y_i)}{\cosh(\lambda r)}(1 - \lambda^2 v_0) \geq \frac{3}{4} \frac{\cosh(\lambda y_i)}{\cosh(\lambda r)}.$$

The first two derivatives of the hyperbolic factor are bounded by a fixed multiple of that factor. The C^0 coefficient perturbation consequently gives $\mathcal{H}_h \mathfrak{B}_i \geq c_2 \mathfrak{B}_i$ for large h.

Let $A_h = C_\epsilon h^{-\alpha+C\epsilon}$ bound the source and let $B_h = C_\epsilon h^{-1+C\epsilon}$ be the coarse boundary bound. On a lateral face $|F| \leq B_h$. On the top face, $M(r) \leq Ce^{-\beta r}$ gives $|F| \leq CB_h e^{-\beta r}$, and on the bottom $F = 0$. A sufficiently large fixed multiple of

$$v_0(s)(A_h + B_h e^{-\beta r}) + B_h \sum_{i=1}^{6} \mathfrak{B}_i(y, s)$$

dominates $|F|$ on every boundary face and has image under $\mathcal{H}_h$ at least $|MH_s|$. Apply the quotient maximum principle to its differences with F and with $-F$. On $\mathcal{Q}^+_{r/2}$, each hyperbolic ratio is at most $2e^{-\lambda r/2}$. The comparison therefore gives

$$\|F\|_{L^\infty(\mathcal{Q}^+_{r/2})} \leq C(A_h + B_h e^{-c_3 r}) \leq C_\epsilon h^{-\alpha+C\epsilon}.$$

The last exponential is smaller than every fixed negative power of h because $r = 2h^\epsilon$ and $\epsilon > 0$ is fixed.

Apply uniform interior Schauder estimates on unit balls and boundary Schauder estimates on unit half-balls at $s = 0$. The bottom data are zero, the coefficients and source have the stated C^{17} bounds, and the retained box has a fixed large buffer from the lateral and top faces. These estimates yield the same bound through fifteen derivatives on a further fixed inner fraction. Reflect the equation and repeat on the negative half-box. The comparison is made separately on the two sides; no derivative of the reflected gauge is taken across $s = 0$. Since F itself is smooth there, the half-box estimates give

$$\|F\|_{C^{15}} \leq C_\epsilon h^{-\alpha+C\epsilon}. \tag{6.19}$$

The norm is taken on the retained inner Fermi box. Retain both this estimate and the unweighted phase bound.

Apply an even normal cutoff to F at depths comparable to h^ϵ. Extend the result to the whole normal line. The cutoff errors are smaller than every fixed negative power of h. Write

$$(-\Delta_y + L_s)F = HM + E, \qquad F = u_0(y)M + F^\perp,$$
$$u_0(y) = \frac{\int F(y, s)M(s)\, ds}{\int M^2\, ds}, \qquad \int F^\perp M\, ds = 0.$$

Through eleven tangential derivatives in normal L^2, we have

$$E = -sb^2 M + O(h^{-1-\alpha+C\epsilon}), \qquad E^{\text{even}} = O(h^{-1-\alpha+C\epsilon}). \tag{6.20}$$

The next curvature term is $h^{-3+C\epsilon}$, which is smaller since $\alpha < 2$. Metric errors multiply (6.19) by $h^{-1+C\epsilon}$. For a nonlinear product, put M on one phase jet using the exact identities

$$M\psi_s = F_s - (M'/M)F, \qquad M\psi_{ss} = F_{ss} - 2(M'/M)F_s + (2(M'/M)^2 - M''/M)F.$$

All coefficients in these expressions and their normal derivatives are bounded. Bound another phase factor by the unweighted phase bound; it contributes $h^{-1+C\epsilon}$. Thus the product has the required size $h^{-1-\alpha+C\epsilon}$. Extra phase factors are harmless. This argument also applies after eleven tangential derivatives by Leibniz's rule. No exponentially small weight is divided out. Passing from a supremum bound to the normal L^2 norm on a box of length $O(h^\epsilon)$ costs only another fixed power $h^{C\epsilon}$.

Put $E_0 = \int EM\,ds / \int M^2\,ds$ and $E^\perp = E - E_0 M$. Orthogonal projection gives

$$(-\Delta_y + L_s)F^\perp = E^\perp, \qquad -\Delta_y u_0 = H + E_0.$$

The normal gap gives an inverse estimate on $M^\perp$, without inverting the kernel. Let $v \perp M$ solve $(-\Delta_y + L_s)v = f \perp M$ and put $\lambda = \inf_{p\in\mathcal{P}} ca^2(2p-1) > 0$. Here is a direct derivation. Put $q_\delta = (\|v\|_2^2 + \delta^2)^{1/2}$. Cauchy–Schwarz in the formula for $\Delta_y q_\delta$ gives $-\Delta_y q_\delta \le \langle v, -\Delta_y v\rangle / q_\delta$. Substitute $-\Delta_y v = f - L_s v$ and use the gap:

$$(-\Delta_y + \lambda)q_\delta \le \|f\|_2 + \lambda\delta^2/q_\delta \le \|f\|_2 + \lambda\delta.$$

Let $\delta \downarrow 0$. Comparison with exponential barriers for $-\Delta_y + \lambda$ now bounds the norm on a smaller box by the source supremum plus a boundary supremum times e^{-cL_h}. The latter is negligible here, because all available boundary derivatives are at most polynomial in h. For example, if A bounds the source and B bounds the boundary norm on $|y_i| < L$, compare with

$$\frac{A}{\lambda} + B\sum_{i=1}^{6}\left(e^{-\kappa(L-y_i)} + e^{-\kappa(L+y_i)}\right), \qquad 0 < \kappa^2 < \lambda.$$

Its image under $-\Delta_y + \lambda$ is at least A, and on every boundary face one summand equals B. On a fixed smaller fractional box the sum is at most $12Be^{-cL}$. The same estimate applies after tangential differentiation, since L_s is independent of y.

To pass from a normal L^2 bound to a value at $s = 0$, commute two additional tangential derivatives and use

$$L_s v = f + \Delta_y v, \qquad \|v\|_{H^2_s} \le C(\|L_s v\|_2 + \|v\|_2).$$

The second estimate follows directly from $L_s = -c\partial_s^2 + V_0(s)$ with bounded V_0; the bound for v' follows by integration by parts or one-dimensional interpolation. Finally, $|v(0)| \le C\|v\|_{H^2_s}$. Thus two more tangential source derivatives suffice for the normal trace. These observations also fix the derivative counts used below. Precisely, if the source and its tangential derivatives through order $k+2$ are bounded by A in L^2_s, the inverse estimate first bounds the corresponding derivatives of v in L^2_s on smaller boxes. The identity $L_s\partial_y^k v = \partial_y^k f + \Delta_y \partial_y^k v$ then bounds $\partial_y^k v$ in H^2_s by CA, apart from the exponentially small lateral term. Evaluation at $s = 0$ is therefore legitimate; a mere normal L^2 bound would not have justified it.

Projection onto $M^\perp$ commutes with even and odd reflection, since M is even. Apply the estimate to the even orthogonal component of (6.20). We have $F(y, 0) = 0$, and the odd component vanishes there. Hence $u_0(y)M(0) = -F^{\perp,\mathrm{even}}(y, 0)$, which gives

$$|\partial_y^k u_0| \le Ch^{-1-\alpha+C\epsilon} \quad (k \le 9).$$

The kernel equation $-\Delta_y u_0 = H + E_0$ gives the same estimate for derivatives of H through order seven.

By (6.16), each positive derivative of b^2 gains $h^{-\sigma}$. Apply the full orthogonal inverse to obtain

$$\|\partial_y^k F\|_{L^2_s} + \|\partial_y^k F\|_{H^2_s} \le Ch^{-2-\sigma+C\epsilon} \quad (1 \le k \le 8). \tag{6.21}$$

All remaining source and kernel terms are smaller, since $1+\alpha > 2+\sigma$.

Differentiate the even remainder tangentially through order six. The odd term $-sb^2M$ again disappears under projection. A differentiated cubic curvature term has size $h^{-3-\sigma+C\epsilon}$. Every horizontal metric or drift term already contains a positive tangential F derivative, even when all further derivatives fall on its coefficient. Equation (6.21), multiplied by the coefficient bound $O(h^{-1+C\epsilon})$, gives the same remainder bound.

For a nonlinear product, first suppose a derivative falls on a phase factor. Put M on that factor and use (6.21). Bound the other phase factor by the unweighted estimate $h^{-1+C\epsilon}$. If all positive derivatives fall on geometric coefficients and a tangential phase jet remains, put M on that jet and use the same improved estimate. If only normal phase jets remain, their geometric dependence is through H_s. After the first kernel estimate, $H_s = O(h^{-2+C\epsilon})$. Its differentiated bounds give the needed extra geometric factor. Together with one weighted phase jet of size $h^{-\alpha+C\epsilon}$ and the unweighted jet bound, this is at most $O(h^{-2-\alpha+C\epsilon})$. That is smaller because $\sigma < \alpha - 1$. Differentiating a flat coefficient depending on ψ adds a phase factor; a flat coefficient independent of ψ depends only on s.

For the separate normal drift, the first kernel estimate and the derivatives of b^2 give $\partial_y^k H_s = O(h^{-2-\sigma+C\epsilon})$ for $1 \le k \le 6$. A derivative on F_s uses (6.21) and the bound for H_s. If all derivatives fall on H_s, use the first bound for F_s instead. The two products are respectively $O(h^{-4-\sigma+C\epsilon})$ and $O(h^{-2-\sigma-\alpha+C\epsilon})$, both smaller than the required remainder. Hence

$$\|\partial_y^k E^{\mathrm{even}}\|_{L^2_s} \le Ch^{-3-\sigma+C\epsilon} \quad (1 \le k \le 6).$$

Project onto the even component again and use the condition at $s = 0$. The same bound holds for positive derivatives of u_0 through order four. The kernel equation gives

$$|\partial_y H| + |\partial_y^2 H| \le Ch^{-3-\sigma+C\epsilon}.$$

Choose the box exponent so that every loss $C\epsilon$ is below $\sigma/2$ and below all earlier positive margins. Connection terms are smaller. We conclude that

$$|H| \le Ch^{-2}, \qquad |\nabla_\Sigma H| + |\nabla_\Sigma^2 H| \le Ch^{-3-\sigma/2}. \tag{6.22}$$

Rescale the graph by h^{-1}. Its $C^{2,\theta}$ norm is bounded. Its prescribed mean hH also has bounded $C^{1,1}$ norm, since first and second derivatives scale by h^2 and h^3. Differentiate the graph equation once and apply interior Schauder estimates. The third-derivative bound gives $|\nabla_\Sigma B| \le Ch^{-2}$. The total fractional shrinkage of the boxes is fixed. The target derivative orders were fixed before ϵ was chosen. The auxiliary interpolation orders are then fixed before $h \to \infty$. Repeat this argument at each inner base. This gives the uniform patch estimates needed for reach differentiation and the envelope Simons estimate.

□

6.2. **Weak reach calculus.** The weak reach calculus and positive-foot transfer adapt [35, Proposition 7.7 and its proof]. The base and tangent-ball estimates control the terms from reach differentiation. At a contact on the constant part of the penalty, call the second point the *receiving foot*. Only these contacts retain a mean-curvature term at the second foot. The next proposition gives the area factor and multiplicity needed to integrate this positive term.

Proposition 6.7 (Weak reach inequality and positive-foot transfer)**.** *For each $\eta_0 > 0$, one can select the penalty parameters in the prescribed order so that the following conclusions hold for all sufficiently large d. The reach d is locally semiconcave on Σ_{reg}. The singular part of its distributional surface Laplacian is a nonpositive measure. Almost everywhere,*

$$|\nabla_\Sigma d| \le C_\Gamma \exp\{C_\Gamma \sqrt{\log(K(d)+1)}\}. \tag{6.23}$$

At an active non-plateau or diagonal contact,

$$\Delta_\Sigma d \le (1 + \eta_0 + C_\Gamma d^{-\gamma}) b^2 d + \mathfrak{b}\,\frac{|\nabla_\Sigma d|^2}{d}, \qquad \mathfrak{b} \le 1 + C_\Gamma d^{-\gamma}. \tag{6.24}$$

If bd is sufficiently small, the same bound can be used with $\mathfrak{b} \le \eta_0 + C_\Gamma bd + C_\Gamma d^{-\gamma}$, and then $|\nabla_\Sigma d| \ge c_\Gamma > 0$. At an active plateau contact, meaning $s \ge 1 - \delta_2$, one has

$$\Delta_\Sigma d \le G_1(|H(z)| + H_{N_y}(y)_+) + \eta_0 \frac{|\nabla_\Sigma d|^2}{d}, \qquad N_y = -e. \tag{6.25}$$

On the regular set with finite reach, there is a Borel choice of an active off-diagonal contact or an active diagonal direction. The choice can be made to select a plateau contact whenever one exists. Any measurable selection of these contacts permits transfer of the positive foot charge to y. The transfer has multiplicity at most one, and its area factor is

$$J_{y\to z} = \frac{|\det(I + \ell B_{N_y}(y))|}{s} \le C_\Gamma, \qquad d(y) \le d(z) = G_1\ell, \qquad b(y) + D(y) \le C_\Gamma / d(z). \tag{6.26}$$

More precisely, $J_{y\to z} \le 1 + C\delta_2 + Cb(y)\ell$ whenever $b(y)\ell$ is bounded by a sufficiently small fixed constant. All constants, including the minimum allowed reach, are selected before the later defect threshold and the collar and exhaustion scales.

Proof. First establish semiconcavity. On a relatively compact convex coordinate ball in a regular patch, $0 < c \le d \le C$. Local graph curvature gives the lower bound. A fixed off-diagonal competitor with positive angle gives the upper bound after the patch is reduced. Competitors with value at most $2C$ have bounded length. Those with length at least $\varepsilon > 0$ satisfy $s \ge \varepsilon/(2C)$, since $\Gamma \ge 1/s$. Fix their second foot. Their values are then smooth functions of the base, with a uniform C^2 bound. This argument needs no regularity at the second foot.

For shorter chords, both points belong to the same local regular graph. In coordinates $X(u)$, write the second point as $X(u + t\omega)$ with $|\omega| = 1$ and small t. The penalty is then $\Gamma = 1/s$. Its reciprocal branch is

$$\frac{|\nu(u) \cdot [X(u + t\omega) - X(u)]|}{|X(u + t\omega) - X(u)|^2}.$$

Here the cancellation at $t = 0$ can be made explicit. Set

$$A(u, t, \omega) = \int_0^1 DX(u + \theta t\omega)\omega \, d\theta, \qquad C(u, t, \omega) = \int_0^1 (1 - \theta) D^2X(u + \theta t\omega)[\omega, \omega] \, d\theta.$$

Then

$$X(u + t\omega) - X(u) = tA(u, t, \omega), \qquad \nu(u) \cdot [X(u + t\omega) - X(u)] = t^2 \nu(u) \cdot C(u, t, \omega).$$

Consequently the reciprocal branch is $|\nu(u) \cdot C(u, t, \omega)|/|A(u, t, \omega)|^2$. The signed quotient $\nu(u) \cdot C(u, t, \omega)/|A(u, t, \omega)|^2$ extends with uniform C^2 bounds through $t = 0$, because the regular graph is smooth and $|A|$ is bounded below. Its absolute value has the same bounds on the nonvanishing sign branches identified below.

Shrink the coordinate ball inside a fixed buffered graph chart, so that all $u + t\omega$ used for short chords remain in that chart. At any base where a branch has cost at most $2C$, the absolute value of this signed quotient is at least $1/(2C)$. Its uniform first-derivative bound therefore supplies a common base neighborhood on which it keeps its sign and remains at least $1/(4C)$. Choose that sign and take the reciprocal there. The resulting upper branch has a common C^2 bound on this neighborhood, including at $t = 0$. For long chords the lower bounds on length and angle give the same common-neighborhood conclusion while the second foot is fixed. Every such function remains an upper bound for d at all bases in its domain; for $t = 0$ this follows by taking genuine chords with $t \to 0$.

For attainment, a minimizing sequence has bounded chord length because $\Gamma \geq 1$. If its second feet stay away from the base, closedness of Σ and the positive-angle bound give an attaining off-diagonal foot. If they approach the base, compactness of (t, ω) and the canceled quotient gives an attaining diagonal direction. Thus every base has an active upper branch on a neighborhood of the same radius, with common first and second derivative bounds. Applying the first-derivative bound once at either endpoint shows that d is locally Lipschitz. If F_u is an active branch at u, its common Hessian bound gives

$$d(u+h) + d(u-h) - 2d(u) \leq F_u(u+h) + F_u(u-h) - 2F_u(u) \leq C_0|h|^2.$$

Thus d is locally semiconcave, with linear modulus, as in [9, Proposition 1.1.3, pp. 2–3]. By [9, Theorem 2.3.1, p. 42], d has an Alexandrov Hessian almost everywhere, and its gradient is locally of bounded variation. Proposition 1.1.3 also bounds its distributional Hessian above by an absolutely continuous matrix measure. Its singular part is therefore negative semidefinite. At an Alexandrov point, each active smooth upper branch has the same first derivative and an upper Hessian for d. The surface metric contracts the singular Hessian with a positive definite matrix. Connection terms contain only locally bounded first derivatives. Thus the singular part of $\Delta_\Sigma d$ is nonpositive, and the smooth upper-branch calculations give its required weak upper bound.

We construct the measurable choice used below. Work first on a compact base piece inside a buffered regular chart, where $0 < c \leq d \leq C$. Restrict to competitors with cost at most $2C$. Their chord lengths are at most $2C$, since $\Gamma \geq 1$. For chords bounded away from zero, the second foot therefore lies in one fixed compact subset of the closed zero set. The positive-angle bound makes their cost continuous on this restricted parameter set. For short chords, use the parameters (t, ω) above and include $t = 0$. The canceled quotient gives a continuous cost at each finite active diagonal direction. Restricting its reciprocal to be at least $1/(2C)$ removes the directions with infinite cost. The parameters then lie in a fixed compact set. The graph chart has a fixed buffer, so all these short chords are defined throughout the compact base piece. Use a disjoint union of these two compact parameter spaces; overlap between their genuine-chord representations is harmless.

The active parameters form a nonempty compact fiber over each base. Their graph is closed, because the cost and d are continuous. Plateau contacts also form a closed subrelation. Indeed, their angle satisfies $s \geq 1 - \delta_2$ and their length is $\ell = d/G_1 \geq c/G_1$. Thus they stay away from the diagonal. Choose the short-chord cutoff smaller than $c/(2G_1)$. The plateau subrelation is then entirely in the compact long-foot parameter space. Its projection onto this compact base piece is closed, and its nonempty fibers are compact.

Here is an elementary selection construction for either compact relation. Fix a countable dense set of parameter points. At stage k, take the first closed ball of radius 2^{-k} for which its intersection with the active fiber and all previously chosen balls is nonempty. Such a ball exists. For each fixed finite list of previous choices, the nonemptiness test is the projection of a closed subset of the product of two compact spaces. That projection is closed. Taking the first successful index therefore gives Borel choices at every stage. The resulting intersections are nested, nonempty and compact. Their diameters tend to zero, so their intersection is one active parameter. The selected ball centers converge to it; hence this parameter is a Borel function of the base. Apply this construction to the plateau relation where its fiber is nonempty. On the complementary Borel set, apply it to the full active relation. This gives the required priority.

Finally cover the regular finite-reach set by countably many such compact chart pieces and assign each base to the first piece containing it. These are disjoint Borel sets. The local choices therefore give a global Borel choice. Decoding a short-chord parameter with $t = 0$ gives its diagonal direction; all other parameters give their second foot and chord. This construction changes neither the reach nor its upper-branch inequalities.

We treat the unmodified penalty first, then diagonal contacts, and finally the transition to the constant part. After these calculations, we transfer the positive mean term to the receiving foot.

For an off-diagonal contact, choose $N = \pm\nu(z)$ with $N \cdot e = s > 0$. The region

$$\{z + \ell' e' : N \cdot e' > 0,\ \ell'\Gamma(N \cdot e') < d\}$$

is zero-free. Translate, rotate, and scale by d. Its boundary becomes a fixed smooth compact embedded radial surface. Near zero it is the sphere $|x|^2 = x_n$, since $\Gamma(s) = 1/s$ there. Elsewhere it is the positive radial graph $1/\Gamma$. It admits inward rolling balls of a uniform radius c_Γ. Indeed, local C^2 graph bounds give small balls. Compactness and embeddedness allow a uniform decrease that keeps them away from the rest of the boundary. Thus the second foot y has a zero-free tangent ball of radius $c_\Gamma d$. Lemma 6.5 makes that foot regular and gives $b(y) + D(y) \le C_\Gamma/d$. Its inward mean satisfies $H_{N_y}(y) \le C_\Gamma d^{-1-\gamma}$.

Assume $s \le 1 - \delta_1$, so that $\Gamma = 1/s$. Put $r = d/2$, $\tau = e - sN$, and $M = I + rB_N$. Then $\ell = 2rs$, and the inward normal at the second foot is $N_y = N - 2se$. Reflect an orthonormal frame v_i of $T_z\Sigma$ across $e^\perp$. This gives $v_i' = v_i - 2(e \cdot v_i)e$ in $T_y\Sigma$. Move z and y along geodesics with these velocities. The chord values touch d from above. Differentiation gives

$$\begin{aligned} \nabla_\Sigma r &= -M\tau/s, \\ \Delta_\Sigma d &\le b^2 d + \left(2 - \frac{2}{1 + r\max_i |\lambda_i|}\right)\frac{|\nabla_\Sigma d|^2}{d} \\ &\quad + \frac{H_{N_y}(y) - (1 - 2s^2)H_N(z) - \ell\tau \cdot \nabla_\Sigma H_N(z)}{s^2}. \end{aligned} \tag{6.27}$$

For the gradient coefficient, let $w = y - z$ and $w_i = v_i' - v_i = -2(e \cdot v_i)e$. Differentiate $|w|^2/2 - rN \cdot w$. It vanishes at contact and is nonnegative for competitors.

With $\tau_i = \tau \cdot v_i$, its first derivative at contact is

$$0 = -s\ell r_i - \ell\tau_i - r\ell(B_N\tau)_i.$$

Division by $s\ell$ gives the first line of (6.27). In the second derivative, the nontrace terms reduce by

$$\begin{aligned} 4|\tau|^2 + 4r\tau \cdot B_N\tau + 4s\tau \cdot \nabla r - 4rs\nabla r \cdot B_N\tau &= -4rs\nabla r \cdot B_N\tau \\ &= 4s^2|\nabla r|^2 + 4s\tau \cdot \nabla r. \end{aligned}$$

Here the first equality uses $M\tau = -s\nabla r$. This keeps the coefficient of the gradient term separate from all mean-curvature estimates.

Its summed second derivative bounds $(\Delta r)s\ell$ by

$$\begin{aligned} &\sum_i |w_i|^2 + (w - rN) \cdot (-H_{N_y}N_y + H_N N) - 2r\sum_i w_i \cdot B_N v_i \\ &\quad - rw \cdot (\nabla H_N - b^2 N) - 2\sum_i r_i(w_i \cdot N + w \cdot B_N v_i). \end{aligned}$$

Insert $w - rN = -rN_y$ and the first derivative identity. Apart from b^2 and the trace terms, the expression reduces to $4s^2|\nabla r|^2 + 4s\tau \cdot \nabla r$. Equation (6.6) gives $0 \le M \le (1 + r\max_i |\lambda_i|)I$. Hence

$$\tau \cdot \nabla r = -\tau \cdot M\tau/s \le -\frac{s|\nabla r|^2}{1 + r\max_i |\lambda_i|}.$$

Divide the summed derivative by rs^2 to obtain (6.27). The identity $\Delta_\Sigma N = \nabla_\Sigma H_N - b^2 N$ determines the trace signs.

The final quotient has a uniform upper bound even for nearly tangent chords. Fix a small $\gamma' > 0$ relative to the gains in Lemmas 6.6 and 6.5. If $s \ge d^{-\gamma'}$, use the upper mean bound at the second foot together with $|H| \le Ch^{-2}$, $|\nabla H| \le Ch^{-3-\gamma}$, and $h = d^{1-o(1)}$ at the base. The quotient is at most $C_\Gamma d^{-1-\gamma''}$ for a fixed $\gamma'' > 0$. If $s < d^{-\gamma'}$, the chord $\ell = ds$ lies in the isolated graph.

Continue the normal orientation between its ends. In tangent-plane coordinates, the displacement is $\ell\tau$. Taylor's formula gives

$$H_{N_y}(y) - H_N(z) - \ell\tau \cdot \nabla H_N(z) = O(\ell^2 h^{-3-\gamma}).$$

The coordinate Hessian has the same bound. Its connection correction contains the controlled gradient. Divide by s^2 and add $2H_N(z)$. This gives the same upper error, after decreasing γ'' if necessary.

In (6.27), the coefficient in parentheses is at most one because $r \max |\lambda_i| \le 1$. It is $O(bd)$ when bd is small. We also have $|\tau| \ge \sqrt{1-(1-\delta_1)^2} > 0$. For sufficiently small bd, the first derivative formula gives $|\nabla d| \ge c_\Gamma$. Otherwise $b^2 d \ge c_\Gamma/d$. In both cases the error $Cd^{-1-\gamma''}$ is bounded by $Cd^{-\gamma''}(b^2 d + |\nabla d|^2/d)$. This proves (6.24) in the unchanged-penalty range.

To estimate the gradient, first assume $\ell \ge ch$. Then $1/s = d/\ell \le Cd/h$ and $|M\tau| \le 2|\tau|$, so $|\nabla d| \le Cd/h$. If $\ell < ch$, the two points lie in the isolated graph. Expand its normal and project onto the base tangent plane. This gives

$$-2s\tau = \ell B_N \tau + O(\ell^2/h^2),$$

using $|\nabla B| \le Ch^{-2}$ and $|B| \le Ch^{-1}$. Indeed, this expansion gives

$$M\tau = \tau + \frac{d}{2} B_N \tau = \tau - \frac{ds}{\ell}\tau + O(d\ell/h^2) = O(sd^2/h^2),$$

because $\ell = ds$. The first line of (6.27) then gives $|\nabla d| = 2|M\tau|/s \le Cd^2/h^2$. Together with the long-chord case, this proves (6.23).

At a diagonal contact, the active constraint is a principal curvature. Choose N so that $S = -B_N$ has largest positive eigenvalue $\lambda = 2/d$. Let v_1 be an active unit eigenvector. Every eigenvalue λ_j of S lies between $-\lambda$ and λ. Take a unit vector v in the top eigenspace V. Along its geodesic with signed length L,

$$\frac{2N \cdot (y-z)}{|y-z|^2} = \lambda + \frac{1}{3}\nabla_v S(v,v) L + O(L^2).$$

This quotient is at most $2/d = \lambda$ for both signs of L, by the defining inequality $d \le |y-z|^2/|N\cdot(y-z)|$. Therefore $\nabla_v S(v,v) = 0$. Codazzi makes ∇S totally symmetric. Polarization then shows that it vanishes on triples from V.

Extend the active eigenvector as a local unit field and touch d from above by $2/S(v_1,v_1)$. In a principal frame, prescribe

$$\langle \nabla_i v_1, v_j \rangle = \frac{\nabla_i S_{1j}}{\lambda - \lambda_j} \quad (\lambda_j < \lambda), \qquad \langle \nabla_i v_1, v_j \rangle = 0 \quad (v_j \in V).$$

A smooth unit extension with these first derivatives exists. Its second derivatives are free except for the unit-length condition. At a top eigenvector, their tangential components disappear from the Laplacian of $S(v_1,v_1)$.

If $a_{ij} = \langle \nabla_i v_1, v_j \rangle$, the vector-variation terms for $j \ne 1$ are

$$4(\nabla_i S_{1j}) a_{ij} + 2(\lambda_j - \lambda) a_{ij}^2.$$

The unit-length constraint supplies the term $-2\lambda a_{ij}^2$. The prescribed first derivative makes this quadratic expression equal to $2|\nabla_i S_{1j}|^2/(\lambda - \lambda_j)$. The remaining second jet has no contribution because the active vector is an eigenvector.

That Laplacian is therefore

$$(\Delta S)_{11} + 2 \sum_{\lambda_j < \lambda} \frac{|\nabla S_{1j}|^2}{\lambda - \lambda_j}.$$

Commuting the Codazzi derivatives and using the Gauss equation gives

$$\Delta S = \nabla^2 \operatorname{tr} S + (\operatorname{tr} S) S^2 - b^2 S \tag{6.28}$$

This follows the calculation of Simons [41, Section 4.2], with the mean-curvature terms retained. Fröhlich–Winklmann give its contracted form in codimension one [21, Lemma 2.1, (5)]. The Ricci contraction is $((\operatorname{tr} S)S - S^2)_{ip}S_{pj}$. The other curvature contraction is $S^3_{ij} - b^2 S_{ij}$. Their sum gives the identity. Further, $\lambda - \lambda_j \le 2\lambda$. Codazzi gives $\nabla_j S_{11} = \nabla_1 S_{j1}$, which vanishes for $v_j \in V$. The denominator $f = S(v_1, v_1)$ consequently satisfies

$$\Delta f \ge -b^2\lambda - C(|\nabla^2 H| + b^2|H|) + |\nabla f|^2/\lambda.$$

For this positive denominator, the reciprocal rules are

$$\nabla(2/f) = -2f^{-2}\nabla f, \qquad \Delta(2/f) = -2f^{-2}\Delta f + 4f^{-3}|\nabla f|^2.$$

At contact $f = \lambda = 2/d$. The positive derivative term left after inserting the preceding lower bound is $2|\nabla f|^2/\lambda^3$. Since $|\nabla d|^2 = 4|\nabla f|^2/\lambda^4$, it equals $|\nabla d|^2/d$. Thus

$$\Delta d \le b^2 d + \frac{|\nabla d|^2}{d} + Cd^2(|\nabla^2 H| + b^2|H|).$$

The final term is bounded by $Cd^{-\gamma}(b^2 d)$. Here we used Lemma 6.6, $h = d^{1-o(1)}$, and $b^2 d \ge 4/d$. This proves (6.24). At a diagonal contact, $bd \ge 2$. Thus the small-bd claim is irrelevant here. Finally, $|\nabla d| \le Cd^2|\nabla B| \le Cd^2/h^2$ proves (6.23).

We still need the range $s \ge 1 - \delta_1$. Put

$$\tau = e - sN, \qquad c_1 = \Gamma - s\Gamma', \qquad Z = c_1 e + \Gamma' N.$$

Variation in the second foot gives its inward normal as $N_y = -Z/|Z|$. Rotate the tangent frame at z by the small plane rotation from N to $Z/|Z|$. This gives v_i' at y. With $w_i = v_i' - v_i$, plane-rotation formulas give

$$w_i = O(|\tau|)N + O(|\tau|^2), \qquad P_{e^\perp} w_i = O(|\tau|^2), \qquad (N - se)\cdot w_i = O(|\tau|^3).$$

The constants are uniform for the fixed penalty. In particular, $|Z| = 1 + O(\delta_1)$ and the rotation stays small. Use the identity rotation at $\tau = 0$. Differentiating $\ell\Gamma(s)$ yields

$$\begin{aligned} \nabla d &= (-c_1 I + \ell\Gamma' B_N)\tau, \\ \Delta d &\le |Z|H_{N_y} + H_N Z\cdot N + \ell\Gamma'\tau\cdot\nabla H_N - \ell\Gamma' s b^2 \\ &\quad + \sum_i \left\{ \frac{c_1}{\ell}|P_{e^\perp} w_i|^2 + 2\Gamma' w_i \cdot B_N v_i + \ell\Gamma''\left[\tau\cdot B_N v_i + \frac{(N - se)\cdot w_i}{\ell}\right]^2 \right\}. \end{aligned} \tag{6.29}$$

To check every term, regard $\Phi(w, N) = |w|\Gamma(N\cdot w/|w|)$ as a function of the two ambient variables. The first variations are $\mathrm{d}\ell = e\cdot \mathrm{d}w$ and $\mathrm{d}s = \mathrm{d}N\cdot e + (N - se)\cdot \mathrm{d}w/\ell$. Write $\mathbf{q} = N - se$. Its first derivatives and Hessian blocks are

$$\begin{aligned} \Phi_w &= Z, & \Phi_N &= \ell\Gamma' e, \\ \Phi_{ww} &= \frac{c_1}{\ell}P_{e^\perp} + \frac{\Gamma''}{\ell}\mathbf{q}\otimes\mathbf{q}, & \Phi_{NN} &= \ell\Gamma'' e\otimes e, \\ \Phi_{wN}[\xi,\zeta] &= \Gamma'\xi\cdot\zeta + \Gamma''(\mathbf{q}\cdot\xi)(e\cdot\zeta). \end{aligned}$$

Along our variations, $N_i = B_N v_i$. The three Γ'' terms combine into the square in (6.29). The term $\Phi_N\cdot\Delta N$ is $\ell\Gamma'(\tau\cdot\nabla H_N - sb^2)$. Finally

$$Z\cdot(-H_{N_y}N_y + H_N N) = |Z|H_{N_y} + H_N Z\cdot N.$$

These identities establish every term and sign in (6.29) without suppressing a second derivative of the moving normal.

The bound $\Gamma''|\tau|^2 \le C\varepsilon_1$ controls all algebraic terms, including $-\ell\Gamma' s b^2$, by

$$(1 + C(\varepsilon_1 + \delta_1))b^2 d + Cb|\tau|^2 + C|\tau|^4/d.$$

The leading coefficient is one up to $O(\delta_1)$ because $-s\Gamma'/\Gamma \le 1 + C\delta_1$. The Γ'' square contributes $C\varepsilon_1 b^2 d$ and $C\varepsilon_1|\tau|^4/d$. The rotation supplies the other two terms. Young's inequality bounds $Cb|\tau|^2$ by a chosen small multiple of $b^2 d$ and a fixed multiple of $|\tau|^4/d$. When bd is small, the gradient formula bounds $|\nabla d|$ below by a fixed multiple of $|\tau|$. Otherwise $|\tau|^4/d$ is at most $C\delta_1^2 b^2 d$. First choose ε_1 and then δ_1 small enough. The full bound is then $(1+\eta_0)b^2 d + \eta_0|\nabla d|^2/d$. Leave room for the errors below by reducing the initial tolerance. Before the plateau, $|\tau|$ has a fixed positive lower bound depending on δ_2. The base and second-foot estimates bound the traces in (6.29) above by $O(d^{-1-\gamma})$. Absorb them as in (6.27). This proves (6.24), including the small-bd estimate. The gradient formula, $\ell \le d$, and $db \le 2\sqrt{m}$ give a uniform bound here. Thus (6.23) also holds.

On the constant part, $\Gamma' = \Gamma'' = 0$. Hence $\nabla d = -G_1\tau$, $Z = G_1 e$, and $N_y = -e$. The remaining algebraic term is $C|\tau|^4/d$, which is bounded by $C\delta_2|\nabla d|^2/d$. The trace terms are $G_1 H_{N_y} + G_1 s H_N \le G_1(H_{N_y,+} + |H_N|)$. Decrease δ_2 to obtain (6.25). At the ends of the ranges the jets agree, since the penalty is smooth. The computations therefore extend to those endpoints.

Integrating these inequalities requires the inverse area factor and no extra multiplicity for the positive plateau mean term.

To parametrize the active bases by their receiving feet, take a plateau pair. It satisfies $z = y + \ell N_y$ and $|N_y \cdot \nu(z)| = s \ge 1 - \delta_2$. Fix the regular sheet through z. The equation $u(y + tN_y(y)) = 0$ has t derivative $\nabla u(z) \cdot N_y \ne 0$ at this pair. The implicit function theorem therefore gives a smooth local parametrization $z(y) = y + \ell(y)N_y(y)$ of the active source sheet. This parametrization may have a singular derivative. Its derivative and its projection onto $T_y\Sigma$ are

$$Dz(y)v = (I + \ell B_{N_y})v + (D\ell \cdot v)N_y, \qquad P_{T_y\Sigma} Dz(y) = I + \ell B_{N_y}.$$

Projection from $T_z\Sigma$ to $T_y\Sigma$ has area factor s. Thus $sJ_{y\to z} = |\det(I + \ell B_{N_y})|$, including points where the determinant vanishes. Since $\ell \le d(z)$ and $b(y) \le C_\Gamma/d(z)$, this bounds $J_{y\to z}$ uniformly. If $\ell b(y)$ is small, then

$$|\det(I + \ell B_{N_y})| \le \prod_{j=1}^{m} (1 + \ell|\lambda_j(y)|) \le 1 + C\ell b(y), \qquad s^{-1} \le 1 + C\delta_2.$$

These formulas give the sharper transfer factor before any incoming charge is integrated.

The reverse chord is normal at y. It is a competitor there of value $G_1\ell$, and hence $d(y) \le d(z)$.

If $H_{N_y} > 0$, the orientation N_y is fixed uniquely. Every base that sends positive charge to this foot lies on the same normal ray. Suppose two bases had lengths $0 < \ell_1 < \ell_2$. At the farther base z_2, the nearer zero z_1 gives a chord in the direction of y with length $\ell_2 - \ell_1$. Its angle penalty is unchanged and its objective is smaller than the chord to y. This contradicts activity. Thus a positive receiving foot has at most one inverse base.

To apply the area formula, cover the regular contact relation by countably many inverse charts. On compact coordinate pieces, the active-parameter relation is closed and compact-valued. Choose a measurable selection on each piece and join them along a countable exhaustion. Decompose the selected pairs into disjoint measurable chart pieces. The inverse area formula uses precisely the Jacobian computed above. Critical images have zero surface measure. The ray argument shows that summing the positive pieces adds no multiplicity. For a nonnegative measurable function $F(z, y)$ supported on selected positive pairs, the formula is

$$\int F(z, y(z))\, \mathrm{d}S_z = \int F(z(y), y) J_{y\to z}\, \mathrm{d}S_y.$$

The right integral is over the selected positive feet. Null critical images are omitted on the left. This proves transfer for every nonnegative weight. □

To join ambient and surface tests, we adapt the collar argument of [35, Corollary 7.8]. We use the tangent-ball and discrepancy bounds proved above for the matched equation. The same control

applies when the second foot, rather than the base, lies near a bad zero. Relative to $X \subset \Sigma$, call a point of $\Sigma_{\mathrm{reg}} \setminus X$ a *good zero*.

Corollary 6.8 (Reach bounds near bad zeros)**.** *For sufficiently large $d(z)$, $D(z) + b(z) \le C_\Gamma / d(z)$. Suppose $X = \{z \in \Sigma : D(z) \ge \delta\}$ for a fixed $\delta > 0$. For each fixed L_0, the reach $d(z)$ has a finite uniform upper bound on the good zeros satisfying* $\operatorname{dist}(z, X) \le 10L_0$. *This bound also applies to a contact reach $d(z)$ whose second foot y satisfies* $\operatorname{dist}(y, X) \le 10L_0$.

Proof. Apply the tangent balls in Lemma 6.1 and the estimates in Lemma 6.5. At an off-diagonal second foot, the proof of Proposition 6.7 gives the same balls, with radius measured by the source reach $d(z)$. For a diagonal contact the second foot is the base, so the first bound applies. By (6.2), D has a positive lower bound depending only on δ, L_0 within distance $10L_0$ of X. Compare this with the upper bound $C_\Gamma / d(z)$ to bound the reach. If X is empty, there is no collar to consider. □

6.3. **Interaction comparison.** We compare discrepancy with neighboring-sheet interaction by the artificial-slope cancellation of [35, Section 8.3]. The reference operator and the cancellation estimates below retain the matched weights. For the next barrier lemma and artificial-layer cancellation proposition, allow any compact interval $\mathcal{P} \Subset (1, \infty)$. Their constants are uniform on that interval. The artificial phase gradients stay near the nonzero planar gradient. For this cancellation, use the reference phase operator and its conjugate:

$$\mathcal{P} = -\Delta_y - c\partial_t^2 + 2c\beta \tanh(at)\partial_t, \qquad \mathcal{H} = -\Delta_y - c\partial_t^2 + ca^2[p^2 - p(p+1)\operatorname{sech}^2(at)],$$

Thus $\mathcal{H}(M\psi) = M\mathcal{P}\psi$. We also need the sharp Dirichlet gap on the positive half-line. Lemma 2.16 transforms the one-dimensional Jacobi form by $\varphi = r_0^{(2-p)/2} F$, where $r_0 = aG$. Its weighted squared norm becomes $\int F^2$. Moreover,

$$\int_{\mathbb{R}} r_0^{p-1} \varphi = a^{p/2} \int_{\mathbb{R}} MF.$$

For $F \in H_0^1(0, \infty)$, its odd extension belongs to $H^1(\mathbb{R})$ and is orthogonal to the even function M. Apply Proposition 2.17 to this extension. Both its energy and its squared norm are twice their half-line values. Therefore

$$\int_0^\infty (c|F'|^2 + V_0 F^2) \ge ca^2(2p-1) \int_0^\infty F^2.$$

First prove this for smooth compactly supported tests and pass to H_0^1 by density. The potential is bounded uniformly on $\mathcal{P}$, so this passage preserves the form. The coefficient is sharp. Indeed, $F_1(t) = M(t)\sinh(at)$ is positive for $t > 0$, vanishes at zero, and lies in $H_0^1(0, \infty)$ because $p > 1$. Using $\mathcal{H}M = 0$ and $M'/M = -pa\tanh(at)$ gives

$$\mathcal{H}(M\sinh(at)) = -cM(a^2 \sinh(at) - 2pa^2 \tanh(at)\cosh(at)) = ca^2(2p-1)F_1.$$

The gap is therefore positive uniformly on the fixed compact interval $\mathcal{P} \Subset (1, \infty)$.

Lemma 6.9 (A phase barrier with small normal growth)**.** *There exist uniform $\eta, \lambda, \omega > 0$, with $2\omega < \min_{p \in \mathcal{P}} \beta_p / 2$, and positive smooth functions v_p on $[0, \infty)$ such that*

$$1 \le v_p(t) \le Ce^{\omega t}, \quad |v_p'| + |v_p''| \le Cv_p, \quad \mathcal{P}v_p \ge \eta v_p + \tfrac{1}{2}.$$

For $A_\lambda(y) = e^{\lambda\sqrt{1+|y|^2}}$, the product $A_\lambda v_p$ remains a strict supersolution for every nondivergence operator whose principal, first-order and zeroth-order coefficients differ sufficiently little, in supremum norm, from those of $\mathcal{P}$. The permitted perturbation and all constants are independent of a finite slab depth.

Proof. Choose $\eta > 0$ well below the uniform half-line gap. For $H_p = -c\partial_t^2 + ca^2[p^2 - p(p+1)\operatorname{sech}^2(at)]$, define

$$W = (H_p - \eta)^{-1} M$$

with zero trace at zero and the energy condition at infinity. The uniform positive Dirichlet gap makes this inverse bounded on L^2. Positivity follows from the gap even though the potential need not be positive at every point. If $(H_p - \eta)W = f \geq 0$ and $W(0) = 0$, test with its negative part. This gives

$$\langle (H_p - \eta)W_-, W_- \rangle \leq 0.$$

The strict Dirichlet gap forces $W_- = 0$. The one-dimensional strong maximum principle gives strict positivity for the nonzero source.

Put $V = W/M$. Then $\mathcal{P}V = 1 + \eta V$. The coefficient of $H_p - \eta$ converges exponentially to $c\beta^2 - \eta$ at infinity. ODE comparison with exponentials gives

$$0 \leq V(t) \leq Ce^{\omega t}, \qquad |V'| + |V''| \leq C(1+V),$$

for each fixed ω larger than $\max_{p\in\mathcal{P}}(\beta_p - \sqrt{\beta_p^2 - \eta/c})$. Variation of constants beyond a fixed large depth gives the same bounds because the coefficients converge exponentially and the parameters range over a compact set. The difference between the two decay rates satisfies

$$\beta_p - \sqrt{\beta_p^2 - \eta/c} = \frac{\eta}{c(\beta_p + \sqrt{\beta_p^2 - \eta/c})} \leq C_{\mathcal{P}}\eta.$$

Choose η small first. Then a single ω can exceed all these differences and still satisfy the stated upper bound. Taking $v_p = 1 + V$ and $\eta < 1/2$ gives $\mathcal{P}v_p = \eta v_p + 1 - \eta$.

The spatial factor satisfies $|\nabla A_\lambda| \leq \lambda A_\lambda$ and $|D^2 A_\lambda| \leq C(\lambda + \lambda^2)A_\lambda$. Choose λ small relative to η. Then $\mathcal{P}(A_\lambda v_p) \geq A_\lambda(\eta v_p/2 + 1/2)$. Every derivative of the product through order two is bounded by $CA_\lambda v_p$. A small uniform change of coefficients therefore preserves $L(A_\lambda v_p) \geq cA_\lambda v_p$. □

Proposition 6.10 (No logarithmic loss in the opposite slopes)**.** *Fix the spatial exponent λ sufficiently small for both the preceding barrier and the unit-block decay constant used below; this final choice is made before imposing the graph estimate. Use the artificial finite-slab construction with the same graph h on its two sides and with corrections $\psi_\pm = 0$ at $t = 0, L$. Suppose its previously established unweighted C^1 and $W^{2,q}_{\mathrm{uloc}}$ smallness tends to zero, and suppose for a number $J > 0$*

$$|Dh(y)| + |D^2 h(y)| \leq C\sqrt{J}\, e^{\lambda\sqrt{1+|y|^2}}, \qquad h(0) = Dh(0) = 0.$$

The unweighted graph coefficients are also sufficiently small. Then, uniformly in $L \geq 2$,

$$|\partial_t \psi_+(0,0) + \partial_t \psi_-(0,0)| \leq CJ.$$

In particular the sum of the two artificial physical inward slopes differs from $2a$ by at most CJ.

Proof. To obtain the finite-slab equations and their signs, use coordinates

$$x = (y, h(y) + \sigma t), \qquad U_\sigma(x) = t + \psi_\sigma(y,t), \qquad \sigma \in \{+1, -1\}.$$

On a negative chamber the sign of the physical solution has first been reversed. Thus U_σ is the positive inward phase on both sides. Its physical gradient is

$$\nabla_x U_\sigma = \big(\nabla_y \psi_\sigma - \sigma(1 + \partial_t\psi_\sigma)Dh,\ \sigma(1 + \partial_t\psi_\sigma)\big).$$

At $\psi_\sigma = 0$ its tangential divergence has the linear term $-\sigma\Delta h$. The exact phase equation

$$\Delta_p U = 2ca\, g(U)(|\nabla U|^p - 1)$$

has linearization $\Delta_y\psi + c\psi_{tt} - 2c\beta g(t)\psi_t$ at $U = t$. Consequently the finite-slab equations are

$$\mathcal{P}\psi_\sigma = -\sigma\Delta h + N_\sigma(\psi_\sigma).$$

The nonlinearities are smooth where the phase gradient is noncritical. Keep the true coefficient of $D^2\psi$ in the principal part. Apply the fundamental theorem of calculus only to the lower-order terms. This gives

$$L_\sigma \psi_\sigma = -\sigma \Delta h + N_\sigma(0),$$

The operator L_σ is uniformly close to $\mathcal{P}$ in supremum norm. This includes the small zeroth-order coefficient arising from $f(t+\psi) - f(t)$. No coefficient depending on $D^2\psi$ is differentiated. Principal coefficients depend only on $Dh, \nabla\psi$. The known unweighted smallness controls all lower coefficients. Further,

$$|N_\sigma(0)| \le C(|Dh|^2 + |Dh|^2 |D^2 h|) \le C\sqrt{J} A_\lambda.$$

Use unweighted smallness on one factor for the final inequality. Apply Lemma 6.9. Division by its positive supersolution gives an equation with positive zeroth-order coefficient, so the maximum principle applies even if the original zeroth-order coefficient changes sign. We obtain

$$|\psi_\sigma(y,t)| \le C\sqrt{J} A_\lambda(y) v_p(t).$$

The corrections vanish on the top and bottom. At horizontal infinity the growing weight dominates each bounded correction. The comparison is therefore uniform in slab depth. The source and solution bounds then give local noncritical $W^{2,q}$ estimates, including at both flat faces. The principal coefficients have uniform Hölder bounds by the known unweighted $W^{2,q}_{\mathrm{uloc}}$ estimates and Sobolev embedding. Thus the local estimates give

$$\|\psi_\sigma\|_{W^{2,q}(B_1(y,t)\cap\{0<t<L\})} \le C\sqrt{J} e^{\lambda\sqrt{1+|y|^2}+\omega t}. \tag{6.30}$$

Their constants are uniform: ellipticity and the unweighted coefficient norms were fixed before the weight. Choose $q > m+1$. Sobolev embedding gives the same pointwise control on first derivatives.

Set $S = \psi_+ + \psi_-$. The terms linear in h cancel. Thus $\mathcal{P}S = F$ with $S = 0$ on the horizontal faces. Each term in $F = N_+(\psi_+) + N_-(\psi_-)$ is at least quadratic in the graph perturbation and the two corrections. More explicitly, on the fixed noncritical neighborhood put

$$E_{1,\sigma} = |\psi_\sigma| + |Dh| + |\nabla\psi_\sigma|, \qquad E_{2,\sigma} = |D^2 h| + |D^2\psi_\sigma|.$$

The nondivergence expansion of the exact slab equation gives

$$|N_\sigma(\psi_\sigma)| \le C\{E_{1,\sigma}^2 + E_{1,\sigma}E_{2,\sigma}\}.$$

Indeed, subtracting $-\sigma\Delta h$ removes the only term linear in the graph. The difference of the principal matrix from its flat value multiplies $D^2\psi_\sigma$ and is $O(|Dh|+|\nabla\psi_\sigma|)$. Differentiating the graph metric contributes $D^2 h$ times a factor vanishing at $Dh = \nabla\psi_\sigma = 0$. The remaining reaction terms have their linear part already included in $\mathcal{P}$. Their remainders have the displayed quadratic bound, since the derivatives of f are uniformly bounded. In particular there is never a product of two second derivatives. Product estimates and (6.30) now yield

$$\|F\|_{L^q(B_1(y,t)\cap\{0<t<L\})} \le CJ e^{2\lambda\sqrt{1+|y|^2}+2\omega t}.$$

At most one factor has second derivatives in L^q. The local C^1 bound controls the other.

Conjugate by writing $\mathcal{H}(MS) = MF$. This removes the slab-depth factor from the bottom derivative estimate. The finite-slab Dirichlet inverse has the same positive gap as the half-line operator. We first bound the solution near each source block. Write $\mathcal{D}_L = \mathbb{R}^m \times (0,L)$ and $\lambda_{\mathrm{gap}} = \min_{p\in\mathcal{P}} ca^2(2p-1) > 0$. For the associated form Q_L, zero extension past $t = L$ and integration of the half-line gap in the horizontal variables give $Q_L(v) \ge \lambda_{\mathrm{gap}}\|v\|_2^2$. Put $c_{\mathrm{ell}} = \min\{1, \min_{p\in\mathcal{P}}(p-1)\} > 0$ and $C_V = \sup|V_0|$. Then

$$c_{\mathrm{ell}}\|\nabla v\|_2^2 \le Q_L(v) + C_V\|v\|_2^2 \le (1 + C_V/\lambda_{\mathrm{gap}}) Q_L(v).$$

Thus the form is coercive in $H_0^1(\mathcal{D}_L)$, uniformly in L. For f_b supported in one unit block, Lax–Milgram gives $w_b \in H_0^1(\mathcal{D}_L)$ solving $\mathcal{H}w_b = f_b$. Testing with w_b gives

$$\lambda_{\mathrm{gap}}\|w_b\|_2^2 \le \langle \mathcal{H}w_b, w_b\rangle = \int_{\mathcal{D}_L} f_b w_b \le \|f_b\|_2\|w_b\|_2.$$

Hence $\|w_b\|_2 \le \lambda_{\mathrm{gap}}^{-1}\|f_b\|_2$. Since V_0 is uniformly bounded, the same identity bounds $\|\nabla w_b\|_2$ by $C\|f_b\|_2$. Take the fixed exponent $q > \max\{2, m+1\}$. The block has bounded volume, so $\|f_b\|_2 \le C\|f_b\|_q$. Local $W^{2,2}$ estimates and Sobolev embedding, repeated on finitely many nested fixed balls or flat half-balls, therefore give

$$\|w_b\|_{W^{2,q}(B_2(x_0)\cap\mathcal{D}_L)} \le C\|f_b\|_{L^q(\mathcal{D}_L)}.$$

Here x_0 is the block center. Use charts of radius less than $1/4$ near a horizontal face. Since $L \ge 2$, one chart cannot meet both faces. The constant principal coefficients and the potential have uniform bounds, including its fixed derivatives. Thus the local estimates and their finite covering constants are independent of L and uniform on $\mathcal{P}$. Sobolev embedding now bounds the solution on the boundary of a fixed padded source block. This starts the exterior comparison. Use the positive gauge v_0 from (6.18), which satisfies $H_p v_0 = 1$. Its upper and lower bounds and its derivative bounds are uniform on $\mathcal{P}$. We prove exponential decay of each unit-block response by a smooth barrier. For a block centered at $x_0 = (y_0, t_0)$ take

$$\rho_{x_0}(y,t) = \sqrt{1 + |y-y_0|^2 + |t-t_0|^2}.$$

Its gradient is bounded and its Hessian is nonnegative. The product rule, $H_p v_0 = 1$, and the bounds on v_0, v_0' give, for a sufficiently small fixed $c_0 > 0$,

$$\mathcal{H}\big(v_0 e^{-c_0\rho_{x_0}}\big) \ge \tfrac{1}{2}e^{-c_0\rho_{x_0}}.$$

The Hessian of ρ_{x_0} supplies a nonnegative term. The possibly negative terms have absolute value at most $C(c_0 + c_0^2)e^{-c_0\rho_{x_0}}$. This positive supersolution also gives the maximum principle despite the negative part of V_0. Local $W^{2,q}$ bounds near the source, followed by comparison outside a padded source block, prove the claimed exponential decay. Local flat-boundary estimates give the same decay for the bottom derivative. These estimates are uniform when the block meets either horizontal face. Imposing zero data at the upper boundary can only decrease the positive response. First truncate the source to finitely many blocks and exhaust the horizontal domain. The same barrier makes the lateral contribution tend to zero. The spatial exponent was chosen so that $2\lambda < c_0/4$. A source block centered at (y,t) has size at most

$$CJe^{2\lambda\sqrt{1+|y|^2}}e^{-(\beta_p - 2\omega)t}.$$

For clarity, index unit blocks by $j \in \mathbb{Z}^m$ and $k \in \mathbb{N}\cup\{0\}$. The contribution of such a block to the bottom $W^{2,q}$ norm near the origin is at most

$$CJ\exp\{-c_1(|j|+k) + 2\lambda\sqrt{1+|j|^2} - (\beta_p - 2\omega)k\},$$

after reducing the fixed exponential decay rate from c_0 to $c_1 > 0$ to compare Euclidean and lattice distances. Choose $2\lambda \le c_1/2$. Since $\sqrt{1+|j|^2} \le 1 + |j|$ and $\beta_p - 2\omega \ge \beta_-/2 > 0$, each summand is bounded by

$$CJe^{-c_1|j|/2}e^{-(c_1+\beta_-/2)k}, \qquad \beta_- := \min_{p\in\mathcal{P}}\beta_p.$$

Both lattice sums converge. Their bounds are independent of the number of depth blocks, and hence of L. The actual source also has bounded unweighted local L^q norms. The block sum therefore gives a bounded zero-trace solution. It agrees with MS: their difference is a bounded homogeneous solution, and the same positive gauge and horizontal exhaustion give uniqueness. The summed local bounds give $\|MS\|_{W^{2,q}(B_1(0,0)\cap\{0<t<L\})} \le CJ$. Since $M(0) = 1$ and $M'(0) = 0$, boundary

Sobolev embedding proves the derivative bound. The physical slope introduces only the factor a, which is uniformly bounded on $\mathcal{P}$. □

Return to the standing interval $\mathcal{P} \Subset (2, \infty)$ for comparison with the actual solution. Combine this cancellation with Theorem 4.14, with its fixed bad threshold and exponent $\alpha > 1$. For R above that theorem's fixed lower bound, a radius-R ball without bad zeros satisfies $D \le CR^{-\alpha}$ at every zero in its middle half. The same theorem gives $|B| \le C/R$ there. Use a further fixed interior fraction when a buffered graph or the full normal estimate is needed. These changes only alter C. Neither area nor the number of sheets is bounded in this argument. For a final threshold $0 < \delta \le \delta_0$, with fixed clean-ball threshold δ_0, set

$$\bar{a}(y) = \begin{cases} |B(y)|, & D(y) < \delta, \\ 1, & D(y) \ge \delta, \end{cases} \qquad J_0(z) = \sup_{y \in \Sigma} e^{-\lambda |y-z|} \bar{a}(y)^2.$$

The constants are independent of δ in this range. A ball with no δ-bad zeros also has no δ_0-bad zeros. The original clean-ball theorem therefore applies.

We will compare the curvature envelope J_0 with the interaction through the reach:

$$D(z) \le C\{J_0(z) + E(z)\}, \tag{6.31}$$

$$E(z) \le C\{J_0(z) + D(z)\} \quad \text{at plateau contacts with } d(z)|\nabla_\Sigma d(z)|^2 \le 1. \tag{6.32}$$

Here $E = e^{-\beta d/G_1}$. The constants are independent of d. Keep the stated restriction on the second inequality. The anisotropic well equation does not provide the unrestricted version.

Use Lemma 3.9 with the exact coefficient of a single face. Apply this barrier to two zero graphs in a cylinder of horizontal radius comparable to R. Suppose their curvature is at most κ and their tangent-plane error is $O(\kappa R^2)$. Assume $\kappa R^N \to 0$ for every fixed N, and assume $D \le CR^{-\alpha}$ at all their zeros. Use inward vertical depth s from the lower graph. Write the upper tangent plane as

$$t_2 = \ell - b \cdot y - ts, \qquad |b|^2 + t^2 = 1, \quad t > 0.$$

The number ℓ is the shortest Euclidean distance from the base to the adjacent graph. The displayed plane is tangent at the minimizing foot. Thus t_2 is a unit-affine distance. Expressing it in graph coordinates introduces only the stated small error. The lower tangent distance differs from s by $O(\kappa R^2)$. Put

$$\delta(y, s) = \ell - b \cdot y - (1 + t)s.$$

On a buffered lower half-chamber with $\delta \ge M_0$,

$$\left| \frac{1-u}{1-g(s)} - 1 \right| \le C\{\varepsilon_0 + e^{-2a\delta(y,s)}\}, \qquad \varepsilon_0 R \longrightarrow 0. \tag{6.33}$$

The same bound holds if either artificial one-sided solution over the base replaces $g(s)$. We may use

$$\varepsilon_0 = CR^{-\alpha} + C_N \kappa^\vartheta R^N + Ce^{-cR}$$

for fixed $\vartheta > 0$ and N.

Choose a fixed depth S in the well range where (3.18) applies. At distance S from each zero graph, fixed-core relative bounds give its planar deficit times $1 + O(R^{-\alpha})$. Replace each graph by an inward parallel tangent plane and subtract the error $O(\kappa R^2)$. On lateral faces, use sign-ball clearing with a larger fixed coefficient C_i. The reaction is increasing in this well range, so comparison with $F(V)$ is valid. Perform it on $\{V < v_*\}$ and choose the clearing threshold below $F(v_*)$. The supersolution already dominates on the added boundary $V = v_*$. This avoids conditions on sums near corners. Choose the horizontal buffer much larger than the maximal depth, which is a small fixed multiple of R. Every lateral exponential, divided by the own-face tail, is then $O(e^{-cR})$ in the comparison box. The own-face coefficient is $1 + O(R^{-\alpha} + \kappa R^2)$. The ratio of the opposite term to it is $O(e^{-2a\delta})$. The lateral terms have relative size $O(e^{-cR})$. This proves the upper bound. Modica's inequality

along the segment from the lower zero gives $u \le g(s)$ in vertical coordinates. This proves the lower bound up to the recorded affine error. The artificial phase is $s + \psi$ with $\|\psi\|_{C^1} \le C\kappa R$. Its local Schauder bounds of fixed order are $C_N \kappa^{\vartheta} R^N$. To derive these from $W^{2,q}_{\mathrm{uloc}}$, interpolate the small C^2 graph norm with the bounded higher core norms. Then apply noncritical Schauder estimates to the artificial phase equation. No derivative estimate at an actual turn enters (6.33).

The reason this comparison preserves the exact own-face coefficient is the identity $1 - g(s) = 2v/(1+v)$ for $v = e^{-2as}$. If the barrier sum is $V = v(1+\xi)$, then

$$\frac{2V/(1+V)}{2v/(1+v)} - 1 = \frac{\xi}{1 + v(1+\xi)}.$$

Here $\xi = O(\varepsilon_0 + e^{-2a\delta(y,s)})$ after the buffer estimate. Thus there is no additional fixed error from replacing $1 - g(s)$ by its leading exponential. This distinction matters when a relative error is integrated over a long gap.

Below the middle band, compare the actual solution with an artificial positive solution v^* over the lower graph. Write $h = 1 - v^*$ and $z = v^* - u = w - h$. Set

$$F_* = (aG(s))^{(p-2)/2} z.$$

Flattening the base gives a nondivergence equation $\mathcal{L}F_* = 0$. In the region $\delta \ge M_0$ its coefficients satisfy

$$\mathcal{L} = -a_{ij}\partial_{ij} + b_i \partial_i + V, \qquad |a_{ij} - a^0_{ij}| + |b_i| + |V - V_0(s)| \le C\{\varepsilon_0 + e^{-2a\delta}\}, \tag{6.34}$$

where

$$a^0 = \mathrm{diag}(I, c), \qquad V_0(s) = ca^2[p^2 - p(p+1)G(s)].$$

Ellipticity and the local Hölder norms are uniform.

We justify (6.34) without differentiating an uncontrolled secant matrix. Equation (6.33), the Modica bound, and $|\nabla h| \ge ch$ show that the divergence secant matrix divided by the local value of h^{p-2} is uniformly elliptic and Hölder on fixed balls. Write the endpoint gradients as ξ, η and put $m_* = \max\{|\xi|, |\eta|\}$. One endpoint has length at least ch, and both have length at most Ch, so $m_* \asymp h$. Reverse the parameter if necessary so that $|\xi| = m_*$. For $0 \le t \le 1/4$,

$$|\xi + t(\eta - \xi)| \ge m_* - t(|\xi| + |\eta|) \ge m_*/2.$$

The standing range here is $\mathcal{P} \Subset (2, \infty)$. Since the whole segment has length at most m_*, this gives

$$\tfrac{1}{4}(m_*/2)^{p-2} \le \int_0^1 |\xi + t(\eta - \xi)|^{p-2}\, dt \le m_*^{p-2}.$$

The eigenvalues of $A_p(v)$ lie between $|v|^{p-2}$ and $(p-1)|v|^{p-2}$. Integration therefore bounds the secant matrix between $ch^{p-2}\,\mathrm{Id}$ and $Ch^{p-2}\,\mathrm{Id}$, uniformly on $\mathcal{P}$. The estimate uses a quarter of the segment near its larger endpoint. It remains valid when another part of the segment passes through zero. Normalized nonlinear $C^{1,\eta}$ estimates control the Hölder norm, even if the segment passes through zero. The local linear gradient estimate for z strengthens (6.33) to

$$|z| + |\nabla z| \le Ch\{\varepsilon_0 + e^{-2a\delta}\}.$$

Choose M_0 large. The actual gradient is then nonzero in the cropped half-gap. Keep $A_p(\nabla w)$ as the principal matrix and use the exact identity

$$(A_p(\nabla w) - A_p(\nabla h)) : D^2 h = \left(\int_0^1 DA_p(\nabla h + \theta \nabla z)\, d\theta : D^2 h \right) \cdot \nabla z.$$

All gradients in this integral are nonzero. The drift contains only $D^2 h$, whose relative bounds are known. Divide by $(aG)^{p-2}$ and conjugate to obtain (6.34). The flat reference becomes exactly $-\Delta_y - c\partial_s^2 + V_0$. Neither $D^2 w$ nor the divergence secant matrix has been differentiated. On the initial core $0 \le s \le S + 2$, use the relative phase bounds and their fixed-core boundary estimates

directly. They give the same coefficient error $O(\varepsilon_0)$. The well comparison starts at S. It is therefore never applied at $q = 1$ or outside the range where the reaction is increasing.

For an explicit check of the conjugation, the flat linearized equation for ζ after division by $(aG)^{p-2}$ is

$$-\Delta_y\zeta - c\zeta_{ss} + 2ca(p-2)g\,\zeta_s + a^{2-p}((2p-1)g^2-1)\zeta = 0.$$

Put $r = p-2$ and $\zeta = (aG)^{-r/2}F$. The logarithmic derivative of the prefactor is rag. The first derivative terms in F cancel, and the new potential is

$$\begin{aligned} ca^2(r^2g^2 - rG) + a^{2-p}((2p-1)g^2-1) &= ca^2((p^2+2)g^2 - (p-2)G - 2) \\ &= ca^2(p^2 - p(p+1)G). \end{aligned}$$

We used $a^{2-p} = 2ca^2$ and $g^2 = 1-G$. This is precisely V_0, including its coefficient at the bottom of the well.

At the bottom, Fermi coordinates provide an additional bound:

$$|a_{ys}(y,s)| \le C\varepsilon_0 s + Ce^{-k\ell}s \qquad (0 \le s \le 1). \tag{6.35}$$

We verify this bound on the whole interval $0 \le s \le 1$. Fix a relative-regularity window of radius three before reducing the defect threshold. At a base $z = z(y)$ put $N = \nu(z)$ and $D_0 = D(z)$. Lemma 2.12, in particular (2.16) and (2.18), gives

$$Z(z+sN) = N + O(D_0), \qquad |D^2U(z+sN)N| \le CD_0 \quad (0 \le s \le 1).$$

Choose the threshold below d_3 and so that $C_{3,0}D_0 < 1/2$. Thus all coefficients and the fixed-order derivatives used here are uniformly bounded on this fixed core. For a unit base tangent vector τ, set

$$V_\tau(s) = d_y[U(z(y) + sN(y))][\tau], \qquad \mathcal{B}\tau = d_yN[\tau].$$

Here $|\mathcal{B}| = |B|$ regardless of the sign convention for B. Since $U = 0$ on the base, $V_\tau(0) = 0$. Differentiation gives

$$V_\tau'(s) = D^2U[N, \tau + s\mathcal{B}\tau] + Z\cdot\mathcal{B}\tau.$$

The first term is $O(D_0)$ on this interval. In the second term, $N\cdot\mathcal{B}\tau = 0$ and $Z - N = O(D_0)$, so it too is $O(D_0)$. Hence $|V_\tau(s)| \le CD_0 s$ and $|V_\tau'(s)| \le CD_0$. The Fermi tangent metric and its inverse are uniformly bounded on the fixed core. The tangential component of Z in these coordinates therefore satisfies the same estimates.

The principal matrix retained in (6.34) is $A_p(\nabla w)/(aG)^{p-2}$. In Fermi coordinates its isotropic part has no mixed normal-tangent entries. The remaining mixed entries are bounded by a uniform constant times the tangential component of Z. The scalar factors are uniformly bounded since $r \ge 1/2$ and q/G is bounded above and below on this core. The fixed-core estimates also bound their normal derivatives. Consequently $|a_{ys}| \le CD_0 s$ and $|\partial_s a_{ys}| \le CD_0$ for $0 \le s \le 1$. The clean-ball bound gives $D_0 \le CR^{-\alpha} \le C\varepsilon_0$. This proves (6.35), including its stated weaker exponentially small allowance, throughout the interval. Retain these exact Fermi coordinates wherever (6.35) is used. Elsewhere use (6.34). The bulk errors from passing between true tangent distances, vertical graph coordinates, and Fermi coordinates are bounded by $C_N\kappa^{\vartheta}R^N$ and already belong to ε_0. They are not substituted for the mixed-entry bound at the bottom and are not multiplied by an unrecorded normal depth.

To construct an upper barrier with the exact exponential rate, let Y_b solve the tilted normal equation

$$\{-c\partial_s^2 + V_0(s) - \beta^2|b|^2\}Y_b = 0, \qquad Y_b(0) = 0, \quad Y_b'(0) = 1. \tag{6.36}$$

For a sufficiently small fixed tilt, uniformly on $\mathcal{P}$,

$$Y_b > 0, \quad Y_b' > 0, \quad Y_b(s) \asymp e^{\mu_b s}\ (s \ge 1), \quad \mu_b = \beta\sqrt{1 - |b|^2/c} \ge \beta t, \quad c_0 \le Y_b'/Y_b \le C(1 + s^{-1}). \tag{6.37}$$

Since $c = p - 1 \geq 1$ and $t^2 = 1 - |b|^2$, the stated rate obeys $\mu_b \geq \beta t$. This comparison includes $p = 2$, where equality holds. For $1 < p < 2$ and sufficiently small nonzero tilt, its direction reverses. The positive half-line Dirichlet gap is $ca^2(2p-1)$. Hence Y_b has no positive zero if $\beta^2|b|^2$ is small. To check the derivative at $b = 0$, use $Y_0 = M\int_0^s M^{-2}$ with its fixed normalization. For $x = as$, positivity of the derivative is equivalent to

$$\int_0^x \cosh^{2p} t\, dt < \frac{\cosh^{2p} x}{p \tanh x}.$$

The difference has derivative $\cosh^{2p} x[1 - 1/(p \sinh^2 x)]$. Its minimum occurs at $x_* = \operatorname{arsinh}(p^{-1/2})$. There, the integral is at most $x_* \cosh^{2p} x_*$, while $x_* < \sqrt{p+1}/p = 1/(p \tanh x_*)$. Therefore $Y_0' > 0$. Continuous dependence on compact intervals and the exponentially convergent tail equation give (6.37) for small tilt. At $b = 0$, the growing exponential has positive coefficient. This coefficient stays bounded away from zero. All choices are uniform on $\mathcal{P}$.

The reference equation has the exact solution

$$F_0(y, s) = e^{-\beta\ell + \beta b\cdot y} Y_b(s)$$

Fix $0 < k < 2a_{\min}$. In the region $\delta \geq M_0$, put

$$w_*(y, s) = -K\varepsilon_0 s - Ke^{-k\delta(y,s)}, \qquad B_* = F_0 e^{w_*}.$$

Choose K first, then M_0, then the large comparison scale. These choices give

$$\mathcal{L}B_* \geq 0, \qquad C^{-1}F_0 \leq B_* \leq F_0. \tag{6.38}$$

The comparison constant is independent of ℓ and R whenever $\varepsilon_0 R \to 0$ and $s \leq CR$.

To check the sign, write $v = (b, 1 + t)$ and $h_b = Y_b'/Y_b$. Then

$$\nabla w_* = -K\varepsilon_0 e_s - Kke^{-k\delta} v, \qquad D^2 w_* = -Kk^2 e^{-k\delta} v \otimes v.$$

The reference operator applied to the barrier, divided by B_*, gives

$$\begin{aligned} &Kk^2 e^{-k\delta}\{|b|^2 + c(1+t)^2\} + 2Kc\varepsilon_0 h_b \\ &\quad + 2Kke^{-k\delta}\{\beta|b|^2 + c(1+t)h_b\} - a^0[\nabla w_*, \nabla w_*]. \end{aligned}$$

The positive terms sum to at least $c_1 K(\varepsilon_0 + e^{-k\delta})(1 + h_b)$. The last term is quadratic in $K(\varepsilon_0 + e^{-k\delta})$. Equation (6.34) bounds the perturbations by

$$C(\varepsilon_0 + e^{-2a\delta})(1 + h_b) + CK(\varepsilon_0 + e^{-k\delta})^2(1 + h_b).$$

This also covers mixed derivatives of F_0 near $s = 0$. Indeed, Y_b''/Y_b is bounded. The only unbounded quotient is $h_b = O(s^{-1})$. The stated order of choices proves (6.38).

At the cropped top $\delta = M_0$, (6.33) bounds the absolute conjugated data by $Ce^{-\beta s}$. There we also have

$$F_0 \geq ce^{-\beta M_0} e^{-\beta s} e^{(\mu_b - \beta t)s} \geq ce^{-\beta M_0} e^{-\beta s}.$$

A fixed multiple of B_* therefore dominates the top data. For the lateral data, use v_0 from (6.18). It satisfies $(-c\partial_s^2 + V_0)v_0 = 1$ and has bounded derivatives and a positive lower bound. For a disk of radius R, take the smooth horizontal distance $l_{\text{side}}(y) = \sqrt{1 + R^2} - \sqrt{1 + |y|^2}$. Its first two derivatives are uniformly bounded. Thus $v_0(s)e^{-\zeta l_{\text{side}}}$ is a supersolution for a small fixed $\zeta > 0$. Division by v_0 gives the maximum principle for $\mathcal{L}$ on the cropped domain.

Split the linear Dirichlet solution into its top and side responses, with zero data on the bottom in both problems. Smoothly assign the corner data to the side response in a fixed lateral collar. The side barrier bounds that response by Ce^{-cR} on a fixed bottom neighborhood. Since the response vanishes on the bottom, local boundary gradient estimates give the same bound for its bottom derivative. For the top response, comparison with both signs of a fixed multiple of B_* is valid.

Both the response and B_* vanish at the bottom, so the one-sided derivative comparison gives the top contribution $Ce^{-\beta\ell}$. Consequently,

$$|\partial_n(v^* - u)(0)| \le Ce^{-\beta\ell} + Ce^{-cR}. \tag{6.39}$$

For a side without a nearby opposite sheet, use the following empty-chamber comparison. Choose fixed $r_e = c_eR$ and $L_e = \tau_eR$, with $r_e \ge 10L_e$. Choose them inside the prepared graph buffer and with $L_e + 2$ below the common artificial top. The empty-chamber hypothesis is that the own graph is the only zero sheet in

$$\{|y| < 2r_e,\ 0 \le s \le 4L_e\},$$

where s is vertical depth above that graph. The fixed fractions used below give exactly this hypothesis. After reversing the sign on a negative chamber, $0 < u < 1$ there. Let v^* be the artificial one-sided solution on the same own graph and put

$$\mathcal{D}_e = B^m_{r_e} \times (0, L_e), \qquad w = 1 - u, \quad h = 1 - v^*, \quad z = w - h.$$

Flatten only the own graph, as in the exact divergence pullback used for (4.24). In these coordinates $z = 0$ at $s = 0$. The artificial solution is constructed in its deeper constant slab before restriction to $\mathcal{D}_e$.

Every point of this retained domain at depth s has distance at least cs from the zero set. The own graph has small slope. Other zeros in the larger comparison window lie either below it or above depth $4L_e$. Zeros outside that window have horizontal or vertical distance at least a fixed multiple of R. The two buffers and $r_e \ge 10L_e$ give the stated lower bound. Choose one fixed large S from one-sided clearing. Then both w and h lie in the same fixed convex well range for $s \ge S$. On the fixed strip $0 \le s \le 2S$, clean-ball preparation gives C^1 convergence to the own planar layer, uniformly in the horizontal base. The artificial estimates give the same convergence for v^*. Choose this fixed strip before its preparation threshold.

The secant equation on $\mathcal{D}_e$ is

$$-\operatorname{div}(\mathbb{M}\nabla z) + bz = 0, \qquad \mu = (w + h)^{p-2}.$$

Its coefficients are the gradient and reaction secants already used above, with the exact own-graph pullback understood. Modica's bound and the artificial phase bound give $|\nabla w| \le Cw$ and $|\nabla h| \le Ch$. Consequently $\mathbb{M} \le C\mu I$ and $|b| \le C\mu$. In the fixed convex well, $b \ge c\mu$. The fixed bottom-strip gap and this well bound are precisely the hypotheses proved for (4.24). Thus

$$Q_e(\zeta) := \int_{\mathcal{D}_e} (\mathbb{M}[\nabla\zeta, \nabla\zeta] + b\zeta^2) \ge \lambda_0 \int_{\mathcal{D}_e} (\mathbb{M}[\nabla\zeta, \nabla\zeta] + \mu\zeta^2)$$

for zero-boundary form tests, with $\lambda_0 > 0$ independent of R. This step uses no lower ellipticity bound for the actual solution on the growing well region.

The full natural energy of the actual difference is at most CR^{m+1}. Indeed, $|z| \le w + h$ and the preceding gradient bounds give pointwise

$$\mathbb{M}[\nabla z, \nabla z] + \mu z^2 \le C(w + h)^p \le C.$$

For a cutoff η that vanishes near the top and lateral faces, ηz also has zero bottom trace. Testing the weak secant equation gives the exact identity

$$Q_e(\eta z) = \int_{\mathcal{D}_e} z^2 \mathbb{M}[\nabla\eta, \nabla\eta].$$

Let $E(d)$ be the natural energy on $\{|y| < r_e - d,\ 0 < s < L_e - d\}$. Choose a cutoff that is one on this set and zero outside the corresponding set with d replaced by $d - \Lambda$, with $|\nabla\eta| \le C/\Lambda$. Coercivity and the upper coefficient bound give

$$E(d) \le C\Lambda^{-2}E(d - \Lambda).$$

Fix Λ so that $C\Lambda^{-2} \le 1/2$. There are at least cR such fixed-width layers between the top or side and a fixed bottom neighborhood of the origin. Iteration therefore gives natural energy at most $CR^{m+1}e^{-c_1R}$ in that neighborhood. There the fixed-core coefficients are uniformly elliptic and Hölder, the weight μ has common positive bounds, and z has zero trace on the smooth bottom. Local boundedness followed by the boundary gradient estimate used in (4.29) yields

$$|\partial_n(v^* - u)(0)| \le CR^{(m+1)/2}e^{-c_1R/2} \le Ce^{-c_e'R}.$$

This proves the claimed coarse Ce^{-cR} contribution after fixing a smaller positive decay rate. It requires no derivative bound at an actual turn or at the growing top. All fractions, the well threshold, the cutoff width and the final decay rate are fixed before the later choice of A_0 and before R tends to infinity.

For the reverse bound, a positive datum is needed near the middle of the gap. Assume $|b|^2\ell \le C_0$ and put $s_c(y) = (\ell - b\cdot y)/(1+t)$. On $|y| \le C\sqrt{\ell}$, the difference $s_c(y) - s_c(0)$ is bounded uniformly. At the cropped top $s = s_c(y) - M_0/(1+t)$, a constant $c_* > 0$ satisfies

$$F_*(y,s) \ge c_*e^{-\beta s} \qquad (|y| \le C\sqrt{\ell}) \tag{6.40}$$

for every sufficiently large gap. This constant may depend on M_0, but not on the gap.

Apply the fixed-band turn preparation in Lemma 3.11 at each local affine bisector in this footprint. Its hypotheses hold uniformly: $|b|^2\ell \le C_0$ gives $b \to 0$, the graph errors tend to zero, and $\varepsilon_0\ell \to 0$ follows from $\ell \le CR$ and $\varepsilon_0 R \to 0$. The polynomial footprint stays inside the buffered chamber. With $h_0 = 2e^{-2as_c(y)}$, that lemma gives the normalized profile $V_p(\tau)/C_p^\infty$, where $\tau = s - s_c(y)$. The cropped top offsets $\tau = -M_0/(1+t)$ lie in one fixed compact interval. The unit bisector-normal coordinate differs from τ by $o(1)$ on these fixed bands because $b \to 0$. The artificial own-face deficit divided by h_0 is $e^{-2a\tau}(1+o(1))$ there.

The first integral gives $V_p'/V_p < 2a$ at every finite positive depth. Thus $e^{-2ar}V_p(r)$ strictly decreases to C_p^∞ as $r \to \infty$, and

$$\frac{V_p(\tau)}{C_p^\infty} > e^{2a|\tau|} \ge e^{-2a\tau}$$

at every finite offset, using evenness. Continuous dependence on p gives a uniform positive gain on the fixed cropped-offset interval. Choose the turn-preparation tolerance smaller than this gain. On the cropped band, $s - s_c(y)$ stays in a fixed bounded interval. Thus

$$h_0(aG(s))^{(p-2)/2} \asymp e^{-2as}e^{-(p-2)as} = e^{-\beta s}.$$

Multiplying the strict deficit gain by this factor gives (6.40), with a constant independent of the gap.

Elsewhere on the cropped top we retain

$$(F_*)_- \le C\varepsilon_{\mathrm{geom}}e^{-\beta s}, \qquad \varepsilon_{\mathrm{geom}} \longrightarrow 0. \tag{6.41}$$

In Fermi coordinates, use the true normal segment of length s from the actual bottom zero. Modica gives $U(x) \le s$, and hence $u(x) \le g(s)$. The artificial phase is $s + \psi$, with $\|\psi\|_\infty \le \varepsilon_{\mathrm{geom}}$ after the recorded coordinate error. Boundedness of the logarithmic derivative of G gives $|g(s+\psi) - g(s)| \le C\varepsilon_{\mathrm{geom}}G(s)$. Therefore

$$(F_*)_- \le C\varepsilon_{\mathrm{geom}}(aG(s))^{(p-2)/2}G(s) \le C\varepsilon_{\mathrm{geom}}e^{-\beta s},$$

which proves (6.41).

Transmit this positive datum to the core on a cylinder whose radius is of order the square root of the gap. For the fixed linear operator, split F_* into the solutions with positive top, negative top, and side data. Give all three zero bottom data. The first solution is nonnegative. Equations (6.38) and (6.41) bound the absolute bottom slope from negative top data by $C\varepsilon_{\mathrm{geom}}e^{-\beta\ell}$. The side contribution is Ce^{-cR}. To make the data compatible at corners, cut off the top data smoothly in

a fixed side collar. Assign the remainder to the side solution. This leaves the central footprint and its side-decay estimate unchanged.

Restrict the positive-top solution to a Fermi cylinder with radius $R_* = c_*\sqrt{\ell}$ and height $L = s_c(0) - C_*$. Choose the fixed number C_* large. Its top then lies a bounded positive distance below the cropped top throughout a slightly larger footprint. We first turn (6.40) into an interior lower bound. The Dirichlet splitting above gives $\mathcal{L}F_+ = 0$ and $F_+ \geq 0$. Choose c_* smaller if needed so that a fixed-size boundary patch above every $|y| \leq R_*$ stays in the larger strict-top footprint. The low-tilt bound makes $s_c(y) - s_c(0)$ uniformly bounded there. After fixing C_*, the face $s = L$ is therefore between two fixed positive distances from the cropped top. Its boundary datum is at least $A = ce^{-\beta L}$ on each patch. The recorded graph-jet bounds give uniformly bounded C^2 coordinates which flatten that top to $d = 0$, with the chamber in $d > 0$. A fixed rescaling makes the patch contain

$$\mathcal{C} = B_2(X_{\text{out}}) \cap \{d > 0\}, \qquad X_{\text{out}} = (0, -1).$$

The pulled-back operator $\widehat{\mathcal{L}}$ is uniformly elliptic and has uniformly bounded lower-order coefficients. Put

$$\mathfrak{b}(X) = \frac{e^{-K|X-X_{\text{out}}|^2} - e^{-4K}}{e^{-K} - e^{-4K}}.$$

On $\mathcal{C}$, one has $1 < |X - X_{\text{out}}| < 2$. Differentiation and ellipticity give

$$\widehat{\mathcal{L}}\mathfrak{b} \leq \frac{e^{-K|X-X_{\text{out}}|^2}}{e^{-K} - e^{-4K}}[-4\lambda_{\text{ell}}K^2 + C(K+1)] \leq 0$$

for one sufficiently large fixed K, where $\lambda_{\text{ell}} > 0$ is the common ellipticity constant. The function $\mathfrak{b}$ vanishes on the spherical boundary and lies between zero and one on $d = 0$. The positive-gauge maximum principle already proved for $\mathcal{L}$ permits comparison with $A\mathfrak{b}$. It gives $F_+ \geq A\mathfrak{b}$, and hence $F_+ \geq cA$ on a fixed interior ball around $(0, 1/2)$. This is the interior starting ball for Harnack's inequality. A uniformly bounded chain of fixed balls connects it to the point on $s = L$ below the patch. The fixed positive margin keeps this chain inside the chamber. Consequently

$$F_+ \geq ce^{-\beta L} \quad \text{on } |y| \leq R_*,\ s = L.$$

The patch sizes, K and the chain length are fixed before the gap grows and are uniform on $\mathcal{P}$. The $o(1)$ error in the transformed top fits inside these fixed margins. The low-tilt condition also gives $\ell - 2s_c(0) = \ell(t-1)/(1+t) = O(|b|^2\ell) = O(1)$. Thus the scales satisfy

$$R_*^2 \asymp L \asymp \ell, \qquad |\ell - 2L| \leq C. \tag{6.42}$$

In this cylinder the coefficient error in (6.34) is bounded by

$$\varepsilon(s) = C\varepsilon_0 + Ce^{-k_0(L-s)-k_0M_0}, \quad \varepsilon_0 L \to 0,$$

The mixed coefficient also vanishes at the bottom as in (6.35).

Use $Y = Y_0$ and define $w(0) = 0$ by

$$w'(s) = KY(s)^{-2}\int_0^s Y(t)^2\{\varepsilon(t)(1 + Y'(t)/Y(t)) + L^{-1}\}\,dt.$$

Then

$$w'' + 2(Y'/Y)w' = K\{\varepsilon(1 + Y'/Y) + L^{-1}\}, \quad 0 \leq w' \leq CK(\varepsilon + L^{-1}), \quad 0 \leq w \leq C_K.$$

The bound for the correction follows directly from the integral. The error function is nondecreasing, and

$$\int_0^s Y(t)^2\frac{Y'(t)}{Y(t)}\,dt = \frac{Y(s)^2}{2}, \qquad Y(s)^{-2}\int_0^s Y(t)^2\,dt \leq C.$$

The second estimate uses the linear behavior at zero and exponential growth beyond one. Thus the right derivative of the correction is finite at zero. Integration gives

$$w(L) \le CK\left(\varepsilon_0 L + \int_0^L e^{-k_0(L-s)-k_0M_0}\,ds + 1\right) \le C_K.$$

The fixed crop makes the normal-error integral bounded independently of the gap.

Let $\phi(y) = (1 - |y|^2/R_*^2)_+^k$ with an integer $k \ge 2$. Once the fixed K is large enough,

$$B_-(y,s) = c_2 e^{-\beta\ell} Y(s) e^{w(s)} \phi(y) \quad \text{satisfies} \quad \mathcal{L}B_- \le 0. \tag{6.43}$$

Choose $c_2 > 0$ after K to fit the top data.

The tangential cutoff introduces no extra power of L. With $d_y = R_*^2 - |y|^2$, its Hessian is exactly

$$\frac{\phi_{ij}}{\phi} = -\frac{2k\delta_{ij}}{d_y} + \frac{4k(k-1)y_iy_j}{d_y^2}.$$

Uniform ellipticity and Young's inequality bound its Hessian and first-order terms above by

$$CR_*^{-2} + C\{|a_{ys}|(Y'/Y + w') + |b_y|\}^2.$$

To justify this estimate also near the side boundary, let $q_s = Y'/Y + w'$ and $v_y = b_y - 2a_{ys}q_s$. The tangential part of $\mathcal{L}B_-/B_-$ is

$$\frac{2k\,\mathrm{tr}(a_{yy})}{d_y} - \frac{4k(k-1)a_{yy}[y,y]}{d_y^2} - \frac{2kv_y\cdot y}{d_y}.$$

For $|y| \le R_*/2$ its upper bound is $CR_*^{-2} + C|v_y|^2$, by Young's inequality. For $|y| \ge R_*/2$, ellipticity gives a negative term $-cR_*^2/d_y^2$. A fixed fraction of this term absorbs both C/d_y and $C|v_y|R_*/d_y$, leaving the same upper bound. This proves the displayed cutoff estimate throughout the open disk. Only continuity and zero boundary values of the cutoff are needed at its boundary; its derivatives are used in the open cylinder.

The negative radial-Hessian term controls the singular cutoff quotients near $|y| = R_*$. For $s \le 1$, bottom vanishing of the mixed coefficient bounds its product with $Y'/Y = O(s^{-1})$ by $C\varepsilon(s)$. For $s \ge 1$ use (6.34) directly. The other normal and potential errors are at most $C\varepsilon(s)(1 + Y'/Y)$. The equation for w supplies the negative term

$$-c_3K\{\varepsilon(s)(1 + Y'/Y) + L^{-1}\}.$$

More explicitly, the normal and potential part of $\mathcal{L}B_-/B_-$ is

$$\begin{aligned}(V - V_0) - (a_{ss} - c)Y''/Y + b_s(Y'/Y + w')\\ - a_{ss}K\{\varepsilon(1 + Y'/Y) + L^{-1}\} - a_{ss}(w')^2.\end{aligned}$$

This identity keeps the favorable square with its actual positive coefficient a_{ss}. The first line is bounded above by $C\varepsilon(1 + Y'/Y) + CK\varepsilon(\varepsilon + L^{-1})$. Choose K to dominate the fixed linear errors, and then enlarge the crop and the gap so that the remaining coefficient error is small. No estimate of w'' by itself is needed near $s = 0$.

Since $R_*^{-2} \asymp L^{-1}$, the same choice also absorbs the tangential cutoff cost. This proves (6.43).

At the top, (6.42) gives $e^{-\beta\ell}Y(L)e^{w(L)} \le Ce^{-\beta L}$. The barrier vanishes on the side and bottom. Compare it with F_+ and take the bottom normal derivative. We obtain

$$\partial_n(v^* - u)(0) \ge ce^{-\beta\ell} - C\varepsilon_{\mathrm{geom}}e^{-\beta\ell} - Ce^{-cR}. \tag{6.44}$$

The constants do not depend on the reach. They may depend on the fixed bound C_0 in the low-tilt condition.

We use a geometric observation before applying the comparison estimates. Let $m \ge 1$, $r > 0$, and write $x = (y,s) \in \mathbb{R}^m \times \mathbb{R}$. Put $\Pi_0 = \{s = 0\}$ and $\pi_0(y,s) = y$. Let Σ be a relatively closed

C^2 embedded hypersurface without boundary in $B_{100r}(0)$. Assume that every tangent plane of Σ makes unsigned angle at most $\arctan(1/20)$ with Π_0. Define

$$\mathcal{K}_{\text{big}} = \{|y| < 2r,\ |s| < 2r\}, \qquad \mathcal{K}_{\text{ret}} = \{|y| < r/4,\ |s| < r/4\}.$$

For every $w \in \Sigma \cap \mathcal{K}_{\text{big}}$ there is a C^2 function f_w on B^m_{4r} whose graph lies in $\Sigma \cap B_{15r}$, contains w, and has $|Df_w| \le 1/20$. Identify functions that agree on the whole disk. The resulting family is finite and strictly ordered by height, say $f_1 < \cdots < f_N$, with $N \ge 0$. It covers every point of $\Sigma \cap \mathcal{K}_{\text{ret}}$. For each adjacent pair, the part of $\{(y, s) : f_i(y) < s < f_{i+1}(y)\}$ inside $\mathcal{K}_{\text{ret}}$ is disjoint from Σ. If $N \ge 1$, the same is true inside $\mathcal{K}_{\text{ret}}$ above f_N and below f_1. No uniform bound on N is asserted.

For a tangent vector (v_y, v_s), the angle assumption gives $|v_s| \le |v_y|/20$. Thus $d\pi_0|_{T\Sigma}$ is invertible and its inverse has operator norm at most $\sqrt{1 + 20^{-2}} < 2$.

Fix $w \in \Sigma \cap \mathcal{K}_{\text{big}}$ and $y \in B^m_{4r}$. Let $h = y - \pi_0 w$. Lift the segment $\pi_0 w + th$, $0 \le t \le 1$, by the tangent vector field

$$X_h(\xi) = \left(d\pi_0|_{T_\xi \Sigma}\right)^{-1} h.$$

This field is C^1 on the C^2 hypersurface. Local existence and uniqueness for its ordinary differential equation therefore apply. Its speed is at most $2|h| \le 12r$. Since $|w| < 3r$, the lift remains in B_{15r}. If it stopped before time one, its bounded speed would give an ambient limit in that ball. Relative closedness puts the limit on Σ, and the local inverse chart continues the lift. Thus the lift exists through time one.

The map $(\xi, h) \mapsto X_h(\xi)$ is jointly C^1 in local coordinates on Σ, since its inverse tangent projection is C^1 and its dependence on h is linear. Fix a target y_0. Its lifted trajectory on $[0, 1]$ is a compact subset of $\Sigma \cap B_{15r}$. Apply the continuous-dependence theorem for ordinary differential equations with a jointly C^1 parameter field. A finite chain of coordinate charts along this trajectory applies that theorem successively on $[0, 1]$. Since the initial point w is fixed and $h = y - \pi_0 w$ depends continuously on y, the endpoint is continuous at y_0. Its projection is y. Choose an inverse chart of $\pi_0|_\Sigma$ at the endpoint for y_0. Endpoint continuity keeps all nearby endpoints in this chart. Their projections determine them uniquely there, so the endpoint section equals this local inverse. The inverse chart is C^2. Since y_0 was arbitrary, the endpoints form a C^2 graph $s = f_w(y)$ on B^m_{4r}. It contains w. The tangent-angle bound gives $|Df_w| \le 1/20$.

If two such graphs meet, the embedded hypersurface has one local inverse chart at the meeting point. Their height functions agree on a neighborhood. Their equality set is also closed in the common disk. Connectedness of the disk implies that they agree everywhere. Distinct graphs never meet. Their height difference has one sign on the connected disk, so the distinct graphs are strictly ordered.

Each graph has an axis height in $(-3r, 3r)$. Indeed, its starting point satisfies $|w_s| < 2r$, $|\pi_0 w| < 2r$, and

$$|f_w(0) - w_s| \le \frac{1}{20}|\pi_0 w| < \frac{r}{10}.$$

The intersection of Σ with the compact axis segment $\{0\} \times [-3r, 3r]$ is compact. Every point of this intersection is isolated on the axis, because the tangent-plane projection is invertible. An infinite sequence of distinct axis points would have an accumulation point there, contradicting its local inverse chart. Hence the axis intersection is finite. Distinct graphs have distinct axis heights, so the graph family is finite. This argument gives no uniform sheet count.

Every point of $\Sigma \cap \mathcal{K}_{\text{ret}}$ was one of the allowed starting points in $\mathcal{K}_{\text{big}}$. Thus it lies on a member of this family. A point of $\Sigma \cap \mathcal{K}_{\text{ret}}$ strictly between two adjacent members would generate a further graph between them, which contradicts adjacency. The same coverage excludes a point in this

box above the highest member or below the lowest member. These are only assertions inside the retained box.

Return to the matched solution and its zero set.

Verify the geometric hypotheses of these comparisons, then combine the two sides of the base sheet. The geometric fractions used below are chosen first. In particular, choose a small graph fraction $\eta_{\mathrm{gr}} > 0$, then $0 < \eta_{\mathrm{int}} < \eta_{\mathrm{gr}}/1000$, and a common artificial slab fraction θ with $2\eta_{\mathrm{int}} < \theta < \eta_{\mathrm{gr}}/32$. Fix the comparison rates after these choices. Put $B = J_0(z) + E(z)$ and $R = A_0 \log(1/B)$. Choose A_0 large after the fixed comparison rates, and then fix λ small enough for Proposition 6.10 and $10\lambda A_0 < 1/2$. Only after these choices do we take B small. There are then no bad zeros in $B_{10R}(z)$, and every zero there has curvature at most $B^{1/4}$. All zeros in this ball are regular because $D < \delta < 1$, and the zero set is relatively closed there. Translate z to zero and rotate its normal into e_n. Use the fixed base plane $\Pi_0 = \{s = 0\}$, with orthogonal projection $\pi_0(y, s) = y$. Put

$$r_{\mathrm{amb}} = R/4, \qquad r_{\mathrm{big}} = \eta_{\mathrm{gr}} R, \qquad \kappa = B^{1/4}.$$

Lemma 3.10 applies at the origin with radius r_{amb}, because its $20r_{\mathrm{amb}} = 5R$ buffer lies in B_{10R}. Choose η_{gr} so that $100r_{\mathrm{big}} < r_{\mathrm{amb}}$. For every zero $w \in B_{100r_{\mathrm{big}}}$, the same buffer lies in $B_{5R}(w) \subset B_{21R/4}(0) \subset B_{10R}(0)$. The angle estimate of that lemma then gives, at every zero $w \in B_{100r_{\mathrm{big}}}$,

$$\angle(T_w\Sigma, \Pi_0) \le C\big(100r_{\mathrm{big}}/r_{\mathrm{amb}} + \kappa r_{\mathrm{amb}}\big).$$

Choose η_{gr} small first and then B small. This angle is below a fixed comparison angle whose tangent is less than $1/20$. In particular the inverse of $d\pi_0|_{T_w\Sigma}$ has operator norm at most two throughout this region.

Apply the common-plane observation with $r = r_{\mathrm{big}}$ and the fixed plane Π_0. The zero set is relatively closed in this ball. At each zero in the larger regular buffer, $u = 0$ and $|\nabla u| > 0$. The matched reaction is smooth on a neighborhood of this value. The local noncritical bootstrap from Section 2.1 therefore gives $u \in C^2$ near that zero. The implicit function theorem makes the zero set a C^2 embedded hypersurface without boundary throughout $B_{100r_{\mathrm{big}}}$. No uniform higher-derivative bound is needed for this geometric application. The preceding angle bound verifies the remaining hypothesis of the observation. Use its boxes $\mathcal{K}_{\mathrm{big}}, \mathcal{K}_{\mathrm{ret}}$ with this value of r. Its graphs $s = f_w(y)$ form a finite ordered family on $B_{4r_{\mathrm{big}}}$ and cover every zero in the retained box. This organizes the projection-extension argument already used in the proof of Lemma 3.10. For a compared chamber whose two boundary graphs and vertical segments lie in $\mathcal{K}_{\mathrm{ret}}$, the zero-free interior fixes the sign of the solution. The regular crossings therefore give opposite phase orientations on its two boundary graphs. If the ordered family has no member on one side, the same observation excludes zeros on that side inside the retained box. The comparisons and the common artificial depth $L = \theta R$ lie strictly inside its vertical buffer. The nearby intercept cutoff is $\eta_{\mathrm{int}} R$, which is smaller still. Zeros outside $\mathcal{K}_{\mathrm{big}}$ are a fixed multiple of R from the smaller comparison boxes.

The curvature formula for a graph of slope at most $1/20$ gives $|D^2 f_w| \le C\kappa$. Hence, after removing its affine part,

$$|f_w(y) - f_w(0) - Df_w(0) \cdot y| \le C\kappa R^2, \qquad |Df_w(y) - Df_w(0)| \le C\kappa R.$$

For every fixed derivative order, the relative core bounds in Lemma 2.12 give uniform higher graph bounds on fixed neighborhoods. Interpolation with $|D^2 f_w| \le C\kappa$ makes their fixed-order nonaffine bounds exponentially small in R, with the usual fixed powers of R. The affine slopes are only required to satisfy the fixed small angle bound; the later nonintersection estimate controls their relative tilt more precisely. Clean-ball preparation gives $D \le CR^{-\alpha}$ on the buffered base and opposite graphs. To check the chamber-scale hypotheses, recall $R = A_0 \log(1/B)$ and the curvature bound $B^{1/4}$. Every compared gap has $\ell \le CR$. Thus, for each fixed $\vartheta > 0$ and finite N,

$$R^{-\alpha}\ell \le CR^{1-\alpha} \longrightarrow 0, \qquad B^{\vartheta/4} R^N \ell \le C B^{\vartheta/4} R^{N+1} \longrightarrow 0, \qquad B^{1/4} R^2 \longrightarrow 0.$$

These are the smallness conditions in (3.20). They also give $\varepsilon_0 R \to 0$ in (6.33).

Nonintersection bounds the tilt of each adjacent sheet with intercept below a small fixed fraction of R. Indeed, on a larger tangent disk the affine gap is positive up to the $O(\kappa R^2)$ graph error. Its smallest value gives $|b| r_{\mathrm{big}} \le \ell + O(\kappa R^2)$. Restrict the comparison disk to a fixed small fraction of that disk. Then $|b \cdot y| \le \ell/4 + o(1)$ there, so the cropped top stays strictly above the base for large gaps. All shrink factors are fixed before the comparison rates and A_0 are chosen. In this smaller buffered box, the chamber between that sheet and the base is zero-free. Here is the empty-cylinder check for every remaining side. Denote the already fixed nearby-intercept cutoff by $\eta_{\mathrm{int}} R$, and keep r_{big} from the common family above. We use its radius-r_{big} disk as the larger ordered disk and retain a further interior fraction for comparison. The family covers every zero in $\mathcal{K}_{\mathrm{ret}}$, including every zero that could enter the cylinders chosen next. For an adjacent graph on a remaining side, write its gap over the own graph as

$$d(y) = d(0) + b_0 \cdot y + O(\kappa R^2), \qquad d(0) \ge \eta_{\mathrm{int}} R.$$

Nonintersection on the larger disk gives $|b_0| r_{\mathrm{big}} \le d(0) + O(\kappa R^2)$. Hence on $|y| \le r_{\mathrm{big}}/4$,

$$d(y) \ge \tfrac{3}{4} d(0) - O(\kappa R^2) \ge \eta_{\mathrm{int}} R/2$$

for large R. A missing graph has no upper restriction. Choose $2r_e \le r_{\mathrm{big}}/4$ and $4L_e < \eta_{\mathrm{int}} R/2$, with $r_e \ge 10 L_e$ and $L_e + 2$ below the common artificial top. These fixed choices are possible after making the intercept fraction small. Ordered graphs now exclude every other zero from the buffered empty chamber used above. Zeros outside the original graph buffer are at distance comparable to R from the retained cylinder. This verifies the geometric input of its coarse transmission estimate. No opposite-face preparation is used on this side. Require $cA_0 > 1$ in the choice above. Then $e^{-cR} = B^{cA_0} \le B$ for small B, with all constants fixed.

Build both artificial slabs by Proposition 4.2 from the same cut-off extension of the graph. Use the same depth $L = \theta R$ on the two sides, with the fixed fraction chosen above. It lies above every cropped comparison box, since the nearby intercepts were restricted to a smaller fixed fraction of R. The coarse comparison on a side without a nearby neighbor also fits below this common artificial top. The cutoff graph satisfies $h(0) = Dh(0) = 0$, $\|Dh\|_\infty \le C\kappa R$, and $\|D^2 h\|_\infty \le C\kappa$, where $\kappa = B^{1/4}$. Absorb this fixed C into the proposition's curvature parameter. Since $R = A_0 \log(1/B)$, its smallness condition $C\kappa R^2 \le \eta_0$ holds for small B, while $R \ge L \ge 2$. With $q_0 > 7$ fixed first, its estimates and uniform local Sobolev embedding give

$$\|\psi_\pm\|_{W^{2,q_0}_{\mathrm{uloc}}} + \|\psi_\pm\|_{C^1} \le C B^{1/4} R \longrightarrow 0, \qquad \|\psi_+ + \psi_-\|_{W^{2,q_0}_{\mathrm{uloc}}} \le C B^{1/2} R^3 \longrightarrow 0.$$

These are unweighted estimates on the entire artificial slabs. They justify the small-coefficient hypotheses for cancellation even though the common depth tends to infinity.

To verify the weighted hypothesis for cancellation, use the curvature bound in the prepared tangent chart:

$$|B(y)| \le \sqrt{J_0(z)}\, e^{\lambda |X(y) - z|/2}.$$

The chart has arbitrarily small slope, so $|X(y) - z| \le (1 + o(1))|y|$. Integrate this curvature bound along radial segments, using $h(0) = Dh(0) = 0$. Absorbing the resulting polynomial factors into the unused half of the exponential gives

$$|Dh(y)| + |D^2 h(y)| \le C_\lambda \sqrt{J_0(z)}\, e^{\lambda \sqrt{1 + |y|^2}}.$$

Multiplication by a cutoff on the buffered radius-R disk preserves this estimate: the additional terms are h/R, Dh/R and h/R^2, and the same integrated bound controls them. The preceding unweighted slab estimates verify the remaining hypotheses of Proposition 6.10. It therefore gives

$$|\partial_n v^*_+(z) + \partial_{-n} v^*_-(z) - 2a| \le C J_0(z).$$

Apply (6.39) on both sides, replacing u by $-u$ in a negative chamber. The fixed small tilt puts each shortest chord on the constant part of the penalty. Thus each true adjacent distance ℓ satisfies

$d \le G_1\ell$ exactly. Both actual inward slopes are $|\nabla u(z)|$. In the prepared core, $1-(|\nabla u(z)|/a)^p$ is comparable to $a-|\nabla u(z)|$. This proves (6.31).

For the restricted lower bound, it remains to consider $E > J_0$ at a plateau contact. Here $d = G_1\ell$ and $E = e^{-\beta\ell}$. Since $B = J_0 + E$ lies between E and $2E$,

$$\beta\ell - \log 2 \le \log(1/B) \le \beta\ell.$$

For small B, the ratio ℓ/R is therefore at most $2/(A_0 \min_{\mathcal{P}} \beta)$. Choose A_0 so large that this is below the fixed intercept fraction used above. The contact lies inside the prepared graph window. It lies on an adjacent graph: otherwise the chord would cross an intermediate zero graph. That earlier intersection has the same incidence and a strictly smaller penalized cost. The plateau gradient is $\nabla_\Sigma d = -G_1 b$. Consequently

$$d|\nabla_\Sigma d|^2 = G_1^3 \ell |b|^2.$$

The assumed bound on the left gives $|b|^2\ell \le C$, as required for the lower transmission estimate. Apply (6.44) on this side. On the other side use the same maximum principle without a positive top patch. It gives a lower slope difference $-C\varepsilon_{\mathrm{geom}} e^{-\beta\ell'} - Ce^{-cR}$ if another sheet exists, and $-Ce^{-cR}$ otherwise. Each such gap satisfies $e^{-\beta\ell'} \le E$. The negative top coefficient tends to zero and is absorbed by the positive side. Artificial cancellation now yields

$$a - |\nabla u(z)| \ge cE - CJ_0 - Ce^{-cR}.$$

The same strict margin $cA_0 > 1$ makes the final term $o(E)$ whenever $E > J_0$, since $B \le 2E$ in this case. Absorb the term and use the slope-discrepancy comparison. This proves (6.32).

If B is bounded below, the upper estimate is immediate. For the lower estimate, the case $E \le J_0$ is immediate too. Otherwise write $B \ge b_*$ for a fixed $0 < b_* < 1$. Then $E > B/2 \ge b_*/2$ and

$$d \le J_* := \frac{G_1}{\min_{\mathcal{P}} \beta} \log(2/b_*).$$

Lemma 6.1 gives $D \ge \delta(J_*) > 0$ by contraposition of its small-defect reach bound. Since $E \le 1$, this proves the lower estimate with a uniform constant. It also covers the arbitrarily short almost-minimizing chords already treated in that lemma.

6.4. **Physical caps.** The preceding comparisons let us cap the normal rays and lift surface tests into the ambient space. Choose a small baseline threshold $\delta > 0$ and set

$$X = \{z \in \Sigma : D(z) \ge \delta\}, \qquad \Sigma_g = \Sigma \setminus X, \qquad s_X(x) = \operatorname{dist}(x, X).$$

When X is empty, Theorem 4.12 gives a planar solution, provided the baseline is below its fixed threshold. The assertions then follow. Hence assume $X \ne \varnothing$. Take the baseline below a fixed universal threshold. We then have $b^2 \le CD \le 1$ on Σ_g. For a rate $\lambda > 0$ to be fixed, define

$$J_0(z) = \sup_{y\in\Sigma} e^{-\lambda|z-y|}\bar{b}(y)^2, \qquad \bar{b}(y) = \begin{cases} b(y), & y \in \Sigma_g, \\ 1, & y \in X. \end{cases}$$

For any two base points, the triangle inequality gives $e^{-\lambda|z-z'|}J_0(z') \le J_0(z) \le e^{\lambda|z-z'|}J_0(z')$. Since $X \ne \varnothing$ and $\bar{b} \le 1$, this proves $0 < J_0 \le 1$ and shows that $\log J_0$ is λ-Lipschitz. Hence $|\nabla_\Sigma \log J_0| \le \lambda$ almost everywhere. Take the baseline below the fixed bad threshold in Theorem 4.14. A ball of radius $s_X(z)/2$ about z contains no point of X, so it also contains no point of that theorem's fixed bad set. When $s_X(z)$ is above a fixed constant, apply both estimates in (4.75) at its center. They give

$$D(z) \le Cs_X(z)^{-\alpha}, \qquad b(z) \le C/s_X(z), \qquad \alpha > 1.$$

The second bound is the theorem's curvature estimate. It is not deduced by taking the square root of the first bound. The theorem uses a fixed threshold, so its constants survive every later decrease of the baseline threshold. In the cap notation, put $\alpha_{\mathrm{cl}} = \alpha$. The argument needs no specific numerical value for this exponent.

For a prepared chamber, let x_{turn} be the point on the own normal ray where the two true distances to its boundary graphs are equal. Write $\epsilon_{\mathrm{ch}} = 1$ in a positive chamber and $\epsilon_{\mathrm{ch}} = -1$ in a negative chamber. Define its middle well deficit and pressure scale by

$$b_{\mathrm{turn}} = 1 - \epsilon_{\mathrm{ch}} u(x_{\mathrm{turn}}), \qquad \Gamma_p = \frac{2^{p-1}}{p}, \qquad \mathcal{P}_{\mathrm{gap}} = \Gamma_p b_{\mathrm{turn}}^p.$$

Put $w = 1 - \epsilon_{\mathrm{ch}} u$. Modica's bound gives

$$|\nabla w| \le a_p w(2 - w), \qquad |\nabla \log w| \le C,$$

with C uniform on $\mathcal{P}$. The true bisector differs from the affine one by $o(1)$ in the prepared window. Integrating this logarithmic bound shows that the two middle deficits differ by a factor $e^{o(1)}$. Lemmas 3.11 and 3.12 therefore give, for the fixed small angle used here,

$$b_{\mathrm{turn}}^p \asymp q(x_{\mathrm{turn}})^p \asymp P_{\mathrm{loc}}, \qquad \mathcal{P}_{\mathrm{gap}} \le CD(z) \le C' I(z).$$

Indeed, $q = b_{\mathrm{turn}}(2 - b_{\mathrm{turn}})$ there, the normalized middle deficit has uniform positive lower and upper bounds, and Γ_p does too on $\mathcal{P}$. All uses of the middle pressure below require only these comparisons. The comparison constant is fixed before choosing the later post-turn offset. On a ball of fixed radius T about x_{turn}, the same logarithmic bound gives

$$e^{-CT} \le w/b_{\mathrm{turn}} \le e^{CT}, \qquad |\nabla(w/b_{\mathrm{turn}})| \le C_T.$$

Thus the normalized deficit, its reciprocal, and its first derivatives have bounds for that radius, uniform on $\mathcal{P}$.

Lemma 6.11 (Logarithmic cap construction)**.** *Given a fixed band width T and an arbitrarily large prescribed exponent M, one can choose constants A_1, $A_0 \gg A_1$, and then $\lambda > 0$, in that order, with the following property. For*

$$J_0(z) = \sup_{y \in \Sigma} e^{-\lambda |z-y|} \bar{b}(y)^2, \qquad \bar{b}(y) = \begin{cases} |B(y)|, & D(y) < \delta, \\ 1, & D(y) \ge \delta, \end{cases}$$

put $L(z) = A_1 \log(1/J_0(z))$ and $R(z) = A_0 \log(1/J_0(z))$ at bases where J_0 is sufficiently small. There is a canonical locally Lipschitz normal cap on each side, equal to one on any prescribed fixed core once J_0 is small enough, which is the product of a far cutoff supported in $|s| < 2L(z)$ and a fixed middle or post-turn cutoff at the first relevant adjacent graph. It has the following properties.

1. *The normal map is nonsingular on its support. The Fermi metric and Jacobian are $1 + o(1)$ as $J_0 \to 0$.*
2. *The base depth has $|\nabla_\Sigma L| \le A_1 \lambda$; on the cap support $\operatorname{dist}(x, \Sigma) \ge |s| - C_T$. For every branch with absolute value at most $q f(d(z))$, its far-cutoff energy is at most $C J_0(z)^M f(d(z))^2$ per unit base area. Fixed Young factors in a product cutoff have the same bound.*
3. *The middle coordinate $t = (r_i - r_j)/2$ is intrinsic. Here r_i, r_j are the true distances to the two boundary graphs. It satisfies $|\nabla_x t| \le 1$ and $|\nabla_\Sigma (t \circ \Phi)| \le C$ on the supported middle band. Each normal ray traverses at most one such band on each side. Its derivative in outward ray arclength is $1 + o(1)$ as the relative angle tends to zero, and in particular is bounded below by a fixed positive constant. Thus the middle-band heights have a uniform Lipschitz bound over fixed base patches, and their padded physical cylinders can have fixed size. A band energy estimated per base is counted once on that side.*
4. *If $x = z + s\nu(z)$, $|s| \le L(z)$, and z is a nearest zero to x, then its cap is exactly one at x.*
5. *The locally defined cutoffs agree on overlapping charts. The normal fibers over a compact physical set are finite, so square roots of sums of squared lifted branches are well-defined. In fact, at most two nonzero branches cover a physical point; if two occur, that point lies within*

a fixed distance of their common prepared turn. Outside the fixed turn bands, $|\nabla U| \geq r_* > 0$ *uniformly on the branch supports.*

Proof. Write $j = J_0(z)$. The envelope is positive in the case under consideration and satisfies $|\nabla_\Sigma \log J_0| \leq \lambda$ almost everywhere. Choose $10\lambda A_0 < 1/2$. For $y \in \Sigma \cap B_{10R(z)}(z)$,

$$\bar{b}(y)^2 \leq e^{10\lambda R(z)} j = j^{1-10\lambda A_0} \leq j^{1/2}.$$

Thus this ball contains no bad zero, and $|B(y)| \leq j^{1/4}$ there. Fix a clean-ball threshold δ_{cl} once and take the baseline $\delta \leq \delta_{\mathrm{cl}}$. Clean-ball preparation with this fixed threshold gives $D \leq CR^{-\alpha_{\mathrm{cl}}}$ on the buffered zero graphs. Its constants do not change if δ is later decreased. Apply Lemma 3.10 on a fixed smaller multiple of R. Its larger buffers lie in $B_{10R}(z)$ after A_0/A_1 is large, all zeros there are regular, and $j^{1/4}R \to 0$. The resulting ordered sheets include the base and every adjacent graph with intercept at most $\Lambda_{\mathrm{obs}}L$. Here $\Lambda_{\mathrm{obs}} \geq 32$ is a sufficiently large fixed observation multiple. Choose it first, and then increase A_0/A_1 so these graphs and all their comparison buffers fit in the window. Their height errors are $Cj^{1/4}R^2 = o(1)$, and (3.19) bounds their relative normal angles by $CA_1/A_0 + o(1)$. Choose A_0/A_1 large enough for all fixed angle accuracies, including those in Lemma 3.12. We also need a uniform lower bound for the adjacent gap lengths. For each fixed G, the local fixed-core consequence in the clean-ball proof applies at every zero in the retained packet. Indeed, Lemma 2.12 and (2.17) give $|\nu(y') - \nu(y)| \leq C_G\sqrt{D(y)}$ for zeros with $|y' - y| \leq G$. The relevant fixed balls lie in $B_{10R}(z)$ for large R, and their base defects are bounded by $CR^{-\alpha_{\mathrm{cl}}} \to 0$. Adjacent oriented normals have scalar product below $-1/2$ once the graph angle is small. Thus no adjacent gap can have bounded length as $j \to 0$. This conclusion is uniform over the retained packet. Apply it with G larger than all the prescribed fixed turn bands, including the one-unit enlargement used below, before choosing the small threshold for j. At the base, $b(z)^2 \leq j$, so $2Lb(z) \leq 2A_1\sqrt{j}\log(1/j) \to 0$. The normal map is therefore nonsingular for $|s| \leq 2L$. Its metric and Jacobian tend uniformly to their product values.

The same lemma gives uniqueness of projection onto each individual graph. Choose a common disk of radius $4r$ inside its graph buffer, with r a small fixed multiple of R. Its slope is at most $1/4$ after choosing the angular accuracy. The curvature bound gives $r\|D^2F\| = o(1)$, and each relevant vertical competitor is at distance $O(L) < r$ from the graph in the smaller observation window. Thus the nearest-foot part of Lemma 3.10 applies. Its strictly convex distance function has one interior critical point. Every normal representation uses that point, so projection is unique on this graph. The two boundary graphs of a chamber may still both have normal preimages of the same physical point.

Take the even far cutoff $\zeta_0(s/L)$, equal to one on $|s| \leq L$ and zero on $|s| \geq 2L$. On a side with a relevant first adjacent graph, let r_j be its true distance and set $r_i = |s|$ and $t = (r_i - r_j)/2$. The own-graph distance is measured in its normal chart. Choose a fixed smooth nonincreasing middle factor in t. For the directional branch, it equals one through $t = T$ and zero for $t \geq T + 1$. If no adjacent graph occurs before $\Lambda_{\mathrm{obs}}L$, use the constant factor one. When a graph enters or leaves that window, its middle band is outside $|s| \leq 2(L + 1)$, with a fixed margin for small j. The two formulas therefore agree in a neighborhood of the change. The order of the graphs is fixed on every common prepared packet.

Define the middle factor first in the zero-free chamber on this side and extend the cap by zero beyond its first adjacent zero. This extension creates no edge singularity. If the first zero lies beyond $2(L + 1)$, the enlarged far factor already vanishes before it. Otherwise choose the small threshold also so that $2(L + 1)b(z) < 1/2$. The unique own projection then remains valid up to that zero. There $r_j = 0$ and $t = r_i/2$; the uniform gap divergence makes this larger than every prescribed cutoff endpoint. Continuity makes the middle factor identically zero on a neighborhood of the endpoint from within the chamber. If no relevant graph exists, the observation-window clearance

keeps both far supports $|s| \le 2L$ and $|s| \le 2(L+1)$ before any zero. Thus both the original cap and its later enlarged cap end before the next zero.

The absolute value in $r_i = |s|$ causes no derivative at the base. Both middle factors are identically one near $s = 0$, and the even far factor is constant there as well.

The cap also covers every nearest-foot representation in its far core. If z is a nearest zero to $x = z + s\nu(z)$ and $|s| \le L(z)$, then the own-sheet distance is $r_i = |s|$ by uniqueness of its projection. Each adjacent-sheet distance satisfies $r_j \ge r_i$, since z is nearest among all zeros. Hence $t \le 0$, where the middle factor is one. If there is no relevant adjacent graph, that factor is one by construction. The far factor is also one on $|s| \le L(z)$, proving the asserted coverage.

Uniqueness of the first graph and of distance projection therefore gives locally Lipschitz cutoffs that agree on chart overlaps.

For $t \le 0$, the own graph is the nearer boundary of the chamber. For $0 < t < T + 1$, the other distance satisfies $r_j = r_i - 2t \ge |s| - 2T - 2$. Every other local sheet lies outside the chamber. Zeros beyond the buffered window are farther away. Hence $\operatorname{dist}(x, \Sigma) \ge |s| - C_T$ throughout the support. Sharp clearing gives

$$q^p \le C(1 + |s|)^{M_0} e^{-2\beta|s|} \quad \text{on } L \le |s| \le 2L.$$

The physical gradient of a lifted base function is $(I + sB)^{-1}\nabla_\Sigma L$. It has size at most $(1 + o(1))A_1\lambda$. Thus $|\nabla(s/L)| \le C/L$ on $|s| \le 2L$. Since $q^2 A_u \le Cq^p \,\mathrm{Id}$, the far-cutoff cost per unit base area is at most $C(1+L)^{M_0} e^{-2\beta L} f(d(z))^2$. Choose A_1 so that $2 \min_{\mathcal{P}} \beta A_1 > M + 1$. The remaining power of j absorbs the logarithmic factor and gives $Cj^M f(d(z))^2$. The area factors are uniformly bounded. To compute the traversal factor, let N_i point along the own ray. Then

$$\frac{dt}{d|s|} = \frac{1 - \nabla r_j \cdot N_i}{2} = \frac{1 + t_{\mathrm{ang}}}{2}, \qquad t_{\mathrm{ang}} = -\nabla r_j \cdot N_i.$$

The last quantity is the cosine of the relative normal angle at the actual projection feet. It tends to one. Thus the traversal factor is uniformly bounded and positive, and each ray crosses the middle band at most once on each side. Both true distance functions are smooth on the prepared chamber and have unit physical gradient. Therefore

$$|\nabla_x t| = \frac{1}{2}|\nabla r_i - \nabla r_j| \le 1.$$

On each side use outward arclength $\rho = |s|$ and $\Phi_\pm(z, \rho) = z \pm \rho\nu(z)$. At fixed ρ, $D_z\Phi_\pm = I \pm \rho B$ on tangent vectors. Its norm is uniformly bounded by the first part of the lemma. Thus $|\nabla_\Sigma(t \circ \Phi_\pm)| \le C$. For any fixed middle level, write its arclength as $\rho = S(z)$. The positive traversal derivative and the implicit-function formula

$$\nabla_\Sigma S = -\frac{\nabla_\Sigma(t \circ \Phi_\pm)}{\partial_\rho(t \circ \Phi_\pm)}$$

give $|\nabla_\Sigma S| \le C$. The map $z \mapsto \Phi_\pm(z, S(z))$ also has bounded derivative, since $I \pm SB$ does. A fixed-size base patch and a fixed-width middle band consequently fit in a physical cylinder of fixed size. The same statement holds for a fixed padding of that cylinder. The lower bound for $I + SB$, unique projection onto each sheet, and the at-most-two-branch property preserve bounded overlap in these patch coverings.

To prove finite fibers, fix z_* with $J_0(z_*) > 0$. The envelope lower bound gives $L(z) \le A_1\lambda|z - z_*| + C_{z_*}$. If $x = z + s\nu(z)$ lies in a fixed compact set and $|s| \le 2L(z)$, then

$$(1 - 2A_1\lambda)|z - z_*| \le C.$$

Choose $2A_1\lambda < 1/2$. All relevant bases and normal parameters are then bounded. Restrict the caps to $J_0 < j_* < 1$. Since $J_0(z) \ge e^{-\lambda \operatorname{dist}(z,X)}$, their limiting bases stay a positive distance from X and are regular. The envelope is continuous, and the bound $2Lb < 1/2$ persists on their closure after j_* is fixed small enough. An accumulating fiber would contradict local invertibility of the normal map at its limit. Every such fiber is therefore finite. For the sharper overlap bound, choose

a preimage with largest L among the finite preimages of one point. Its R window contains all other bases. Every ray length is at most twice the largest L, and A_0/A_1 is large. All contributing sheets are therefore ordered graphs in one flat window. A capped ray ends a fixed distance beyond the bisector. Since every gap is much wider than the turn band, it never crosses the next zero. Each contributing base must lie on one of the two boundary graphs of the chamber containing the point. Distance projection onto either graph is unique. Thus there are at most two preimages. Their supports overlap only within the common fixed turn band.

We finish by proving the off-turn gradient bound. Reflect a negative chamber, and use outward depth s on the positive side. Put $w = 1 - u$ and $w_0(s) = 1 - g_p(s)$. On each fixed bottom core, Lemma 2.12 gives the required gradient comparison from the small defect at its zero base. For the remaining well part, let ε_R denote the chamber error in (3.20), with zero defects $CR^{-\alpha_{\mathrm{cl}}}$ and curvature $j^{1/4}$. The chosen clean-ball exponent satisfies $\alpha_{\mathrm{cl}} > 1$. Since $R = A_0 \log(1/j)$ and $L = (A_1/A_0)R$, we have $R^{-\alpha_{\mathrm{cl}}}L = (A_1/A_0)R^{1-\alpha_{\mathrm{cl}}} \to 0$. Every fixed polynomial factor in R is also absorbed by $j^{\vartheta/4}$. Thus $\varepsilon_R L \to 0$. If an adjacent graph is present, the logistic own-face comparison of Lemma 3.9 gives, on enlarged unit balls before a fixed crop of the middle,

$$\left|\frac{w}{w_0} - 1\right| \le C(\varepsilon_R + e^{-cT_{\mathrm{crop}}}).$$

True and affine distances differ by $o(1)$ on these packets, so the same estimate holds in the own tangent coordinates. Use the normalized interpolation implication proved above. It requires only this relative value estimate and the bounded-source equation, with no noncriticality assumption. First choose T_{crop} large, then take j small. It puts the phase gradient in a fixed neighborhood of the own unit direction.

Here is the separate value comparison when no relevant graph is present. Retain a smaller fixed fraction of the horizontal disk. For a positive affine gap $d - b \cdot y$ on a larger radius-r disk, nonintersection and the $o(1)$ graph error give $|b|r \le d + o(1)$. On its central radius-$r/4$ disk that gap is at least $3d/4 - o(1)$. The ordered graph construction and the choice of Λ_{obs} therefore give a zero-free cylinder of depth $8L$ on the retained packet when no adjacent intercept is at most $\Lambda_{\mathrm{obs}}L$. Zeros outside the larger graph buffer are farther away because $R/L = A_0/A_1$ is fixed and large. Choose a fixed clearing depth S with a fixed extra buffer. The region $S \le s \le 8L - S$ then lies in the common small well range. Preparation gives the own-face exponential coefficient $1 + O(\varepsilon_R)$ there. Apply Lemma 3.9 with

$$V = (1 + C\varepsilon_R)e^{-2as} + C_{\mathrm{far}}e^{-2a(8L-s)} + V_{\mathrm{lat}}, \qquad F(V) = \frac{2V}{1+V}.$$

Choose the bounded constant C_{far} so that $V \ge v_*$ on the far face $s = 8L - S$. Choose the lateral exponentials V_{lat} in the larger box in the same way. Comparison is on $\{V < v_*\}$; on its added boundary clearing gives $w \le F(v_*)$. The near-face ordering follows from its prepared coefficient. The positive-part weak comparison is therefore valid, even if the actual gradient vanishes. On the retained packet and for $S \le s \le 2(L+1) + 1$,

$$\frac{C_{\mathrm{far}}e^{-2a(8L-s)}}{e^{-2as}} \le Ce^{-cL}, \qquad \frac{V_{\mathrm{lat}}}{e^{-2as}} \le Ce^{-cR}.$$

The second estimate follows by choosing A_0/A_1 large enough for the fixed horizontal padding. The exact logistic ratio, combined with Modica's lower comparison along the true own normal, consequently gives

$$1 - C\varepsilon_R \le \frac{w}{w_0} \le 1 + C(\varepsilon_R + e^{-cL} + e^{-cR}).$$

The graph-coordinate error is already included in ε_R. Applying the same normalized interpolation implication on enlarged unit balls proves the gradient comparison uniformly through $2(L+1)$ on

this empty side. Thus clearing has supplied only the boundary values; the relative comparison has supplied the gradient conclusion.

On every fixed post-turn interval separated from the turn, Lemma 3.11 gives the corresponding comparison in C^1. The limiting phase-gradient magnitude is

$$\frac{|V_p'(t)|}{2aV_p(t)} = \left(1 - V_p(t)^{-p}\right)^{1/p} > 0 \qquad (t \neq 0).$$

Its minimum on each such compact interval is positive uniformly on $\mathcal{P}$. Choose its fixed-band accuracy below half that minimum. Enlarge the excluded turn band to contain the central interval and the endpoint of the enlarged post-turn cap. All the remaining branch supports have $|\nabla U| \geq r_* > 0$, with one common r_*. No ellipticity at the turn was used. □

Lemma 6.12 (Arbitrarily small post-turn cost for the directional branch)**.** *For every $\eta > 0$, the fixed choices in Lemma 6.11 can be made so that, for*

$$h_z(x) = \frac{q(x) + \nabla u(x) \cdot \nu(z)/a}{2}, \qquad 0 \leq h_z \leq q,$$

the post-turn cutoff cost on the two sides of a base is at most $\eta I(z) f(d(z))^2$ per unit base area, plus the far-cap error $C J_0^M f(d(z))^2$. The constants and all normal caps are fixed before f.

Proof. Apply Lemma 3.12 at a fixed crop T_0. This fixes C_{pr} with $P_{\mathrm{loc}} \leq C_{\mathrm{pr}} I(z)$. Choose a later offset T. For the normalized one-dimensional turn, on $T < t < T + 1$,

$$q^p \leq C P_{\mathrm{loc}} e^{2\beta T}, \qquad h_z/q \leq C e^{-2\beta T}.$$

To see the second exponent directly, the turn first integral gives the magnitude of the normalized phase gradient as $(1 - V_p(t)^{-p})^{1/p}$. After the turn its orientation is opposite to the initial normal. Hence the limiting value of h_z/q is

$$\frac{1 - (1 - V_p(t)^{-p})^{1/p}}{2} \leq C V_p(t)^{-p} \leq C e^{-2\beta t} \qquad (t \geq T),$$

once T is large. Indeed, $V_p(t)^{-p} \leq 1/2$ there, so the mean value theorem bounds the derivative of $(1 - x)^{1/p}$ on $0 \leq x \leq 1/2$ uniformly in p. The asymptotic constant in $V_p(t) \asymp e^{2at}$ is uniform on the compact exponent interval. These are computations for the limiting turn; the finite-chamber errors are included next.

A small relative angle ϑ changes the projection of the opposite normal onto $\nu(z)$ by $C\vartheta^2$. The bisector coordinate and outward ray arclength differ by the factor $(1 + t_{\mathrm{ang}})/2 = 1 + O(\vartheta^2)$. Require $\vartheta^2(T + 2)$ to be small. The resulting change in the exponential on this band is then bounded. Lemma 3.11, with a chosen fixed-band C^1 accuracy, gives

$$\text{one-side cutoff cost} \leq C P_{\mathrm{loc}} e^{2\beta T} (e^{-2\beta T} + \vartheta^2 + \varepsilon_T)^2 f(d(z))^2.$$

The metric, traversal, and unit-width derivative factors are bounded. First choose T large enough for the first contribution to be below $\eta/(8C_{\mathrm{pr}})$. Then choose the angle and ε_T small enough for the other terms. Enforce these choices by A_0/A_1 and a small curvature threshold. There is at most one band per side. Reducing the initial tolerance makes their sum at most $\eta I(z) f(d(z))^2$. Fix the Young factor for the product cutoff before making these choices. Lemma 6.11 bounds the remaining far-cutoff cost. Reflection $u \mapsto -u$, $\nu \mapsto -\nu$ gives the negative-side estimate. The pressure constant was fixed before T. All caps are fixed before the amplitude f. □

Lemma 6.13 (Raw bad-point mass)**.** *Let $\mathcal{P} \Subset (1, \infty)$, $p \in \mathcal{P}$, and let u be an entire matched weak solution with $-1 < u < 1$. Let $\mathcal{X}$ be a maximal unit-separated subset of $X = \{u = 0, D \geq \delta\}$, and*

$N_R = \sum_{z\in\mathcal{X}} \chi_R(z)^2$, *where* $\chi_R(x) = \exp[-\sqrt{1+|x-x_c|^2}/R]$ *and* $R \geq 1$. *There are fixed* $c > 0$ *and* $L_{\mathrm{bad}} < \infty$ *such that*

$$\int_{\cup_{z\in\mathcal{X}} B_{c\delta^{1/\alpha_{\mathrm{reg}}}}(z)} q^{p+1} D\,\chi_R^2 \geq c\delta^{L_{\mathrm{bad}}} N_R, \qquad L_{\mathrm{bad}} = 1 + n/\alpha_{\mathrm{reg}},$$

where $\alpha_{\mathrm{reg}} > 0$ *is a uniform local Hölder exponent for* ∇u. *The balls are disjoint when* δ *is sufficiently small.*

Proof. We first justify the uniform gradient modulus on the full exponent range in the statement. For this step the flux may be singular at zero when $p < 2$, so we use an approximation. Write $p_- = \min \mathcal{P}$, $p_+ = \max \mathcal{P}$, and set $\delta_* = \min\{1, p_- - 1\} > 0$ and $g_* = \max\{1, p_+ - 1\}$. For $0 < \epsilon \leq 1$ define

$$A_{\epsilon,p}(\xi) = (\epsilon^2 + |\xi|^2)^{(p-2)/2}\xi, \qquad g_{\epsilon,p}(t) = t(\epsilon^2 + t^2)^{(p-2)/2}.$$

These fluxes are C^1 at every gradient, and

$$\delta_* \leq \frac{t g'_{\epsilon,p}(t)}{g_{\epsilon,p}(t)} = 1 + (p-2)\frac{t^2}{\epsilon^2 + t^2} \leq g_*.$$

The eigenvalues of $DA_{\epsilon,p}(\xi)$ lie between $\delta_* g_{\epsilon,p}(|\xi|)/|\xi|$ and $g_* g_{\epsilon,p}(|\xi|)/|\xi|$, with their continuous values at zero. Thus the structural constants in [28, Theorem 3.2 and the following paragraph, p. 303] are uniform in p and ϵ. More explicitly, use

$$\widetilde{A}_{\epsilon,p} = \frac{A_{\epsilon,p}}{\delta_* g_{\epsilon,p}(1)}, \qquad \widetilde{g}_{\epsilon,p} = \frac{g_{\epsilon,p}}{g_{\epsilon,p}(1)}.$$

The lower ellipticity coefficient is then one, the upper coefficient is g_*/δ_*, and $\widetilde{g}_{\epsilon,p}(1) = 1$. The logarithmic growth bounds are unchanged. Also $g_{\epsilon,p}(1)$ stays in a fixed positive compact interval, so dividing a bounded source by $\delta_* g_{\epsilon,p}(1)$ keeps its bound uniform. The coefficient-continuity condition has zero left side because the flux is autonomous. In particular, it satisfies the stronger version in the paragraph after the theorem, which applies to bounded weak solutions. Only a bounded source is needed here.

Fix z and put $H(x) = f_p(u(x))$ on $B_2(z)$, so $|H| \leq 1$. Minimize

$$\int_{B_2(z)} \left\{ \frac{(\epsilon^2 + |\nabla v|^2)^{p/2} - \epsilon^p}{p} - H(x)v \right\} dx \quad \text{over } v - u \in W_0^{1,p}(B_2(z)).$$

The integrand is strictly convex in the gradient. It is coercive since its first term is at least $(|\nabla v|^p - 1)/p$. The direct method and Poincaré's inequality give a unique minimizer u_ϵ, with $-\operatorname{div} A_{\epsilon,p}(\nabla u_\epsilon) = H$. We need a bound on its oscillation independent of p, ϵ, z. Choose $S \geq 1$ uniformly so that $g_{\epsilon,p}(S) \geq 2/n$ for all these parameters. This is possible because $g_{\epsilon,p}(S) \geq c_{\mathcal{P}} S^{p_- - 1}$. Writing $r = |x - z|$, define

$$b_\epsilon(r) = \int_r^2 g_{\epsilon,p}^{-1}(t/n)\, dt.$$

Then $0 \leq b_\epsilon \leq 2S$, its boundary value is zero, and $A_{\epsilon,p}(\nabla b_\epsilon) = -(x - z)/n$. Consequently $-\operatorname{div} A_{\epsilon,p}(\nabla b_\epsilon) = 1$. Weak comparison with $1 + b_\epsilon$ and $-1 - b_\epsilon$ gives $|u_\epsilon| \leq 1 + 2S$.

The cited theorem now gives one interior Hölder exponent and bound for ∇u_ϵ, independent of p, ϵ, z. Its bounded-weak-solution version applies since the coefficient continuity is autonomous and H is bounded. An interior difference quotient along a short segment bounds $|\nabla u_\epsilon|$ by its oscillation and this Hölder seminorm. Thus the full $C^{1,\alpha_{\mathrm{reg}}}$ norm on $B_1(z)$ is uniform. As $\epsilon \downarrow 0$ with p fixed, coercivity gives weak $W^{1,p}$ compactness and strong L^p compactness on $B_2(z)$. The boundary class is weakly closed. The lower bound

$$\frac{(\epsilon^2 + |\xi|^2)^{p/2} - \epsilon^p}{p} \geq \frac{|\xi|^p - \epsilon^p}{p}$$

gives the lower energy bound. Testing minimality against each fixed competitor and using dominated convergence gives the reverse bound. Every limit therefore minimizes $\int(|\nabla v|^p/p - Hv)$ with boundary value u. The original solution u is this minimizer: its weak equation is the Euler equation of that convex functional. Strict monotonicity of the flux gives uniqueness. Interior C^1 compactness then passes the uniform gradient Hölder bound to u.

Proposition 2.1 keeps q uniformly positive on a fixed ball about any zero. On that ball $D = 1-(|\nabla u|/(a_p q))^p$ inherits the same uniform Hölder modulus, after reducing α_{reg} if necessary. Indeed, a_p stays in a fixed positive compact interval, q is bounded below, and $t \mapsto t^p$ has a uniform Lipschitz bound on every fixed bounded interval when $p \in \mathcal{P} \Subset (1,\infty)$. A sufficiently small $r_\delta = c\delta^{1/\alpha_{\mathrm{reg}}}$ therefore gives $q^{p+1}D \geq c\delta$ throughout each bad ball. Unit separation makes these balls disjoint. Since $|\nabla \log \chi_R| \leq R^{-1}$, their weights are comparable to their center weights.

On each small bad ball the squared weight is at least e^{-2r_δ} times its center value, so

$$\int_{B_{r_\delta}(z)} q^{p+1} D\chi_R^2 \geq c\delta r_\delta^n \chi_R(z)^2 = c\delta^{1+n/\alpha_{\mathrm{reg}}}\chi_R(z)^2.$$

The exponent is the product of the defect lower bound and the ball volume. For an infinite net, sum first over finite subsets and use monotone convergence. The source integral is finite because $q^{p+1}D \leq 1$ and χ_R^2 is integrable. Thus $N_R < \infty$; when $X \neq \varnothing$, maximality makes the net nonempty and gives $N_R > 0$.

□

Apply the branch calculation from Proposition 3.13. Write

$$\mathcal{H} = D^2U, \quad \mathcal{S} = \operatorname{tr}(A\mathcal{H}A\mathcal{H}), \quad e = \nabla s, \quad T = De, \quad \varphi = (1 + Z\cdot e)/2, \quad h = q\varphi.$$

In particular, (3.29) gives, for a compact tube amplitude η,

$$\begin{aligned}\langle L_u h, h\eta^2\rangle = &-\frac{1}{2}\int q^{p+1}D\varphi\eta^2 - \frac{a^{p-2}}{2}\int q^p\varphi\operatorname{tr}(A\mathcal{H}T)\eta^2 \\ &+ \frac{a^{p-2}}{2}\int q^p ATZ\cdot\nabla(\varphi\eta^2).\end{aligned} \tag{6.45}$$

Here T is the distance Hessian. The identity has the weighted weak meaning established in Section 2. Proposition 2.9 and the negative-power estimate proved after it justify its use through a normal turn. The critical-set approximation is the same as in (3.31); it needs no unweighted Hessian estimate.

The far depth $L(z)$ now depends on the base. Put $D_0 = D(z)$ and take a compact base amplitude g. In the negative-power test used for (3.31), choose

$$\zeta = D_0^{1/4} g(z)\widehat{\Theta}(z,s),$$

where $0 \leq \Theta \leq \widehat{\Theta} \leq 1$. Choose the enlarged cap by replacing the far depth L by $L+1$ and moving the end of the middle cutoff outward by one fixed unit. Use a nonincreasing far profile, so the new cap dominates the old one. Its derivatives have the same bounds. For small j_*, $2(L+1)b < 1/2$, and the enlarged middle cap still ends before the next zero. It is therefore supported in an open nonsingular tube. The estimates (3.30) still hold on these tubes: Lemma 6.11 gives $\operatorname{dist}(x,\Sigma) \geq |s| - C_T$, and Theorem 2.20 gives the same initial-normal discrepancy bound. The two reciprocal-weight exponents are $-\beta + O(\xi)$; the exponent for $q^{p+1}D/D_0$ is $-2a + O(\xi)$. Choose the losses uniformly small on $\mathcal{P}$. Consequently

$$\int q^{p+1}DJ\,ds \leq C_T D_0, \qquad \int_{|s|>P} q^p (D/D_0)^{1/2} J\,ds \leq C_T e^{-cP}.$$

In particular the upper comparison $I \leq C_T D_0$ is independent of a later enlargement of the fixed core P; its lower comparison already holds on a unit core.

The derivative of $D_0^{1/4}$ is controlled by $|\nabla_\Sigma D_0| \le C D_0$. At a post-turn cap, Lemma 3.12 bounds its pressure by CD_0, with a constant fixed before the cap offset. The source contribution is at most $C_T D_0^{1/2} P_{\mathrm{loc}} \le C_T D_0$. At the far cap, the lifted gradient of L is bounded, so $|\nabla(s/(L+1))| \le C/(L+1)$. The far-cap calculation in Lemma 6.11 is unchanged by this fixed extension. It replaces the error $\delta^{Q'}$ in (3.31) by $J_0(z)^M$. We obtain

$$\int q^p \mathcal{S}(D_0/D)^{1/2} g^2 \widehat{\Theta}^2 \le C_T \int_\Sigma D_0 g^2 + C_T \int_\Sigma D_0 |\nabla_\Sigma g|^2 + C_T \int_\Sigma J_0^M g^2. \tag{6.46}$$

Tube integrals use pullback Euclidean measure. The inequality may also be summed over abstract tubes. As in (3.31), it introduces no unweighted base-area term.

Here is the tail calculation, including the far-cap error. Put

$$M_g = \int D_0 g^2, \qquad E_g = \int D_0 |\nabla_\Sigma g|^2, \qquad F_g = \int J_0^M g^2.$$

With $\theta = 1/2$, insert reciprocal factors into the first current:

$$\begin{aligned}
&\int_{|s|>P} q^p |T| \sqrt{\mathcal{S}}\, g^2 \Theta^2 \\
&\le \left(\int q^p \mathcal{S} (D_0/D)^{1/2} g^2 \widehat{\Theta}^2 \right)^{1/2} \left(\int_{|s|>P} q^p |T|^2 (D/D_0)^{1/2} g^2 \frac{\Theta^4}{\widehat{\Theta}^2} \right)^{1/2} \\
&\le C_T e^{-cP/2} \sqrt{(M_g + E_g + F_g) M_g}.
\end{aligned}$$

The quotient is defined as zero where $\widehat{\Theta} = 0$ and is at most one. We used $|T| \le Cb$, $b^2 \le CD_0$, and the reciprocal tail estimate for the second factor. More explicitly, that factor squared is bounded by

$$C \int_\Sigma b(z)^2 g(z)^2 \left[\int_{|s|>P} q^p (D/D_0)^{1/2} J \, ds \right] dS_z \le C_T e^{-cP} M_g.$$

Thus the apparent reciprocal power of D_0 has disappeared before Young's inequality is applied. In particular the cap error is not multiplied by D_0^{-1}. The elementary inequalities

$$\sqrt{M_g(M_g + E_g)} \le M_g + E_g, \qquad 2\sqrt{M_g F_g} \le M_g + F_g, \qquad E_g \le \int |\nabla_\Sigma g|^2$$

show that this current is an arbitrarily small multiple of $\int D_0 g^2 + \int |\nabla_\Sigma g|^2$, plus $C_T F_g$, once P is chosen after T. The other absolute current bounds are $Cq^p |T|^2 g^2$ and $Cq^p |T| |g| |\nabla g|$. For example, $\nabla\varphi = (\mathcal{H}e + TZ)/2$. The term with $\mathcal{H}$ satisfies $|ATZ \cdot \mathcal{H}e| \le C|T|\sqrt{\mathcal{S}}$, and the term with TZ is bounded by $C|T|^2$. These inequalities follow by writing $A = r^{p-2}(\mathrm{Id} + (p-2) n \otimes n)$: its normalized matrix and its inverse have bounded norms for $p \in \mathcal{P}$. They remain valid in the weighted sense at $r = 0$. Clearing and $2b|g||\nabla_\Sigma g| \le b^2 g^2 + |\nabla_\Sigma g|^2$ give the same conclusion. Post-turn derivatives cost at most $C_T b P_{\mathrm{loc}} g^2 \le C_T \sqrt{\delta}\, D_0 g^2$; far derivatives cost $C J_0^M g^2$. Thus every current outside the fixed core is accounted for. Splitting an integral into $|s| \le P$ and $|s| > P$ involves no integration by parts at $s = \pm P$, and creates no trace term. The actual cap vanishes at the outer boundary of the enlarged tube.

These tail estimates extend Proposition 3.13 to the variable envelope caps, with far-cap error $C \int J_0^M g^2$. For the fixed-core mass and kinetic coefficient, relative regularity on $|s| \le P$ gives

$$Z - e = O_P(D_0), \quad U - s = O_P(D_0), \quad \varphi = 1 + O_P(D_0), \quad \nabla\varphi = O_P(D_0).$$

Apply the fixed-ray transport argument in Proposition 3.13, with the constant ambient vector $e_0 = \nu(z)$. It gives $\mathcal{H}(z + se_0) = \mathcal{H}(z) + O_P(D_0)$ on $|s| \le P$. The Fermi normal equals e_0 along this ray. Derivatives of its extension remain in the separate currents through $T = De$. At the base,

the tangential block of $\mathcal{H}$ is $r(z)B(z) = B(z) + O(D_0 b)$. Also $T = B(I + sB)^{-1} = B + O_P(b^2)$ on the tangential block and $Te = 0$. Since $b^2 \le CD_0$, these formulas give

$$\operatorname{tr}(A\mathcal{H}T) = b^2 + O_P(D_0^{3/2}), \qquad ATZ = AT(Z - e) = O_P(bD_0).$$

Finally $J = 1 + O_P(b)$ and $q^p = G^p(1 + O_P(D_0))$. Inserting these formulas in the zeroth-order terms of (6.45) gives the normalized mass

$$\frac{1}{2}I + \frac{J_{p,P}}{2J_p}b^2 + E_P(z), \qquad |E_P(z)| \le C_P D_0^{3/2}, \qquad J_{p,P} = \int_{-P}^{P} G^p.$$

For example, the Jacobian error in the first mass is bounded by $C_P bD_0 \le C_P D_0^{3/2}$. The first-order current satisfies $bD_0|g||\nabla g| \le \eta D_0 g^2 + C_\eta D_0^2 |\nabla g|^2$. For the kinetic term, a lifted tangential gradient is orthogonal to e, while $A = \mathrm{Id} + (p-2)e \otimes e + O_P(D_0)$. Its core coefficient is therefore $a^{p-2}J_{p,P}(1 + O_P(\sqrt{D_0}))$. The kinetic tail is at most $C_T e^{-cP}$. Thus the full coefficient tends to $\sigma_q = a^{p-2}J_p$. The two limits have different roles: enlarge P to make $J_{p,P}/J_p$ close to one and to control tails, then decrease the defect threshold to make $|E_P(z)|/D_0$ small. In the local envelope lemma below this smallness also follows by decreasing j_*, since its clean-ball bound forces $D_0 \to 0$. Lemma 6.12 controls the explicit post-turn derivative. Its Young factor is fixed before its accuracy is chosen.

Lemma 6.14 (Physical surface form with variable envelope caps)**.** *Fix $\tau \in (0, 1/10)$ and $M > 2$. There are a finite core P, cap constants $A_1, A_0, \lambda > 0$, and a threshold $j_* > 0$ such that*

$$(1-\tau)\int_{\Sigma_g} \frac{I + b^2}{2}\psi^2 \le (1+\tau)\int_{\Sigma_g} |\nabla_\Sigma \psi|^2 + C_{\tau,M}\int_{\Sigma_g} J_0^M \psi^2 \tag{6.47}$$

for every compactly supported C^1 test in $\{J_0 < j_*\} \cap \Sigma_g$. The constants are independent of every subsequent decrease of the baseline threshold and of every later reach amplitude. In particular, after relabeling the tolerance,

$$(1-\tau)\int b^2\psi^2 \le (1+\tau)\int |\nabla_\Sigma \psi|^2 + C_{\tau,M}\int J_0^M \psi^2. \tag{6.48}$$

Proof. Use the caps from Lemma 6.11. Its fixed-threshold clean-ball estimate and curvature bound give

$$\epsilon_0 \le CR^{-\alpha_{\mathrm{cl}}} + C_N J_0^{\vartheta/4} R^N, \qquad \epsilon_0 R \longrightarrow 0.$$

Here ϵ_0 is the coefficient error in (3.20), and $\alpha_{\mathrm{cl}} > 1$. The radius is $R = A_0 \log(1/J_0)$, the curvature is at most $J_0^{1/4}$, and every relevant gap is $O(R)$. The chamber-scale calculation for (6.33) therefore applies with B replaced by J_0. For small j_*, all chamber, turn, and pressure comparisons apply. The same choice ensures the fixed-core relative bounds and curvature-mass comparison. These choices use the fixed clean-ball threshold, so their constants survive every later decrease of the baseline δ.

For a compact base amplitude ψ, sum the squares of $h_z \Theta_z \psi(z)$ over the normal fiber and take the square root. This definition is independent of chart labels. Lemma 6.11 gives finite fibers and at most two active branches. The positive minimum of J_0 on the compact base support bounds L there, so the physical support is compact. Proposition 2.5 gives local form membership of each u_i. In a nonsingular inverse chart, the own normal field is smooth. Weighted multiplication therefore gives form membership of h_z and of its product with the Lipschitz cap and amplitude. Choose the open tube slightly larger than the capped support. On a smaller inverse chart, a smooth compact cutoff makes each local branch extend by zero as a form-domain function. Lemma 2.10 then gives local form membership of the root-sum test and bounds its form by the sum of branch forms. A finite cover of its compact support makes the test globally admissible. This is the same assembly used in Proposition 3.13. It applies through turns and adds no derivative of a chart label.

To distinguish the two uses of localization, the auxiliary smooth chart cutoff is one on the smaller chart where membership is being checked. The actual test remains the intrinsic norm of its normal branches. We do not insert these chart cutoffs into its definition or into its energy estimate. On the smaller chart its weak gradient is therefore exactly the norm-gradient formula, without a partition of unity error. These local gradients agree on overlaps.

For this admissible test, apply (6.45) and (6.46) to every branch. The positive core mass is

$$\sigma_q\left(\frac{1}{2}I+\frac{J_{p,P}}{2J_p}b^2+o_P(D)\right)\psi^2.$$

The tail, base-current, and post-turn costs are arbitrarily small multiples of $D\psi^2+|\nabla_\Sigma\psi|^2$, plus $CJ_0^M\psi^2$. The tangential coefficient is at most $(1+o(1))\sigma_q$. Fix the pressure crop and its constant first. Then fix the tangential Young factor, post-turn offset and accuracy, and current tolerances. For the coefficient-one conclusion, first apply Corollary 3.5 with loss $\tau/2$. It gives a fixed window P_I and a defect threshold. Choose the common core $P\ge P_I$ large enough for all tail errors and for $J_{p,P}/J_p$ to be as close to one as needed. The window restrictions in the physical calculation are lower bounds, so this additional enlargement is permitted. Since $D\ge 0$ along the normal ray by Proposition 2.1, the physical mass satisfies $I_P\ge I_{P_I}$. Reduce j_* later so that its clean-ball bound puts D_0 below both the mass-comparison threshold and the threshold for this fixed core. Then $I_P\ge(1-\tau/2)b^2$ on the whole support. Choose the far-cap exponent and A_1 next. Choose A_0/A_1, then λ, and finally j_*. The bound $I\ge c_I D$ absorbs every small defect cost into the retained mass. Divide by σ_q to obtain (6.47). For (6.48), carry out the preceding form estimate with initial loss $\tau/4$, and use the common-window comparison just obtained. Pointwise,

$$(1-\tau/4)\frac{I_P+b^2}{2}\ge(1-\tau/4)^2b^2\ge(1-\tau)b^2.$$

The tangential coefficient $1+\tau/4$ is at most $1+\tau$. This proves (6.48) with fixed constants before any later decrease of the baseline threshold. □

6.5. **Global stability.** We use the sum-of-squares gluing construction of [35, Section 9.2] to combine the capped surface tests with a global ambient partition. The branch energies and the gradient variance are evaluated in the physical weighted form below. On $r>0$, let $n=Z/r$ and $A=A_p(Z)=r^{p-2}(I+(p-2)n\otimes n)$. Set $A=0$ at $r=0$. The physical stability matrix is $A_u=a^{p-2}q^{p-2}A$. We use Modica's inequality and its equality case. We also use sharp clearing, fixed-order relative bounds, clean-ball preparation, and initial-normal propagation with an arbitrarily small rate loss. Lemmas 3.11, 3.12, 6.11, and 6.12 supply the tubes, turn preparation, pressure comparison, and caps. All these inputs were proved without the fourth-moment estimate.

For the global partition, write $H_\Sigma=H$. Keep X,Σ_g,s_X from Section 6, at the small baseline threshold δ to be fixed below. Choose a maximal unit-separated net $\mathcal{X}$ in X and set

$$\chi_R(x)=\exp\left(-\frac{\sqrt{1+|x-x_c|^2}}{R}\right),\qquad N_R=\sum_{x\in\mathcal{X}}\chi_R(x)^2.$$

Use $I=I_P$ from (2.1), with the finite core P selected below. A prescribed finite lower bound for P is allowed. Choose this core before δ, and choose the physical caps to contain it as in Lemmas 6.11 and 6.14. Thus all retained mass terms below use the same window. The uniform comparisons are $c_ID(z)\le I(z)\le C_ID(z)$ and $b^2\le CI$ on good zeros. Keep the envelope J_0 and its source $\bar b$ from Section 6. The cap depth is $L(z)=A_1\log(1/J_0(z))$ and the preparation radius is $A_0\log(1/J_0(z))$. Require $A_0\gg A_1$ and take λA_0 small. The empty-bad-set case has already been classified. Hence this envelope is positive at every base.

Theorem 6.15 (Oriented physical gluing with collar reserve)**.** *For every $\tau>0$, prescribed finite $P_{\min}>0$, and fixed $M>2+L_{\mathrm{bad}}$, one can choose a finite $P\ge\max\{1,P_{\min}\}$, constants $A_1,A_0,\lambda>$*

0, and a threshold $\delta_* > 0$, with the following property. For each $0 < \delta \le \delta_*$, form X, Σ_g, J_0, and a maximal unit-separated bad net at this threshold. There is a finite $L_{\mathrm{gl}}(\delta)$ such that every $L_0 \ge L_{\mathrm{gl}}(\delta)$ is allowed. For each such choice there are positive constants c_1, c_0, C, R_0 for which the following holds. Let $f : \Sigma_g \to [0,1]$ be locally Lipschitz, equal to one on $s_X \le 10L_0$, and have finite exponentially weighted Dirichlet energy. Then for $R \ge R_0$,

$$(1-\tau)\int_{\Sigma_g\cap\{s_X\ge 4L_0\}} \frac{I+b^2}{2} f^2\chi_R^2 + c_1\int_{\Sigma_g\cap\{s_X\le 4L_0\}} D\chi_R^2 + c_0 N_R \le (1+\tau)\int_{\Sigma_g}\chi_R^2|\nabla_\Sigma f|^2 + \frac{C}{R^2}\int_{\Sigma_g}\chi_R^2 f^2 + \varepsilon_{\mathrm{cap}}\int_{\Sigma_g}\chi_R^2 J_0^2. \tag{6.49}$$

The coefficient c_1 has a positive lower bound fixed before δ and L_0. The bad-net coefficient satisfies $c_0 \ge c\delta^{L_{\mathrm{bad}}}$, where $c > 0$ is fixed before δ, L_0 and L_{bad} is the exponent in Lemma 6.13. The cap coefficient has the form

$$\varepsilon_{\mathrm{cap}} = C(C_0\delta)^{M-2}. \tag{6.50}$$

The constants in (6.50) are fixed before δ, L_0, and f. The spatial-cutoff constant in (6.49) and the radius threshold R_0 may depend on δ, L_0, but not on f. In particular, decreasing δ makes the cap coefficient smaller than any prescribed fixed multiple of c_0. This choice does not use the eventual compact multiplier flux. The good set and bad net are formed at the chosen threshold and are then kept fixed when L_0, f, and the exhaustion radius are chosen.

Proof of Theorem 6.15. It is enough to treat $0 < \tau < 1/10$, since decreasing τ strengthens the estimate. If X is empty, Theorem 4.12 makes the solution planar, and the conclusion is immediate. Otherwise X is nonempty, and the nonplanar equality case gives $D > 0$. Fix R for which the displayed weighted Dirichlet integral is finite. If that integral is infinite, the estimate already holds.

Choose $0 \le \vartheta(s_X/L_0) \le \pi/2$, equal to zero for $s_X \le 2L_0$ and to $\pi/2$ for $s_X \ge 3L_0$. Bound its derivative by $O(L_0^{-1})$. Put $\alpha = \cos\vartheta$ and $b_0 = \sin\vartheta$. Choose $\mu(z)$ zero for $s_X(z) \le L_0/2$ and one for $s_X(z) \ge L_0$. Let Θ_z be the cap in Lemma 6.11. Define the branches by

$$T_0(x) = q(x)\chi_R(x)\alpha(x), \qquad T_z(\Phi(z,s)) = h_z(\Phi(z,s))\Theta_z(z,s)\chi_R(z)\mu(z)f(z)b_0(\Phi(z,s)).$$

Sum their squares over the normal preimages and take the square root. Lemma 6.11 gives compatible local definitions and finite fibers, with at most two lifted branches. Use its compact root-sum construction, adding the ambient branch T_0. The continued normal is smooth on each lifted chart. Weighted multiplication therefore preserves the local form domain of h_z. The caps, distance switch, and amplitude f are locally Lipschitz, so their product rules use only first weak derivatives. Lemma 2.10 bounds the assembled form above by the sum of branch forms, and the area formula gives the lifted integrals. Initially multiply the assembled test by a common compact physical cutoff. We keep all geometric factors fixed and approximate this compact test in the natural form norm from Section 2. The global exhaustion is justified at the end of the proof.

The logarithmic cap also localizes the physical switch. The lower bound $J_0(z) \ge e^{-\lambda s_X(z)}$ implies

$$|s| \le 2L(z) \le 2A_1\lambda s_X(z)$$

on supported branches. Since s_X is one-Lipschitz,

$$(1-2A_1\lambda)s_X(z) \le s_X(\Phi(z,s)) \le (1+2A_1\lambda)s_X(z).$$

Choose $2A_1\lambda < 1/10$. If the physical point is in the switch $2L_0 \le s_X \le 3L_0$, these inequalities put its base in

$$S_{\mathrm{sw}} = \{3L_0/2 \le s_X(z) \le 4L_0\}.$$

There $f = \mu = 1$. In the actual lifted kinetic term, the derivative of μ is zero: its whole tube has $b_0 = 0$. In the auxiliary estimate (6.46), this derivative costs only $CL_0^{-2}\int_{s_X\le L_0} D\chi_R^2$. The good-collar reserve absorbs that term. Therefore retain $b_0(x)^2$ in both the kinetic energy and the current

containing $ATZ\cdot\nabla(\chi_R\mu f)$ until the base derivative is split. Its $\nabla\mu$ term then vanishes exactly. Only afterward estimate the ∇f and $\nabla\chi_R$ terms using $b_0^2\le 1$. An unweighted whole-branch Dirichlet estimate for $\chi_R\mu f$ cannot be used before this split.

The chosen switch has gradient supported in

$$\Omega_{\mathrm{sw}}=\{2L_0\le s_X\le 3L_0\},\qquad |\nabla\vartheta|\le C/L_0.$$

For the active normal preimages write

$$\beta_i=\varphi_i\Theta_i,\qquad \varphi_i=\frac{1+Z\cdot e_i}{2},\qquad \mathcal{B}=\sum_i\beta_i^2,\qquad \mathcal{G}=\sum_i\beta_i\nabla\beta_i,\qquad \rho^2=\alpha^2+b_0^2\mathcal{B}.$$

Temporarily use a common physical weight $\chi_R(x)$ for all branches; the difference from $\chi_R(z)$ is handled below. Then $\mu=f=1$ on this region, and the root-sum test is $q\rho\chi_R$. The zeroth-order part of the ground-state identity cancels because $\rho^2=\alpha^2+b_0^2\sum_i\beta_i^2$. Before multiplying by the common spatial weight, the precise remainder is

$$\begin{aligned}&Q(q\rho)-\langle L_uq,q\alpha^2\rangle-\sum_i\langle L_u(q\beta_i),q\beta_ib_0^2\rangle\\&\quad=\int q^2\left[|\nabla\rho|_{A_u}^2-b_0^2\sum_i|\nabla\beta_i|_{A_u}^2-2b_0\mathcal{G}\cdot A_u\nabla b_0\right].\end{aligned}$$

This is first a local identity with compact tests; the common spatial cutoff is restored by the product rule. It follows by expanding each $\langle L_u(q\beta_i),q\beta_ib_0^2\rangle$ using $L_uq=-q^pD$. In detail, the product identity for one branch is

$$\begin{aligned}\langle L_u(q\beta_i),q\beta_ib_0^2\rangle&=\langle L_uq,q\beta_i^2b_0^2\rangle\\&\quad+\int q^2\left(b_0^2|\nabla\beta_i|_{A_u}^2+2b_0\beta_i\nabla\beta_i\cdot A_u\nabla b_0\right).\end{aligned}$$

Summing it and subtracting from $Q_u(q\rho)=\langle L_uq,q\rho^2\rangle+\int q^2|\nabla\rho|_{A_u}^2$ gives the displayed remainder, since the coefficients of L_uq add to ρ^2.

Let $A=A_p(Z)$. Apply the exact ground-state identity to each branch. Dividing the remaining partition term by $a^{p-2}q^p$ gives

$$\mathcal{R}=-b_0^2\sum_i|\nabla\beta_i|_A^2+\frac{b_0^4}{\rho^2}|\mathcal{G}|_A^2-\frac{2b_0\alpha}{\rho^2}\mathcal{G}\cdot A\nabla\vartheta \tag{6.51}$$

$$+\frac{b_0^2\alpha^2(\mathcal{B}-1)^2}{\rho^2}|\nabla\vartheta|_A^2. \tag{6.52}$$

Here

$$\rho\nabla\rho=\alpha b_0(\mathcal{B}-1)\nabla\vartheta+b_0^2\mathcal{G}.$$

Subtracting the branch currents contributes $-2\alpha b_0\mathcal{G}\cdot A\nabla\vartheta$. Combining it with the cross term in $|\nabla\rho|_A^2$ gives $-2\alpha b_0\mathcal{G}\cdot A\nabla\vartheta/\rho^2$, as in (6.51). For its upper bound, matrix Cauchy–Schwarz gives

$$|\mathcal{G}|_A^2\le\mathcal{B}\sum_i|\nabla\beta_i|_A^2,\qquad \frac{b_0^4\mathcal{B}}{\rho^2}\le b_0^2.$$

Thus the first two terms have nonpositive sum. Also $\rho^2-2\alpha b_0\sqrt{\mathcal{B}}=(\alpha-b_0\sqrt{\mathcal{B}})^2\ge 0$, so

$$\frac{2\alpha b_0}{\rho^2}|A\mathcal{G}|\le\frac{2\alpha b_0\sqrt{\mathcal{B}}}{\rho^2}\left(\sum_i|A\nabla\beta_i|^2\right)^{1/2}\le\sum_i|A\nabla\beta_i|.$$

Finally, $b_0^2\alpha^2/\rho^2 \le 1$ and $A \le C\,\mathrm{Id}$. These bounds account for the entire switch square and give

$$\mathcal{R} \le C\left[\sum_i |A\nabla\beta_i||\nabla\vartheta| + (\mathcal{B}-1)^2|\nabla\vartheta|^2\right]. \tag{6.53}$$

Regularize the square root to extend the bound through $\rho = 0$. The upper estimate contains no reciprocal power of α, b_0, or a branch amplitude. The calculation is made on the positive-weight region, using the ordinary weak derivatives there. The form gradients extend it across critical points as in Lemma 2.10. Where $\rho = 0$, every component of the norm has zero form gradient almost everywhere. Thus the regularization has no additional limiting term.

We integrate this remainder over Ω_{sw}, with bases in the already fixed collar S_{sw}. The required estimate is

$$\int_{\Omega_{\mathrm{sw}}} q^p\mathcal{R}\chi_R^2 \le o_{L_0}(1)\int_{S_{\mathrm{sw}}} D(z)\chi_R(z)^2\,dS_z + o_{L_0}(1)N_R + \frac{C_{L_0}}{R}N_R. \tag{6.54}$$

The last term accounts for the difference between physical and base spatial weights. Choose L_0 first and then take R large. This adds no new power to the L_0 budget.

To integrate the switch error, use the good-surface area bound (6.1). It applies to the present set $\Sigma_g = \{z \in \Sigma : D(z) < \delta\}$, with constants fixed by $\mathcal{P}$ and the upper threshold δ_*. For $s \ge 1$, the bound $K(s) \le 1 + Cs$ gives

$$\mathcal{H}^6(\Sigma_g \cap B_s(x)) \le CK(s)s^6 \le C(1+s)^7.$$

For $0 < s < 1$, apply the same estimate on $B_1(x)$ and use $B_s(x) \subset B_1(x)$. Thus this polynomial area bound holds at every radius. Its constant is independent of later decreases of δ and requires no finite-density assumption.

Since $u \in C^1(\mathbb{R}^7)$, the zero set Σ is closed. On Σ, the identity $D = 1 - |\nabla u|^p/a_p^p$ shows that D is continuous, so $X = \{z \in \Sigma : D(z) \ge \delta\}$ is closed. If $X = \varnothing$, the collar and the net are empty, so the collar estimate has both sides zero. Suppose now $X \ne \varnothing$. For each y, a distance-minimizing sequence in X is bounded; a convergent subsequence has its limit in X and gives a nearest point. If $s_X(y) \le C_0L_0$, choose a nearest point of the closed set X and then a point $x \in \mathcal{X}$ within distance less than one of it. Maximal unit separation gives this latter covering property. Hence the collar is contained in $\bigcup_{x\in\mathcal{X}} B_{C_0L_0+1}(x)$. For the spatial weight, put $\psi(y) = \sqrt{1+|y-x_c|^2}$. Then

$$\nabla\psi(y) = \frac{y-x_c}{\sqrt{1+|y-x_c|^2}}, \qquad |\nabla\psi(y)| \le 1.$$

Integrating this derivative on the segment from x to y gives $|\psi(y) - \psi(x)| \le |y-x|$. Since $\chi_R = e^{-\psi/R}$, the weights satisfy

$$\chi_R(y)^2 \le e^{2|x-y|/R}\chi_R(x)^2.$$

Write $r_0 = C_0L_0 + 1$. For $L_0 \ge 1$, $r_0 \le (C_0+1)L_0$. Thus $R \ge (C_0+1)L_0$ makes the weight ratio on each $B_{r_0}(x)$ at most e^2. The ambient volume and the surface bound above give

$$|B_{r_0}(x)| + \mathcal{H}^6(\Sigma_g \cap B_{r_0}(x)) \le \omega_7 r_0^7 + C(1+r_0)^7 \le C_{C_0}L_0^7.$$

Multiplying by $e^2\chi_R(x)^2$ bounds the two weighted integrals on this ball by $C_{C_0}L_0^7\chi_R(x)^2$. The balls $B_{1/2}(x)$ with $x \in \mathcal{X} \cap B_r(y)$ are disjoint and lie in $B_{r+1/2}(y)$. Volume comparison on any finite subcollection gives $\#(\mathcal{X} \cap B_r(y)) \le (2r+1)^7$. Thus the net is locally finite and countable, by exhaustion with integer-radius balls. This bound is independent of δ and L_0. Sum over the net. The integrands are nonnegative, so Tonelli's theorem permits the countable sum and overlap only increases the upper bound. This proves, for $R \ge CL_0$ with a sufficiently large fixed C,

$$\int_{\{s_X \le C_0L_0\}} \chi_R^2\,dx + \int_{\Sigma_g\cap\{s_X\le C_0L_0\}} \chi_R^2\,dS \le C_{C_0}L_0^7N_R. \tag{6.55}$$

The constants depend on the fixed threshold and the charts. They are independent of L_0, R and of the number of sheets.

To separate accurate and inaccurate short rays, fix C_1 with $2C_1 \min_{\mathcal{P}} \beta > 8$. Then set $M_0 = C_1 \log L_0$. Choose $\xi > 0$ small enough for the preceding current tolerances and for

$$2C_1\xi < \frac{1}{2}, \qquad C_1(2\min_{\mathcal{P}} \beta - \xi) > 7. \tag{6.56}$$

Apply Theorem 2.20 with rate $2\beta + \xi/2$. Call a depth-$|s|$ ray accurate if

$$B_z(s) = D(z)e^{(2\beta+\xi)|s|} \le b_\xi$$

where $b_\xi > 0$ is a small fixed number. Propagation keeps D small on the entire segment. The identities

$$D^2U\, Z = -\frac{\nabla D}{p|Z|^{p-2}}, \qquad |D^2U| \le C\sqrt{D}, \qquad |\nabla D| \le CD$$

give, for the fixed initial normal e,

$$|D^2U\, e| \le CD + C\sqrt{D}\,|Z - e|.$$

Integrate and apply Gronwall to obtain

$$|Z - e| + |U - s| + |D^2U\, e| \le C_\xi D(z)e^{(2\beta+\xi/2)|s|}.$$

The integral of $\sqrt{D}$ is at most $C\sqrt{b_\xi}$, so the Gronwall constant is fixed. For the continued Fermi normal, $De\, e = 0$ and $|De| \le Cb(z)$. Thus

$$|\varphi - 1| + |\nabla\varphi| \le C_\xi B_z(s), \qquad q^p \le CG(s)^p. \tag{6.57}$$

To prove uniqueness of the active branch at an accurate point, suppose two caps overlap there. The point belongs to a common fixed turn band $|t| \le T_*$. Its limiting discrepancy is $V_p(t)^{-p}$, and hence

$$d_* := \min_{\substack{p\in\mathcal{P}\\ |t|\le T_*}} V_p(t)^{-p} > 0.$$

Choose preparation accuracy so that the discrepancy on this band is at least $d_*/2$. An accurate ray gives $D(x) \le C_\xi b_\xi$ instead. The choice $b_\xi < d_*/(4C_\xi)$ excludes a second branch, regardless of whether that branch satisfies its own accuracy test. The same lower discrepancy bound holds on the whole supported middle-cap transition. Hence an accurate short point lies outside that transition. Its middle cap is one.

No short switch ray reaches its far cap. To bound the envelope there, put $d = s_X(z)$ and split its defining supremum at $|y - z| = d/2$. For a nearer zero y, one has $s_X(y) \ge d/2$. The fixed-threshold clean-ball estimate and $b^2 \le CD$ give $\bar{b}(y)^2 \le Cd^{-\alpha_{\mathrm{cl}}}$. For a farther zero, the uniform bound for $\bar{b}^2$ gives a contribution at most $Ce^{-\lambda d/2}$. Consequently, on S_{sw},

$$J_0(z) \le Cd^{-\alpha_{\mathrm{cl}}} + Ce^{-\lambda d/2} \le CL_0^{-\alpha_0}, \qquad \alpha_0 = \alpha_{\mathrm{cl}} > 0.$$

This exponent and these constants use the fixed clean-ball threshold, so they are unchanged by later decreases of the baseline δ. Choose $A_1\alpha_0 > 4C_1$ in the cap construction. Then $L(z) = A_1 \log(1/J_0(z)) > 2M_0$ once L_0 is large.

At an accurate short point there is one branch with $\beta = \varphi$. Equation (6.57) gives

$$q^p(|A\nabla\beta| + (\mathcal{B} - 1)^2) \le C_\xi D(z)e^{\xi|s|}.$$

For an inaccurate short point on a supported ray, clearing and the cap distance bound give

$$q^p \le C_\xi e^{-(2\beta-\xi)|s|}, \qquad D(z) > b_\xi e^{-(2\beta+\xi)|s|}.$$

Divide these inequalities to obtain

$$q^p \le C_\xi D(z)e^{2\xi|s|}. \tag{6.58}$$

The only extra rate is the chosen small loss 2ξ. We justify the common off-turn derivative bound. The cap construction gives $r(x_0) \ge r_* > 0$ at every supported point outside the fixed turn bands. To obtain a whole elliptic neighborhood, first use the weak phase equation on arbitrary compact sets,

including critical points. For a compact smooth η, test the original weak equation with $a^{1-p}q^{1-p}\eta$. This is admissible because $q > 0$ on its compact support. Since $\nabla q = -2auqZ$ and $a^{1-p} = 2ac$, the product rule gives

$$\int |Z|^{p-2}Z \cdot \nabla\eta + 2ac \int u(r^p - 1)\eta = 0.$$

Thus $\Delta_p U = 2ac\,u(r^p - 1)$ weakly on the whole space. Modica's bound in Proposition 2.1 gives $|\nabla U| \le 1$. Hence $V = U - U(x_0)$ is bounded by two on $B_2(x_0)$, and its weak equation has a common bounded right side. The bounded-source estimate [28, Theorem 3.2, p. 303] gives a uniform $C^{1,\alpha_{\mathrm{reg}}}$ bound for V on $B_1(x_0)$. Its constant is independent of $U(x_0)$ and of the supported point. Choose a common $r_{\mathrm{ell}} > 0$ so that the resulting gradient modulus gives $r \ge r_*/2$ on $B_{r_{\mathrm{ell}}}(x_0)$.

The shared noncritical bootstrap at the start of Section 2 makes u and U smooth on this ball. There the equation is

$$A_p(\nabla V) : D^2V = 2ac\tanh\big(a[U(x_0) + V]\big)(|\nabla V|^p - 1).$$

Its principal coefficients and right side have common $C^{0,\alpha_{\mathrm{reg}}}$ bounds and uniform ellipticity. All derivatives of the shifted hyperbolic tangent are bounded independently of $U(x_0)$. Interior Schauder estimates on a smaller fixed ball therefore give $|D^2U(x_0)| \le C$. For the continued Fermi normal, $De = B(I + sB)^{-1}$ on tangent vectors and $De\,e = 0$. The cap metric bound $|sB| < \gamma < 1$ gives $|De| \le Cb(z) \le C$. Since $\varphi = (1 + Z \cdot e)/2$, we obtain

$$|\nabla\varphi| \le \tfrac{1}{2}\big(|D^2U| + |De|\,|Z|\big) \le C, \qquad |A\nabla\varphi| \le C.$$

The last step uses $|A_p(Z)\xi| \le (p-1)|\xi|$ for every vector ξ when $r \le 1$. This proves the required bound uniformly off the turn bands. The far cap is inactive. Derivatives of the middle cap occur only in a fixed turn band. Also $|\mathcal{B} - 1| \le C$. Equation (6.58) therefore controls the short inaccurate part of (6.53).

Use the area formula and add the accurate and inaccurate short contributions off the turn bands. Their sum is at most

$$\frac{CM_0}{L_0} e^{2\xi M_0} \int_{S_{\mathrm{sw}}} D(z)\chi_R(z)^2\, dS_z. \tag{6.59}$$

The square term has a further factor L_0^{-1}. Discarding it only enlarges this bound.

The same collar discrepancy controls every supported turn band, including the long ones. Cover the base by small fixed graph patches with bounded overlap. A patch meets at most one turn band on each side. The normal derivative of the middle coordinate is bounded below, and its tangential derivative is bounded above. Its height therefore varies by a bounded amount across each patch. A padded physical cylinder of fixed size contains the corresponding pieces of the turn.

Use the middle deficit b_{turn} defined before Lemma 6.11. On each padded cylinder normalize by $v = (1 - \epsilon_{\mathrm{ch}}u)/b_{\mathrm{turn}}$. The logarithmic well-deficit estimate recorded before Lemma 6.11 gives common bounds for v, v^{-1} and ∇v. The fixed padding lies in the prepared chamber buffer. These bounds are uniform on $\mathcal{P}$.

Writing $b = b_{\mathrm{turn}}$ in this paragraph, the normalized equation is

$$\Delta_p v = F_b(v), \qquad F_b(t) = (1 - bt)t^{p-1}(2 - bt)^{p-1}.$$

Both F_b and F_b' are uniformly bounded on the fixed range of v. Use the local weak Jacobi identity in Proposition 2.5, with reaction $-F_b$. Put $A_v = A_p(\nabla v)$ and $v_i = \partial_i v$. Choose a smooth ζ which is one on the inner cylinder and has compact support in the padded cylinder. That proposition places ζv_i in the natural weighted form space. Multiplication by ζ therefore makes $\zeta^2 v_i$ an admissible test,

including through critical gradients. The weak identity and matrix Cauchy–Schwarz give

$$\int \zeta^2 A_v[\nabla v_i, \nabla v_i] = -2\int \zeta v_i A_v \nabla v_i \cdot \nabla\zeta - \int F_b'(v)\zeta^2 v_i^2$$
$$\le \frac{1}{2}\int \zeta^2 A_v[\nabla v_i, \nabla v_i] + 2\int v_i^2 A_v[\nabla\zeta, \nabla\zeta] + C\int \zeta^2 v_i^2.$$

Absorb the first term and sum over the coordinate directions. The eigenvalue bounds $|\nabla v|^{p-2}I \le A_v \le (p-1)|\nabla v|^{p-2}I$ then give

$$\int \zeta^2 |\nabla v|^{p-2}|D^2 v|^2 \le C\int \left(|\nabla v|^p |\nabla\zeta|^2 + |\nabla v|^2\zeta^2\right) \le C.$$

This invokes the weighted identity through critical gradients, with no ordinary unweighted Hessian assumption. Since $u = \epsilon_{\mathrm{ch}}(1-bv)$, the phase gradient is $Z = -\epsilon_{\mathrm{ch}}\nabla v/[av(2-bv)]$. Put $F(t) = 1/[at(2-bt)]$. Then

$$\mathcal{H} = -\epsilon_{\mathrm{ch}}(F(v)D^2 v + F'(v)\nabla v \otimes \nabla v).$$

The factors F, F' are bounded and $r \asymp |\nabla v|$ on the cylinder. Since $p > 2$ and $|\nabla v|$ is bounded,

$$\mathcal{S} \le Cr^{2p-4}|\mathcal{H}|^2 \le C|\nabla v|^{p-2}|D^2 v|^2 + C|\nabla v|^{2p}.$$

We obtain $\int \mathcal{S} \le C$ for the weighted phase source and

$$|A\nabla\varphi| \le C\sqrt{\mathcal{S}} + C|De|.$$

Cauchy–Schwarz and $q^p \le Cb_{\mathrm{turn}}^p$ imply

$$\int_{\mathrm{patch}} q^p(|A\nabla\beta| + 1) \le Cb_{\mathrm{turn}}^p.$$

All cap derivatives are bounded here. The coarse pressure comparison above gives $b_{\mathrm{turn}}^p \le CD(z)$ at the source; no exact stress identity is required. On the fixed source patch, $|\nabla_\Sigma D| \le CD$ gives $D(z_0) \le CD(z)$ for its chosen reference base z_0 and every base z in the patch. Thus the fixed normalization obeys $b_{\mathrm{turn}}^p \le CD(z)$ throughout that patch. A fixed-radius graph patch has a uniform positive area, so its integral of D controls b_{turn}^p. Bounded overlap and Jacobians yield the total estimate

$$\frac{C}{L_0}\int_{S_{\mathrm{sw}}} D(z)\chi_R(z)^2\, dS_z. \tag{6.60}$$

This uses no pointwise Hessian bound at the turn. Alternatively, clearing bounds a long turn by $Ce^{-(2\beta-\xi)M_0}$ per patch and gives the long-ray estimate below.

Sharp clearing controls the long rays and the switch points with no active branch. First, the cap construction gives $|sB(z)| < \gamma < 1$ on every supported ray. Thus

$$\|(I + sB)^{\pm 1}\| \le C, \qquad 0 < J(z,s) \le C.$$

The bounds do not depend on cap length. Away from turn bands, $|A\nabla\varphi| \le C$. The far-cutoff derivative is $O(1/L(z))$ because $|\nabla_\Sigma L| \le A_1\lambda$. All other cutoff derivatives have fixed bounds. Hence $|A\nabla\beta| \le C$ on the long pieces.

Sharp clearing gives

$$\int_{|s|>M_0} q^p J\, ds \le C_\xi e^{-(2\beta-\xi)M_0}.$$

Absorb its polynomial factor into the fixed exponent loss ξ. Integrating to infinity avoids a factor proportional to $L(z)$. Combine this with (6.55). Even after discarding the helpful switch factors L_0^{-1}, L_0^{-2}, the long contribution outside turn bands is bounded by

$$CL_0^7 e^{-(2\beta_{\min}-\xi)M_0} N_R, \qquad \beta_{\min} = \min_{\mathcal{P}} \beta_p. \tag{6.61}$$

Choose L_0 so large that $M_0 \le L_0/2$ and $L(z) > 2M_0$ on S_{sw}. The latter follows from the preceding envelope bound and $A_1\alpha_0 > 4C_1$. Suppose a switch point x has no active preimage and $\operatorname{dist}(x,\Sigma) \le M_0$. Choose a nearest zero z; it exists because Σ is closed and nonempty. The distance to X is one-Lipschitz, so

$$2L_0 - M_0 \le s_X(z) \le 3L_0 + M_0.$$

The inequality $M_0 \le L_0/2$ puts z in S_{sw}. In particular $s_X(z) > 0$, so $z \notin X$ and $z \in \Sigma_g$. Thus this nearest zero is regular. Orthogonality at this regular nearest foot gives $x = z + s\nu(z)$ with $|s| \le M_0 < L(z)$. The nearest-foot coverage property of Lemma 6.11 therefore gives $\Theta_z(x) = 1$. Nonplanarity and the equality case of Modica's bound give $D(x) > 0$. Thus $|Z(x)| < 1$ and $\varphi_z(x) = (1 + Z(x)\cdot\nu(z))/2 > 0$. This zero supplies a nonzero capped branch, a contradiction. Every uncovered switch point is therefore farther than M_0 from Σ. Clearing and the ambient volume bound in (6.55) control its $(\mathcal{B}-1)^2 q^p$ contribution by (6.61).

The rate choices in (6.56) give

$$\frac{CM_0}{L_0} e^{2\xi M_0} \le C(\log L_0) L_0^{-1+2C_1\xi} \longrightarrow 0,$$

for short rays and

$$CL_0^7 e^{-(2\beta_{\min}-\xi)M_0} = CL_0^{7-C_1(2\beta_{\min}-\xi)} \longrightarrow 0.$$

for long rays. Add the turn-band estimate to prove (6.54). Increase L_0 while keeping the baseline bad set fixed. These choices work on every compact interval in $(2,\infty)$. No finite upper restriction on the exponent is needed.

We now replace the common physical weight by the base weights. The following exact calculation keeps the signed Hessian terms unchanged. On an inverse normal chart, set

$$w_i(x) = \frac{\chi_R(z_i(x))}{\chi_R(x)}, \qquad \widetilde{\beta}_i = w_i\beta_i, \qquad \widetilde{\mathcal{B}} = \sum_i \widetilde{\beta}_i^2.$$

On the switch, $|x - z_i(x)| \le CL_0$, while the derivative of the projection $x \mapsto z_i(x)$ is uniformly bounded by the Fermi metric estimate. Hence, for $R \ge CL_0$,

$$|\log w_i| \le CL_0/R, \qquad |w_i - 1| \le CL_0/R, \qquad |\nabla w_i| \le C/R.$$

The actual root-sum is exactly $q\chi_R\sqrt{\alpha^2 + b_0^2\widetilde{\mathcal{B}}}$. Apply (6.51) with $\widetilde{\beta}_i$ in place of β_i. The two nonpositive terms remain nonpositive. For the remaining upper bound,

$$\begin{aligned}
\sum_i |A\nabla\widetilde{\beta}_i| &\le (1 + CL_0/R)\sum_i |A\nabla\beta_i| + C/R,\\
|\,\widetilde{\mathcal{B}} - \mathcal{B}\,| &\le CL_0/R,\\
(\widetilde{\mathcal{B}} - 1)^2 &\le (\mathcal{B}-1)^2 + CL_0/R.
\end{aligned}$$

Here at most two branches occur and $0 \le \beta_i \le 1$. The turn estimate and the off-turn bound give

$$\int_{\Omega_{\mathrm{sw}}} q^p \chi_R^2 \left(1 + \sum_i |A\nabla\beta_i|\right) \le C_{L_0} N_R.$$

Indeed, integrate the off-turn part along the normal rays using clearing, and use the fixed-patch estimate at turns; then apply (6.55). The displayed perturbations therefore cost at most $C_{L_0}R^{-1}N_R$.

For completeness, the product rule accounts for the spatial gradients as well. For any branch v and Lipschitz factors w, k, the exact identity is

$$\langle L_u(vw), vwk^2\rangle - \langle L_u v, vw^2k^2\rangle = \int v^2\left(k^2|\nabla w|^2_{A_u} + 2wk\,\nabla w\cdot A_u\nabla k\right).$$

It follows by writing $Q(vwk)$ in the two ways that first differentiate v and that first differentiate vw. Take $v = q\beta_i$, $w = w_i$, and $k = \chi_R b_0$. The second pairing on the left has exactly the desired

base weight $\chi_R(z_i)^2$. Since $v^2 A_u \le Cq^p\,\mathrm{Id}$, the added integrand on the switch is bounded by $Cq^p\chi_R^2(R^{-2}+(RL_0)^{-1})$. It has the required bound by (6.55).

There is one common-weight cross term in the norm formula. If $\widetilde{\rho}^2=\alpha^2+b_0^2\widetilde{\mathcal{B}}$, it is

$$2q^2\chi_R\alpha b_0(\widetilde{\mathcal{B}}-1)\nabla\vartheta\cdot A_u\nabla\chi_R+q^2\widetilde{\rho}^2|\nabla\chi_R|_{A_u}^2.$$

This follows from $\widetilde{\rho}\nabla\widetilde{\rho}-b_0^2\sum_i\widetilde{\beta}_i\nabla\widetilde{\beta}_i=\alpha b_0(\widetilde{\mathcal{B}}-1)\nabla\vartheta$. It has the same bound. These are identities for local form densities; we split their integrals at the switch without a new integration by parts on its boundary. Outside the switch the original branch estimate uses $\chi_R(z)$ directly. This proves the last term in (6.54). Choose L_0 first and R afterward; no uniform bound on C_{L_0} as $L_0\to\infty$ is needed.

We still need the physical factor $b_0(x)^2$ in the branch identity. Freezing it at the base over the fixed core costs $C_P L_0^{-1}\int_{S_{\mathrm{sw}}} D\chi_R^2$. Outside the core, apply (6.46) to the $|T|\sqrt{\mathcal{S}}$ and $|T|^2$ terms. Do not differentiate b_0 here. Its absolute value is at most one, so the whole-ray error is a small multiple of $\int D_0\mu^2 f^2\chi_R^2$ plus the allowed Dirichlet and cap costs. The exterior mass and good-collar reserve provide this defect mass everywhere. Split the base derivative of $\chi_R\mu f$ while retaining b_0^2. There is then no actual $\nabla\mu$ charge. Differentiating b_0 in (6.45) adds the explicit current

$$a^{p-2}\int q^p b_0\varphi\eta^2 ATZ\cdot\nabla b_0.$$

Denote this current by $\mathfrak{C}_{b_0}$. Its support is Ω_{sw}, where $|\nabla b_0|\le C/L_0$. Every supported preimage satisfies $|s|\le 2A_1\lambda s_X(z)$. The preceding comparison of physical and base distances therefore puts its base in S_{sw}. In particular $\mu(z)=f(z)=1$ there. During a compact exhaustion we may take

$$\eta_z(\Phi(z,s))=\Theta_z(z,s)\chi_R(z)\mu(z)f(z)\kappa_K(\Phi(z,s)),\qquad 0\le\kappa_K\le 1.$$

Thus $\eta_z^2\le\chi_R(z)^2$ on this support. The Fermi Jacobian is bounded, and at most two branches cover a physical point. Moreover $|s|\le CL_0$ here, so the physical and base weights are comparable when $R\ge CL_0$. The area formula reduces the absolute current to

$$|\mathfrak{C}_{b_0}|\le\frac{C}{L_0}\int_{S_{\mathrm{sw}}}\chi_R(z)^2\int_{\Phi(z,s)\in\Omega_{\mathrm{sw}}}q^p|ATZ|J(z,s)\,ds\,dS_z.$$

All constants in this reduction are independent of K.

At an accurate short point, $Te=0$ and the propagated estimate above give

$$|ATZ|\le Cb(z)D(z)e^{(2\beta+\xi/2)|s|},\qquad q^p\le Ce^{-2\beta|s|}.$$

Since $b(z)$ is uniformly bounded, their product is at most $C_\xi D(z)e^{2\xi|s|}$. At an inaccurate short point outside the turn bands, use (6.58) and $|ATZ|\le Cb(z)\le C$. Integrating over $|s|\le M_0$ bounds these contributions by

$$\frac{CM_0}{L_0}e^{2\xi M_0}\int_{S_{\mathrm{sw}}}D(z)\chi_R(z)^2\,dS_z.$$

For the short turn pieces, use the fixed padded patches in the weighted Caccioppoli calculation leading to (6.60). On such a patch $q^p\le Cb_{\mathrm{turn}}^p$ and $|ATZ|\le C$. The pressure comparison gives $b_{\mathrm{turn}}^p\le CD(z)$ on its source patch. The base weights vary by a bounded factor there. The positive fixed patch area and bounded overlap therefore give the additional bound

$$\frac{C}{L_0}\int_{S_{\mathrm{sw}}}D(z)\chi_R(z)^2\,dS_z.$$

This uses only the weighted turn calculation and its normalized deficit bounds. No pointwise phase Hessian is needed at a turn.

On every supported long piece, including its turn bands, $|A| \le C$, $|Z| \le 1$ and $|T| \le Cb(z) \le C$. The product ATZ has the same bounded extension at a critical gradient, since $p > 2$. Sharp clearing and the cap distance bound therefore give directly

$$\int_{|s|>M_0} q^p |ATZ| J \, ds \le C_\xi e^{-(2\beta-\xi)M_0}.$$

Integrating to infinity removes any dependence on the cap length. Apply (6.55) to the lifted base weights. Combining all pieces yields

$$|\mathfrak{C}_{b_0}| \le C \left(\frac{M_0}{L_0} e^{2\xi M_0} + \frac{1}{L_0} \right) \int_{S_{\mathrm{sw}}} D(z) \chi_R(z)^2 \, dS_z + \frac{C}{L_0} L_0^7 e^{-(2\beta_{\min}-\xi)M_0} N_R.$$

With $M_0 = C_1 \log L_0$, the first coefficient tends to zero by $2C_1\xi < 1/2$. The net coefficient tends to zero by $C_1(2\beta_{\min} - \xi) > 7$. Hence this current is an $o_{L_0}(1)$ collar-defect and net charge. The estimates are uniform in the compact cutoff and pass to the finite form-norm exhaustion used below. The ground-state identity for $h\Theta$ also contains the cap-partition cross

$$2 \int b_0 \Theta h^2 A_u \nabla\Theta \cdot \nabla b_0.$$

At a post-turn cap its absolute value is at most $C_T L_0^{-1} \mathcal{P}_{\mathrm{gap}}$ per base. This is a small collar-defect charge. There is no far cap on short switch rays. On long rays, the same exponentially small net estimate controls this cross directly. No Young split leaving an unweighted partition square is needed. This accounts for every physical normal partition current.

The ambient branch is unmodified on the disjoint bad-net balls, where $\alpha = 1$. Lemma 6.13 gives the raw mass $c\delta^{L_{\mathrm{bad}}} N_R$ there. To make the collar reserve uniform, fix a normal depth $\rho > 0$ independently of L_0. Choose it so that the tubes $|s| < \rho$ over all good zeros are jointly injective, with Jacobians bounded above and below. This is possible by Lemma 6.1 after fixing a sufficiently small good-zero threshold. On these tubes the core comparisons give

$$q^{p+1} D \asymp D_0, \qquad \varphi \ge \tfrac{1}{2}, \qquad \Theta = 1.$$

To check joint injectivity, take the fixed two-sided tangent-ball radius $r_{\mathrm{tb}} > 2\rho$ furnished by that lemma. For any other zero z', its chord bound gives

$$|z + s\nu(z) - z'|^2 \ge s^2 + (1 - |s|/r_{\mathrm{tb}})|z - z'|^2.$$

Thus z is the unique nearest zero when $|s| < \rho$. This proves joint injectivity, as well as local nonsingularity. For bases in the collar, $f = 1$. Whenever b_0 is nonzero on such a core, its base has $s_X > L_0$, so also $\mu = 1$. Thus the coefficient of the raw defect density supplied by the ambient and lifted branches is bounded below by

$$\alpha(x)^2 + \tfrac{1}{2} b_0(x)^2 \varphi(x) \ge \tfrac{1}{4} \{\alpha(x)^2 + b_0(x)^2\} = \tfrac{1}{4}.$$

The lifted and physical spatial weights differ by a fixed bounded factor on this core, since $|s| < \rho$ and $|\nabla \log \chi_R| \le R^{-1}$. Reducing the numerical reserve preserves a uniform positive lower bound. Where $\mu \ne 1$, the whole core has $b_0 = 0$ and the ambient term alone gives the same bound. Integration over $|s| < \rho$ therefore supplies a fixed multiple of $D_0 \chi_R(z)^2$ per unit base area, without a collar-width factor.

Allocate the negative sources pointwise before integrating. Choose $\epsilon > 0$ small relative to τ before any cap or collar choice. Earmark an ϵ-fraction of each nonnegative raw source density for the good collar. Earmark a second ϵ-fraction of the ambient density for the bad balls. At least a

$(1-2\epsilon)$-fraction of each density remains, even where the two reserve regions overlap. Thus each allocated part has its own budget. The fixed-core and bad-ball estimates give

$$c_1 \geq c\epsilon\rho > 0, \qquad c_0 \geq c\epsilon\delta^{L_{\mathrm{bad}}}.$$

Uniform upper and lower bounds for σ_q are included in these constants. In particular, c_1 is independent of L_0.

Only a 2ϵ-fraction of the raw $I/2$ mass has been used; the curvature part of the exterior core mass is unchanged. Choose ϵ small enough relative to τ and then make the core and tail errors small enough that the remaining exterior coefficient is at least $(1-\tau)(I+b^2)/2$. For $s_X(z) \geq 4L_0$, the whole cap satisfies $s_X(x) \geq (1-2A_1\lambda)4L_0 > 3L_0$, so $b_0 = 1$ there. Thus its core has precisely the exterior coefficient. The same allocation works on the level set $s_X = 4L_0$ even if that level has positive surface measure: it supplies both the exterior term and the separately reserved collar term. This proves that c_1 can be fixed before L_0, as required.

First fix the relative tolerances and the finite constants for one-neighbor pressure comparison. Fix the tangential Young factors. Then choose the post-turn offset and preparation accuracy. Use $\mathcal{P}_{\mathrm{gap}} \leq CD_0$, with C fixed before the offset, to place cap costs below the required D_0 tolerances. Fix C_1 and the loss ξ in (6.56). Also make the losses small enough for the reciprocal-weight decay estimates above. The offset and these losses now fix the constants in the whole-ray estimates. Choose their Young tolerances and then enlarge the normal core to control weighted-current and kinetic tails. Require also $P \geq \max\{1, P_{\min}\}$. Every core restriction here is a lower bound: the profile fractions increase to one, and the weighted-current and kinetic tails tend to zero. The relative estimates are then applied on this fixed core by decreasing their discrepancy threshold. The same core can be used in Lemma 6.14, whose proof has these same lower-window restrictions. The constants may depend on the selected finite P; no estimate at all larger windows is asserted. With the prescribed $M > 2 + L_{\mathrm{bad}}$, choose A_1 large enough for this cap exponent and for $A_1\alpha_0 > 4C_1$. Next choose $A_0 \gg A_1$. Finally take λ small enough for angular preparation and $2A_1\lambda < 1/10$.

The two-radius envelope constants and the radius R_{env} in Lemma 7.2 are now fixed independently of δ, L_0, f. That lemma uses only relative regularity, the regular-chart area bound, and exponential convolution. These inputs are already available here; its proof does not use global gluing or moment closure. Choose $\delta_* > 0$ so that every $0 < \delta \leq \delta_*$ satisfies the core and preparation bounds. Also require $C(C_0\delta)^{M-2}$ to be below the prescribed tolerances and the required multiple of $\delta^{L_{\mathrm{bad}}}$. For each such threshold, construct its bad set, its maximal unit-separated net, and its envelope. These objects are constructed anew if the threshold changes. Their estimates use the fixed clean-ball threshold and the same chart constants, so the preceding bounds hold uniformly for all these choices.

The choice of the cap power has a precise role. For fixed earlier constants and a fixed allocated source fraction $\epsilon > 0$,

$$\frac{C(C_0\delta)^{M-2}}{c\epsilon\delta^{L_{\mathrm{bad}}}} = \frac{CC_0^{M-2}}{c\epsilon}\delta^{M-2-L_{\mathrm{bad}}} \longrightarrow 0.$$

The strict inequality for the cap power makes its error smaller than a prescribed fraction of the bad reserve. No constant in this ratio changes when the threshold decreases.

Keep this bad set fixed and enlarge L_0. The envelope bound on supported bases follows from two separate contributions. Where $\mu \neq 0$, one has $s_X(z) \geq L_0/2$. Good zeros satisfy $\bar{b}(y)^2 = b(y)^2 \leq C\delta$, with the fixed relative-curvature constant. Bad zeros satisfy $\bar{b}(y) = 1$ and $|y-z| \geq s_X(z)$. Hence

$$J_0(z) \leq \max\{C\delta, e^{-\lambda L_0/2}\}.$$

The constant $C_0 = \max\{C, 1\}$ is fixed before δ. Choose L_0 also so large that $e^{-\lambda L_0/2} \leq \delta$. Then $J_0 \leq C_0\delta$ on supported bases. All logarithmic windows are prepared. Every sufficiently large L_0 now works. Indeed, the short-ray coefficient is bounded by $C(\log L_0)L_0^{-1+2C_1\xi}$, the turn and fixed-core freezing coefficients tend to zero like C/L_0 and C_P/L_0, and the long-ray net coefficient

is bounded by $CL_0^{7-C_1(2\beta_{\min}-\xi)}$. The exponent in the first bound is negative, and that in the last bound is negative by (6.56). After fixing δ, choose $L_{\mathrm{gl}}(\delta)$ so that all these coefficients are below their assigned collar and bad-net reserves for every $L_0 \geq L_{\mathrm{gl}}(\delta)$. Include also $e^{-\lambda L_0/2} \leq \delta$, $M_0 \leq L_0/2$, and the fixed preparation bounds in this lower threshold. The positive reserve fractions remain fixed as L_0 increases. Take $R_0 \geq R_{\mathrm{env}}$ large afterward to absorb $C_{L_0}R^{-1}N_R$ and the ambient spatial-cutoff cost. Thus R_0 may depend on the selected δ, L_0. The amplitude f may be chosen last. No constant from its completed weight affects an earlier parameter.

The total far-cap cost is $C\int J_0^M\chi_R^2\mu^2 f^2$. On its support, $0 \leq \mu f \leq 1$ and $J_0 \leq C_0\delta$, so

$$J_0^M\mu^2 f^2 \leq (C_0\delta)^{M-2}J_0^2.$$

This gives the final term of (6.49), with the constant fixed before δ. Expanding the lifted spatial cutoff gives $CR^{-2}\int_{\Sigma_g}\chi_R^2 f^2$. The ambient cutoff also has a cost. Since $q^2A_u \leq Cq^p\,\mathrm{Id} \leq C\,\mathrm{Id}$ and α is supported in $s_X \leq 3L_0$, (6.55) bounds that cost by $CL_0^7R^{-2}N_R$. Take R_0 large after δ, L_0 so that this term and $C_{L_0}R^{-1}N_R$ use only a fixed fraction of the bad-net reserve. The remaining coefficient still satisfies $c_0 \geq c\delta^{L_{\mathrm{bad}}}$. The actual kinetic contribution from $\nabla\mu$ vanishes. Its auxiliary D-weighted cost is absorbed by the collar reserve. The only remaining base derivative is $(1+\tau)\int\chi_R^2|\nabla_\Sigma f|^2$.

To justify the global stability limit, apply scalar stability first to compact assembled tests. The polynomial area bound makes their exponentially weighted zeroth-order norms finite. The extra factor $L(z)$ grows at most linearly, so the lifted zeroth-order integral is finite too. We verify the global gradient norm directly from the Jacobi equation. Fix $x_* \in X$ and put $a_* = 2A_1\lambda < 1/10$. On a supported fiber, $|s| \leq a_*s_X(z) \leq a_*|z - x_*|$. Thus $\chi_R(z)$ is bounded by a constant times a slower ambient exponential $\eta(x) = e^{-\kappa\sqrt{1+|x-x_*|^2}}$, with $0 < \kappa < ((1+a_*)R)^{-1}$. For each Cartesian derivative u_i, test its weak Jacobi equation with $u_i\eta^2$, first adding a compact cutoff. The bounds for u_i, A_u, and W_p'' and Young's inequality give

$$\int\eta^2|\nabla u_i|_{A_u}^2 \leq C\int(\eta^2 + |\nabla\eta|^2) < \infty.$$

The compact cutoffs are removed by Fatou's lemma. Since $|De| \leq Cb \leq C$ on supported branches, the product rule for $u_e = e_iu_i$ controls its weighted gradient by these integrals and a zeroth-order term. The same holds for $h = (q + a^{-1}u_e)/2$. Finite fiber multiplicity permits summation over all branches. For the lifted derivative of f, the bounded Fermi metric gives

$$h_z^2|\nabla(f\circ\pi_z)|_{A_u}^2 \leq Cq^p|\nabla_\Sigma f(z)|^2,$$

where π_z is the own normal projection. Clearing gives $\int q^pJ\,ds \leq C$ on supported rays, so the area formula bounds its cost by $C\int_{\Sigma_g}\chi_R^2|\nabla_\Sigma f|^2$. The other amplitude derivatives are bounded and have integrable exponential weights. Hence the assembled test has a finite global weighted gradient norm. These bounds establish admissibility; they do not change the coefficients in the quantitative estimate above.

There is also no omitted boundary trace at a bad zero or at the end of a normal chart. The factor μ is zero on an open collar of X. On its nonzero support $J_0 < j_*/2$, so the cap construction takes place strictly inside its allowed envelope region. Each cap is supported inside an enlarged open nonsingular tube and vanishes at its outer boundary. Local zero extensions and products are therefore precisely the form-domain operations supplied by the previous weighted calculus. For the exhaustion, choose common physical cutoffs with derivatives bounded by C/K. Write the assembled test as $\mathcal{T}$ and the common compact cutoff as κ_K.

Then

$$\|\mathcal{T} - \kappa_K\mathcal{T}\|_{L^2} \longrightarrow 0,$$

and the product rule gives

$$\int |\nabla(\mathcal{T}-\kappa_K\mathcal{T})|^2_{A_u} \le 2\int_{\{|x|>K\}} |\nabla\mathcal{T}|^2_{A_u} + \frac{C}{K^2}\int_{\{|x|>K\}} \mathcal{T}^2 \longrightarrow 0.$$

The first limit follows from absolute continuity of an integrable function; the second uses $A_u \le C\,\mathrm{Id}$ and $\mathcal{T}\in L^2$. Thus the compact tests converge in the natural form norm. The potential term also converges, since $W_p''(u)$ is bounded. Closedness and density pass stability to the assembled limit. There is no residual energy from the artificial outer boundary. For example, its mixed term is bounded by

$$2\left(\int_{\{|x|>K\}} |\nabla\mathcal{T}|^2_{A_u}\right)^{1/2}\left(\int \mathcal{T}^2|\nabla\kappa_K|^2_{A_u}\right)^{1/2} \longrightarrow 0,$$

and the pure cutoff term tends to zero by the same bounds. The retained surface masses are integrable because D, b^2, I, and J_0 are bounded on good zeros and the weighted surface area is finite. The net sum is finite by ambient packing. The current terms are absolutely integrable by the reciprocal and turn estimates above. Dominated convergence therefore passes the estimated integrals to the same limit; nonnegative masses could also be passed by Fatou's lemma.

All geometric parameters and R stay fixed during this limit. There is no need to smooth f, the distance function, and the caps independently; the local Lipschitz product rules already give their form-domain test. Every retained mass and error coefficient is therefore unchanged. This proves (6.49).

□

7. Curvature moments

The fourth-moment and reach-multiplier argument adapts [35, Section 9]. We use the weighted physical forms and discrepancy estimates proved above. Fix $\mathcal{P} \Subset (2,\infty)$ and keep the notation of Section 2. Write $G = G_p$ for the profile and $I = I_P$ for the normal mass. The common window is chosen below. As in the preceding section, $n = 7$ and $m = 6$. Constant wells and planar layers satisfy all estimates below, so we exclude them. The constants are uniform on $\mathcal{P}$.

Use clean-ball preparation with the fixed exponent $1 < \alpha < 2\gamma < 2$ from (4.74). The same constants work for every sufficiently small baseline threshold. Namely, $X = \{u = 0, D \ge \delta\}$ and $s_X(z) \ge L$ imply $D(z) \le CL^{-\alpha}$. At good zeros, relative regularity gives

$$b^2 + |H| \le CD, \qquad |\nabla_\Sigma B| \le C\sqrt{D}.$$

Choose the mean, curvature–mass, and physical tolerances first, including the fixed finite-window tolerances for radial interpolation. That subsection's local mass and capacity calculations assume a prepared gap and use only the earlier analytic and chamber estimates. Their finite window and recovery accuracies do not depend on the fourth-moment conclusion. The radial argument later uses that conclusion to find bases satisfying those local hypotheses.

Here are sufficient numerical choices for the two physical-form margins. Write η_{ph} for the larger of the final tolerances in (6.48) and (6.49), after relabeling their initial accuracies. Set

$$\theta = \frac{1}{m} = \frac{1}{6}, \qquad t = \frac{\theta}{16}, \qquad \omega = 10^{-3}, \qquad 0 < \eta_{\mathrm{ph}} \le \min\left\{\frac{\theta}{64(1+\theta)}, \frac{\omega}{16}, \frac{1}{100}\right\}.$$

The Codazzi–Kato calculation in Section 7.2, with $\epsilon_K = m = 6$, gives

$$|\nabla B|^2 \ge (1+\theta)|\nabla b|^2 - \frac{11}{36}|\nabla H|^2, \qquad \theta = \frac{1}{6}.$$

This applies to the smooth regular zero hypersurface with its Euclidean induced metric. The mean-gradient error is retained. Lemma 7.1 chooses its averaging radius before its local defect threshold.

The clean-collar argument in Proposition 7.4 then assigns this error any fixed small tolerance. Its auxiliary collar uses the same bad set and net. For its stability test these choices give

$$\frac{(1+\eta_{\mathrm{ph}})(1+t)}{1-\eta_{\mathrm{ph}}} \le 1+\frac{\theta}{8} < 1+\frac{\theta}{4}.$$

Indeed, $(1+\eta)/(1-\eta) \le 1+4\eta$ for $\eta \le 1/2$. The margin relative to the coefficient $1+\theta/4$ used in the Simons argument is therefore at least $\theta/8 > 0$. For the reach multiplier choose $\varepsilon_I \le 1/100$ and fix the mean tolerances so that $\epsilon_{\mathrm{mean}} \le \omega/16$. Later choose $\epsilon_\delta = CC_I\delta \le \omega/16$ and the collar derivative cost at most $\omega I/8$. The factor $\Lambda_\delta = (1+\eta_{\mathrm{ph}})/(1-\epsilon_\delta)$ then satisfies $\Lambda_\delta - 1 < \omega/4$. The algebra in (7.32), together with the cutoff-error accounting in the proof of Proposition 7.7, retains an exterior coefficient at least $.196 - 7\omega/8$. Its margin above the required 4ω is

$$\frac{49}{250} - \frac{39}{8}\omega > 0.$$

The curvature allocation in Lemma 7.6 uses only the indicated quarter of the small-contact drift term and leaves the separate $Y_R/2$ reserve. It causes no further loss in this exterior budget. Thus the two physical tolerances can be chosen at the same time. These numerical choices and the compact-interval lower bound for κ are uniform on $\mathcal{P}$ and precede the common window and final threshold.

Take the cap power larger than $2 + L_{\mathrm{bad}}$, where L_{bad} is the fixed power in the positive bad-net coefficient. The mean and mass-comparison requirements give a finite lower bound $P_{\min}$ for the common window. Use this lower bound in the initial choice of Theorem 6.15. Its proof permits the selected core to be used also in Lemma 6.14. The cofinal-window statement of Theorem 3.7 applies at this selected fixed window. Its proof combines the pointwise nonlinear estimate and Lemma 3.6 at that same window. Decrease their common local discrepancy threshold before choosing the baseline. The lower curvature–mass comparison persists because the mass integrand is nonnegative. For this step, include the window P_{mass} supplied by Corollary 3.5 in the initial $P_{\min}$, and retain its threshold δ_{mass} in the final baseline choice. The identical-normal integral comparison and the final fixed-P upper constant are checked with (7.12). Fix the resulting weighted mean constant in (3.16), the physical accuracies, logarithmic cap constants, and envelope rate $\lambda > 0$ before selecting the final baseline threshold. Fix also the corresponding radius R_{env} from Lemma 7.2 at this stage. Every later outer-radius threshold is taken at least R_{env}. The later use of gluing will make this joint choice explicit.

7.1. **Mean derivatives and curvature envelopes.**

Lemma 7.1 (Averaged mean derivative)**.** *For every $\eta > 0$ there are $S \ge 10$ and $\delta > 0$ such that, if $u(z) = 0$ and $D(z) < \delta$, the zeros in $B_{4S}(z)$ lie on one regular graph over the tangent disk $B^m_{4S}(z)$, with slope at most $1/10$, and*

$$|\nabla_\Sigma H(z)|^2 \le \frac{\eta}{S^m}\int_{B_S(z)\cap\Sigma} D(w)^2\, dS_w.$$

The radius is chosen first; the defect threshold is chosen afterward.

Proof. We use the averaging argument of [35, proof of Lemma 8.2], with the normalized limits and normal potential of the matched equation. Use the normalized-limit construction in the proof of Theorem 3.4. For a sequence with $0 < \epsilon_j = D_j(0) \to 0$, take a subsequence with $p_j \to p \in \mathcal{P}$. Translate the zeros to the origin and rotate their normals into e_s. These entire matched solutions take values in $[-1, 1]$. Their centered zeros exclude both constant wells. Remark 2.2 therefore gives $-1 < u_j < 1$ everywhere. This verifies the strict-range hypothesis of the normalized-limit construction. Use the phase $U_j = a_{p_j}^{-1} \operatorname{arctanh} u_j$ with its actual exponent. The construction gives

smooth local limits

$$U_j \to s, \qquad D_j/\epsilon_j \to \rho > 0, \qquad D^2U_j/\sqrt{\epsilon_j} \to T(y), \quad Te_s = 0.$$

The tangential tensor T is the Hessian of a harmonic function. The positive function $F = G^{p/2}\rho$ solves

$$\left[-\Delta_y - c\partial_{ss} + ca^2\{p^2 - p(p-1)G(s)\}\right] F = pG^{p/2}|T(y)|^2.$$

The normal potential is bounded below by $ca^2p > 0$, uniformly on $\mathcal{P}$. Separate the even source contribution from the homogeneous part as in Theorem 3.1. Write $\mathscr{L}_D$ for the displayed operator. On $\Omega_k = (-k,k)^m \times (-k,k)$, let $F^{(k)}$ solve the zero-boundary problem with source $pG^{p/2}|T|^2$. The index k denotes this exhaustion, not the nonlinear sequence. If $V(s) = ca^2\{p^2 - p(p-1)G(s)\}$, the bilinear form is

$$\mathfrak{a}_k(v,w) = \int_{\Omega_k} (\nabla_y v \cdot \nabla_y w + c\, v_s w_s + V vw).$$

It is coercive on $H_0^1(\Omega_k)$ because $c > 0$ and $V \geq ca^2p > 0$. The source is smooth and bounded on each fixed box, so the Dirichlet problem has a unique weak solution. Testing with the negative part proves $F^{(k)} \geq 0$. The difference $F^{(k)} - F$ solves the homogeneous equation and has nonpositive boundary trace. Testing with its positive part proves $F^{(k)} \leq F$. The same comparison on Ω_k gives $F^{(k)} \leq F^{(k+1)}$, since $F^{(k+1)} \geq 0$ on its boundary. These arguments use Sobolev traces on the boxes and do not require smoothness at their corners.

Their increasing limit F_{src} is locally finite and smooth by interior estimates. The operator, source, and boxes are even in s, so $F_{\mathrm{src}}(y,-s) = F_{\mathrm{src}}(y,s)$. Both $(F_{\mathrm{src}})_s$ and $\nabla_y(F_{\mathrm{src}})_s$ therefore vanish at $s = 0$.

Subtract the source part by putting $F_{\mathrm{hom}} = F - F_{\mathrm{src}}$. It is nonnegative and solves the homogeneous equation. If it vanishes identically, the derivative estimate below is immediate. Otherwise the strong maximum principle makes it positive, and we normalize it at the origin. The normalized positive homogeneous solutions form a compact convex set in the locally smooth topology. Indeed, Harnack chains bound each solution on every fixed compact set by its value at the origin. Interior estimates then bound all derivatives there. A diagonal subsequence converges locally smoothly, and its limit has value one at the origin. The strong maximum principle makes the limit positive. Thus the normalization is preserved under limits. The extreme-ray argument in the proof of Theorem 3.1 applies to this normalized cone, with no growth hypothesis. Its required horizontal translation comparison follows from Harnack chains with constants independent of the base point: the principal matrix is constant and the bounded potential depends only on s. Its positive extremals have the form $e^{\xi\cdot y}h(s)$. We need the spectral and derivative bounds in the physical normal coordinate. Put $x = as$ and $f(x) = h(x/a)$. The separated equation becomes

$$-f'' + \{\mu^2 - p(p-1)\operatorname{sech}^2 x\}f = 0, \qquad \mu^2 = p^2 - \frac{|\xi|^2}{ca^2}.$$

Apply the compact-test ground-state identity from the proof of Theorem 3.1. Positivity of f makes the form of $-\partial_x^2 + \mu^2 - p(p-1)\operatorname{sech}^2 x$ nonnegative, with no growth condition on f. Factorization (3.3) has the zero mode $w(x) = \operatorname{sech}^{p-1} x$. Choose ϑ_T equal to one on $[-T,T]$, zero outside $[-2T,2T]$, and with $|\vartheta_T'| \leq C/T$. The first-order factor annihilates w, so its value on $\vartheta_T w$ is exactly $\vartheta_T' w$. The compact-test calculation therefore gives

$$0 \leq \int_{\mathbb{R}} |\vartheta_T' w|^2\, dx + [\mu^2 - (p-1)^2] \int_{\mathbb{R}} \vartheta_T^2 w^2\, dx.$$

The first term is at most $CT^{-2}\int w^2$ and tends to zero. The second integral tends to $\int w^2 > 0$, since $p > 1$. Letting $T \to \infty$ gives $\mu^2 \geq (p-1)^2$. Because $\mu^2 = p^2 - |\xi|^2/(ca^2)$, this means $|\xi|^2 \leq ca^2[p^2 - (p-1)^2] = ca^2(2p-1)$. The decay of w and the positive lower bound for its squared integral are uniform when p ranges over $\mathcal{P}$.

For the derivative bound, the normal equation gives $h'' = [ca^2\{p^2 - p(p-1)G\} - |\xi|^2]h/c \le (pa)^2 h$. Put $k = pa$ and apply variation of constants to $\widetilde{h}(t) = h(-t)$. Its initial derivative is $-h'(0)$, and the forcing $\widetilde{h}'' - k^2\widetilde{h}$ is nonpositive. The kernel $\sinh(k(t-r))/k$ is nonnegative for $0 \le r \le t$. If $h'(0) > pa\,h(0)$, this yields

$$0 < h(-t) \le h(0)\cosh(pa\,t) - \frac{h'(0)}{pa}\sinh(pa\,t) \qquad (t \ge 0).$$

The right side is negative for large t, a contradiction. Reflection gives the opposite bound. Hence $|h'(0)| \le pa\,h(0)$. Both pa and the spectral endpoint are bounded uniformly on $\mathcal{P}$, which gives the stated uniform constant.

For each $\zeta > 0$ choose S large enough that

$$C|\xi| \le \zeta \fint_{B^m_{S/2}} e^{\xi\cdot y}\,dy$$

over the whole spectral range, uniformly for $p \in \mathcal{P}$. Indeed, the average is at least one by symmetry and Jensen's inequality. For $|\xi| \le r_0$ first choose $Cr_0 \le \zeta$. For $r_0 \le |\xi| \le r_1$, where r_1 is the uniform spectral upper bound, a cap occupying a fixed fraction of the ball gives the lower bound $c_m e^{S|\xi|/4}$ for the average. Increasing S handles this remaining compact interval. For an extremal and any horizontal unit vector e, the left side of the desired signed inequality is $\pm(\xi\cdot e)h'(0)$. The uniform bound on $|h'(0)|/h(0)$ therefore proves both signs of that inequality. For fixed e, evaluation of this derivative and averaging on the fixed disk are continuous linear functionals on the normalized compact cone. The compact-convex argument from Theorem 3.1 extends the inequality from its extreme points to every normalized positive homogeneous solution. Undo the normalization and then take the supremum over unit vectors e. Adding F_{src} changes no derivative on the left and only increases the nonnegative average on the right. Thus

$$|\nabla_y F_s(0,0)| \le \zeta \fint_{B^m_{S/2}} F(y,0)\,dy.$$

The exact zero-level identity is $H = cD_\nu/(p|\nabla U|^p)$. Pull it back by the zero graph $X_j(y) = (y, h_j(y))$. The normalized compactness gives $h_j \to 0$ smoothly, $\nu_j \to e_s$, and $\nabla_y \nu_j = O(\sqrt{\epsilon_j})$ on each fixed disk. Also $\nabla D_j = O(\epsilon_j)$, $D^2 D_j = O(\epsilon_j)$, and $\nabla_y(1 - D_j)^{-1} = O(\epsilon_j)$. For example, differentiating its numerator gives

$$\partial_i\Big[(D_j)_{\nu_j}(X_j(y))\Big] = D^2 D_j(X_j(y))[\partial_i X_j, \nu_j] + \nabla D_j(X_j(y))\cdot\partial_i\nu_j.$$

After division by ϵ_j, the second term tends to zero, and the first tends to $\partial_i\rho_s$. The differentiated denominator contributes $O(\epsilon_j^2)$ before division. Since the graph metric tends to the Euclidean metric, it follows that

$$\frac{\nabla_{\Sigma_j} H_j(0)}{\epsilon_j} \longrightarrow \frac{c}{p}\nabla_y\rho_s(0,0) = \frac{c}{p}\nabla_y F_s(0,0).$$

The last equality uses $G(0)^{p/2} = 1$ and $(G^{p/2})'(0) = 0$.

Cauchy–Schwarz yields

$$|\nabla_y F_s(0,0)|^2 \le C_m\zeta^2 S^{-m}\int_{B^m_{S/2}} \rho(y,0)^2\,dy.$$

Choose ζ so that, after the uniformly bounded factor $(c/p)^2$, this coefficient is strictly smaller than η. On the fixed ball, graph area tends to Lebesgue measure and $F(y,0) = \rho(y,0)$. The smaller flat disk lies in the limiting surface ball $B_S \cap \Sigma$, and its integral is positive because $\rho(0) = 1$. The spectral choice of S used only $\mathcal{P}$ and ζ, so this radius is fixed before choosing a defect threshold. If no threshold worked at this radius, take a contrary sequence and use the limits above. The strict limiting inequality gives a contradiction. The zero-defect case is a planar layer and satisfies

the conclusion directly. Keep this radius S fixed and decrease the threshold once more. Apply Lemma 2.12 on $B_{6S}(z)$. Rotate the tangent plane so that its normal is e_s and translate z to zero. At the base, $Z(0) = (1 - D(0))^{1/p} e_s$. The Hessian bound and integration along segments in this fixed ball give

$$\sup_{B_{6S}} |Z - e_s| \le C_S \sqrt{D(0)}, \qquad \sup_{B_{6S}} |U - s| \le C_S \sqrt{D(0)}.$$

The constants here may depend on the already fixed S. Choose $\delta \le d_{6S}$ small enough that these bounds give $U_s \ge 9/10$, $|\nabla_y U|/U_s \le 1/10$, and $|U - s| < S$. The cylinder $B^m_{4S} \times [-4S, 4S]$ is contained in B_{6S}. For every $y \in B^m_{4S}$, the endpoint values $U(y, -4S)$ and $U(y, 4S)$ therefore have opposite signs. The positive vertical derivative gives exactly one zero $s = h(y)$. The implicit function theorem and $\nabla h = -\nabla_y U / U_s$ give the asserted slope bound. Every zero in B_{4S} lies on this graph. The threshold depends on S, but the area bound for this graph is at most $\sqrt{1 + 10^{-2}}\, |B^m_{4S}|$. Its constant divided by S^m therefore depends only on the dimension. This is the constant used in the later Fubini estimate.

□

Weighted area bounds let us integrate the local estimates. Keep $X, \Sigma_g, s_X, J_0, \bar{b}, \chi_R, \mathcal{X}$, and N_R from Section 6, with the baseline threshold $0 < \delta \le \delta_{\text{univ}}$. Unmarked surface integrals are over Σ_g. The empty-bad-set case has already been classified, so $X \neq \varnothing$. As in Section 6, decrease the universal threshold so that $b^2 \le CD \le 1$ on Σ_g. Then $0 < J_0 \le 1$ and $|\nabla_\Sigma \log J_0| \le \lambda$ almost everywhere. Lemma 6.13 gives $N_R < \infty$. Put

$$K(R) = 1 + \sup_x M_R(u; x), \quad A_R = \int_{\Sigma_g} \chi_R^2 b^4, \quad Z_R = \int_{\Sigma_g} \chi_R^2 D^2, \quad V_R = K(R) R^{m-4}.$$

Use the normalized density M_R and the growth bounds from Section 6:

$$K(t) \le C(t/R) K(R) \quad (t \ge R \ge 1), \qquad K(R) \le 1 + CR.$$

Fixed regular charts give

$$\int_{\Sigma_g} \chi_R^2 \, dS \le C K(R) R^m, \qquad \int_{\Sigma_g \cap \{s_X \le L\}} \chi_R^2 \, dS \le C_L N_R \tag{7.1}$$

for fixed L and all large R. For the first bound, cover the good zeros by uniform regular graph balls. On a smaller ball the phase is noncritical and stays in a fixed interior range. Integration over a fixed transverse segment then bounds the graph area by the energy on a fixed enlargement. A maximal separated choice has bounded ambient overlap. Summing gives the unweighted area bound on each ambient ball. To include the weight, decompose the surface into the ambient annuli $jR \le |z - x_c| < (j+1)R$, $j = 0, 1, \ldots$. The weight is at most e^{-2j} there, while

$$\mathcal{H}^m(\Sigma_g \cap B_{(j+1)R}(x_c)) \le C K((j+1)R)((j+1)R)^m \le C K(R) R^m (j+1)^{m+1}.$$

The convergent series $\sum_{j \ge 0} e^{-2j} (j+1)^{m+1}$ proves the first inequality in (7.1).

For the collar bound, use the covering from (6.55). If $s_X(z) \le L$, choose a point of X within distance $L+1$ and then a net point $x \in \mathcal{X}$ within distance one. Thus the collar lies in $\bigcup_{x \in \mathcal{X}} B_{L+2}(x)$. For z in this ball and $R \ge L+2$, $\chi_R(z)^2 \le e^2 \chi_R(x)^2$. Using the polynomial area bound on each ball and summing gives the second inequality in (7.1). Overcounting is harmless because the integrand is nonnegative.

Thus every weighted integral below is finite before moment closure. Bounded global density is not needed.

Lemma 7.2 (Supremum envelope with a small defect coefficient)**.** *There is a finite radius $R_{\text{env}} = R_{\text{env}}(\mathcal{P}, \lambda) \ge 1$ such that, for every $\eta > 0$, a finite constant C_η satisfies the following estimate for*

every $R \geq R_{\mathrm{env}}$:

$$\int_{\Sigma_g} \chi_R^2 J_0^2 \leq C_\eta A_R + \eta Z_R + C_\eta N_R. \tag{7.2}$$

The constants are independent of the completed reach weight, the bad gluing collar, and any sufficiently small baseline threshold δ. They may depend on the fixed envelope rate λ.

Proof. At small-defect zeros, relative regularity gives common-radius charts of bounded area and

$$|\nabla_\Sigma B| \leq C\sqrt{D}.$$

Take a radius-$r < 1/4$ disk about a zero y whose unit chart is good. Relative Harnack and integration of the gradient estimate give

$$b(y)^4 \leq Cr^{-m} \int_{B_r(y)\cap\Sigma} b(w)^4 \, dS_w + Cr^4 D(y)^2.$$

For example, $b(y) \leq b(w) + Cr\sqrt{D(y)}$ on this disk. Raise this bound to the fourth power and average. Keep the second term at the base. Average that term separately on a unit chart using Harnack:

$$D(y)^2 \leq C \int_{B_1(y)\cap\Sigma} D(w)^2 \, dS_w.$$

This second averaging constant is independent of r.

Square the exponential envelope and take the supremum over y. Extend the local integrals to the full exponential convolution. Good bases more than a fixed unit distance from X contribute at most

$$C_r \int_{\Sigma_g} e^{-c\lambda|z-w|} b(w)^4 \, dS_w + Cr^4 \int_{\Sigma_g} e^{-c\lambda|z-w|} D(w)^2 \, dS_w.$$

Choose one fixed rate $\nu = c\lambda > 0$ for both the good-base convolutions and the bad-net kernel below. It is independent of the averaging radius r, the tolerance η, and the baseline threshold. Both averaging steps and the transfer to a net point move the center by a bounded distance. The triangle inequality therefore changes only a bounded exponential factor, not the rate. Fix

$$R_{\mathrm{env}} = \max\{1, 4/\nu\}.$$

Chart constants remain uniform as the threshold decreases. The pointwise energy bound and the same chart comparison give $\mathcal{H}^m(\Sigma_g \cap B_t(x)) \leq C(1+t)^n$ for every center and $t > 0$. Consequently,

$$\sup_{w\in\mathbb{R}^n} \int_{\Sigma_g} e^{-\nu|z-w|} \, dS_z \leq C \sum_{j=0}^{\infty} e^{-\nu j}(j+2)^n =: C_\nu < \infty.$$

The estimate is uniform in the center and in the baseline threshold. The logarithmic gradient bound for the spatial weight gives, for $R \geq 4/\nu$,

$$\chi_R(z)^2 e^{-\nu|z-w|} \leq \chi_R(w)^2 e^{-(\nu-2/R)|z-w|} \leq \chi_R(w)^2 e^{-\nu|z-w|/2}.$$

Tonelli's theorem therefore bounds each nonnegative convolution by $C_{\nu/2}$ times its weighted source integral. Applying this first to b^4 and then to D^2 gives $C_r A_R + Cr^4 Z_R$. In particular, the second term retains its small factor r^4; this integration introduces no factor r^{-m}.

The remaining bases are bad zeros, where $\overline{b} = 1$, and good zeros within a fixed unit collar of X. Choose, for each such base, a net point within a fixed distance. The triangle inequality bounds its exponential by a fixed multiple of the exponential centered at that net point. Thus the squared envelope of these bounded values is at most $C\sum_{x\in\mathcal{X}} e^{-c\lambda|z-x|}$. A maximal unit-separated net has bounded overlap at every fixed scale. Integrating over good chart area and transferring the spatial weight costs $C_r N_R$. Choose r small enough that the coefficient of Z_R is below η. The two distinct averaging radii keep the factor r^{-m} off the small coefficient r^4. □

Lemma 7.3 (Decay of the envelope outside the fixed bad set)**.** *At every zero z with $s_X(z) \geq L$, for L sufficiently large,*

$$D(z) \leq d_L := CL^{-\alpha}, \qquad J_0(z) \leq j_L := CL^{-\alpha} + e^{-\lambda L/2} \leq C_\lambda L^{-\alpha}. \tag{7.3}$$

The constants are independent of the baseline threshold $0 < \delta \leq \delta_{\mathrm{univ}}$.

Proof. Use Theorem 4.14 with its original threshold δ_0, constants and exponent $\alpha > 1$. Its hypotheses are exactly the present scalar-stable entire matched solution in $\mathbb{R}^7$, with $-1 < u < 1$ and $p \in \mathcal{P}$. Keep the fixed universal ceiling $\delta_{\mathrm{univ}} \leq \delta_0$. For every later baseline $0 < \delta \leq \delta_{\mathrm{univ}}$, a ball without δ-bad zeros also has no δ_0-bad zero. If $s_X(z) \geq L$, the ball $B_{L/2}(z)$ is clean for both thresholds. For $L \geq 4R_0$, where R_0 is the theorem's fixed radius threshold, its defect bound gives $D(z) \leq CL^{-\alpha}$.

Split the envelope for $J_0(z)$ into $|y - z| \leq L/2$ and its complement. In the first part, $s_X(y) \geq L/2 > 0$. Such a zero is good, so $\bar{b}(y) = b(y)$. The ball $B_{L/4}(y)$ is clean for the original threshold as well. The same theorem gives $D(y) \leq CL^{-\alpha}$ there. The fixed small-threshold curvature bound (2.19) therefore gives $b(y)^2 \leq CD(y) \leq CL^{-\alpha}$. In the complement, $\bar{b}(y)^2 \leq 1$ and $e^{-\lambda|y-z|} \leq e^{-\lambda L/2}$. Taking the supremum proves the second bound in (7.3). All radius thresholds and constants come from the original clean-ball theorem and the fixed analytic bounds. They remain independent of the later baseline δ. □

7.2. **The Simons estimate.** We combine Simons' identity with the physical surface form (6.48). Its tests have compact support in $\{J_0 < j_*\}$. We retain its cap error, with the fixed power $M > 2$. The envelope estimate will control this error after the tests are placed far enough from the bad set.

For a hypersurface the normal bundle is flat. The proof of Fröhlich–Winklmann [21, Lemma 2.4, (13)–(15)] therefore gives, for every fixed $\epsilon_K > 0$,

$$|\nabla B|^2 \geq \left(1 + \frac{2}{m + \epsilon_K}\right) |\nabla b|^2 - C_{m,\epsilon_K} |\nabla H|^2.$$

The tensor algebra makes this estimate explicit.

At a point with $b > 0$, choose an orthonormal geodesic frame that diagonalizes B, with eigenvalues $\lambda_1, \ldots, \lambda_m$. Write

$$T_{ijk} = \nabla_k B_{ij}.$$

Symmetry of B and the Euclidean Codazzi equation make T symmetric in all three indices. For each fixed k, set $a_i^{(k)} = T_{iik}$ and $h_k = \nabla_k H = \sum_i a_i^{(k)}$. Differentiating $b^2 = \sum_i \lambda_i^2$ at the chosen point gives

$$\partial_k b = \frac{1}{b} \sum_i \lambda_i a_i^{(k)}, \qquad |\partial_k b|^2 \leq \sum_i |a_i^{(k)}|^2.$$

In the full sum $\sum_{i,j,k} T_{ijk}^2$, a component with all three indices equal occurs once. A component with exactly two equal indices occurs three times, and a component with three distinct indices occurs six times. Discarding only the last, nonnegative group therefore gives

$$|\nabla B|^2 \geq \sum_k \left(|a_k^{(k)}|^2 + 3 \sum_{i \neq k} |a_i^{(k)}|^2 \right).$$

There is no double counting here: the index k is the index occurring only once in each term with two equal indices.

For a fixed $\epsilon_K > 0$, use $a_k^{(k)} = h_k - \sum_{i \neq k} a_i^{(k)}$ and Young's inequality:

$$|a_k^{(k)}|^2 \leq (m - 1 + \epsilon_K) \sum_{i \neq k} |a_i^{(k)}|^2 + \left(1 + \frac{m-1}{\epsilon_K}\right) |h_k|^2.$$

For example, this follows by applying $(x+y)^2 \le (1+t)x^2+(1+t^{-1})y^2$ with $x = \sum_{i\neq k} a_i^{(k)}$, $y = -h_k$, $t = \epsilon_K/(m-1)$, and then Cauchy–Schwarz. Since $m = 6$, the denominator $m-1$ is positive. The exact algebraic difference is

$$|a_k^{(k)}|^2 + 3\sum_{i\neq k}|a_i^{(k)}|^2 - \left(1+\frac{2}{m+\epsilon_K}\right)\sum_i |a_i^{(k)}|^2$$
$$= \frac{2}{m+\epsilon_K}\left((m-1+\epsilon_K)\sum_{i\neq k}|a_i^{(k)}|^2 - |a_k^{(k)}|^2\right).$$

Combining this identity with the multiplicity estimate above, and summing over k, proves

$$|\nabla B|^2 \ge \left(1+\frac{2}{m+\epsilon_K}\right)|\nabla b|^2 - \frac{2}{m+\epsilon_K}\left(1+\frac{m-1}{\epsilon_K}\right)|\nabla H|^2.$$

Choose $\epsilon_K = m$ and set $\theta = 1/m$. This gives the required strict dimensional margin, without assuming $H = 0$. On $\{b = 0\}$, use the almost-everywhere weak-gradient convention. The Codazzi–Kato bound and Simons' identity are geometric and impose no condition on p. The moment estimate below still uses the physical form and preparation in the standing range $p > 2$.

Proposition 7.4 (An arbitrarily small defect coefficient)**.** *For every $\varepsilon > 0$ there are constants $C_\varepsilon, R_\varepsilon$ such that, uniformly on $p \in \mathcal{P}$, in x_c, and over all sufficiently small baseline thresholds,*

$$A_R \le \varepsilon Z_R + C_\varepsilon(N_R + V_R) \qquad (R \ge R_\varepsilon). \tag{7.4}$$

The baseline bad set and its net are kept fixed during the proof. The constants are independent of every completed reach amplitude or compact multiplier flux.

Proof. Use the preceding Codazzi–Kato bound, with its fixed dimensional margin $\theta > 0$. Keep the chosen cap power $M > 2$. Set $t = \theta/16$ and choose

$$0 < \tau < \min\left\{\frac{1}{10}, \frac{\theta}{32+3\theta}\right\}$$

Reduce τ further to meet any already prescribed positive physical tolerance. Do this before choosing the physical cap constants and their common normal window. Then $\tau(32+3\theta) < \theta$, and hence

$$\frac{(1+\tau)(1+\theta/16)}{1-\tau} \le 1+\theta/8 < 1+\theta/4.$$

This preserves the required strict dimensional margin. The cap constants, envelope rate, and C_1 in (7.2), with its tolerance $\eta = 1$, are now fixed.

The physical form also applies to compact H^1 tests in its open regular domain. Approximate such a test strongly in H^1 by smooth tests with support in the same domain. The metric and the two zeroth-order coefficients are bounded on the compact support, so every term converges. This permits the locally Lipschitz norm of the curvature in the test below.

Choose a Lipschitz cutoff m_L with $0 \le m_L \le 1$, vanishing on $s_X \le L$ and equal to one on $s_X \ge 2L$, with $|\nabla_\Sigma m_L| \le C/L$ almost everywhere. Such a cutoff is obtained by composing a fixed smooth cutoff with s_X/L. Set $v = \chi_R m_L$. We choose L large below. Each compact approximation of bv is then supported in $\{J_0 < j_*\}$. The codimension-one case of [21, Lemma 2.1, (5)] gives

$$\frac{1}{2}\Delta_\Sigma b^2 = |\nabla B|^2 + B_{ij}\nabla_i\nabla_j H + H\,\mathrm{tr}(B^3) - b^4.$$

This is also the contraction of (6.28) with B, since that tensor identity is unchanged by reversing B and H together. The source convention has the opposite signs for both quantities, so its quadratic

and quartic terms agree with ours. All derivatives are covariant for the induced metric. Integrate against v^2 and use Codazzi, $\nabla_i B_{ij} = \nabla_j H$, to obtain

$$\int v^2|\nabla B|^2 = \int v^2 b^4 - \int v^2 H \operatorname{tr}(B^3) + \int v^2|\nabla H|^2 + 2\int vB_{ij}(\nabla_j H)(\nabla_i v) - 2\int vB_{ij}(\nabla_k B_{ij})(\nabla_k v).$$

Write $Q_v = \int b^2|\nabla v|^2$. For any $\epsilon_0 > 0$, Young's inequality gives

$$2\int |v|b|\nabla B||\nabla v| \le \epsilon_0 \int v^2|\nabla B|^2 + \epsilon_0^{-1} Q_v,$$
$$2\int |v|b|\nabla H||\nabla v| \le \int v^2|\nabla H|^2 + Q_v.$$

Lemma 2.12, specifically (2.20), gives $|H| \le CD$ after the fixed unit-scale defect threshold has been imposed. Since $|\operatorname{tr}(B^3)| \le b^3$, the cubic term is at most $C\int v^2 D b^3$ in absolute value.

Move the ϵ_0 term to the left and apply the Kato bound. The resulting gradient coefficient is $(1-\epsilon_0)(1+\theta)$. For example, $\epsilon_0 \le \theta/[2(1+\theta)]$ keeps this coefficient at least $1+\theta/2$. This gives

$$(1+\theta')\int v^2|\nabla b|^2 \le \int v^2 b^4 + C\int v^2(|\nabla H|^2 + Db^3) + C\int b^2|\nabla v|^2 \tag{7.5}$$

with $\theta' = \theta/2 > 0$. To apply (6.48) to bv, use

$$|\nabla(bv)|^2 \le (1+t)v^2|\nabla b|^2 + (1+t^{-1})b^2|\nabla v|^2.$$

The initial choice of t and τ bounds the gradient coefficient by $1+\theta'/2$. Thus

$$\int v^2 b^4 \le (1+\theta'/2)\int v^2|\nabla b|^2 + C\int b^2|\nabla v|^2 + C_{\tau,M}\int J_0^M b^2 v^2. \tag{7.6}$$

Insert (7.6) in (7.5). Subtracting the common gradient term leaves $(\theta'/2)\int v^2|\nabla b|^2$ on the left. This controls the gradient integral by the three error terms. Inserting that bound back in (7.6) controls the fourth moment as well:

$$\int v^2(b^4 + |\nabla b|^2) \le C\int v^2(|\nabla H|^2 + Db^3) + C\int b^2|\nabla v|^2 + C_*\int J_0^M b^2 v^2. \tag{7.7}$$

These constants are independent of L, δ and the desired coefficient ε. The cap estimate below uses the full A_R. Its omitted collar contribution is charged to N_R before absorption.

For the mean-derivative term and each $\epsilon_H > 0$, Lemma 7.1 first fixes a radius S and then a local threshold so that

$$|\nabla_\Sigma H(z)|^2 \le \frac{\epsilon_H}{S^m}\int_{\Sigma\cap B_S(z)} D(w)^2\, dS_w.$$

All constants in that lemma are uniform on the exponent interval. Choose $L > 10S$ with $d_{L/2}$ below this threshold. Fubini, relative chart area, and comparison of spatial weights give

$$\int v^2|\nabla H|^2 \le C\epsilon_H Z_R \tag{7.8}$$

for $R \ge CS$. To see that the Fubini constant is independent of S, fix an averaging point w and choose one contributing center z_0, if one exists. Every other center z satisfies $|z - z_0| < 2S$. The single graph in $B_{4S}(z_0)$ therefore contains all contributing centers. Its slope is at most $1/10$, so their area is at most CS^m, with C independent of S. Also $L > 10S$ ensures that every averaging point is still in Σ_g. The area bound cancels the averaging denominator; also $\chi_R(z)^2/\chi_R(w)^2 \le e^{2S/R}$. For

the cubic term, $b^2 \le CD$ gives $Db^3 \le CD^{5/2}$. On the support of v, (7.3) bounds the remaining factor $D^{1/2}$ by $\sqrt{d_L}$. Hence

$$\int v^2 D b^3 \le C\sqrt{d_L}\, Z_R. \tag{7.9}$$

The cap error must keep its clean-collar gain. On $\operatorname{supp} v$,

$$J_0^M b^2 \le C d_L j_L^{M-2} J_0^2.$$

The independent envelope estimate therefore gives

$$C_* \int J_0^M b^2 v^2 \le a_L(C_1 A_R + Z_R + C_1 N_R), \qquad a_L = CC_* d_L j_L^{M-2} \le C_{\lambda,\tau,M} L^{-\alpha(M-1)} \to 0. \tag{7.10}$$

The pointwise comparison $J_0^M \ll D$ is unnecessary. Smallness comes from the clean collar and one application of the integrated envelope bound.

For the spatial mask, $|\nabla \chi_R| \le \chi_R/R$ and $|\nabla m_L| \le C/L$ give

$$\int b^2 |\nabla v|^2 \le CR^{-2} \int \chi_R^2 b^2 + CL^{-2} \int_{\{s_X \le 2L\}} \chi_R^2 b^2.$$

The second term is at most $C_L N_R$ by (7.1) and $b^2 \le 1$. For the first term, Cauchy–Schwarz and the weighted area bound give

$$R^{-2} \int \chi_R^2 b^2 \le R^{-2} A_R^{1/2} (CK(R) R^m)^{1/2} \le \epsilon' A_R + C_{\epsilon'} K(R) R^{m-4}.$$

Thus for fixed $\epsilon' > 0$,

$$\int b^2 |\nabla v|^2 \le C_L N_R + \epsilon' A_R + C_{\epsilon'} V_R. \tag{7.11}$$

The missing collar is controlled before absorption:

$$0 \le A_R - \int v^2 b^4 = \int \chi_R^2 (1 - m_L^2) b^4 \le \int_{\{s_X \le 2L\}} \chi_R^2 \le C_L N_R.$$

This uses the plateau of the mask and the pointwise curvature bound, rather than the moment being proved.

Insert (7.8)–(7.11) in (7.7), and add the missing collar estimate. First choose ϵ' so that its A_R coefficient is below $1/4$. Choose ϵ_H so that the mean-derivative contribution is below $\varepsilon Z_R/4$. Then enlarge L, leaving X and its net fixed. Require $C_1 a_L < 1/4$ and make the sum of the cubic and cap coefficients of Z_R smaller than $\varepsilon/4$. The total A_R coefficient on the right is then at most $1/2$, and the total Z_R coefficient is at most $\varepsilon/2$. Moving the former to the left proves (7.4). Only the allowed coefficient of N_R grows with L. Finally take $R_\varepsilon \ge R_{\mathrm{env}}$ above all the fixed averaging and collar radii. This includes the convolution radius in Lemma 7.2 at every use of (7.10). None of these choices depends on the baseline threshold. For the form tests, first use compact spatial cutoffs and replace b by $(b^2 + \epsilon^2)^{1/2}$ where needed. The relative derivative bounds and (7.1) make all weighted terms integrable. They permit the compact cutoffs to tend to one and then $\epsilon \downarrow 0$. □

7.3. **A weighted reach inequality.** The Simons estimate leaves a defect term to control. We adapt the long-plateau multiplier and positive-foot accounting of [35, Sections 9.3–9.4]. The reach weight below preserves the mean-curvature margin in the matched physical form. Use the penalized reach d from Proposition 6.7. Our global application concerns a scalar-stable entire solution in $\mathbb{R}^7$ which is neither a well nor a layer. By monotonicity (5.28), M_∞ exists, with infinity allowed. Theorem 5.25 therefore gives $M_\infty = \infty$. Thus it belongs to the global branch in which the signed physical gluing theorem applies. All fixed accuracy requirements below are included when its penalty is first chosen. In particular, require $\eta_0 < 1/1600$, as needed for the radial construction, and set

$$\kappa = \beta/G_1, \qquad 0 < \kappa_- \le \kappa \le \kappa_+ < \infty, \qquad S_{\mathrm{ph}} = (I + b^2)/2,$$

The tolerances for the common window ensure

$$I \geq (1-\varepsilon_I)b^2, \qquad c_I D \leq I \leq C_I D, \tag{7.12}$$

We use (3.16). Its coefficient $C_{\mathcal{P}}$ is fixed before the baseline threshold. It need not be small.

Keep $\omega = 10^{-3}$ and the physical tolerance η_{ph} fixed above. Choose $\varepsilon_I \leq 1/100$. The mean budget (7.32) then retains at least $.196I$. Use gluing tolerance $\tau \leq \eta_{\mathrm{ph}}$ and the mean and geometric accuracies in the preceding numerical budget. They preserve its margin above $4\omega I$ and the separate small-contact gradient reserve.

Fix the desired small $\varepsilon_c > 0$ first. Use an a priori lower bound κ_- valid for all admissible penalties; the initial choice $G_1 \leq 2$ permits $\kappa_- \leq \min_{p\in\mathcal{P}} \beta_p/2$. Choose $\zeta > 0$ with $2\zeta/\kappa_- \leq \varepsilon_c$. Since $G_1\kappa = \beta$, this gives

$$\frac{2G_1(\kappa/4+\zeta)}{\beta} \leq \frac{1}{2} + \varepsilon_c, \tag{7.13}$$

with the chosen ε_c. The ratio on the left is exactly $1/2 + 2\zeta/\kappa$. Choose the plateau width δ_2 and then $c_3 > 0$. Make the receiving-foot Jacobian as close to one as needed whenever $b(y)d(z) \leq c_3$. Next choose $\varepsilon_t > 0$ small enough that this Jacobian, the eventual spatial-weight ratio, and $\kappa/4+\varepsilon_t$ together stay below $\kappa/4+\zeta$. The choices are uniform on $\mathcal{P}$.

For nonplateau contacts with small bd, choose $c_2 > 0$ and the penalty accuracy so that, above a sufficiently large reach threshold,

$$B_{\mathrm{dr}} \leq \frac{1}{4}, \qquad (1+\eta_0+Cd^{-\gamma})b^2 d \leq \frac{1}{4}\frac{|\nabla_\Sigma d|^2}{d} \quad \text{if } bd \leq c_2. \tag{7.14}$$

Indeed, (6.24) gives $B_{\mathrm{dr}} \leq \eta_0 + C_\Gamma c_2 + C_\Gamma d^{-\gamma}$ and $|\nabla d| \geq c_\Gamma > 0$. Choose η_0, c_2 before that threshold. At plateau contacts, $B_{\mathrm{dr}} \leq \eta_0$.

Fix the constant C in B_* at least as large as the error constant in (6.24) before choosing $t_{\min}$. Choose $t_{\min}$ large enough for the weight below, for $t_{\min}c_\Gamma^2 > 1$, and for

$$.79(1+\eta_0+Ct_{\min}^{-\gamma}) < \frac{4}{5}, \qquad \frac{.79C_\Gamma}{t_{\min}} < \kappa/4+\zeta. \tag{7.15}$$

In the second condition enlarge C_Γ to cover outgoing charges too. The fixed $\gamma > 0$ is the geometric reach error exponent. It need not be the half-strip transmission exponent.

We now apply the parameter ranges in Theorem 6.15. The mean accuracy was fixed before the common window. Include the lower bound from Theorem 3.7 in $P_{\min}$. Its cofinal-window statement then gives (3.16) at the selected fixed gluing window after its local threshold is decreased. Thus its constant $C_{\mathcal{P}}$ and the mass-comparison constant C_I are already fixed. The curvature–mass comparison has a separate fixed-window input. For the initially fixed ε_I, Corollary 3.5 supplies a finite window P_{mass} and a positive threshold δ_{mass}, uniformly on $\mathcal{P}$, such that

$$I_{P_{\mathrm{mass}}}(z) \geq (1-\varepsilon_I)b(z)^2 \quad \text{if } D(z) < \delta_{\mathrm{mass}}.$$

Include P_{mass} in $P_{\min}$ before the gluing window is selected, so its chosen value satisfies $P \geq P_{\mathrm{mass}}$. For this comparison the normal is the same initial $\nu(z)$ on both windows, and the positive normalization σ_q^{-1} depends only on the actual exponent, not on the window. Proposition 2.1 gives $q^{p+1}D \geq 0$ along the entire ray. Consequently the physical definition (2.1) gives

$$I_P(z) \geq I_{P_{\mathrm{mass}}}(z) \geq (1-\varepsilon_I)b(z)^2.$$

Keep the final baseline δ strictly below δ_{mass}, in addition to the thresholds required at the selected fixed P for the mean and physical estimates. Then every good zero has the needed local hypothesis. No small-defect condition at all zeros of the entire solution is used in this application. The integral comparison itself needs neither an injective normal tube nor preparation on the added part of the larger window. The bounds $c_I D \leq I_P \leq C_I D$ in (7.12) come from Lemma 3.6 at this selected P. Its upper constant C_I is fixed for that window before choosing δ; it is not inherited unchanged from

P_{mass}. Its positive lower constant may be kept from the unit window. Choose the final baseline threshold below δ_* from gluing and all the local thresholds above. Require every good reach to be at least $t_{\min}$. Require also that $CC_I\delta$ absorb the mean-gradient cost in (7.34). Impose the cap-absorption condition from the closure argument. Only then form this threshold's bad set and net, and keep them fixed.

Choose $L_0 \geq L_{\mathrm{gl}}(\delta)$ so large that $C_{\mathcal{P}}L_0^{-2}I$ has the assigned mean-budget coefficient. The gluing theorem allows every collar above this lower threshold, with the same positive lower bound for its good-collar reserve and the same lower bound $c\delta^{L_{\mathrm{bad}}}$ for its bad-net reserve. Choose its radius threshold after this collar. This gives

$$(1-\tau)\int_{s_X\geq 4L_0}\chi_R^2 f^2 S_{\mathrm{ph}} + c_{\mathrm{col}}\int_{s_X\leq 4L_0} D\chi_R^2 + c_{\mathrm{gl}}N_R \\ \leq (1+\tau)T_R + CR^{-2}\int_{\Sigma_g}\chi_R^2 f^2 + \varepsilon_{\mathrm{cap}}\int_{\Sigma_g}\chi_R^2 J_0^2, \qquad T_R = \int_{\Sigma_g}\chi_R^2|\nabla_\Sigma f(d)|^2. \tag{7.16}$$

Here c_{col} has a fixed positive lower bound, $c_{\mathrm{gl}} \geq c\delta^{L_{\mathrm{bad}}}$, and $\varepsilon_{\mathrm{cap}} \leq C\delta^{M-2}$. The raw cap term is still present. Every test below satisfies the gluing hypotheses.

Corollary 6.8 supplies a finite bound $t_* \geq t_{\min}$ for reaches based in $s_X \leq 10L_0$. It also bounds source reaches whose positive receiving foot is in that collar. The weight will satisfy

$$f = 1 \quad \text{on every reach based in } \{s_X \leq 10L_0\}. \tag{7.17}$$

Fix the bad set, collar, and t_* before choosing the compact flux. If a smaller Simons tolerance is later needed, enlarge only an auxiliary collar of the same X. Keep this gluing collar and its positive bad-net coefficient unchanged.

The next one-dimensional construction uses only $C, \gamma > 0$ and a positive compact range for κ. It imposes no further exponent restriction. Its application to the reach uses the preceding $p > 2$ estimates.

Lemma 7.5 (Explicit reach weight)**.** *Let $\zeta_0 = .01$ and $k = 1.53$. The threshold $t_{\min}$ above can be fixed so large that the following holds. Given any finite $t_* \geq t_{\min}$ and any $\varepsilon_3, \eta_f > 0$, there exist $t_0 > t_*$ and a positive nonincreasing locally Lipschitz function $f : [t_{\min}, \infty) \to (0, 1]$, equal to one on $[t_{\min}, t_*]$, for which*

$$B_*(t) = 1 + Ct^{-\gamma} + \zeta_0\mathbf{1}_{\{t\geq t_0\}}, \qquad p_*(t) = \exp\int_{t_{\min}}^t \frac{B_*(s)}{s}\,ds, \qquad P_*(t) = \frac{1}{p_*(t)}\int_t^\infty p_*(s)f'(s)^2\,ds$$

satisfy

$$-P_*' = f'^2 + B_*P_*/t \quad \text{almost everywhere}, \tag{7.18}$$

$$tP_*(t) \leq .79f(t)^2 \quad (t < t_0), \tag{7.19}$$

$$P_*(t) \leq (\kappa/4 + \varepsilon_t)f(t)^2, \qquad t^3 f(t)^2 \leq \varepsilon_3 \quad (t \geq t_0), \tag{7.20}$$

$$\sup_{[t_{\min}, t_*]} P_* \leq \eta_f. \tag{7.21}$$

The scalar endpoints and power exponents can be chosen commonly for $\kappa \in [\kappa_-, \kappa_+]$, with the other data and targets fixed. This gives a family $f = f_\kappa$ with the stated exponential tail. The constants for the completed family are uniform on that compact rate interval. In particular, for a finite such constant,

$$e^{-\kappa t} \leq f(t)^2 \leq C_f e^{-\kappa t}, \qquad \Psi(t) := \int_t^\infty P_*(s)\,ds \leq C_f f(t)^2, \qquad \sup_{t\geq t_{\min}}\frac{f(t)^2}{P_*(t)} < \infty. \tag{7.22}$$

The limiting ratio is $P_*/f^2 \to \kappa/4$.

Proof. The integrating factor can be written explicitly as

$$p_*(t) = \frac{t}{t_{\min}} \exp\left[\frac{C}{\gamma}(t_{\min}^{-\gamma} - t^{-\gamma})\right] \max\{1, t/t_0\}^{\zeta_0}.$$

This formula isolates the small factor from the decaying drift and the additional power that starts at the late threshold.

Set $F_0 = \exp[C/(\gamma t_{\min}^{\gamma})]$ and require

$$F_0 < 1.001, \qquad (F_0 - 1)\kappa_+/4 < \varepsilon_t/4, \qquad t_{\min} > 2k/\kappa_-.$$

Make these choices before choosing δ. Choose $t_s \geq t_*$ and use

$$f(t) = \begin{cases} 1, & t \leq t_s, \\ (t/t_s)^{-\vartheta}, & t_s < t < t_p, \\ f(t_p)(t/t_p)^{-k}, & t_p < t < t_e, \\ f(t_e)e^{-\kappa(t-t_e)/2}, & t \geq t_e, \end{cases} \tag{7.23}$$

The free parameters will be chosen in the order $t_s, \vartheta, t_p, t_0, t_e$, with $t_s < t_p < t_0 < t_e$.

For later calculations, write $Q(t) = P_*(t)/f(t)^2$. Splitting its defining integral at any $T > t$ gives the exact identity

$$Q(t) = \int_t^T \frac{p_*(s)}{p_*(t)} \frac{f'(s)^2}{f(t)^2}\, ds + \frac{p_*(T)}{p_*(t)} \frac{f(T)^2}{f(t)^2} Q(T).$$

Thus a later interval contributes to an earlier one through the second, explicitly controlled term.

For $s \geq t$,

$$\frac{p_*(s)}{p_*(t)} \leq F_0 (s/t)^{1+\zeta_0}.$$

For $t \geq t_e$,

$$Q(t) = \frac{\kappa^2}{4} \int_0^\infty \frac{p_*(t+r)}{p_*(t)} e^{-\kappa r}\, dr.$$

The integrating factor is increasing, so $Q(t) \geq \kappa/4$. Since $t_e > t_0$, the logarithmic derivative of p_* on $[t, \infty)$ is at most

$$a_t := \frac{1 + \zeta_0 + Ct^{-\gamma}}{t}.$$

Consequently, if $a_t < \kappa$,

$$\frac{\kappa}{4} \leq Q(t) \leq \frac{\kappa^2}{4} \int_0^\infty e^{-(\kappa - a_t)r}\, dr = \frac{\kappa^2/4}{\kappa - a_t}.$$

This proves

$$Q(t) = \kappa/4 + o_{t\to\infty}(1)$$

uniformly for $\kappa \in [\kappa_-, \kappa_+]$. In fact, when $a_t \leq \kappa_-/2$, subtracting $\kappa/4$ from the preceding upper bound gives the explicit estimate

$$0 \leq Q(t) - \frac{\kappa}{4} \leq \frac{\kappa a_t}{4(\kappa - a_t)} \leq \frac{a_t}{2}.$$

In particular, $Q_e := Q(t_e)$ is uniformly bounded and can be made arbitrarily close to $\kappa/4$ by enlarging t_e.

On the principal power range $t_p \leq t \leq t_e$, use the flux splitting identity with $T = t_e$. The part before t_e and the terminal tail are estimated separately. Using $f'(s)^2 = k^2 f(s)^2/s^2$, we obtain

$$\begin{aligned} Q(t) &\leq F_0 \frac{k^2}{t} \int_1^{t_e/t} u^{-1-(2k-\zeta_0)}\, du + F_0 Q_e (t_e/t)^{1+\zeta_0-2k} \\ &\leq \frac{F_0 A}{t} + F_0 Q_e (t_e/t)^{-2.05}, \qquad A = \frac{k^2}{2k - \zeta_0}. \end{aligned}$$

Here

$$2k - \zeta_0 = 3.05, \qquad A = \frac{2.3409}{3.05} = 0.767508\ldots, \qquad F_0 A < 0.77.$$

The preceding tail estimate gives $Q_e = \kappa/4 + o(1)$. Once t_0 is fixed, the terminal contribution multiplied by t tends uniformly to zero on $[t_p, t_0]$ as $t_e \to \infty$. Require that product to be at most .005. On $[t_0, t_e]$ the full ratio is bounded by $F_0(A/t_0 + Q_e)$. Increase t_0 and then t_e to make this at most $\kappa/4 + \varepsilon_t$, uniformly in p. The small factor $F_0 - 1$ was chosen for this step. No special bound such as $\kappa \le \sqrt{2}$ is needed.

On the slow bridge, before the extra ζ_0 drift starts, direct integration gives

$$\frac{tP_*(t)}{f(t)^2} \le F_0 \left[\frac{\vartheta}{2} \{1 - (t/t_p)^{2\vartheta}\} + (t/t_p)^{2\vartheta} \frac{t_p P_*(t_p)}{f(t_p)^2} \right].$$

On the slow bridge, direct integration of the squared derivative produces

$$\frac{\vartheta^2}{t} \int_1^{t_p/t} u^{-1-2\vartheta}\, du = \frac{\vartheta}{2t} \{1 - (t/t_p)^{2\vartheta}\}.$$

This term is small with the bridge exponent. The tail enters separately through the endpoint flux, which is why lengthening the bridge can make the flux small on the whole fixed collar.

For $t \le t_s$, the derivative of f vanishes, so the same flux identity gives $P_*(t) = p_*(t_s) P_*(t_s)/p_*(t)$. Since the extra drift has not started, this implies $tP_*(t) \le F_0 t_s P_*(t_s)$. For $\vartheta < 1$, this and the bridge bound give the common early bound

$$F_0^2 (F_0 A + .005) < .775 < .79.$$

More explicitly,

$$\sup_{[t_{\min}, t_*]} P_* \le \frac{F_0^2}{t_{\min}} \left[\frac{\vartheta}{2} + .775 (t_s/t_p)^{2\vartheta} \right].$$

The first term is reduced by choosing the bridge exponent small. For that fixed positive exponent, lengthening the bridge reduces the second term. An explicit sufficient choice is

$$0 < \vartheta < \min \left\{ 1, k, \frac{\eta_f t_{\min}}{2F_0^2} \right\}, \qquad \frac{t_p}{t_s} > \max \left\{ 2, \left(\frac{3.1 F_0^2}{\eta_f t_{\min}} \right)^{1/(2\vartheta)} \right\}.$$

Including the outside factor $F_0^2/t_{\min}$, the two summands are then each less than $\eta_f/4$. This proves (7.21). The preceding estimates used only the uniform bound on $t_p Q(t_p)$. They are therefore preserved by the later choices of t_0 and t_e.

The late amplitude is controlled by the strict inequality $2k > 3$. On the power interval it has the exact form

$$t^3 f(t)^2 = t_p^3 f(t_p)^2 (t/t_p)^{3-2k} = t_p^3 f(t_p)^2 (t/t_p)^{-0.06}.$$

It can therefore be made smaller than the prescribed late target by increasing the start of the late range. On the exponential tail, its logarithmic derivative is $3/t - \kappa \le 0$ once the final endpoint is larger than $3/\kappa_-$. The final endpoint cannot spoil the earlier amplitude bound.

Choose t_e last, imposing the terminal smallness conditions and $t_e > 3/\kappa_-$. The exponential tail preserves the bound for $t^3 f^2$. Keep t_s, ϑ, t_p fixed afterward.

Before t_e we have $-f'/f \le k/t < \kappa/2$. Hence $f^2 e^{\kappa t}$ increases until t_e and is constant afterward. Its value at $t_{\min}$ is at least one. This proves both exponential comparisons in (7.22). The integral defining P_* includes a nonzero exponential tail, so P_* is positive and continuous. All transition points can be chosen uniformly on $\mathcal{P}$. We next record the lower and primitive bounds, including their dependence on this completed construction.

For $t \le t_e$, positivity and monotonicity of p_* give

$$P_*(t) \ge \frac{p_*(t_e)}{p_*(t)} P_*(t_e) \ge \frac{\kappa_-}{4} f(t_e)^2 > 0.$$

This is an explicit lower bound on the common compact construction interval. On the tail, $P_*/f^2 \ge \kappa_-/4$. Since $f \le 1$, these two lower bounds give explicitly

$$\sup_{t \ge t_{\min}} \frac{f(t)^2}{P_*(t)} \le \frac{4}{\kappa_- f(t_e)^2}.$$

For the primitive, the early and late estimates give the uniform upper ratio

$$Q(t) \le Q_+ := \max\left\{ \frac{0.79}{t_{\min}}, \frac{\kappa_+}{4} + \varepsilon_t \right\}.$$

If $t \ge t_e$, direct integration of the exponential gives $\Psi(t) \le Q_+ f(t)^2/\kappa_-$. If $t \le t_e$, split its defining integral at t_e and use $f \le 1$ on the compact interval:

$$\Psi(t) \le Q_+ \left(t_e - t_{\min} + \frac{f(t_e)^2}{\kappa_-} \right) \le Q_+ \left(\frac{t_e - t_{\min}}{f(t_e)^2} + \frac{1}{\kappa_-} \right) f(t)^2.$$

The last inequality uses $f(t)^2 \ge f(t_e)^2$. This proves the bound for Ψ with an explicit finite constant. It is uniform in κ after the common endpoints have been chosen; it need not remain bounded as the compact-flux target η_f is sent to zero. To record this explicitly, put

$$f_e = (t_p/t_s)^{-\vartheta} (t_e/t_p)^{-k} > 0.$$

This value is common to the family. A common completed constant can be taken at least as large as

$$\max\left\{ 1,\ f_e^2 e^{\kappa_+ t_e}, \frac{4}{\kappa_- f_e^2},\ Q_+, \frac{k}{t_{\min}} + \frac{\kappa_+}{2} \right\},$$
$$Q_+ \left(\frac{t_e - t_{\min}}{f_e^2} + \frac{1}{\kappa_-} \right).$$

These bounds control the exponential comparison, the upper and lower flux ratios, the primitive Ψ/f^2, and $|f'|/f$. They depend only on the common construction data and the fixed targets.

To obtain the ODE, differentiate the defining tail integral:

$$(p_* P_*)' = -p_* f'^2, \qquad P_*' = -f'^2 - \frac{p_*'}{p_*} P_* = -f'^2 - \frac{B_*}{t} P_*.$$

This is (7.18) almost everywhere. The functions p_* and P_* are continuous and locally absolutely continuous. On each compact scalar interval, f'^2, B_*/t, and P_* are bounded. Equation (7.18) therefore gives a locally bounded derivative of P_*, so P_* is locally Lipschitz. Its finitely many derivative corners produce no atoms. □

We integrate against this weight while retaining the gradient reserve. First choose a Lipschitz exterior cutoff $0 \le m_0 \le 1$ with bounded gradient. Let it vanish on $s_X \le 5L_0$ and equal one on $s_X \ge 6L_0$. Use the measurable selector from Proposition 6.7, first on the closed plateau contact relation and then on its complement. This chooses a plateau contact whenever one exists and leaves the same reach function and contact geometry unchanged. Almost everywhere the selected inequality is

$$\Delta_\Sigma d \le A_{\mathrm{src}} + B_{\mathrm{dr}} |\nabla_\Sigma d|^2/d.$$

For definiteness, take the drift to be the following measurable upper bound:

$$B_{\mathrm{dr}} = \begin{cases} \eta_0, & \text{at selected plateau contacts,} \\ \eta_0 + C_\Gamma b d + C_\Gamma d^{-\gamma}, & \text{at selected nonplateau contacts with } bd \le c_2, \\ 1 + C_\Gamma d^{-\gamma}, & \text{at all remaining contacts.} \end{cases}$$

Replacing a drift by a larger one preserves the upper reach inequality. This choice is nonnegative and bounded on compact regular patches. Its second row is at most $1/4$ by (7.14). Since $c_2 \le 1$ and Lemma 6.1 gives $bd \ge 2$ at every active diagonal contact, all diagonal contacts belong to the last row. The distributional inequality has the same upper bound because the singular part of $\Delta_\Sigma d$ is nonpositive.

Multiply the measure inequality by $P_*(d)m_0\chi_R^2$. Writing $w = m_0\chi_R^2$, integration by parts gives

$$\int wP_*(d)\Delta_\Sigma d = -\int wP_*'(d)|\nabla d|^2 - \int P_*(d)\nabla d\cdot\nabla w.$$

Using (7.18) and moving the geometric drift to the left yields

$$\begin{aligned}\int m_0\chi_R^2 f'(d)^2|\nabla d|^2 + \int m_0\chi_R^2(B_* - B_{\mathrm{dr}})P_*(d)\frac{|\nabla d|^2}{d}\\ \le \int m_0\chi_R^2 A_{\mathrm{src}}P_*(d) + \int P_*(d)\nabla d\cdot\nabla(m_0\chi_R^2).\end{aligned} \tag{7.24}$$

Here is the justification for this test. With a compact outer cutoff, $P_*(d)m_0\chi_R^2$ is a nonnegative compact Lipschitz function supported away from X. Approximate it on regular patches by nonnegative smooth functions, uniformly and in local $W^{1,1}$. Local semiconcavity makes $\Delta_\Sigma d$ a Radon measure, so uniform convergence passes the measure pairing. The reach gradient is bounded on compact regular patches, so $W^{1,1}$ convergence passes the gradient pairing. The chain rule $\nabla P_*(d) = P_*'(d)\nabla d$ then gives (7.24). This calculation retains the full Laplacian measure, including its nonpositive singular part. At a corner value of P_*', the level-set property gives $\nabla d = 0$ almost everywhere on that level. Hence the coefficient corners cause no extra term in the chain rule.

At the scalar left endpoint one may extend the two Lipschitz functions constantly to smaller arguments. The compositions are unchanged because every good reach is at least the lower threshold. The gradient of the reach vanishes almost everywhere on the level equal to that threshold, so this extension makes no extra term.

Since $f = 1$ for reaches based in $s_X \le 10L_0$, its surface gradient vanishes almost everywhere there. The first term of (7.24) is therefore exactly $\int_{\Sigma_g}\chi_R^2|\nabla_\Sigma f(d)|^2$.

Define

$$\mathcal{R} = \{\text{selected plateau contacts}\}\cup\{\text{selected nonplateau contacts with } bd \le c_2\},$$

$$Y_R = \int_{\mathcal{R}} m_0\chi_R^2 P_*(d)\frac{|\nabla d|^2}{d}.$$

Keep the full drift surplus in (7.24). Write

$$G_R = \int m_0\chi_R^2(B_* - B_{\mathrm{dr}})\frac{P_*(d)}{d}|\nabla d|^2,$$

$$Y_R^{\mathrm{sm}} = \int_{\substack{\text{selected nonplateau contacts}\\ bd\le c_2}} m_0\chi_R^2\frac{P_*(d)}{d}|\nabla d|^2.$$

On this small-curvature set, $B_* - B_{\mathrm{dr}} \ge 3/4$. On selected plateau contacts it is at least $1-\eta_0 > 1/2$. On all remaining contacts it is nonnegative, by the choice of C in B_*. Consequently

$$G_R - \frac{1}{4}Y_R^{\mathrm{sm}} \ge \frac{1}{2}Y_R.$$

The subtraction on the left is reserved for the local curvature source. We perform it once, when the full source estimate below is inserted into (7.24). This includes reaches in the initial plateau of f.

Consider the last term of (7.24). Its ∇m_0 part is at most

$$C_{\delta,L_0,t_*}\eta_f N_R.$$

Indeed, on this fixed collar, $d \le t_*$, and (6.23) bounds $|\nabla d|$ uniformly. Also $P_* \le \eta_f$, and the weighted area is $C_{L_0} N_R$. For the spatial derivative use $\nabla \Psi(d) = -P_*(d)\nabla d$:

$$\int m_0 P_*(d) \nabla d \cdot \nabla \chi_R^2 = \int \Psi(d)\{m_0 \Delta_\Sigma \chi_R^2 + \nabla m_0 \cdot \nabla \chi_R^2\}.$$

For $r_c = \sqrt{1 + |x - x_c|^2}$, the ambient Hessian is

$$D^2(\chi_R^2) = \chi_R^2 \left(\frac{4}{R^2} dr_c \otimes dr_c - \frac{2}{R} D^2 r_c \right).$$

The second term is negative semidefinite. Tracing over the tangent space gives at most $4mR^{-2}\chi_R^2$. The normal mean-curvature term is kept separately in the next bound.

Choose the good-zero threshold below the fixed unit-scale threshold in Lemma 2.12. Its bound (2.20) gives $|H| \le CD$, uniformly on $\mathcal{P}$. Since $\Delta_\Sigma x = -H\nu$,

$$\Delta_\Sigma \chi_R^2 \le CR^{-2}\chi_R^2 + CR^{-1} D\chi_R^2.$$

The whole cutoff error is therefore at most

$$C_{\delta, L_0, t_*} \eta_f N_R + C_f R^{-2} \int_{\Sigma_g} \chi_R^2 f(d)^2 + C_f R^{-1} \int_{\{s_X \ge 4L_0\}} \chi_R^2 Df(d)^2 + C_{\delta, L_0, f} R^{-1} N_R. \tag{7.25}$$

Only first derivatives of the Lipschitz mask m_0 occur. To justify exhaustion, use $K(t) \le 1 + Ct$ in (6.23). Its bound for $|\nabla d|$ grows more slowly than every positive power of d. The completed weight satisfies $|f'| \le C_f f$ and $P_* \le C_f f^2 \le C_f e^{-\kappa d}$. Consequently

$$|\nabla_\Sigma f(d)|^2 + P_*(d)|\nabla_\Sigma d| + \frac{P_*(d)}{d} |\nabla_\Sigma d|^2 \le C_f$$

almost everywhere on Σ_g. The weighted area estimate (7.1) therefore makes these terms integrable with χ_R^2, for each fixed R. The source is integrable too: its nonplateau part is bounded by a constant times $b^2 dP_*$, and the receiving-foot estimate (6.26) bounds the plateau source by $C(|H| + d^{-1})P_*$. All these functions are bounded after the weight is completed. Compact outer cutoffs with derivatives $O(K^{-1})$ now have vanishing boundary errors as $K \to \infty$. The area estimate and exponential spatial weight also justify the integration involving $\Psi(d)$. In particular, $f(d)$ has finite spatially weighted Dirichlet energy and is admissible for signed gluing. The reach and its active contacts remain fixed throughout; only the compact test functions are approximated.

Lemma 7.6 (All mean charges use one admissible test)**.** *Fix a small $\zeta > 0$. All estimates in this lemma hold for $R \ge R_{\mathrm{geo}} \ge 1$, where the geometric radius threshold is uniform on $\mathcal{P}$. All geometric and lower-reach parameters can be chosen before δ, L_0, f so that the positive outgoing and incoming mean charges in the weak reach inequality are bounded by*

$$2G_1(\kappa/4 + \zeta) \int |H| \mu^2 \chi_R^2 f(d)^2 + C\varepsilon_3 \int \chi_R^2 b^4 + C_{\delta, L_0, t_*} \eta_f N_R + \mathcal{E}_R, \tag{7.26}$$

where $\mu = 0$ on $s_X \le 4L_0$, $\mu = 1$ on $s_X \ge 5L_0$, $|\nabla \mu| \le C/L_0$. Here $\mathcal{E}_R$ is bounded by the spatial-cutoff expression in (7.25) together with $C_f e^{-c\sqrt{R}} K(R) R^m$. Let $\mathcal{C}_R^{\mathrm{np}}$ be the full local curvature charge, including diagonal contacts:

$$\mathcal{C}_R^{\mathrm{np}} = \int_{\text{selected nonplateau or diagonal contacts}} m_0 \chi_R^2 (1 + \eta_0 + Cd^{-\gamma}) b^2 dP_*(d).$$

It satisfies the accounting bound

$$\mathcal{C}_R^{\mathrm{np}} \le \frac{1}{4} Y_R^{\mathrm{sm}} + \frac{4}{5} \int_{s_X \ge 4L_0} b^2 \chi_R^2 f(d)^2 + C\varepsilon_3 \int \chi_R^2 b^4. \tag{7.27}$$

Thus the last two terms bound the residual source after the indicated part of the drift surplus is used.

Proof. Use the uniform collar bound $t_* = t_*(\mathcal{P}, \delta, L_0)$ from Corollary 6.8. It bounds every relevant solution and exponent; it is not chosen as a maximum for the current solution. First remove outgoing charges based in $s_X \le 10L_0$ and incoming charges received there. For an outgoing charge, $d \le t_*$. For an incoming one, the source reach is at most t_*, so its source lies in a fixed enlargement of the bad collar. At a good source use $|H| \le CD$; at a receiving foot use the tangent-ball bound $H_{N_y,+} \le C/d(z)$. The source reach is at least $t_{\min}$, so both charges are uniformly bounded before multiplying by P_*. The fixed-collar source-area bound and (7.21) then apply. The total cost is $C_{\delta,L_0,t_*}\eta_f N_R$, with a constant fixed before the weight. No Jacobian transfer near a bad zero is used.

The remaining outgoing charges have $\mu = 1$. Their early ratio is at most $.79/t_{\min}$. Increase the lower reach threshold to put it below $\kappa/4 + \zeta$ uniformly on $\mathcal{P}$. The late ratio is bounded by (7.20).

For incoming charges, remove $d(z) > \sqrt{R}$ at the source. The tangent-ball estimate $H_{N_y}(y)_+ \le C/d(z)$ and $P_*(d) \le C_f e^{-\kappa d}$ bound their total by $C_f e^{-c\sqrt{R}} K(R) R^m$, using (7.1). The remaining pairs have $\ell \le d(z) \le \sqrt{R}$ and

$$\frac{\chi_R(z)^2}{\chi_R(y)^2} \le e^{2/\sqrt{R}} = 1 + o_R(1). \tag{7.28}$$

The map (6.26) has positive-charge multiplicity one and $d(y) \le d(z)$. Hence $f(d(z)) \le f(d(y))$. Its early Jacobian is bounded by a fixed C_Γ. Choose $t_{\min}$ large enough that $C_\Gamma.79/t_{\min}$ meets the same coefficient. For late contacts with $b(y)d(z) \le c_3$, first choose the angle tolerance and c_3 so that the Jacobian is close to one. Then choose ε_t small. In both retained ranges, the transferred mean density is bounded by

$$G_1(\kappa/4 + \zeta)|H(y)|\chi_R(y)^2 f(d(y))^2. \tag{7.29}$$

All other late contacts cost

$$C\frac{f(d(z))^2}{d(z)} \le C\varepsilon_3 d(z)^{-4} \le Cc_3^{-4}\varepsilon_3 b(y)^4. \tag{7.30}$$

The retained feet are outside $10L_0$, where $\mu = 1$. Use the positive-charge area formula from (6.26), with its multiplicity one, also for the discarded large-curvature feet in (7.30). Their Jacobian is bounded by C_Γ, and (7.28) transfers the spatial weight. Thus their total is bounded by $C\varepsilon_3 \int \chi_R^2 b^4$. Add the two retained nonnegative mean charges and enlarge the domain to obtain the first integral in (7.26). Extending through the transition of μ only increases it.

On selected nonplateau contacts with $bd \le c_2$, (7.14) bounds the source density by $\frac{1}{4}P_*|\nabla d|^2/d$. Their full contribution, both early and late, is therefore at most $Y_R^{\mathrm{sm}}/4$. At the other early nonplateau or diagonal contacts, $dP_* \le .79f^2$. The first condition in (7.15) therefore gives the coefficient $4/5$ in (7.27). For late contacts with $bd > c_2$, use $P_* \le Cf^2$ and $d^3 f^2 \le \varepsilon_3$. The source obeys

$$(1 + \eta_0 + Cd^{-\gamma})b^2 dP_*(d) \le C\varepsilon_3 b^2 d^{-2} \le Cc_2^{-2}\varepsilon_3 b^4. \tag{7.31}$$

These three disjoint ranges prove (7.27) for the original selector. When this bound is used in (7.24), subtract $Y_R^{\mathrm{sm}}/4$ from G_R once. The remaining surplus is at least $Y_R/2$, as proved above. The mean charges and the two residual curvature terms are then the only source terms left on the right. This is the reserve used in the multiplier below. □

The numerical reserve comes from

$$S_{\mathrm{ph}} - \left[\frac{4}{5}b^2 + \frac{1}{4}\mu^2(I-b^2)\right] = \left(\frac{1}{2} - \frac{\mu^2}{4}\right) I - \left(\frac{3}{10} - \frac{\mu^2}{4}\right) b^2$$
$$\geq \frac{1/5 - \varepsilon_I/2}{1-\varepsilon_I} I \qquad (0 \leq \mu \leq 1). \tag{7.32}$$

The lower bound uses $b^2 \leq I/(1-\varepsilon_I)$. The coefficient of b^2 being subtracted is nonnegative for every $0 \leq \mu \leq 1$, so this substitution has the stated direction. The identity applies without a mean mask on the curvature charge and without separating plateau and nonplateau rows. For example, $\varepsilon_I \leq 1/100$ leaves more than $.196I$. Keep $\omega = 10^{-3}$ fixed above. The fixed losses below satisfy the preceding numerical budget, so the exterior reserve after gluing and the T_R absorption exceeds $4\omega I$.

Proposition 7.7 (Strengthened reach multiplier)**.** *For every compact $\mathcal{P} \Subset (2,\infty)$ the weak reach inequality, the signed physical gluing form, and (3.16) give the following estimate: for every $\varepsilon_4 > 0$ one may choose the four-range weight and a finite radius threshold $R_f \geq 1$ so that, for every $R \geq R_f$,*

$$c_D \int Df(d)^2\chi_R^2 + c_N N_R + c_Y Y_R \leq \varepsilon_4 \int b^4\chi_R^2 + \varepsilon_{\mathrm{cap}} \int J_0^2\chi_R^2$$
$$+ C_f R^{-2} \int f(d)^2\chi_R^2 + C_f K(R) R^{m-4}. \tag{7.33}$$

The constants c_D, c_Y have fixed positive lower bounds after the local tolerances are fixed; $c_N \geq c\delta^{L_{\mathrm{bad}}}$. The cap coefficient satisfies $\varepsilon_{\mathrm{cap}} \leq C\delta^{M-2}$ and is kept explicit until the independent moment comparisons have been used. No new error of the form CN_R is produced by the mean estimate. The weight family, its completed constants, and R_f may all be chosen uniformly for $p \in \mathcal{P}$, after the common local parameters, threshold, collar and targets have been fixed.

Proof. Set $\varphi = \mu\chi_R f(d)$. We first justify its use in (3.16) at the already fixed common window. The preceding exhaustion bounds give $|\nabla_\Sigma f(d)| \leq C_f$ almost everywhere. Together with $0 \leq \mu, f \leq 1$, the bounded gradient of μ, and $|\nabla\chi_R| \leq \chi_R/R$, this gives

$$|\varphi|^2 + |\nabla_\Sigma\varphi|^2 \leq C_{f,L_0,R}\chi_R^2.$$

Fix R while taking this limit. The polynomial good-area bound and the exponential spatial weight give $\int_{\Sigma_g}\chi_R^2 < \infty$ before moment closure. On the prepared support, $I \leq C_I D$, $b^2 \leq CD$, and $|H| \leq CD$ are bounded. Consequently all the zeroth-order terms in the mean form and $\int I|\nabla\varphi|^2$ are finite.

Choose $0 \leq \theta_N \leq 1$ smooth in the ambient space, equal to one on $B_N(x_c)$ and zero outside $B_{2N}(x_c)$, with $|\nabla\theta_N| \leq C/N$. Put $\varphi_N = \theta_N\varphi$. The mask μ vanishes on $s_X \leq 4L_0$. The zero set is closed, so the support of φ_N is a compact subset of the prepared regular sheet. The reach is locally Lipschitz there. Thus φ_N is a compact H^1 test, and the H^1 extension in Theorem 3.7 applies.

The integrable bound above shows that $\varphi_N \to \varphi$ strongly in H^1. For the extra derivative,

$$\int_{\Sigma_g} I\varphi^2|\nabla_\Sigma\theta_N|^2 \leq \frac{C}{N^2}\int_{\Sigma_g\cap(B_{2N}(x_c)\setminus B_N(x_c))} I\chi_R^2 f(d)^2 \longrightarrow 0.$$

The same estimate without I and the tails of $|\varphi|^2 + |\nabla\varphi|^2$ prove the asserted strong convergence. Since I is bounded, it also gives convergence of the weighted Dirichlet term. Dominated convergence passes the I, b^2, and $|H|$ integrals. Apply the mean form to φ_N and let $N \to \infty$, with P, δ, L_0, f, R fixed. The form constant is unchanged in this limit. We have therefore proved the same weighted

mean inequality for φ before any absorption below. With $A = 2G_1(\kappa/4+\zeta)/\beta = 1/2 + O(\zeta)$, the mean charge is at most

$$A\int \mu^2\chi_R^2 f^2[(1/2+\varepsilon)I - (1/2-\varepsilon)b^2] + AC_{\mathcal{P}}\int I|\nabla(\mu\chi_R f)|^2.$$

Choose the fixed $\zeta, \varepsilon, \varepsilon_I$ small. The first term then differs from the mean term in (7.32) by an arbitrarily small multiple of $\int_{s_X\geq 4L_0} If^2\chi_R^2$.

To see this directly, put $r = 2\zeta/\kappa$. The pointwise difference is

$$\frac{r}{2}(I-b^2) + (1/2+r)\varepsilon(I+b^2).$$

Use $|I-b^2| \leq I+b^2 \leq [1+(1-\varepsilon_I)^{-1}]I$. For example, one can take

$$\epsilon_{\text{mean}} = \left[1+\frac{1}{1-\varepsilon_I}\right]\left[\frac{\zeta}{\kappa_-} + \left(\frac{1}{2}+\frac{2\zeta}{\kappa_-}\right)\varepsilon\right].$$

Choose the fixed mean tolerances so that $\epsilon_{\text{mean}} \leq \omega/16$. These fixed tolerances precede the threshold, collar, and completed weight.

For the gradient error, keep I in the last integral. The estimate $f = 1$ holds wherever $\nabla\mu \neq 0$. It bounds the gradient error by

$$CC_I\delta T_R + CR^{-2}\int If^2\chi_R^2 + \frac{C}{L_0^2}\int_{4L_0<s_X<5L_0} I\chi_R^2. \tag{7.34}$$

Choose the baseline threshold after fixing $P, C_{\mathcal{P}}$. This makes the first coefficient arbitrarily small. Move its multiple of T_R to the left of the weak reach inequality before inserting that inequality into (7.16). The factor $(1+\tau)/(1-CC_I\delta)$ is then as close to one as needed. The strict reserve in (7.32) survives.

One can keep the comparison coefficient explicit. With $\epsilon_\delta = CC_I\delta$, it is

$$\Lambda_\delta = \frac{1+\tau}{1-\epsilon_\delta}, \qquad \Lambda_\delta - 1 = \frac{\tau+\epsilon_\delta}{1-\epsilon_\delta}.$$

Choose $\tau \leq \omega/16$, and then decrease the threshold until $\epsilon_\delta \leq \omega/16$. Thus $\Lambda_\delta - 1 < \omega/4$. Write

$$C_\mu = \frac{4}{5}b^2 + \frac{1}{4}\mu^2(I-b^2).$$

The budget gives $0 \leq C_\mu \leq S_{\text{ph}} \leq 2I$. Hence the exterior mass left after multiplying the reach inequality by Λ_δ is at least

$$\begin{aligned}(1-\tau)S_{\text{ph}} - \Lambda_\delta C_\mu - \Lambda_\delta\epsilon_{\text{mean}}I \\ \geq [.196 - 2\tau - 2(\Lambda_\delta - 1) - \Lambda_\delta\epsilon_{\text{mean}}]I.\end{aligned}$$

This is a fixed positive reserve. The mask derivative is supported where the exterior mass is retained and the reach amplitude equals one. Its cost can therefore be absorbed by enlarging the fixed collar.

The third term of (7.34) lies in $s_X \geq 4L_0$, where physical mass has full strength and $f = 1$. Choose the minimum collar width from the fixed mean constant so that $C/L_0^2 \leq \omega/16$. Its cost after multiplication by $\Lambda_\delta < 2$ is at most $\omega I/8$. Together with $\epsilon_{\text{mean}} \leq \omega/16$, the preceding choices leave an exterior coefficient greater than $.196 - 7\omega/8 > 4\omega$. The exterior I reserve therefore absorbs this term directly. Neither area nor bad-net bounds are needed. The second term is an allowed spatial error because $I \leq C_I\delta$.

The exterior and collar reserves can now be combined without a weight-dependent lower bound. The exterior coefficient is at least 4ω, and the good-collar coefficient is at least $c_{\text{col},*} > 0$. Their sum is therefore at least

$$\min\{4\omega c_I, c_{\text{col},*}\}\int Df(d)^2\chi_R^2.$$

The exterior uses $I \geq c_I D$; the collar uses $f = 1$. The two regions cover the good surface, including their common level set.

Choose η_f so that compact-flux collar charges use at most one quarter of the fixed bad-net reserve. Choose ε_3 so that late curvature charges cost at most $\varepsilon_4 \int b^4 \chi_R^2$. The four-range weight meets both choices.

The remaining inverse-radius terms are the original spatial-cutoff costs. After completing the weight, enlarge R to absorb $C_f R^{-1} \int D f^2 \chi_R^2$ and $C_{\delta, L_0, f} R^{-1} N_R$ in their positive reserves. Long contacts cost at most $C_f e^{-c\sqrt{R}} K(R) R^m$. For large R this is bounded by $C_f K(R) R^{m-4}$. This proves (7.33), still with its raw cap term.

The test and exhaustion argument leading to (7.25) also applies here. The extra mask μ has bounded gradient in a fixed good collar, where $f = 1$, and I is bounded. Thus the mean form introduces only the costs in (7.34). Throughout this argument, the reach and its measurable contact selector stay fixed. Only the compact test functions are approximated; the resulting tests are admissible for signed gluing. □

7.4. **Fourth-moment bounds.** The envelope estimate and the reach multiplier together control the fourth moments. Keep the quantities A_R, Z_R, V_R defined above, and set

$$F_R = \int_{\Sigma_g} \chi_R^2 D f(d)^2, \qquad E = e^{-\kappa d}.$$

Keep the exterior mask m_0 and selected-contact set $\mathcal{R}$. The retained term is exactly

$$Y_R = \int_{\mathcal{R}} m_0 \chi_R^2 \frac{P_*(d)}{d} |\nabla_\Sigma d|^2, \tag{7.35}$$

$$\mathcal{R} = \{\text{selected plateau contacts}\} \cup \{\text{selected nonplateau contacts with } bd \leq c_2\}.$$

It does not include the discarded collar or all nonplateau branches. Retain the same scalar-stable entire matched solution, with wells and layers already excluded. Its unbounded density was verified in the classification argument preceding (7.16). Thus the interaction comparisons (6.31) and (6.32) apply and give the bounds we need:

$$D \leq C(J_0 + E), \tag{7.36}$$

$$E \leq C(J_0 + D) \quad \text{on plateau contacts with } d|\nabla d|^2 \leq 1. \tag{7.37}$$

Their constants precede the reach-weight construction.

Close the envelope estimate before fixing the final threshold. For these integrated comparisons take $R \geq R_{\mathrm{env}}$ from Lemma 7.2. Any further radius thresholds used below are included when the final outer radius is chosen. Let $C_{\mathrm{int}} \geq 1$ be the constant in (7.36). Multiplication by $D \geq 0$ and Young's inequality give

$$D^2 \leq C_{\mathrm{int}} D J_0 + C_{\mathrm{int}} D E,$$
$$C_{\mathrm{int}} D J_0 \leq \tfrac{1}{2} D^2 + \tfrac{1}{2} C_{\mathrm{int}}^2 J_0^2.$$

Thus

$$D^2 \leq C_{\mathrm{int}}^2 J_0^2 + 2 C_{\mathrm{int}} D E.$$

Put $C_{\mathrm{cmp}} = \max\{1, C_{\mathrm{int}}^2, 2C_{\mathrm{int}}\}$ and $\mathcal{J}_R = \int \chi_R^2 J_0^2$. Use $E \leq f^2$ from (7.22), and apply Lemma 7.2 with tolerance $\eta_{\mathrm{env}} > 0$. Integration gives

$$\begin{aligned} Z_R &\leq C_{\mathrm{cmp}} \mathcal{J}_R + 2 C_{\mathrm{int}} F_R \\ &\leq C_{\mathrm{cmp}} C_{\eta_{\mathrm{env}}} (A_R + N_R) + C_{\mathrm{cmp}} \eta_{\mathrm{env}} Z_R + 2 C_{\mathrm{int}} F_R. \end{aligned}$$

Here the only estimate involving the weight was $E \leq f^2$, with coefficient one. Choose $\eta_{\mathrm{env}} \leq 1/(2C_{\mathrm{cmp}})$. Moving the resulting term to the left allows the choice

$$C_0 = \max\{1,\ 2C_{\mathrm{cmp}} C_{\eta_{\mathrm{env}}},\ 4C_{\mathrm{int}}\}.$$

We obtain

$$Z_R \le C_0(A_R + F_R + N_R). \tag{7.38}$$

The constant C_0 is independent of the weight and gluing collar. The integral F_R covers the whole good region, including its good collar. Theorem 6.15 retains the good-collar defect, so no factor depending on L_0 is needed here.

Proposition 7.4 gives, for every $\eta > 0$,

$$A_R \le \eta Z_R + C_\eta(N_R + V_R). \tag{7.39}$$

Choose a fixed Simons tolerance $\eta_S < 1/(4C_0)$ and write its bound as $A_R \le \eta_S Z_R + H_S(N_R + V_R)$. Inserting this bound in (7.38) puts $C_0\eta_S Z_R$ on the right. Moving it to the left is legal because $1 - C_0\eta_S > 3/4$. This absorption uses no completed reach weight and no lower bound for its flux.

Fix the mean-derivative preparation and auxiliary Simons collar for the chosen tolerance η_S. They are independent of the later bad-net coefficient. The clean-ball estimate keeps the same constants for smaller baseline thresholds: a ball without zeros satisfying $D \ge \delta$ is also clean for the original fixed threshold. Thus all these constants can be fixed before the last decrease of δ.

Combining (7.38) with (7.4) gives

$$Z_R \le C_F F_R + C_N N_R + C_V V_R, \tag{7.40}$$

and an estimate of the same form for A_R. In particular,

$$A_R + Z_R \le C'_F F_R + C'_N N_R + C'_V V_R, \tag{7.41}$$

with fixed finite constants independent of the completed weight. To specify them, enlarge H_S to be at least one and take

$$C_F = \frac{C_0}{1 - C_0\eta_S}, \quad C_N = \frac{C_0(H_S + 1)}{1 - C_0\eta_S}, \quad C_V = \frac{C_0 H_S}{1 - C_0\eta_S}.$$

The Simons estimate gives $A_R + Z_R \le (1 + \eta_S)Z_R + H_S(N_R + V_R)$. We may therefore use

$$C'_F = (1 + \eta_S)C_F,$$
$$C'_N = (1 + \eta_S)C_N + H_S, \qquad C'_V = (1 + \eta_S)C_V + H_S.$$

Their construction uses only the already fixed interaction and envelope constants and the fixed Simons tolerance. It does not use the later gluing collar or any lower bound for the compact flux.

The cap power $M > 2 + L_{\mathrm{bad}}$, logarithmic geometry constants, and envelope rate were fixed earlier. The preceding comparisons are uniform for smaller thresholds. Their constants are therefore fixed before selecting the final threshold. The envelope estimate with one fixed tolerance bounds the raw cap error by

$$\varepsilon_{\mathrm{cap}} \int \chi_R^2 J_0^2 \le C\varepsilon_{\mathrm{cap}}(A_R + Z_R + N_R), \qquad \varepsilon_{\mathrm{cap}} \le C\delta^{M-2}. \tag{7.42}$$

The bad-net coefficient in Proposition 7.7 satisfies $c_N^{\mathrm{mult}} \ge c\delta^{L_{\mathrm{bad}}}$. We use (7.41) to choose δ so that the cap costs consume only fixed fractions of the defect and bad-net reserves. The inequality $M - 2 > L_{\mathrm{bad}}$ makes this possible without changing the preceding constants. The following choices show the dependencies. Take fixed positive numbers d_*, y_*, n_* below the uniform lower bounds supplied by the multiplier reserves, so that

$$c_D \ge d_*, \qquad c_Y \ge y_*, \qquad c_N^{\mathrm{mult}} \ge n_*\delta^{L_{\mathrm{bad}}}.$$

These numbers are fixed before the final threshold, collar, and weight. Choose fixed $C_J, C_{\mathrm{cap}} \ge 1$ which bound, respectively, the envelope coefficient in the cap error and the coefficient in $\varepsilon_{\mathrm{cap}} \le C_{\mathrm{cap}}\delta^{M-2}$. Increase C_J if needed to cover the fixed coefficient of the same cap term in the multiplier.

All these constants precede the final decrease of δ. Let $\delta_{\mathrm{loc}} > 0$ lie below the finitely many earlier local thresholds. Put

$$\delta_D = \left(\frac{d_*}{8C_J C_{\mathrm{cap}} C'_F} \right)^{1/(M-2)},$$

$$\delta_N = \left(\frac{n_*}{8C_J C_{\mathrm{cap}} (C'_N + 1)} \right)^{1/(M-2-L_{\mathrm{bad}})}.$$

Choose $0 < \delta < \min\{\delta_{\mathrm{loc}}, \delta_D, \delta_N\}$. Both powers are positive because $M - 2 > L_{\mathrm{bad}}$. This gives

$$C_J C_{\mathrm{cap}} \delta^{M-2} C'_F < d_*/8, \qquad C_J C_{\mathrm{cap}} \delta^{M-2} (C'_N + 1) < n_* \delta^{L_{\mathrm{bad}}}/8.$$

In particular, the cap-to-bad-net ratio is bounded by a fixed constant times $\delta^{M-2-L_{\mathrm{bad}}}$. This is why the cap power was fixed before the threshold. Keep this threshold and its bad net fixed from now on.

Now fix L_0 and take the uniform collar bound $t_* = t_*(\mathcal{P}, \delta, L_0)$. Choose ε_4 in Proposition 7.7 using (7.41) and the uniform positive reserves. For example, require

$$0 < \varepsilon_4 < \min\left\{ \frac{d_*}{8C'_F}, \frac{n_* \delta^{L_{\mathrm{bad}}}}{8C'_N} \right\}.$$

This choice is made after δ and the collar are fixed. Then choose ε_3 and the compact-flux target η_f. Complete the four-range weight with these targets. Its range endpoints and bridge parameters can be chosen uniformly for $\kappa \in [\kappa_-, \kappa_+]$: the construction uses only these fixed rate bounds and the already fixed geometric constants. No parameter is chosen from a solution-dependent supremum. Dividing by fixed positive coefficients if needed, the multiplier reads

$$\begin{aligned} c_D F_R + c_N^{\mathrm{mult}} N_R + c_Y Y_R \le{}& \varepsilon_4 A_R + C\varepsilon_{\mathrm{cap}} (A_R + Z_R + N_R) \\ &+ C_f R^{-2} \int_{\Sigma_g} \chi_R^2 f(d)^2 + C_f V_R. \end{aligned} \tag{7.43}$$

The choices of δ and ε_4 above absorb the cap and curvature terms while retaining fixed fractions of the positive reserves. Indeed, (7.41) bounds these terms by

$$\begin{aligned} &(\varepsilon_4 C'_F + C_J \varepsilon_{\mathrm{cap}} C'_F) F_R \\ &\quad + (\varepsilon_4 C'_N + C_J \varepsilon_{\mathrm{cap}} (C'_N + 1)) N_R \\ &\quad + (\varepsilon_4 + C_J \varepsilon_{\mathrm{cap}}) C'_V V_R. \end{aligned}$$

The first coefficient is less than $c_D/4$, and the second is less than $c_N^{\mathrm{mult}}/4$. Moving these terms to the left leaves

$$\frac{3}{4} c_D F_R + \frac{3}{4} c_N^{\mathrm{mult}} N_R + c_Y Y_R \le C_f R^{-2} \int_{\Sigma_g} \chi_R^2 f(d)^2 + C_f V_R.$$

The terms containing V_R only enlarge the final finite constant. The weight is now fixed. Its constants may enter the later spatial-cutoff estimate. Choosing the last Young tolerance changes neither δ nor the collar nor f.

Lemma 7.8 (Interaction-square comparison for a completed weight)**.** *For the completed f, P_* there is a finite constant C_f such that the following estimate holds for every $R \ge \max\{R_{\mathrm{env}}, 6L_0 + 2\}$:*

$$\int_{\Sigma_g} \chi_R^2 E^2 \le C_f (A_R + Z_R + N_R + Y_R). \tag{7.44}$$

No lower bound for the compact flux uniform over different weights is asserted or needed.

Proof. The fixed collar $s_X \le 6L_0$ has weighted area bounded by $C_{L_0}N_R$, and $E \le 1$. It therefore contributes at most $C_{L_0}N_R$ to the left side of (7.44). Outside this collar, $m_0 = 1$. We divide this exterior region according to the selected contact. At nonplateau contacts with $bd > c_2$, exponential decay gives

$$E = e^{-\kappa d} \le C_\kappa d^{-2} \le C_{\kappa,c_2} b^2.$$

Squaring and integrating bounds this part by a constant times A_R. At plateau contacts with $d|\nabla d|^2 \le 1$, (7.37) gives $E^2 \le C(J_0^2 + D^2)$. Lemma 7.2, used with one fixed tolerance, bounds its integral by $C(A_R + Z_R + N_R)$.

The remaining plateau contacts have $d|\nabla d|^2 > 1$. This inequality also holds at nonplateau contacts with $bd \le c_2$: improved drift gives $|\nabla d| \ge c_\Gamma$, while $d \ge t_{\min}$ and $t_{\min}c_\Gamma^2 > 1$. These are precisely the remaining exterior cases, and they lie in $\mathcal{R}$. Their retained gradient density satisfies

$$\frac{P_*(d)}{d}|\nabla d|^2 \ge \frac{P_*(d)}{d^2}.$$

Define

$$C_f^* := \sup_{t \ge t_{\min}} \frac{t^2 e^{-2\kappa t}}{P_*(t)}.$$

Use the comparisons stated in Lemma 7.5. Put

$$C_{0,f} = \sup_{t \ge t_{\min}} \frac{f(t)^2}{P_*(t)} < \infty.$$

Since $e^{-\kappa t} \le f(t)^2$, they imply directly

$$\frac{t^2 e^{-2\kappa t}}{P_*(t)} \le C_{0,f} t^2 e^{-\kappa t} \le \frac{4C_{0,f}}{e^2\kappa^2} \le \frac{4C_{0,f}}{e^2\kappa_-^2}.$$

The middle bound follows by differentiating $t^2 e^{-\kappa t}$; its maximum on $[0,\infty)$ occurs at $t = 2/\kappa$. For the common completed family in Lemma 7.5, its exported compact-rate bounds give a common finite $C_{0,f}$. Thus C_f^* is finite and uniform on $\mathcal{P}$ after the construction data and targets have been fixed. Thus $E^2 \le C_f^* P_*(d)|\nabla d|^2/d$ on this last set. This is precisely a part of Y_R, with its original mask. The disjoint cases prove (7.44). □

For the remaining spatial-cutoff error, use $f^2 \le C_f E$. Cauchy–Schwarz, the good-area bound, and Lemma 7.8 give

$$R^{-2}\int \chi_R^2 f^2 \le C_f K(R)^{1/2} R^{m/2-2}(A_R + Z_R + N_R + Y_R)^{1/2}.$$

For any later choice of $\epsilon > 0$, Young's inequality gives

$$R^{-2}\int \chi_R^2 f^2 \le \epsilon(A_R + Z_R + N_R + Y_R) + C_{f,\epsilon} V_R. \tag{7.45}$$

The weight and all its constants are now fixed uniformly on $\mathcal{P}$. After the cap and curvature absorptions, denote the remaining coefficients by $\widetilde{c}_D, \widetilde{c}_N, \widetilde{c}_Y > 0$. Use their positive uniform lower bounds. Let Q_f be a uniform upper bound for the coefficient of the spatial cutoff in (7.43). Choose ϵ so small that

$$Q_f \epsilon C_F' < \widetilde{c}_D/4, \qquad Q_f\epsilon(C_N' + 1) < \widetilde{c}_N/4, \qquad Q_f\epsilon < \widetilde{c}_Y/4.$$

Then (7.41) and (7.45) absorb this last error. To check this last absorption directly, (7.41) gives

$$A_R + Z_R + N_R + Y_R \le C_F' F_R + (C_N' + 1)N_R + Y_R + C_V' V_R.$$

Thus each of the three preceding choices of ϵ pays at most one quarter of the corresponding positive reserve. The term containing V_R only enlarges the final finite constant. Neither δ, the gluing collar, nor f changes. After these absorptions,

$$F_R + N_R + Y_R \le C_f V_R.$$

Substitution in (7.41) then gives

$$A_R + Z_R + N_R + Y_R \le C_f K(R) R^{m-4}. \tag{7.46}$$

Finally, Lemma 7.8 gives

$$\boxed{\int_{\Sigma_g} \chi_R^2(|B|^4 + D^2 + e^{-2\kappa d}) + N_R \le CK(R)R^2.} \tag{7.47}$$

Choose the final outer-radius threshold after all these parameters. Take it above R_{env}, the Simons threshold R_{η_S}, the multiplier threshold R_f, $6L_0 + 2$, and the fixed averaging and collar radii. Every preceding integrated comparison then applies for every larger R. The collar bound, weight parameters, absorption tolerances, and this radius threshold use data uniform on each compact exponent interval. This makes all constants finite and uniform. The final constant is independent of the solution, center, and sufficiently large outer radius. No bounded-density hypothesis was used.

8. Radial rigidity and proof of the main theorem

For the radial argument, retain the standing solution from Sections 6–7: $u : \mathbb{R}^7 \to (-1, 1)$ is a scalar-stable entire weak solution of the matched equation. We work in the unbounded-density branch. The finite-density branch was classified in Theorem 5.25. Thus a nonplanar stable solution would have to belong to this branch. Write $m = 6$ and fix $\mathcal{P} \Subset (2, \infty)$. Keep the notation and profile constants from Section 2. Write $G = G_p$ and $A_{\mathrm{ph}} = A_p(Z)$. Then $\sigma_E = a^2\sigma_q$. Profile constants and local estimate constants are uniform on $\mathcal{P}$. Large-radius choices may also depend on the fixed solution through its unbounded function K. In sequence estimates, every power and profile uses the actual exponent p_j. These standing assumptions apply to the radial estimates below. The proof of the main theorem states its own solution and dimension hypotheses.

Use the common finite window P selected in Section 7, including the tolerances needed for the interpolation mass $I = I_P$ below. At that initial selection, Theorem 3.4 permitted this larger fixed window after decreasing its defect threshold. The lower mass comparison persisted because the integrand defining I_P is nonnegative. The corresponding physical core in Lemma 6.14 was chosen before applying Theorem 6.15. All mass-comparison constants and thresholds were then fixed before the final baseline threshold was selected. We now keep that window, the good set Σ_g, and the bad net fixed. Only the radii R and B will vary in the limits below.

8.1. **The radial amplitude.** We use the fourth-moment bound in a radial test to exclude unbounded energy density. The common radial test and final area comparison adapt [35, Section 10]. The mass and capacity coefficients below are computed for the matched equation.

Set $\vartheta = 1/100$. In the initial choice of the angular reach penalty, require $\eta_0 < \vartheta/16$. The plateau-width losses must also fit the strict inequalities below. These requirements, and the finite-window tolerances used in the interpolation step, are included among the initial choices for Section 7. We then apply its fourth-moment theorem with those choices. The reach, the good set, the bad net, and the normal window are fixed from this point onward. They will not depend on an exhaustion radius.

Write

$$\kappa = \frac{\beta}{G_1}, \qquad K(R) = 1 + \sup_{x \in \mathbb{R}^7} M_R(u; x), \qquad E = e^{-\kappa d}.$$

Here G_1 is the plateau value of the angular reach penalty. Since $p \in \mathcal{P} \Subset (2, \infty)$ and the penalty is fixed, κ has fixed positive upper and lower bounds. The fourth-moment theorem gives, at every center and for every R above its fixed radius threshold,

$$\int_{\Sigma_g} \chi_R^2 (b^4 + D^2 + E^2)\, dS + N_R \le CK(R)R^{m-4}. \tag{8.1}$$

We may take $0 \le \chi_R \le 1$ to be a smooth ball cutoff which equals one on $B_R(x_c)$ and vanishes outside $B_{2R}(x_c)$. Its corresponding net term is $N_R = \sum_{z\in\mathcal{X}} \chi_R(z)^2$. To justify this version, let $\widehat{\chi}_R$ be the exponential cutoff used in (7.47). On $B_{2R}(x_c)$ it has a fixed positive lower bound: for $R \ge 1$, $\widehat{\chi}_R \ge e^{-\sqrt{5}}$. Thus $\chi_R^2 \le C\widehat{\chi}_R^2$, both on the surface and at the net points. Applying (7.47) proves (8.1). The same comparison gives estimates on fixed enlargements of balls and on dyadic annuli.

The good-area bound (6.1) gives $\mathcal{H}^m(\Sigma_g \cap B_R(x)) \le CK(R)R^m$. The density bounds from Section 6 give

$$1 \le K(R) \le 1 + CR, \qquad K(s) \le C(s/R)K(R) \quad (s \ge R \ge 1). \tag{8.2}$$

Their pointwise-energy and covering proof was given at the start of Section 6. It uses the six-dimensional normalization of M_R. The normalizing constant σ_E has uniform positive upper and lower bounds on $\mathcal{P}$.

By the earlier monotonicity formula, $M_R(u;x)$ is nondecreasing in R at every center. Hence K is nondecreasing as well. In the unbounded-density branch, $K(R) \to \infty$ along the full large-radius limit. The planar layer density calculation gives $K(R) \le 2$ for every planar layer and every radius. Such a layer is therefore excluded in this branch. We work here until a contradiction is obtained.

For $K(R) > 2$, the number $K(R)-1$ is strictly larger than $K(R)/2$. The definition of a supremum therefore supplies a center x_R with

$$M_R(u;x_R) > K(R)/2.$$

Translate this center to the origin. No maximum of the density function is required. All preceding estimates are uniform in the chosen center.

Define

$$\rho(x) = (R^2 + |x|^2)^{1/2}, \qquad \tau = \log\rho, \qquad h = \rho^{-2}, \qquad \phi = h^2 = \rho^{-4}, \qquad L = \frac{R}{\sqrt{K(R)}}.$$

The upper bound in (8.2) gives $c\sqrt{R} \le L \le R$ for $R \ge 1$. Thus $L \to \infty$, whereas $L/R = K(R)^{-1/2} \to 0$. Since $s \mapsto (\log s)/s$ is decreasing for $s > e$,

$$\sup_{\rho\ge R} \frac{L + \log\rho}{\rho} \le \frac{1}{\sqrt{K(R)}} + \frac{\log R}{R} \longrightarrow 0. \tag{8.3}$$

No rate of divergence for K is used here.

Set

$$q_* = \frac{3}{2}\kappa, \qquad c_{\mathrm{rad}} = \frac{5}{4\kappa}, \qquad y = \frac{q_* G_1}{\beta} = \frac{3}{2}. \tag{8.4}$$

The interpolation calculation below involves the function

$$F(y) = \frac{y^2}{(1+y)^{2+2/y}}.$$

At the chosen value,

$$F(3/2)^3 = \frac{16\,3^6}{5^{10}} < \frac{1}{3^6}, \qquad 5^{10} = 9\,765\,625 > 8\,503\,056 = 16\,3^{12}.$$

Hence $F(3/2) < 1/9$. We record the numerical criterion needed later. For every $\gamma \in [0,1]$,

$$[1 - \gamma F(3/2)]^{-1} < \frac{9}{8} < \frac{23}{20}.$$

The gap between the last two numbers is $1/40$. Thus coefficients sufficiently close to the first bound can meet the intermediate bound $23/20$. More precisely, suppose a positive number C_{pre} and $0 \le \zeta \le 1/70$ satisfy $\kappa C_{\mathrm{pre}} \le 23/20$. Set $C_{\mathrm{eff}} = C_{\mathrm{pre}}/(1-\zeta)$. Then

$$\kappa C_{\mathrm{eff}} \le \frac{7}{6}, \qquad C_{\mathrm{eff}} q_* \le \frac{7}{4}. \tag{8.5}$$

Indeed, $(23/20)(70/69) = 7/6$ and $q_*/\kappa = 3/2$. This is a conditional numerical statement. The interpolation ratio and the actual effective coefficient are introduced after their estimates are proved in Sections 8.2 and 8.3. The radial amplitude and its preparation below use only q_*, c_{rad} and the scalar weights. They do not use an interpolation estimate or either actual coefficient.

The independent tail requirement also has a strict margin:

$$q_*\left(\frac{2}{\kappa} - c_{\mathrm{rad}}\right) = \frac{9}{8} > 1, \qquad 0 < \frac{q_*}{\kappa} = \frac{3}{2} < 2. \tag{8.6}$$

The first inequality will make the outer annular sum converge. The second allows interpolation with the previously controlled E^2 moment. These numerical identities hold uniformly in p. Their application uses preparation and physical interpolation in the standing range $p > 2$.

Choose a fixed smooth function $\eta : \mathbb{R} \to [0,1]$ which is one on $(-\infty, -1]$ and zero on $[1, \infty)$. Put

$$f_*(t) = \exp\left[-q_* \int_t^1 \eta(s)\, ds\right].$$

Then $0 < f_* \le 1$ and

$$(\log f_*)' = q_*\eta, \qquad f_*(t) = 1 \quad (t \ge 1), \qquad f_*(t) = C_* e^{q_* t} \quad (t \le -1),$$

where $C_* = \exp[q_*(1 - \int_{-1}^1 \eta)] > 0$. All fixed-order derivatives and constants have uniform bounds for $p \in \mathcal{P}$. For $t > 0$, define

$$\begin{aligned} b_*(t) &= \left(t^{1-\vartheta} - L^{1-\vartheta}\right)_+^{1/(1-\vartheta)}, \\ W_*(\tau, t) &= t^{-\vartheta} f_*(c_{\mathrm{rad}}\tau - b_*(t)), \\ \mathcal{F}(\tau, t) &= \int_t^\infty W_*(\tau, v)\, dv. \end{aligned} \tag{8.7}$$

The scalar b_* is unrelated to the curvature norm $b = |B|$. For $t > L$, differentiating $b_*^{1-\vartheta} = t^{1-\vartheta} - L^{1-\vartheta}$ gives

$$b_*'(t) = \left(\frac{b_*(t)}{t}\right)^\vartheta, \qquad 0 < b_*'(t) \le 1. \tag{8.8}$$

For $t \le L$, $b_*(t) = 0$. In particular, $b_*(t)/t \to 1$ as $t \to \infty$. The exponential tail of f_* then proves convergence of the integral defining $\mathcal{F}$.

Write

$$\lambda_*(\tau, t) = (\log f_*)'(c_{\mathrm{rad}}\tau - b_*(t)), \qquad 0 \le \lambda_* \le q_*.$$

We suppress the two arguments after setting $t = d$. The elementary derivative identities are

$$\frac{\partial_t W_*}{W_*} = -\frac{\vartheta}{t} - \lambda_* b_*', \qquad \frac{\partial_\tau W_*}{W_*} = c_{\mathrm{rad}}\lambda_*, \qquad \partial_t \mathcal{F} = -W_*. \tag{8.9}$$

For large R, $c_{\mathrm{rad}}\tau > 2$ on $\rho \ge R$. Thus $f_*' = 0$ whenever $b_*(t)$ is sufficiently near zero. Although b_* itself need not be twice differentiable at L, the factor $f_*(c_{\mathrm{rad}}\tau - b_*(t))$ is identically one in a neighborhood of that point. Near $t = L$ we therefore have $W_*(\tau, t) = t^{-\vartheta}$ and

$$\mathcal{F}(\tau, t) = \mathcal{F}(\tau, L) + \int_t^L v^{-\vartheta}\, dv.$$

These formulas give all derivatives used below at L. Whenever a formula contains λ_*/b_*', its value is defined to be zero where $\lambda_* = 0$. This extension is regular near L, because the numerator vanishes on an entire neighborhood.

A useful change of variable also explains the size of the primitive. For $v > L$, put $w = b_*(v)$. By (8.8), $v^{-\vartheta}\,dv = w^{-\vartheta}\,dw$. Consequently,

$$\mathcal{F}(\tau,t) = \begin{cases} \dfrac{L^{1-\vartheta} - t^{1-\vartheta}}{1-\vartheta} + \displaystyle\int_0^\infty w^{-\vartheta} f_*(c_{\mathrm{rad}}\tau - w)\,dw, & 0 < t \le L, \\ \displaystyle\int_{b_*(t)}^\infty w^{-\vartheta} f_*(c_{\mathrm{rad}}\tau - w)\,dw, & t > L. \end{cases} \tag{8.10}$$

The integrable singularity at $w = 0$ uses $\vartheta < 1$. Splitting the integral at $c_{\mathrm{rad}}\tau + 1$ and using the exponential tail after that point gives

$$0 < \mathcal{F}(\tau,t) \le C(L + \tau + 1)^{1-\vartheta}, \qquad 0 \le \partial_\tau \mathcal{F} \le c_{\mathrm{rad}} q_* \mathcal{F}. \tag{8.11}$$

For example, the part before the split is at most $(c_{\mathrm{rad}}\tau + 1)^{1-\vartheta}/(1-\vartheta)$. The remaining part is bounded by a uniform exponential integral. For every fixed k, the formula for f_* gives $|f_*^{(k)}| \le C_k f_*$, with constants uniform on $\mathcal{P}$. The same integrable bound therefore justifies each required τ derivative under the integral sign on compact τ intervals. In particular, the primitive stays finite even when $t \le L$. The first term in (8.10) is then the contribution of the interval $[t, L]$, on which $f_* = 1$. This term will later supply the lower bound of order $L^{1-\vartheta}$ on bases with $d \le L/2$.

Define

$$\mathcal{Z}_R = \{z \in \Sigma : D(z) \ge \rho(z)^{-1/50}\}, \qquad r_H(x) = \rho(x)^{1/200}.$$

The ambient gradient satisfies

$$|\nabla \rho| \le 1, \qquad |\nabla r_H| \le \epsilon_R := \frac{1}{200} R^{-199/200} \longrightarrow 0.$$

This gives quantitative comparability of nearby radii. If $B_{Ar_H(z)}(z)$ meets $B_{Ar_H(z')}(z')$, then

$$|r_H(z) - r_H(z')| \le A\epsilon_R(r_H(z) + r_H(z')), \qquad \frac{r_H(z)}{r_H(z')} \le \frac{1 + A\epsilon_R}{1 - A\epsilon_R}.$$

Here A is any fixed constant and R is large enough that $A\epsilon_R < 1$.

Choose a maximal disjoint family $\{B_{r_H(z_i)/10}(z_i)\}_i$ with $z_i \in \mathcal{Z}_R$. For any $z \in \mathcal{Z}_R$, maximality gives a selected ball meeting $B_{r_H(z)/10}(z)$. Their radii differ by $1 + o_R(1)$. For large R, the triangle inequality therefore gives

$$B_{r_H(z)}(z) \subset B_{2r_H(z_i)}(z_i).$$

The enlarged balls $B_{3r_H(z_i)}(z_i)$ have bounded overlap. Indeed, all those containing one point have comparable radii; their disjoint one-tenth balls lie in one fixed enlargement. Comparing Euclidean volumes bounds their number by a constant depending only on dimension. The family is locally finite: if $B_{3r_H(z_i)}(z_i)$ meets a fixed ball B_M, then $|z_i| \le M + 3(R^2 + |z_i|^2)^{1/400}$. The rightmost term grows sublinearly in $|z_i|$, so these centers and their radii are bounded. Their one-tenth balls have radius at least $R^{1/200}/10$. Only finitely many disjoint balls of that size fit in the resulting bounded region.

For each selected ball, take a smooth function ζ_i which is zero on $B_{2r_H(z_i)}(z_i)$ and one outside $B_{3r_H(z_i)}(z_i)$, with $0 \le \zeta_i \le 1$ and $|D^k\zeta_i| \le C_k r_H(z_i)^{-k}$ for $k = 1, 2$. Define the locally finite product

$$m_H = \prod_i \zeta_i.$$

It is a smooth ambient function. It vanishes on every $B_{r_H(z)}(z)$ with $z \in \mathcal{Z}_R$ and is one outside $\bigcup_{z\in\mathcal{Z}_R} B_{4r_H(z)}(z)$. If the rough set is empty, take $m_H \equiv 1$. At each point, only a bounded number of factors differ from one. The product rule for a second derivative contains the second derivatives of these factors and their pairwise first-derivative products. Their radii are comparable to $\rho^{1/200}$, so bounded overlap gives, on $\rho \asymp s$,

$$|\nabla m_H| \le C s^{-1/200}, \qquad |D^2 m_H| \le C s^{-2/200}. \tag{8.12}$$

With the convention $B = d\nu$, restriction to a regular zero sheet gives

$$\Delta_\Sigma m_H = \operatorname{tr}_{T\Sigma} D^2 m_H - H\nu \cdot \nabla m_H.$$

Thus the surface derivative terms are controlled by (8.12) and the earlier bound for H.

Choose an ambient outer cutoff m_B, equal to one on B_B, zero outside B_{2B}, with derivative bounds C/B and C/B^2. Keep $B > 3R$ finite for now. The same cutoff is used in every surface and physical test. Define

$$\begin{aligned} m_c &= m_H m_B, \qquad V = h\sqrt{W_*(\tau, d)}, \qquad \psi = m_c V, \\ \mathcal{L} &= \int_\Sigma I\psi^2 \, dS, \qquad T_r = 1 - |\nabla_\Sigma \rho|^2 = \frac{R^2 + (x \cdot \nu)^2}{\rho^2} \geq R^2/\rho^2. \end{aligned} \tag{8.13}$$

For the last identity, use $\nabla_\Sigma \rho = x^T/\rho$ and $|x^T|^2 = |x|^2 - (x \cdot \nu)^2$. We also have $\nabla_\Sigma \tau = \rho^{-1} \nabla_\Sigma \rho$ and $\nabla_\Sigma h = -2h \nabla_\Sigma \tau$.

Every point of $\operatorname{supp}\psi \cup \operatorname{supp}\nabla_\Sigma \psi$ lies outside the removed rough-zero neighborhoods. In particular its defect is less than $\rho^{-1/50}$. For $R^{-1/50} < \delta$, these supports lie in the already fixed good set. Apply Lemma 3.6 on the already fixed window P, after its fixed-core defect threshold is met. Its upper comparison is $I_P \leq C_P D$. For the normalization, relative regularity on that window gives $D(z + s\nu(z)) \leq C_P D(z)$ and $0 \leq q \leq 1$. Also $\sigma_q = a^{p-2} J_p$ is continuous and positive on the compact exponent interval. Thus the definition gives directly

$$I_P(z) \leq \frac{2PC_P}{\min_{p \in \mathcal{P}} \sigma_q} D(z).$$

This verifies that one finite constant C_I works uniformly after P is fixed. Since the supported zero defects are at most $R^{-1/50}$, the required threshold holds for all large R. We obtain

$$I \leq C_I D \leq C_I \rho^{-1/50} \leq C_I R^{-1/50} \quad \text{on } \operatorname{supp}\psi \cup \operatorname{supp}\nabla_\Sigma \psi. \tag{8.14}$$

The reach is finite, positive, and locally Lipschitz on the regular zero set in this unbounded-density branch, by Lemma 6.1 and the preceding reach calculus. On each compact subset of the supports it therefore stays in a compact subinterval of $(0, \infty)$. The chain rule for locally Lipschitz functions makes ψ an admissible compactly supported H^1 surface test. Extend the integrands by zero across the removed collar. No derivative crosses the possibly singular part of the zero set.

For every fixed J, choose R so large that $R^{-1/50} < \delta(J)$ in Lemma 6.1. Then $d > J$ everywhere on the supports. Thus their reaches tend uniformly to infinity as $R \to \infty$. Equivalently, there is a lower bound $d_{\min}(R) \to \infty$ for the reaches on all these supports, uniformly in the finite outer radius B.

We record the resulting weight comparison, since it is what the switching argument needs. At a fixed τ, if $d_{\min}(R) \leq t_1 \leq t_2$, then

$$\begin{aligned} 0 \leq \log \frac{W_*(\tau, t_1)}{W_*(\tau, t_2)} &= \int_{t_1}^{t_2} \left(\frac{\vartheta}{v} + \lambda_*(\tau, v) b_*'(v) \right) dv \\ &\leq \left(q_* + \frac{\vartheta}{d_{\min}(R)} \right) (t_2 - t_1). \end{aligned} \tag{8.15}$$

Here $b_*' = 0$ on $v < L$ and $0 \leq b_*' \leq 1$ everywhere. Also, for every fixed $t > 0$,

$$\left| \log \frac{W_*(\tau_1, t)}{W_*(\tau_2, t)} \right| \leq c_{\mathrm{rad}} q_* |\tau_1 - \tau_2|.$$

For fixed reach t, these formulas give

$$|\nabla_x \log(h\sqrt{W_*(\tau, t)})| \leq \frac{2 + c_{\mathrm{rad}} q_*/2}{\rho}.$$

At points separated by $O(\log \rho)$, the radii are comparable. Integrating this bound along the joining segment changes the logarithm by $O((\log \rho)/\rho) = o_R(1)$. Thus the amplitude changes by a factor $1 + o_R(1)$. The bound is uniform in t, including the exponential tail of the weight.

Let z be a supported zero and abbreviate $\rho_0 = \rho(z)$. There is a fixed $c_0 > 0$ such that $B_{c_0\rho_0^{1/200}}(z)$ contains no rough zero. To see this, suppose a rough zero w lay sufficiently close to z. The Lipschitz estimate for r_H would give $r_H(w) = (1+o_R(1))r_H(z)$, so its removed ball $B_{r_H(w)}(w)$ would contain z in its interior, contradicting the support condition. Hence all zeros in this ball have

$$D \le C\rho_0^{-1/50}.$$

The relative regularity estimates from Section 2 give, on these zeros, $|B| \le C\rho_0^{-1/100}$ and the analogous bounds for each required fixed-order surface derivative.

Take $S = c_1\rho_0^{1/200}$, with $0 < c_1 < c_0/100$, and put $\epsilon_0 = C\rho_0^{-1/100}$. Every radius-$20S$ ball centered within S of z stays in the rough-free ball. Its zero set is relatively closed, smooth, embedded, and without boundary, because all its zeros have defect less than one. Also $\epsilon_0 S \to 0$. Lemma 3.10 therefore supplies tangent graph disks of radius at least S, with normal variation at most $C\epsilon_0 S$ and height above the tangent plane at most $C\epsilon_0 S^2$. This checks the continuation and buffer hypotheses at every nearby base used below. For clarity, the two relevant quantities have different sizes: $\epsilon_0 S = O(\rho_0^{-1/200})$ tends to zero, whereas $\epsilon_0 S^2 = O(1)$ need not. The polynomial disk is used to continue and compare sheets. The smaller logarithmic disks below are what give the absolute height error tending to zero that the pressure theorem requires.

Use one ordered family for all these windows. Choose a fixed $\eta > 0$ sufficiently small and put $r_{\mathrm{fam}} = \eta S$. Require $100\eta < 1$. At every zero $w \in B_{100r_{\mathrm{fam}}}(z)$, Lemma 3.10, with buffered radius S, gives

$$\angle(T_w\Sigma, T_z\Sigma) \le C(100\eta + \epsilon_0 S).$$

The radius-$20S$ buffers at both points remain in the rough-free ball, as checked above. Choose η first so that the first term is small, and then take R large. This angle is below $\arctan(1/20)$. The zero set is relatively closed in this ball. For large R, Lemma 2.12 applies at every zero by the uniform defect bound above. It gives smoothness near those zeros. The implicit function theorem therefore makes the zero set a C^2 embedded hypersurface without boundary in the ball. The common-plane graph observation therefore applies with $r = r_{\mathrm{fam}}$ and base plane $T_z\Sigma$. It gives ordered graphs over $B_{4r_{\mathrm{fam}}}$ that contain every zero in the retained box with horizontal and vertical half-width $r_{\mathrm{fam}}/4$. It also gives the adjacent-member and missing-member alternatives in this box. The box contains every fixed log-power horizontal disk with normal depth $O(\log \rho_0)$, for all large R. Thus a third graph cannot enter a pressure window between two adjacent members. We use the same family for the half-tubes below. Its fixed small slope bound is enough for this coverage argument. The following sharper estimate supplies the errors tending to zero on the logarithmic comparison disks.

Now take two nearby zero bases separated by $\ell = O(\log \rho_0)$. They lie within distance S for large R. Apply (3.19) with buffered radius S and curvature bound ϵ_0. For the acute angle α between their tangent planes, it gives

$$\sin\alpha \le C\left(\frac{\ell}{S} + \epsilon_0 S\right) \le C\rho_0^{-1/200}(1 + \log\rho_0). \tag{8.16}$$

This estimate also covers patches from the same connected zero sheet, whether their selected graph pieces meet or remain disjoint.

After choosing their normals with positive mutual scalar product, both sheets can consequently be written over the first tangent plane. On any disk of radius $(\log \rho_0)^A$, where A is fixed, their slopes and height errors relative to the two parallel base planes are bounded by

$$C_A\rho_0^{-1/200}(1 + \log\rho_0)^{C_A}.$$

For example, the height error is bounded by the initial angle times $(\log \rho_0)^A$, plus $C\epsilon_0(\log \rho_0)^{2A}$. The earlier higher derivative estimates give the corresponding fixed-order bounds. Every such logarithmic disk is contained in the polynomial disk for large R. This proves the quantitative graph preparation required below.

We still need a lower bound for an adjacent gap. Let C_{prop} be the exponential constant in (6.2), and choose a fixed $c > 0$ with $2cC_{\mathrm{prop}} \le 1/100$. Since $D(z) \le \rho_0^{-1/50}$, that estimate gives

$$D(x) \le C\rho_0^{-1/100} \quad \text{for } x \in B_{2c\log \rho_0}(z).$$

This entire ball is noncritical for large R. By (2.31), $|D^2U| \le C\rho_0^{-1/200}$ on a slightly smaller ball. To apply that fixed-radius estimate on a growing ball, use its phase-shifted unit-ball form at each point of the smaller ball. The constants do not depend on the phase value. The propagated bound for D therefore gives the same Hessian bound at every such point, without inserting a constant depending on $\log \rho_0$. Integrating along line segments yields

$$|\nabla U(x) - \nabla U(z)| \le C|x - z|\rho_0^{-1/200},$$
$$|U(x) - \nabla U(z) \cdot (x - z)| \le C|x - z|^2\rho_0^{-1/200}.$$

The value $U(z)$ is zero. Since $\nabla U(z) = |\nabla U(z)|\nu(z)$ and $|\nabla U(z)| \to 1$, the derivative in the direction $\nu(z)$ is at least $1/2$ on a smaller logarithmic ball. Every vertical fiber there therefore has at most one zero. To prove existence on each fiber as well, take a cylinder $|y| < r$, $|t| < r$ in the coordinates $x = z + y + t\nu(z)$, where $r = c_2 \log \rho_0$ and $c_2 > 0$ is sufficiently small. On its upper and lower faces,

$$U(z + y \pm r\nu(z)) = \pm|Z(z)|r + O(r^2\rho_0^{-1/200}).$$

The error is $o(r)$, while $|Z(z)| \to 1$, so these signs are strictly positive and negative. The intermediate value theorem gives a zero on each fiber, and the positive vertical derivative makes that zero unique. The implicit function theorem gives a smooth graph containing every zero in the cylinder. In particular, no second sheet can enter its fixed smaller ball.

The normal rays from z stay inside this smaller ball for a fixed multiple of $\log \rho_0$, so adjacent normal gaps are at least $c' \log \rho_0$. The small angle in (8.16) compares the normal and vertical gaps by a factor $1 + o_R(1)$. The vertical gaps have the same lower bound. In the physical interpolation step, normal half-tubes are truncated at depth $M \log \rho_0$, with fixed M. A middle interaction seen by two such half-tubes can therefore occur only for a gap of length at most $C_M \log \rho_0$. If the gap is longer, its middle region is absent from this test; only its one-sided core and its truncated tail remain. The tail is controlled by the explicit half-tube estimate below. Thus every gap for which a two-sided pressure calculation is used satisfies

$$c' \log \rho_0 \le \ell \le C_M \log \rho_0.$$

Together with the graph estimates above, these are exactly the preparation hypotheses used in the next subsection. All errors tend to zero uniformly for $\rho_0 \ge R$.

We specify the half-tubes and the tail estimate used here. Fix a finite middle-band width $T > 0$ before increasing R. At a supported zero z, let $L_z = M \log \rho(z)$, with fixed M, and use either inward normal side $x = z + sN(z)$, $0 \le s \le L_z + 1$. The polynomial common-family box above contains this window and all its fixed buffers. Rotate coordinates so that $z = 0$, $N(z) = e_n$, and the own graph has height $f_0(0) = 0$ and slope $Df_0(0) = 0$. Reversing the normal direction only reverses the order of the same family. For nearest-foot estimates put $r_{\mathrm{near}} = 16(L_z + 1)$. The common graphs are defined over $B_{4r_{\mathrm{near}}}$ for large R, since $r_{\mathrm{near}} = o(r_{\mathrm{fam}})$. Their slopes are at most $1/20$ and their Hessians at most $C\epsilon_0$, with $\epsilon_0 r_{\mathrm{near}} \to 0$. For $x = (0, s)$, $0 \le s \le L_z + 1$, the own graph has $|f_0(0) - s| \le L_z + 1 < r_{\mathrm{near}}$. If an adjacent graph has axis height $H \le 4L_z$, it has $|H - s| \le 5(L_z + 1) < r_{\mathrm{near}}$ as well. The nearest-foot part of Lemma 3.10 applies to each of these graphs on this convex disk. Points outside the disk are farther than its axis point, so the same

foot minimizes over the whole common graph. The own foot is z, because $x - z$ is normal there. Thus its distance is exactly s. For an observed neighbor, write s_j for its true distance on the first chamber. These projections are unique and smooth there. The own normal map is nonsingular. Its Jacobian satisfies

$$J(z, s) = \det(I + sB_N(z)) = 1 + O(\epsilon_0(L_z + 1)).$$

The inverse tangential metric is bounded by two for large R. These are local assertions on each prepared graph.

Order the common family at the normal axis. If its first member on the chosen side has axis height $H < 4L_z$, it is the first neighbor. Use its true inward distance s_j in the zero-free chamber between it and the own graph, and set $t = (s - s_j)/2$. Multiply the half-tube by a fixed smooth factor equal to one for $t \le T$ and zero for $t \ge T + 1$. The preceding gap lower bound gives $H \ge c \log \rho_0 \to \infty$. At the first zero $s = H$ this factor is therefore already zero. It vanishes on a neighborhood of that zero as well. The branch is defined in the first chamber and extended by zero beyond this boundary. This adds no derivative cost. Equivalently, if s_j is continued as the unsigned distance beyond that graph, then $s_j \le s - H$ for $s \ge H$, so $t \ge H/2 > T + 1$ and the same formula is zero there. If no axis neighbor occurs before $4L_z$, use factor one on the retained depth interval. Also multiply by a smooth depth cutoff equal to one for $s \le L_z$ and zero for $s \ge L_z + 1$. When a neighbor enters or leaves the observation window, its middle band is beyond the depth cutoff, so the definitions agree. To verify this agreement quantitatively, abbreviate $L = L_z$ and suppose that an observed neighbor has $3L \le H < 4L$. Write its graph as f_j, so $f_j(0) = H$ and $\operatorname{Lip} f_j \le 1/20$. For $x = (0, s)$ and any point $(y, f_j(y))$ of this graph,

$$H - s \le |f_j(y) - s| + \frac{1}{20}|y| \le \sqrt{1 + 20^{-2}}\, |(y, f_j(y)) - x|.$$

The same nearest-foot bounds remain valid for $0 \le s \le L + 2$: $|H - s| \le 4L < r_{\text{near}}$ and $s < r_{\text{near}}$, and the common graph disk retains its full buffer. Taking the infimum over the neighbor graph therefore gives

$$s_j \ge \frac{H - s}{\sqrt{1 + 20^{-2}}} \ge \frac{2L - 2}{\sqrt{1 + 20^{-2}}}, \qquad t = \frac{s - s_j}{2} \le -\frac{L}{4} \quad (0 \le s \le L + 2),$$

for all sufficiently large L; $L \ge 12$ suffices. Since $T > 0$ is fixed, $\chi_T(t)$ is identically one on this whole interval, with a strict margin from its transition range. This includes a fixed neighborhood of the retained depth cutoff.

Near a base where $H = 4L_z$, this neighbor stays in the interior of the same common-family box. Its regular graph is transverse to the own normal axis. The implicit function theorem therefore continues its axis intersection and height H continuously as the base varies on the own sheet. The ordering prevents another graph from entering between this neighbor and the own sheet. Also $L_z = M \log \rho(z)$ is continuous. Thus in a base neighborhood of the observation threshold the observed side satisfies $H > 3L_z$, where the calculation above makes its middle factor one. The unobserved side has factor one by definition. The two formulas agree on an open neighborhood in both base and depth variables. All derivatives of this middle factor vanish there, so changing the observation alternative adds no switching or boundary term. The choice is uniform over $\rho(z) \ge R$, because $\inf_{\rho \ge R} M \log \rho \to \infty$ and the slope and buffer bounds are uniform. For a compact base amplitude ψ the corresponding physical half-tube branch is

$$\eta_i(z + sN(z)) = \psi(z)\, \zeta(s - L_z)\, \chi_T(t),$$

where $\chi_T = 1$ in the case with no neighbor. Here ζ is the stated depth cutoff. This describes the branch used for the tail estimates; the two base amplitudes are joined across a middle band in the interpolation construction below.

We now verify the distance to the full zero set. The own zero gives $\operatorname{dist}(x,\Sigma)\le s$. Any nearest zero w consequently satisfies

$$|w-z|\le |w-x|+|x-z|\le 2s\le 2(L_z+1).$$

Such a zero lies in the retained common-family box. The segment from x to w stays in this convex box, and its horizontal projection stays in the common graph disk. If a neighbor is observed and the branch is nonzero, x lies in the first chamber. A segment to a zero on or below the own graph meets that graph, so its length is at least s. A segment to a zero on or beyond the adjacent graph meets that graph, so its length is at least s_j. There is no third family member between these two graphs. Hence every nearest zero has distance at least $\min\{s,s_j\}$. Where the middle factor is nonzero, $t<T+1$ gives $s_j\ge s-2T-2$.

If no neighbor is observed, a potentially nearer zero above the own graph would have an axis height H_w satisfying

$$0<H_w\le |w_n|+\tfrac{1}{20}|w'|\le 2(1+\tfrac{1}{20})(L_z+1)<4L_z$$

for large R. The adjacent member at or below that height would then have been observed, a contradiction. Zeros on or below the own graph are blocked by the same segment argument. Thus the own distance s is the distance to the full zero set in this case. Both alternatives prove

$$\operatorname{dist}(z+sN(z),\Sigma)\ge s-C_T.$$

If there is no nearby neighbor, the own graph gives the same bound. Sharp clearing from Lemma 3.8 therefore yields, for every fixed small $\xi>0$,

$$q(z+sN(z))\le C_{T,\xi}e^{-(2a-\xi)s}.$$

Together with the bounded Jacobian, these are precisely the clearing and geometric hypotheses of Corollary 2.23. The unbounded-density solution cannot be a layer, so its $D_0=D(z)$ is positive by the gradient equality case. That corollary supplies its normalized discrepancy tail bounds. The same ray propagation and clearing calculation gives

$$\int_{s>P} q^{p+1}DJ\,ds\le CD_0e^{-cP},\qquad \int_{s>P} q^pJ\,ds\le Ce^{-(2\beta-p\xi)P},$$

where the integration is over the retained half-tube. For the first bound, use $D\le CD_0e^{(2\beta+\varepsilon)s}$ and choose $\varepsilon+(p+1)\xi<2a$ uniformly on $\mathcal{P}$. All these constants are independent of the far depth and base.

Finally, $|\nabla_\Sigma L_z|\le M/\rho(z)$. The physical derivative of the lifted depth cutoff is therefore uniformly bounded. Since $q^2A_u\le Cq^pI$, its far-cutoff energy per unit base area is at most

$$Ce^{-(2\beta-p\xi)L_z}\psi(z)^2.$$

The tangential energy beyond a fixed depth P is at most $Ce^{-(2\beta-p\xi)P}|\nabla_\Sigma\psi(z)|^2$. The derivative of the middle factor is kept as a middle-band cost; it is evaluated by the capacity calculation below. These estimates define and control the truncated tails even when no two-sided middle band is present.

8.2. **Mass and physical interpolation.** To use the surface amplitude in ambient stability, we extend it between neighboring sheets as in [35, Sections 10.6–10.7]. The normal conductance and defect mass are computed here for the matched equation. There are two relevant coefficients. The first is the defect mass available on a fixed normal core. The second is the energy needed to join two different amplitude values across a gap. We compute both coefficients and then compare them.

The local inputs for these estimates are Modica, relative phase bounds, clearing, and prepared-gap comparison. For $p\ge 2$, that comparison follows from positive well secants and natural-energy lateral decay. It does not require ellipticity at a turn.

We state the physical local computation for a sequence. Let $N_j \to \infty$ and $p_j \in \mathcal{P} \Subset (2, \infty)$. Let $u_j : \mathbb{R}^n \to (-1, 1)$ be entire weak solutions of

$$-\Delta_{p_j} u_j = u_j(1 - u_j^2)^{p_j - 1},$$

with the ambient dimension fixed as above. Use their C^1 representatives supplied by local regularity. Proposition 2.1 gives $|\nabla u_j| \le a_{p_j}(1 - u_j^2)$. This supplies the gradient bound used in the physical mass and conductance calculations. The independent scalar profile and resistance formulas below retain their explicitly stated $p > 1$ ranges. Suppose two adjacent graphs in $\{u_j = 0\}$ enclose a positive phase of u_j, with gap

$$c_0 \log N_j \le \ell_j \le C_0 \log N_j.$$

Put the lower base at the origin and its normal along e_s. On a horizontal disk of any fixed power of ℓ_j, assume that the graph errors relative to $s = 0$ and $s = \ell_j$, the slopes, and the zero defects are bounded by $N_j^{-\zeta} \ell_j^C$, for some fixed $\zeta > 0$. The fixed power C may increase when the requested disk increases. Also assume the preceding fixed-core preparation and clearing estimates on both sheets. The radial preparation argument establishes these properties at the supported bases.

We check the chamber hypotheses before using the turn and pressure estimates. Choose a fixed $k > 1$ and use the buffered horizontal radius $R_{\mathrm{ch},j} = \ell_j^k$. Lemma 2.12 gives $b^2 \le CD$ on these small-defect graphs. Thus, after reducing the positive exponent, the defect bound $\delta_{0,j}$ and curvature bound κ_j satisfy

$$\delta_{0,j} + \kappa_j \le C N_j^{-\zeta_1} \ell_j^C \qquad (\zeta_1 > 0).$$

Let N_{ch} and $\theta_{\mathrm{ch}} > 0$ be the fixed derivative exponents in (3.20). Since $\ell_j \asymp \log N_j$, a negative power of N_j dominates every fixed power of ℓ_j. Consequently

$$\kappa_j R_{\mathrm{ch},j}^2 \to 0, \qquad \kappa_j^{\theta_{\mathrm{ch}}} R_{\mathrm{ch},j}^{N_{\mathrm{ch}}} \ell_j \to 0, \qquad \delta_{0,j} \ell_j \to 0.$$

Also $\ell_j / R_{\mathrm{ch},j} \to 0$ and $e^{-c R_{\mathrm{ch},j}} \ell_j \to 0$. These are all the smallness and buffer conditions in (3.20). The preparation on every fixed power of the gap supplies the larger disk needed before restriction. All these limits are uniform on $\mathcal{P}$.

In this local computation, P denotes the finite mass window. The gap pressure will be written P_j.

Let

$$\Pi_p = \frac{4^p}{2p} \exp\{-\gamma_E - \psi_{\mathrm{dig}}(1 - 1/p)\}, \qquad P_j = (\Pi_{p_j} + o(1)) e^{-\beta_j \ell_j}.$$

The quantity P_j is the actual middle pressure supplied by the prepared-gap comparison. In the formula for Π_p, γ_E is Euler's constant and ψ_{dig} is the digamma function. The pressure asymptotics are (4.21)–(4.22). Lemma 3.11 gives the normalized turn profile. Write $I_{p,P} = I_P$ when the exponent must be shown. This is the mass in (2.1).

Lemma 8.1 (A finite-core one-neighbor mass)**.** *For every $\varepsilon > 0$ there is a finite $P_* = P_*(\mathcal{P}, \varepsilon) \ge 1$ with the following property. For every preceding prepared-gap sequence and every fixed $P \ge P_*$, the following bounds hold for all sufficiently large j:*

$$I_{p_j,P}(0) \ge (1 - \varepsilon) \frac{2\beta_j}{\sigma_{E,j}} P_j,$$

$$I_{p_j,P}(0) \ge (1 - \varepsilon) \frac{2\beta_j \Pi_{p_j}}{\sigma_{E,j}} e^{-\beta_j \ell_j}.$$

The window lower bound depends only on $\mathcal{P}$ and ε. The index threshold may depend on the prepared sequence and the chosen fixed window. The two bounds use a smaller intermediate tolerance and the asymptotic formula for P_j.

Proof. For readability, suppress j and the subscript p in intermediate differential equations. Every coefficient still uses the actual exponent p_j.

Let e be the constant initial normal. Put $W_e = q - u_e/a = q(1 - Z_e) \geq 0$, where nonnegativity follows from Modica's inequality. Equation (2.9) and the Jacobi identity (2.5) from Proposition 2.5 give the following positive supersolution on every noncritical corridor:

$$[-\operatorname{div}(A_u \nabla) + cq^{p-2}(2 - q)]W_e \geq 0. \tag{8.17}$$

Comparison will use only noncritical regions. There the constant vector e makes u_e a Jacobi field. Thus, with $L_u = -\operatorname{div}(A_u\nabla) + W_p''(u)$, the earlier identity for q gives $L_u W_e = -q^p D$. The potential in (8.17) differs from $W_p''(u)$ by pq^{p-1}. Convexity gives $|Z|^p \geq 1 + p(Z_e - 1)$, or $D \leq p(1 - Z_e)$, and hence

$$-q^p D + pq^{p-1} W_e = q^p[p(1 - Z_e) - D] \geq 0.$$

This calculation does not use pointwise Jacobi regularity at the turn.

Apply the preceding conjugation (2.24)–(2.25). The conjugated quantity $F = q^{(p-2)/2} W_e$ therefore satisfies

$$\mathscr{H}_u^+ F := -\operatorname{div}(A_{\mathrm{ph}} \nabla F) + V_u^+ F \geq 0, \qquad V_u^+ = ca^2\{p^2 - p(p-1)q + (p-2)[p - cq]D\}. \tag{8.18}$$

To check the added potential in this conjugation, the term pq^{p-1} becomes $pa^{2-p}q = 2pca^2 q$. Thus the earlier coefficient $-p(p+1)q$ becomes $-p(p-1)q$; the remaining discrepancy term is unchanged. The inequalities $q \leq 1$, $D \geq 0$, and $p > 2$ give $p^2 - p(p-1)q \geq p$ and $(p-2)[p - cq]D \geq 0$. Thus $V_u^+ \geq ca^2 p > 0$, uniformly on $\mathcal{P}$. The maximum principle therefore has a positive potential bound independent of the gap length. At the planar profile the operator is

$$H_p^+ = -\Delta_y - c\partial_s^2 + ca^2[p^2 - p(p-1)G(s)].$$

For the corridor bound, fix large T and set $L_j = \ell_j/2$, $S_j = L_j - T$. Use a horizontal disk of radius ℓ_j^k, with fixed $k > 1$, inside a fixed larger buffered disk. All preparation and top-data estimates are taken on that larger disk before restriction. On $-P_0 < s < S_j$ the nondivergence coefficients of $\mathscr{H}_u^+$ differ from those of H_p^+ by a function $E_j(s)$ satisfying

$$\int_{-P_0}^{S_j} E_j(s)\, ds \leq o_j(1) + Ce^{-\kappa_0 T}, \qquad \sup E_j \leq o_j(1) + Ce^{-\kappa_0 T}, \tag{8.19}$$

Here P_0 is fixed and $\kappa_0 > 0$ is a uniform corridor decay constant; this local decay constant is unrelated to the reach exponent $\kappa = \beta/G_1$.

To derive (8.19), put $s_0 = A_0 \log \ell_j$ and place the comparison faces at $s = s_0$ and $s = \ell_j - s_0$. We check the propagation at this growing depth. Fix A_0 before increasing j. On a slightly larger buffered disk, write the two graphs as $s = f_{-,j}(y)$ and $s = f_{+,j}(y)$. Their preparation gives

$$\|f_{-,j}\|_{C^1} + \|f_{+,j} - \ell_j\|_{C^1} + \sup_{\text{graph bases}} D_j \leq \delta_{\mathrm{gr},j} := CN_j^{-\zeta}\ell_j^{C_0}.$$

Here the disk power, C_0 and $\zeta > 0$ are fixed. We use bases above the comparison disk and retain its unit neighborhood in this larger disk. Let $z_{\pm,j}(y) = (y, f_{\pm,j}(y))$ and let $n_{\pm,j}(y)$ point into the positive chamber. These normals are $Z_j/|Z_j|$ at their bases. Put $\beta_+ = \max_{p\in\mathcal{P}} \beta_p$ and use propagation loss one. Uniformly at all these bases,

$$D_j(z_{\pm,j})e^{(2\beta_j+1)(s_0+2)} \leq CN_j^{-\zeta}\ell_j^{C_0+(2\beta_+ +1)A_0} \longrightarrow 0.$$

Indeed, $\ell_j \asymp \log N_j$, so a negative power of N_j dominates every fixed power of ℓ_j. Thus the smallness hypothesis in (2.37) holds on the whole segment $0 \leq t \leq s_0 + 2$, with one index threshold for all the bases. That estimate gives, for a fixed finite C_1,

$$|U_j(z_{\pm,j} + tn_{\pm,j}) - t| \leq CN_j^{-\zeta}\ell_j^{C_1} \qquad (0 \leq t \leq s_0 + 2).$$

The initial normals need not be vertical. Set $x_{-,j}(y,t) = (y,t)$ and $x_{+,j}(y,t) = (y, \ell_j - t)$. The graph height and slope bounds give

$$|x_{\pm,j}(y,t) - (z_{\pm,j}(y) + t n_{\pm,j}(y))| \le C\delta_{\mathrm{gr},j}(1+t).$$

All these points remain in the larger comparison buffer. Proposition 2.1 gives $|\nabla U_j| \le 1$, so the phase is globally 1-Lipschitz. It follows that, after enlarging the fixed logarithmic power, uniformly through the same depth,

$$|U_j(x_{\pm,j}(y,t)) - t| \le \eta_j := C N_j^{-\zeta'} \ell_j^C \longrightarrow 0, \qquad \zeta' > 0.$$

Finally, $|\partial_t \log(1 - g_{p_j}(t))| \le 2 \max_{p\in\mathcal{P}} a_p$. Composition with $u_j = g_{p_j}(U_j)$ therefore gives

$$1 - u_j(x_{\pm,j}(y,t)) = (1 - g_{p_j}(t))(1 + O(\eta_j)) \qquad (0 \le t \le s_0 + 2).$$

In particular, both horizontal comparison faces have this relative error at $t = s_0$. All constants may depend on the already fixed A_0 and disk power, but are uniform in the horizontal base and in $p_j \in \mathcal{P}$. For large j, choose constants $b_- \le b_+$ in the convex well so that both actual face values lie between them and

$$1 - b_\pm = (1 - g_{p_j}(s_0))(1 + O(\eta_j)).$$

Use slightly larger fixed constants in this error to obtain the required ordering on both faces. Clearing keeps the intervening chamber in the convex well where f_p is decreasing.

We construct the symmetric comparison profile for each boundary value $b \in \{b_-, b_+\}$. Suppress j briefly, and set $q_b = 1 - b^2$ and $d = \ell - 2s_0$. If $Q \in (0, q_b)$ is the value of $1 - v^2$ at its central turn, the first integral prescribes the half-travel time

$$T_b(Q) = \frac{1}{2a} \int_Q^{q_b} \frac{dq}{\sqrt{1-q}\,(q^p - Q^p)^{1/p}}.$$

This function is continuous for $0 < Q < q_b$. The endpoint singularity has exponent $1/p < 1$, so it is integrable. It tends to zero as $Q \uparrow q_b$ and to infinity as $Q \downarrow 0$. The intermediate value theorem therefore gives $T_b(Q) = d/2$. Invert the increasing branch defined by this integral and reflect it about the midpoint. The result is a C^1 weak solution on the interval, with boundary value b and one central positive turn. Its flux is continuously differentiable through the turn and satisfies the one-dimensional equation there as well. The reaction f_p is decreasing on this well interval. Testing the difference of two solutions with its positive part therefore gives uniqueness and comparison by boundary values. In particular, the two profiles just constructed are ordered. Their deficits satisfy the relative gradient bound $|v'| \le a(1 - v^2)$ by the first integral. Thus (4.19) compares them with the actual solution on the smaller cylinder. It leaves only the lateral error $O(\ell_j^C e^{-c\ell_j^k})$.

We now quantify the positions of their virtual zeros. Define

$$c_p = \int_1^\infty \left\{ (t^p - 1)^{-1/p} - \frac{1}{t} \right\} dt.$$

The integral is finite and uniformly bounded on $\mathcal{P}$. Near one its singularity is uniformly integrable. At infinity the integrand is $O(t^{-p-1})$ uniformly on this compact interval. For $Q/q_b \le 1/2$, rescaling $q = Qt$ in the travel time gives

$$2aT_b(Q) = \log(q_b/Q) + c_p + O(q_b) + O((Q/q_b)^p).$$

For the $O(q_b)$ term, use $|(1-q)^{-1/2} - 1| \le Cq$ and $\int_Q^{q_b} q(q^p - Q^p)^{-1/p} dq \le C q_b$. The tail of the defining integral for c_p gives the last term. All constants are uniform on $\mathcal{P}$. Since $d \to \infty$, the ratio Q/q_b tends to zero, and this formula applies for large j. It implies

$$Q \asymp q_b e^{-ad} \asymp e^{-a\ell}.$$

The last comparison uses $q_b = G(s_0)(1 + O(\eta_j))$.

Extend the increasing branch backwards from b to zero. The required travel time is

$$\tau_b(Q) = a^{-1} \int_0^b \left((1-v^2)^p - Q^p\right)^{-1/p} dv.$$

Write $U_b = a^{-1} \operatorname{arctanh} b$. The boundary-data estimate gives $|U_b - s_0| \le C\eta_j$. For $0 \le v \le b$, expand the factor $[1-(Q/(1-v^2))^p]^{-1/p}$ around one. Since $Q/q_b \to 0$,

$$\begin{aligned} 0 \le \tau_b(Q) - U_b &\le CQ^p \int_0^b (1-v^2)^{-p-1} dv \\ &\le C(Q/q_b)^p \le Ce^{-pa(\ell-2s_0)}. \end{aligned}$$

The lower virtual zero is $s_0 - \tau_b(Q)$, and the upper one is its reflection about $\ell/2$. Their distances from 0 and ℓ are therefore bounded by $C\eta_j + Ce^{-pa(\ell-2s_0)}$. For every prescribed fixed K, the exponential term is $O(\ell_j^{-K})$, uniformly on $\mathcal{P}$. Choose A_0 large enough for the well-face errors of that order, then let j grow. This proves the claimed quantitative shifts $O(N_j^{-\zeta'}\ell_j^C) + O(\ell_j^{-K})$. The lateral error divided by the minimum deficit has exponent $-c\ell_j^k + O(\ell_j)$, so it is smaller than every fixed power of ℓ_j^{-1}. Thus it does not change these phase errors. The one-dimensional first integral now gives, before $L_j - T$,

$$|U_{1\mathrm{D}}(s) - s| \le C\{N_j^{-\zeta'}\ell_j^C + \ell_j^{-K} + e^{-\kappa_0(L_j-s)}\}.$$

Here is the phase-delay computation behind this estimate. For the one-dimensional comparison profile, write q_m for its value of $1-u^2$ at the turn. Its first integral is

$$\frac{c}{p}|u'|^p = \frac{q^p - q_m^p}{2p}.$$

Between its lower zero and its positive turn, the phase derivative is

$$U' = (1-(q_m/q)^p)^{1/p}, \qquad 0 \le 1 - U' \le C(q_m/q)^p$$

whenever $q \ge C_T q_m$, with fixed $C_T > 1$. In the same region U' has a fixed positive lower bound. Integrating with U as the variable gives

$$0 \le s - U(s) \le Cq_m^p \int_0^{U(s)} G(v)^{-p}\, dv \le C\left(\frac{q_m}{q(s)}\right)^p.$$

In this display the zero of the comparison profile is placed at $s = 0$; a displaced virtual zero adds its already estimated shift. For the last bound, split at $v = 1$ and use the exponential growth of $G(v)^{-p}$ on the positive half-line. The bound is uniform on $\mathcal{P}$. Now $q_m \asymp e^{-a\ell_j} = e^{-2aL_j}$, while the gradient bound gives $q(s) \ge G(s)$ after this harmless shift. Thus the delay is at most $Ce^{-2\beta(L_j-s)}$. This gives the stated error with a positive uniform rate. The fixed segment $[-P_0, 0]$ is controlled separately by the previous fixed-window relative estimates.

The whole-strip comparison (4.19) transfers this bound to the actual phase in C^0. Its graph and clearing hypotheses hold on the buffered logarithmic disks just checked in (3.20). To pass from the well deficit to the phase, use

$$\frac{d}{dw}\left[a^{-1}\operatorname{arctanh}(1-w)\right] = -\frac{1}{a\,w(2-w)}.$$

The compared deficits are uniformly comparable to the planar deficit on the cropped corridor. The additive lateral error, divided by their minimum, still tends to zero faster than every power of ℓ_j^{-1}: its exponential is $-c\ell_j^k + O(\ell_j)$, and $k > 1$. Thus the comparison indeed gives a small *phase* error, rather than only a small unscaled error in u. On unit balls, divide the well deficit by its planar value. The normalized equations have uniform $C^{1,\alpha}$ bounds. Interpolation gives C^1 closeness with a fixed positive power of the C^0 error. For large T, it follows that $|\nabla U - e_s|$ is small. The phase

equation is then uniformly elliptic. Here is the equation which justifies the subsequent derivative estimate. Since $\nabla u = aq\nabla U$ and $\nabla q = -2auq\nabla U$, the original equation gives

$$A_p(\nabla U) : D^2U = 2ac\tanh(aU)(|\nabla U|^p - 1).$$

Set $v = U - s$ and

$$\mathfrak{b}(U, \nabla v) = 2ac\tanh(aU)\int_0^1 p|e_s + t\nabla v|^{p-2}(e_s + t\nabla v)\,dt.$$

Then v satisfies the linear equation, with the actual solution used in its coefficients,

$$A_p(\nabla U) : D^2v - \mathfrak{b}(U, \nabla v)\cdot\nabla v = 0.$$

On each padded unit ball these coefficients have uniform Hölder bounds, and $A_p(\nabla U)$ has a uniform positive ellipticity constant. Interior Schauder estimates therefore bound the C^2 norm of v on the smaller ball by its C^0 norm on the larger ball. This controls both A_{ph} and the drift $\operatorname{div} A_{\mathrm{ph}}$ in the nondivergence form of $\mathscr{H}_u^+$. The final error has form $N_j^{-\zeta''}\ell_j^C + \ell_j^{-K'} + Ce^{-\kappa_0(L_j - s)}$, with $K' > 2$ chosen beforehand. Integrating gives (8.19). All enlarged unit balls stay inside the padded logarithmic cylinder. Thus ellipticity near the actual turn has not been assumed.

The integration in the preceding step is explicit. The corridor has length $O(\ell_j)$, so its first two error terms contribute

$$CN_j^{-\zeta''}\ell_j^{C+1} + C\ell_j^{1-K'} = o_j(1).$$

The remaining term satisfies

$$\int_{-P_0}^{L_j - T} e^{-\kappa_0(L_j - s)}\,ds \le \kappa_0^{-1}e^{-\kappa_0 T}.$$

These estimates are uniform in the horizontal variable. Their integral form prevents a small coefficient error from accumulating over the growing corridor.

The positive solution of $H_p^+ h = 0$ that decays at $-\infty$ and has right growing coefficient one at $+\infty$ is

$$h_+(s) = \frac{2^p}{J_p}G(s)^{-p/2}\int_{-\infty}^{s} G(t)^p\,dt, \qquad h_+(s) \sim e^{\beta s} \quad (s \to +\infty). \tag{8.20}$$

To verify this formula, put $f_0 = G^{-p/2}$. Since $G'/G = -2ag$ and $g' = aG$,

$$\frac{f_0'}{f_0} = pag, \qquad \frac{f_0''}{f_0} = pa^2G + p^2a^2g^2 = a^2[p^2 - p(p-1)G].$$

Thus f_0 solves the normal equation. Reduction of order gives $f_0(s)\int_{-\infty}^{s} f_0(t)^{-2}\,dt$ as the solution which decays at the left end. At the right end, $G(s)^{-p/2} \sim 2^{-p}e^{\beta s}$ and $\int_{-\infty}^{s} G^p \to J_p$. This proves the normalization.

The corresponding solution with a zero at $-P_0$ and the same right growing coefficient is

$$h_{P_0}(s) = \frac{2^p}{\int_{-P_0}^{\infty} G^p}G(s)^{-p/2}\int_{-P_0}^{s} G(t)^p\,dt.$$

Thus $h_{P_0} \uparrow h_+$ on compact intervals as $P_0 \to \infty$. Both h_{P_0} and its first derivative are positive to the right of $-P_0$.

For a direct check of the monotone convergence, write $J(s) = \int_{-\infty}^{s} G^p$ and $a_0 = \int_{-\infty}^{-P_0} G^p$. The factor which varies with P_0 is $(J(s) - a_0)/(J_p - a_0)$. Its derivative with respect to a_0 is $(J(s) - J_p)/(J_p - a_0)^2 < 0$. Positivity of h_{P_0}' also follows from the ODE: its positive initial derivative increases while the solution is positive.

The next barrier preserves the relative coefficient as well as the exponential rate. Take $E_j \geq 0$ dominating the coefficient errors. Choose T large and then j large so that $c - CE_j \geq c/2$. Solve for ϕ_j in

$$-(c - CE_j)\phi_j'' + CE_j\phi_j' + (ca^2[p^2 - p(p-1)G] + CE_j)\phi_j = 0, \quad \phi_j(-P_0) = 0, \quad \phi_j(S_j) = 1.$$

Smooth upper approximations may be chosen with the same integral and supremum bounds in (8.19). The ODE maximum principle makes ϕ_j positive. Its positive initial derivative and the equation then make it increasing and convex. These signs make it a subsolution for every actual corridor operator: the worst principal, drift, and potential errors are precisely the three added terms in its ODE.

Here is a direct estimate for the change in the coefficient. Let $\widetilde{\phi}_j$ solve the perturbed initial-value problem with the same initial derivative as h_{P_0}. After division by the leading coefficient, write the equations as

$$\widetilde{\phi}_j'' = b_j^{\mathrm{ode}}\widetilde{\phi}_j' + v_j\widetilde{\phi}_j, \qquad h_{P_0}'' = v_0 h_{P_0}.$$

The construction gives

$$0 \leq b_j^{\mathrm{ode}} \leq CE_j, \qquad 0 \leq v_j - v_0 \leq CE_j, \qquad 0 < \mu_0^2 \leq v_0 \leq \mu_1^2,$$

for fixed positive μ_0, μ_1. The Wronskian satisfies

$$(h_{P_0}\widetilde{\phi}_j' - h_{P_0}'\widetilde{\phi}_j)' = b_j^{\mathrm{ode}} h_{P_0}\widetilde{\phi}_j' + (v_j - v_0)h_{P_0}\widetilde{\phi}_j \geq 0.$$

It starts at zero. Thus the logarithmic derivatives obey

$$r_j = \frac{\widetilde{\phi}_j'}{\widetilde{\phi}_j} \geq r_0 = \frac{h_{P_0}'}{h_{P_0}}.$$

Their difference $w = r_j - r_0$ solves

$$w' + (r_j + r_0 - b_j^{\mathrm{ode}})w = b_j^{\mathrm{ode}} r_0 + (v_j - v_0). \tag{8.21}$$

Comparison with constant-potential equations gives

$$\mu_0 \coth(\mu_0(s + P_0)) \leq r_0(s) \leq \mu_1 \coth(\mu_1(s + P_0)).$$

This comparison follows by differentiating the Wronskian of h_{P_0} and $\sinh(\mu(s + P_0))/\mu$, and then taking $\mu = \mu_0$ or μ_1.

The variation kernel for (8.21) is

$$K_j(t, s) = \exp\left[-\int_t^s (r_j + r_0 - b_j^{\mathrm{ode}})\, dr\right].$$

Since $r_j \geq r_0$ and $b_j^{\mathrm{ode}} \leq CE_j$, it is at most $e^{C\int E_j} \exp(-2\int_t^s r_0)$. Integrating the lower comparison for r_0 therefore bounds it by

$$C\left[\frac{\sinh(\mu_0(t + P_0))}{\sinh(\mu_0(s + P_0))}\right]^2 \qquad (-P_0 < t < s).$$

The constant includes $\exp(C\int E_j)$, which is bounded by (8.19). Its integral in s is explicit. If $x = t + P_0 > 0$, then

$$\int_x^\infty \left[\frac{\sinh(\mu_0 x)}{\sinh(\mu_0 v)}\right]^2 dv = \frac{1 - e^{-2\mu_0 x}}{2\mu_0} \leq \min\{x, (2\mu_0)^{-1}\}.$$

Since $r_0(t) \leq C(1 + (t + P_0)^{-1})$, this factor controls the apparent singularity in the forcing term. Fubini's theorem now gives

$$0 \leq \int_{-P_0}^{S_j} w(s)\, ds \leq C \int_{-P_0}^{S_j} E_j(s)\, ds.$$

The solutions have the same first-order zero at $-P_0$. To check the endpoint in the variation formula, put $x = s + P_0$ and let $k_0 > 0$ be their common initial derivative. Smoothness of the chosen ODE coefficients gives

$$h_{P_0} = k_0 x + O(x^3), \qquad \widetilde{\phi}_j = k_0 x + \tfrac{1}{2} b_j^{\mathrm{ode}}(-P_0) k_0 x^2 + O(x^3).$$

Consequently $r_0 = x^{-1} + O(x)$ and $r_j = x^{-1} + b_j^{\mathrm{ode}}(-P_0)/2 + O(x)$. The difference w is bounded. The homogeneous variation factor from a point $x = \delta$ to a fixed positive x is $O(\delta^2)$, so its product with $w(\delta)$ tends to zero. No endpoint term survives when $\delta \downarrow 0$. Normalize $\widetilde{\phi}_j$ at S_j. Then

$$\log \frac{\phi_j(s) h_{P_0}(S_j)}{h_{P_0}(s)} = -\int_s^{S_j} w(t)\, dt.$$

In particular,

$$e^{-C\int E_j} \frac{h_{P_0}(s)}{h_{P_0}(S_j)} \le \phi_j(s) \le \frac{h_{P_0}(s)}{h_{P_0}(S_j)}. \tag{8.22}$$

This bound is uniform on every fixed compact interval contained in $(-P_0, \infty)$, independently of the upper endpoint S_j.

To control the lateral boundary, fix a small $\omega > 0$. The positive potential and the uniform coefficient bounds make

$$B_j(y) = \exp\Big(\omega\big[\sqrt{1 + |y|^2} - \sqrt{1 + \ell_j^{2k}}\big]\Big)$$

a strict supersolution of $\mathscr{H}_u^+$. Since $0 \le \phi_j \le 1$, the subsolution $\phi_j - B_j$ is nonpositive on the side and bottom and at most one on the top. Compare F with $\inf_y F(y, S_j)(\phi_j - B_j)$. On a fixed core the lateral error, relative to the top datum, is $O(e^{-c\ell_j^k})$, which is smaller than $e^{-\beta S_j}$ because $k > 1$.

Write $\mathfrak{a}_j = 1 - u_j(0, L_j)$ for the middle well deficit, and recall $\Gamma_p = 2^{p-1}/p$. Prepared comparison gives $P_j = (\Gamma_p + o(1))\mathfrak{a}_j^p$ and, for fixed T,

$$\frac{F(y, S_j)}{\mathfrak{a}_j^{p/2}} \longrightarrow 2^{p/2} V_p(T)^{(p-2)/2} \left[V_p(T) - \frac{V_p'(T)}{2a}\right]$$

uniformly on the whole retained logarithmic disk. Here $V_p(0) = 1$, $V_p' > 0$ on $(0, \infty)$, and $V_p' = 2a(V_p^p - 1)^{1/p}$. If $C_p^\infty = \lim_{T\to\infty} e^{-2aT} V_p(T)$, planar-tail matching gives $\mathfrak{a}_j C_p^\infty = (2 + o(1)) e^{-a\ell_j}$. Since

$$V - (V^p - 1)^{1/p} = (p^{-1} + o(1)) V^{1-p},$$

the top coefficient satisfies

$$\frac{\inf_y F(y, S_j)}{h_{P_0}(S_j)} = (1 + o_T(1) + o_j(1)) \frac{\mathfrak{a}_j^p}{p}.$$

Here $j \to \infty$ is taken with T, P_0 fixed. To check the coefficient, the numerator is asymptotic to

$$\frac{2^{p/2}}{p} \mathfrak{a}_j^{p/2} (C_p^\infty)^{-p/2} e^{-\beta T},$$

whereas the denominator is asymptotic to $e^{\beta L_j} e^{-\beta T}$. Substituting $e^{-\beta L_j} \sim (\mathfrak{a}_j C_p^\infty/2)^{p/2}$ leaves exactly $\mathfrak{a}_j^p/p$.

The transmission loss is $e^{-C\int E_j}$, which is at least $e^{-C(o_j(1) + e^{-\kappa_0 T})}$ by (8.19). For any fixed P_0 and core, first make the limiting error small by choosing T large. Then take j large, with T fixed. The lateral error is negligible relative to $\mathfrak{a}_j^p$ because $k > 1$. Combining these bounds gives, on the fixed core,

$$F(s) \ge (1 - \varepsilon) \frac{\mathfrak{a}_j^p}{p} h_{P_0}(s). \tag{8.23}$$

We may decrease ε before choosing these fixed parameters.

The core bound is for W_e; the mass estimate requires D. The inequality $W_e/q \geq D/p$ alone does not give the needed lower bound. For every $P \geq 1$, the unit-core estimate gives $I_{p,P}(0) \geq c_I D(0)$. Since $P_j = (\Gamma_p + o(1))\mathfrak{a}_j^p$, a sufficiently large fixed multiple of $\mathfrak{a}_j^p$ for $D(0)$ already proves the lemma. For example, choose

$$C_* \geq \frac{4}{c_I} \max_{p \in \mathcal{P}} \frac{2\beta\Gamma_p}{\sigma_E}.$$

The case $D(0) > C_*\mathfrak{a}_j^p$ then follows from the unit-core estimate, once j is large. It remains to treat $D(0) \leq C_*\mathfrak{a}_j^p$, with C_* fixed uniformly on $\mathcal{P}$. For each fixed P, the normal-ray estimate 2.20 applies on $[-P, P]$ once j is large, because $D(0) \to 0$. Its accurate-ray estimates give $D(s) \leq C_P D(0)$, $|Z(s) - e| \leq C_P D(0)$, and $U(s) - s = O_P(D(0))$. With $n = Z/|Z|$, this gives $|n(s) - e| \leq C_P D(0)$. Hence $1 - n \cdot e = |n - e|^2/2 = O_P(D(0)^2)$, and

$$\frac{W_e}{q} = 1 - Z_e = \frac{D}{p} + O_P(D(0)^2).$$

The displayed expansion follows from $|Z| = (1-D)^{1/p} = 1 - D/p + O(D^2)$ and $Z_e = |Z|(n \cdot e)$. Both remainders are uniform on each fixed core. Since $F = q^{p/2}(W_e/q)$ and $D(0)^2 = O(\mathfrak{a}_j^{2p}) = o(\mathfrak{a}_j^p)$ in this case, (8.23) gives $D(s) \geq (1 - \varepsilon)\mathfrak{a}_j^p G^{-p/2} h_{P_0}(s) - o(\mathfrak{a}_j^p)$ uniformly on the fixed core.

Multiplying the core lower bound by q^{p+1} gives

$$\int_{-P}^{P} q^{p+1} D \, ds \geq (1 - \varepsilon)\mathfrak{a}_j^p \int_{-P}^{P} G^{(p+2)/2} h_{P_0} \, ds - o(\mathfrak{a}_j^p).$$

The exponent is $p + 1 - p/2 = (p + 2)/2$. Here $q = G + O_P(D(0))$ on the fixed core, and $D(0) = O(\mathfrak{a}_j^p)$, so this replacement contributes only $o(\mathfrak{a}_j^p)$. The finite-window choice is a model calculation, independent of the prepared sequence. Symmetry and $\int_{\mathbb{R}} G = 2/a$ give the exact integral

$$\int_{\mathbb{R}} G^{(p+2)/2} h_+ \, ds = \frac{2^p}{J_p} \int_{\mathbb{R}} G(s) \int_{-\infty}^{s} G(t)^p \, dt \, ds = \frac{2^p}{a}.$$

Here is the symmetry calculation. Put $A(s) = \int_{-\infty}^{s} G(t)^p \, dt$ and $J = \int_{\mathbb{R}} G(s)A(s) \, ds$. Evenness gives $A(-s) = J_p - A(s)$. Substitution of $-s$ for s therefore gives $J = J_p \int G - J$, so $J = (J_p/2)(2/a) = J_p/a$.

We make the uniform choices explicit. Put

$$a_- := \min_{p \in \mathcal{P}} a_p > 0, \quad p_- := \min \mathcal{P}, \quad p_+ := \max \mathcal{P}, \quad J_- := \min_{p \in \mathcal{P}} J_p > 0, \quad \mathfrak{m}_- := \min_{p \in \mathcal{P}} \frac{2^p}{a_p} > 0.$$

Let $\eta = \min\{\varepsilon/100, 1/100\}$ and use loss at most η in the preceding fixed-core lower bound. Since $0 \leq A(s) \leq J_p$ and $G(s) \leq 4e^{-2a_-|s|}$, the model integrand satisfies

$$0 \leq G^{(p+2)/2} h_+(s) = \frac{2^p}{J_p} G(s)A(s) \leq 2^{p_+ + 2} e^{-2a_-|s|}.$$

Thus its two tails outside $[-P, P]$ have integral at most $2^{p_+ + 2} a_-^{-1} e^{-2a_- P}$. Choose $P_* \geq 1$, using only these model constants, so that this bound at $P = P_*$ is at most $\eta\mathfrak{m}_-$. The mass on that interval is then at least $(1 - \eta)2^p/a_p$, uniformly on $\mathcal{P}$. Set $P = P_*$ in the preceding core estimate.

The second truncation is uniform as well. For $P_0 > P_*$ put $a_0 = \int_{-\infty}^{-P_0} G^p$. Its common tail bound is

$$0 \leq a_0 \leq \frac{4^{p_+}}{2p_- a_-} e^{-2p_- a_- P_0}.$$

The exact formulas for the two normal solutions give, on $[-P_*, P_*]$,

$$G^{(p+2)/2}(h_+ - h_{P_0}) = \frac{2^p G(s) a_0 [J_p - A(s)]}{J_p(J_p - a_0)} \geq 0.$$

Once $a_0 \le J_-/2$, its integral is at most $2^{p_+ +2}a_0/(a_-J_-)$, because $\int G \le 2/a_-$. Choose $P_0 > P_* + 1$ so that this bound is at most $\eta\mathfrak{m}_-$ for all $p \in \mathcal{P}$. This choice also depends only on $\mathcal{P}$ and ε. It gives a truncated model mass at least $(1-2\eta)2^p/a_p$. Choose the crop and then the sequence index as in the preceding transmission argument. The remaining $o_j(1)$ errors are made smaller than their assigned η fractions at that index. These finitely many losses are smaller than the stated tolerance. The core integral therefore satisfies

$$\int_{-P}^{P} q^{p+1} D \, ds \ge (1-\varepsilon)\frac{2^p}{a}\mathfrak{a}_j^p.$$

The exact coefficient identity is $2^p/a = (2\beta/a^2)\Gamma_p$. Use $P_j/(\Gamma_p\mathfrak{a}_j^p) \to 1$ and $P_j e^{\beta_j \ell_j}/\Pi_{p_j} \to 1$. Taking the intermediate losses smaller than the prescribed tolerance gives both stated lower bounds after division by $\sigma_q = \sigma_E/a^2$. The choices are made in this order: a smaller internal tolerance, then the model window P_*, then P_0, then T, and finally j. In particular P_* is fixed before any prepared sequence is given. For every larger fixed P, Modica's bound gives $D \ge 0$ along the whole normal ray. Hence $I_{p,P} \ge I_{p,P_*}$, and the same mass lower bounds hold on that larger window. All remaining convergence errors affect only the sufficiently large index for each prepared sequence. A sequence of failing radial bases and exponents would itself be a prepared sequence, with this same model window, and would contradict the estimate just proved. □

For this computation, write $\mathfrak{a}$ for the positive well deficit at the middle of a prepared gap. It is a scalar amplitude, not the curvature $b = |B|$. Define

$$\Gamma_p = \frac{2^{p-1}}{p}.$$

This coefficient follows from the actual midpoint stress. With the phase reflected if necessary, normalized turn preparation gives $1 - u = \mathfrak{a}V_p + o(\mathfrak{a})$ in C^1 on fixed bands. At the middle, $V_p(0) = 1$ and $V_p'(0) = 0$; all normalized tangential derivatives also tend to zero. Consequently $|\nabla u|/\mathfrak{a} \to 0$ there. For the physical normal stress

$$\mathsf{p} = W_p(u) + \frac{|\nabla u|^p}{p} - |\nabla u|^{p-2}u_s^2$$

at that point, the kinetic terms are $o(\mathfrak{a}^p)$. The potential satisfies

$$\frac{W_p(1-\mathfrak{a})}{\mathfrak{a}^p} = \frac{(2-\mathfrak{a})^p}{2p} \longrightarrow \Gamma_p.$$

Thus $\mathsf{p} = (\Gamma_p + o(1))\mathfrak{a}^p$. The same leading relation holds for the scalar stress $W_p(u) - (p-1)|\nabla u|^p/p$. All powers use the actual p_j in a sequence, and these limits are uniform on $\mathcal{P}$.

We check the distance coordinate used for physical switching. The graph preparation gives buffered graph disks of radius much larger than the gap, with slopes tending to zero and $\ell\|D^2 f\| \to 0$. For an observation point in a fixed middle band, a vertical competitor is at distance $O(\ell)$. Its minimizing foot on either graph therefore lies in the interior of the larger disk. The horizontal Hessian of half the squared distance to a graph is

$$I + Df \otimes Df + (f - s)D^2 f.$$

On a common convex buffered disk, $|f - s| \le C\ell$. Hence this Hessian is at least $(1 - C\ell\|D^2 f\|)I \ge I/2$ for large j. The graph height and curvature errors have the required bounds on each such logarithmic disk. Hence each foot is unique. The normal-coordinate map is nonsingular because $\ell\|B\| \to 0$. The two true inward distances s_i, s_j are smooth on a neighborhood of the middle band.

In the parallel reference coordinates, put $\tau = s - \ell/2$. The graph height error, the squared slope times ℓ, and the curvature times ℓ^2 all tend to zero under the recorded $N_j^{-\zeta}\ell_j^C$ preparation.

Distance comparison with the two reference planes gives, on every fixed middle band,

$$s_i = s + o(1), \quad s_j = \ell - s + o(1), \qquad \nabla s_i = e_s + o(1), \quad \nabla s_j = -e_s + o(1).$$

Thus the intrinsic middle coordinate

$$t = \frac{s_i - s_j}{2}$$

satisfies $t = \tau + o(1)$ and $\nabla t = e_s + o(1)$. Along an inward normal ray from sheet i its traversal derivative is

$$\frac{dt}{ds_i} = \frac{1 - \nabla s_j \cdot \nabla s_i}{2} = 1 + o(1) > \frac{1}{2}.$$

It therefore parametrizes the band once. Its inverse derivative tends to one. The normal Jacobian is also $1 + o(1)$, since $|s_i B_i| = o(1)$. These facts justify both the coordinate change and the physical measure in the energy integral. Changing the affine middle point to $t = 0$ moves it by $o(1)$. Normalized C^1 convergence and $V_p(0) = 1$ show that this changes $\mathfrak{a}$ by a factor $1 + o(1)$ and preserves the stress expansion.

We approximate optimal normal interpolation by smooth switches on fixed finite bands in this coordinate. The normalized physical conductance converges to

$$\mathfrak{a}^{-p} q^2 A_u[\nabla t, \nabla t] \longrightarrow C_p(t) := 4c V_p(t)^2 |V_p'(t)|^{p-2}.$$

To see the coefficient, use $q = 2\mathfrak{a} V_p + o(\mathfrak{a})$, $\nabla u = -\mathfrak{a} V_p' e_s + o(\mathfrak{a})$, and $\nabla t = e_s + o(1)$, with a consistent orientation of t. The normal eigenvalue of the linearized p-Laplace matrix is $c|\nabla u|^{p-2}$. Multiplying it by q^2 gives exactly $4c\mathfrak{a}^p V_p^2 |V_p'|^{p-2}$. For the varying-exponent sequence, keep p in the standing compact subset of $(2, \infty)$. On a fixed band, the normalized gradients are uniformly bounded. The map $(p, \xi) \mapsto A_p(\xi)$ is jointly continuous on this bounded set, including at $\xi = 0$. The normalized C^1 convergence therefore gives the stated L^1 conductance convergence, uniformly in the base, through the turn. At the fixed endpoint $p = 2$, this passage uses $A_2 = I$. For $1 < p < 2$, normalized C^1 convergence alone does not control the singular linearized coefficient.

For a scalar conductance $C(t) > 0$ almost everywhere, a switch from one to zero has energy $\int C|\mathcal{U}'|^2$. Cauchy–Schwarz gives

$$1 = \left| \int \mathcal{U}' \right|^2 \le \left(\int C|\mathcal{U}'|^2 \right) \left(\int C^{-1} \right).$$

If $0 < \int C^{-1} < \infty$, its reciprocal is the optimal capacity. Indeed, the primitive of $-C^{-1} / \int C^{-1}$ is an absolutely continuous switch with the required end values. Substitution gives equality in Cauchy–Schwarz. This minimizing derivative can be unbounded. We therefore approximate it by bounded smooth derivatives before constructing the ambient test.

For the one-dimensional profile, the resistance is explicit for every $p > 1$:

$$\begin{aligned} \int_{\mathbb{R}} C_p(t)^{-1} \, dt &= \frac{1}{2c(2a)^{p-1}} \int_1^\infty V^{-2} (V^p - 1)^{-(p-1)/p} \, dV \\ &= \frac{1}{2c(2a)^{p-1}} = \frac{a}{2^{p-1}}. \end{aligned}$$

The factor two comes from the evenness of V_p. On $t > 0$, use $dt = dV / [2a(V^p - 1)^{1/p}]$. For the remaining integral, put $z = V^{-p}$. The powers of z cancel:

$$V^{-2} (V^p - 1)^{-(p-1)/p} \, dV = -\frac{1}{p} (1 - z)^{-(p-1)/p} \, dz.$$

The resulting integral is $p^{-1} \int_0^1 (1 - z)^{-(p-1)/p} \, dz = 1$. The last equality uses $2ca^p = 1$. The optimal capacity is consequently

$$\mathfrak{a}^p \left(\int_{\mathbb{R}} C_p^{-1} \right)^{-1} = \frac{2^{p-1}}{a} \mathfrak{a}^p = (1 + o(1)) \frac{\beta}{a^2} \mathfrak{p}. \tag{8.24}$$

Here p is the actual gap pressure normalized above.

We derive the model properties needed for uniform choices of bands and smoothing parameters. These properties of V_p hold on every compact interval $[p_-, p_+] \Subset (1, \infty)$. Its defining inverse integral is

$$\mathcal{T}_p(v) = \frac{1}{2a_p} \int_1^v (z^p - 1)^{-1/p}\, dz, \qquad \mathcal{T}_p(V_p(t)) = t \quad (t \geq 0).$$

Near $z = 1$, the integrand is bounded by $C(z-1)^{-1/p_-}$, which is integrable. Away from one it is jointly continuous in (p, z). Dominated convergence therefore gives joint continuity of $\mathcal{T}_p(v)$ on compact parameter and value intervals. For $v \geq 2$,

$$\mathcal{T}_p(v) \geq \frac{1}{2a_+} \log(v/2), \qquad a_+ = \max_{p \in [p_-, p_+]} a_p.$$

Thus bounded t gives a common bound for its inverse $V_p(t)$. Any convergent parameter-time sequence has a subsequence of inverse values converging to the unique inverse at the limit. This proves joint continuity of $(p, t) \mapsto V_p(t)$. The first integral gives

$$V_p'(t) = 2a_p(V_p(t)^p - 1)^{1/p} \quad (t \geq 0).$$

It proves joint continuity of V_p' as well, including at zero. Even reflection gives the assertions for negative t.

The time $\mathcal{T}_p(2)$ is uniformly bounded. Past a common such time, $V_p \geq 2$ and

$$0 < \lambda_- := \min_{p \in [p_-, p_+]} 2a_p(1 - 2^{-p})^{1/p} \leq \frac{V_p'}{V_p} \leq 2a_+.$$

Integration gives common positive exponential lower and upper bounds for V_p outside a fixed interval. It also gives $cV_p \leq |V_p'| \leq CV_p$ there. Consequently the resistance density has a common exponential tail:

$$C_p(t)^{-1} \leq Ce^{-p_-\lambda_-|t|} \quad \text{outside a fixed interval.}$$

The coefficient $C_p = 4(p-1)V_p^2|V_p'|^{p-2}$ is jointly continuous on each compact (p, t) set with $p > 2$, including at the turn. Here the exponent $p - 2$ has a positive lower bound. These are the model facts used below for the standing physical range $\mathcal{P} \Subset (2, \infty)$.

For any $\varepsilon > 0$, first choose a finite width T_*. Then choose a smooth switch $\mathcal{U}_p$, equal to one left of $-T_*$ and zero right of T_*, with $\mathcal{U}_p(-t) = 1 - \mathcal{U}_p(t)$ and

$$\int_{-T_*}^{T_*} C_p \mathcal{U}_p'^2 \leq (1 + \varepsilon)\frac{2^{p-1}}{a}.$$

Choose T_* so large that the resistance on $[-T_*, T_*]$ has inverse at most $(1 + \varepsilon/4)2^{p-1}/a$. This choice is uniform on $\mathcal{P}$, because V_p grows exponentially with uniform positive rates outside a fixed neighborhood of the turn. Near the turn, the ODE gives

$$V_p(t) - 1 \asymp |t|^{p/(p-1)}, \qquad |V_p'(t)| \asymp |t|^{1/(p-1)}, \qquad C_p(t) \asymp |t|^{(p-2)/(p-1)}.$$

The comparison constants are uniform on $\mathcal{P}$. For example, integrate $V_p' = 2a[p(V_p - 1) + O((V_p - 1)^2)]^{1/p}$ near $V_p = 1$ to obtain the first two relations. If $p_+ = \max \mathcal{P}$, then $\alpha_+ = (p_+ - 2)/(p_+ - 1) < 1$. Consequently $C_p^{-1} \leq C|t|^{-\alpha_+}$ on $0 < |t| \leq 1$. This proves uniform integrability at the turn. Put $R_\epsilon = \int_{-T_*}^{T_*} (C_p + \epsilon)^{-1}$ and first use the switch with derivative $-(C_p + \epsilon)^{-1}/R_\epsilon$. Its energy is

$$\frac{\int_{-T_*}^{T_*} C_p(C_p + \epsilon)^{-2}}{R_\epsilon^2} \longrightarrow \left(\int_{-T_*}^{T_*} C_p^{-1} \right)^{-1}.$$

The integrand in the numerator is bounded by C_p^{-1}, so dominated convergence applies. Uniformity in p can be checked without interchanging two unspecified limits. Given $\delta > 0$, the integral over $|t| < \delta$ is bounded by $C\delta^{1-\alpha_+}$, independently of p and ϵ. On $\delta \leq |t| \leq T_*$, the conductance has a

common positive lower bound and depends continuously on (p,t). The convergence there is uniform. First decrease δ and then decrease ϵ. The same argument applies to R_ϵ. The limiting resistance has a common positive lower bound, so division by its square preserves uniformity. Write the negative of this derivative as an even nonnegative density $\mathfrak{g}_{p,\epsilon}$ of integral one. Its ordinary supremum is finite after $\epsilon > 0$ is fixed. Multiply it by an even cutoff which vanishes near $\pm T_*$, and divide by the resulting integral. The removed intervals can be chosen so short that this changes its weighted energy by an arbitrarily small amount. Next convolve with an even nonnegative smooth kernel, whose support is smaller than the removed endpoint strips. This preserves nonnegativity, evenness, integral one, and support inside $(-T_*, T_*)$. Weighted energy converges because C_p is continuous and bounded on the fixed interval. For the fixed positive ϵ, the densities $\mathfrak{g}_{p,\epsilon}$ are jointly continuous in (p,t) on $\mathcal{P} \times [-T_*, T_*]$. Indeed, C_p is jointly continuous there, including at the turn, since $\min \mathcal{P} > 2$. The normalization R_ϵ has a common positive lower bound. Hence these densities have a common supremum bound and a common modulus of continuity. One endpoint cutoff and one mollification radius therefore give the required energy accuracy for the whole family. Denote the resulting smooth density by $\mathfrak{g}_p$ and define $\mathcal{U}_p(t) = \int_t^\infty \mathfrak{g}_p(v)\,dv$. Its derivative $-\mathfrak{g}_p$ has a common ordinary supremum bound. Evenness and integral one give the required reflection identity. Fix the width and regularization before taking the large-radius limit. The switch then has a bounded ordinary derivative, so the L^1 conductance convergence suffices for the actual energy limit through the turn. For a negative gap, replace u by $-u$. The equation, q, D, and A_u are unchanged under this reflection. Thus the same mass and capacity estimates apply on either side of every sheet.

Use the common radial amplitude in (8.13). Here d is the reach, G_1 its plateau penalty, and W_* the common weight. The cutoff m_H removes the polynomial rough neighborhood, and m_B is the outer cutoff. The logarithmic slope of W_* in d is at most $q_* + o_R(1)$. Denote cutoff costs by $\mathcal{E}_{R,B}$. The error convention is (8.29). The bounds (8.30)–(8.32) are proved in the next subsection directly from the fourth moment and the scalar weight. They do not use the interpolation proposition below. Thus those forward references cause no circularity. Every calculation is first made at finite B. The outer limit $B \to \infty$ is then taken before $R \to \infty$.

Supported clean bases have the polynomial preparation on every needed logarithmic window. In such a window the zero graphs are ordered. On either normal side of a sheet there is therefore at most one first neighboring sheet. A gap between two successive graphs has one middle band. These are the only bands used in the construction.

We justify uniformity when these local computations are used on all supported bases. Fix internal coefficient tolerances strictly smaller than the final assigned loss budgets. Choose the common finite window P above the lower bound in Lemma 8.1. Fix also the finite switch band T_*, its regularization, and all required powers of the logarithmic comparison radius. These choices use only the tolerances and $\mathcal{P}$; they precede R and the finite outer cutoff B.

Suppose a normalized mass or capacity bound used below failed uniformly for $\rho \geq R$. There would be $R_j \to \infty$ and violating contributing gaps, with supported bases z_j and $N_j := \rho(z_j) \geq R_j$. Allow the entire solutions and exponents to vary, and pass to $p_j \to p_* \in \mathcal{P}$. Translate z_j to zero, rotate its own normal to e_s, and reflect the solution if the chamber is negative. These operations preserve the entire matched equation, strict range, and Modica bound. The preparation proved in the preceding subsection gives fixed constants $c_0, C_M > 0$ with

$$c_0 \log N_j \leq \ell_j \leq C_M \log N_j.$$

On every required fixed power of this gap, the graph, jet and zero-defect errors are bounded by $C_A N_j^{-\zeta}(1 + \log N_j)^{C_A}$, with one positive ζ and constants independent of the selected bases and exponents. Fixed-core preparation, clearing, adjacency, and all comparison buffers hold there as well. For a nearest-gap mass used to pay a contributing side, choose any nearest adjacent graph on either normal side of the selected base. Such a graph exists because the contributing band supplies an adjacent graph at distance $O(\log N_j)$. The lower gap bound proved above applies on both sides.

The nearest gap is no larger than the contributing gap. After adjusting c_0, C_M once to account for the $o_R(1)$ difference between normal and shortest distances, its length is therefore between $c_0 \log N_j$ and $C_M \log N_j$. Both feet and every required fixed logarithmic power disk lie in the same polynomially clean neighborhood. They inherit the graph, jet, adjacency and buffer estimates just checked. If this nearest chamber is negative, reflect u_j; this leaves q, D and A_u unchanged. Thus Lemma 8.1 applies at the selected base with the same fixed mass window, even when its nearest chamber lies on the other side from the contributing band. The selected configurations are precisely a prepared-gap sequence of the class used in Lemma 8.1 and in (8.24).

Apply those results to this sequence, at the already fixed P, T_* and regularizations. Their sequence-dependent index threshold is sufficient: beyond that one index, no selected configuration can violate the fixed coefficient tolerance. This contradicts the chosen sequence of violations. Every power and normalization retains the actual exponent p_j. The positive factors $a_p, \beta_p, \Pi_p, \sigma_q$ have uniform upper and lower bounds on $\mathcal{P}$, so passing to a compact exponent subsequence creates no additional rate condition. Thus the chosen mass lower bounds and capacity upper bounds hold simultaneously at all supported bases once R exceeds one threshold for these fixed finite choices. The additional preparation errors are $o_R(1)$ uniformly over those bases. The finite-window and regularization losses remain inside their chosen fixed tolerances. The argument also permits arbitrary finite B_j, since preparation and the local coefficients use N_j and do not depend on B_j. No common convergence rate over arbitrary prepared sequences has been assumed. On a side with no contributing middle band, only the one-sided core and cap estimates are used; no upper bound is imposed on such a gap.

We need the reach–gap identity only at a contributing band. Such a band gives a competitor of length $O(\log \rho)$, and hence $d = O(\log \rho)$. A diagonal competitor costs at least a fixed multiple of reciprocal curvature. The curvature is a negative power of ρ, so this cost is much larger than $\log \rho$. Every active competitor of competitive cost is therefore an off-diagonal chord lying in the polynomially clean neighborhood.

An off-diagonal chord could still join the base to another point on its *own* local graph. We must exclude that case as well. Write ϵ_0 for the curvature bound on the clean graph disk; here $\epsilon_0 \le C\rho^{-1/100}$. If y belongs to the source graph and $r = |y - z| = O(\log \rho)$, its height above $T_z\Sigma$ satisfies

$$|\nu(z) \cdot (y - z)| \le C\epsilon_0 r^2.$$

Thus the incidence cosine is $s = |\nu(z) \cdot (y - z)|/r \le C\epsilon_0 r$. For $s > 0$, the inequality $\Gamma(s) \ge 1/s$ gives

$$r\Gamma(s) \ge r/s \ge (C\epsilon_0)^{-1} \gg \log \rho.$$

If $s = 0$, the cost is infinite and the same exclusion holds. Every competitive chord lies in this graph disk because $\Gamma \ge 1$ and its cost is $O(\log \rho)$. Consequently a competitive off-diagonal chord must join two distinct local graphs. This exclusion is needed before minimizing their Euclidean separation.

For two parallel planes at distance ℓ, a chord with normal component $s \in (0, 1]$ has length ℓ/s and penalized cost $\ell\Gamma(s)/s$. Moreover,

$$\frac{d}{ds}\frac{\Gamma(s)}{s} = \frac{\Gamma'(s)}{s} - \frac{\Gamma(s)}{s^2} < 0.$$

Outside the fixed angular plateau this cost exceeds the normal cost by a fixed positive fraction. The graph errors on the logarithmic window tend uniformly to zero. Comparison with a normal-line intersection therefore forces every minimizing chord into the plateau when R is large.

The shortest Euclidean chord to a neighboring graph is normal to that graph at its endpoint. Its incidence with the source normal also lies in the plateau, because both slopes tend to zero. Every penalty is at least G_1. The lower bound for penalized cost and this Euclidean competitor

consequently give the exact identity

$$d(z_i) = G_1 \ell_i,$$

where ℓ_i is the shortest distance from the base to an adjacent graph. A chord to a nonadjacent graph would cross a closer graph along the same direction, and its cost would strictly decrease at that first crossing. It cannot minimize. The distance ℓ_i and the length along the initial normal line differ by $o_R(1)$ on these prepared windows. We assert this identity only where a band contributes to the test. A useful consequence is that the reach of a contributing base is controlled by its nearest adjacent gap, even if the band under consideration lies on its other side. For a band on side ij, let ℓ_{ij} be the distance from the fixed source base to its partner graph. Normal-foot geometry gives

$$s_i + s_j = \ell_{ij} + o_R(1), \qquad d(z_j) \le G_1(\ell_{ij} + o_R(1)).$$

uniformly on the band. The Jacobians are $1 + o_R(1)$, and all additive errors are uniform for $\rho \ge R$. These facts come from geometry, independently of the capacity computation. Here are the projection estimates behind these statements. Fix a contributing base and use its tangent plane as the horizontal plane. Write the partner as $s = f(y)$, with $f(0) = \ell_N$, where ℓ_N is the normal-line intersection length. On every required logarithmic disk, preparation gives

$$|Df| \le \epsilon_\rho, \qquad |D^2 f| \le \kappa_\rho, \qquad \epsilon_\rho \le C\rho^{-c}(1+\log\rho)^C, \quad \kappa_\rho \le C\rho^{-c}(1+\log\rho)^C$$

for some fixed $c > 0$. Both ℓ_N and each contributing ray length are $O(\log\rho)$. For $x = (0, s)$ on the own normal ray, the partner foot $y(s)$ satisfies

$$y + (f(y) - s)Df(y) = 0.$$

Its Jacobian in y is $I + Df \otimes Df + (f - s)D^2 f = I + o_R(1)$. It is invertible with inverse operator norm at most two. The vertical competitor and the larger graph buffer place the minimizer in the interior of this convex disk; the positive Jacobian also gives uniqueness. The equation implies $|y(s)| \le C\ell_N \epsilon_\rho$. Consequently, uniformly on each fixed middle band,

$$s_j(x) = \ell_N - s + O(\epsilon_\rho^2 \ell_N).$$

Indeed, $|f(y) - \ell_N| \le \epsilon_\rho |y|$, and the horizontal part of the distance contributes at most $C|y|^2/\ell_N$ there. Apply the same calculation at $s = 0$ to get $\ell_{ij} = \ell_N + O(\epsilon_\rho^2 \ell_N)$. Thus the shortest and normal-line gaps differ by $o_R(1)$, and $s_i + s_j = \ell_{ij} + o_R(1)$ on the band. All errors are a negative power of ρ times a fixed logarithmic power, so they are uniform for $\rho \ge R$.

Differentiating the foot equation gives

$$y'(s) = \left[I + Df \otimes Df + (f - s)D^2 f\right]^{-1} Df, \qquad |y'(s)| \le 2\epsilon_\rho. \tag{8.25}$$

The physical partner foot $(y(s), f(y(s)))$ therefore moves by at most $C\epsilon_\rho$ times the ray length traversed. Since $dt/ds = 1 + o_R(1)$, its displacement across the fixed recovered band is $O_{T_*}(\epsilon_\rho)$. The normal Jacobians are $1 + O(\kappa_\rho \log\rho)$. The chord from the own base to this partner foot has length $\ell_{ij} + o_R(1)$ and incidence in the fixed angular plateau. It is an admissible receiving-foot competitor, so $d(z_j) \le G_1(\ell_{ij} + o_R(1))$. This proves the required distance, displacement, and measure estimates in the actual normal coordinates.

The associated multiplicative errors remain attached to the mass or surface Dirichlet term until the finite-cutoff absorption. They are not being asserted to have an absolutely summable integral without those terms.

Proposition 8.2 (Physical interpolation with unit mass)**.** *Under these inputs, put* $\mathcal{L} = \int_\Sigma I_{p,P}\psi^2$ *and* $y = q_* G_1/\beta$*. For every fixed* $\varepsilon > 0$*, after all finite normal windows and regularizations are chosen and* R *is increased, physical stability yields*

$$\left[1 - \left(\frac{1}{2} + \varepsilon\right)\frac{y^2}{(1+y)^{2+2/y}} - \varepsilon\right]\mathcal{L} \le (1+\varepsilon)\int_\Sigma |\nabla_\Sigma \psi|^2 + \mathcal{E}_{R,B}. \tag{8.26}$$

In particular the effective switching ratio may be taken at most one, with fixed strict room for all finite Young and geometric factors. No coefficient $(p-1)$ is lost in the leading tangential energy, and no normal phase current occurs.

Proof. Use the exact physical ground-state identity

$$0 \le Q_u(q\eta) = \int q^2 A_u[\nabla\eta, \nabla\eta] - \int q^{p+1} D\eta^2. \tag{8.27}$$

This is the ground-state calculation behind (2.10): test (2.9) with $q\eta^2$ and retain the exact value of $Q_u(q\eta)$ before using stability. The physical coefficient denoted by A there is A_u here; both equal $A_p(\nabla u) = a^{p-2}q^{p-2}A_{\mathrm{ph}}$. Thus that calculation gives the sign and coefficient in (8.27). It remains exact for compact Lipschitz η by approximation. For $p > 2$, A_u is continuous at critical gradients and bounded on compact sets. The potential $W_p''(u)$ is bounded there as well. On a single-projection region, $z_i(x)$ is the nearest normal foot on its zero graph. Put $\widetilde{\eta} = \psi(z_i(x))$. On a middle band use

$$\widetilde{\eta} = \mathcal{U}_p(t)\psi(z_i(x)) + (1 - \mathcal{U}_p(t))\psi(z_j(x)).$$

Here the two sheet distances are nonnegative in their common gap, and $t = (s_i - s_j)/2$. Thus $t < 0$ on the side nearest sheet i, as required by the choice $\mathcal{U}_p = 1$ on the left. The middle band has $|t| < T_*$; all gaps under consideration are longer than $2T_* + 2P$ once R is large. Consequently distinct middle bands are disjoint, and each lies outside the fixed cores. Interchanging the labels replaces t by $-t$. The identity $\mathcal{U}_p(-t) = 1 - \mathcal{U}_p(t)$ therefore leaves the formula unchanged. A normal foot is determined by the actual sheet, so changing graph coordinates also leaves the formula unchanged. At the two ends of the middle band the switch is constant; hence the formula agrees with the adjacent single-projection formula. These facts establish compatibility on all overlaps. Multiply by a depth cutoff, expressed in the distance to the nearest zero, equal to one through $M \log \rho$ and zero one unit later. On a side whose middle band is beyond this depth, use only the single-projection formula and terminate it at that cutoff. In particular, no upper bound on the gap is needed on such a side. A middle band meeting the cutoff support has gap at most $2M \log \rho + O(1)$ and has the prepared comparison used above. Choose the auxiliary bad boundary inside the portion killed by m_H. The killed polynomial collar is much wider than any interacting logarithmic window. Thus every relevant foot at that boundary has $m_H = 0$. If a band has a nonzero amplitude, one supported foot gives a buffered polynomially clean neighborhood containing both feet and the band. The foot maps used there are therefore regular. Extend the resulting function by zero at the auxiliary boundary.

At fixed R, B, the base amplitude is locally Lipschitz on its compact regular support. The foot maps have bounded derivatives on the prepared tubes. The fixed recovered switch is Lipschitz even at a critical turn. The formulas agree on overlaps, and the depth cutoff is Lipschitz because distance to the zero set is Lipschitz. They therefore define a compact ambient Lipschitz function η. Its support has normal depth $O(\log(R + B))$ at finite B. Consequently $q\eta \in H_c^1$. Ordinary smooth approximation in H^1 is valid for the bounded form coefficients. Apply it at these fixed parameters and pass to the limit before increasing B or R. No unregularized derivative of the turn profile is used.

The disjoint fixed cores contain no middle band, and the depth cutoff is one there. They satisfy $\eta = \psi(z)$. The fixed-window definition of $I_{p,P}$ and the tube Jacobian $1 + o_R(1)$ give the lower bound $(1 - o_R(1))\sigma_q\mathcal{L}$ for their negative mass in (8.27). The source outside these cores is nonnegative and may be discarded. This gives the mass coefficient one.

For the tangential energy, consider a fixed core. There the tangential restriction of $q^2 A_u$ tends to $a^{p-2}G^p I$. The inverse metric and Jacobian tend to the identity. Beyond a fixed depth, $A_{\mathrm{ph}} \le cI$ and clearing give a uniformly integrable bound for q^p. Make the tail small first, then use core

convergence. The integrated coefficient is therefore

$$\int_{\mathbb{R}} a^{p-2} G(s)^p \, ds + o_R(1) = \sigma_q + o_R(1).$$

Only the tangential eigenvalues enter this limit. The larger normal eigenvalue c does not multiply the leading surface Dirichlet term. In a middle band, write $\mathfrak{a}_{ij}$ for that gap's middle well deficit. Convexity bounds the two differentiated base amplitudes by their separate energies. Their total coefficient is $O(\mathfrak{a}_{ij}^p) = o_R(1)$, since all contributing gaps tend uniformly to infinity. Separate the switch derivative with a small fixed Young factor. The reciprocal factor on the band's tangential cost is fixed first; the coefficient $O(\mathfrak{a}_{ij}^p)$ then tends uniformly to zero as $R \to \infty$. It therefore changes no leading tangential coefficient.

Use the proved uniform clearing rate for the far-depth cutoff. Write $c_{\mathrm{cl}} > 0$ for a rate in (6.3). Since $q^p = 2pW_p(u) \le 2pe_p(u)$, it gives $q^p(x) \le Ce^{-c_{\mathrm{cl}} \operatorname{dist}(x,\Sigma)}$. On a cap the nearest-zero depth is $M \log \rho$, up to a fixed middle-band offset and the $o_R(1)$ coordinate errors just proved. The base and physical radii differ by $O(\log \rho)$. For large R, the distance is therefore at least $(M/2) \log \rho$. Put $c_{\mathrm{cap}} = c_{\mathrm{cl}}/2$. The derivative of the depth cutoff is uniformly bounded, and

$$q^p \le C\rho^{-c_{\mathrm{cap}} M}$$

there. This estimate uses no specified sharp decay rate.

The cap energy is bounded by a dyadic sum of $Cs^{3-c_{\mathrm{cap}} M}(1+\log s)^C$, $s \ge R$. The power three is $m+1-4$ at $m=6$: the coarse good-area bound is Cs^{m+1} and $h^2 = s^{-4}$, while $W_* \le C$. The tube metrics are uniformly bounded on the logarithmic windows. Fix the radial half-tube depth constant M before R, B, larger than every earlier fixed depth requirement and such that $c_{\mathrm{cap}} M \ge 13$. This choice leaves the physical caps and all constants in the preceding fourth-moment estimate fixed. The exponent is then at most -10. Its dyadic sum tends to zero, even with any fixed logarithmic power. It is $o_R(\mathcal{A})$ uniformly in finite B. This pays caps on long unpaired sides and caps meeting a middle band. All constants and the chosen M are uniform on $\mathcal{P}$.

The bad-collar part of a middle band is negligible by (8.30). More precisely, remove all bands having either foot where $m_H \ne 1$ before making the sharp switching estimate. Their two feet are separated by $O(\log \rho)$, which is smaller than the polynomial collar scale. A fixed enlargement of that collar therefore contains both feet. Its shell estimate pays the bounded switch derivatives and all cutoff differences there. On the remaining bands both rough cutoffs equal one. The switch has bounded derivatives, A_{ph} is bounded, and tube widths add only a fixed logarithmic power of $1 + \log \rho$.

For the sharp switching estimate, first freeze the uncut amplitudes across each fixed recovered turn band. In a normal parametrization from one foot, (8.25) bounds the other foot's displacement across this band by $C_{T_*}\epsilon_\rho$. This is a negative power of ρ times a fixed logarithmic power. Both contributing reaches are $O(\log \rho)$. Equations (6.23) and (8.7) give

$$|\nabla_\Sigma d| \le Ce^{C\sqrt{\log\log \rho}}, \qquad |\nabla_\Sigma \log V| \le C(1 + |\nabla_\Sigma d|).$$

Thus the amplitudes vary by a relative $o_R(1)$ on the band, uniformly for $\rho \ge R$. The recovered width was fixed before this limit. If the choice of larger amplitude changes within the band, the frozen amplitudes differ by only a relative $o_R(1)$. The squared switch difference then costs $o_R(1)$ times the core mass, by the capacity and mass comparisons. For the leading pure switch cost, parametrize each band by a point on its middle graph and the transverse coordinate t. The projections of the middle graph onto its two sheets are local diffeomorphisms, with Jacobian $1+o_R(1)$. For injectivity, follow a normal ray from sheet i. Along this ray, $\partial_s t = (1 - \nabla s_j \cdot \nabla s_i)/2 = 1 + o_R(1) > 1/2$, because the two inward normals are almost opposite. Thus the ray meets the middle graph $t = 0$ at most once. The same argument applies from sheet j. Freeze both amplitudes at these two middle feet. Assign the whole transverse segment to the foot i with larger frozen uncut amplitude V_i, breaking ties by the sheet order. This is a measurable partition of a nonnegative integral; no derivative

of the assignment is taken. The change of variables to the assigned sheet has multiplicity one. This is what permits the later counting of at most two side charges at each base. In the following band calculation, V_i, V_j denote the frozen values. The freezing error is included in $o_R(1)$. Put $r = V_j/V_i \le 1$ and separate the outer cutoff by

$$|m_{B,i}V_i - m_{B,j}V_j|^2 \le (1+\epsilon_s)m_{B,i}^2V_i^2(1-r)^2 + C_{\epsilon_s}V_i^2|m_{B,i} - m_{B,j}|^2.$$

The second term is an outer-cutoff error. It is nonzero only near the outer transition, where $\rho \asymp B$. There $|m_{B,i} - m_{B,j}| \le C\ell_{ij}/B = O(\log B/B)$. After multiplication by the squared amplitude and integration, its bound is $C_R(1+\log B)^C B^{-\epsilon_{\mathrm{out}}}$ by (8.32). Thus it tends to zero at fixed R; no uncut mass is used.

For the first term set $u = \ell_{ij} - \ell_i \ge 0$. The radial factors at the two feet differ by $1 + o_R(1)$. The reach is $d(z_i) = G_1\ell_i$, while the receiving reach is at most $G_1(\ell_{ij} + o_R(1))$. Integrating the bound on the logarithmic slope of W_* in its second variable therefore gives

$$r \ge (1 - o_R(1)) \exp[-(q_*G_1/2 + o_R(1))u].$$

Let P_{ij} be this band's pressure and P_i the pressure in a nearest adjacent gap at the assigned foot. Fix a mass-window tolerance ϵ_m and a switch tolerance ϵ_{sw}. The smooth capacity bound (8.24) gives the side coefficient $(1 + \epsilon_{\mathrm{sw}} + o_R(1))\beta P_{ij}/a^2$. Lemma 8.1 gives $\sigma_q I_{p,P}(z_i) \ge (1-\epsilon_m)2\beta P_i/a^2$ for all large R, with the chosen fixed window. Both pressures have the same leading constant Π_p, so their asymptotic formulas give $P_{ij}/P_i = (1 + o_R(1))e^{-\beta(\ell_{ij}-\ell_i)} = (1 + o_R(1))e^{-\beta u}$. If $m_{B,i} = 0$, the first term is zero and needs no division. Otherwise, divided by $\sigma_q I_{p,P}(z_i)m_{B,i}^2V_i^2$, the cost of this side is at most

$$\left(\frac{1}{2} + O(\epsilon_m + \epsilon_{\mathrm{sw}} + \epsilon_s) + o_R(1)\right) e^{-\beta u}(1 - e^{-q_*G_1u/2})^2 + o_R(1).$$

The convergence is uniform even for unbounded u. On bounded u use the amplitude ratio. On $u > U_0$ the normalized cost is at most $Ce^{-\beta U_0}$, since $0 \le r \le 1$; the proposed upper bound has the same tail bound. Increase U_0 before the final increase of R.

In summing the switch costs, each band is assigned to only one of its two feet. Each foot has at most two sides. If both sides are assigned to that foot, a nearest-gap side has $u = 0$ and contributes only $o_R(1)$ times its mass. If fewer sides are assigned there, at most one side can carry a positive leading charge. Thus the positive supremum is paid at most once. The $o_R(1)$ contributions from the at most two sides are absorbed into the mass tolerance. To compute the positive supremum, set $v = e^{-q_*G_1u/2}$. The function becomes $v^{2/y}(1-v)^2$ on $[0,1]$. Its endpoint values vanish. On $(0,1)$ its logarithmic derivative is

$$\frac{2}{yv} - \frac{2}{1-v}.$$

It is positive before $v = (1+y)^{-1}$ and negative afterward. This is the unique maximizer, and therefore

$$\sup_{u\ge 0} e^{-\beta u}(1 - e^{-q_*G_1u/2})^2 = \frac{y^2}{(1+y)^{2+2/y}}.$$

The mass-window tolerance, switch recovery, and Young factors are fixed before increasing R. The geometric and profile errors are then uniform over $\rho \ge R$ and in finite B. To keep the resulting losses separate, let $\delta_0, \dots, \delta_3$ be the four internal losses in the mass, tangential energy, switch coefficient, and residual mass error. The preceding estimates give the following inequality at each finite B. Take the internal losses as fixed nonnegative upper budgets after the finite windows are chosen. They hold once the two large-radius thresholds below are met. Define the interpolation ratio by $\Gamma_{\mathrm{int}} := 1/2 + \delta_2$, where $\delta_2 \ge 0$ is the internal switch loss just specified. This is a coefficient in the

proved switching estimate, distinct from the angular reach penalty.

$$(1-\delta_0)\mathcal{L} \le (1+\delta_1)\int_\Sigma |\nabla_\Sigma \psi|^2 + \big[(1/2+\delta_2)F(y)+\delta_3\big]\mathcal{L} + \mathcal{E}_{R,B}. \tag{8.28}$$

Here division by σ_q preserves the error convention, because σ_q has uniform positive lower and upper bounds. All profile and tube-metric errors multiplying $\mathcal{L}$ or the Dirichlet integral have been included in $\delta_0, \dots, \delta_3$. Only the separately estimated depth, rough-collar, and outer-cutoff terms are included in $\mathcal{E}_{R,B}$. This distinction is needed because we have not yet proved that either uncut principal integral is finite. Choose the fixed tolerances first and then R so that $\delta_0 + \delta_3 \le \varepsilon$ and $\delta_1, \delta_2 \le \varepsilon$. Moving the displayed mass terms to the left proves (8.26) at finite B. In particular, the defined ratio satisfies $\Gamma_{\mathrm{int}} \le 1$ after these choices. The later radial argument may therefore remove the outer cutoff before increasing R; no uncut mass bound was used here. The exact physical identity already includes every normal term, so no additional phase-current term must be supplied. □

8.3. **The radial contradiction.** We apply physical interpolation and the weak reach inequality to the same amplitude and cutoffs. For the test in (8.13), set

$$\mathcal{A} = K(R)L^{1-\vartheta}.$$

An error $\mathcal{E}_{R,B}$ means one satisfying

$$|\mathcal{E}_{R,B}| \le o_R(\mathcal{A}) + o_{B\to\infty;R}(1). \tag{8.29}$$

The first term is uniform for finite B. More explicitly, it is bounded by $\varepsilon_R\mathcal{A}$ with $\varepsilon_R \to 0$ independent of B. The second term may have constants depending on R, but tends to zero as $B \to \infty$ for each fixed R. Absorb mass and Dirichlet errors before either limit. A small coefficient multiplying $\mathcal{L}$ is not, by itself, an error of form (8.29). No step assumes finiteness of the uncut mass $\mathcal{L}$. In the shell estimates below, an unmarked surface integral over $\{\rho \asymp s\}$ means an integral over $\Sigma_g \cap \{s \le \rho < 2s\}$, or a subset of this shell. Fixed enlargements of the shell only change the constant. By the initial threshold choice, $d > 1$ throughout Σ_g, so $W_* \le 1$ there. Integrals with a factor m_c are over the regular part of Σ. The integrand is zero on the removed neighborhood of the other zeros. At finite B, this is a compactly supported test on a smooth surface. The dependence on B will always be shown before it is removed.

The fourth-moment bound controls errors from rough zeros and transferred curvature. On $\rho \asymp s$, choose a separated fixed-scale net in $\mathcal{Z}_R$. Here $\mathcal{Z}_R$ is the rough-zero set used for the radial amplitude. First count the net points within a fixed unit collar of X. Each lies within a fixed distance of a point of $\mathcal{X}$. Ambient packing and the N_s term of (8.1) bound their number by $CK(s)s^{m-4}$, after a fixed enlargement of the shell. Every remaining point is outside that fixed collar of X. Choose a common disk radius smaller than the collar width and the regular-chart radius, so that its disk lies in Σ_g. Relative Harnack preserves $D \ge cs^{-1/50}$ on a smaller disk. Its surface area has a fixed positive lower bound, so its D^2 integral is at least $cs^{-2/50}$. The disks have bounded overlap, so (8.1) counts these centers too. Thus the whole net has at most $CK(s)s^{m-4+2/50}$ points. Enlarge its balls by $Cs^{1/200}$. Let $r_s = s^{1/200}$. A ball of radius Cr_s is covered by Cr_s^n unit balls. Each has bounded ambient volume. Its good surface area is bounded by energy in a fixed enlargement, by the regular-chart argument in (7.1); the pointwise energy bound then gives a uniform constant. This applies to all good sheets in the ball together. A bound for the area of just one graph would not suffice here. Hence both the ambient volume and good surface area of the moving rough collar are at most

$$CK(s)s^{m-4+2/50+n/200}.$$

Every collar-supported term, including logarithmic tube factors, is therefore bounded on the shell by

$$CK(s)s^{-2+2/50+n/200}(1+\log s)^C(L+\log s)^{1-\vartheta}. \tag{8.30}$$

The factor s^{-4} comes from ϕ. In the Dirichlet expansion, first keep amplitude-gradient terms in the principal energy by a fixed Young inequality. The remaining cutoff terms contain only W_* or $\mathcal{F}$, bounded by $C(L+\log s)^{1-\vartheta}$, and fixed powers of the tube logarithms. The cutoff derivatives are bounded; their negative powers of s only help. Since $-2+2/50+7/200=-1.925<-1$, the covering estimate (8.2) makes the dyadic sum of (8.30), divided by $\mathcal{A}$, vanish. It is at most

$$CR^{-1.925}(1+\log R)^C\sum_{j\ge0}2^{-0.925j}(1+j)^C\left(1+\frac{\log R+j}{L}\right)^{1-\vartheta},$$

which tends to zero because $L\ge c\sqrt{R}$.

The curvature sources and their large-curvature receiving terms satisfy

$$CK(s)e^{C\sqrt{\log(K(s)+1)}}s^{-2}(L+\log s)^{3-\vartheta}. \tag{8.31}$$

We verify this directly in the contact calculation below. Call the bound in (8.31) $Q(s)$. Its dyadic sum is negligible: using $L\ge c\sqrt{R}$ gives

$$\frac{\sum_{j\ge0}Q(2^jR)}{\mathcal{A}}\le C\frac{L^2}{R^2}e^{C\sqrt{\log(K(R)+1)}}\sum_{j\ge0}2^{-j}e^{C\sqrt{j+1}}(1+j)^C\longrightarrow0.$$

To check the estimate, write $s=2^jR$. The covering inequality gives $K(s)\le C2^jK(R)$ and hence

$$\sqrt{\log(K(s)+1)}\le C+\sqrt{\log(K(R)+1)}+C\sqrt{j}.$$

Also $\log R/L\to0$, so $(L+\log s)/L\le C(1+j)$ for R large. The remaining series is finite: for every fixed C, one has $C\sqrt{j}+C\log(1+j)\le(\log2)j/2$ for large j. Finally $L^2/R^2=1/K(R)$, and

$$\frac{e^{C\sqrt{\log(K(R)+1)}}}{K(R)}\longrightarrow0.$$

For example, when K is sufficiently large, $C\sqrt{\log(K+1)}\le\frac12\log K$. Thus the argument works even when $K(R)$ diverges very slowly. In a transferred term $b(y)^4d(z)^3$, keep the source reach until its weight has been bounded. Do not replace it by the possibly smaller reach at y.

Summable outer tails allow the outer cutoff to be removed at a fixed inner radius. We can make the numerical margin explicit. Set

$$\epsilon_{\mathrm{tail}}=\frac{1}{48},\qquad\zeta=\frac{q_*(1-\epsilon_{\mathrm{tail}})}{\kappa}=\frac{47}{32},\qquad\epsilon_{\mathrm{out}}=2\zeta-c_{\mathrm{rad}}q_*-1=\frac{1}{16}.$$

In particular, $0<\zeta<2$ and $\epsilon_{\mathrm{out}}>0$. For this fixed ϵ_{tail},

$$b_*(t)\ge(1-\epsilon_{\mathrm{tail}})t-C_{\mathrm{tail},L}.$$

For example, put

$$T_{\mathrm{tail}}=\frac{L}{[1-(1-\epsilon_{\mathrm{tail}})^{1-\vartheta}]^{1/(1-\vartheta)}},\qquad C_{\mathrm{tail},L}=(1-\epsilon_{\mathrm{tail}})T_{\mathrm{tail}}.$$

For $t\ge T_{\mathrm{tail}}$, the formula for $b_*(t)/t$ gives the inequality without the subtractive constant. For $t<T_{\mathrm{tail}}$, its right-hand side is nonpositive. Hence, for fixed R,L and on $\rho\asymp s$,

$$W_*\le C_Rs^{c_{\mathrm{rad}}q_*}E^\zeta,\qquad\zeta=\frac{q_*(1-\epsilon_{\mathrm{tail}})}{\kappa}<2.$$

Hölder, (8.1), and the good-area estimate give

$$\int_{\rho\asymp s}E^\zeta\le\Big(CK(s)s^{m-4}\Big)^{\zeta/2}(CK(s)s^m)^{1-\zeta/2}\le CK(s)s^{m-2\zeta}.$$

Since $c_{\mathrm{rad}} q_* = 15/8$, the stated value of ϵ_{out} is positive. Using $K(s) \le Cs$ gives

$$\int_{\rho \asymp s} W_* \, dS \le C_R s^{m-\epsilon_{\mathrm{out}}}. \tag{8.32}$$

The constants may depend on R because we take $B \to \infty$ first. Indeed, the exponent before using $K(s) \le Cs$ is $m + c_{\mathrm{rad}} q_* - 2\zeta$; the extra factor s changes it to $m - \epsilon_{\mathrm{out}}$. This is the reason for the strict inequality in (8.6).

An outer-cutoff error without mean curvature is bounded by $C_R(1+\log B)^C B^{-\epsilon_{\mathrm{out}}}$. Its radial and cutoff derivatives contribute $B^{-6} = B^{-m}$, and $\mathcal{F}/W_* \le C_R(1+\log B)$ on that shell. Restricting the ambient cutoff to the surface also produces a mean term. Use $|H| \le CD$, $W_* \le C$, and

$$\int_{\rho \asymp B} D^2 \le C B^{m-3}, \qquad \int_{\rho \asymp B} W_*^2 \le C_R B^{m-\epsilon_{\mathrm{out}}}.$$

Then Cauchy–Schwarz and the factor B^{-5} give

$$C_R(1+\log B) B^{-5+(m-3)/2+(m-\epsilon_{\mathrm{out}})/2} = C_R(1+\log B) B^{-(1+\epsilon_{\mathrm{out}})/2}.$$

Both errors vanish for fixed R.

Discard also the reach tail $d > M_0(L+\tau)$, with fixed large M_0. Exponential decay of f_* absorbs the fixed polynomial powers of d, ρ, and tube logarithms below. For uniformity, use $K(d) \le 1 + Cd$ in (6.23). Then $|\nabla d|^2 \le C e^{C\sqrt{\log(d+1)}}$ is bounded by any fixed positive power of d after changing the constant. For completeness, choose M_0 so large that

$$\left(1 - M_0^{-(1-\vartheta)}\right)^{1/(1-\vartheta)} \ge \frac{1}{2}.$$

Then $b_*(d) \ge d/2$ whenever $d \ge M_0 L$. Also require $M_0 \ge 4 \max_{p \in \mathcal{P}} c_{\mathrm{rad}} + 4$. For large R, this puts the discarded tail in the exact exponential range of f_*. On that tail, the exponential part of f_* therefore gives

$$W_* \le C d^{-\vartheta} \rho^{c_{\mathrm{rad}} q_*} e^{-q_* d/2} \le C d^{-\vartheta} e^{-q_* d/4} e^{-q_* M_0 L/4} \rho^{c_{\mathrm{rad}} q_* - q_* M_0/4}.$$

The factor $e^{-q_* d/4}$ absorbs every fixed power of d, including the bound for $|\nabla d|$. Increasing M_0 makes the remaining power of ρ arbitrarily negative, uniformly for $p \in \mathcal{P}$. The resulting shell sums are bounded by $Ce^{-cL} R^{-A}$ for any prescribed fixed A, after changing the constants. They are therefore $o_R(\mathcal{A})$. The same argument applies to the finite number of derivatives of the weight used below. One may first insert a smooth finite reach cutoff, argue at finite B, and remove that cutoff with these bounds. We can therefore use

$$d \le M_0(L+\tau) = o(\rho). \tag{8.33}$$

in every geometric error. The rough-collar and curvature bounds, together with (8.32), cover derivatives of m_H, m_B and transferred cutoff differences. For example, a transferred outer cutoff changes by at most Cd/B. After (8.33), its squared cost is at most $C_R(1+\log B)^2 B^{-\epsilon_{\mathrm{out}}}$.

We distinguish these summable errors from changes in the leading coefficients. Errors in the tube metric that multiply a mass or a Dirichlet integral remain, respectively, $o_R(1)\mathcal{L}$ or $o_R(1)\int |\nabla_\Sigma \psi|^2$. Their small coefficients are uniform in finite B and are retained until the finite-cutoff absorption below. We do not assert that such terms satisfy (8.29) before that absorption.

Differentiating the common weight gives the positive terms needed in the reach comparison. Uniformly on the support as $R \to \infty$,

$$\mathcal{F}_\tau = (1+o_R(1)) c_{\mathrm{rad}} W_* \left(\frac{d}{c_{\mathrm{rad}}\tau \vee b_*(d)}\right)^{\vartheta}, \qquad \frac{\mathcal{F}}{W_*} \ge \frac{1}{q_* + o_R(1)}, \tag{8.34}$$

$$\frac{\mathcal{F} + \mathcal{F}_\tau}{W_*} \le C(L+\tau), \qquad \mathcal{F} \le C(L+\tau)^{1-\vartheta}.$$

Unmarked weights are evaluated at (τ, d). To verify the first identity, use $b_*' = (b_*/t)^\vartheta$ on $t > L$, which gives $t^{-\vartheta}dt = b_*^{-\vartheta}db_*$. Hence

$$\mathcal{F}_\tau = c_{\mathrm{rad}} \int_{b_*(d)}^\infty v^{-\vartheta} f_*'(c_{\mathrm{rad}}\tau - v)\, dv,$$

The lower limit is zero when $d \le L$. Put $v_0 = b_*(d)$ and $A = c_{\mathrm{rad}}\tau$. The nonnegative measure $f_*'(A - v)\, dv$ on $[v_0, \infty)$ has mass $f_*(A - v_0)$, because $f_*(-\infty) = 0$. Let $v_* = A \vee v_0$. If $v_0 \le A + 1$, its support begins at $A - 1$ and its tail beyond $A + 1$ is exponential. Its mass is bounded below by $f_*(-1) > 0$. If $v_0 > A+1$, its normalization is exactly an exponential probability density starting at v_0. Thus its normalized first moment of $|v - v_*|$ is uniformly bounded. These bounds are uniform on $\mathcal{P}$: q_* has a positive lower bound, and $f_*(-1)$ has a positive lower bound there. On $|v - v_*| \le v_*/2$, the mean value theorem bounds the relative change of $v^{-\vartheta}$ by $C|v - v_*|/v_*$. On the complement, the exponential tail gives the same order of error. Also $v \ge A - 1$ wherever the integrand is nonzero, so the possible singularity at zero never enters this estimate. Since $v_* \ge A \to \infty$, we obtain

$$\frac{\int_{v_0}^\infty v^{-\vartheta} f_*'(A - v)\, dv}{v_*^{-\vartheta} f_*(A - v_0)} = 1 + O(\tau^{-1}).$$

Replacing $v^{-\vartheta}$ by its value at v_* therefore costs a relative $o_R(1)$, uniformly in the lower limit. This proves the first part of (8.34). Also,

$$-\partial_t \log W_*(\tau, t) = \frac{\vartheta}{t} + \lambda_* b_*'(t) \le \frac{\vartheta}{t} + q_*.$$

Integrating forward from d gives $\mathcal{F}/W_* \ge (q_* + \vartheta/d)^{-1}$. The minimum supported reach tends to infinity. For the upper bounds, the part below L is at most $L^{1-\vartheta}/(1 - \vartheta)$. Above L, the same change of variable gives

$$\int_L^\infty W_*(\tau, t)\, dt = \int_0^\infty v^{-\vartheta} f_*(c_{\mathrm{rad}}\tau - v)\, dv \le C(1 + \tau)^{1-\vartheta}.$$

This proves the asserted upper bound for $\mathcal{F}$. To bound its ratio to $W_*(\tau, d)$, split at $b_* = c_{\mathrm{rad}}\tau + 2$. Before this point, $d \le C(L + \tau)$ and $f_*(c_{\mathrm{rad}}\tau - b_*(d)) \ge f_*(-2) > 0$. The bound for $\mathcal{F}$ just proved therefore gives $\mathcal{F}/W_* \le C(L + \tau)^{1-\vartheta} d^\vartheta \le C(L + \tau)$. After this point, put $v_0 = b_*(d) > c_{\mathrm{rad}}\tau + 2$. The exponential formula gives

$$\int_{v_0}^\infty v^{-\vartheta} f_*(c_{\mathrm{rad}}\tau - v)\, dv \le \frac{1}{q_*} v_0^{-\vartheta} f_*(c_{\mathrm{rad}}\tau - v_0).$$

Dividing by W_* leaves at most $C(d/v_0)^\vartheta$. The identity $d^{1-\vartheta} = v_0^{1-\vartheta} + L^{1-\vartheta}$ and $v_0 \ge c\tau$ bound this by $C(L + \tau)$. For $\mathcal{F}_\tau/W_*$, the first estimate in (8.34) gives the same bound, using $d \le C(L + \tau)$ before the split and the same identity afterward.

Set

$$J_1 = \int_{b_*(d)}^\infty v^{-\vartheta-1} f_*'(c_{\mathrm{rad}}\tau - v)\, dv.$$

The integrand vanishes for $v < c_{\mathrm{rad}}\tau - 1$, so $J_1 \le C\mathcal{F}_\tau/\tau$. This also gives convergence at a zero lower limit. For the second τ derivative, integrate by parts in v:

$$\int_{v_0}^\infty v^{-\vartheta} f_*''(A - v)\, dv = v_0^{-\vartheta} f_*'(A - v_0) - \vartheta \int_{v_0}^\infty v^{-\vartheta-1} f_*'(A - v)\, dv.$$

When $v_0 = 0$, the boundary term is zero, since $f_*'(A - v)$ vanishes near $v = 0$. When $v_0 > 0$, it is $W_*\lambda_*/b_*'$. This proves, together with direct d derivatives,

$$\mathcal{F}_d = -W_*, \qquad \mathcal{F}_{dd} = W_*\left(\frac{\vartheta}{d} + \lambda_* b_*'\right), \qquad \mathcal{F}_{\tau d} = -c_{\mathrm{rad}} W_*\lambda_*,$$
$$\mathcal{F}_{\tau\tau} = c_{\mathrm{rad}}^2\left(\frac{W_*\lambda_*}{b_*'} - \vartheta J_1\right).$$

At $b_*' = 0$ the quotient is zero by (8.7). To see the square that appears in the chain rule, collect its three terms containing λ_*:

$$c_{\mathrm{rad}}^2\frac{W_*\lambda_*}{b_*'}|\nabla\tau|^2 - 2c_{\mathrm{rad}}W_*\lambda_*\nabla\tau\cdot\nabla d + W_*\lambda_* b_*'|\nabla d|^2$$
$$= \frac{W_*\lambda_*}{b_*'}|c_{\mathrm{rad}}\nabla\tau - b_*'\nabla d|^2.$$

Here and below the gradients in surface calculations are tangential. When $b_*' = 0$, all three terms are zero and the identity uses the stated zero extension. The weak chain rule therefore gives

$$\Delta_\Sigma\mathcal{F} \geq \mathcal{F}_\tau\Delta_\Sigma\tau - c_{\mathrm{rad}}^2\vartheta J_1|\nabla_\Sigma\tau|^2 - W_*(\Delta_\Sigma d)_{\mathrm{upper}}$$
$$+ \frac{\vartheta W_*}{d}|\nabla_\Sigma d|^2 + \frac{W_*\lambda_*}{b_*'}|\nabla_\Sigma(c_{\mathrm{rad}}\tau - b_*(d))|^2. \tag{8.35}$$

Semiconcavity makes the singular part of $\Delta_\Sigma d$ nonpositive. Multiplication by $\mathcal{F}_d = -W_* < 0$ produces a nonnegative measure. We may discard it in (8.35). On compact regular patches, smooth approximation of a semiconcave function proves the chain rule. Equivalently, combine its almost-everywhere Hessian with this signed singular measure. No unsigned boundary current has been omitted.

The contact sources must be separated without losing the favorable gradient reserve. Fix the measurable active-contact selector whose existence is proved in Proposition 6.7, with the plateau priority specified there. Keep this selector fixed. Every receiving foot of a supported source is regular. Moreover, (6.26) gives $D(y) \leq C_\Gamma/d(z)$. The minimum source reach tends to infinity, so these feet all belong to the same fixed good set Σ_g for large R. The fourth-moment estimate may therefore be used after transfer.

Choose fixed $\eta_b > 0$ in the small-curvature regime of (6.24) so that

$$C_\Gamma\eta_b \leq \vartheta/16, \qquad 2\eta_b^2/c_\Gamma^2 \leq \vartheta/16.$$

Decrease it further if needed to control the receiving Jacobian. For large R, all supported reaches satisfy $C_\Gamma d^{-\gamma} \leq \vartheta/16$. At a nonplateau contact with $bd \leq \eta_b$, use the improved drift bound and $|\nabla_\Sigma d| \geq c_\Gamma$. The two adverse terms satisfy

$$(1 + \eta_0 + C_\Gamma d^{-\gamma})b^2 d \leq \frac{2\eta_b^2}{c_\Gamma^2}\frac{|\nabla_\Sigma d|^2}{d} \leq \frac{\vartheta}{16d}|\nabla_\Sigma d|^2,$$
$$\mathfrak{b} \leq \eta_0 + C_\Gamma\eta_b + C_\Gamma d^{-\gamma} \leq \frac{3\vartheta}{16}.$$

Their sum uses at most $\vartheta/4$ of the gradient coefficient in (8.35). At least $3\vartheta/4$ therefore remains. At a plateau contact the drift is at most $\eta_0 < \vartheta/16$, so at least $15\vartheta/16$ remains. Keep its mean sources separate. At the other contacts the whole adverse source is paid as an error, so the full positive gradient coefficient remains there. We may therefore take $c_1 = 1/2$ in the integrated estimate below.

On the other nonplateau contacts, $bd > \eta_b$. Equations (6.23), (8.2), and (8.33) bound their curvature and drift sources on $\rho \asymp s$ by

$$C_{\eta_b}e^{C\sqrt{\log(K(s)+1)}}b^4 d^3.$$

Indeed, $b^2 d \le \eta_b^{-2} b^4 d^3$ and $|\nabla d|^2/d \le \eta_b^{-4} b^4 d^3 |\nabla d|^2$. Moreover,

$$\sup_{t>0} t^3 W_*(\log s, t) \le C(L + \log s)^{3-\vartheta}.$$

For $t \le C(L + \log s)$, this follows from $f_* \le 1$. For larger t, $b_*(t)$ is a fixed positive fraction of t, and the exponential tail of f_* gives the same bound. Multiply the source bound by $\phi m_c^2 W_*$ and integrate. The fourth moment gives (8.31). This pays the full adverse source and leaves the positive gradient reserve.

Treat large-curvature receiving feet similarly. If $b(y)d(z) > \eta_b$, the foot bounds give

$$H_{N_y}(y)_+ \le \frac{C}{d(z)} \le C\eta_b^{-4} b(y)^4 d(z)^3.$$

The relevant area factor is $J_{y\to z}$ from (6.26): a surface element at the receiving foot is multiplied by this factor when it is transported back to the source. It is uniformly bounded, and a receiving foot has at most one positive incoming charge. Thus the area formula bounds the transferred integral by

$$C \int_{\text{receiving feet}} b(y)^4 \sup_{t>0} [t^3 W_*(\tau(y), t)] \rho(y)^{-4}\, dS(y).$$

We used $|y - z| = d(z)/G_1 = o(\rho(z))$ to compare the radial factors; the logarithmic derivative of W_* in τ is bounded. Apply the preceding supremum to get (8.31) at the receiving point. This calculation keeps the source value $t = d(z)$ until the supremum is taken. It does not require comparability of $d(y)$ and $d(z)$.

At a small-curvature foot the area factor is at most $1 + \epsilon_{\text{geo}}$. The fixed loss $\epsilon_{\text{geo}} > 0$ can be made arbitrarily small through the plateau and η_b choices. The radial factors differ by $1 + o_R(1)$, using (8.33). Also $d(y) \le d(z)$. Since W_* decreases in its second variable, the source weight is bounded by the receiving weight, with only $1 + o_R(1)$ in the radial argument. Positive incoming multiplicity is one.

If the receiving point is in the moving rough collar, integrate over that collar itself and use (8.30). This keeps its width unchanged. Outside it, $m_H(y) = 1$, whereas $m_H(z) \le 1$. For the outer cutoff and any fixed $\epsilon_B > 0$, the Lipschitz bound gives

$$m_B(z)^2 \le (1 + \epsilon_B) m_B(y)^2 + C_{\epsilon_B} \frac{d(z)^2}{B^2}.$$

Indeed, $|m_B(z) - m_B(y)| \le C|z - y|/B \le C d(z)/B$, and Young's inequality separates this difference from $m_B(y)$. The squared difference is supported in a fixed enlargement of the outer shell and is paid by (8.32) after reach truncation. Thus both retained charges use the same compact amplitude ψ. They are at most

$$(2G_1 + o_{\text{tol}}(1) + o_R(1)) \int_\Sigma |H|\psi^2 + o_R(1)\mathcal{L} + \mathcal{E}_{R,B}. \tag{8.36}$$

The symbol $o_{\text{tol}}(1)$ records only fixed geometric and Young losses. Make them small before increasing R.

Before integrating the weak reach inequality against the compact radial test, compute the surface Laplacians. Our convention $B = d\nu$ gives $\Delta_\Sigma x = -H\nu$. Since $\rho^2 = R^2 + |x|^2$,

$$\nabla_\Sigma \rho = \frac{x^T}{\rho}, \qquad |\nabla_\Sigma \rho|^2 = 1 - T_r, \qquad \Delta_\Sigma(\rho^2) = 2m - 2Hx \cdot \nu.$$

Consequently

$$\Delta_\Sigma \rho = \frac{m - 1 + T_r - Hx \cdot \nu}{\rho}.$$

For later use, the chain rule gives, for every constant $\alpha > 0$,

$$\frac{\Delta_\Sigma \rho^{-\alpha}}{\rho^{-\alpha}} = \frac{\alpha[\alpha - m + 2 - (\alpha+2)T_r + Hx\cdot\nu]}{\rho^2}. \tag{8.37}$$

The logarithm follows from $\Delta \log\rho = \rho^{-1}\Delta\rho - \rho^{-2}|\nabla\rho|^2$. Taking $m = 6$ and $\alpha = 4$ gives

$$\Delta_\Sigma \tau = \frac{4 + 2T_r - Hx\cdot\nu}{\rho^2}, \qquad \frac{\Delta_\Sigma\phi}{\phi} = \frac{-24T_r + 4Hx\cdot\nu}{\rho^2}. \tag{8.38}$$

Integrate (8.35) against ϕm_c^2 and transfer the surface Laplacian to this compact test. Shell estimates already control derivatives of m_c. Ignoring only cutoff derivatives for this display, the exact radial combination is

$$\mathcal{F}_\tau \Delta_\Sigma \tau - \mathcal{F}\frac{\Delta_\Sigma\phi}{\phi} = \frac{4\mathcal{F}_\tau + (2\mathcal{F}_\tau + 24\mathcal{F})T_r - (\mathcal{F}_\tau + 4\mathcal{F})Hx\cdot\nu}{\rho^2}.$$

Thus the identities (8.38) leave $4\mathcal{F}_\tau + 24\mathcal{F}T_r$ positive. The extra $2\mathcal{F}_\tau T_r$ is also nonnegative. The J_1 term costs $O(1/\tau)\mathcal{F}_\tau/\rho^2$. The other radial mean terms are bounded by

$$C\sup_{\rho\ge R}\frac{L + \log\rho}{\rho}\int_\Sigma I\psi^2 = o_R(1)\mathcal{L}.$$

Here $|H| \le CD \le CI$ on the supported sheets, by fixed-window relative regularity and mass comparison. More precisely, apply (2.20) with a fixed unit radius at each supported zero. This is a local estimate for the entire matched solution; it requires only that the defect at that zero be below its fixed threshold. The bound $D(z) \le R^{-1/50}$ on the support meets that threshold for all large R, uniformly on $\mathcal{P}$, and gives $|H(z)| \le CD(z)$. The common window satisfies $P \ge 1$. The lower comparison in Lemma 3.6 therefore gives $c_I D(z) \le I_P(z)$ after its fixed-window threshold is met. The same support bound meets this threshold as well. Hence $|H| \le Cc_I^{-1} I_P$ there, with a constant independent of R, B, the base point, and the actual $p \in \mathcal{P}$. This uses the previously proved pointwise mean-curvature bound; no estimate for $\Delta_\Sigma I_P$ is needed for this step.

Restrict only the constant W_*/ρ^2 lower contribution to $d \ge \kappa^{-1}\tau$. On the omitted set $E^2 \ge \rho^{-2}$ and $W_* \le C$. Its shell mass is at most

$$Cs^{-6}s^2\int_{\rho\asymp s} E^2 \le CK(s)s^{-2}.$$

Indeed,

$$\sum_{j\ge0} CK(2^jR)(2^jR)^{-2} \le CK(R)R^{-2}\sum_{j\ge0}2^{-j} = o_R(\mathcal{A}).$$

Thus the constant lower contribution can be extended to the omitted set at this error cost. Keep the positive $24\mathcal{F}T_r/\rho^2$ term everywhere. On the retained part, $d/(c_{\rm rad}\tau) \ge 4/5$ and $b_*(d) \le d$. Since $(4/5)^{1/100} > 0.99$, (8.34) gives $\mathcal{F}_\tau \ge (0.99 + o_R(1))c_{\rm rad}W_*$. With $c_1 = 1/2$ as above, we therefore obtain

$$\begin{aligned}\int_\Sigma \phi m_c^2 W_*\left[\frac{(3.96 - o_R(1))c_{\rm rad} + 24(\mathcal{F}/W_*)T_r}{\rho^2} + \frac{c_1\vartheta}{d}|\nabla_\Sigma d|^2 + \frac{\lambda_*}{b_*'}|\nabla_\Sigma(c_{\rm rad}\tau - b_*(d))|^2\right]\\ \le (2G_1 + o_{\rm tol}(1) + o_R(1))\int_\Sigma |H|\psi^2 + o_R(1)\mathcal{L} + \mathcal{E}_{R,B}.\end{aligned} \tag{8.39}$$

Every bound holds at finite B, before taking the outer limit.

Use the weighted mean estimate (3.16) with the single compact amplitude $\psi = m_H m_B h\sqrt{W_*}$. Only now discard the favorable negative b^2 term. Equation (8.14) yields

$$2G_1\int_\Sigma |H|\psi^2 \le \frac{1+2\epsilon}{\kappa}\mathcal{L} + C_{\mathcal{P}}R^{-1/50}\int_\Sigma |\nabla_\Sigma\psi|^2. \tag{8.40}$$

Both cutoff derivatives retain this small factor. We never apply a mean estimate to a contact-set indicator or an amplitude without cutoffs.

We also fix the small losses in the mean estimate and the reach coefficient in the initial joint choice, before fixing the common window and final baseline threshold. Use (3.16) with loss at most $\epsilon/4$. Choose the fixed geometric and Young losses so that their contribution $o_{\mathrm{tol}}(1)$ in (8.39) satisfies $o_{\mathrm{tol}}(1)/(2G_1) \le \epsilon/2$. For $0 < \epsilon \le 1$, their combined leading mass coefficient is therefore at most

$$\frac{1}{\kappa}(1+\epsilon/2)^2 \le \frac{1+2\epsilon}{\kappa}.$$

Thus these fixed losses are already included in the explicit numerator below. The envelopes a_R, b_R will contain only errors that vanish when R and $K(R)$ tend to infinity. Include the window and regularization losses in the same ϵ. The interpolation (8.26) gives, for the same ψ,

$$[1 - \Gamma_{\mathrm{int}} F(y) - \epsilon - o_R(1)]\mathcal{L} \le (1+\epsilon)\int_\Sigma |\nabla_\Sigma \psi|^2 + \mathcal{E}_{R,B}, \qquad \Gamma_{\mathrm{int}} \le 1.$$

With the outer cutoff still present, insert interpolation and (8.40) in (8.39). We define an admissible coefficient explicitly. After the finite windows and tolerances are fixed, let $a_R, b_R \ge 0$ bound the remaining coefficient errors, uniformly in finite B and for all bases with $\rho \ge R$. Choose a_R to include the numerator errors from the reach and mean estimates, and b_R to include the remaining loss in the interpolation mass coefficient. Increase a_R to cover the coefficient of the additional $o_R(1)\mathcal{L}$ term. The local preparation errors tend uniformly to zero as R grows. The remaining radial scale error is controlled by $K(R)^{-1/2} + \log R/R$ in (8.3). Thus, for every prescribed accuracy, these envelopes are small whenever $R \ge R_0$ and $K(R) \ge K_0$, with common thresholds for the fixed compact exponent data and finite windows. For our fixed solution both conditions hold eventually. This does not require a common divergence rate for all functions K.

Take the tolerances and these two thresholds so that $1 - \Gamma_{\mathrm{int}} F(y) - \epsilon - b_R > 0$, and define

$$C_{\mathrm{pre}} := \frac{(1+2\epsilon)(1+\epsilon) + a_R}{\kappa[1 - \Gamma_{\mathrm{int}} F(y) - \epsilon - b_R]} + C_{\mathcal{P}} R^{-1/50}. \tag{8.41}$$

Then the full right side of (8.39) is at most

$$C_{\mathrm{pre}} \int_\Sigma |\nabla_\Sigma \psi|^2 + \mathcal{E}_{R,B}.$$

The $o_R(1)\mathcal{L}$ term has been included in the numerator. It is not discarded without a bound for its mass. The denominator has a fixed positive lower bound after these choices. Interpolation therefore controls each such mass term at finite B. Every local error is bounded by its supremum on $\rho \ge R$. Inner-radius shell errors are bounded by their full infinite dyadic sums, independently of B. Outer-cutoff errors tend to zero as $B \to \infty$ at fixed R. Thus no uncut mass bound is used in this substitution. For each fixed finite ϵ, the coefficient has the bound

$$\limsup_{R\to\infty} \kappa C_{\mathrm{pre}} \le \frac{(1+2\epsilon)(1+\epsilon)}{1 - \Gamma_{\mathrm{int}} F(y) - \epsilon}. \tag{8.42}$$

The limit is along the fixed solution, for which $K(R) \to \infty$. Since $\Gamma_{\mathrm{int}} \le 1$, the right side is at most

$$\frac{(1+2\epsilon)(1+\epsilon)}{1 - F(y) - \epsilon} \longrightarrow \frac{1}{1 - F(y)} \qquad (\epsilon \downarrow 0).$$

For $y = 3/2$ this last number is less than $9/8$. Choose the finite tolerances first, and then require $R \ge R_0$ and $K(R) \ge K_0$ so that

$$\kappa C_{\mathrm{pre}} \le \frac{23}{20}. \tag{8.43}$$

This bound is uniform in finite B. Its constants and the thresholds for the pair $(R, K(R))$ are uniform on $\mathcal{P}$. The final radius for a particular solution may depend on that solution.

Keep the outer cutoff finite during the subtraction. Remove the spatial cross term with the exact identity

$$\int_\Sigma |\nabla_\Sigma(hm_c\sqrt{W_*})|^2 = \int_\Sigma \phi|\nabla_\Sigma(m_c\sqrt{W_*})|^2 - \int_\Sigma h\Delta_\Sigma h\, m_c^2 W_*. \tag{8.44}$$

For clarity, put $g = m_c\sqrt{W_*}$. The cross term and the $|\nabla h|^2$ term combine exactly as

$$|\nabla(hg)|^2 = h^2|\nabla g|^2 + \nabla h\cdot\nabla(hg^2),$$
$$\int \nabla h\cdot\nabla(hg^2) = -\int h\Delta_\Sigma h\, g^2.$$

The support is compact and avoids the removed zero set, so integration by parts has no boundary contribution. Approximation gives the same identity for the locally Lipschitz amplitude used here. The radial identity and logarithmic derivative are

$$-\frac{\Delta_\Sigma h}{h} = \frac{4+8T_r-2Hx\cdot\nu}{\rho^2}, \qquad \nabla_\Sigma \log W_* = \lambda_*\nabla_\Sigma(c_{\rm rad}\tau - b_*(d)) - \frac{\vartheta}{d}\nabla_\Sigma d. \tag{8.45}$$

Let $0<\delta_c\le 1/10$ be fixed. The product rule and Young's inequality give

$$\phi|\nabla(m_c\sqrt{W_*})|^2 \le \frac{1+\delta_c}{4}\phi m_c^2 W_*|\nabla\log W_*|^2 + C_{\delta_c}\phi W_*|\nabla m_c|^2.$$

The second term is one of the rough-collar or outer-cutoff errors already estimated. Write $J_{R,B} = \int_\Sigma|\nabla_\Sigma\psi|^2$. The radial mean term is bounded by $CR^{-1}\mathcal{L}$, since $|x\cdot\nu|\le\rho$ and $|H|\le CI$ on the support. This is still a multiple of the finite cutoff mass, rather than an error of (8.29). Thus the Dirichlet identity has the form

$$J_{R,B} \le \mathcal{D}_{R,B} + \alpha_R\mathcal{L} + \mathcal{E}_{R,B}, \qquad \alpha_R\longrightarrow 0,$$

where, explicitly,

$$\mathcal{D}_{R,B} = \int_\Sigma \phi m_c^2 W_*\left[\frac{4+8T_r}{\rho^2} + \frac{1+\delta_c}{4}|\nabla_\Sigma\log W_*|^2\right].$$

All terms in this definition are nonnegative. Require $b_R\le\epsilon$ and $1-F(y)-2\epsilon>0$. Since $\Gamma_{\rm int}\le 1$, interpolation gives $\mathcal{L}\le C_{\rm mass}J_{R,B}+\mathcal{E}_{R,B}$ with

$$C_{\rm mass} := \frac{1+\epsilon}{1-F(y)-2\epsilon}.$$

This is a fixed finite coefficient. Substitute this bound before removing any cutoff:

$$(1-C_{\rm mass}\alpha_R)J_{R,B} \le \mathcal{D}_{R,B}+\mathcal{E}_{R,B}.$$

Take $\alpha_R\ge 0$ as the upper envelope of the mean mass error. For large R, require $C_{\rm mass}\alpha_R\le 1/70$. The denominator is positive. Define the final coefficient by

$$C_{\rm eff} := \frac{C_{\rm pre}}{1-C_{\rm mass}\alpha_R}.$$

Equation (8.43) now gives

$$\kappa C_{\rm eff} \le \frac{23}{20}\frac{70}{69} = \frac{7}{6}, \qquad C_{\rm eff}q_* \le \frac{7}{4}.$$

These are the actual bounds in (8.5). The discarded mean term was controlled by the finite-cutoff mass before division. No assertion about the uncut mass is needed.

Compare the coefficients before subtracting. First,

$$0.99c_{\rm rad} = \frac{99}{80\kappa} > \frac{7}{6\kappa} \ge C_{\rm eff}.$$

Thus the constant radial coefficient in (8.39) exceeds $4C_{\rm eff}W_*/\rho^2$ even with the fixed losses. Next, (8.34) gives $W_*\le(q_*+o_R(1))\mathcal{F}$. The remaining coefficient of $\mathcal{F}T_r/\rho^2$ is at least

$$24-8C_{\rm eff}q_*-o_R(1) \ge 10-o_R(1).$$

Finally put $Y = \nabla_\Sigma(c_{\mathrm{rad}}\tau - b_*(d))$. For $0 < \delta_Y \le 1/10$,

$$\frac{1}{4}\left|\lambda_* Y - \frac{\vartheta}{d}\nabla_\Sigma d\right|^2 \le \frac{1+\delta_Y}{4}\lambda_*^2|Y|^2 + C_{\delta_Y}\frac{\vartheta^2}{d^2}|\nabla_\Sigma d|^2.$$

As $0 \le \lambda_* \le q_*$ and $0 \le b_*' \le 1$, multiplying the first term by C_{eff} allows absorption into λ_*/b_*':

$$\frac{C_{\mathrm{eff}}(1+\delta_c)(1+\delta_Y)\lambda_*^2/4}{\lambda_*/b_*'} \le (1+\delta_c)(1+\delta_Y)\frac{C_{\mathrm{eff}}q_*}{4} \le \frac{121}{100}\frac{7}{16} < \frac{3}{5} < 1.$$

Interpret this ratio as zero for $\lambda_* = 0$. The remaining $C\vartheta^2 d^{-2}|\nabla_\Sigma d|^2$ is absorbed by $c_1\vartheta d^{-1}|\nabla_\Sigma d|^2$, since the minimum supported reach diverges. These strict margins allow all fixed Young factors to be chosen before R grows.

Here is the subtraction as an inequality between finite quantities. Let $\mathcal{P}_{R,B}$ be the full left side of (8.39). The mean estimate and interpolation above give $\mathcal{P}_{R,B} \le C_{\mathrm{pre}}J_{R,B} + \mathcal{E}_{R,B}$. The finite-cutoff absorption above therefore gives

$$\mathcal{P}_{R,B} \le C_{\mathrm{eff}}\mathcal{D}_{R,B} + \mathcal{E}_{R,B}.$$

The coefficient checks give a pointwise lower bound for the difference of the two nonnegative integrands. For example, the constant radial reserve before the vanishing errors is

$$3.96c_{\mathrm{rad}} - 4C_{\mathrm{eff}} \ge \frac{1}{\kappa}\left(\frac{99}{20} - \frac{14}{3}\right) = \frac{17}{60\kappa}.$$

For R large, we may retain half of it, half of the $10 - o_R(1)$ reserve in the T_r term, and fixed fractions of the two gradient reserves. Consequently

$$\begin{aligned}\frac{17}{120\kappa}\int_\Sigma \phi m_c^2\frac{W_*}{\rho^2} + 5\int_\Sigma \phi m_c^2\mathcal{F}\frac{T_r}{\rho^2} + \frac{c_1\vartheta}{2}\int_\Sigma \phi m_c^2\frac{W_*}{d}|\nabla_\Sigma d|^2 \\ + \frac{2}{5}\int_\Sigma \phi m_c^2 W_*\frac{\lambda_*}{b_*'}|Y|^2 \le \mathcal{E}_{R,B}.\end{aligned} \tag{8.46}$$

The coefficients are fixed independently of B. Dropping all but the second term and dividing by its fixed coefficient proves

$$\int_\Sigma \phi m_c^2\mathcal{F}\frac{T_r}{\rho^2} \le o_R(\mathcal{A}) + o_{B\to\infty;R}(1). \tag{8.47}$$

The subtraction used only finite compact integrals. Its coefficients and shell-error bounds are uniform in B.

At the selected center, the energy is at least $cK(R)R^m$ in B_R. We transfer a fixed part to good zeros to bound the same integral below. Choose a maximal separated fixed-scale net in the whole zero set. The gradient bound gives $|\nabla u| \le \max_{p\in\mathcal{P}} a_p$. Choose $r_0 > 0$ such that $r_0 \max_{p\in\mathcal{P}} a_p \le 1/2$. Then $|u| \le 1/2$ in $B_{r_0}(z)$ whenever $u(z) = 0$. Consequently

$$E_p(u; B_{r_0}(z)) \ge |B_{r_0}| \min_{p\in\mathcal{P},\ |t|\le 1/2} W_p(t) =: e_0 > 0.$$

Choose the separation of the zero net to be a fixed fraction of r_0. These balls have bounded overlap. Summing their energy lower bounds and using bounded overlap bounds their number by the energy on a fixed enlargement of the ball. Equation (8.2) therefore bounds their number in B_{R+j+2} by

$$CK(R)R^m(1 + j/R)^{m+1}.$$

Clearing bounds the energy in B_R at distance at least P_0 from all zeros by

$$CK(R)R^m\sum_{j\ge P_0}(1+j/R)^{m+1}(j+2)^n e^{-cj}.$$

For this estimate, assign each point measurably to a nearest net point and an integer distance shell. The shell of index j about one net point has volume at most $C(j+2)^n$. The net covers Σ at a fixed distance, so clearing still gives Ce^{-cj} on that shell, after changing the constants. Choose a fixed

P_0 so that the displayed tail is a small fraction of the initial energy, uniformly for $R \geq 1$. This is possible since $(1 + j/R)^{m+1} \leq (1 + j)^{m+1}$ and the series has an exponential tail.

Remove also the fixed $2P_0$ enlargement of the moving rough collar. Its volume is $o(K(R)R^m)$ by the same rough-set count. The bounded energy density makes its energy negligible. Every remaining point has a nearest zero $z \in B_{2R}$ within distance P_0, and this zero satisfies $m_H(z) = 1$. Its reach tends uniformly to infinity. Lemma 6.3 therefore makes these depth-P_0 normal tubes jointly injective for large R. Nearest-zero projection is normal: minimizing the squared distance along a regular sheet gives orthogonality to its tangent space. The elementary reach bound gives $|B(z)| \leq C/d(z)$. For the supported bases, $d(z) \to \infty$ uniformly. Therefore, for $|t| \leq P_0$ and all large R,

$$2^{-m} \leq \det(I + tB(z)) \leq 2^m.$$

If $\mathcal{B}_R$ is the set of these bases, injectivity and the area formula give

$$\begin{aligned} E_p(u; \text{remaining region}) &\leq \int_{\mathcal{B}_R} \int_{-P_0}^{P_0} e_p(u(z + t\nu(z))) \det(I + tB(z))\, dt\, dS(z) \\ &\leq CP_0\, \mathcal{H}^m(\mathcal{B}_R). \end{aligned}$$

Here P_0 is fixed, and the pointwise energy bound supplies C. Thus a fixed fraction of the original energy forces the stated lower area bound. Hence

$$\mathcal{H}^m(\Sigma \cap B_{2R} \cap \{m_H = 1\}) \geq cK(R)R^m. \tag{8.48}$$

To bound the area with large reach, apply Remark 6.2 with reach threshold $L/2$. Its jointly injective two-sided tubes and uniform positive Jacobian give

$$\mathcal{H}^m(\Sigma \cap B_{2R} \cap \{d > L/2\}) \leq \frac{CR^{m+1}}{L} = o(K(R)R^m).$$

The ratio to $K(R)R^m$ is $R/(LK(R)) = K(R)^{-1/2}$. Thus (8.48) shows that

$$S_R = \Sigma \cap B_{2R} \cap \{m_H = 1,\ d \leq L/2\}$$

has area at least $cK(R)R^m$. For $B > 3R$ this set satisfies $m_c = 1$, $\rho \leq \sqrt{5}R$, and $T_r \geq 1/5$. Moreover, $b_*(t) = 0$ on $[L/2, L]$ and $f_*(c_{\mathrm{rad}}\tau) = 1$ for large R. Hence

$$\mathcal{F}(\tau, d) \geq \int_{L/2}^{L} t^{-\vartheta}\, dt \geq cL^{1-\vartheta}, \qquad \frac{\phi T_r}{\rho^2} \geq cR^{-6} = cR^{-m}.$$

The left side of (8.47) is therefore at least $cK(R)L^{1-\vartheta} = c\mathcal{A}$, uniformly for finite $B > 3R$. First take $B \to \infty$ at fixed R. Then let $R \to \infty$. We reach the impossible conclusion $c\mathcal{A} \leq o_R(\mathcal{A})$.

We have proved the following without restricting the growth rate of the density.

Proposition 8.3. *A stable entire matched solution in $\mathbb{R}^7$ satisfying the established preparation, reach, and fourth-moment estimates cannot have unbounded energy density. The conclusion holds for every fixed $p > 2$, uniformly on compact exponent intervals.*

8.4. **Proof of the main theorem.**

Proof of Theorem 1.1. Fix $p > 2$ and a compact interval $\mathcal{P} \Subset (2, \infty)$ containing it. Let $v : \mathbb{R}^7 \to [-1, 1]$ be scalar-stable. The only constant solutions are $-1, 0, 1$. Remark 2.2 applies to v: any contact with 1 or -1 makes it the corresponding constant well.

For the zero constant, $A_p(0) = 0$ and $W_p''(0) = -1$, so

$$Q_0(\phi) = -\int \phi^2 < 0 \qquad (0 \neq \phi \in C_c^1).$$

It is unstable. Any remaining solution therefore has strict range $(-1, 1)$. Lemma 2.3 gives a nonempty zero set: if there were no zero, clearing could be applied on arbitrarily large balls about each fixed point and would force a well.

If $\sup_R M_R(v; 0) < \infty$, the finite-density classification in Theorem 5.25 gives a planar transition. Otherwise $M_R(v; 0) \to \infty$, and hence $K(R)$ is unbounded. The monotonicity formula (5.28), applied at every center, shows that $K(R)$ is nondecreasing, so in this case $K(R) \to \infty$. The reach and fourth-moment results from Sections 6 and 7 apply on one fixed good set and bad net. Proposition 8.3 then gives a contradiction. This proves the stable classification in $\mathbb{R}^7$.

Let $1 \le k < 7$, and let v be a stable solution on $\mathbb{R}^k$. Write $x \in \mathbb{R}^k$, $y \in \mathbb{R}^{7-k}$, and define $\widetilde{v}(x, y) = v(x)$. Fubini's theorem in the weak equation shows that $\widetilde{v}$ solves the same equation in $\mathbb{R}^7$. For $\xi \in \mathbb{R}^k$, the diffusion matrix has the block form

$$A_p^{(7)}((\xi, 0)) = \begin{pmatrix} A_p^{(k)}(\xi) & 0 \\ 0 & |\xi|^{p-2} I_{7-k} \end{pmatrix}.$$

The formula remains valid at $\xi = 0$, since $p > 2$. For every $\Phi \in C_c^1(\mathbb{R}^7)$, it follows that

$$Q_{\widetilde{v}}(\Phi) = \int_{\mathbb{R}^{7-k}} Q_v(\Phi(\cdot, y))\, dy + \int_{\mathbb{R}^7} |\nabla v(x)|^{p-2} |\nabla_y \Phi|^2\, dx\, dy \ge 0.$$

Thus the lift is stable. Apply the seven-dimensional classification. If it is a layer $g_p(\omega \cdot (x, y) - t_0)$, then

$$0 = \partial_{y_j} \widetilde{v} = \omega_{k+j} g_p'(\omega \cdot (x, y) - t_0).$$

Since $g_p' > 0$, every passive component of ω vanishes. The resulting plane and profile lie in the original k-dimensional space.

For strict-range monotone solutions, the local monotonicity-to-stability calculation is valid for every $p > 1$. On each compact test support, strict monotonicity keeps the gradient nonzero and the values away from the wells. The endpoint-profile theorem has the same exponent range. Here we use them with the established stable classification and endpoint-limit argument in the range $p > 2$. We now consider $u : \mathbb{R}^N \to (-1, 1)$ with $u_{x_N} > 0$. It suffices to work in dimension eight. When $N < 8$, add passive coordinates and place the original increasing direction last. The extension still solves the equation and is strictly increasing in x_8. The same derivative argument as above will remove the passive components of its final planar normal.

Write $x = (x', x_8) \in \mathbb{R}^7 \times \mathbb{R}$, and set $w = u_{x_8} > 0$. On every compact set, $|\nabla u| \ge w$ has a positive minimum. The local noncritical bootstrap from Section 2.1 therefore makes u smooth there. Differentiating gives

$$-\operatorname{div}(A_p(\nabla u)\nabla w) + W_p''(u) w = 0.$$

For a compactly supported smooth test ϕ, the positive minimum of w on its support makes ϕ^2/w an admissible test. Its gradient is $2\phi\nabla\phi/w - \phi^2\nabla w/w^2$. Testing the differentiated equation and completing the square gives

$$Q_u(\phi) = \int_{\mathbb{R}^8} A_p(\nabla u) \left(\nabla\phi - \frac{\phi}{w}\nabla w\right) \cdot \left(\nabla\phi - \frac{\phi}{w}\nabla w\right) dx \ge 0. \tag{8.49}$$

To extend the inequality to every $\phi \in C_c^1$, approximate ϕ in C^1 by smooth tests whose supports lie in one fixed compact neighborhood. Both $A_p(\nabla u)$ and $W_p''(u)$ are bounded on that neighborhood. Hence the two terms of Q_u converge, and $Q_u(\phi) \ge 0$ follows. Thus u is scalar-stable. This is the positive-Jacobi-field calculation in [16, p. 785]. Its use here needs only local positivity of w on the support of each test.

Monotonicity and boundedness give pointwise limits

$$v_\pm(x') = \lim_{t \to \pm\infty} u(x', t) \in [-1, 1].$$

Consider a sequence $t_j \to +\infty$ and the translates $u_j(x', s) = u(x', s + t_j)$. The translation-uniform local compactness from Section 2.1 gives a subsequence converging in $C^1_{\mathrm{loc}}(\mathbb{R}^8)$. At each fixed (x', s), monotonicity gives $u_j(x', s) \to v_+(x')$. Hence the limit is independent of s. The continuous flux $\xi \mapsto |\xi|^{p-2}\xi$ and the reaction converge locally uniformly. Passing to the weak equation shows that $(x', s) \mapsto v_+(x')$ solves the equation in $\mathbb{R}^8$. Testing with $\eta(x')\zeta(s)$, where $\int \zeta = 1$, gives the matched equation for v_+ in $\mathbb{R}^7$. The same proof applies at $-\infty$.

The translates u_j are stable. Since $p > 2$, the maps $A_p(\xi)$ and $W_p''(t)$ are continuous also at $\xi = 0$ and $t = \pm 1$. Thus on the support of each fixed test, both form coefficients converge uniformly. They are uniformly bounded there, so integration passes the quadratic-form inequality to the translated limit in $\mathbb{R}^8$. Let v denote either endpoint, choose $\eta \in C^1_c(\mathbb{R}^7)$ and a nonzero $\zeta \in C^1_c(\mathbb{R})$, and put $\zeta_T(s) = \zeta(s/T)$. The block form of $A_p((\nabla v, 0))$ gives

$$\begin{aligned} 0 &\le Q_{v(x')}(\eta\zeta_T) \\ &= \left(\int_{\mathbb{R}} \zeta_T^2\right) Q_v(\eta) + \left(\int_{\mathbb{R}} |\zeta_T'|^2\right) \int_{\mathbb{R}^7} |\nabla v|^{p-2}\eta^2. \end{aligned}$$

Now

$$\int \zeta_T^2 = T \int \zeta^2, \qquad \int |\zeta_T'|^2 = T^{-1} \int |\zeta'|^2.$$

Divide by the first factor and let $T \to \infty$. The extra term is

$$\frac{1}{T^2} \frac{\int |\zeta'|^2}{\int \zeta^2} \int_{\mathbb{R}^7} |\nabla v|^{p-2}\eta^2,$$

which tends to zero. Therefore $Q_v(\eta) \ge 0$. Both v_- and v_+ are stable in $\mathbb{R}^7$.

The stable classification makes each endpoint a constant well or a planar layer. Each therefore depends on at most two orthogonal linear coordinates: for a planar layer take its normal and any perpendicular unit direction; for a constant take any two directions. The two endpoints need not initially use the same pair.

Use the extension

$$\widehat{W}_p(t) = \frac{|1 - t^2|^p}{2p}, \qquad t \in \mathbb{R}.$$

It is C^2 for $p > 2$. In particular it lies in $C^{1,\alpha}_{\mathrm{loc}}(\mathbb{R}) \cap C^{1,1}_{\mathrm{loc}}((-1, 1))$ for $0 < \alpha < 1$. It is positive away from ± 1 and vanishes at those two points. The formula

$$\widehat{W}_p'(t) = -t \operatorname{sgn}(1 - t^2)|1 - t^2|^{p-1}$$

shows that its only critical points are $-1, 0, 1$. For $0 \le \theta \le 1$,

$$\widehat{W}_p(-1 + \theta) = \widehat{W}_p(1 - \theta) = \frac{\theta^p(2 - \theta)^p}{2p},$$

so explicitly

$$\frac{1}{2p}\theta^p \le \widehat{W}_p(\pm(1 - \theta)) \le \frac{2^{p-1}}{p}\theta^p.$$

For $0 < \theta \le 1/2$,

$$\widehat{W}_p'(-1 + \theta) = -\widehat{W}_p'(1 - \theta) = (1 - \theta)(2 - \theta)^{p-1}\theta^{p-1},$$

$$\frac{1}{2}\left(\frac{3}{2}\right)^{p-1} \theta^{p-1} \le \widehat{W}_p'(-1 + \theta) \le 2^{p-1}\theta^{p-1}.$$

Finally, on the interior of the wells,

$$W_p''(t) = (1 - t^2)^{p-2}((2p - 1)t^2 - 1) > 0 \quad \text{when} \quad |t| > (2p - 1)^{-1/2}.$$

Choose

$$0 < \theta_* < \min\left\{\frac{1}{2}, \min_{p \in \mathcal{P}} \left[1 - (2p - 1)^{-1/2}\right]\right\}.$$

Then W_p' is increasing on $(-1, -1+\theta_*) \cup (1-\theta_*, 1)$. The displayed well bounds have common positive lower constants and finite upper constants on $\mathcal{P}$. They can be enlarged or reduced to meet the convention $0 < c < 1 < C$ in that theorem. All the double-well hypotheses of [18, Theorem 1.1] hold. Farina–Valdinoci's theorem gives $v_- = -1$, $v_+ = 1$, and one-dimensionality in dimension eight.

Write $u(x) = g(\omega \cdot x)$ with $|\omega| = 1$. Since $\partial_{x_8} u > 0$, the component ω_8 is nonzero. Reverse the normal if necessary, so that $\omega_8 > 0$ and $g' > 0$. The endpoint limits imply $g(\pm\infty) = \pm 1$. The one-dimensional equation is

$$(|g'|^{p-2} g')' = W_p'(g).$$

Multiplying by g' gives

$$\frac{d}{dt}\left(\frac{p-1}{p}|g'|^p - W_p(g)\right) = 0.$$

The constant is zero. Indeed, $g' > 0$ and $\int_{\mathbb{R}} g' = 2$, so there are sequences tending to either infinity along which $g' \to 0$. The potential tends to zero there. We conclude that

$$\frac{p-1}{p}|g'|^p = W_p(g).$$

Thus equality holds in (2.2) at every point of the nonconstant strict-range solution. The equality case of Proposition 2.1 identifies the profile, as in (2.3). With the increasing orientation already chosen, it is

$$g(t) = \tanh(a_p(t - t_0)).$$

When passive coordinates were added, their normal components vanish by the same derivative calculation. In the original dimension the increasing direction therefore has $\omega_N > 0$. This proves the stated formula. □

All parameters were chosen on a fixed compact subset of $(2, \infty)$. The theorem applies to every finite $p > 2$. It does not require uniform windows, thresholds, or constants as $p \to \infty$.

Status and AI-use disclosure. This is early work. We used ChatGPT 6 Astra for mathematical exploration, calculations, proof strategies, and candidate proofs. Codex helped revise the manuscript, check references, and examine selected arguments and exponent ranges. Our related preprints helped guide this study. The classical geometric strategy adapted here comes from OpenAI's manuscript [35]. The earlier methods used with it are credited at their points of use. We asked ChatGPT 6 Astra to examine the classical manuscript and our related preprints and to develop the weighted adaptations for the matched equation. It helped propose constructions and draft their proofs. The authors checked the overall logic and the examples in the text. The results have not been peer reviewed. We post this work to share the claims and method early and to invite feedback. Each author is responsible for the full content, including the mathematical statements, proofs, references, and any errors.

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